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(54) **BCMA SPECIFIC VCAR COMPOSITIONS
AND METHODS FOR USE**

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None

See application file for complete search history.

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(57) **ABSTRACT**

Disclosed are VHH chimeric antigen receptors (VCARs),
VCAR transposons encoding VCARs of the disclosure, cells
modified to express VCARs of the disclosure, as well as
methods of making and methods of using the same for
adoptive cell therapy.

79 Claims, 30 Drawing Sheets

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FIGURE 1

Weight Change

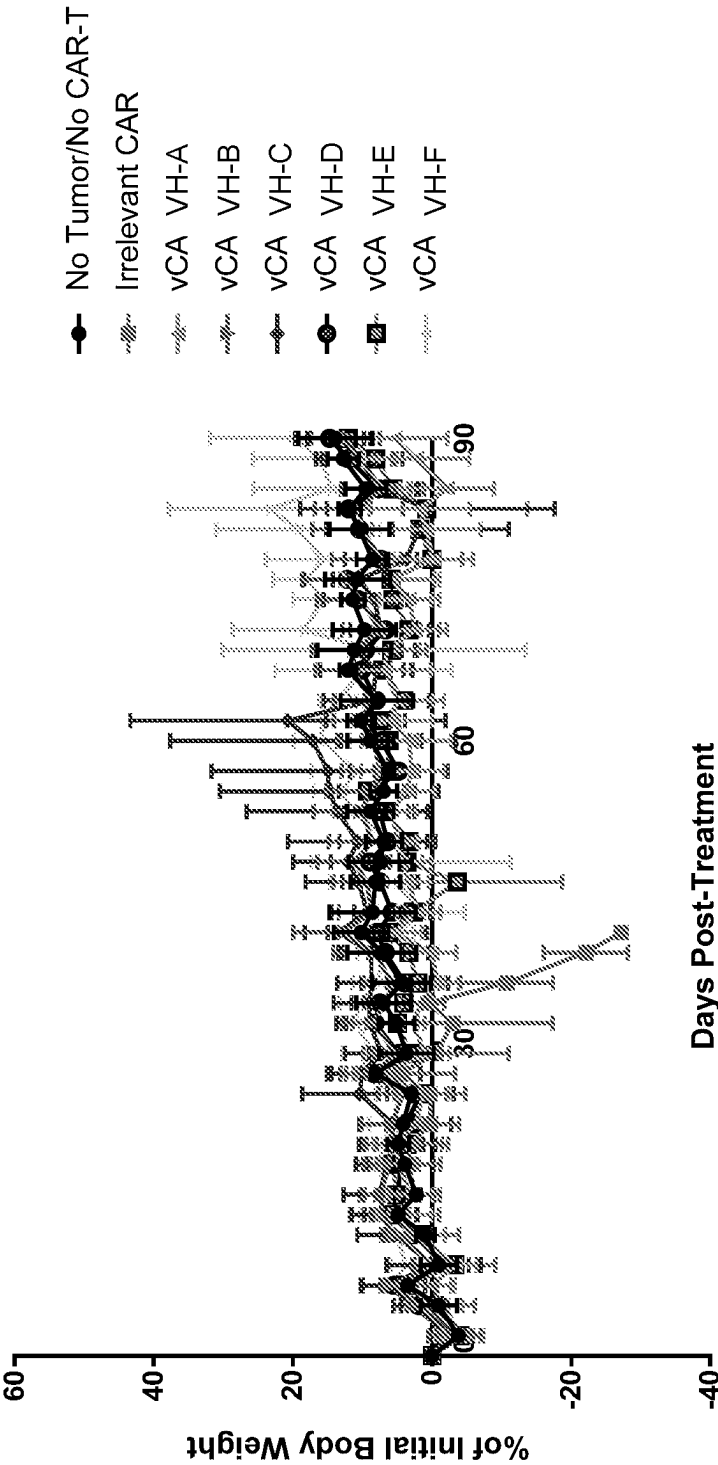


FIGURE 2

Tumor Burden

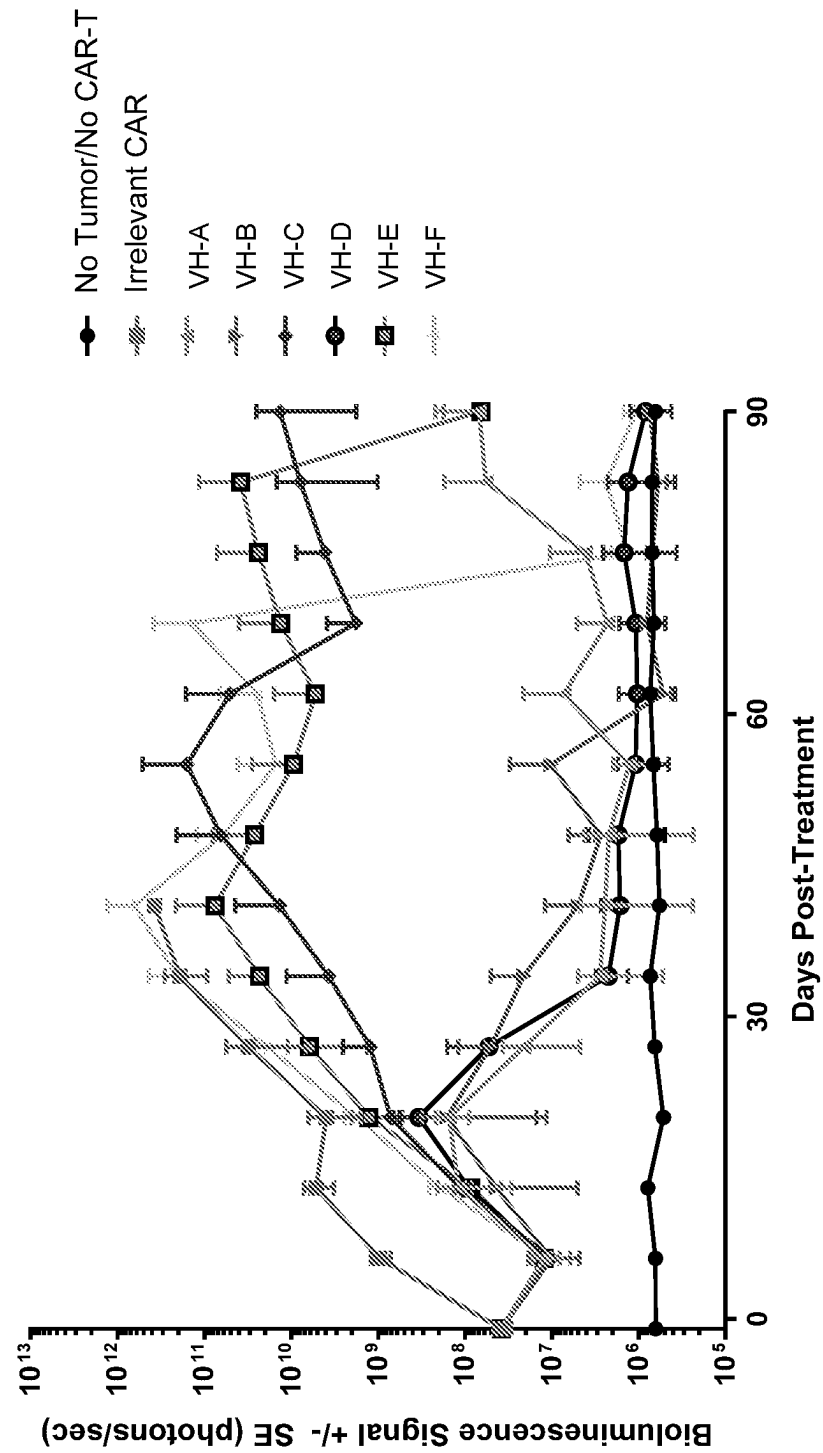


FIGURE 3

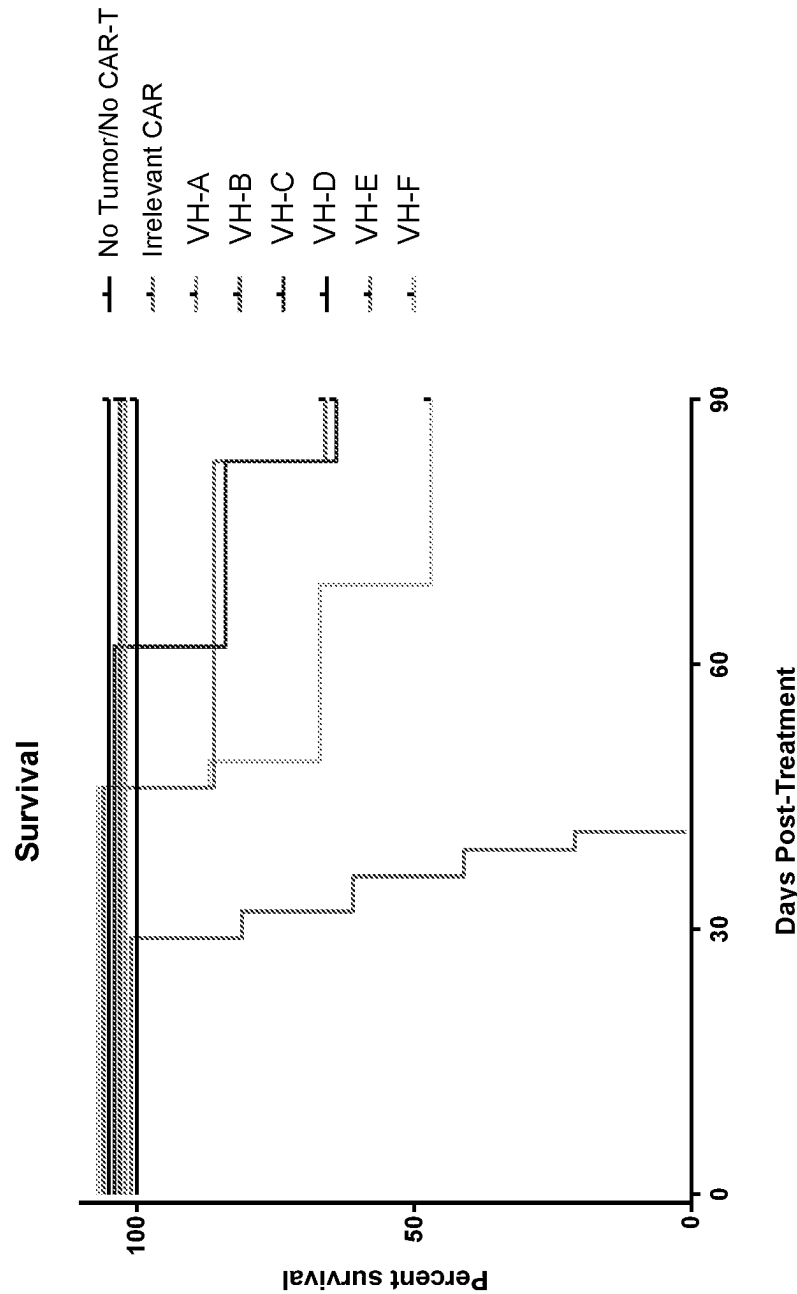


FIGURE 4

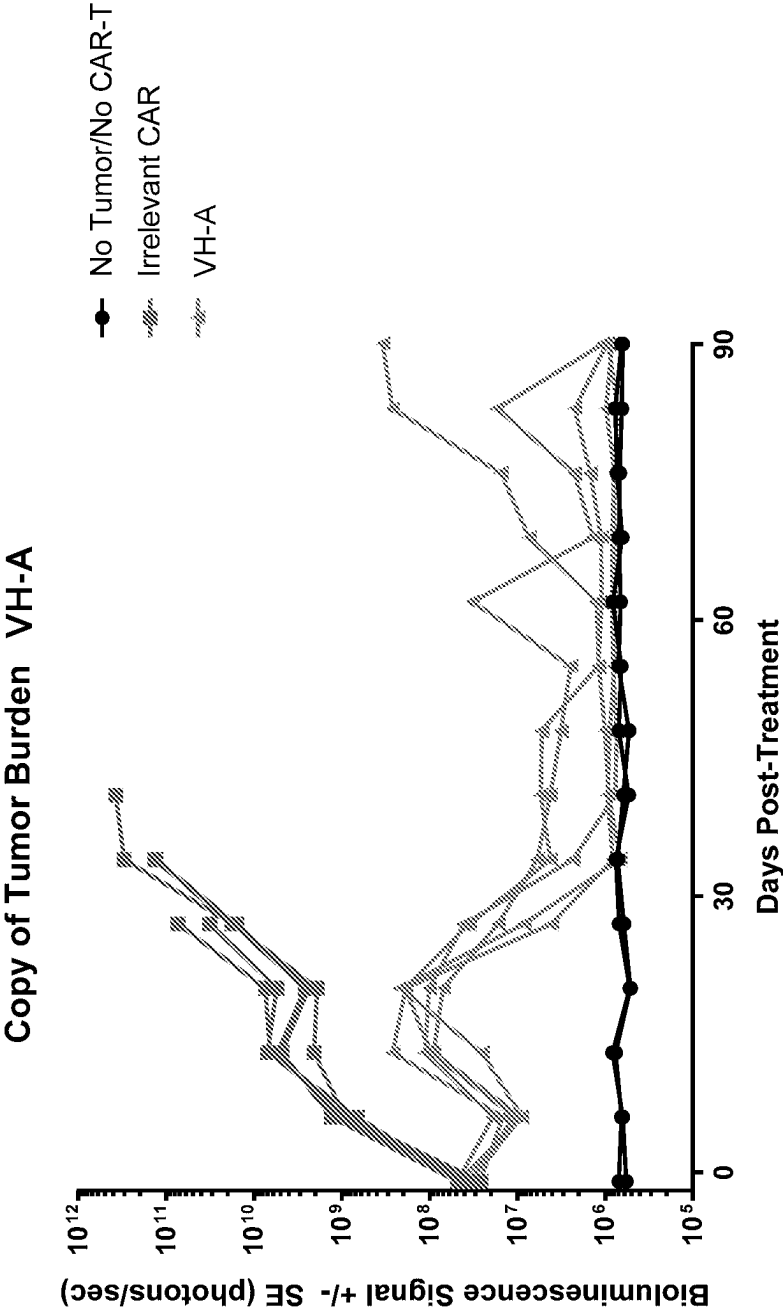


FIGURE 5

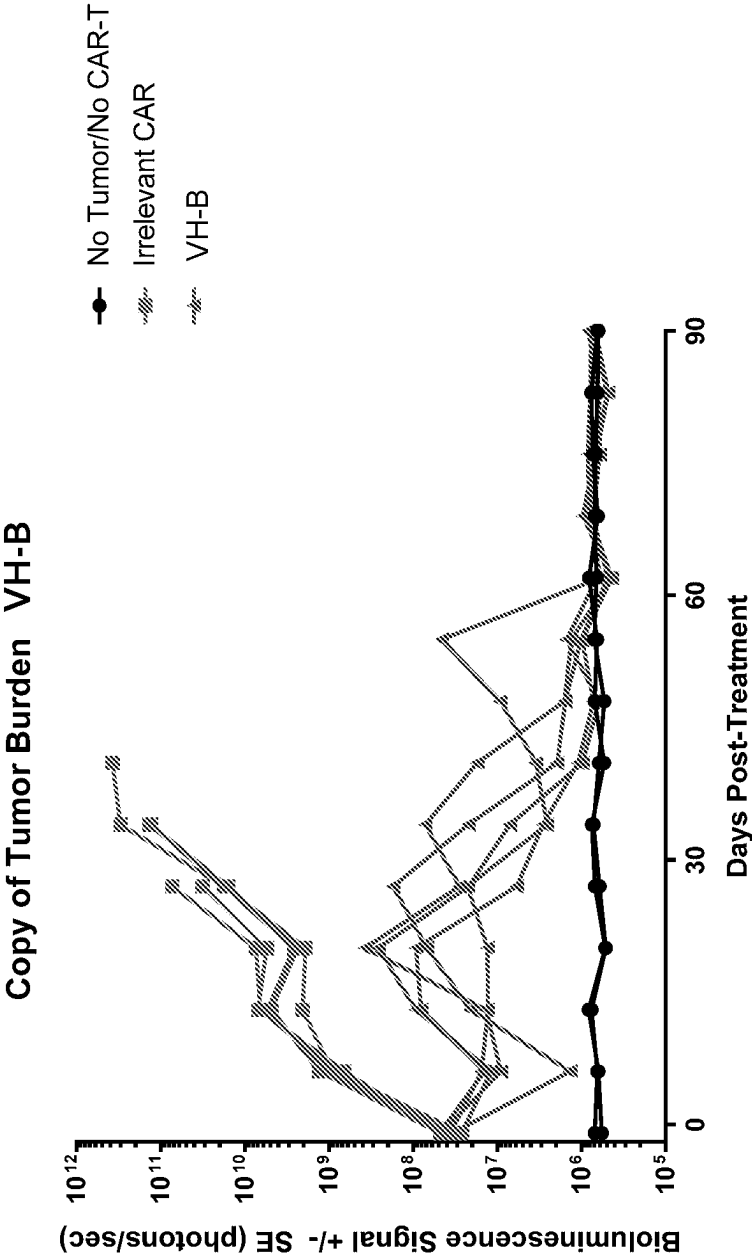


FIGURE 6

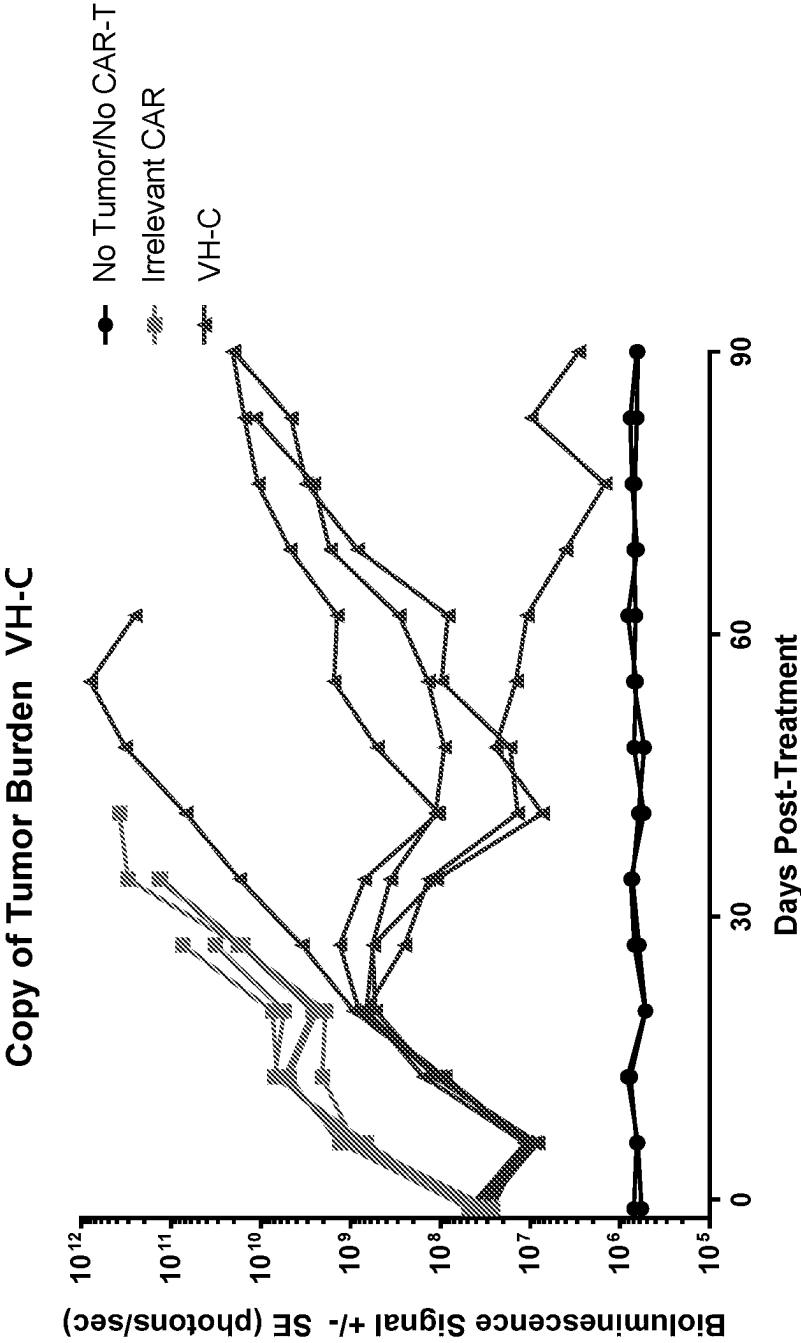


FIGURE 7

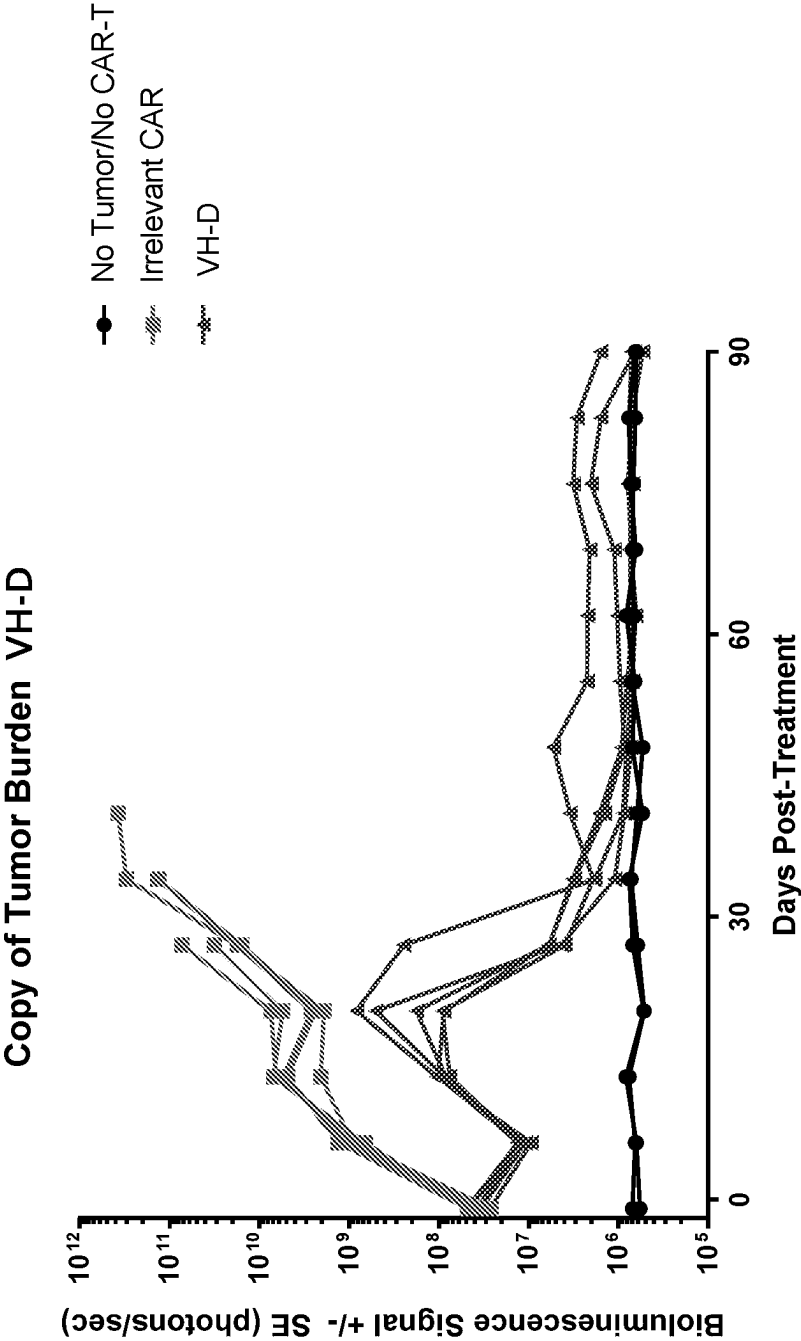


FIGURE 8

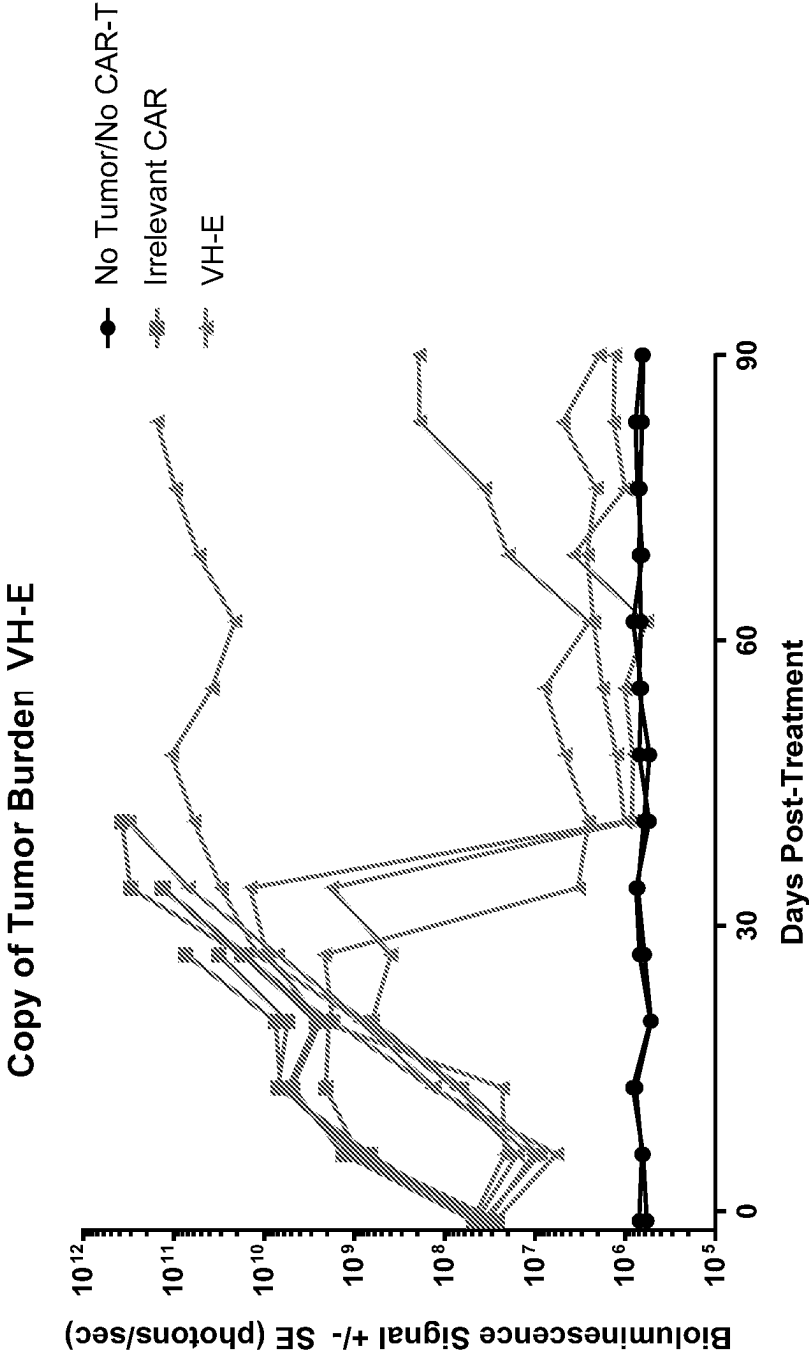


FIGURE 9

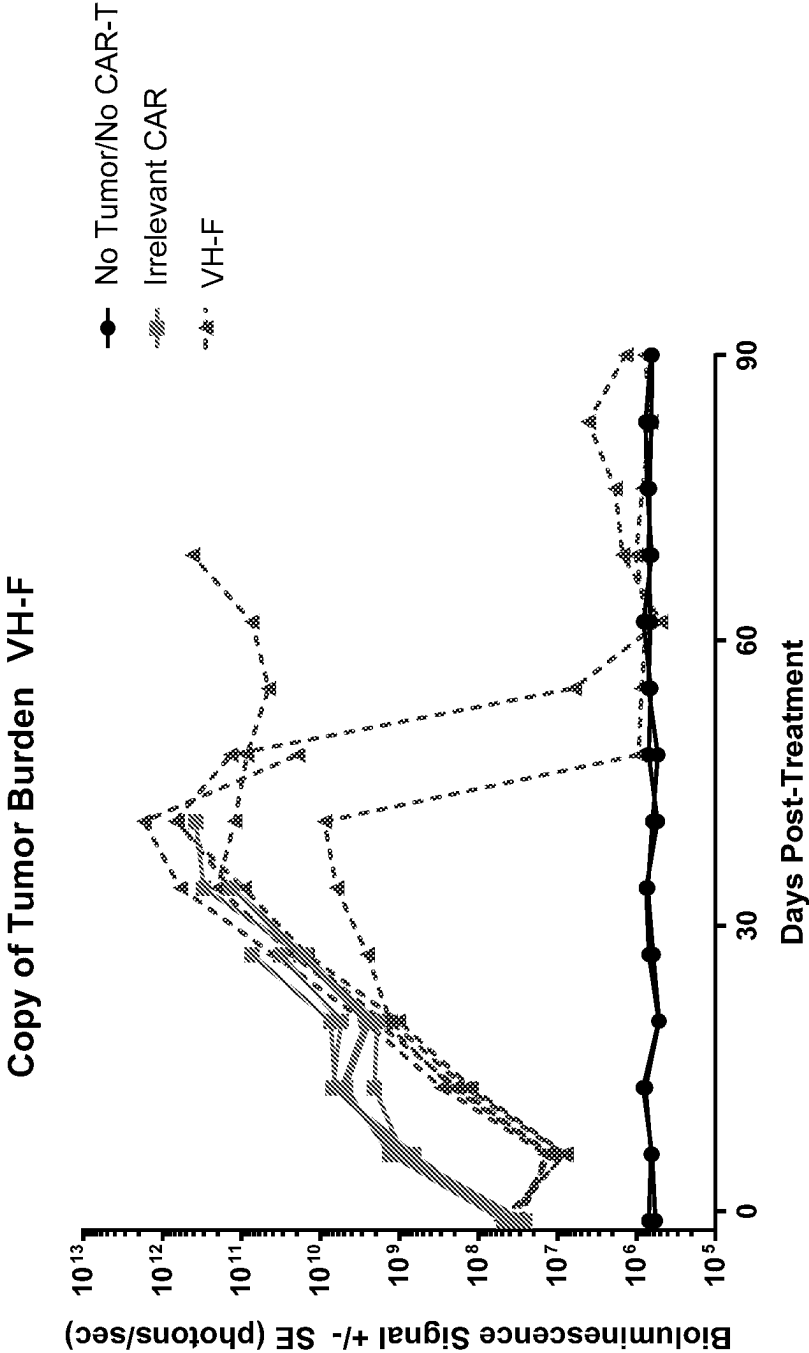


FIGURE 10A

Consensus_seq_aa_FR1_FR4	FR1	CDR1	FR2	CDR2	FR3	CDR3	FR4
EVQLLESGGGLVQPGGSLTISCAAS GFTE SNYAMN WVRQAPGKLEWVSG IIGSGGAT YYADSVKGRFTISRDNSKNTLYLQMN SLR AEDTAIYYC VKDWNTMITE RGQGT LVTV SS (SEQ ID NO: 18050)	EVQLLESGGG LVQPGGSLTL SCAAS (SEQ ID NO: 18065)	GFTFSN YA (SEQ ID NO: 180 56)	MNWVRQAPG KLEWVSG (SEQ ID NO: 18065)	IIGSGAT T(SEQ ID NO: 18059)	YYADSVKGRFTIS RDNSKNTLYLQMN SLRAEDTAIYYC (SEQ ID NO: 18073)	VKDWNTTM ITE (SEQ ID NO: 18064)	RGQGT LV TVSS (SEQ ID NO: 1807 9)
EVQLLESGGGLVQPGGSLRISCAAS GFTE SNYAM TWIRQAPGKLEWVSG ITCDGGST FYADSVKGRFTISRDNSKNTLYLQMN SLR AEDTAIYYC VKDWNTMITE RGQGT LVTV SS (SEQ ID NO: 18051)	EVQLLESGGG LVQPGGSLRL SCAAS (SEQ ID NO: 18066)	GFTFSN YA (SEQ ID NO: 180 56)	MTWIRQAPG KLEWVSG (SEQ ID NO: 18071)	ITGDGGS T(SEQ ID NO: 18060)	FYADSVKGRFTIS RDNSKNTLYLQMN SLRAEDTAIYYC (SEQ ID NO: 18074)	VKDWNTTM ITE (SEQ ID NO: 18064)	RGQGT LV TVSS (SEQ ID NO: 1807 9)
EVQLLESGGGLVQPGGSLRISCAAS GFTE SSYAM NWVRQAPGKLEWVSG IIGSGGAT YYADSVKGRFTISRDNSKNTLYLQMN SLR AEDTAIYYC VKDWNTMITE RGQGT LVTV SS (SEQ ID NO: 18052)	EVQLLESGGG LVQPGGSLRL SCAAS (SEQ ID NO: 18066)	GFTFSN YA (SEQ ID NO: 180 57)	MNWVRQAPG KLEWVAG (SEQ ID NO: 18072)	IIGSGGS T (SEQ ID NO: 18061)	YYADSVKGRFTIS RDNSKNTLYLQMN SLRAEDTAIYYC (SEQ ID NO: 18075)	VKDWNTTM ITE (SEQ ID NO: 18064)	RGQGT LV TVSS (SEQ ID NO: 1807 9)
EVQLLESGGGLVQPGGSLRISCAAS GFTE SSYAM NWVRQAPGKLEWVSG ISGGGGST YYADSVKGRFTISRDNSKNTLYLQMN SLR AEDTAIYYC VKDWNTMITE RGQGT LVTV SS (SEQ ID NO: 18053)	EVQLLESGGG LAQPGGSLRL SCAAS (SEQ ID NO: 18067)	GFTFSN YA (SEQ ID NO: 180 57)	MNWVRQAPG KLEWVSG (SEQ ID NO: 18071)	ISGGGS T (SEQ ID NO: 18061)	YYADSVKGRFTIS RDNSKNTLYLQMN SLRAEDTAIYYC (SEQ ID NO: 18076)	VKDWNTTM ITE (SEQ ID NO: 18064)	RGQGT LV TVSS (SEQ ID NO: 1807 9)
EVQLLESGGGLVQPGGSLRISCAAS GFTE TNYAM NWVRQAPGKLEWVSG ISGGGGST YYADSVKGRFTISRDNSKNTLYLQMN SLR AEDTAIYYC VKDWNTMITE RGQGT LVTV SS (SEQ ID NO: 18054)	EVQLLESGGG LVQPGSLRL SCAAS (SEQ ID NO: 18068)	GFTFTN YA (SEQ ID NO: 180 58)	MNWVRQAPG KLEWVSG (SEQ ID NO: 18070)	ISGGGS T (SEQ ID NO: 18062)	YYADSVKGRFTIS RDNSKNTLYLQMN SLRAEDTAIYYC (SEQ ID NO: 18077)	VKDWNTTM ITE (SEQ ID NO: 18064)	RGQGT LV TVSS (SEQ ID NO: 1807 9)
EVQLLESGGGLVQPGGSLRISCAAS GFTE SNYAM NWVRQAPGKLEWVSG IVGGGGTS YYADSVKGRFTISRDNSKNTLYLQMN SLR AEDTAIYYC VKDWNTMITE RGQGT LVTV SS (SEQ ID NO: 18055)	EVQLLESGGG LVQPGSLRL SCAAS (SEQ ID NO: 18069)	GFTFSN YA (SEQ ID NO: 18056)	MNWVRQAPG KLEWVSG (SEQ ID NO: 18070)	IVGGGGT S (SEQ ID NO: 18063)	YYADSVKGRFTIS RDNSKNTLYLQMN SLRAEDTAIYYC (SEQ ID NO: 18078)	VKDWNTTM ITE (SEQ ID NO: 18064)	RGQGT LV TVSS (SEQ ID NO: 18079)

VH-B

VH-D

VH-A

VH-E

VH-F

VH-C

VH-B	EVQLLESGGCLQPGISLV	20	ISCAASGFTFSYANMWR	30	YANMWRQAPCGLEWV	40	CGCTTAYADSV	50	CRFTISDN	60	SKNTLQ	70	NSLRAEDTA	80	YVCV	90	KDNTMI	100	TERG	110	QCTLVTVSS
VH-D	EVQLLESGGCLQPGISLV	20	ISCAASGFTFSYANMWR	30	YANMWRQAPCGLEWV	40	CGCTTAYADSV	50	CRFTISDN	60	SKNTLQ	70	NSLRAEDTA	80	YVCV	90	KDNTMI	100	TERG	110	QCTLVTVSS
VH-A	EVQLLESGGCLQPGISLV	20	ISCAASGFTFSYANMWR	30	YANMWRQAPCGLEWV	40	CGCTTAYADSV	50	CRFTISDN	60	SKNTLQ	70	NSLRAEDTA	80	YVCV	90	KDNTMI	100	TERG	110	QCTLVTVSS
VH-E	EVQLLESGGCLQPGISLV	20	ISCAASGFTFSYANMWR	30	YANMWRQAPCGLEWV	40	CGCTTAYADSV	50	CRFTISDN	60	SKNTLQ	70	NSLRAEDTA	80	YVCV	90	KDNTMI	100	TERG	110	QCTLVTVSS
VH-F	EVQLLESGGCLQPGISLV	20	ISCAASGFTFSYANMWR	30	YANMWRQAPCGLEWV	40	CGCTTAYADSV	50	CRFTISDN	60	SKNTLQ	70	NSLRAEDTA	80	YVCV	90	KDNTMI	100	TERG	110	QCTLVTVSS
VH-C	EVQLLESGGCLQPGISLV	20	ISCAASGFTFSYANMWR	30	YANMWRQAPCGLEWV	40	CGCTTAYADSV	50	CRFTISDN	60	SKNTLQ	70	NSLRAEDTA	80	YVCV	90	KDNTMI	100	TERG	110	QCTLVTVSS

FIGURE 10B

FIGURE 11

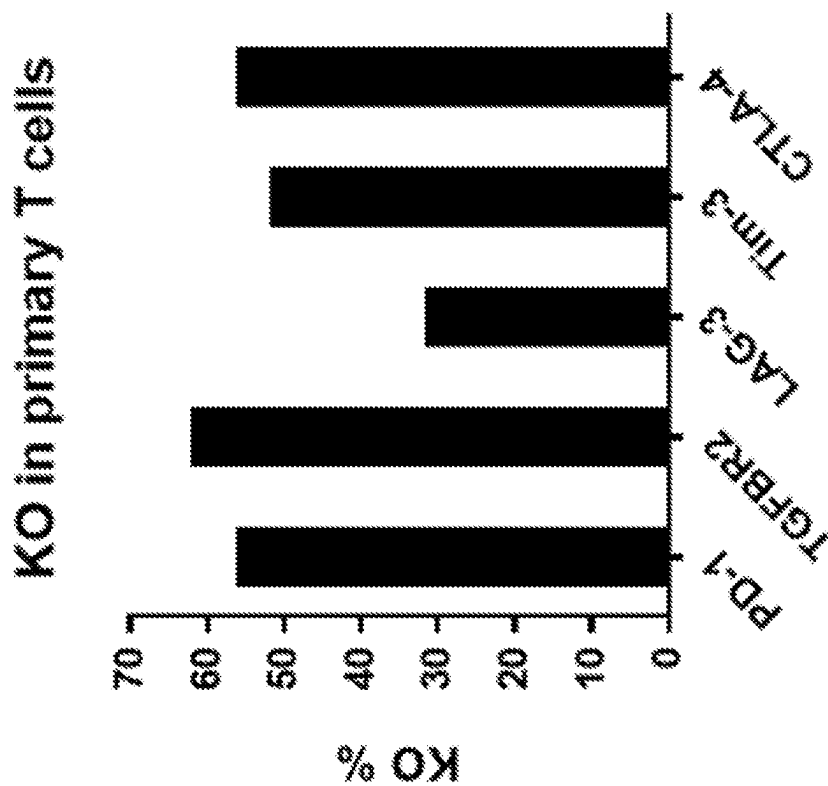


FIGURE 12

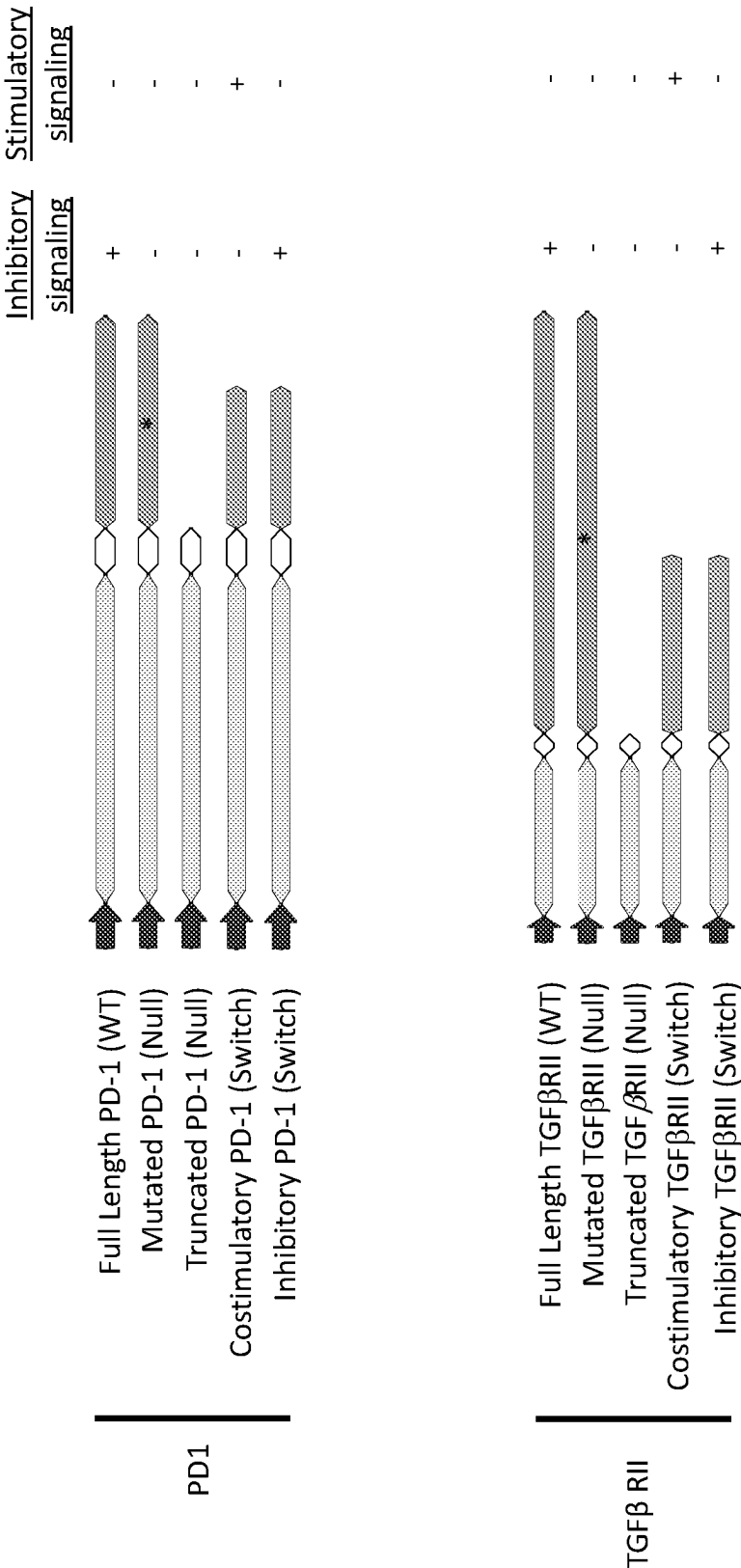


FIGURE 13

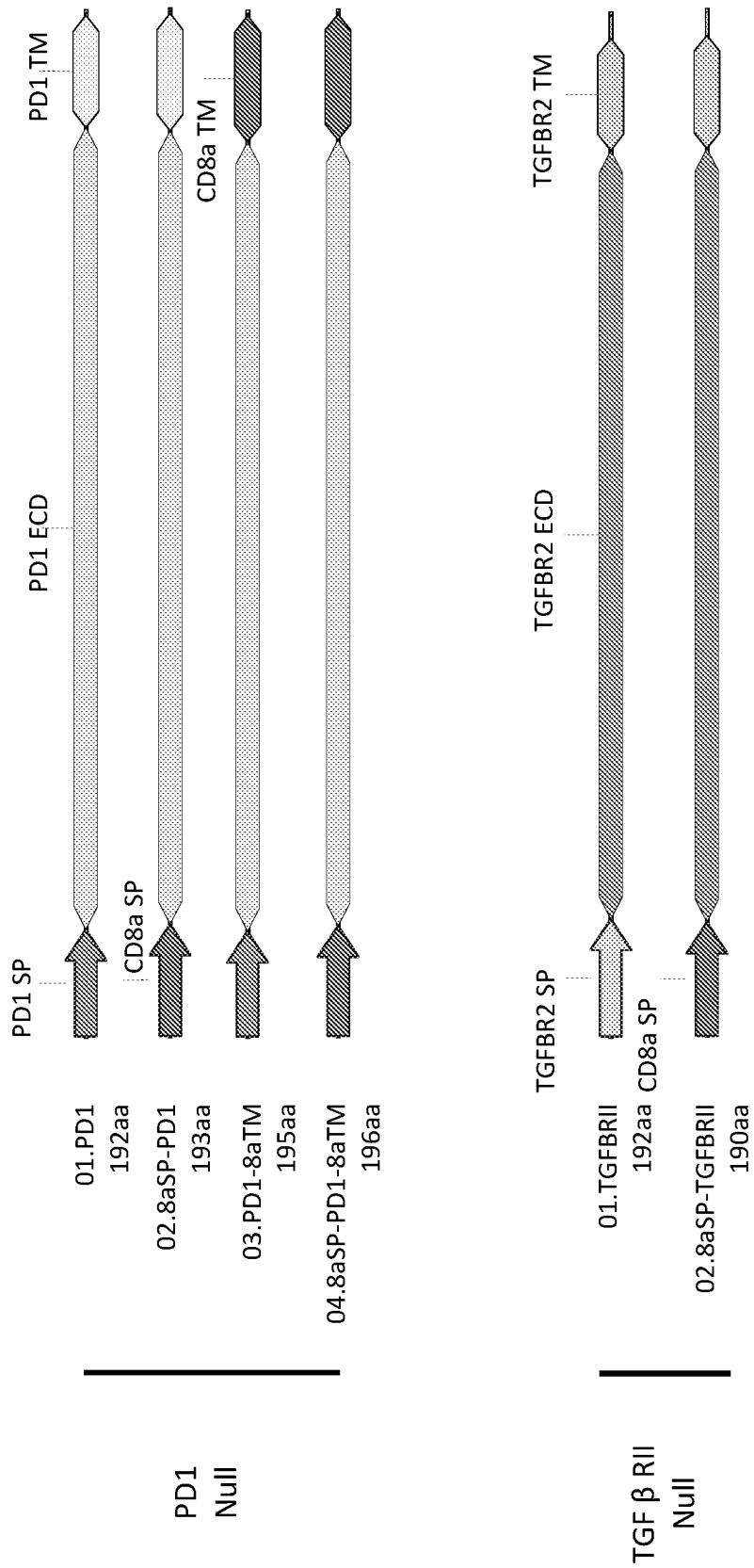


FIGURE 14

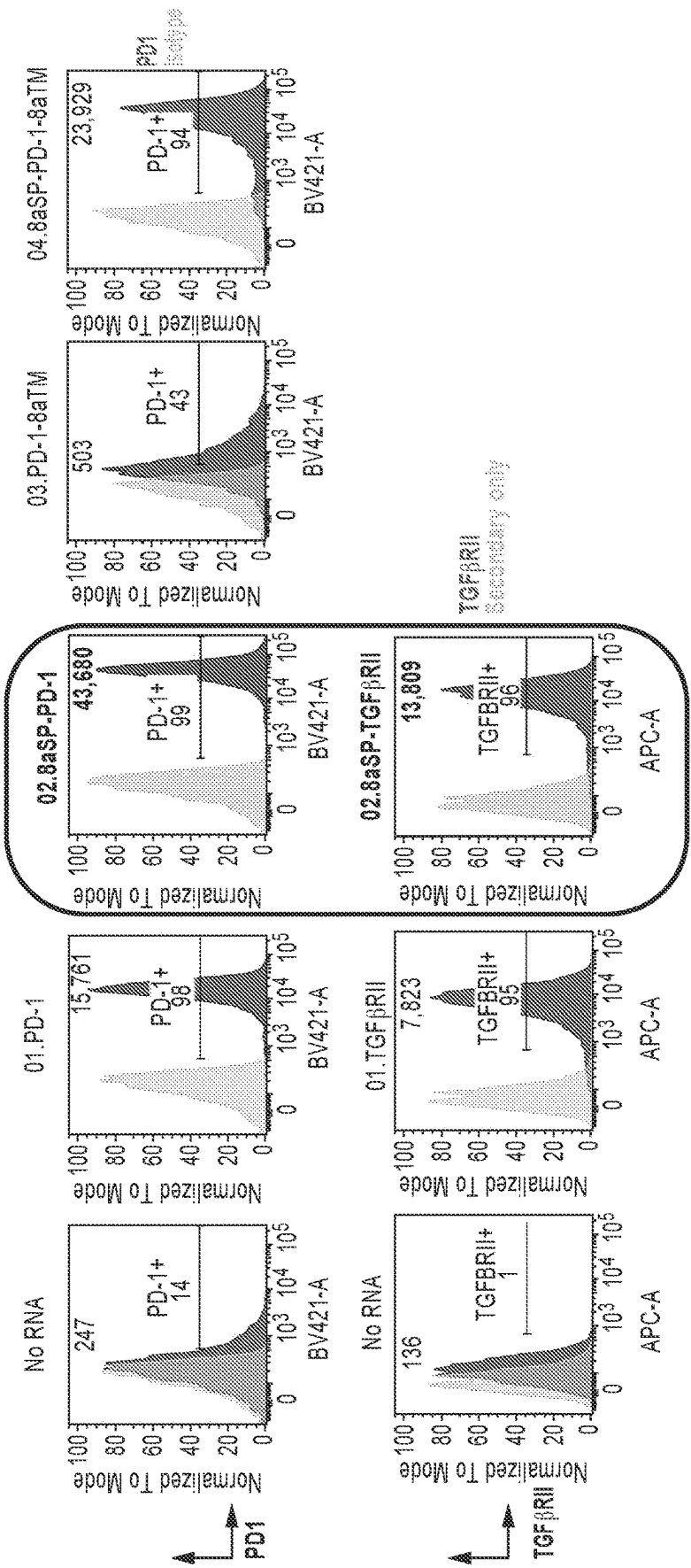
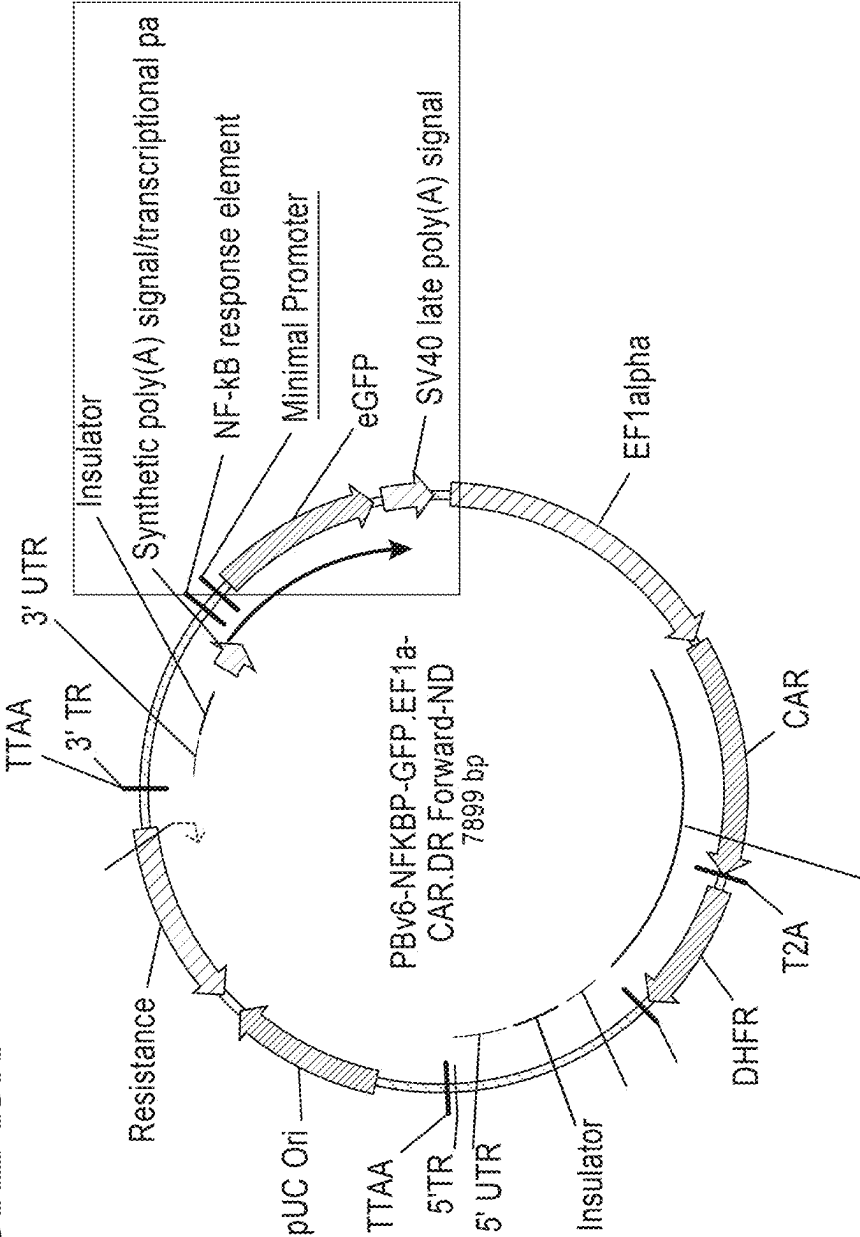
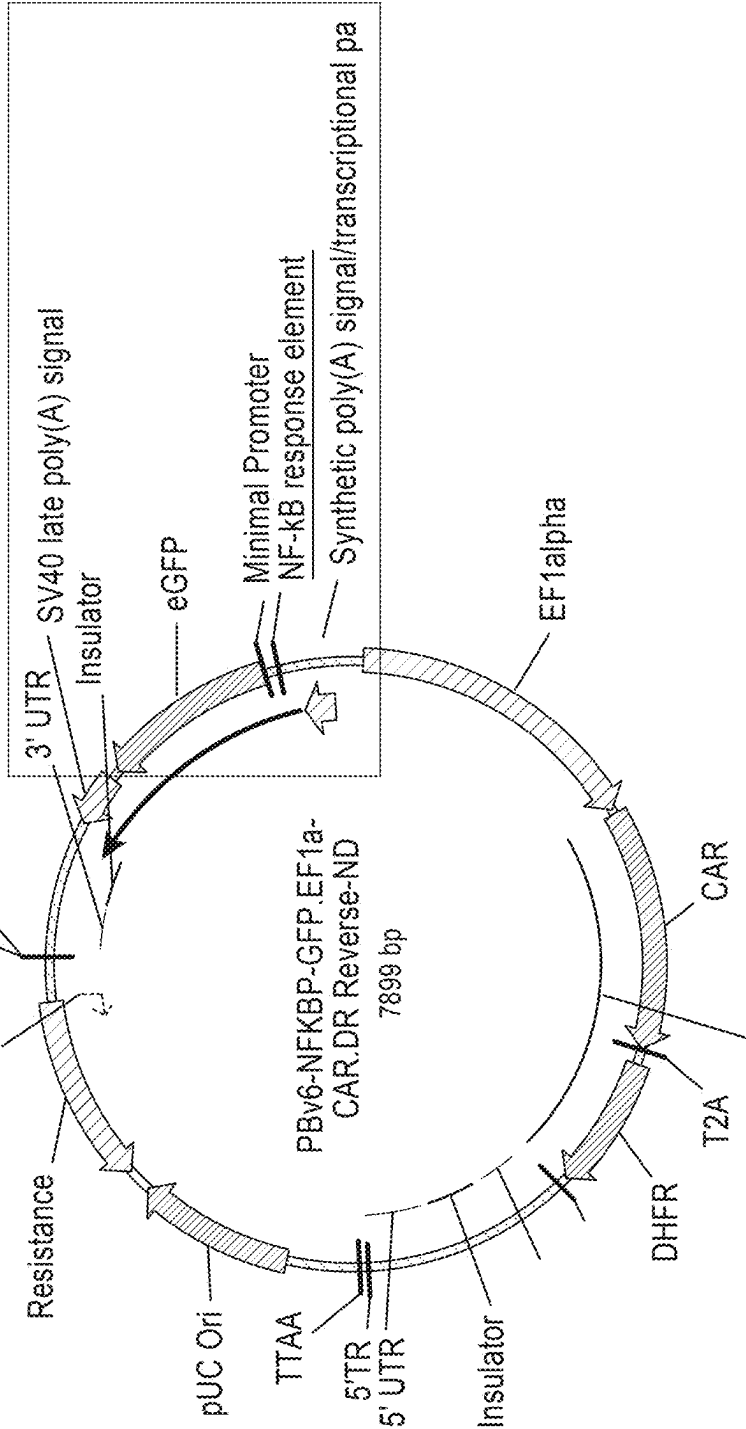


FIGURE 15A



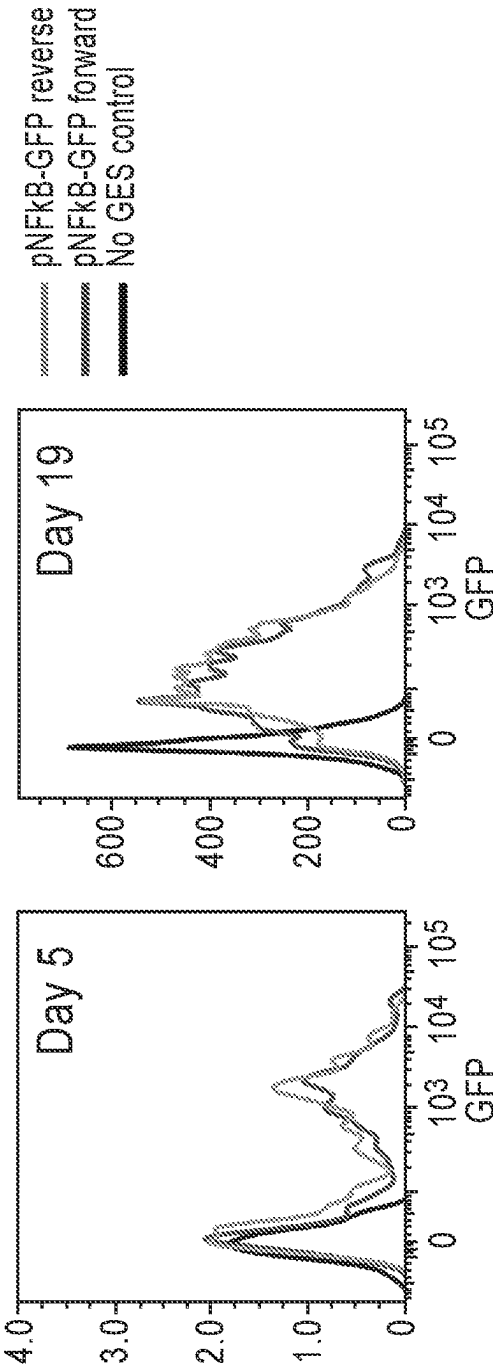
pNFKB-GFP & EF1a-CAR in the same direction

FIGURE 15B



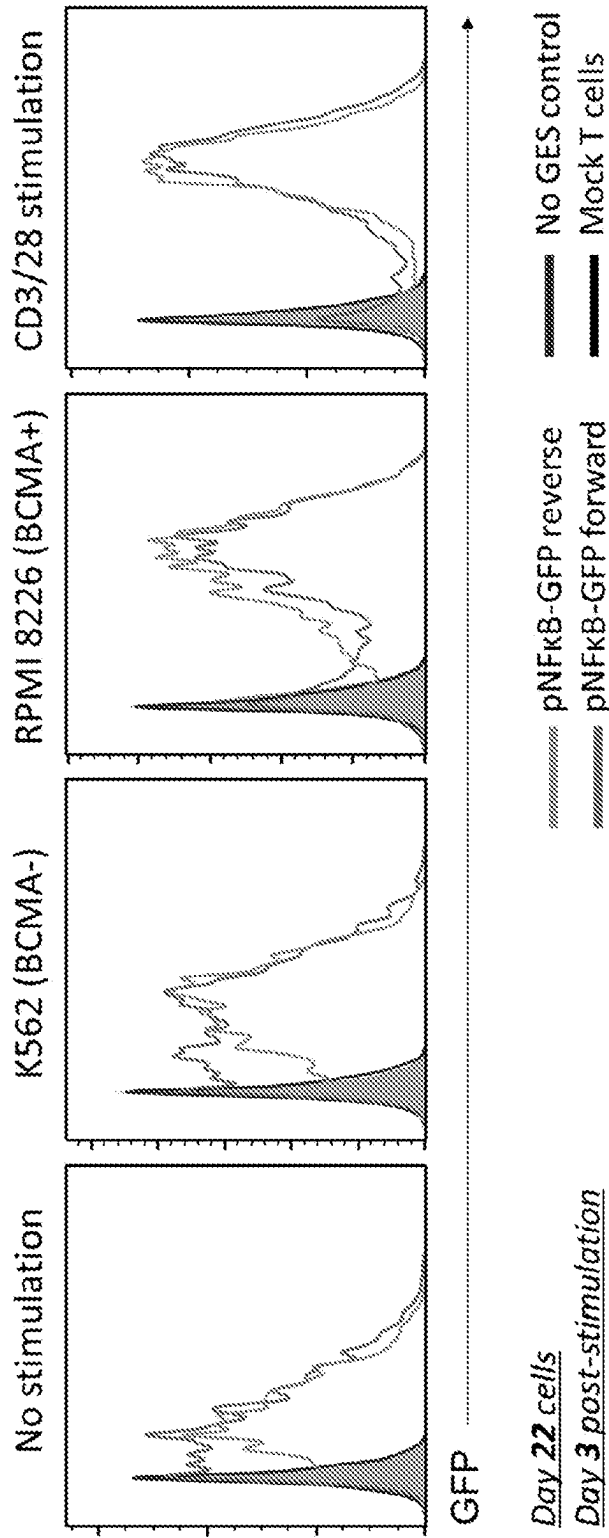
pNFKB-GFP & EF1a-CAR in the opposite direction

FIGURE 16

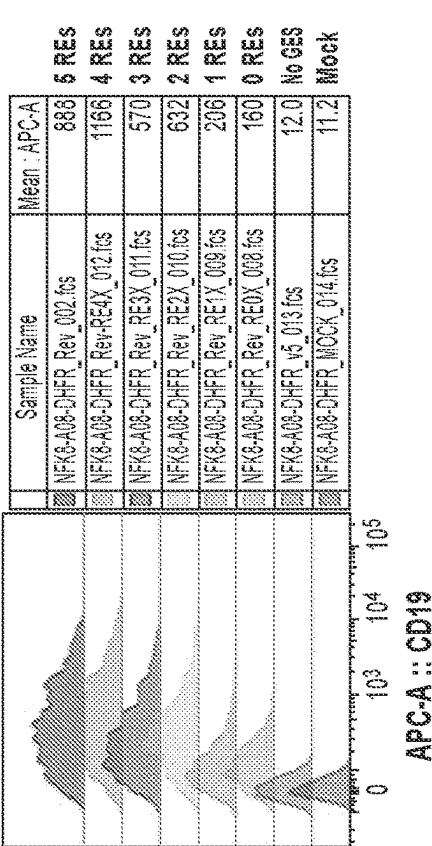


- At Day 5-post T cell activation (Day 5), T cells are proliferating and are highly stimulated; GFP expression is strong since NFkB activity is high
- By Day 19, T cells are almost fully resting; GFP expression is significantly lower than Day 5 (~1/8 MFI), since NFkB activity is low
- GFP expression is still observed at Day 19, which may be due to the long half life of GFP protein (~30hr), or, basal level of NFkB activity through TCR, CAR, cytokine, growth factor, etc...

FIGURE 17



CD8+ T Cells



CD4+ T Cells

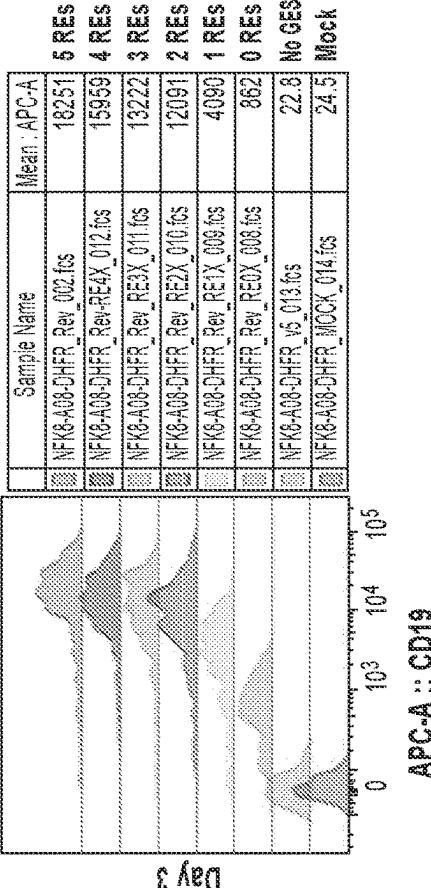
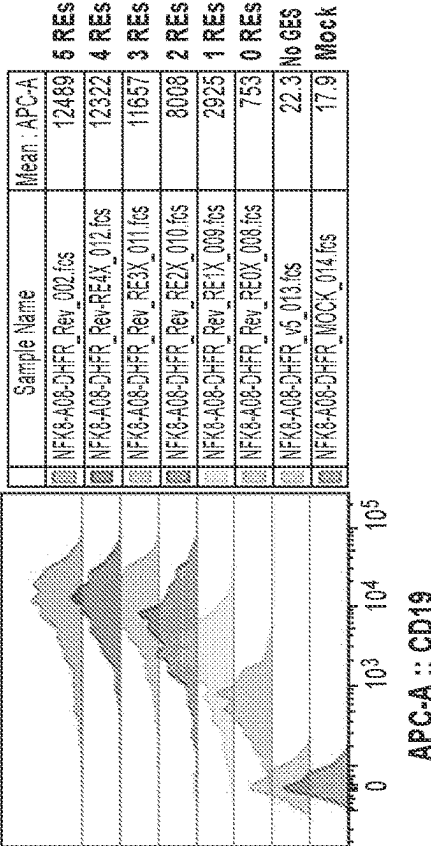
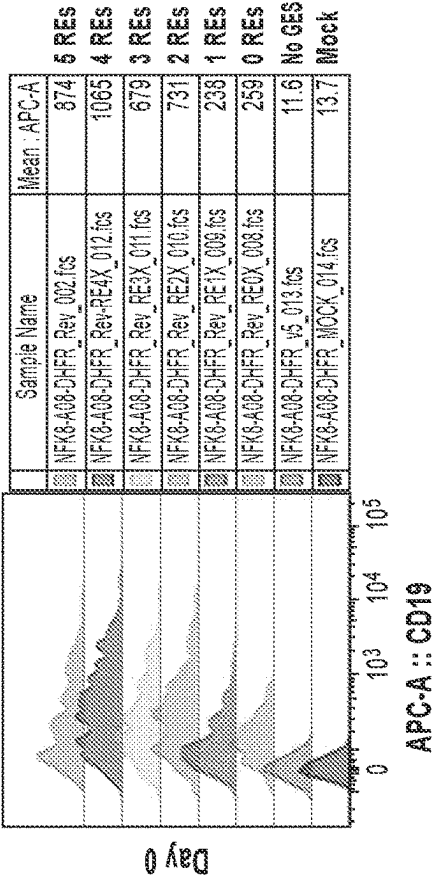
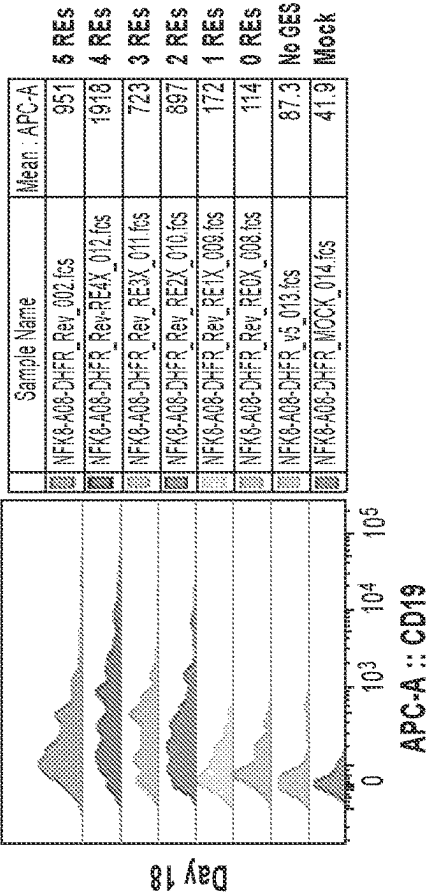


FIGURE 18 (cont.)

CD4+ T Cells



CD8+ T Cells

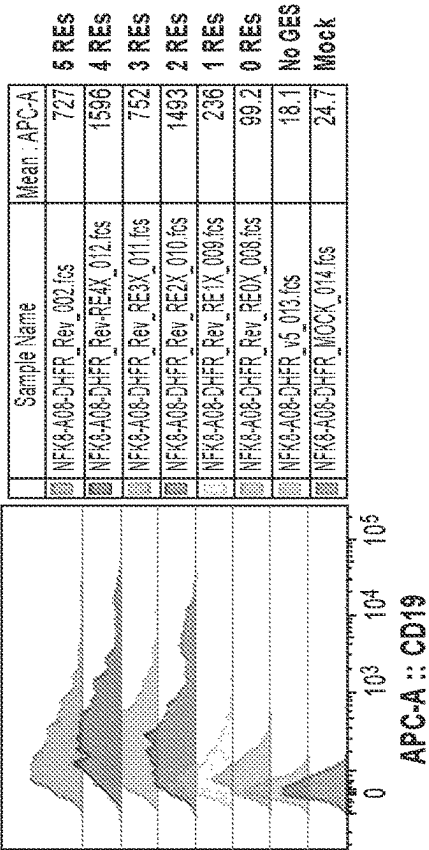


FIGURE 18 (cont.)

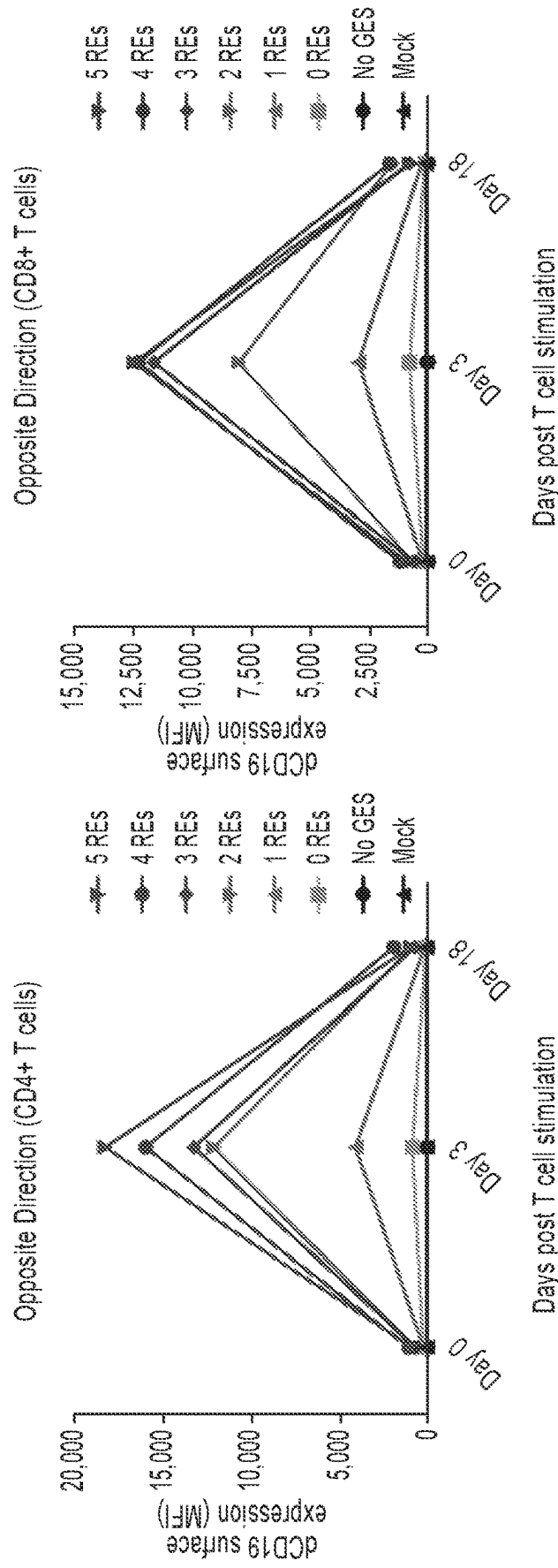


FIGURE 19

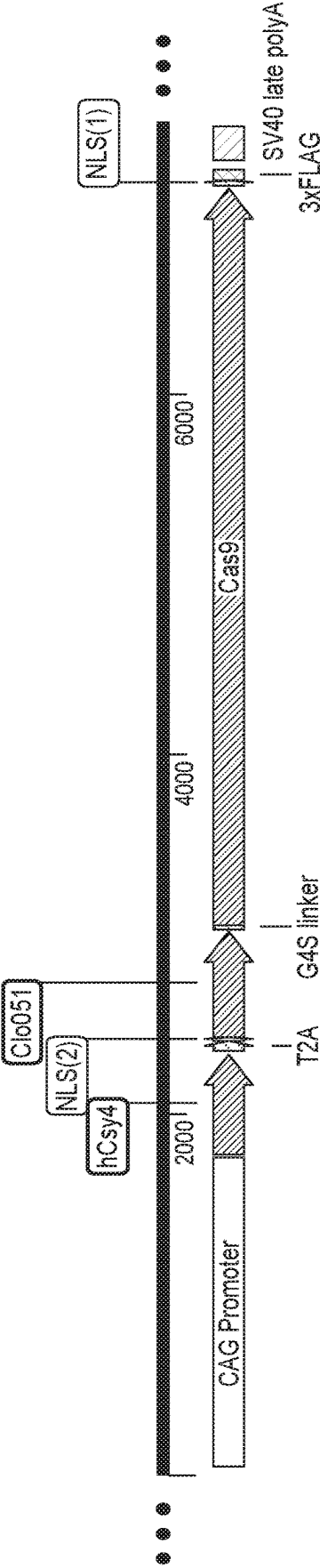
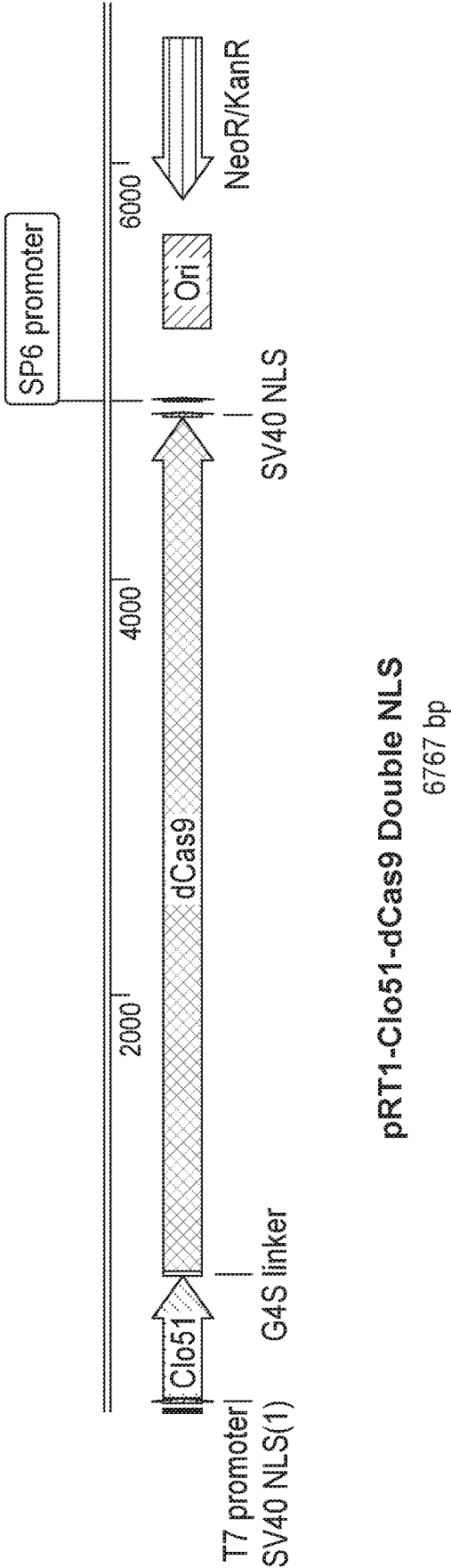


FIGURE 20



pRT1-Clo51-dCas9 Double NLS

6767 bp

FIGURE 21

Cas-CLOVER Fusion Protein Comparison
Embodiment 1 vs Embodiment 2

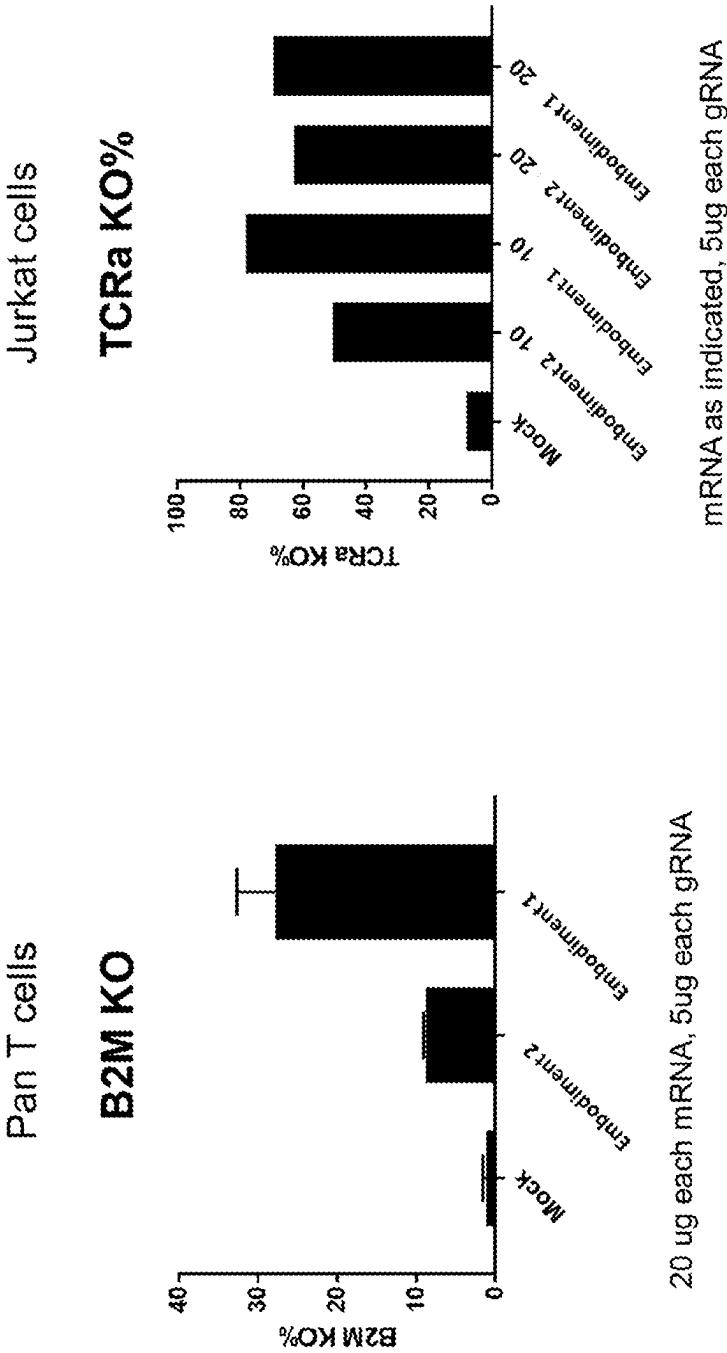


FIGURE 22

Cas-CLOVER Fusion Protein Comparison

Embodiment 1 vs Embodiment 2

mRNA

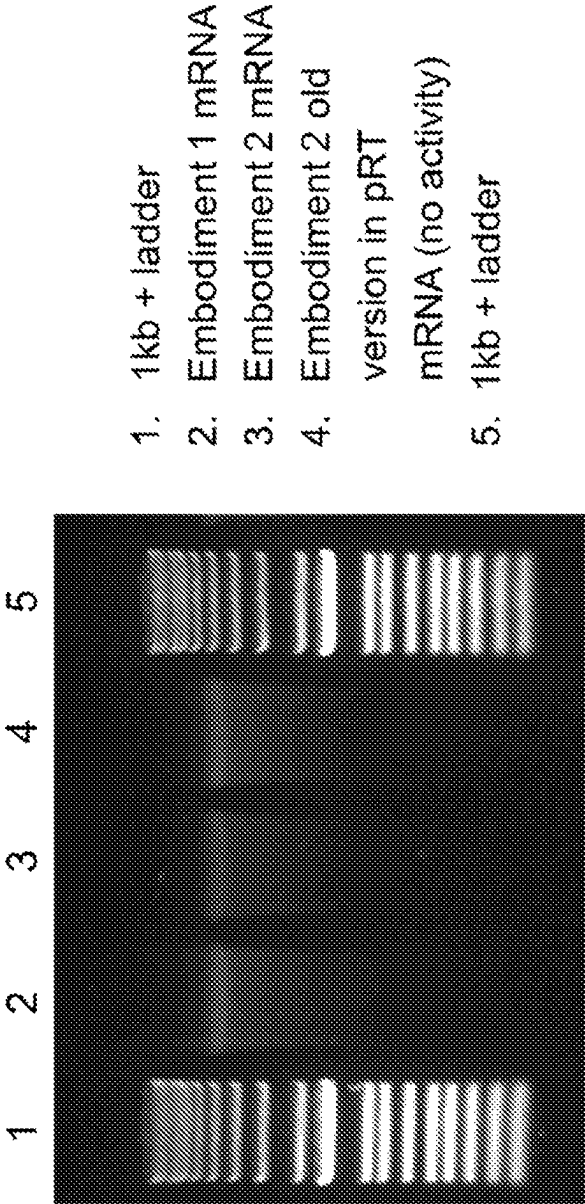


FIGURE 23

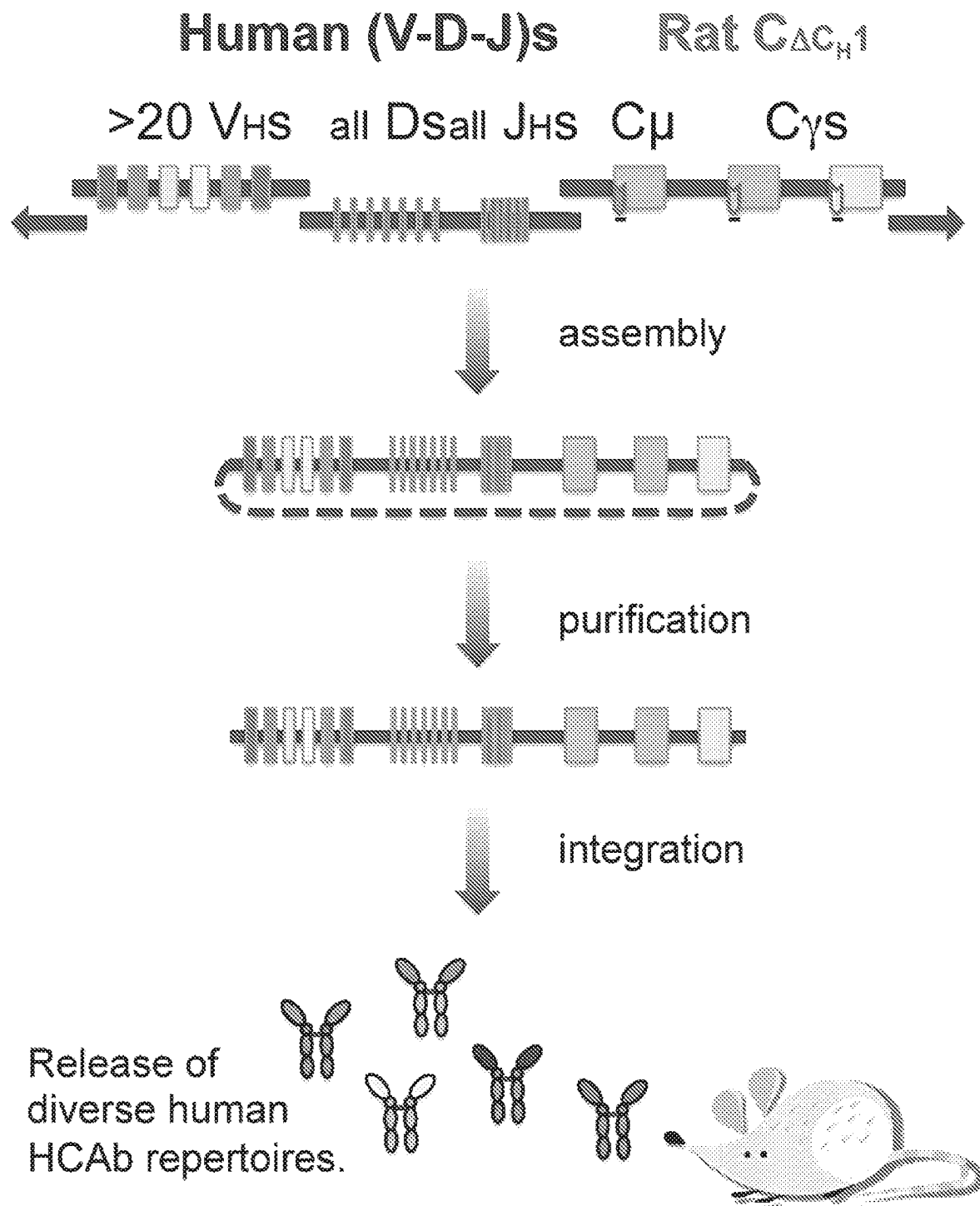


FIGURE 24

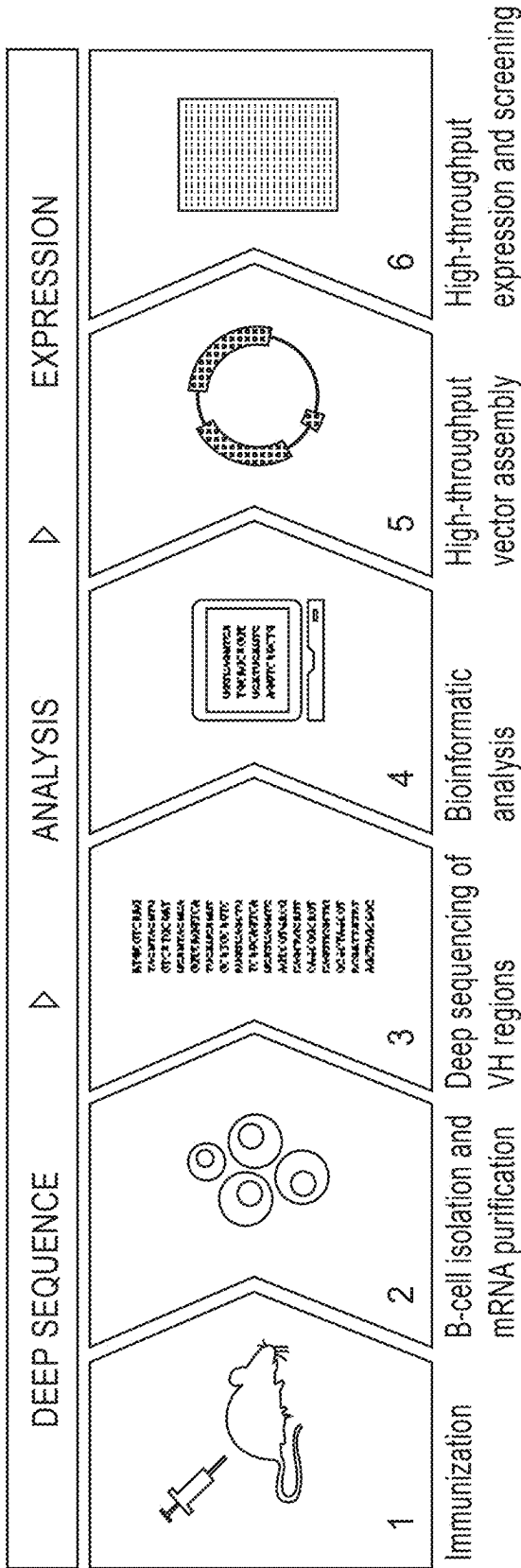


FIGURE 25

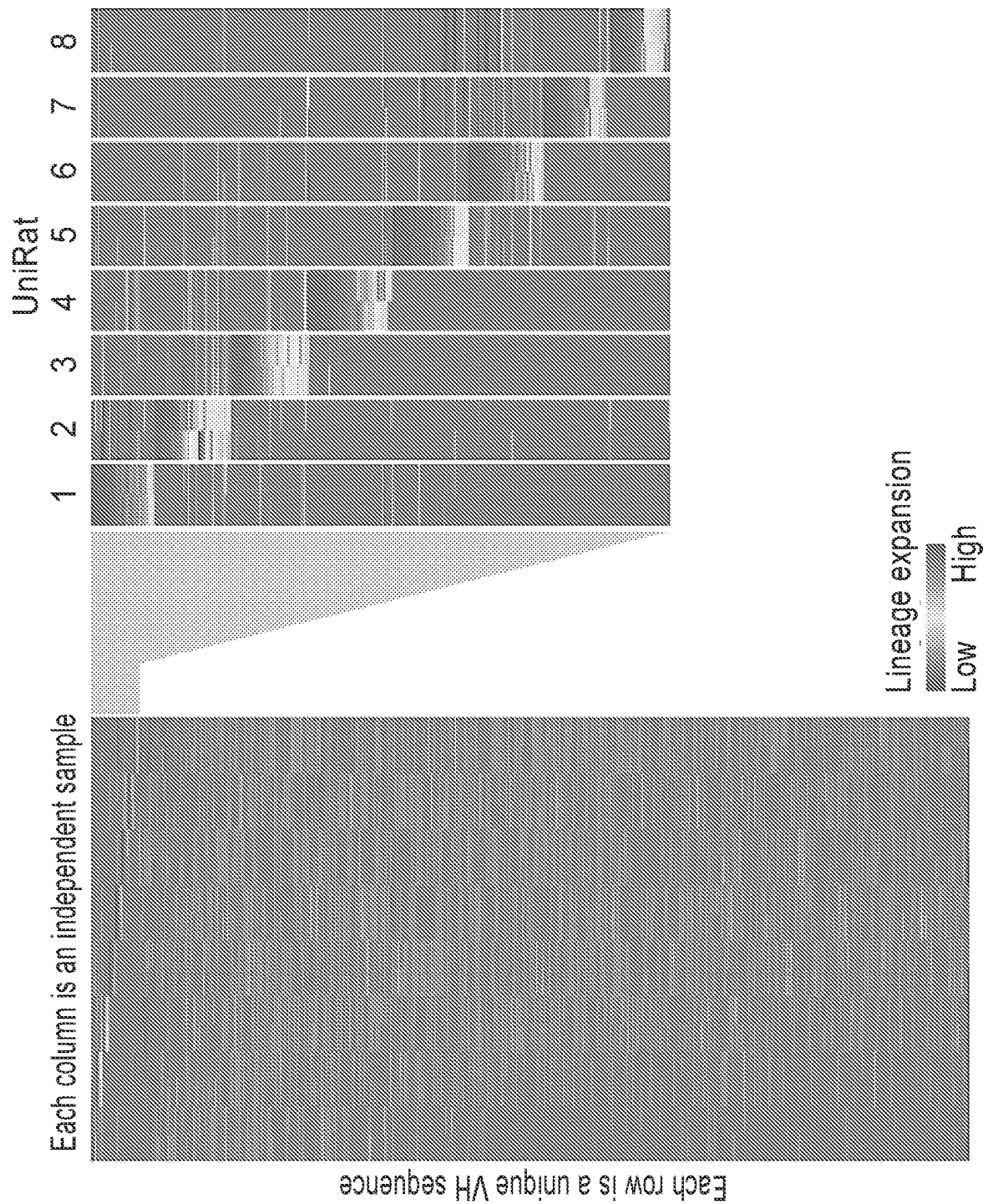
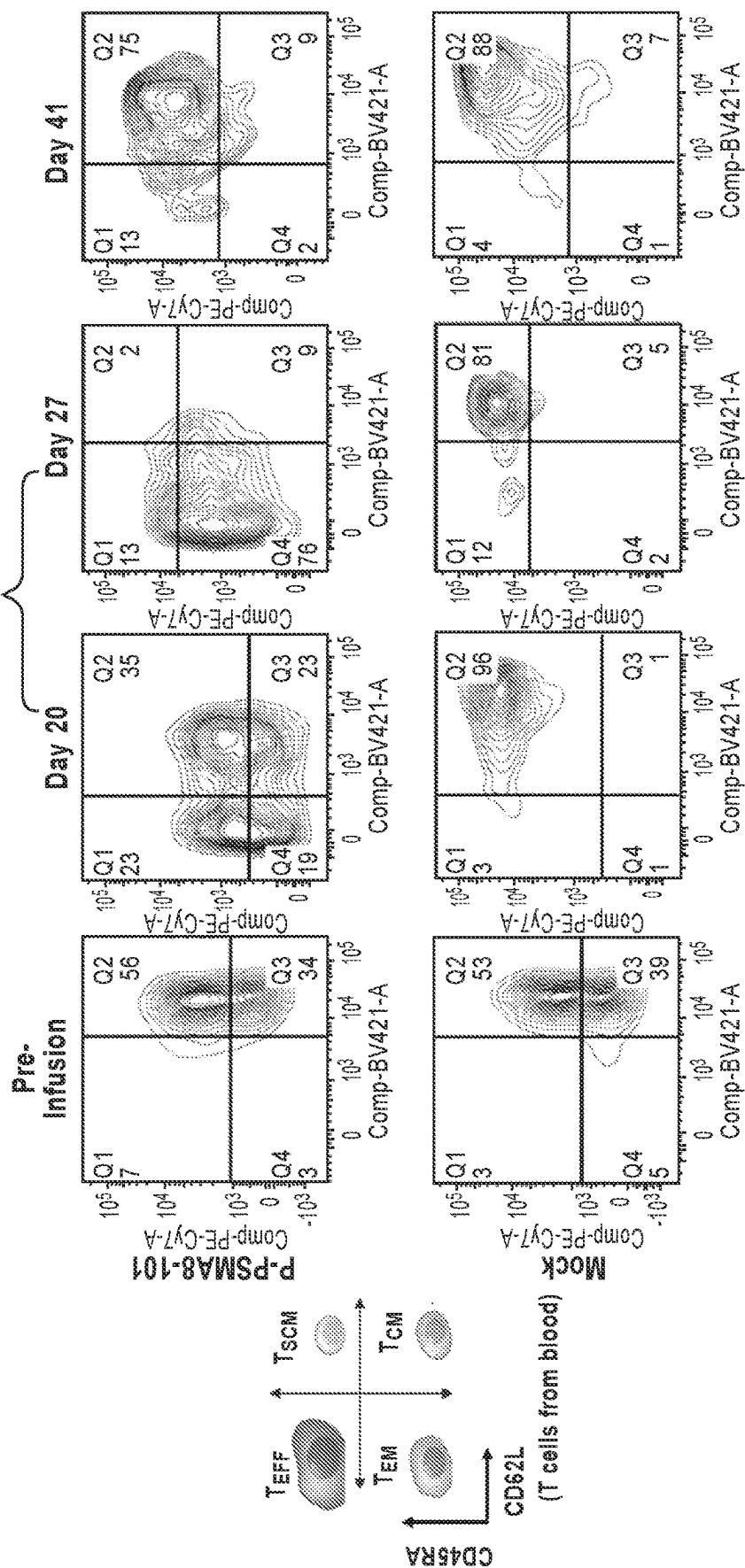


FIGURE 26

No tumor by caiper (Day 20-26)



1

**BCMA SPECIFIC VCAR COMPOSITIONS
AND METHODS FOR USE****RELATED APPLICATIONS**

This application is a U.S. National Phase Application, filed under 35 U.S.C. § 371 of International Patent Application No. PCT/US2018/066936, filed Dec. 20, 2018, which claims the benefit of provisional applications U.S. Ser. No. 62/608,571, filed Dec. 20, 2017 and U.S. Ser. No. 62/608,894, filed Dec. 21, 2017, the contents of which are herein incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

The disclosure is directed to molecular biology, and more, specifically, to chimeric antigen receptors, and to transposons containing one or more VCARs, as well as methods of making and using the same.

INCORPORATION OF SEQUENCE LISTING

The contents of the text file named "POTH-034_N01US_SegListing_ST25_R.txt", which was created on Jun. 10, 2020, and is 54 MB in size are hereby incorporated by reference in their entirety.

BACKGROUND

There has been a long-felt but unmet need in the art for a method of directing the specificity of an immune cell without using traditional antibody sequences or fragments thereof. The disclosure provides a superior chimeric antigen receptor.

SUMMARY

The disclosure provides a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one single domain antibody; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the single domain antibody comprises a human or humanized sequence. In some embodiments, the single domain antibody comprises a non-naturally occurring sequence. In some embodiments, the single domain antibody comprises a recombinant or chimeric sequence. In some embodiments, the single domain antibody comprises a VHH or a sequence encoding a VHH. In some embodiments, the single domain antibody comprises a VH or a sequence encoding a VH. In some embodiments, the single domain antibody comprises a VH of the disclosure or a sequence encoding a VH of the disclosure.

The disclosure provides a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VHH; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. As used throughout the disclosure, a CAR comprising a VHH is referred to as a VCAR. In certain embodiments, the antigen recognition region may comprise two VHHs to produce a bi-specific or tandem VCAR. In certain embodiments, the antigen recognition region may comprise three VHHs to produce a tri-specific VCAR.

In certain embodiments of the VCARs of the disclosure, the ectodomain may further comprise a signal peptide.

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Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain. In certain embodiments, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain.

In certain embodiments of the VCARs of the disclosure, the VHH comprises or consists of the amino acid sequence

(VH-A; SEQ ID NO: 18000)

malpvatalllplalllhaarpevqllesggglvqpqgsrlrlscaa

sgftfssyamnwwrqapgkglewvagiigsggstyyadsvkgrfs

isrdnsknldldlqmnsraedtavyyvcvkdwnnttmitergggtlv

tvssttppaprpptpaptiasqplslrpeacrpaaaggavhtrgld

facdiyiwaplagtcgvlllslvitlyckrgrkkllyifkqpfmr

pvqttgeedgcsrpfpeeeeggcelrvkfrsadaapaykqgnql

ynelnlgrreeydvldkrrgrdpemggkprnrnpqeglynelqkd

kmaeyseigmkgenrgkghdglyqglstatkdydalhmqalp

pr

or the VHH comprises or consists of the nucleic acid sequence

(VH-A; SEQ ID NO: 18001)

atggctctgctgtgacagctctgctgtgctctgctgtgctgt

cttcatgcggcgcgcctgaagttcagctgctgaaatctggcgga

ggcctgggtcaacctggcggaatctctgagactgagctgtgccgc

agcggcttcaccttttagcagctacgccatgaactgggtccgacag

gcccctggcaaaggactggaatgggtggcggaatcatcggcagc

ggcggcagcacatattacgcgattctgtgaaggcgcttcagc

atcagcgggacaacagcaagaacacctggacctgcagatgaac

agcctgagagccgaggataccgcggtgtactactgcgtgaaggat

tggaacaccaccatgatcaccgagagagggcagggcacactggtc

accgtgtcctctacaacaacacggcgccctcggcctccaacacca

gtcctacaatcgcgagtcagccctgtctctcagaccgaagcc

tgtagacctgctgctggcgagctgtgcataccagaggactggat

ttcgctgcgacatctacatctgggctcctctggtggcacatgc

ggagttttgctgctgagcctggtcatccctgtactgtaagaga

ggcaggaagaagctgctgtatatctcaagcagccctcatgaga

cccgtgcagaccacacaggaggaggagcggctgctctgtaggttc

ccagaggaggaggaggaggatgagagctgcgcgtgaagttagc

cggtcgcgatgcacctgcatacaaggaggacagaaccagctg

tataacgagctgaatctgggcccggagagaggagtacgagctgtg

gataagaggcgggcgggaccccgagatgggaggcaagcctcgg

agaaagaaccacaggaggcctgtacaatgagctgcaaaaggac

aagatggccgaggcctattctgagatcgccatgaaggagagagg

3

-continued

cgccggggcaaggacacgatggcctgtaccagggcctgagcacc
gccacaaaggacacctatgatgccctgcacatgcaggccctgccc
cctagatga.

In certain embodiments of the VCARS of the disclosure,
the VHH comprises or consists of the amino acid sequence

(VH-B; SEQ ID NO: 18002)

malpvtalllplalllhaarpevqllesggglvqpqgslltscas
sgftfsnyamnvrqapgkglewvsgliggattyyadsvkgrft
isrdnsntlnlqmnsraedtaiyycvkdwnntmitergggtlv
tvssttppaprpptpaptiasqplslrpeacrpaaggavhtrgld
facdiyiwaplagtcgvlllslvitlyckrgrklllyifkqpfmr
pvqttqeedgcscrfeeeeeggcclrvkfsrsadapaykqggnql
ynelnlgrreeydvldkirgrdpemggkprknpqeglynelqkd
kmaeyseigmkgerrrgkghdglyqglstatkdydalhmqalp
pr

or the VHH comprises or consists of the nucleic acid
sequence

(VH-B; SEQ ID NO: 18003)

atggctctgcctgtgacagctctgctgctgcctctggctctgctt
cttcctgcggcgccctgaagtccagctgcttgaatctggcgga
ggcctgggttaacctggcggtatctgacactgagctgtgccgcc
agcggcttcaccttcagcaactacgccatgaactgggtccgacag
gccccgggcaaaaggccttgaatgggtgtccggcatcattggctct
ggcgccaccacctactacgccgattctgtgaagggcagattcacc
atcagccgggacaacagcaagaacacctgaacctgcagatgaac
agcctgagagccgaggacaccgccatctactactgcgtgaaggac
tggaacaccacctatgatcaccgagagagggcagggcacactggtc
accgtgtcctctacaacaacacggcgccctcgccctccaacacca
gctcctacaatcgcgagtcagccccctgtctctcagaccggaagcc
tgtagacctgctgctggcgagctgtgcataccagaggactggat
ttcgctgcgacatctacatctgggtcctctggctggcacatgc
ggagttttgctgctgagcctggatcatcaccctgtactgtaagaga
ggcaggaagaagctgctgtatatcttcaagcagcccttcattgaga
cccggtgcagaccacacaggaggaggcggctgctcttgtaggttc
ccagaggaggaggaggaggatgcgagctgcgcgtgaagtttagc
cggtcgccgatgcacctgcatacaagcagggacagaaccagctg
tataacgagctgaatctgggcccggagagaggagtacgacgtgctg
gataagaggcggggccgggaccccgagatgggaggcaagcctcgg
agaaagaaccacaggaggcctgtacaatgagctgcaaaaggac
aagatggccgaggcctattctgagatcggcataagggagagagg

4

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cgccggggcaaggacacgatggcctgtaccagggcctgagcacc
gccacaaaggacacctatgatgccctgcacatgcaggccctgccc
cctagatga.

In certain embodiments of the VCARS of the disclosure,
the VHH comprises or consists of the amino acid sequence

(VH-C; SEQ ID NO: 18004)

malpvtalllplalllhaarpevqllesggglvqpqgsllriscas
sgftfsnyamnvrqapgkglewvsgligggtssyadsvrgrft
isrdnsntlnlqmnsraedtavycvkdwnntmitergggtlv
tvssttppaprpptpaptiasqplslrpeacrpaaggavhtrgld
facdiyiwaplagtcgvlllslvitlyckrgrklllyifkqpfmr
pvqttqeedgcscrfeeeeeggcclrvkfsrsadapaykqggnql
ynelnlgrreeydvldkirgrdpemggkprknpqeglynelqkd
kmaeyseigmkgerrrgkghdglyqglstatkdydalhmqalp
pr

or the VHH comprises or consists of the nucleic acid
sequence

(VH-C; SEQ ID NO: 18005)

atggctctgcctgtgacagctctgctgctgcctctggctctgctt
tcttcctgcggcgccctgaagttcagctgcttgaatctggcg
gaggcctgggtcagcctggcgaatctctgagactgagctgtgcc
gccagcggcttcaccttcagcaactacgccatgaactgggtccg
acaggccccctggcaaaaggccttgaatgggtgtccggaatcgttg
gcggcgaggacacaagctactacgccgattctgtgcggggcaga
ttcaccatcagccgggacaacagcaagaacacctgtacctgca
gatgaacagcctgagagccgaggacaccgccgtgtactactgag
tgaaggactggaacaccacctatgatcaccgagagaggccagggc
acactgggtcaccgtgtcctctacaacaacacggcgccctcgcc
tccaacaccagctcctacaatcgcgagtcagccccctgtctctca
gacccgaagcctgtagacctgctgctggcgagctgtgcatacc
agaggactggatttcgcctgcgacatctacatctgggtcctct
ggctggcacatgcggagttttgctgctgagcctggatcatcacc
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag
cagcccttcattgagaccctgcagaccacacaggaggaggacgg
ctgctcttgtaggttcccagaggaggaggaggaggatgcgagc
tgccgctgaagtttagccggtccgcgatgcacctgcatacaag
cagggacagaaccagctgtataacgagctgaatctgggcccggag
agaggagtacgagctgctggataagaggcggggccgggaccccg
agatgggaggcaagcctcggaagaagaaccacagaggaggcctg
tacaatgagctgcaaaaggacaagatggccgaggcctattctga
gatcgccatgaaggaggagaggcgccggggcaaggacacgatg

5

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gacctgtaccagggcctgagcaccgccacaaaggacacctatgat
 gccctgcacatgcaggccctgccccctagatga.

In certain embodiments of the VCARS of the disclosure,
 the VHH comprises or consists of the amino acid sequence

(VH-D; SEQ ID NO: 18006)
 malpvtalllplalllhaarpevqllesggglvqpggslrlsca
 asgftfsnyamtwirqapgkglewsvgitgdggstfyadsvkgr
 ftisrdnskntlylqmnsdraedtavyyvcwdnttmiterggg
 tlvtssttppaprpptpaptiasqplslrpeacrpaaggavht
 rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk
 qpfmrvqttqeedgcscrfeeeeeggcclrvkfsrsadapayk
 qgqnqlynelnlgrreeydvldkrrgrdpemggkprknpqegl
 ynelqkdkaeyseigmkgerrrgkghdglyqglstatkdyd
 alhmqalppr

or the VHH comprises or consists of the nucleic acid
 sequence

(VH-D; SEQ ID NO: 18007)
 atggctctgcctgtgacagctctgctgctgcctctggctctgcttc
 ttcatcgccgcgcctgaagttcagctgcttgaatctggcggagg
 cctggttcaacctggcggatctctgagactgagctgtgccgccagc
 ggcttcaccttcagcaattacgcctgacctggatcagacaggccc
 ctggcaaaggcctggaatgggtgtccggaattacaggcgacggcg
 cagcaccttttacgccgattctgtgaaggcgagattcaccatcagc
 cgggacaacagcaagaacaccctgtacctgcagatgaacagcctga
 gagccgaggacacgcgcgtgtactactgcgtgaaggactggaacac
 caccatgatcaccgagagaggccagggcacactggtcaccgtgtcc
 tctacaacaacaccggcgccctcgccctccaacaccagctcctaca
 tcgcgagtcagccctgtctctcagaccgaagcctgtagacctgc
 tgcctggcggagctgtgcataccagaggactggatttcgcctgcgac
 atctacatctgggtcctctggtggcacatgcggagttttgctgc
 tgagcctggtcatcacctgtactgtaagagaggcaggaagaagct
 gctgtatatcttcaagcagcccttcagagaccgtgcagaccaca
 caggaggaggacggctgctctttaggttccagaggaggaggagg
 gaggatgcgagctgcgcgtgaagtttagccggtccgccgatgcacc
 tgcatacaagcaggacagaaccagctgtataacgagctgaatctg
 ggccggagagaggagtacgacgtgctggataagaggcggggccggg
 accccgagatgggaggcaagcctcgagagaagaaccacaggaggg
 cctgtacaatgagctgcaaaaggacaagatggccgaggcctattct

6

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gagatcggcatgaaggagagaggcgccggggcaaggacacgatg
 gcctgtaccagggcctgagcaccgccacaaaggacacctatgatgc
 cctgcacatgcaggccctgccccctagatga.

In certain embodiments of the VCARS of the disclosure,
 the VHH comprises or consists of the amino acid sequence

(VH-E; SEQ ID NO: 18008)
 malpvtalllplalllhaarpevqllesggglapggslrlscaa
 sgftfssyamnwirqapgkglewsvgisgsggstyyadsvkgrft
 isrdnskntvylqmnsdraedtavyyvcwdnttmitergggtlv
 tvssttppaprpptpaptiasqplslrpeacrpaaggavhtrgld
 facdiyiwaplagtcgvllslvitlyckrgrkkllyifkqpfmr
 pvqttqeedgcscrfeeeeeggcclrvkfsrsadapaykqgqnq
 ynelnlgrreeydvldkrrgrdpemggkprknpqeglynelqkd
 kmaeyseigmkgerrrgkghdglyqglstatkdydalhmqalp
 pr

or the VHH comprises or consists of the nucleic acid
 sequence

(VH-E; SEQ ID NO: 18009)
 atggcactgcctgtgacagccctgctgctgctctggccctgct
 gctgcacgcagcacggcccgagggtcagctgctggagtcggag
 gaggcctggcccagcctggcggcagcctgaggtgtcctgcgcc
 gcctctggcttcacctttagctcctacgccatgaactggatcag
 acaggccctggcaaggcctggagtggtgtccggcatctccg
 gctctggaggctctacatactatgccgacagcgtgaaggccgg
 ttcaccatcagcagagataaactccaagaataaccgtgtacctca
 gatgaactctctgcccggcaggacaccgcctgtactattgcg
 tgaaggattggaataaccacaatgatcacagagaggggcccagggc
 accctggtgacagtgtctagaccacaaccctgccccagacc
 tcccacaccgcccctaccatcgcgagtcagccactgtccctgc
 ggctgaggcctgccggcccgccggcgaggagcagtgacacaca
 cggggcctggactttgcctgtgacatctacatatgggcaccact
 ggcaggaacctgcggcgtgctgctgctgagcctggtcatcacc
 tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag
 cagcccttcagagaccgtgcagaccacacaggaggaggacgg
 ctgctctttaggttccagaggaggaggaggaggatgcgagc
 tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag
 cagggacagaaccagctgtataacgagctgaatctgggccggag
 agaggagtacgacgtgctggataagaggcggggccgggacccc
 agatgggaggcaagcctcgagagaagaaccacaggaggccctg
 tacaatgagctgcaaaaggacaagatggccgaggcctattctga

7

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gatcgccatgaagggagagagcgccggggcaagggacacgatg
gcctgtaccaggcgctgagcaccgccacaaaggacacctatgat
gcctgcacatgcaggccctgccccctagatga.

In certain embodiments of the VCARs of the disclosure, the VHH comprises or consists of the amino acid sequence

(VH-F; SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglvqpgsrslrlsca
asgftftnyamnvrqapgkglewvsgisggggstyadsvkgr
ftisrdnskntlylqmnsraedtavyyvcvkdwnntmiterggg
tlvtvssttpprptpaptiasqplslrpeacrpaaggavht
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk
qpfmrpvqttqeedgcscrfeeeggcelrvkfsrsadapayk
qqgnqlynelnlgrreeydvldkrrgrdpemggkprknpqegl
ynelqdkmaeayseigmkgerrrgkghdglyqglstatkdttyd
alhmqalppr

or the VHH comprises or consists of the nucleic acid sequence

(VH-F; SEQ ID NO: 18011)

atggcactgcctgtgacagccctgctgctgcctctggccctgct
gctgcacgcagcagcgcccgaggtgcagctgctggagtctggag
gaggcctggtgcagcccgccggtccctgagactgtcttgccg
gccagcggttcacctttacaaactacgccatgaattgggtg
gcagggccctggcaagggcctggagtggtgtctggcatcagcg
gaggaggaggcagcacctactatgcagactccgtgaagggcagg
ttccacctctcccgcgataaactctaagaatacactgtacctcca
gatgaacagcctgagggcagaggacaccccgctgtactattg
tgaaggattggaataccacaatgatcacagagaggggacagggc
accctggtgacctgagcagcaccacaaccctgccccagacc
tcccacacccgccctaccatcgcgagtcagccactgtccctgc
ggcctgaggcctgcggcccgccgcccggcgagcagtgacaca
cggggcctggactttgctgtgacatctacatatgggcaccact
ggcaggaacctgcggcgctgctgctgagcctggtcatcacc
tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag
cagcccttcatgagaccctgcagaccacacaggaggaggacgg
ctgctctgtaggttcccagaggaggaggaggatgcgagc
tgcgcgtaagtttagccggtcccgcatgcacctgcatacaag
cagggacagaaccagctgtataacgagctgaatctgggcccggag
agaggagtacgactgctggataagaggcggggccgggaccccg
agatgggaggcaagcctcgagaaagaacccacaggagggcctg
tacaatgagctgcaaaaggacaagatggccgaggcctattctga
gatcgccatgaagggagagagggcgccggggcaaggacacgatg

8

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gcctgtaccaggcgctgagcaccgccacaaaggacacctatgat
gcctgcacatgcaggccctgccccctagatga.

The disclosure provides a chimeric antigen receptor (CAR) comprising: (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. As used throughout the disclosure, a CAR comprising a VH is referred to as a VCAR. In certain embodiments, the antigen recognition region may comprise two VHs to produce a bi-specific or tandem VCAR. In certain embodiments, the antigen recognition region may comprise three VHs to produce a tri-specific VCAR. In certain embodiments, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain. In certain embodiments, the ectodomain may further comprise a signal peptide. Alternatively, or in addition, in certain embodiments, the ectodomain may further comprise a hinge between the antigen recognition region and the transmembrane domain.

In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH comprises a human or a humanized sequence.

In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH comprises a non-naturally occurring sequence.

In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH is not naturally occurring.

In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH comprises a recombinant or chimeric sequence.

In certain embodiments of the VCARs of the disclosure, including those comprising an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH, the VH is produced by an in vitro procedure of affinity selection and recombination.

In certain embodiments of the VCARs of the disclosure, the VH comprises or consists of the amino acid sequence

(VH-A; SEQ ID NO: 18000)

malpvtalllplalllhaarpevqllesggglvqpgslrlsca
asgftfssyamnvrqapgkglewvagiisgggstyyadsvkgr
fsisrdnskntldlqmnsraedtavyyvcvkdwnntmiterggg
tlvtvssttpprptpaptiasqplslrpeacrpaaggavht
rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk
qpfmrpvqttqeedgcscrfeeeggcelrvkfsrsadapayk
qqgnqlynelnlgrreeydvldkrrgrdpemggkprknpqegl

9

-continued

ynelqkdkmaeyseigmkgerrrgkghdglyqglstatktdtyd

alhmqalppr

or the VH comprises or consists of the nucleic acid sequence 5

(VH-A; SEQ ID NO: 18001)

atggctctgctgtgacagctctgctgctgctctggctctgct

tcttcatgcggcgccctgaagttcagctgcttgaatctggcg

gaggcctggttcaacctggcggatctctgagactgagctgtgcc

gccagcggcttcacctttagcagctacgccatgaactgggtccg

acagggccctggcaaaggactggaatgggtggcgggaatcatcg

gcagcggcggcagcacatattacgccgattctgtgaaggccgc

ttcagcatcagccgggacaacagcaagaacacccctggacctgca

gatgaacagcctgagagccgagataaccgccgtgtactactgcg

tgaaggattggaacaccaccatgatcaccgagagagggcagggc

acactgggtcacctgtcctctacaacaacacggcgccctcgcc

tccaacaccagctcctacaatcgcgagtcagccctgtctctca

gacccgaagcctgtagacctgctgctggcgagctgtgcatacc

agaggactggatttcgcctgcgacatctacatctgggtcctct

ggctggcacatgcggagttttgtgctgagcctgggtcatcacc

tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag

cagcccttcagagaccctgcagaccacacaggaggaggacgg

ctgctcttgtaggttcccagagaggaggaggaggatgcgagc

tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag

cagggacagaaccagctgtataacgagctgaatctgggcccggag

agaggagtacgacgtgctggataagaggcggggccgggaccccg

agatgggaggcaagcctcggagaaagaacccacaggaggccctg

tacaatgagctgcaaaaggacaagatggccgaggcctattctga

gatcggcatgaaggagagaggcgccggggcaaggacacgatg

gcctgtaccaggcctgagcaccgccacaaaggacacctatgat

gccctgcacatgcaggccctgccccctagatga.

In certain embodiments of the VCARS of the disclosure,
the VH comprises or consists of the amino acid sequence

(VH-B; SEQ ID NO: 18002)

malpvtalllplalllhaarpevqllesggglvqpggsltlsc

asgtttsnyamnwwrqpapkglewsgngsgatttyadsvkgrt

tisrdnsntlnlqmnsraedtaiyycvkdwnntmitergggt

lvtvssttpprptpaptiasqplslrpeacrpaaggavhtr

gldfacdiyiwaplagtcgvllslvitlyckrgrklllyifkq

pfmrpvqttqeedgcscrpfpeeeeggcelrvkfssadapaykq

ggnqlynelnlgrreeydvldkrrgrdpemggkprrknpqegly

nelqkdkmaeyseigmkgerrrgkghdglyqglstatktdtyda

alhmqalppr

10

or the VH comprises or consists of the nucleic acid sequence

(VH-B; SEQ ID NO: 18003)

atggctctgctgtgacagctctgctgctgctctggctctgct

tcttcatgcggcgccctgaagttcagctgcttgaatctggcg

gaggcctggttcaacctggcggatctctgacactgagctgtgcc

gccagcggcttcaccttcagcaactacgccatgaactgggtccg

acagggccctggcaaaggccttgaatgggtgtccggcatcattg

gctctggcgccaccacactactacgccgattctgtgaaggcgaga

ttcaccatcagccgggacaacagcaagaacacccctgaacctgca

gatgaacagcctgagagccgaggacaccgccatctactactgcg

tgaaggactggaacaccaccatgatcaccgagagaggccagggc

acactgggtcacctgtcctctacaacaacacggcgccctcgcc

tccaacaccagctcctacaatcgcgagtcagccctgtctctca

gacccgaagcctgtagacctgctgctggcgagctgtgcatacc

agaggactggatttcgcctgcgacatctacatctgggtcctct

ggctggcacatgcggagttttgtgctgagcctgggtcatcacc

tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag

cagcccttcagagaccctgcagaccacacaggaggaggacgg

ctgctcttgtaggttcccagaggaggaggaggaggatgcgagc

tgcgcgtgaagtttagccggtccgccgatgcacctgcatacaag

cagggacagaaccagctgtataacgagctgaatctgggcccggag

agaggagtacgacgtgctggataaaggcggggccgggaccccg

agatgggaggcaagcctcggagaaagaacccacaggaggccctg

tacaatgagctgcaaaaggacaagatggccgaggcctattctga

gatcggcatgaaggagagaggcgccggggcaaggacacgatg

gcctgtaccaggcctgagcaccgccacaaaggacacctatgat

gccctgcacatgcaggccctgccccctagatga.

In certain embodiments of the VCARS of the disclosure,
the VH comprises or consists of the amino acid sequence

(VH-C; SEQ ID NO: 18004)

malpvtalllplalllhaarpevqllesggglvqpggslrlsca

asgfttsnyamnwwrqpapkglewsgivgggtsyyadsvrgr

ftisrdnsntlnlqmnsraedtavyyvcvkdwnntmiterggg

tlvtvssttpprptpaptiasqplslrpeacrpaaggavht

rgldfacdiyiwaplagtcgvllslvitlyckrgrklllyifk

qpfmrpvqttqeedgcscrpfpeeeeggcelrvkfssadapayk

qqnqlynelnlgrreeydvldkrrgrdpemggkprrknpqegl

ynelqkdkmaeyseigmkgerrrgkghdglyqglstatktdtyd

alhmqalppr

11

or the VH comprises or consists of the nucleic acid sequence

(VH-C; SEQ ID NO: 18005)

atggctctgctgtgacagctctgctgctgctctgctctgct 5
 tcttcatgcggcgcgccctgaagttcagctgcttgaatctggcg
 gaggcctggttcagctggcgaatctctgagactgagctgtgcc
 gccagcggttcaccttcagcaactacgccatgaactgggtccg 10
 acaggccctggcaaaggccttgaatgggtgtccggaatcgttg
 gcggcgaggcacaagctactacgccgattctgtgcggggcaga
 ttaccatcagccgggacaacagcaagaacacctgtacctgca 15
 gatgaacagcctgagagccgaggacaccgctgtactactgcg
 tgaaggactggaacaccaccatgatcaccgagagagccagggc
 aactggtcacctgtcctctacaacaacaccggcgccctcgcc 20
 tccaacaccagctcctacaatcgcgagtcagccctgtctctca
 gaccgaagcctgtagacctgctgctggcgagctgtgcatacc
 agaggactggatttcgctgcgacatctacatctgggctcctct 25
 ggctggcacatgcggagttttgctgctgagcctggtcatcacc
 tgtactgtaagagaggcaggaagaagctgctgtatatcttcaag
 cagcccttcagagaccgtgcagaccacagaggaggagcgg
 ctgctctttaggttcccagaggaggaggaggagatgcgagc 30
 tgcgctgaagttagccggtccgccgatgcacctgcatacaag
 cagggacagaaccagctgtataacgagctgaatctggccggag
 agaggagtacgagctgctgataagaggcgggccgggaccccg
 agatgggaggcaagcctcgagaaagaacccacaggaggcctg 35
 tacaatgagctgcaaaaggacaagatggccgaggcctattctga
 gatcgcatgaaggagagaggcgccggggcaaggacacgatg
 gcctgtaccagggcctgagcaccgccacaaaggacacctatgat 40
 gccctgcacatgcaggccctgccccctagatga .

In certain embodiments of the VCARS of the disclosure, the VH comprises or consists of the amino acid sequence

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(VH-D; SEQ ID NO: 18006)

malpvtalllplalllhaarpevqllesggglvqpgslrlsca
 asgftfnyamtwirqapgkglewsgitgdggstfyadsvkgr 55
 ftisrdnskntlylqmnsdraedavyyvcvkdwnntmiterggg
 tlvtvssttppaprpptpaptiasqplslrpeacrpaaggavht
 rgldfacdiyiwaplagtcgvllslvitlyckrgrkkllyifk 60
 qpfmrpvqttqeedgcscrfeeeeeggcclrvkfssadapayk
 qggnqlynelnlgrreeydvldkrrgrdpemggkprknpqegl
 ynelqkdmaeyseigmkgerrrgkghdglyqglstatkdyd
 alhmqalppr 65

12

or the VH comprises or consists of the nucleic acid sequence

(VH-D; SEQ ID NO: 18007)

atggctctgctgtgacagctctgctgctgctctgctctgct 5
 ttcttcatgcggcgcgccctgaagttcagctgcttgaatctgg
 cggaggcctggttcaacctggcggtctctgagactgagctgt
 gccgccagcggttcaccttcagcaattacgccatgacctgga 10
 tcagacaggccctggcaaaggcctggaatgggtgtccggaat
 tacaggcgacggcggcagcaccttttacgccgattctgtgaag
 ggcagattcaccatcagccgggacaacagcaagaacacctgt 15
 acctgcagatgaacagcctgagagccgaggacaccgctgtga
 ctactgctgaaggactggaacaccaccatgatcaccgagaga
 ggccagggcacactgggtcacctgtcctctacaacaacaccgg 20
 cgctcgccctccaacaccagctcctacaatcgcgagtcagcc
 cctgtctctcagaccggaagcctgtagacctgctgctggcgga
 gctgtgcataccagaggactggatttcgctgcgacatctaca 25
 tctgggctcctctggctggcacatgcggagttttgctgctgag
 cctggtcatcaccctgtactgtaagagaggcaggaagaagctg
 ctgtatatcttcaagcagcccttcagagaccctgcagacca 30
 cacaggaggaggacggctgctctttaggttcccagaggagga
 ggaggaggagatgcgagctgcgctggaagtttagccggtccgcc
 gatgcacctgcatacaagcagggacagaaccagctgtataacg 35
 agctgaatctgggcccggagagaggagtacgacgtgctggataa
 gaggcggggccgggaccccgagatgggaggcaagcctcggaga
 aagaacccacaggaggcgctgtacaatgagctgcaaaaggaca 40
 agatggccgaggcctattctgagatcgccatgaaggagagag
 gcgcccgggcaagggacacgatggcctgtaccagggcctgagc
 accgccacaaaggacacctatgatccctgcacatgcaggccc 45
 tgcctcctagatga .

In certain embodiments of the VCARS of the disclosure, the VH comprises or consists of the amino acid sequence

50

(VH-F; SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglvqpgslrlsc
 aasgftfnyamnvrqapgkglewsgisggggstyyadsvk 55
 grftisrdnskntlylqmnsdraedavyyvcvkdwnntmiter
 gqgtlvtvssttppaprpptpaptiasqplslrpeacrpaagg
 avhtrgldfacdiyiwaplagtcgvllslvitlyckrgrkkl 60
 lyifkqpfmrpvqttqeedgcscrfeeeeeggcclrvkfssaa
 dapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr
 knpqqglynelqkdmaeyseigmkgerrrgkghdglyqgl
 tatkdtydalhmqalppr 65

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or the VH comprises or consists of the nucleic acid sequence

(VH-E; SEQ ID NO: 18009)

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atggcactgcctgtgacagccctgctgctgctctggccctg
ctgctgcacgcagcacggcccgaggtgcagctgctggagtcc
ggaggaggcctggcccagcctggcggcagcctgaggtgtcc
tgcgcgcctctggcttcacctttagctcctacgccatgaac
tggatcagacagggccctggcaaggcctggagtgggtgtcc
ggcatctccggtcttgaggctctacatactatgccgacagc
gtgaaggggcgggttcaccatcagcagagataaactccaagaat
accgtgtacctccagatgaactctctcgggcgaggacacc
gccgtgtactattgctgaaggattggaataccacaatgatc
acagagagggggccaggccacctggtagctgtctagcacc
acaacccctgccccagacctccacacccgccccctaccatc
gcgagtccagcactgtccctgcggtcctgaggcctgccggccc
gccgccggcgagcagtgacacacggggcctggactttgcc
tgtgacatctacatatgggcaccactggcaggaacctgcggc
gtgctgctgctgagcctggtcatcacctgtactgtaagaga
ggcaggaagaagctgctgtatatcttcaagcagcccttcag
agaccctgacagaccacacaggaggaggagcggctgctctgt
agggtcccagaggaggaggaggaggatgagctgcgcgtg
aagtttagccggtccgcatgacctgcatacaagcaggga
cagaaccagctgtataacgagctgaatctggcgagagag
gagtacgagctgctggataaagaggcgggcgggaccccgag
atgggagggaagcctcgagaaagaacccacaggaggcctg
tacaatgagctgaaaaggacaagatggcgaggcctattct
gagatcgcatgaaggagagagggcggggcaaggacac
gatggcctgtaccaggcctgagcaccgccacaaaggacacc
tatgatgccctgcacatgcaggccctgccccctagatga.

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In certain embodiments of the VCARS of the disclosure, the VH comprises or consists of the amino acid sequence

(VH-F; SEQ ID NO: 18010)

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malpvtalllplalllhaarpevqllesgglyqgrldscasgftft
nyamnwyrgapkglewysgisggggstyyadsvkgrftisrdsnkntly
lqmnsrlaedtavyyvkdwnntmitergggtlvtssttpaprpptpa
ptiasqplslrpeacrpaaggavhtrglfdacdiyiwaplgtcgvlills
lvitlyckrgrklllyifkqpfmrpvqttgeedgcscrfeeeggcelr
vkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr
knpqeglynelqkdkmaeyseigmkgerrrgkghdglyqglstatkdy
dalhmqalppr

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14

or the VH comprises or consists of the nucleic acid sequence

(VH-F; SEQ ID NO: 18011)

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atggcactgcctgtgacagccctgctgctgctctggccctgctgctgca
cgcagcacggcccgaggtgcagctgctggagtctggaggaggcctggtagc
agcccgccgggtccctgagactgtcttgccggccagcggttcaccttt
acaaactacgccatgaattgggtgcggcagggccctggcaaggcctgga
gtgggtgtctggcatcagcgaggaggaggcagcacctactatgcagact
ccgtgaaggggcaggttcaccatctcccgcgataaactctaagaatacactg
tacctccagatgaacagcctgagggcagaggacaccccgctgtactattg
cgtgaaggattggaataaccacaatgatcacagagaggggacaggggacccc
tggtgaccgtgagcagcaccacaacccctgccccagacctccacacccc
gccccctaccatcgcgagtcagccactgtccctgcggtcctgaggcctgccg
gcccccgccggcgagcagtgacacacggggcctggactttgcctgtg
acatctacatatgggcaccactggcaggaacctgcggcgtgctgctgctg
agcctggtcatcacctgtactgtaagagaggcaggaagaagctgctgta
tatcttcaagcagcccttcagtagaccctgacagaccacacaggaggagg
acggctgctctttaggttcccagaggaggaggaggaggatgagcagctg
cgctgaagttagccggtccgcatgacactgcatacaagcaggggaca
gaaccagctgtataacgagctgaatctggcgagagaggagtacgacg
tgctggataaagggcgggggcgggaccccgagatgggaggcaagcctcgg
agaaagaacccacaggaggggcctgtacaatgagctgcaaaaggacaagat
ggccgaggcctattctgagatcgcatgaaggagagaggcgccgggggca
agggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacc
tatgatgccctgcacatgcaggccctgccccctagatga.

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In certain embodiments of the VCARS of the disclosure, the VCAR comprises a single domain antibody, VHH, VH or a combination thereof. In some embodiments, the single domain antibody, VHH or VH comprises or consists of a recombinant sequence and/or a chimeric sequence. In some embodiments, the single domain antibody, VHH or VH comprises or consists of a human sequence and/or a humanized sequence.

In certain embodiments of the VCARS of the disclosure, the VCAR comprises a single domain antibody. In some embodiments, the single domain antibody is a VHH or a VH antibody. In some embodiments, the VH antibody is a UniDab antibody. In some embodiments, VH antibody is not a fragment of a naturally occurring monoclonal antibody.

In certain embodiments of the VCARS of the disclosure, the signal peptide may comprise a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR signal peptide. In certain embodiments of the VCARS of the disclosure, the signal peptide may comprise a sequence encoding a human CD8α signal peptide. The human CD8α signal peptide may comprise an amino acid sequence comprising MALPVTALLLPLALLLHAARP (SEQ ID NO: 18012). The human CD8α signal peptide may comprise an amino acid sequence comprising MALPVTALLLPLALLLHAARP (SEQ ID NO: 18012) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the an amino acid sequence comprising MALPVTALLLPLALLLHAARP (SEQ ID NO: 18012).

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The human CD8 α signal peptide may be encoded by a nucleic acid sequence comprising atggcactgccagtcaccgccctgctgctcctgctgctgctgctgcacgcagctagacca (SEQ ID NO: 18013).

In certain embodiments of the VCARS of the disclosure, the transmembrane domain may comprise a sequence encoding a human CD2, CD3 δ , CD3 ϵ , CD3 γ , CD3 ζ , CD4, CD8 α , CD19, CD28, 4-1BB or GM-CSFR transmembrane domain. In certain embodiments of the VCARS of the disclosure, the transmembrane domain may comprise a sequence encoding a human CD8 α transmembrane domain. The CD8 α transmembrane domain may comprise an amino acid sequence comprising IYIWAPLAGTCGVLLSLVITLYC (SEQ ID NO: 18014) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising IYIWAPLAGTCGVLLSLVITLYC (SEQ ID NO: 18014). The CD8 α transmembrane domain may be encoded by the nucleic acid sequence comprising atctacattgggcacactggccgggacgtgtggagtgtgctgctgctgagcctggatcacactgtactgc (SEQ ID NO: 18015).

In certain embodiments of the VCARS of the disclosure, the endodomain may comprise a human CD3 ζ endodomain.

In certain embodiments of the VCARS of the disclosure, the at least one costimulatory domain may comprise a human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. In certain embodiments of the VCARS of the disclosure, the at least one costimulatory domain may comprise a CD28 and/or a 4-1BB costimulatory domain. The CD3zeta costimulatory domain may comprise an amino acid sequence comprising RVKFSRSADAPAYKQGQNQLYNELNLGRREEY-DVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHGDLGYQGLSTATKDTYDALHMQALPPR (SEQ ID NO: 18016) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising RVKFSRSADAPAYKQGQNQLYNELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQEGLYNELQKDKMAEAYSEIGMKGERRRGKGHGDLGYQGLSTATKDTYDALHMQALPPR (SEQ ID NO: 18016). The CD3zeta costimulatory domain may be encoded by the nucleic acid sequence comprising cgcgtgaagtttagtcgacagcagatgccccagcttaacaacaggacagacaggctgtataacgagctgaatctggccgccgagag gaatatgacgtgctgga-taaggcggagaggacgcgaccccgaaatgggaggaagccagcgcca-aaaacctcaggaaggcctgtat aacgagctgcagaaggacaaaatggcagaagcctattgagatcggcatgaagggggagcgcagcagaggaagggcac-gatgg gctgtaccaggagcactgagcaccgccacaagacacatgatgctctgcatatgcaggcactgctccaagg (SEQ ID NO: 18017). The 4-1BB costimulatory domain may comprise an amino acid sequence comprising KRGRKKLLYIFKQPFMRPVQ-TTQEEDGCSCRFPEEEEEGGCEL (SEQ ID NO: 18018) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising KRGRKKLLYIFKQPFMRPVQTTQEEDGCSCRFPEEEEEGGCEL (SEQ ID NO: 18018). The 4-1BB costimulatory domain may be encoded by the nucleic acid sequence comprising aagagaggcaggaagaaactgctgtatatatttcaaacagccctcatgcgcc-cgtgcagactaccaggaggagagcagggtgctcctgt cgattccagaggaaggaaggcgggtgtgagctg (SEQ ID NO: 18019). The 4-1BB costimulatory domain may be located between the transmembrane domain and the CD28 costimulatory domain.

In certain embodiments of the VCARS of the disclosure, the hinge may comprise a sequence derived from a human CD8 α , IgG4, and/or CD4 sequence. In certain embodiments of the VCARS of the disclosure, the hinge may comprise a sequence derived from a human CD8 α sequence. The hinge

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may comprise a human CD8 α amino acid sequence comprising TTTAPRPPPTAPTASQPLSLRPEACR-PAAGGAVHTRGLDFACD (SEQ ID NO: 18020) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising TTTAPRPPPTAPTASQPLSLRPEACRPAAGGAVHTRGLDFACD (SEQ ID NO: 18020). The human CD8 α hinge amino acid sequence may be encoded by the nucleic acid sequence comprising

(SEQ ID NO: 18021)

actaccacaccagcaccctagaccaccaactccagctccaaccatgcgcag

tcagccctgagctgtagacgtgaggcctgcaggccagctgcaggaggag

ctgtgcacaccaggggcctggacttcgctgcgcac.

VHHs and/or VCARS of the disclosure may bind an antigen with at least one affinity selected from a K_D of less than or equal to 10^{-9} M, less than or equal to 10^{-10} M, less than or equal to 10^{-11} M, less than or equal to 10^{-12} M, less than or equal to 10^{-13} M, less than or equal to 10^{-14} M, and less than or equal to 10^{-15} M. The K_D may be determined by surface plasmon resonance.

The disclosure provides an anti-BCMA VCAR. The disclosure provides a composition comprising the VCAR of the disclosure and at least one pharmaceutically acceptable carrier.

The disclosure provides a transposon comprising the VCAR of the disclosure.

Transposons of the disclosure may comprise a selection gene for identification, enrichment and/or isolation of cells that express the transposon. Exemplary selection genes encode any gene product (e.g., transcript, protein, enzyme) essential for cell viability and survival. Exemplary selection genes encode any gene product (e.g., transcript, protein, enzyme) essential for conferring resistance to a drug challenge against which the cell is sensitive (or which could be lethal to the cell) in the absence of the gene product encoded by the selection gene. Exemplary selection genes encode any gene product (e.g., transcript, protein, enzyme) essential for viability and/or survival in a cell media lacking one or more nutrients essential for cell viability and/or survival in the absence of the selection gene. Exemplary selection genes include, but are not limited to, neo (conferring resistance to neomycin), DHFR (encoding Dihydrofolate Reductase and conferring resistance to Methotrexate), TYMS (encoding Thymidylate Synthetase), MGMT (encoding O(6)-methylguanine-DNA methyltransferase), multidrug resistance gene (MDR1), ALDH1 (encoding Aldehyde dehydrogenase 1 family, member A1), FRANCF, RAD51C (encoding RAD51 Paralog C), GCS (encoding glucosylceramide synthase), and NKX2.2 (encoding NK2 Homeobox 2).

Transposons of the disclosure may comprise an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the ligand binding region may be a multimeric ligand binding region. Inducible proapoptotic polypeptides of the disclosure may also be referred to as an "iC9 safety switch". In certain embodiments, transposons of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and

(c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, transposons of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, transposons of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand-binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the amino acid sequence of the ligand binding region that comprise a FK506 binding protein 12 (FKBP12) polypeptide may comprise a modification at position 36 of the sequence. The modification may be a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising GVQVETISPGDGRITFPKRQTCVVHYTGMLDGGK-VDSSRDRNPKPFKMLGKQEVIRG WEEGVAQMSV-GQRAKLTISPDYAYGATGHPGIIPPHATLVFDVELL-LE (SEQ ID NO: 18022). In certain embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising GGGGTCCAGGTCGAGACTATTTACCA-GGGGATGGGCGAACATTTCCAAAAAGGGG CCA-GACTTGCGTCGTGCATTACACCGGGATGCTGGAG-GACGGGAAGAAAGTGGACA GCTCCAGGGATCG-CAACAAGCCCTTCAAGTTTCATGCTGGGAAAGCAG-GAAGTGATC CGAGGATGGGAGGAAGGCGTGGC-ACAGATGTCAGTCGGCCAGCGGGCCAAACTGA CCATTAGCCCTGACTACGCTTATGGAGCAACAGGC-CACCCAGGGATCATTCCCCCTC ATGCCACCCTGG-TCTTCGAT GTGGAAGTCTGAAGCTGGAG (SEQ ID NO: 18023). In certain embodiments, the induction agent specific for the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V) comprises AP20187 and/or AP1903, both synthetic drugs.

In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 18024) or a nucleic acid sequence comprising GGAG-GAGGAGGATCC (SEQ ID NO: 18025). In certain embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

In certain embodiments of the truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. Alternatively, or in addition, in certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 of the sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(SEQ ID NO: 18026)

GFGDVGALES LRGNADLAYILSMEPCGHCLIIINN VNF CRESGLRTRTGSN

5 IDCEKLRRRFSSSLHFMVEVKGDLTAKKMLALLELAQQDHGALDCCVVVI

LSHGCQASHLQFPGAVYGTGDCPVSVKIVNIFNGTSCPSLGGKPKLFFI

10 QACGGEQKDHGFEVASTSPEDES PGSNPEPDATPFQEGRLRTFDQLDAISS

LPTPSDIFVSYSTFPFVSWRDPKSGSWYVETLDDIFEQWAHSEDLQSL

15 LRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS

a nucleic acid sequence comprising

(SEQ ID NO: 18027)

20 TTTGGGGACGTGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGATCTGGC

TTACATCCTGAGCATGGAACCTGCGCCACTGTCTGATCATTAAACATG

TGAACCTCTGCAGAGAAAGCGGACTGCGAACACGAGCTGGCTCCAATATT

25 GACTGTGAGAAGCTGCGGAGAAGTTCTCTAGTCTGCACTTTATGGTCGA

AGTGAAAGGGGATCTGACCGCCAAGAAATGGTGTGGCCCTGCTGGAGC

TGGCTCAGCAGGACCATGGAGCTCTGGATTGCTGCGTGGTCTGATCCTG

30 TCCCACGGGTGCCAGGCTTCTCATCTGCAGTTCCCGGAGCAGGTACGG

AACAGACGGCTGTCTGTGACGCTGGAGAAGATCGTCAACATCTTCAACG

GCACCTTCTGCCCTAGTCTGGGGGAAAGC AAAACTGTTCTTTATCCAG

35 GCCTGTGGCGGGAAACAGAAAGATCACGGCTTCGAGGTGGCCAGCACCAG

CCCTGAGGACGAATCACAGGAGCAACCTGAACAGATGCAACTCCAT

TCCAGGAGGGACTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTG

40 CCCACTCCTAGTGACATTTTCGTGTCTTACAGTACCTTCCAGGCTTTGT

CTCATGCGCGATCCCAAGTCAGGAGCTGGTACGTGGAGACTGGAGC

ACATCTTTGAACAGTGGGCCATTGAGAGGACCTGCAGAGCCTGCTGCTG

CGAGTGGCAAACGCTGTCTCTGTGAAGGCATCTACAAACAGATGCCCGG

45 GTGCTTCAATTTCTGAGAAAGAACTGTTCTTTAAGACTTCC.

In certain embodiments of the inducible proapoptotic polypeptides, wherein the polypeptide comprises a truncated caspase 9 polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(SEQ ID NO: 18028)

55 GVQVETISPGDGRITFPKRQTCVVHYTGMLDGGKVDSSRDRNPKPFKM

LGKQEVIRGWEEGVAQMSVQRAKLTISPDYAYGATGHPGIIPPHATLVF

DVELLLEGGGGSGFGDVGALES LRGNADLAYILSMEPCGHCLIIINN VNF

CRESGLRTRTGSNIDCEKLRRRFSSSLHFMVEVKGDLTAKKMLALLELAQ

60 QDHGALDCCVVVILSHGCQASHLQFPGAVYGTGDCPVSVKIVNIFNGTS

CPSLGGKPKLFFIQACGGEQKDHGFEVASTSPEDES PGSNPEPDATPFQE

GLRTFDQLDAISSLPTPSDIFVSYSTFPFVSWRDPKSGSWYVETLDDIF

65 EQWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS

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or the nucleic acid sequence comprising

(SEQ ID NO: 18029)
 ggggtccaggtcgagactatttaccaggggatggcgaaacatttccaaa
 aagggggccagacttcgctcgtgattacaccgggatgctggaggacggga
 agaaagtggacagctccagggatcgcaacaagcccttcaagttcatgctg
 ggaaagcaggaagtgatccgagggatgggaggaaggcgtggcacagatgct
 agtcggccagcgggccaactgaccattagccctgactacgcttatggag
 caacaggccacccagggatcattccccctcatgccaccctggtcttcgat
 gtggaactgctgaagctggaggaggaggaggatccgatttggggacgt
 gggggccctggagtctctcgaggaaatgccgatctggtttacatcctga
 gcatggaaccctgcccactgtctgatcattaacaatgtgaacttctgc
 agagaaagcggactcgcaacacggactggctccaatattgactgtgagaa
 gctcgaggagaaggttctctagtctgcactttatggtcgaagtgaagggg
 atctgaccgccaagaaaatggtgctggccctgctggagctggctcagcag
 gaccatggagctctggattgctgctggtgctgatcctgtcccacgggtg
 ccaggcttctcatctgcagttccccggagcagtgtaggaacagacggct
 gtctgtcagcgtggagaagatcgtcaacatcttcaacggcacttcttgc
 cctagtctgggggaaagccaaactgttctttatccaggcctgtggcgg
 ggaacagaaagatcacggtctcgaggtggccagcaccagcctgaggagc
 aatcaccagggagcaaccctgaaccagatgcaactccattccaggaggga
 ctgaggacctttgaccagctggatgctatctcaagcctgccactcctag
 tgacattttctgtcttacagtaccttccaggctttgtctcatggcgcg
 atcccaagtcaggagctggtacgtggagacactggacgacatcttgaa
 cagtgggcccatctagaggacctgcagagcctgctgctgcgagtgga
 cgctgtctctgtgaaggcgcattacaaacagatgccgggtgcttcaatt
 ttctgagaagaaactgttctttaagacttcc.

Transposons of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more VHH(s) or VCAR(s) of the disclosure and a selection gene of the disclosure. Transposons of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more VHH(s) or VCAR(s) of the disclosure and an inducible proapoptotic polypeptide of the disclosure. Transposons of the disclosure may comprise at least two self-cleaving peptide(s), a first self-cleaving peptide located, for example, upstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure and a second first self-cleaving peptide located, for example, downstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure.

The at least one self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence com-

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prising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagagggaagggaagcctgctgacctgtgga-gacgtggaggaaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

Transposons of the disclosure may comprise a first and a second self-cleaving peptide, the first self-cleaving peptide located, for example, upstream of one or more VHH(s) or VCAR(s) of the disclosure the second self-cleaving peptide located, for example, downstream of the one or more VHH(s) or VCAR(s) of the disclosure. The first and/or the second self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagagggaagggaagcctgctgacctgtggagacgtggag-gaaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence

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comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

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Transposons of the disclosure may comprise piggyBac transposons. In certain embodiments of this method, the transposon is a plasmid DNA transposon with a sequence encoding the chimeric antigen receptor flanked by two cis-regulatory insulator elements. In certain embodiments, the transposon is a piggyBac transposon. Transposase enzymes of the disclosure may include piggyBac transposases or compatible enzymes. In certain embodiments, and, in particular, those embodiments wherein the transposon is a piggyBac transposon, the transposase is a piggyBac or a Super piggyBac (SPB) transposase. In certain embodiments, and, in particular, those embodiments wherein the transposase is a Super piggyBac (SPB) transposase, the sequence encoding the transposase is an mRNA sequence.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme. The piggyBac (PB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14487)

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1  MGSSLDDEHI  LSALLQSDDE  LVGEDSDSEI  SDHVSDDVQ  SDTEEFIDE  VHEVQPTSSG
61  SEILDEQNV I  EQPGSSLASN  RILTLPORTI  RGKNKHCWST  SKSTRRSRVS  ALNIVRSQRG
121 PTRMCRNIYD  PLLCFKLFFT  DEIISEIVKW  TNAEISLKRR  ESMGTATFRD  TNEDEIYAFF
181 GILVMTAVRK  DNHMSTDDL F  DRSLSMVYVS  VMSRDRDFDL  IRCLRMDDKS  IRPTLRENDV
241 FTPVRKIWDL  FIHQCIQNYT  PGAHLTIDEQ  LLGFRGRCPF  RMYIPNPKSK  YGKILMMCD
301 SGTKYMINGM  PYLGRGTQTN  GVPLGEYVVK  ELSKPVHGSC  RNITCDNWFT  SIPLAKNLLQ
361 EPYKLTIVGT  VRSNKREIPE  VLKNSRSRPV  GTSMFCDGDP  LTLVSYKPKP  AKMVYLLSSC
421 DEDASINEST  GKPQMVMYYN  QTKGGVDTLD  QMCSVMTCSR  KTNRWPBALL  YGMINIACIN
481 SFIIYSHNVS  SKGEKVQSRK  KPMRNLYMSL  TSSPMRKRLE  APTLKRYLRD  NISNILPNEV
541 PGTSDDSTEE  PVMKRTYCT  YCPSKIRKKA  NASCKKCKKV  ICREHNIDMC  QSCF.

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The disclosure provides a composition comprising the transposon the disclosure. In certain embodiments, the composition may further comprise a plasmid comprising a sequence encoding a transposase enzyme. The sequence encoding a transposase enzyme may be an mRNA sequence.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at one or more of positions 30, 165, 282, or 538 of the sequence:

(SEQ ID NO: 14487)

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1  MGSSLDDEHI  LSALLQSDDE  LVGEDSDSEI  SDHVSDDVQ  SDTEEFIDE  VHEVQPTSSG
61  SEILDEQNV I  EQPGSSLASN  RILTLPORTI  RGKNKHCWST  SKSTRRSRVS  ALNIVRSQRG
121 PTRMCRNIYD  PLLCFKLFFT  DEIISEIVKW  TNAEISLKRR  ESMGTATFRD  TNEDEIYAFF
181 GILVMTAVRK  DNHMSTDDL F  DRSLSMVYVS  VMSRDRDFDL  IRCLRMDDKS  IRPTLRENDV
241 FTPVRKIWDL  FIHQCIQNYT  PGAHLTIDEQ  LLGFRGRCPF  RMYIPNPKSK  YGKILMMCD
301 SGTKYMINGM  PYLGRGTQTN  GVPLGEYVVK  ELSKPVHGSC  RNITCDNWFT  SIPLAKNLLQ
361 EPYKLTIVGT  VRSNKREIPE  VLKNSRSRPV  GTSMFCDGDP  LTLVSYKPKP  AKMVYLLSSC
421 DEDASINEST  GKPQMVMYYN  QTKGGVDTLD  QMCSVMTCSR  KTNRWPBALL  YGMINIACIN

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481 SFIIYSHNV SKGEKVQSRK KPMRNLYMSL TSSFMKRRL APTLKRYLRD NISNILPNEV
541 PGTSDSDSTEE PVMKKRTYCT YCPSKIRKKA NASCKKCKKV ICREHNIDMC QSCF.

In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at two or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at three or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at each of the following positions 30, 165, 282, and 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the amino acid substitution at position 30 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 165 of the sequence of SEQ ID NO: 14487 is a substitution of a serine (S) for a glycine (G). In certain embodiments, the amino acid substitution at position 282 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 538 of the sequence of SEQ ID NO: 14487 is a substitution of a lysine (K) for an asparagine (N).

In certain embodiments of the methods of the disclosure, the transposase enzyme is a Super piggyBac (SPB) transposase enzyme. In certain embodiments, the Super piggyBac (SPB) transposase enzymes of the disclosure may comprise or consist of the amino acid sequence of the sequence of SEQ ID NO: 14487 wherein the amino acid substitution at position 30 is a substitution of a valine (V) for an isoleucine (I), the amino acid substitution at position 165 is a substitution of a serine (S) for a glycine (G), the amino acid substitution at position 282 is a substitution of a valine (V) for a methionine (M), and the amino acid substitution at position 538 is a substitution of a lysine (K) for an asparagine (N). In certain embodiments, the Super piggyBac (SPB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

The disclosure provides a vector comprising the VCAR of the disclosure. In certain embodiments, the vector is a viral vector. The vector may be a recombinant vector.

Viral vectors of the disclosure may comprise a sequence isolated or derived from a retrovirus, a lentivirus, an adenovirus, an adeno-associated virus or any combination thereof. The viral vector may comprise a sequence isolated or derived from an adeno-associated virus (AAV). The viral vector may comprise a recombinant AAV (rAAV). Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure comprise two or more inverted terminal repeat (ITR) sequences located in cis next to a sequence encoding a VHH or VCAR of the disclosure. Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure include, but are not limited to all serotypes (e.g., AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, and AAV9). Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure include, but are not limited to, self-complementary AAV (scAAV) and AAV hybrids containing the genome of one serotype and the capsid of another serotype (e.g., AAV2/5, AAV-DJ and AAV-DJ8). Exemplary adeno-associated viruses and recombinant adeno-associated viruses of the disclosure include, but are not limited to, rAAV-LK03.

Viral vectors of the disclosure may comprise a selection gene. The selection gene may encode a gene product essential for cell viability and survival. The selection gene may encode a gene product essential for cell viability and survival when challenged by selective cell culture conditions. Selective cell culture conditions may comprise a compound harmful to cell viability or survival and wherein the gene product confers resistance to the compound. Exemplary selection genes of the disclosure may include, but are not limited to, neo (conferring resistance to neomycin), DHFR (encoding Dihydrofolate Reductase and conferring resistance to Methotrexate), TYMS (encoding Thymidylate Synthetase), MGMT (encoding O(6)-methylguanine-DNA methyltransferase), multidrug resistance gene (MDR1), ALDH1 (encoding Aldehyde dehydrogenase 1 family, member A1), FRANCE, RAD51C (encoding RAD51 Paralog C),

(SEQ ID NO: 14484)

1 MGSSLDDEHI LSALLQSDDE LVGEDSDSEV SDHVEDDVQ SDTEEFIDE VHEVQPTSSG
61 SEILDEQNI EQPGSSLASN RILTLPORTI RGKNKHCWST SKSTRSRVS ALNIVRSQRG
121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR ESMTSATFRD TNEDEIYAFF
181 GILVMTAVRK DNHMSTDDLF DRSLSMVYVS VMSRDRFDLF IRCLRMDDKS IRPTLRENDV
241 FTPVRKIWL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF RVYIPNPKPSK YGIKILMMCD
301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ
361 EPYKLTIIVGT VRSNKREIPE VLKNSRSRPV GTSMFCFDGP LTLVSYPKPK AKMVYLLSSC
421 DEDASINEST GKPQVMVYYN QTKGGVDTLD QMCSVMTCSR KTNRWPALL YGMINIACIN
481 SFIIYSHNV SKGEKVQSRK KPMRNLYMSL TSSFMKRRL APTLKRYLRD NISNILPKEV
541 PGTSDSDSTEE PVMKKRTYCT YCPSKIRKKA NASCKKCKKV ICREHNIDMC QSCF.

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GCS (encoding glucosylceramide synthase), NKX2.2 (encoding NK2 Homeobox 2) or any combination thereof.

Viral vectors of the disclosure may comprise an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the ligand binding region may be a multimeric ligand binding region. Inducible proapoptotic polypeptides of the disclosure may also be referred to as an "iC9 safety switch". In certain embodiments, viral vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, viral vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the amino acid sequence of the ligand binding region that comprise a FK506 binding protein 12 (FKBP12) polypeptide may comprise a modification at position 36 of the sequence. The modification may be a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising GVQVETISPGDGRITPFRGQTCVVHYTGMLDQGGK-KLE (SEQ ID NO: 18022). In certain embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising GGGGTCCAGGTCGAGACTATTTACCA-GGGGATGGGCGAACATTTCACAAAAGGGG CCA-GAGCTGCGTCGTGCATTACACCGGGATGCTGGAG-GACGGGAAGAAAGTGGACA GCTCCAGGGATCG-CAACAAGCCCTTCAAGTTCATGCTGGGAAAGCAG-GAAGTGATC CGAGGATGGGAGGAAGGCGTGGCA-CAGATGTCAGTCGGCCAGCGGGCCAAACTGA CCATTAGCCCTGACTACGCTTATGGAGCAACAGGC-CACCCAGGGATCAATCCCCCTC ATGCCACCCTGG-TCTTCGAT GTGGAAGTCTGAAGCTGGAG (SEQ ID NO: 18023). In certain embodiments, the induction agent specific for the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V) comprises AP20187 and/or AP1903, both synthetic drugs.

In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 18024) or a nucleic acid sequence comprising GGAG-GAGGAGGATCC (SEQ ID NO: 18025). In certain embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

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In certain embodiments of the truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. Alternatively, or in addition, in certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 the sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(SEQ ID NO: 18026)

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GFGDVGALSLRGNADLAYILSMEPCGHCLIIINNVCRESGLRTRTGSN
IDCEKLRRRFFSSLHFMVEVKGDLTAKKMVLALLELAQQDHGALDCCVVVI
LSHGCQASHLQFPGAVYGTGDCPVSEKIVNIFNGTSCPSLGGKPKLFFI
QACGGEQKDHGFEVASTSPEDSPGSNPEPATPFQELRTFDQLDAISS
LPTPSDIFVSYSTFPGFVSWRDPKSGSWYVETLDDIFEQWAHSEDLQSL
LRVANAVSVKGIYKQMPGCFNFLRKKLFFKTS
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or a nucleic acid sequence comprising

(SEQ ID NO: 18027)

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TTTGGGGACGTGGGGCCCTGGAGTCTCTGCGAGGAATGCCGATCTGGC
TTACATCCTGAGCATGGAACCTGCGGCCACTGTCTGATCATTAAACATG
TGAACCTCTGCAGAGAAAGCGGACTGCGAACACGAGCTGGCTCCAATATT
GACTGTGAGAAGCTGCGGAGAAGTTCTCTAGTCTGCACCTTTATGGTCGA
AGTGAAAGGGGATCTGACCGCCAAGAAATGGTCTGGCCCTGCTGGAGC
TGGCTCAGCAGGACCATGGAGCTCTGGATTGCTGCGTGGTCTGATCCTG
TCCACGGGTGCCAGGCTTCTCATCTGCAGTTCCCGGAGCAGTGATACGG
AACAGACGGCTGTCTGTGAGCGTGGAGAAGATCGTCAACATCTTCAACG
GCACCTTCTTGCCCTAGTCTGGGGGAAAGCCAAACTGTTCTTTATCCAG
GCCTGTGGCGGGAACAGAAAGATCACGGCTTCGAGGTGGCAGCAGCAG
CCCTGAGGACGAATCACCAGGAGCAACCTGAACAGATGCAACTCCAT
TCCAGGAGGGACTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTG
CCCACTCCTAGTGACATTTTCGTGTCTTACAGTACCTTCCAGGCTTTGT
CTCATGGCGCGATCCCAAGTCAGGAGCTGGTACGTGGAGACTGGAGC
ACATCTTTGAACAGTGGGCCATTGAGAGACCTGCAGAGCCTGCTGCTG
CGAGTGGCAAACGCTGTCTCTGTGAAGGCATCTACAAACAGATGCCCGG
GTGCTTCAATTTTCTGAGAAAGAACTGTTCTTTAAGACTTCC.
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In certain embodiments of the inducible proapoptotic polypeptides, wherein the polypeptide comprises a truncated

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caspase 9 polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(SEQ ID NO: 18028)
 GVQVETISPGDGRTPFKRGQTCVVHYTGMLGDKKVDSSRDNRKPKFKML
 GKQEVIRGWEEGVAQMSVGQRAKLITSPDYAYGATGHPGIIPPHATLVFD
 VELLKLEGGGSGFGDVGALSLRGNADLAYILSMEPCGHCLIINNPNFC
 RESGLRTRTGSNIDCEKLRFRSSLHFMVEVKGDLTAKKMLALLELAQQ
 DHGALDCCVVVILSHGCQASHLQFPAGVYGTGCPVSVKIVNIFNGTSC
 PSLGGKPKLFFIQACGGEQKDHGFEVASTSPEDSPGSNPEPDATPFQEG
 LRTFDQLDAISSLPSTDFVSYSTFPGFVSWRDPKSGSWYVETLDDIFE
 QWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFRKKLFFKTS

or the nucleic acid sequence comprising

(SEQ ID NO: 18029)
 ggggtccaggctcgagactatctaccaggggatggcgaaactttccaaa
 aagggggcagacttcgctcgtgcatcaccgggatgctggaggacggga
 agaaagtggacagctccagggatcgcaacaagcccttcaagttcatgctg
 ggaaagcaggaagtgatccgaggatgggaggaagcgctggcacagatgtc
 agtcggccagcgggcaaaactgaccattagccctgactacgcttatggag
 caacaggccaccagggatcattccccctcatgccaccctggtcttcgat
 gtggaactgctgaagctggaggaggaggagatccgatttggggacgt
 gggggccctggagtctctcgaggaaatgccgatctggcttacatcctga
 gcatggaaccctcgccgactgtctgatcattaacaatgtgaacttctgc
 agagaaagcggactcgcaacacggactggctccaatattgactgtgagaa
 gctgcggaagaaggttctctagtctgcaatttatggtcgaagtgaagggg
 atctgaccgccaaagaaatggtgctggccctgctggagctggctcagcag
 gaccatggagctctggattgctgctggtgctgacccctgtcccacgggtg
 ccaggcttctcatctgcagttccccggagcagtgtaacggaacagacggct
 gtctgtcagcgtggagaagatcgtaacatcttcaacggcacttcttgc
 cctagtctgggggaaagccaaactgttctttatccaggcctgtggcg
 ggaacagaaagatcacggcttcgaggtggccagcaccagccctgaggacg
 aatcaccaggagcaaccctgaaccagatgcaactccattccaggaggga
 ctgaggacctttgaccagctggatgctatctcaagcctgccactcctag
 tgacattttcgtgtcttacagtaccttcccaggctttgtctcatggcgcg
 atcccaagtcaggagctggtacgtggagacactggacgacatctttaa
 cagtgggccattcagaggacctgcagagcctgctgctgagtgagtgaaa
 cgctgtctctgtgaaggcatctacaaacagatgcccggtgcttcaatt
 ttctgagaagaactgttctttaagacttcc.

Viral vectors of the disclosure may comprise at least one self-cleaving peptide. In some embodiments, the vector may comprise at least one self-cleaving peptide and wherein a self-cleaving peptide is located between a CAR and a selection gene. In some embodiments, the vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located upstream of a CAR and a second self-cleaving peptide is located downstream of a

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CAR. Viral vectors of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more of a VCAR, VCAR or VCAR of the disclosure and an inducible proapoptotic polypeptide of the disclosure.

5 Viral vectors of the disclosure may comprise at least two self-cleaving peptide(s), a first self-cleaving peptide located, for example, upstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure and a second first self-cleaving peptide located, for example, downstream
 10 or immediately upstream of an inducible proapoptotic polypeptide of the disclosure. The self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A
 15 peptide may comprise an amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030). A GSG-
 20 T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031). A
 25 GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagagggaagggaagcctgctgactgtgga-gacgtggaggaaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a
 30 sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID
 35 NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP
 40 (SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGD-
 45 VESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising
 50 ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence
 55 comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

The disclosure provides a vector comprising the VCAR of the disclosure. In certain embodiments, the vector is a nanoparticle. Exemplary nanoparticle vectors of the disclosure include, but are not limited to, nucleic acids (e.g., RNA, DNA, synthetic nucleotides, modified nucleotides or any combination thereof), amino acids (L-amino acids, D-amino acids, synthetic amino acids, modified amino acids, or any combination thereof), polymers (e.g., polymersomes), micelles, lipids (e.g., liposomes), organic molecules (e.g.,

carbon atoms, sheets, fibers, tubes), inorganic molecules (e.g., calcium phosphate or gold) or any combination thereof. A nanoparticle vector may be passively or actively transported across a cell membrane.

Nanoparticle vectors of the disclosure may comprise a selection gene. The selection gene may encode a gene product essential for cell viability and survival. The selection gene may encode a gene product essential for cell viability and survival when challenged by selective cell culture conditions. Selective cell culture conditions may comprise a compound harmful to cell viability or survival and wherein the gene product confers resistance to the compound. Exemplary selection genes of the disclosure may include, but are not limited to, neo (conferring resistance to neomycin), DHFR (encoding Dihydrofolate Reductase and conferring resistance to Methotrexate), TYMS (encoding Thymidylate Synthetase), MGMT (encoding O(6)-methylguanine-DNA methyltransferase), multidrug resistance gene (MDR1), ALDH1 (encoding Aldehyde dehydrogenase 1 family, member A1), FRANCE, RAD51C (encoding RAD51 Paralog C), GCS (encoding glucosylceramide synthase), NKX2.2 (encoding NK2 Homeobox 2) or any combination thereof.

Nanoparticle vectors of the disclosure may comprise an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the ligand binding region may be a multimeric ligand binding region. Inducible proapoptotic polypeptides of the disclosure may also be referred to as an "iC9 safety switch". In certain embodiments, nanoparticle vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, nanoparticle vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, nanoparticle vectors of the disclosure may comprise an inducible caspase polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments of the inducible proapoptotic polypeptides, inducible caspase polypeptides or truncated caspase 9 polypeptides of the disclosure, the ligand binding region may comprise a FK506 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the amino acid sequence of the ligand binding region that comprise a FK506 binding protein 12 (FKBP12) polypeptide may comprise a modification at position 36 of the sequence. The modification may be a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising GVQVETISPGDGRTPKRGQ-TCVVHYTGMLEDGKKVDSSRDNRNPKFKFMLGKQEVIRG WEEGVAQMSVQRAKLTI SPDYAYGATGHPGIIPPHATLVFDVELLKE (SEQ ID NO: 18022). In certain embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising GGGGTCCAGGTTCGAGACTATTTACACAGGGGATGGGCGAACATTTCCAA-
AAAGGGG CCAGACTTGCGTCGTGCATTACACCGG-
GATGCTGGAGGACGGGAAGAAAGTGGACA GCTC-

CAGGGATCGCAACAAGCCCTTCAAGTTCATGCT-
GGGAAAGCAGGAAGTGATC CGAGGATGGGAG-
GAAGGCGTGGCACAGATGTCAGTCGCCAGCGG-
GCCAAACTGA CCATTAGCCCTGACTACGCTTATG-
GAGCAACAGGCCACCCAGGGATCATTCCCCCTC
ATGCCACCCTGGTCTTCGAT GTGGAAGTGT-
GAAGCTGGAG (SEQ ID NO: 18023). In certain embodi-
ments, the induction agent specific for the ligand binding
region may comprise a FK506 binding protein 12 (FKBP12)
polypeptide having a substitution of valine (V) for phenyl-
alanine (F) at position 36 (F36V) comprises AP20187 and/or
AP1903, both synthetic drugs.

In certain embodiments of the inducible proapoptotic
polypeptides, inducible caspase polypeptides or truncated
caspase 9 polypeptides of the disclosure, the linker region is
encoded by an amino acid comprising GGGGS (SEQ ID
NO: 18024) or a nucleic acid sequence comprising GGAG-
GAGGAGGATCC (SEQ ID NO: 18025). In certain embodi-
ments, the nucleic acid sequence encoding the linker does
not comprise a restriction site.

In certain embodiments of the truncated caspase 9 poly-
peptides of the disclosure, the truncated caspase 9 polypep-
tide is encoded by an amino acid sequence that does not
comprise an arginine (R) at position 87 of the sequence.
Alternatively, or in addition, in certain embodiments of the
inducible proapoptotic polypeptides, inducible caspase
polypeptides or truncated caspase 9 polypeptides of the
disclosure, the truncated caspase 9 polypeptide is encoded
by an amino acid sequence that does not comprise an alanine
(A) at position 282 the sequence. In certain embodiments of
the inducible proapoptotic polypeptides, inducible caspase
polypeptides or truncated caspase 9 polypeptides of the
disclosure, the truncated caspase 9 polypeptide is encoded
by an amino acid comprising

(SEQ ID NO: 18026)
GFGDVGALSLRGNADLAYILSMEPCGHCLIIINNVCRESGLRTRTGSN
IDCEKLRRRFSSSLHFMVEVKGDLTAKKMVLALLELAQQDHGALDCCVVVI
LSHGCGQASHLQFPGAVYGTGDCPVSVEKIVNIFNGTSCPSLGGKPKLFFI
QACGGEQKDHGFVASTSPEDSPGSNPEPDATPFQGLRFTDQLDAISS
LPTPSDIFVSYSTFPFVSWRDPKSGSWYVETLDDIFEQWAHSEDLQSL
LRVANAVSVKGIYKQMPGCFNPLRKLKLFKTS

or a nucleic acid sequence comprising

(SEQ ID NO: 18027)
TTTGGGGACGTGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGATCTGGC
TTACATCCTGAGCATGGAACCTGCGGCCACTGTCTGATCATTAAACATG
TGAAGTCTGCGAGAGAAAGCGGACTGCGAACACGGACTGGCTCCAAATATT
GACTGTGAGAAGCTGCGGAGAAGTTCTCTAGTCTGCACTTTATGGTCTGA
AGTGAAAGGGGATCTGACCGCCAAGAAATGGTGTGGCCCTGTCTGGAGC
TGGCTCAGCAGGACCATGGAGCTCTGGATTGCTGCGTGGTCTGTATCTCTG
TCCCACGGGTGCCAGGCTTCTCATCTGCGAGTTCCCGGAGCAGTGATACGG
AACAGACGGCTGTCTGTCTGCGTGGAGAAGATCGTCAACATCTTCAACG
GCATTCTTGCCCTAGTCTGGGGGAAAGCCAAACTGTCTTTATCCAG
GCCTGTGGCGGGGAACAGAAAGATCACGGCTTCGAGGTGGCCAGCACCAG

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CCCTGAGGACGAATCACCAGGGAGCAACCTGAACCAGATGCAACTCCAT
 TCCAGGAGGGACTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTG
 CCCACTCCTAGTGACATTTTCGTGTCTTACAGTACCTTCCAGGCTTTGT
 CTCATGGCGCGATCCCAAGTCAGGGAGCTGGTACGTGGAGACACTGGACG
 ACATCTTTGAACAGTGGGCCCATTCAGAGGACCTGCAGAGCCTGCTGCTG
 CGAGTGGCAAACGCTGTCTCTGTGAAGGCATCTACAAACAGATGCCCGG
 GTGCTTCAATTTTCTGAGAAAGAACTGTTCTTTAAGACTTCC.

In certain embodiments of the inducible proapoptotic polypeptides, wherein the polypeptide comprises a truncated caspase 9 polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(SEQ ID NO: 18028)

GVQVETISPGDGRTPFKRGQTCVVHYTGMLDGGKKVDSSRDNRKPFKFML
 GKQEVIRGWEEGAQMSVQGRKLITISPDYAYGATGHPGIIPPHATLVFD
 VELLKLEGGGSGFGDVGALSLRGNADLAYILSMEPCGHCLIIINNINFC
 RESGLRTRTGSNIDCEKLRRRFSLLHFMVEVKGDLTAKKMLALLELAQQ
 DHGALDCCVVVILSHGCQASHLQFPAGVYGTGCPVSVKIVNIFNGTSC
 PSLGGKPKLFFIQACGGEQKDHGFEVASTSPEDSPGSNPEPDATPFQEG
 LRTFDQLDAISSLPTPSDIFVSYSYTFPGFVSWRDPKSGSWYVETLDDIFE
 QWAHSEDLQSLLRVANAVSVKGIYKQMPGCFNLRKLLFFKTS

or the nucleic acid sequence comprising

(SEQ ID NO: 18029)

gggggtccaggctcgagactatttcaccaggggatggcgaaacatttccaaa
 aaggggccagacttcgctgctgattacaccgggatgctggaggacggga
 agaaagtggacagctccagggatcgcaacaagcccttcaagttcatgctg
 ggaaagcaggaagtgatccgaggatgggaggaagcgctggcacagatgtc
 agtcggccagcgggccaactgaccattagccctgactacgcttatggag
 caacaggccaccagggatcattccccctcatgccaccctggtcttcgat
 gtggaactgctgaagctggaggaggaggagatccggatttggggacgt
 gggggccctggagtctctgcgaggaaatgccgatctggttacatcctga
 gcatggaaacctgcgccactgtctgatcattaacaatgtgaacttctgc
 agagaaagcggactcgcaacacggactggctccaatattgactgtgagaa
 gctgcgagagaaggttctctagtctgcactttatggtcgaagtgaagggg
 atctgaccgccaaagaaatggtgctggccctgctggagctggctcagcag
 gaccatggagctctggattgctgctggtcgatcctgtcccacgggtg
 ccaggcttctcatctgcagttccccggagcagtgtaacggaacagacggct
 gtctgtcagcgtggagaagatcgtaacatcttcaacggcattcttgc
 cctagtctgggggaaagccaaactgttctttatccaggcctgtggcg
 ggaacagaaagatcacggcttcgaggtggccagcaccagcctgaggacg
 aatcaccagggagcaacctgaaccagatgcaactccattccaggaggga
 ctgaggacctttgaccagctggatgctatctcaagcctgccactcctag

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tgacattttctgtgtcttacagtaccttcccaggctttgtctcatggcgcg
 atcccaagtcaggagctggtacgtggagacactggagacatctttgaa
 5 cagtgggccattcagaggacctgcagagcctgctgctgcagtgggcaaa
 cgctgtctctgtgaaggcattctacaacagatgccgggtgcttcaatt
 ttctgagaagaaactgttctttaagacttcc.

10 Nanoparticle vectors of the disclosure may comprise at least one self-cleaving peptide. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a self-cleaving peptide is located between a VCAR and the nanoparticle. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located upstream of a VCAR and a second self-cleaving peptide is located downstream of a VCAR. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located between a VCAR and the nanoparticle and a second self-cleaving peptide is located downstream of the VCAR. In some embodiments, the nanoparticle vector may comprise at least one self-cleaving peptide and wherein a first self-cleaving peptide is located between a VCAR and the nanoparticle and a second self-cleaving peptide is located downstream of the VCAR, for example, between the VCAR and a selection gene.

Nanoparticle vectors of the disclosure may comprise at least one self-cleaving peptide(s) located, for example, between one or more VHH(s) or VCAR(s) of the disclosure and an inducible proapoptotic polypeptide of the disclosure. Nanoparticle vectors of the disclosure may comprise at least two self-cleaving peptide(s), a first self-cleaving peptide located, for example, upstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure and a second first self-cleaving peptide located, for example, downstream or immediately upstream of an inducible proapoptotic polypeptide of the disclosure. The self-cleaving peptide may comprise, for example, a T2A peptide, GSG-T2A peptide, an E2A peptide, a GSG-E2A peptide, an F2A peptide, a GSG-F2A peptide, a P2A peptide, or a GSG-P2A peptide. A T2A peptide may comprise an amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising EGRGSLTTCGDVEENPGP (SEQ ID NO: 18030). A GSG-T2A peptide may comprise an amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGEGRGSLTTCGDVEENPGP (SEQ ID NO: 18031). A GSG-T2A peptide may comprise a nucleic acid sequence comprising ggatctggagggaagggaagcctgctgacgtgtggagacgtggagaaaccaggacca (SEQ ID NO: 18032). An E2A peptide may comprise an amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising QCTNYALLKLAGDVESNPGP (SEQ ID NO: 18033). A GSG-E2A peptide may comprise an amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGQCTNYALLKLAGDVESNPGP (SEQ ID NO: 18034). An F2A peptide may comprise an amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP

(SEQ ID NO: 18035) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising VKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18035). A GSG-F2A peptide may comprise an amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGVKQTLNFDLLKLAGDVESNPGP (SEQ ID NO: 18036). A P2A peptide may comprise an amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising ATNFSLLKQAGDVEENPGP (SEQ ID NO: 18037). A GSG-P2A peptide may comprise an amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038) or a sequence having at least 70%, 80%, 90%, 95%, or 99% identity to the amino acid sequence comprising GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 18038).

The disclosure provides a composition comprising a vector of the disclosure.

The disclosure provides a cell comprising a VCAR of the disclosure. The disclosure provides a cell comprising a transposon of the disclosure. In certain embodiments, the cell comprising a VCAR, a transposon, or a vector of the disclosure may express a VCAR on the cell surface. The cell may be any type of cell. Preferably, the cell is an immune cell. The immune cell may be a T-cell, a Natural Killer (NK) cell, a Natural Killer (NK)-like cell, a Cytokine Induced Killer (CIK) cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell or an umbilical cord blood (UCB) derived T-cell. Preferably, the immune cell is a T-cell. The T-cell may be an early memory cell, a stem-like T-cell, a T_{SCM} -like cell, a T_{SCM} or a T_{CM} . The T-cell may be a T_{SCM} . The cell may be an artificial antigen presenting cell, which, optionally, may be used to stimulate and expand a modified immune cell or T cell of the disclosure. The cell may be a tumor cell, which, optionally, may be used as an artificial or modified antigen presenting cell.

Modified cells of the disclosure that may be used for adoptive therapy may be autologous or allogeneic.

The disclosure provides a method for expressing a VCAR on the surface of a cell, comprising: (a) obtaining a cell population; (b) contacting the cell population to a composition comprising a VCAR of the disclosure or a sequence encoding the VCAR, under conditions sufficient to transfer the VCAR across a cell membrane of at least one cell in the cell population, thereby generating a modified cell population; (c) culturing the modified cell population under conditions suitable for integration of the transposon; and (d) expanding and/or selecting at least one cell from the modified cell population that express the VCAR on the cell surface.

In certain embodiments of this method of expressing a VCAR, the cell population may comprise leukocytes and/or CD4+ and CD8+ leukocytes. The cell population may comprise CD4+ and CD8+ leukocytes in an optimized ratio. The optimized ratio of CD4+ to CD8+ leukocytes does not naturally occur in vivo. The cell population may comprise a tumor cell.

In certain embodiments of this method of expressing a VCAR, a transposon or vector comprises the VCAR or the sequence encoding the VCAR.

In certain embodiments of this method of expressing a VCAR, the conditions sufficient to transfer the sequence encoding the VCAR across a cell membrane of at least one cell in the cell population comprise nucleofection.

In certain embodiments of this method of expressing a VCAR, wherein the conditions sufficient to transfer the sequence encoding the VCAR across a cell membrane of at least one cell in the cell population comprise at least one of an application of one or more pulses of electricity at a specified voltage, a buffer, and one or more supplemental factor(s). In certain embodiments, the buffer may comprise PBS, HBSS, OptiMEM, BTXpress, Amaxa Nucleofector, Human T cell nucleofection buffer or any combination thereof. In certain embodiments, the one or more supplemental factor(s) may comprise (a) a recombinant human cytokine, a chemokine, an interleukin or any combination thereof; (b) a salt, a mineral, a metabolite or any combination thereof; (c) a cell medium; (d) an inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof; and (e) a reagent that modifies or stabilizes one or more nucleic acids. The recombinant human cytokine, the chemokine, the interleukin or any combination thereof may comprise IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L or any combination thereof. The salt, the mineral, the metabolite or any combination thereof may comprise HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl2, Na2HPO4, NaH2PO4, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, CINa, Glucose, Ca(NO3)2, Tris/HCl, K2HPO4, KH2PO4, Polyethyleneimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, or any combination thereof. The cell medium may comprise PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium or any combination thereof. The inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof comprise inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 β (GSK-3 β) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK or any combination thereof. The reagent that modifies or stabilizes one or more nucleic acids comprises a pH modifier, a DNA-binding protein, a lipid, a phospholipid, CaPO4, a net neutral charge DNA binding peptide with or without a NLS sequence, a TREX1 enzyme or any combination thereof.

In certain embodiments of this method of expressing a VCAR, the conditions suitable for integration of the VCAR or a sequence encoding the VCAR of the disclosure comprise at least one of a buffer and one or more supplemental

factor(s). In certain embodiments, a transposon or vector of the disclosure comprise the VCAR or a sequence encoding the VCAR of the disclosure. In certain embodiments, the buffer may comprise PBS, HBSS, OptiMEM, BTXpress, Amaxa Nucleofector, Human T cell nucleofection buffer or any combination thereof. In certain embodiments, the one or more supplemental factor(s) may comprise (a) a recombinant human cytokine, a chemokine, an interleukin or any combination thereof; (b) a salt, a mineral, a metabolite or any combination thereof; (c) a cell medium; (d) an inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof, and (e) a reagent that modifies or stabilizes one or more nucleic acids. The recombinant human cytokine, the chemokine, the interleukin or any combination thereof may comprise IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L or any combination thereof. The salt, the mineral, the metabolite or any combination thereof may comprise HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl2, Na2HPO4, NAH2PO4, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, CINa, Glucose, Ca(NO3)2, Tris/HCl, K2HPO4, KH2PO4, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, or any combination thereof. The cell medium may comprise PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium or any combination thereof. The inhibitor of cellular DNA sensing, metabolism, differentiation, signal transduction, one or more apoptotic pathway(s) or combinations thereof comprise inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3β (GSK-3β) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK or any combination thereof. The reagent that modifies or stabilizes one or more nucleic acids comprises a pH modifier, a DNA-binding protein, a lipid, a phospholipid, CaPO4, a net neutral charge DNA binding peptide with or without a NLS sequence, a TREX1 enzyme or any combination thereof.

In certain embodiments of this method of expressing a VCAR, the expansion and selection steps occur sequentially. The expansion may occur prior to selection. The expansion may occur following selection, and, optionally, a further (i.e. second) selection may occur following expansion.

In certain embodiments of this method of expressing a VCAR, the expansion and selection steps may occur simultaneously.

In certain embodiments of this method of expressing a VCAR, the expansion may comprise contacting at least one cell of the modified cell population with an antigen to stimulate the at least one cell through the VCAR, thereby generating an expanded cell population. The antigen may be presented on the surface of a substrate. The substrate may have any form, including, but not limited to a surface, a well, a bead or a plurality thereof, and a matrix. The substrate may further comprise a paramagnetic or magnetic component. In certain embodiments of this method of expressing a VCAR, the antigen may be presented on the surface of a substrate, wherein the substrate is a magnetic bead, and wherein a magnet may be used to remove or separate the magnetic beads from the modified and expanded cell population. The antigen may be presented on the surface of a cell or an artificial antigen presenting cell. Artificial antigen presenting cells of the disclosure may include, but are not limited to, tumor cells and stem cells.

In certain embodiments of this method of expressing a VCAR, wherein the transposon or vector comprises a selection gene and wherein the selection step comprises contacting at least one cell of the modified cell population with a compound to which the selection gene confers resistance, thereby identifying a cell expressing the selection gene as surviving the selection and identifying a cell failing to express the selection gene as failing to survive the selection step.

In certain embodiments of this method of expressing a VCAR, the expansion and/or selection steps may proceed for a period of 10 to 14 days, inclusive of the endpoints.

The disclosure provides a composition comprising the modified, expanded and selected cell population of the methods of the disclosure.

The disclosure provides a method of treating cancer in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a tumor cell. In certain embodiments, the tumor cell may be a malignant tumor cell. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

The disclosure provides a method of treating an autoimmune condition in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on an autoimmune cell of the subject. In certain embodiments, the autoimmune cell may be a lymphocyte that specifically binds to a self-antigen on a target cell of the subject. In certain embodiments, the autoimmune cell may be a B lymphocyte (i.e. a B cell). In certain embodiments, the autoimmune cell may be a T lymphocyte (i.e. a T cell). In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

The disclosure provides a method of treating an infection in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a cell comprising an infectious agent, a cell in communication with an infectious

agent or a cell exposed to an infection agent. In some embodiments, a cell in communication with an infectious agent may be in air communication (e.g., the infectious agent is airborne or inhaled) or fluid communication (e.g., the infectious agent is carried in an aqueous or a biological fluid) with the infectious agent. The infectious agent causing the infection of the host cell may be a bacterium, a virus, a yeast, or a microbe. The infectious agent may induce in the cell or the cell's host organism (the subject), exemplary conditions including, but not limited to, a viral infection, an immunodeficiency condition, an inflammatory condition and a proliferative disorder. In certain embodiments, the infection causes tuberculosis, microencephaly, neurodegeneration or malaria. In certain embodiments, the infection causes microencephaly in a fetus of the subject. In certain embodiments, wherein the infection causes microencephaly in a fetus of the subject, the infectious agent is a virus and wherein the virus is a Zika virus. In certain embodiments, the immunodeficiency condition is acquired immune deficiency syndrome (AIDS). In certain embodiments, the proliferative disorder is a cancer. In certain embodiments, the cancer is cervical cancer and wherein the infectious agent is a human papilloma virus (HPV). In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

The disclosure provides a method of treating a mast cell disease in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a mast cell. In certain embodiments, the VCAR specifically binds to an antigen on a mast cell of the subject. In certain embodiments, the mast cell disease may include, but is not limited to, disorders associated with an excessive proliferation of mast cells, disorders associated with mast cells having abnormal activity, and disorders associated with both abnormal numbers of mast cells and abnormal mast cell activity. Exemplary disorders associated with an excessive proliferation of mast cells include, but are not limited to, mastocytosis, cutaneous mastocytosis (e.g., urticaria pigmentosa or maculopapular cutaneous mastocytosis), systemic mastocytosis (including mast cell leukaemia), and localized mast cell proliferations. Exemplary disorders associated with mast cells having abnormal activity, include, but are not limited to, mast cell activation syndrome (MCAS) or mast cell activation disorder (MCAD), allergic disease (including anaphylaxis), asthma, inflammatory disease (including autoimmune related inflammation of, for example, joint tissues, arthritis, etc.), or any combination thereof. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic. The disclosure provides a method of treating a degenerative disease in a subject in need thereof, comprising administering to the subject a composition of the disclosure, wherein the VCAR specifically binds to an antigen on a deleterious cell or an aged cell. In certain embodiments, the VCAR specifically binds to an antigen on a deleterious cell or an aged cell of the subject. In certain embodiments, the degenerative disease may include, but is

not limited to, a neurodegenerative disorder, a metabolic disorder, a vascular disorder and aging. Exemplary neurodegenerative disorders include, but are not limited to, disorders associated with a loss of a function or efficacy of one or more of a neuron, a glial cell or a microglia. Exemplary neurodegenerative disorders include, but are not limited to, disorders associated with an accumulation of one or more of a signaling molecule, a protein, or a prion that interferes with a function or decreases an efficacy of one or more of a neuron, a glial cell or a microglia. Exemplary metabolic disorders include, but are not limited to, disorders associated with mitochondrial disorders, interruptions of the electron transport chain, interruptions of cellular production of ATP, a loss of a function or a decreased efficacy of one or more mitochondria of one or more of a neuron, a glial cell or a microglia. Exemplary metabolic disorders include, but are not limited to, disorders associated with a loss of circulating blood flow or a decreased blood flow to a neuron, a glial cell or a microglia (e.g., a stroke); a transient or permanent state of hypoxia in a neuron, a glial cell or a microglia (for example, sufficient to release free radicals in a cell); a loss of circulating CNS or a decreased CNS flow to a neuron, a glial cell or a microglia during a sleep state of the subject sufficient to decrease efficacy of removal of a waste product of a neuron, a glial cell or a microglia during that sleep state. Exemplary aging disorders include, but are not limited to, disorders associated with an increased shortened or shortened telomeres on one or more chromosomes of a neuron, a glial cell or a microglia; a loss of a function or a decreased efficacy of telomerase in a neuron, a glial cell or a microglia; or a loss of a function or a decreased efficacy of a DNA repair mechanism in a neuron, a glial cell or a microglia. In certain embodiments, the deleterious cell or the aged cell interferes with a function or decreases an efficacy of another cell in a network comprising the deleterious cell or the aged cell and the targeted removal of the deleterious cell or the aged cell improves or restores a function or increases an efficacy of the network. In certain embodiments, the deleterious cell or the aged cell may transform the function or efficacy of a second cell and the targeted removal of the deleterious cell or the aged cell prevents the transformation of the second cell. In certain embodiments, the degenerative disease is a neurodegenerative disorder and the deleterious cell or the aged cell is a stem cell, an immune cell, a neuron, a glia or a microglia. In certain embodiments, the degenerative disease is a metabolic disorder and the deleterious cell or the aged cell is a stem cell, a somatic cell, a neuron, a glia or a microglia. In certain embodiments, the degenerative disease is a vascular disorder and the deleterious cell or the aged cell is a stem cell, a somatic cell, an immune cell, an endothelial cell, a neuron, a glia or a microglia. In certain embodiments, the degenerative disease is aging and the deleterious cell or the aged cell is an oocyte, a sperm, a stem cell, a somatic cell, an immune cell, an endothelial cell, a neuron, a glia or a microglia. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be autologous. In certain embodiments, comprising administering to the subject the composition comprising a modified cell or cell population of the disclosure, the cell or cell population may be allogeneic.

The disclosure provides a method of modifying a cell therapy in a subject in need thereof, comprising administering to the subject a composition comprising a cell comprising a transposon or vector of the composition comprising an inducible proapoptotic polypeptide, wherein apoptosis may be selectively induced in the cell by contacting the cell with

an induction agent. In certain embodiments, the cell is autologous. In certain embodiments, the cell is allogeneic. In certain embodiments of this method, the cell therapy is an adoptive cell therapy. In certain embodiments of this method, modifying the cell therapy comprises a termination of the cell therapy. In certain embodiments of this method, modifying the cell therapy comprises a depletion of a portion of the cells provided in the cell therapy. In certain embodiments, the method further comprises the step of administering an inhibitor of the induction agent to inhibit modification of the cell therapy, thereby restoring the function and/or efficacy of the cell therapy.

Methods of modifying a cell therapy of the disclosure may be used to terminate or dampen a therapy in response to, for example, a sign of recovery or a sign of decreasing disease severity/progression, a sign of disease remission/cessation, and/or the occurrence of an adverse event. Cell therapies of the disclosure may be resumed by inhibiting the induction agent should a sign or symptom of the disease reappear or increase in severity and/or an adverse event is resolved.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a plot showing weight change versus days post treatment for VCARs of the disclosure.

FIG. 2 is a plot showing tumor burden versus days post treatment in for VCARs of the disclosure.

FIG. 3 is a plot showing percent survival versus days post treatment for VCARs of the disclosure.

FIG. 4 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-A (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).

FIG. 5 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-B (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).

FIG. 6 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-C (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).

FIG. 7 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-D (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).

FIG. 8 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-E (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).

FIG. 9 is a plot of tumor burden as shown by bioluminescence signal versus days post treatment of VH-F (triangles), an irrelevant CAR (squares) or no tumor/no CAR-T (circles).

FIG. 10A is a table providing consensus sequences of framework and CDR sequences for exemplary VHs of the disclosure.

FIG. 10B is an alignment of exemplary VHs of the disclosure. From top to bottom, the sequences correspond with the consensus sequences of the frameworks VH-B (SEQ ID NO: 18050), VH-D (SEQ ID NO: 18051), VH-A (SEQ ID NO: 18052), VH-E (SEQ ID NO: 18053), VH-F (SEQ ID NO: 18054) and VH-C (SEQ ID NO: 18055).

FIG. 11 is a bar graph depicting the knock out efficiency of checkpoint signaling proteins on armored T-cells. Primary human T-cells are typically in resting state when isolated from normal healthy donors. Cas-CLOVER was used to

knockout the checkpoint receptors, PD-1, TGFBR2, LAG-3, TIM-3 and CTLA-4. Percent knock-out is shown on the y-axis. Gene editing resulted in 30-70% loss of protein expression at the cell surface as measured by flow cytometry.

FIG. 12 are schematic diagrams of wildtype, null and switch receptors and their effects on intracellular signaling, either inhibitory or stimulatory, in primary T-cells. Binding of the wildtype inhibitory receptor expressed endogenously on a T-cell with its endogenous ligand results in transmission of an inhibitory signal which, in part, reduces T-cell effector function. However, mutation (Mutated null) or deletion (Truncated null) of the intracellular domain (ICD) of a checkpoint receptor protein, such as PD1 (top panel) or TGFBR2 (bottom panel), reduces or eliminates its signaling capability when cognate ligand(s) is bound. Thus, expression of engineered mutated or truncated null receptors on the surface of modified T cells results in a competition with endogenously-expressed wildtype receptors for binding of the free endogenous ligand(s), effectively reducing or eliminating delivery of inhibitory signals by endogenously-expressed wildtype receptors. Specifically, any binding by a mutated or null receptor sequesters the endogenous ligand(s) from binding the wildtype receptor and results in dilution of the overall level of checkpoint signaling effectively delivered to the modified T-cell, thereby reducing or blocking checkpoint inhibition and functional exhaustion of the modified T cells. A switch receptor is created by replacement of the wildtype ICD with an ICD from either a co-stimulatory molecule (such as CD3z, CD28, 4-1B) or a different inhibitory molecule (such as CTLA4, PD1, Lag3). In the former case, binding of the endogenous ligand(s) by the modified switch receptor results in the delivery of a positive signal to the T-cells, thereby helping to enhance stimulation of the modified T cell and potentially enhance target tumor cell killing. In the latter case, binding of the endogenous ligand(s) by the modified switch receptor results in the delivery of a negative signal to the T-cells, thereby eliminating stimulation of the modified T cell and potentially reducing target tumor cell killing. The signal peptide (purple arrow), extracellular domain (ECD) (bright green), transmembrane domain (yellow), intracellular signaling domain (ICD)(orange), and replacement ICD (green) are displayed in the receptor diagrams. "*" indicates a mutated ICD. "+" indicates the presence of a checkpoint signal. "-" indicates the absence of a checkpoint signal.

FIG. 13 is a schematic diagram showing the design of PD1 and TGFBR2 null receptors. The signal peptide domain (SP), transmembrane domain (TM) and extracellular domain (ECD) of truncated null receptors for PD1 (top panel) and TGFBR2 (bottom panel) are shown. The first of the top four molecules is the wildtype PD-1 receptor, which encodes the wildtype PD-1 SP and TM. For the PD1 null receptor, replacement of PD1 wildtype SP or TM domain (green; light green) with the SP or TM domain of a human T cell CD8 α receptor (red) is depicted. The second molecule encodes the CD8 α SP along with the native PD-1 TM, the third encodes the wildtype PD-1 SP and the alternative CD8 α TM, and the fourth encodes both the alternative CD8 α SP and TM. Similarly, for the null receptor of TGFBR2, replacement of the wildtype TGFBR2 SP (pink) with a SP domain of a human T cell CD8 α receptor (red). The names of the constructs and the amino acid lengths (aa) of each construct protein is listed on the left of the diagram.

FIG. 14 is a series of histograms depicting the expression of the PD1 and TGFBR2 null Receptors on the surface of primary human T cells determined by flow cytometry. Each

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of the six truncated null constructs from FIG. 13 were expressed on the surface of primary human T cells. T cells were stained with either anti-PD1 (top; blue histograms) or anti-TGFBRII (bottom; blue histograms), or isotype control or secondary only (gray histograms). Cells staining positive for PD-1 or TGFβRII expression were gated (frequency shown above gate) and mean fluorescence intensity (MFI) value is displayed above each positive histogram. The names of the null receptor constructs are depicted above each plot. Both null receptor gene strategies, replacement of the wild-type SP with the alternative CD8α were successfully expressed. 02.8aSP-PD-1 and 02.8aSP-TGFβRII resulted in the highest level of expression at the T-cell surface. 02.8aSP-PD-1 null receptor exhibited an MFI of 43,680, which is 177-fold higher than endogenous T cell PD-1 expression and 2.8-fold higher than the wildtype PD-1 null receptor. 02.8aSP-TGFβRII null receptor exhibited an MFI of 13,809, which is 102-fold higher than endogenous T cell TGFβRII expression and 1.8-fold higher than the wildtype TGFβRII null receptor. Replacement of wildtype SP with the alternative CD8α SP for both PD1 and TGRBRII results in enhanced surface expression of the null or Switch receptor, which helps to maximize checkpoint inhibition or co-stimulation, respectively, upon binding of the endogenous ligand(s).

FIG. 15A-B is a pair of schematic diagrams depicting NF-KB inducible vectors for expression in T-cells. Two T cell activation NF-KB inducible vectors were developed; one with the gene expression system (GES) in the forward orientation (A) and the other in the complementary direction (B), both preceding the constitutive EF1a promoter. These vectors also direct expression of a CAR molecule and a DHFR selection gene, separated by a T2A sequence. Both the conditional NF-KB inducible system and the EF1a directed genes are a part of a piggyBac transposon that can be permanently integrated into T cells using electroporation (EP). Once integrated into the genome, the T cells will constitutively express the CAR on the membrane surface and the DHFR within the cell, while expression of the NF-KB inducible gene, GFP, will be expressed to the highest level only upon T cell activation.

FIG. 16 is a pair of graphs depicting NF-KB inducible expression of GFP inactivated T cells. T cells were nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mitein gene under control of an EF1a promoter along with the absence (No GES control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFkB-GFP forward) or reverse orientation (pNFkB-GFP reverse). Cells were cultured in the presence of methotrexate selection until the cells were almost completely resting (Day 19) and GFP expression was assessed at Day 5 and Day 19. At Day 5, all T cells are proliferating and highly stimulated, with cells harboring the NF-KB inducible expression cassette producing high levels of GFP due to strong NFkB activity. The No GES control cells did not express detectable levels of GFP. By Day 19, the GES T cells were almost fully resting and GFP expression was significantly lower than Day 5 (~1/8 MFI), since NFkB activity is lower. GFP expression is still observed at Day 19, which may due to the long half-life of GFP protein (~30 hr), or, basal level of NFkB activity through, for example, a TCR, a CAR, a cytokine receptor, or a growth factor receptor signal.

FIG. 17 is a series of graphs depicting anti-BCMA CAR-mediated activation of NF-KB inducible expression of GFP in presence of BCMA+ tumor cells. T cells were either unmodified (Mock T cells) or nucleofected with a piggyBac

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vector expressing an anti-BCMA CAR and a DHFR mitein gene under control of an EF1a promoter along with the absence (No GES control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFkB-GFP forward) or reverse orientation (pNFkB-GFP reverse). All cells were cultured for 22 days, either with or without methotrexate selection (Mock T cells), until the cells were almost completely resting. Cells were then stimulated for 3 days in the absence (No stimulation) or presence of BCMA- (K562), BMCA+(RPMI 8226), or positive control anti-CD3 anti-CD28 activation reagent (CD3/28 stimulation). GFP expression was undetectable under all conditions with the No GES control or Mock T cells. However, while pNFkB-GFP forward- and reverse-transposed cells exhibited little GFP expression over the No stimulation control when cultured with BCMA- K562 cells, they both demonstrated dramatic upregulation of gene expression either in the presence of BCMA+ tumor cells or under positive control conditions. Little difference in GFP expression was observed between the pNFkB-GFP forward- and reverse-transposed cells that were cocultured with BCMA+ tumor cells.

FIG. 18 is a series of graphs demonstrating that the Expression level of inducible gene can be regulated by number of response elements preceding the promoter T cells were nucleofected with a piggyBac vector encoding an anti-BCMA CARTyrin followed by a selection gene, both under control of a human EF1a promoter. Further, vectors either additionally encoded the conditional NF-KB inducible gene expression system driving expression of a truncated CD19 protein (dCD19) and included a number of NFkB response elements (RE) varying from 0-5, no GES (No GES), or received an electroporation pulse but no piggyBac nucleic acid (Mock). Data are shown for only the GES in the reverse (opposite) direction/orientation. All cells were cultured for 18 days and included selection for piggyBac-modified T cells using methotrexate addition. Cells were then stimulated for 3 days using anti-CD3 anti-CD28 bead activation reagent and dCD19 surface expression was assessed by FACS at Days 0, 3 and 18, and data are shown as FACS histograms and MFI of target protein staining. Surface dCD19 expression was detected at low levels at Day 0 in all T cells transposed with vectors encoding the GES. At 3 days post-stimulation, dramatic upregulation of dCD19 expression was observed for all T cells expressing the GES, with a greater fold increase in surface expression in those with higher numbers of REs. Thus, surface dCD19 expression was directly proportional with the number of REs encoded in the GES. No dCD19 was detected on the surface of T cells that did not harbor the GES: No GES and Mock controls.

FIG. 19 is a schematic depiction of the Csy4-T2A-Clo051-G4Slinker-dCas9 construct map (Embodiment 2).

FIG. 20 is a schematic depiction of the pRT1-Clo051-dCas9 Double NLS construct map (Embodiment 1).

FIG. 21 is a pair of graphs comparing the efficacy of knocking out expression of either B2M in Pan T-cells (left) or the α-chain of the T-cell Receptor in Jurkat cells (right) for either Embodiment 1 (pRT1-Clo051-dCas9 Double NLS, as shown in FIG. 20) or Embodiment 2 (Csy4-T2A-Clo051-G4Slinker-dCas9, as shown in FIG. 19) of a Cas-Clover fusion protein of the disclosure. For the right-hand graph, the fusion protein is provided at either 10 μg or 20 μg, as indicated.

FIG. 22 is a photograph of a gel electrophoresis analysis of mRNA in the presence of each of Embodiment 1 (pRT1-Clo051-dCas9 Double NLS, as shown in FIG. 20) and

Embodiment 2 (Csy4-T2A-Clo051-G4Slinker-dCas9, as shown in FIG. 19). As shown, both are effective at knocking down mRNA expression.

FIG. 23 is a diagram of the UniRat™ Human Heavy Chain Antibody production platform.

FIG. 24 is a diagram of the pipeline of immunization, B-cell isolation, mRNA purification, Next Generation Sequencing (NGS), bioinformatics analysis, high-throughput vector assembly and high-throughput expression and screening used to identify the full antigen-specific repertoire of heavy-chain antibodies generated by the UniRat™ after immunization. Unique gene assembly methods convert the antibody repertoire sequence information into large collections of fully human heavy chain antibodies that can be screened for a variety of functions.

FIG. 25 is a pair of heat maps showing how next generation sequencing (NGS) analysis reveals expanded VH sequence lineages. The degree of red (high) or blue (low) in the heatmap indicates expanded VH sequence families due to immunization. Some highly ranked VH sequences are unique to a single animal. Other highly ranked VH sequences appear in more than one animal suggesting convergent selection of those sequence families.

FIG. 26 is a series of flow cytometry plots showing that a CAR-T population expressing P-PSMA-101 and comprising a mixture of T_{SCM}/T_{CM} give rise to CAR+ T_{CM} , T_{EM} and Teff to attack solid tumor. After solid tumor elimination, a population of CAR-T+ T_{SCM} persists. Although demonstrated with a CARTyrin, this principle shown here applies to VCAR+ T_{SCM} and VCAR+ T_{CM} populations of the disclosure. Specifically, a murine xenograft model using a luciferase-expressing LNCaP cell line (LNCaP.luc) injected subcutaneously (SC) into NSG mice was utilized to assess in vivo anti-tumor efficacy of a CAR (P-PSMA5-101 and P-PSMA8-101) at a 'stress' dose (4×10^6) total CAR-T cells. For these in vivo studies, all CAR-T cells were produced using PB delivery of either the P-PSMA5-101 or P-PSMA8-101 plasmid using the Poseida manufacturing process. Mice were injected in the axilla with LNCaP and treated when tumors were established ($100-300 \text{ mm}^3$ by caliper measurement). Mice were treated with a 'stress' dose (4×10^6) of P-PSMA-101 by IV injection in order to tease out any possible differences in efficacy between the PSMA5 and the PSMA8 CARs. Anti-tumor activity was evaluated by survival, CD8+ T cell expansion and detection in the blood, tumor volume assessment by caliper measurement, and bioluminescence of LNCaP tumor. P-PSMA5-101 and P-PSMA8-101 at a 'stress' dose demonstrated significantly enhanced anti-tumor efficacy and survival in comparison to the T cells (no CAR) control mice against established SC LNCaP.luc solid tumors in NSG mice. Specifically, there was no survival in T cells (no CAR) control animals, 25% survival in the P-BCMA-101 treated group, 75% survival in the P-PSMA5-101 treated group, and 100% survival in animals treated with a 'stress' dose of P-PSMA8-101. In the peripheral blood, P-PSMA5-101 and P-PSMA8-101 expanded and gave rise to differentiated effector CARTyrin+ T-cells that were concomitant with a decrease in tumor burden below detectable caliper and bioluminescent imaging limits. These cells then contracted, yet persisted in the peripheral blood.

DETAILED DESCRIPTION

The disclosure provides chimeric antigen receptors (CARs) comprising at least one VHH (VCAR). Chimeric antigen receptors of the disclosure may comprise more than

one VHH. For example, a bi-specific VCAR may comprise two VHHs that specifically bind two distinct antigens.

VHH proteins of the disclosure specifically bind to an antigen. Chimeric antigen receptors of the disclosure comprising one or more VHHs that specifically bind an antigen may be used to direct the specificity of a cell, (e.g., a cytotoxic immune cell) towards the specific antigen.

The disclosure provides chimeric antigen receptors (CARs) an antigen recognition region comprising a single domain antibody (VCARs). In some embodiments, the single domain antibody is a VHH antibody. In some embodiments, the single domain antibody is a VH antibody.

Chimeric antigen receptors of the disclosure may comprise a signal peptide of human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR. A hinge/spacer domain of the disclosure may comprise a hinge/spacer/stalk of human CD8α, IgG4, and/or CD4. An intracellular domain or endodomain of the disclosure may comprise an intracellular signaling domain of human CD3ζ and may further comprise human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. Exemplary transmembrane domains include, but are not limited to a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain.

The disclosure provides genetically modified cells, such as T cells, NK cells, hematopoietic progenitor cells, peripheral blood (PB) derived T cells (including T cells from G-CSF-mobilized peripheral blood), umbilical cord blood (UCB) derived T cells rendered specific for one or more antigens by introducing to these cells a VCAR of the disclosure. Cells of the disclosure may be modified by electrotransfer of a transposon encoding a VCAR of the disclosure and a plasmid comprising a sequence encoding a transposase of the disclosure (preferably, the sequence encoding a transposase of the disclosure is an mRNA sequence).

VHs of the Disclosure

The disclosure provides chimeric antigen receptors (CARs) comprising a single domain antibody (VCARs). In some embodiments, the single domain antibody comprises a VH. In some embodiments, the VH is isolated or derived from a human sequence. In some embodiments, VH comprises a human CDR sequence and/or a human framework sequence and a non-human or humanized sequence (e.g., a rat Fe domain). In some embodiments, the VH is a fully humanized VH. In some embodiments, the VH is neither a naturally occurring antibody nor a fragment of a naturally occurring antibody. In some embodiments, the VH is not a fragment of a monoclonal antibody. In some embodiments, the VH is a UniDab™ antibody (TeneoBio).

In some embodiments, the VH is fully engineered using the UniRat™ (TeneoBio) system and "NGS-based Discovery" to produce the VH. Using this method, the specific VH are not naturally-occurring and are generated using fully engineered systems. The VH are not derived from naturally-occurring monoclonal antibodies (mAbs) that were either isolated directly from the host (for example, a mouse, rat or human) or directly from a single clone of cells or cell line (hybridoma). These VHs were not subsequently cloned from said cell lines. Instead, VH sequences are fully-engineered using the UniRat™ system as transgenes that comprise human variable regions (VH domains) with a rat Fc domain, and are thus human/rat chimeras without a light chain and are unlike the standard mAb format. The native rat genes are knocked out and the only antibodies expressed in the rat are from transgenes with VH domains linked to a Rat Fc

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(UniAbs). These are the exclusive Abs expressed in the UniRat. Next generation sequencing (NGS) and bioinformatics are used to identify the full antigen-specific repertoire of the heavy-chain antibodies generated by UniRat™ after immunization. Then, a unique gene assembly method is used to convert the antibody repertoire sequence information into large collections of fully-human heavy-chain antibodies that can be screened in vitro for a variety of functions. In some embodiments, fully humanized VH are generated by fusing the human VH domains with human Fcs in vitro (to generate a non-naturally occurring recombinant VH antibody). In some embodiments, the VH are fully humanized, but they are expressed in vivo as human/rat chimera (human VH, rat Fc) without a light chain. Fully humanized VHs are expressed in vivo as human/rat chimera (human VH, rat Fc) without a light chain are about 80 kDa (vs 150 kDa).

VCARs of the disclosure may comprise at least one VH of the disclosure. In some embodiments, the VH of the disclosure may be modified to remove an Fc domain or a portion thereof. In some embodiments, a framework sequence of the VH of the disclosure may be modified to, for example, improve expression, decrease immunogenicity or to improve function.

Exemplary VCARS of the Disclosure

In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

VH-A:

(SEQ ID NO: 18000)
malpvtalllplalllhaarpevqllesggglvqpqgslrlscaasgftf
ssyamnwrvqapqgkglewvagiisggstyyadsvkgrfsisrdnskn1
dlqmns1raedtavyycvkdwnntmtergggtlvtvssttpprptp
aptiasqplslrpeacrpaaggavhtrgldfacdiyiwaplagtcgvll1
slvitlyckrgrkkllyifkqpfmrpvqttqeedgcscrffeeeggcel
rvkfsrsadapaykqgnqlynelnlgrreeydvldkrrgrdpemggkpr
rknpgelynelqkdkmaeyseigmkgerrrgkghdglyqglstatkdt
ydahmqalppr.

In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

VH-A:

(SEQ ID NO: 18001)
atggctctgcctgtgacagctctgctgctgcctctggctctgcttctca
tgccgcgccctgaagttcagctgcttgaatctggcgaggcctggttc
aacctggcgagatctctgagactgagctgtgcgcagcaggcttcacctt
agcagctacgcatgaactgggtccgacaggccctggcaaggactgga
atgggtggcgggaatcatcggcagcggcggcagcacatattacgccgatt
ctgtgaagggcgcgttcagcatcagccgggacaacagcaagaacacctg
gacctgcagatgaacagcctgagagccgaggataccgcggtgtactactg
cgtgaagatttgaacaccaccatgatcaccgagagaggccagggcacac

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tggtcaccgtgtcctctacaacaacacggcgccctcgccctcaacacca
gtcctacaatcgcgagtcagccctgtctctcagaccgaagcctgtag
5 acctgctgctggcgagctgtgcataccagaggactggatttcgcctgcg
acatctacatctgggctcctctggctggcacatgaggagttttgctgctg
agcctgggtcatcacctgtactgtaagagaggcaggaagaagctgctgta
10 tatcttcaagcagcccttcatgagaccgtgcagaccacacaggaggagg
acggctgctctttagtgggtccagaggaggaggaggaggatgctgagctg
cgctgaagtttagcgggtccgcatgacacgtgcatacaagcagggaca
15 gaaccagctgtataacgagctgaatctgggcccagagaggagtacgacg
tgctggataagaggcgggggccgggaccccgagatggagggaagcctcgg
agaaagaacccacaggaggcgctgtacaatgagctgcaaaaggacaagat
ggccgaggcctattctgagatcgccatgaaggagagaggcgccggggca
20 agggacacgatggcctgtaccaggcctgagcaccgccacaaggacacc
tatgatgcctgcacatgcaggccctgcccttagatga.

In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

30 VH-B:

(SEQ ID NO: 18002)
malpvtalllplalllhaarpevqllesggglyqpqgslrlscaasgftf
snyamnwrvqapqgkglewysgiisgagtttyadsvkgrftisrdnskn1
35 nlqmns1raedtaiyyvcvkdwnntmtergggtlvtvssttpprptp
aptiasqplslrpeacrpaaggavhtrgldfacdiyiwaplagtcgvll1
slvitlyckrgrkkllyifkqpfmrpvqttqeedgcscrffeeeggcel
40 rvkfsrsadapaykqgnqlynelnlgrreeydvldkrrgrdpemggkpr
rknpgelynelqkdkmaeyseigmkgerrrgkghdglyqglstatkdt
ydahmqalppr.

45 In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

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VH-B

(SEQ ID NO: 18003)
atggctctgcctgtgacagctctgctgctgcctctggctctgcttctca
55 tgccgcgccctgaagttcagctgcttgaatctggcgaggcctggttc
aacctggcgagatctctgacactgagctgtgcgcagcaggccttcacctc
agcaactacgcatgaactgggtccgacaggccctggcaaggccttga
atgggtgtccggcatcattggctctggcgccaccacctactacgccgatt
60 ctgtgaagggcagattcaccatcagccgggacaacagcaagaacacctg
aacctgcagatgaacagcctgagagccgaggacaccgccatctactactg
cgtgaaggactggaacaccaccatgatcaccgagagaggccagggcacac
65 tggtcaccgtgtcctctacaacaacacggcgccctcgccctcaacacca

47

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gctcctacaatcgcgagtcagccctgtctctcagaccgaagcctgtag
 acctgctgctggcggagctgtgcataccagaggactggatttcgctgcg
 acatctacatctgggctcctctggctggcacatgcggagttttgctgctg
 agcctggtcatcacctgtactgtaagagaggcaggaagaagctgctgta
 tatcttcaagcagccatcatgagaccgtgcagaccacacaggaggagga
 cggtgctctttaggttcccagaggaggaggaggaggatgcgagctgc
 gcgtgaagtttagccggtccgccgatgcacctgcatacaagcagggacag
 aaccagctgtataacgagctgaatctgggcccggagagaggagtacgacgt
 gctggataagaggcggggccgggaccccagatgggaggcaagcctcgga
 gaaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagatg
 gccgaggcctattctgagatcggcatgaaggagagaggcgcggggcaa
 gggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacct
 atgatgccctgcacatgcaggccctgccccctagatgac.

In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

VH-C:

(SEQ ID NO: 18004)

malpvtalllplalllhaarpevqllesggglvqpgeslriscasgftf
 snyamnwvrqapkgglewsvlgvggtsyyadsvrgrftisrdnsknt1
 ylmnslraedavyyvckdwnttmitergggtlvtvssttpprptp
 aptiasqplslrpeacrpaaggavhtrglfdacdiyiwaplagtcgvll1
 slvitlyckrgrklllyifkqpfmrpvqttqeedgcsrfpeeeeggcel
 rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr
 rknppqeglynelqkdkmaeyseigmkgerrrgkghdglyqglstatkdt
 ydalhmqalppr.

In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of

VH-C:

(SEQ ID NO: 18005)

atggctctgcctgtgacagctctgctgctgcctctggctctgcttctca
 tgcggcgccctgaagttcagctgctgtaacttgccggaggcctggttc
 agcctggcgaatctctgagactgagctgtgccgccagcgatcaccttca
 gcaactacgccatgaactgggtccgacaggccccctggcaaggccttgaa
 tgggtgtccggaatcggtggcggcgaggacacaagctactacgccgattc
 tgtgcggggcagattcaccatcagccgggacaacagcaagaacacctgt
 acctgcagatgaacagcctgagagccgaggacaccgctgtactactgc
 gtgaaggactggaacaccaccatgatcaccgagagaggccagggcacact
 ggtcaccgtgtcctctacaacaacaccggcgctcgccctccaacaccag

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ctcctacaatcgcgagtcagccctgtctctcagaccgaagcctgtaga
 cctgctgctggcggagctgtgcataccagaggactggatttcgctgcga
 5 catctacatctgggctcctctggctggcacatgcggagttttgctgctga
 gcctgggtcatcacctgtactgtaagagaggcaggaagaagctgctgtat
 atcttcaagcagccatcatgagaccgtgcagaccacacaggaggaggac
 10 ggctgctctttaggttcccagaggaggaggaggaggatgcgagctgcg
 cgtgaagtttagccggtccgccgatgcacctgcatacaagcagggacaga
 accagctgtataacgagctgaatctgggcccggagagaggagtacgacgtg
 15 ctggataagaggcggggccgggaccccagatgggaggcaagcctcgga
 aaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagatgg
 ccgaggcctattctgagatcggcatgaaggagagaggcgcggggcaaag
 20 ggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacct
 tgatgccctgcacatgcaggccctgccccctagatga.

In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

VH-D:

(SEQ ID NO: 18006)

malpvtalllplalllhaarpevqllesggglvqpgeslriscasgftf
 snyamtwirqapkgglewsvlgvggtsyyadsvrgrftisrdnsknt1
 ylmnslraedavyyvckdwnttmitergggtlvtvssttpprptp
 35 aptiasqplslrpeacrpaaggavhtrglfdacdiyiwaplagtcgvll1
 slvitlyckrgrklllyifkqpfmrpvqttqeedgcsrfpeeeeggcel
 rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr
 40 rknppqeglynelqkdkmaeyseigmkgerrrgkghdglyqglstatkdt
 ydalhmqalppr.

In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

VH-D:

(SEQ ID NO: 18007)

atggctctgcctgtgacagctctgctgctgcctctggctctgcttctca
 tgcggcgccctgaagttcagctgctgtaacttgccggaggcctggttc
 55 aacctggcggatctctgagactgagctgtgccgccagcggttcaccttc
 agcaattacgccatgacctggatcagacaggccccctggcaaggcctgga
 atgggtgtccggaattacaggcgacggcgagcaccttttacgccgatt
 ctgtgaagggcagattcaccatcagccgggacaacagcaagaacacctg
 60 tacctgcagatgaacagcctgagagccgaggacaccgctgtactactg
 cgtgaaggactggaacaccaccatgatcaccgagagaggccagggcacac
 tggtcaccgtgtcctctacaacaacaccggcgctcgccctccaacacca
 65 gctcctacaatcgcgagtcagccctgtctctcagaccgaagcctgtag

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acctgctgctggcggagctgtgcataccagaggactggatttcgcctgcg
 acatctacatctgggtcctctggtggcacatgcggagttttgctgctg
 agcctggtcatcaccctgtactgtaagagaggcaggaagaagctgctgta
 tatcttcaagcagccatcatgagaccctgcagaccacacaggaggagga
 cggctgctcttgtaggttcccagaggaggaggaggaggatgcgagctgc
 gcgtagaagtttagccggtccgcccgtgcacctgcatacaagcagggacag
 aaccagctgtataacgagctgaatctgggcccagagaggagtagcagct
 gctggataagaggcggggccgggaccccgagatgggaggcaagcctcgga
 gaaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagatg
 gccgaggcctattctgagatcgcatgaaggagagaggcgccggggcaa
 gggacacgatggcctgtaccaggcctgagcaccgccacaaaggacacct
 atgatgcctgcacatgcaggcctgccccctagatga.

In some embodiments of the VCARS of the disclosure, the VCAR comprises at least one of an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

VH-E:

(SEQ ID NO: 18008)

malpvtalllplalllhaarpevqllesggglaqpggslrlscaasgftf
 ssyamnwirqapgkglewvsgisggstyyadsvkgrftisrdnsknv
 ylmnslraedtavvyckdwnttmtergggtltvvssttppaprpptp
 aptiasqplslrpeacrpaaggavhtrglfdacdiyiwaplagtcgvlll
 slvitlyckrgrkkllyifkqpfmrpvqttqeedgcsrfpeeeeggcel
 rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr
 rknpqeglynelqkdkmaeyseigmkgerrrrgkghdglyqglstatkdt
 ydalhmqalppr.

In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

VH-E

(SEQ ID NO: 18009)

atggcactgcctgtgacagccctgctgctgcctctggcctgctgctgca
 cgcagcacggcccaggtgcagctgctggagtcggaggaggcctggccc
 agcctggcggcagcctgaggctgtcctgcgcgcctctggttccacctt
 agctcctacgccatgaactggatcagacaggccctggcaaggcctgga
 gtgggtgtccggcatctccggtctgaggctctacatactatgccgaca
 gcgtgaaggggccggttcaccatcagcagagataaactccaagaataccgtg
 tacctccagatgaactctctgcgggcccaggacaccgcgtgtactattg
 cgtgaaggattggaataaccacaatgatcacagagagggggcagggcaccc
 tggtagcagtgctagcaccacaacccctgccccagacctccacacccc
 gccctaccatcgcgagtcagccactgtccctgcggcctgaggcctgcgg

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gcccccccgccgggagcagtgacacacggggcctggactttgctgtg
 acatctacatattgggcaccactggcaggaacctgcggcgctgctgctg
 5 agcctggtcatcaccctgtactgtaagagaggcaggaagaagctgagtat
 atcttcaagcagccctcatgagaccctgcagaccacacaggaggagga
 cggctgactttaggttcccagaggaggaggaggaggatgcgagagcgc
 10 gtgaagtttagccggtccgcccgtgcacctgcatacaagcagggacagaa
 ccagagtataacgagctgaatctgggcccagagaggagtagcagctgct
 ggataagaggcggggccgggaccccgagatgggaggcaagcctcgagaa
 15 agaaccacaggaggcctgtacaatgagagcaaaaggacaagatggccg
 aggcctattctgagatcgcatgaaggagagaggcgccggggcaggga
 cacgatggcctgtaccaggcctgagcaccgccacaaaggacacctatga
 20 tgccagcacatgcaggccagccccctagatga.

In some embodiments of the VCARS of the disclosure, the VCAR at least one of comprises an amino acid sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the amino acid sequence of:

VH-F:

(SEQ ID NO: 18010)

malpvtalllplalllhaarpevqllesggglvqpggrslrlscaasgftf
 30 tnyamnwvraqpgkglewvsgisggstyyadsvkgrftisrdnsknv
 ylmnslraedtavvyckdwnttmtergggtltvvssttppaprpptp
 aptiasqplslrpeacrpaaggavhtrglfdacdiyiwaplagtcgvlll
 35 slvitlyckrgrkkllyifkqpfmrpvqttqeedgcsrfpeeeeggcel
 rvkfsrsadapaykqggnqlynelnlgrreeydvldkrrgrdpemggkpr
 rknpqeglynelqkdkmaeyseigmkgerrrrgkghdglyqglstatkdt
 40 ydalhmqalppr.

In some embodiments of the VCARS of the disclosure, the amino acid sequence of the VCAR is encoded by a nucleotide sequence comprising a sequence of at least 95%, at least 96%, at least 97%, at least 98% or at least 99% identity to the nucleotide sequence of:

VH-F

(SEQ ID NO: 18011)

atggcactgcctgtgacagccctgctgctgcctctggcctgctgctgca
 cgcagcacggcccaggtgcagctgctggagtcggaggaggcctgggtgc
 agccccggcggtccctgagactgtcttgccgcgcagcggcttcacctt
 55 acaactacgccatgaattgggtgcggcaggccctggcaaggcctgga
 gtgggtgtctggcatcagcggaggaggaggcagcactactatgcagact
 ccgtgaagggcaggttcaccatctcccgcatgaactctaagaatacactg
 60 tacctccagatgaacagcctgagggcagaggacaccgcgtgtactattg
 cgtgaaggattggaataaccacaatgatcacagagagggggcagggcaccc
 tggtagcctgagcagcaccacaacccctgccccagacctccacacccc
 65 gccctaccatcgcgagtcagccactgtccctgcggcctgaggcctgcgg
 gcccgcccgccgggagcagtgacacacggggcctggactttgctgtg

51

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acatctacatatgggcaccactggcaggaacctgcggcgtgctgctgctg
 agcctggtcatcacccgtactgtaagagaggcaggaagaagctgctgta
 tatcttcaagcagcccttcatgagaccctgcagaccacacaggaggagg
 acggctgctctttaggttcccagaggaggaggaggaggatgctgagctg
 cgctgaagtttagccggtccgcatgacactgcatacaagcagggaca
 gaaccagctgtataacgagctgaatctgggcccggagagaggagtacgacg
 tgctggataagaggcggggccgggaccccgagatgggaggcaagcctcgg
 agaaagaacccacaggaggcctgtacaatgagctgcaaaaggacaagat
 ggccgaggcctattctgagatcgccatgaaggagagaggcgcggggca
 agggacacgatggcctgtaccaggcctgagcaccgccacaaggacacc
 tatgatgccagcacatgacggccagcccccctagatga.

In some embodiments of the VCARs of the disclosure, the VCAR comprises a sequence encoding VH-A, VH-B, VH-C, VH-D, VH-E, or VH-F. In some embodiments of the VCARs of the disclosure, the VCAR comprises two sequences encoding en VH-A, VH-B, VH-C, VH-D, VH-E, or VH-F.

Immune and Immune Precursor Cells

In certain embodiments, immune cells of the disclosure comprise lymphoid progenitor cells, natural killer (NK) cells, T lymphocytes (T-cell), stem memory T cells (T_{SCM} cells), central memory T cells (T_{CM}), stem cell-like T cells, B lymphocytes (B-cells), myeloid progenitor cells, neutrophils, basophils, eosinophils, monocytes, macrophages, platelets, erythrocytes, red blood cells (RBCs), megakaryocytes or osteoclasts.

In certain embodiments, immune precursor cells comprise any cells which can differentiate into one or more types of immune cells. In certain embodiments, immune precursor cells comprise multipotent stem cells that can self renew and develop into immune cells. In certain embodiments, immune precursor cells comprise hematopoietic stem cells (HSCs) or descendants thereof. In certain embodiments, immune precursor cells comprise precursor cells that can develop into immune cells. In certain embodiments, the immune precursor cells comprise hematopoietic progenitor cells (HPCs). Hematopoietic Stem Cells (HSCs)

Hematopoietic stem cells (HSCs) are multipotent, self-renewing cells. All differentiated blood cells from the lymphoid and myeloid lineages arise from HSCs. HSCs can be found in adult bone marrow, peripheral blood, mobilized peripheral blood, peritoneal dialysis effluent and umbilical cord blood.

HSCs of the disclosure may be isolated or derived from a primary or cultured stem cell. HSCs of the disclosure may be isolated or derived from an embryonic stem cell, a multipotent stem cell, a pluripotent stem cell, an adult stem cell, or an induced pluripotent stem cell (iPSC).

Immune precursor cells of the disclosure may comprise an HSC or an HSC descendent cell. Exemplary HSC descendent cells of the disclosure include, but are not limited to, multipotent stem cells, lymphoid progenitor cells, natural killer (NK) cells, T lymphocyte cells (T-cells), B lymphocyte cells (B-cells), myeloid progenitor cells, neutrophils, basophils, eosinophils, monocytes, and macrophages.

HSCs produced by the methods of the disclosure may retain features of “primitive” stem cells that, while isolated or derived from an adult stem cell and while committed to

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a single lineage, share characteristics of embryonic stem cells. For example, the “primitive” HSCs produced by the methods of the disclosure retain their “stemness” following division and do not differentiate. Consequently, as an adoptive cell therapy, the “primitive” HSCs produced by the methods of the disclosure not only replenish their numbers, but expand in vivo. “Primitive” HSCs produced by the methods of the disclosure may be therapeutically-effective when administered as a single dose. In some embodiments, primitive HSCs of the disclosure are CD34+. In some embodiments, primitive HSCs of the disclosure are CD34+ and CD38-. In some embodiments, primitive HSCs of the disclosure are CD34+, CD38- and CD90+. In some embodiments, primitive HSCs of the disclosure are CD34+, CD38-, CD90+ and CD45RA-. In some embodiments, primitive HSCs of the disclosure are CD34+, CD38-, CD90+, CD45RA-, and CD49f+. In some embodiments, the most primitive HSCs of the disclosure are CD34+, CD38-, CD90+, CD45RA-, and CD49f+.

In some embodiments of the disclosure, primitive HSCs, HSCs, and/or HSC descendent cells may be modified according to the methods of the disclosure to express an exogenous sequence (e.g., a chimeric antigen receptor or therapeutic protein). In some embodiments of the disclosure, modified primitive HSCs, modified HSCs, and/or modified HSC descendent cells may be forward differentiated to produce a modified immune cell including, but not limited to, a modified T cell, a modified natural killer cell and/or a modified B-cell of the disclosure.

T Cells

Modified T cells of the disclosure may be derived from modified hematopoietic stem and progenitor cells (HSPCs) or modified HSCs.

Unlike traditional biologics and chemotherapeutics, modified-T cells of the disclosure possess the capacity to rapidly reproduce upon antigen recognition, thereby potentially obviating the need for repeat treatments. To achieve this, in some embodiments, modified-T cells of the disclosure not only drive an initial response, but also persist in the patient as a stable population of viable memory T cells to prevent potential relapses. Alternatively, in some embodiments, when it is not desired, modified-T cells of the disclosure do not persist in the patient.

Intensive efforts have been focused on the development of antigen receptor molecules that do not cause T cell exhaustion through antigen-independent (tonic) signaling, as well as of a modified-T cell product containing early memory T cells, especially stem cell memory (T_{SCM}) or stem cell-like T cells. Stem cell-like modified-T cells of the disclosure exhibit the greatest capacity for self-renewal and multipotent capacity to derive central memory (T_{CM}) T cells or T_{CM} like cells, effector memory (T_{EM}) and effector T cells (T_E), thereby producing better tumor eradication and long-term modified-T cell engraftment. A linear pathway of differentiation may be responsible for generating these cells: Naïve T cells (T_N) $\rightarrow$ T_{SCM} $\rightarrow$ T_{CM} $\rightarrow$ T_{EM} $\rightarrow$ T_E $\rightarrow$ T_{TE} , whereby T_N is the parent precursor cell that directly gives rise to T_{SCM} , which then, in turn, directly gives rise to T_{CM} , etc. Compositions of T cells of the disclosure may comprise one or more of each parental T cell subset with T_{SCM} or T_{CM} cells being the most abundant (e.g., T_{SCM} $\rightarrow$ T_{CM} $\rightarrow$ T_{EM} $\rightarrow$ T_E $\rightarrow$ T_{TE}).

In some embodiments of the methods of the disclosure, the immune cell precursor is differentiated into or is capable of differentiating into an early memory T cell, a stem cell like T-cell, a Naïve T cells (T_N), a T_{SCM} , a T_{CM} , a T_{EM} , a T_E ,

or a T_m. In some embodiments, the immune cell precursor is a primitive HSC, an HSC, or a HSC descendent cell of the disclosure.

In some embodiments of the methods of the disclosure, the immune cell is an early memory T cell, a stem cell like T-cell, a Naïve T cells (T_N), a T_{SCM}, a T_{CM}, a T_{EM}, a T_E, or a T_{TE}.

In some embodiments of the methods of the disclosure, the immune cell is an early memory T cell.

In some embodiments of the methods of the disclosure, the immune cell is a stem cell like T-cell.

In some embodiments of the methods of the disclosure, the immune cell is a T_{SCM}.

In some embodiments of the methods of the disclosure, the immune cell is a T_{CM}.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of an early memory T cell. In certain embodiments, the plurality of modified early memory T cells comprises at least one modified stem cell-like T cell. In certain embodiments, the plurality of modified early memory T cells comprises at least one modified T_{SCM}. In certain embodiments, the plurality of modified early memory T cells comprises at least one modified T_{CM}.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a stem cell-like T cell. In certain embodiments, the plurality of modified stem cell-like T cells comprises at least one modified T_{SCM}. In certain embodiments, the plurality of modified stem cell-like T cells comprises at least one modified T_{CM}.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a stem memory T cell (T_{SCM}). In certain embodiments, the cell-surface markers comprise CD62L and CD45RA. In certain embodiments, the cell-surface markers comprise one or more of CD62L, CD45RA, CD28, CCR7, CD127, CD45RO, CD95, CD95 and IL-2RP. In certain embodiments, the cell-surface markers comprise one or more of CD45RA, CD95, IL-2Rβ, CCR7, and CD62L.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a central memory T cell (T_{CM}). In certain embodiments, the cell-surface markers comprise one or more of CD45RO, CD95, IL-2RP, CCR7, and CD62L.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%,

20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a naïve T cell (T_N). In certain embodiments, the cell-surface markers comprise one or more of CD45RA, CCR7 and CD62L.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of an effector T-cell (modified TEFF). In certain embodiments, the cell-surface markers comprise one or more of CD45RA, CD95, and IL-2RP.

In some embodiments of the methods of the disclosure, the methods modify and/or the methods produce a plurality of modified T cells, wherein at least 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between of the plurality of modified T cells expresses one or more cell-surface marker(s) of a stem cell-like T cell, a stem memory T cell (T_{SCM}) or a central memory T cell (T_{CM}).

In some embodiments of the methods of the disclosure, a buffer comprises the immune cell or precursor thereof. The buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the immune cell or precursor thereof, including T-cells. In certain embodiments, the buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the primary human T cells prior to the nucleofection. In certain embodiments, the buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the primary human T cells during the nucleofection. In certain embodiments, the buffer maintains or enhances a level of cell viability and/or a stem-like phenotype of the primary human T cells following the nucleofection. In certain embodiments, the buffer comprises one or more of KCl, MgCl₂, ClNa, Glucose and Ca(NO₃)₂ in any absolute or relative abundance or concentration, and, optionally, the buffer further comprises a supplement selected from the group consisting of HEPES, Tris/HCl, and a phosphate buffer. In certain embodiments, the buffer comprises 5 mM KCl, 15 mM MgCl₂, 90 mM ClNa, 10 mM Glucose and 0.4 mM Ca(NO₃)₂. In certain embodiments, the buffer comprises 5 mM KCl, 15 mM MgCl₂, 90 mM ClNa, 10 mM Glucose and 0.4 mM Ca(NO₃)₂ and a supplement comprising 20 mM HEPES and 75 mM Tris/HCl. In certain embodiments, the buffer comprises 5 mM KCl, 15 mM MgCl₂, 90 mM ClNa, 10 mM Glucose and 0.4 mM Ca(NO₃)₂ and a supplement comprising 40 mM Na₂HPO₄/NaH₂PO₄ at pH 7.2. In certain embodiments, the composition comprising primary human T cells comprises 100 μl of the buffer and between 5×10⁶ and 25×10⁶ cells. In certain embodiments, the composition comprises a scalable ratio of 250×10⁶ primary human T cells per milliliter of buffer or other media during the introduction step.

In some embodiments of the methods of the disclosure, the methods comprise contacting an immune cell of the disclosure, including a T cell of the disclosure, and a T-cell expansion composition. In some embodiments of the methods of the disclosure, the step of introducing a transposon and/or transposase of the disclosure into an immune cell of the disclosure may further comprise contacting the immune cell and a T-cell expansion composition. In some embodiments, including those in which the introducing step of the methods comprises an electroporation or a nucleofection step, the electroporation or a nucleofection step may be

performed with the immune cell contacting T-cell expansion composition of the disclosure.

In some embodiments of the methods of the disclosure, the T-cell expansion composition comprises, consists essentially of or consists of phosphorus; one or more of an octanoic acid, a palmitic acid, a linoleic acid, and an oleic acid; a sterol; and an alkane.

In certain embodiments of the methods of producing a modified T cell of the disclosure, the expansion supplement comprises one or more cytokine(s). The one or more cytokine(s) may comprise any cytokine, including but not limited to, lymphokines. Exemplary lymphokines include, but are not limited to, interleukin-2 (IL-2), interleukin-3 (IL-3), interleukin-4 (IL-4), interleukin-5 (IL-5), interleukin-6 (IL-6), interleukin-7 (IL-7), interleukin-15 (IL-15), interleukin-21 (IL-21), granulocyte-macrophage colony-stimulating factor (GM-CSF) and interferon-gamma (INF). The one or more cytokine(s) may comprise IL-2.

In some embodiments of the methods of the disclosure, the T-cell expansion composition comprises human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid, nicotinamide, 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD), diisopropyl adipate (DIPA), n-butyl-benzenesulfonamide, 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester, palmitic acid, linoleic acid, oleic acid, stearic acid hydrazide, oleamide, a sterol and an alkane. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid, palmitic acid, linoleic acid, oleic acid and a sterol. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of between 0.9 mg/kg to 90 mg/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; oleic acid at a concentration of 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; and a sterol at a concentration of about 0.1 mg/kg to 10 mg/kg, inclusive of the endpoints. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of about 9 mg/kg, palmitic acid at a concentration of about 2 mg/kg, linoleic acid at a concentration of about 2 mg/kg, oleic acid at a concentration of about 2 mg/kg and a sterol at a concentration of about 1 mg/kg. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of between 6.4 μ mol/kg and 640 μ mol/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.7 μ mol/kg and 70 μ mol/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.75 μ mol/kg and 75 μ mol/kg, inclusive of the endpoints; oleic acid at a concentration of between 0.75 μ mol/kg and 75 μ mol/kg, inclusive of the endpoints; and a sterol at a concentration of between 0.25 μ mol/kg and 25 μ mol/kg, inclusive of the endpoints. In certain embodiments of this method, the T-cell expansion composition further comprises one or more of octanoic acid at a concentration of about 64 μ mol/kg, palmitic acid at a concentration of about 7 μ mol/kg, linoleic acid at a concentration of about 7.5 μ mol/kg, oleic acid at a concentration of about 7.5 μ mol/kg and a sterol at a concentration of about 2.5 μ mol/kg.

In certain embodiments, the T-cell expansion composition comprises one or more of human serum albumin, recombi-

nant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement to produce a plurality of expanded modified T-cells, wherein at least 2% of the plurality of modified T-cells expresses one or more cell-surface marker(s) of an early memory T cell, a stem cell-like T cell, a stem memory T cell (TscM) and/or a central memory T cell (TcM). In certain embodiments, the T-cell expansion composition comprises or further comprises one or more of octanoic acid, nicotinamide, 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD), diisopropyl adipate (DIPA), n-butyl-benzenesulfonamide, 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester, palmitic acid, linoleic acid, oleic acid, stearic acid hydrazide, oleamide, a sterol and an alkane. In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid, palmitic acid, linoleic acid, oleic acid and a sterol (e.g., cholesterol). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of between 0.9 mg/kg to 90 mg/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; oleic acid at a concentration of 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; and a sterol at a concentration of about 0.1 mg/kg to 10 mg/kg, inclusive of the endpoints (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of about 9 mg/kg, palmitic acid at a concentration of about 2 mg/kg, linoleic acid at a concentration of about 2 mg/kg, oleic acid at a concentration of about 2 mg/kg, and a sterol at a concentration of about 1 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of about 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of about 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of between 6.4 μ mol/kg and 640 μ mol/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.7 μ mol/kg and 70 μ mol/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.75 μ mol/kg and 75 μ mol/kg, inclusive of the endpoints; oleic acid at a concentration of between 0.75 μ mol/kg and 75 μ mol/kg, inclusive of the endpoints; and a sterol at a concentration of between 0.25 μ mol/kg and 25 μ mol/kg, inclusive of the endpoints. In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of about 64 μ mol/kg, palmitic acid at a concentration of about 7 μ mol/kg, linoleic acid at a concentration of about 7.5 μ mol/kg, oleic acid at a concentration of about 7.5 μ mol/kg and a sterol at a concentration of about 2.5 μ mol/kg. In certain embodiments, the T-cell expansion composition comprises one or more of octanoic acid at a concentration of about 63.75 μ mol/kg, palmitic acid at a concentration of about 7.27 μ mol/kg, linoleic acid at a concentration of about 7.57 μ mol/kg, oleic acid at a concentration of about 7.56 μ mol/kg and a sterol at a concentration of about 2.61 μ mol/kg. In certain embodi-

ments, the T-cell expansion composition comprises octanoic acid at a concentration of about 63.75 $\mu\text{mol/kg}$, palmitic acid at a concentration of about 7.27 $\mu\text{mol/kg}$, linoleic acid at a concentration of about 7.57 $\mu\text{mol/kg}$, oleic acid at a concentration of 7.56 $\mu\text{mol/kg}$ and a sterol at a concentration of 2.61 $\mu\text{mol/kg}$.

As used herein, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of phosphorus, an octanoic fatty acid, a palmitic fatty acid, a linoleic fatty acid and an oleic acid. In certain embodiments, the media comprises an amount of phosphorus that is 10-fold higher than may be found in, for example, Iscove's Modified Dulbecco's Medium (IMDM); available at ThermoFisher Scientific as Catalog number 12440053).

As used herein, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, Iscove's MDM, and an expansion supplement at 37° C. Alternatively, or in addition, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following elements: boron, sodium, magnesium, phosphorus, potassium, and calcium. In certain embodiments, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following elements present in the corresponding average concentrations: boron at 3.7 mg/L, sodium at 3000 mg/L, magnesium at 18 mg/L, phosphorus at 29 mg/L, potassium at 15 mg/L and calcium at 4 mg/L.

As used herein, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following components: octanoic acid (CAS No. 124-07-2), nicotinamide (CAS No. 98-92-0), 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD) (CAS No. 126-86-3), diisopropyl adipate (DIPA) (CAS No. 6938-94-9), n-butyl-benzenesulfonamide (CAS No. 3622-84-2), 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester (CAS No. 84-69-5), palmitic acid (CAS No. 57-10-3), linoleic acid (CAS No. 60-33-3), oleic acid (CAS No. 112-80-1), stearic acid hydrazide (CAS No. 4130-54-5), oleamide (CAS No. 3322-62-1), sterol (e.g., cholesterol) (CAS No. 57-88-5), and alkanes (e.g., nonadecane) (CAS No. 629-92-5). In certain embodiments, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following components: octanoic acid (CAS No. 124-07-2), nicotinamide (CAS No. 98-92-0), 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD) (CAS No. 126-86-3), diisopropyl adipate (DIPA) (CAS No. 6938-94-9), n-butyl-benzenesulfonamide (CAS No. 3622-84-2), 1,2-benzenedicarboxylic acid, bis(2-methylpropyl)

ester (CAS No. 84-69-5), palmitic acid (CAS No. 57-10-3), linoleic acid (CAS No. 60-33-3), oleic acid (CAS No. 112-80-1), stearic acid hydrazide (CAS No. 4130-54-5), oleamide (CAS No. 3322-62-1), sterol (e.g., cholesterol) (CAS No. 57-88-5), alkanes (e.g., nonadecane) (CAS No. 629-92-5), and phenol red (CAS No. 143-74-8). In certain embodiments, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following components: octanoic acid (CAS No. 124-07-2), nicotinamide (CAS No. 98-92-0), 2,4,7,9-tetramethyl-5-decyn-4,7-diol (TMDD) (CAS No. 126-86-3), diisopropyl adipate (DIPA) (CAS No. 6938-94-9), n-butyl-benzenesulfonamide (CAS No. 3622-84-2), 1,2-benzenedicarboxylic acid, bis(2-methylpropyl) ester (CAS No. 84-69-5), palmitic acid (CAS No. 57-10-3), linoleic acid (CAS No. 60-33-3), oleic acid (CAS No. 112-80-1), stearic acid hydrazide (CAS No. 4130-54-5), oleamide (CAS No. 3322-62-1), phenol red (CAS No. 143-74-8) and lanolin alcohol.

In certain embodiments, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following ions: sodium, ammonium, potassium, magnesium, calcium, chloride, sulfate and phosphate.

As used herein, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, and an expansion supplement at 37° C. Alternatively, or in addition, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following free amino acids: histidine, asparagine, serine, glutamate, arginine, glycine, aspartic acid, glutamic acid, threonine, alanine, proline, cysteine, lysine, tyrosine, methionine, valine, isoleucine, leucine, phenylalanine and tryptophan. In certain embodiments, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following free amino acids in the corresponding average mole percentages: histidine (about 1%), asparagine (about 0.5%), serine (about 1.5%), glutamine (about 67%), arginine (about 1.5%), glycine (about 1.5%), aspartic acid (about 1%), glutamic acid (about 2%), threonine (about 2%), alanine (about 1%), proline (about 1.5%), cysteine (about 1.5%), lysine (about 3%), tyrosine (about 1.5%), methionine (about 1%), valine (about 3.5%), isoleucine (about 3%), leucine (about 3.5%), phenylalanine (about 1.5%) and tryptophan (about 0.5%). In certain embodiments, the terms "supplemented T-cell expansion composition" or "T-cell expansion composition" may be used interchangeably with a media comprising one or more of the following free amino acids in the corresponding average mole percentages: histidine (about 0.78%), asparagine (about 0.4%), serine (about 1.6%), glutamine (about 67.01%), arginine (about 1.67%), glycine (about 1.72%), aspartic acid (about 1.00%), glutamic acid (about 1.93%), threonine (about 2.38%), alanine (about 1.11%), proline (about 1.49%), cysteine (about 1.65%), lysine (about 2.84%), tyrosine (about 1.62%),

methionine (about 0.85%), valine (about 3.45%), isoleucine (about 3.14%), leucine (about 3.3%), phenylalanine (about 1.64%) and tryptophan (about 0.37%).

As used herein, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of human serum albumin, recombinant human insulin, human transferrin, 2-Mercaptoethanol, Iscove’s MDM, and an expansion supplement at 37° C. Alternatively, or in addition, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of phosphorus, an octanoic fatty acid, a palmitic fatty acid, a linoleic fatty acid and an oleic acid. In certain embodiments, the media comprises an amount of phosphorus that is 10-fold higher than may be found in, for example, Iscove’s Modified Dulbecco’s Medium (IMDM); available at ThermoFisher Scientific as Catalog number 12440053).

In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid, palmitic acid, linoleic acid, oleic acid and a sterol (e.g., cholesterol). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of between 0.9 mg/kg to 90 mg/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; oleic acid at a concentration of 0.2 mg/kg to 20 mg/kg, inclusive of the endpoints; and a sterol at a concentration of about 0.1 mg/kg to 10 mg/kg, inclusive of the endpoints (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 9 mg/kg, palmitic acid at a concentration of about 2 mg/kg, linoleic acid at a concentration of about 2 mg/kg, oleic acid at a concentration of about 2 mg/kg, and a sterol at a concentration of about 1 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of about 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of about 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of 9.19 mg/kg, palmitic acid at a concentration of 1.86 mg/kg, linoleic acid at a concentration of 2.12 mg/kg, oleic acid at a concentration of about 2.13 mg/kg, and a sterol at a concentration of 1.01 mg/kg (wherein mg/kg=parts per million). In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of between 6.4 μmol/kg and 640 μmol/kg, inclusive of the endpoints; palmitic acid at a concentration of between 0.7 μmol/kg and 70 μmol/kg, inclusive of the endpoints; linoleic acid at a concentration of between 0.75 μmol/kg and 75 μmol/kg, inclusive of the endpoints; oleic

acid at a concentration of between 0.75 mol/kg and 75 mol/kg, inclusive of the endpoints; and a sterol at a concentration of between 0.25 mol/kg and 25 mol/kg, inclusive of the endpoints. In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 64 mol/kg, palmitic acid at a concentration of about 7 μmol/kg, linoleic acid at a concentration of about 7.5 μmol/kg, oleic acid at a concentration of about 7.5 μmol/kg and a sterol at a concentration of about 2.5 μmol/kg.

In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 63.75 μmol/kg, palmitic acid at a concentration of about 7.27 μmol/kg, linoleic acid at a concentration of about 7.57 μmol/kg, oleic acid at a concentration of about 7.56 mol/kg and a sterol at a concentration of about 2.61 μmol/kg. In certain embodiments, the terms “supplemented T-cell expansion composition” or “T-cell expansion composition” may be used interchangeably with a media comprising one or more of octanoic acid at a concentration of about 63.75 μmol/kg, palmitic acid at a concentration of about 7.27 μmol/kg, linoleic acid at a concentration of about 7.57 μmol/kg, oleic acid at a concentration of 7.56 mol/kg and a sterol at a concentration of 2.61 μmol/kg.

In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T_{SCM} and/or a T_{CM}) of the disclosure, the method comprises contacting a modified T cell and an inhibitor of the PI3K-Akt-mTOR pathway. Modified T-cells of the disclosure, including modified stem cell-like T cells, T_{SCM} and/or T_{CM} of the disclosure, may be incubated, cultured, grown, stored, or otherwise, combined at any step in the methods of the procedure with a growth medium comprising one or more inhibitors a component of a PI3K pathway. Exemplary inhibitors a component of a PI3K pathway include, but are not limited to, an inhibitor of GSK3β such as TWS119 (also known as GSK 3B inhibitor XII; CAS Number 601514-19-6 having a chemical formula $C_{18}H_{14}N_4O_2$). Exemplary inhibitors of a component of a PI3K pathway include, but are not limited to, bb007 (BLUEBIRDBIO™). Additional Exemplary inhibitors of a component of a PI3K pathway include, but are not limited to, an allosteric Akt inhibitor VIII (also referred to as Akti-1/2 having Compound number 10196499), ATP competitive inhibitors (Orthosteric inhibitors targeting the ATP-binding pocket of the protein kinase B (Akt)), Isoquinoline-5-sulfonamides (H-8, H-89, and NL-71-101), Azepane derivatives (A series of structures derived from (–)-balanol), Aminofurazans (GSK690693), Heterocyclic rings (7-azaindole, 6-phenylpurine derivatives, pyrrolo[2,3-d]pyrimidine derivatives, CCT128930, 3-aminopyrrolidine, anilinothiazole derivatives, spiroindoline derivatives, AZD5363, ipatasertib (GDC-0068, RG7440), A-674563, and A-443654), Phenylpyrazole derivatives (AT7867 and AT13148), Thiophenecarboxamide derivatives (Afuresertib (GSK2110183), 2-pyrimidyl-5-amidothiophene derivative (DC120), uprosertib (GSK2141795)), Allosteric inhibitors (Superior to orthosteric inhibitors providing greater specificity, reduced side-effects and less toxicity), 2,3-diphenylquinoxaline analogues (2,3-diphenylquinoxaline derivatives, triazolo[3,4-f][1,6]naphthyridin-3 (2H)-one derivative (MK-2206)), Alkylphospholipids (Edelfosine (1-O-octadecyl-2-O-methyl-rac-glycero-3-phosphocholine, ET-18-OCH₃), ilmofosine (BM 41.440), miltefosine (hexadecylphosphocholine, HePC), perifosine

(D-21266), erucylphosphocholine (ErPC), erufosine (ErPC3, erucylphosphohomocholine), Indole-3-carbinol analogues (Indole-3-carbinol, 3-chloroacetylindole, diindolylmethane, diethyl 6-methoxy-5,7-dihydroindolo [2,3-b] carbazole-2,10-dicarboxylate (SR13668), OSU-A9), Sulfonamide derivatives (PH-316 and PHT-427), Thiourea derivatives (PIT-1, PIT-2, DM-PIT-1, N-[(1-methyl-1H-pyrazol-4-yl)carbonyl]-N'-(3-bromophenyl)-thiourea), Purine derivatives (Triciribine (TCN, NSC 154020), triciribine mono-phosphate active analogue (TCN-P), 4-amino-pyrido[2,3-d]pyrimidine derivative API-1, 3-phenyl-3H-imidazo[4,5-b]pyridine derivatives, ARQ 092), BAY 1125976, 3-methyl-xanthine, quinoline-4-carboxamide and 2-[4-(cyclohexa-1,3-dien-1-yl)-1H-pyrazol-3-yl]phenol, 3-oxo-tirucallic acid, 3 α - and 3 β -acetoxy-tirucallic acids, acetoxy-tirucallic acid, and irreversible inhibitors (antibiotics, Lactoquinomycin, Frenolicin B, kalafungin, medermycin, Boc-Phe-vinyl ketone, 4-hydroxynonenal (4-HNE), 1,6-naphthyridinone derivatives, and imidazo-1,2-pyridine derivatives).

In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T_{SCM} and/or a T_{CM}) of the disclosure, the method comprises contacting a modified T cell and an inhibitor of T cell effector differentiation. Exemplary inhibitors of T cell effector differentiation include, but are not limited to, a BET inhibitor (e.g., JQ1, a hienotriazolodiazepine) and/or an inhibitor of the BET family of proteins (e.g., BRD2, BRD3, BRD4, and BRDT).

In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T_{SCM} and/or a T_{CM}) of the disclosure, the method comprises contacting a modified T cell and an agent that reduces nucleocytoplasmic Acetyl-CoA. Exemplary agents that reduce nucleocytoplasmic Acetyl-CoA include, but are not limited to, 2-hydroxy-citrate (2-HC) as well as agents that increase expression of Acss1.

In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T_{SCM} and/or a T_{CM}) of the disclosure, the method comprises contacting a modified T cell and a composition comprising a histone deacetylase (HDAC) inhibitor. In some embodiments, the composition comprising an HDAC inhibitor comprises or consists of valproic acid, Sodium Phenylbutyrate (NaPB) or a combination thereof. In some embodiments, the composition comprising an HDAC inhibitor comprises or consists of valproic acid. In some embodiments, the composition comprising an HDAC inhibitor comprises or consists of Sodium Phenylbutyrate (NaPB).

In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T_{SCM} and/or a T_{CM}) of the disclosure, the activation supplement may comprise one or more cytokine(s). The one or more cytokine(s) may comprise any cytokine, including but not limited to, lymphokines. Exemplary lymphokines include, but are not limited to, interleukin-2 (IL-2), interleukin-3 (IL-3), interleukin-4 (IL-4), interleukin-5 (IL-5), interleukin-6 (IL-6), interleukin-7 (IL-7), interleukin-15 (IL-15), interleukin-21 (IL-21), granulocyte-macrophage colony-stimulating factor (GM-CSF) and interferon-gamma (INF). The one or more cytokine(s) may comprise IL-2.

In certain embodiments of the methods of producing a modified T cell (e.g., a stem cell-like T cell, a T_{SCM} and/or a T_{CM}) of the disclosure, the activation supplement may comprise one or more activator complexes. Exemplary and nonlimiting activator complexes may comprise a monomeric, dimeric, trimeric or tetrameric antibody complex that binds one or more of CD3, CD28, and CD2. In some

embodiments, the activation supplement comprises or consists of an activator complex that comprises a human, a humanized or a recombinant or a chimeric antibody. In some embodiments, the activation supplement comprises or consists of an activator complex that binds CD3 and CD28. In some embodiments, the activation supplement comprises or consists of an activator complex that binds CD3, CD28 and CD2.

Natural Killer (NK) Cells

In certain embodiments, the modified immune or immune precursor cells of the disclosure are natural killer (NK) cells. In certain embodiments, NK cells are cytotoxic lymphocytes that differentiate from lymphoid progenitor cells.

Modified NK cells of the disclosure may be derived from modified hematopoietic stem and progenitor cells (HSPCs) or modified HSCs.

In certain embodiments, non-activated NK cells are derived from CD3-depleted leukopheresis (containing CD14/CD19/CD56+ cells).

In certain embodiments, NK cells are electroporated using a Lonza 4D nucleofector or BTX ECM 830 (500V, 700 usec pulse length, 0.2 mm electrode gap, one pulse). All Lonza 4D nucleofector programs are contemplated as within the scope of the methods of the disclosure.

In certain embodiments, 5 $\times$ 10⁶ cells were electroporated per electroporation in 100 μ L P3 buffer in cuvettes. However, this ratio of cells per volume is scalable for commercial manufacturing methods.

In certain embodiments, NK cells were stimulated by co-culture with an additional cell line. In certain embodiments, the additional cell line comprises artificial antigen presenting cells (aAPCs). In certain embodiments, stimulation occurs at day 1, 2, 3, 4, 5, 6, or 7 following electroporation. In certain embodiments, stimulation occurs at day 2 following electroporation.

In certain embodiments, NK cells express CD56.

B Cells

In certain embodiments, the modified immune or immune precursor cells of the disclosure are B cells. B cells are a type of lymphocyte that express B cell receptors on the cell surface. B cell receptors bind to specific antigens.

Modified B cells of the disclosure may be derived from modified hematopoietic stem and progenitor cells (HSPCs) or modified HSCs.

In certain embodiments, HSPCs are modified using the methods of the disclosure, and then primed for B cell differentiation in presence of human IL-3, Flt3L, TPO, SCF, and G-CSF for at least 3 days, at least 4 days, at least 5 days, at least 6 days or at least 7 days. In certain embodiments, HSPCs are modified using the methods of the disclosure, and then primed for B cell differentiation in presence of human IL-3, Flt3L, TPO, SCF, and G-CSF for 5 days.

In certain embodiments, following priming, modified HSPC cells are transferred to a layer of feeder cells and fed bi-weekly, along with transfer to a fresh layer of feeders once per week. In certain embodiments, the feeder cells are MS-5 feeder cells.

In certain embodiments, modified HSPC cells are cultured with MS-5 feeder cells for at least 7, 14, 21, 28, 30, 33, 35, 42 or 48 days. In certain embodiments, modified HSPC cells were cultured with MS-5 feeder cells for 33 days.

Transposition Systems

Exemplary transposon/transposase systems of the disclosure include, but are not limited to, piggyBac transposons and transposases, piggyBac-like transposons and transpos-

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ases, Sleeping Beauty transposons and transposases, Hel-raiser transposons and transposases and Tol2 transposons and transposases.

The piggyBac transposase recognizes transposon-specific inverted terminal repeat sequences (ITRs) on the ends of the transposon, and moves the contents between the ITRs into TTAA chromosomal sites. The piggyBac transposon system has no payload limit for the genes of interest that can be included between the ITRs. In certain embodiments, and, in particular, those embodiments wherein the transposon is a piggyBac transposon, the transposase is a piggyBac or a Super piggyBac (SPB) transposase. In certain embodiments, and, in particular, those embodiments wherein the transposase is a Super piggyBac (SPB) transposase, the sequence encoding the transposase is an mRNA sequence.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme. The piggyBac (PB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14487)

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1 MGSSLDDDEHI LSALLQSDDE LVGEDSDSEI
SDHVS EDDVQ SDTEEFIDE VHEVQPTSSG
61 SEILDEQNVI EQPGSSLASN RILTLPQRTI
RGKNKHCWST SKSTRRSRVS ALNIVRSQRG
121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW
TNAEISLKRR ESMTGATFRD TNEDEIYAFF
181 GILVMTAVRK DNHMSTDDLF DRSLSMVYVS
VMSRDRFDL IRLRMDDKS IRPTLRENDV
241 FTPVRKIIDL FIHQCIQNYT PGAHLTIDEQ
LLGFRGRCPF RMYIPNKPSK YGIKILMMCD
301 SGT KYMINGM PYLGRGTQTN GVPLGEYYVK
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ
361 EPYKLTIVGT VRSNKREIPE VLKNSRSRPV
GTSMFCDGP LTLVSYKPKP AKMVYLLSSC
421 DEDASINEST GKPQVMYYN QTKGGVDTLD
QMCSVMTC SR KTRWPMALL YGMINIACIN
481 SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL
TSSFMKRLE APTLKRYLRD NISNILPNEV
541 PGTSDDSTEE PVMKKRTYCT YCPSKIRKKA
NASCKCKKV ICREHNIDMC QSCF.

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In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at one or more of positions 30, 165, 282, or 538 of the sequence:

(SEQ ID NO: 14487)

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1 MGSSLDDDEHI LSALLQSDDE LVGEDSDSEI
SDHVS EDDVQ SDTEEFIDE VHEVQPTSSG

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61 SEILDEQNVI EQPGSSLASN RILTLPQRTI
RGKNKHCWST SKSTRRSRVS ALNIVRSQRG
121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW
TNAEISLKRR ESMTGATFRD TNEDEIYAFF
181 GILVMTAVRK DNHMSTDDLF DRSLSMVYVS
VMSRDRFDL IRLRMDDKS IRPTLRENDV
241 FTPVRKIIDL FIHQCIQNYT PGAHLTIDEQ
LLGFRGRCPF RMYIPNKPSK YGIKILMMCD
301 SGT KYMINGM PYLGRGTQTN GVPLGEYYVK
ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ
361 EPYKLTIVGT VRSNKREIPE VLKNSRSRPV
GTSMFCDGP LTLVSYKPKP AKMVYLLSSC
421 DEDASINEST GKPQVMYYN QTKGGVDTLD
QMCSVMTC SR KTRWPMALL YGMINIACIN
481 SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL
TSSFMKRLE APTLKRYLRD NISNILPNEV
541 PGTSDDSTEE PVMKKRTYCT YCPSKIRKKA
NASCKCKKV ICREHNIDMC QSCF.

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In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at two or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at three or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac (PB) transposase enzyme that comprises or consists of an amino acid substitution at each of the following positions 30, 165, 282, and 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the amino acid substitution at position 30 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 165 of the sequence of SEQ ID NO: 14487 is a substitution of a serine (S) for a glycine (G). In certain embodiments, the amino acid substitution at position 282 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 538 of the sequence of SEQ ID NO: 14487 is a substitution of a lysine (K) for an asparagine (N).

In certain embodiments of the methods of the disclosure, the transposase enzyme is a Super piggyBac (SPB) transposase enzyme. In certain embodiments, the Super piggyBac (SPB) transposase enzymes of the disclosure may comprise or consist of the amino acid sequence of the sequence of SEQ ID NO: 14487 wherein the amino acid substitution at position 30 is a substitution of a valine (V) for an isoleucine (I), the amino acid substitution at position 165 is a substitution of a serine (S) for a glycine (G), the amino acid substitution at position 282 is a substitution of a valine (V) for a methionine (M), and the amino acid substitution at

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position 538 is a substitution of a lysine (K) for an asparagine (N). In certain embodiments, the Super piggyBac (SPB) transposase enzyme may comprise or consist of an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14484)

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1  MGSSLDDEHI  LSALLQSDDE  LVGEDSDSEV
    SDHVSEDDVQ  SDTEEAFFIDE  VHEVQPTSSG
61  SEILDEQNVII  EQPGSSLASN  RILTLPQRTI
    RGKNKHCWST  SKSTRRSRVS  ALNIVRSQRG
121 PTRMCRNIYD  PLLCFKLFFT  DEIIEIVKW
    TNAEISLKRR  EMSATSATFRD  TNEDEIYAFF
181 GILVMTAVRK  DNHMSTDDL  FDRSLSMVYVS
    VMSRDRFDFL  IRCRLMDDKS  IRPTLRENDV
241 FTPVRKIWDL  FIIHQCIQNYT  PGAHLTIDEQ
    LLGFRGRCPF  RVYIPNKPSK  YGIKILMMCD
301 SGTKYMINGM  PYLGRGTQTN  GVPLGEYYVK
    ELSKPVHGSC  RNITCDNWFT  SIPLAKNLLQ
361 EPYKLTIVGT  VRSNKREIPE  VLKNSRSRPV
    GTSMFCDPDP  LTLVSYKPKP  AKMVYLLSSC
421 DEDASINEST  GKPQVMYYN  QTKGGVDTLD
    QMCSVMTCSR  KTNRWPMALL  YGMINIACIN
481 SFIIYSHNVS  SKGEKVQSRK  KFMRLNLYMSL
    TSSFMKRKLE  APTLKRYLRD  NISNILPKEV
541 PGTSDDSTEE  PVMKKRTYCT  YCPSKIRKKA
    NASCKKCKKV  ICREHNIDMC  QSCF.
  
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In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 3, 46, 82, 103, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 258, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 486, 503, 552, 570 and 591 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 46, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 485, 503, 552 and 570. In certain embodiments, the amino acid substitution at position 3 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for a serine (S). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an alanine (A). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 82 of SEQ ID NO: 14487 or

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SEQ ID NO: 14484 is a substitution of a tryptophan (W) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 119 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for an arginine (R). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) a cysteine (C). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 185 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 187 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for an alanine (A). In certain embodiments, the amino acid substitution at position 200 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 207 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a valine (V). In certain embodiments, the amino acid substitution at position 209 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a valine (V). In certain embodiments, the amino acid substitution at position 226 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a methionine (M). In certain embodiments, the amino acid substitution at position 235 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a leucine (L). In certain embodiments, the amino acid substitution at position 240 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 241 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 243 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a proline (P). In certain embodiments, the amino acid substitution at position 258 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a

leucine (L). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a proline (P). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine for a proline (P). In certain embodiments, the amino acid substitution at position 315 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for an arginine (R). In certain embodiments, the amino acid substitution at position 319 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a threonine (T). In certain embodiments, the amino acid substitution at position 327 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 328 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a cysteine (C). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 421 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for the aspartic acid (D). In certain embodiments, the amino acid substitution at position 436 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a valine (V). In certain embodiments, the amino acid substitution at position 456 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a methionine (M). In certain embodiments, the amino acid substitution at position 470 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 485 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a serine (S). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a methionine (M). In certain embodiments, the amino acid substitution at position 552 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a glutamine (Q). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a glutamine (Q).

In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac transposase enzyme may

comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at two, three, four, five, six or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac transposase enzyme may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 194 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 372 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for an arginine (R). In certain embodiments, the amino acid substitution at position 375 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a lysine (K). In certain embodiments, the amino acid substitution at position 450 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for an aspartic acid (D). In certain embodiments, the amino acid substitution at position 509 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a serine (S). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487. In certain embodiments, including those embodiments wherein the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, the piggyBac transposase enzyme may further comprise an amino acid substitution at positions 372, 375 and 450 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, and a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487. In certain embodiments, the piggyBac transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487 and a substitution of an asparagine (N) for an aspartic acid (D) at position 450 of SEQ ID NO: 14487.

The sleeping beauty transposon is transposed into the target genome by the Sleeping Beauty transposase that recognizes ITRs, and moves the contents between the ITRs into TA chromosomal sites. In various embodiments, SB

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transposon-mediated gene transfer, or gene transfer using any of a number of similar transposons, may be used in the compositions and methods of the disclosure.

In certain embodiments, and, in particular, those embodiments wherein the transposon is a Sleeping Beauty transposon, the transposase is a Sleeping Beauty transposase or a hyperactive Sleeping Beauty transposase (SB100X).

In certain embodiments of the methods of the disclosure, the Sleeping Beauty transposase enzyme comprises an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

```
(SEQ ID NO: 14485)
1  MGKSKEISQD LRKKIVDLHK SGSSLGAISK
    RLKVPRSSVQ TIVRKYKHHG TTQPSYRSGR
61  RRVLSRPRDER TLVRKVQINP RTTAKDLVKM
    LEETGTKVSI STVKRVLYRH NLKGRSARKK
121 PLLQNRHKKA RLRFATAHGD KDRTFWRNVL
    WSDETKIELF GHNDHRYVWR KKGGEACKPKN
181 TIPTVKHGGG SIMLWGCFAA GGTGALHKID
    GIMRKENYVD ILKQHLKTSV RKLKLGKRWV
241 FQMDNDPKHT SKVVAKWLD NKVKVLEWPS
    QSPDLNPIEN LWAEKKRVR ARRPTNLTQL
301 HQLCQEEWAK IHPTYCGKLV EGYPKRLTQV
    KQFKGNATKY.
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In certain embodiments of the methods of the disclosure, the hyperactive Sleeping Beauty (SB100X) transposase enzyme comprises an amino acid sequence at least 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

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(SEQ ID NO: 14486)
1  MGKSKEISQD LRKRIVDLHK SGSSLGAISK
    RLAVPRSSVQ TIVRKYKHHG TTQPSYRSGR
61  RRVLSRPRDER TLVRKVQINP RTTAKDLVKM
    LEETGTKVSI STVKRVLYRH NLKGRSARKK
121 PLLQNRHKKA RLRFATAHGD KDRTFWRNVL
    WSDETKIELF GHNDHRYVWR KKGGEACKPKN
181 TIPTVKHGGG SIMLWGCFAA GGTGALHKID
    GIMDAVQYVD ILKQHLKTSV RKLKLGKRWV
241 FQHDNDPKHT SKVVAKWLD NKVKVLEWPS
    QSPDLNPIEN LWAEKKRVR ARRPTNLTQL
301 HQLCQEEWAK IHPNYCGKLV EGYPKRLTQV
    KQFKGNATKY.
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The Helraiser transposon is transposed by the Helitron transposase. Helitron transposases mobilize the Helraiser transposon, an ancient element from the bat genome that was active about 30 to 36 million years ago. An exemplary Helraiser transposon of the disclosure includes Helibat1, which comprises a nucleic acid sequence comprising:

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(SEQ ID NO: 17006)
1  TCCTATATAA TAAAGAGAA ACATGCAAAT TGACCATCCC TCCGCTACGC TCAAGCCACG
61  CCCACCAGCC AATCAGAAGT GACTATGCAA ATTAACCCAA CAAAGATGGC AGTTAAATTT
121 GCATACGCAG GTGTCAAGCG CCCAGGAGG CAACGGCGGC CGCGGGCTCC CAGGACCTTC
181 GCTGGCCCCG GGAGGCGAGG CCGGCCGCGC CTAGCCACAC CCGCGGCTC CCGGGACCTT
241 CGCCAGCAGA GAGCAGAGCG GGAGAGCGGG CGGAGAGCGG GAGGTTTGA GGA CTG GCA
301 GAGCAGGAGG CCGCTGGACA TAGAGCAGAG CGAGAGAGAG GGTGGCTTG AGGCGCTGGC
361 TCCCTCTGTC ACCCCAGCTT CCTCATCACA GCTGTGAAA CTGACAGCAG GGAGGAGGAA
421 GTCCACCCCC CACAGAATCA GCCAGAATCA GCCGTTGGTC AGACAGCTCT CAGCGGCCTG
481 ACAGCCAGGA CTCTCATTCA CTGCATCTC AGACCGTGAC AGTAGAGAGG TGGGACTATG
541 TCTAAAGAAC AACTGTTGAT ACAACGTAGC TCTGCAGCCG AAAGATGCCG GCGTTATCGA
601 CAGAAAATGT CTGCAGAGCA ACGTGCCTCT GATCTTGAAA GAAGCGGCGC CCGCAACAG
661 AATGTATCTG AAGAGCAGCT ACTGGAAAA CGTCGCTCTG AAGCCGAAAA ACAGCGGCGT
721 CATCGACAGA AAATGTCTAA AGACCAACGT GCCTTTGAAG TTGAAAGAAG GCGGTGGCGA
781 CGACAGAATA TGTCTAGAGA ACAGTCATCA ACAAGTACTA CCAATACCGG TAGGA ACTGC
841 CTTCTCAGCA AAAATGGAGT ACATGAGGAT GCAATTCTCG AACATAGTTG TGGTGAATG
901 ACTGTTTCGAT GTGAATTTTG CCTATCACTA AATTTCTCTG ATGAAAAACC ATCCGATGGG
961 AAATTTACTC GATGTTGTAG CAAAGGGAAA GTCTGTCCAA ATGATATACA TTTTCCAGAT
1021 TACC CGCAT ATTTAAAAAG ATTAATGACA AACGAAGATT CTGACAGTAA AAATTT CATG
1081 GAAAATATTC GTTCCATAAA TAGTTCTTTT GCTTTTGCTT CCATGGGTGC AAATATTGCA
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1141 TCGCCATCAG GATATGGGCC ATACTGTTTT AGAATACACG GACAAGTTTA TCACCGTACT
 1201 GGAACTTTAC ATCCTTCGGA TGGTGTTCCT CGGAAGTTTG CTCAACTCTA TATTTTGAT
 1261 ACAGCCGAAG CTACAAGTAA AAGATTAGCA ATGCCAGAAA ACCAGGGCTG CTCAGAAAGA
 1321 CTCATGATCA ACATCAACAA CCTCATGCAT GAAATAAATG AATTAACAAA ATCGTACAAG
 1381 ATGCTACATG AGGTAGAAAA GGAAGCCCAA TCTGAAGCAG CAGCAAAAGG TATTGCTCCC
 1441 ACAGAAGTAA CAATGGCGAT TAAATACGAT CGTAACAGTG ACCCAGGTAG ATATAATTCT
 1501 CCCCCTGTAA CCGAGGTTGC TGTCATATTC AGAAACGAAG ATGGAGAACC TCCTTTTGAA
 1561 AGGGACTTGC TCATTCATTG TAAACCAGAT CCCAATAATC CAAATGCCAC TAAAATGAAA
 1621 CAAATCAGTA TCCTGTTTCC TACATTAGAT GCAATGACAT ATCCTATTCT TTTCCACAT
 1681 GGTGAAAAAG GCTGGGGAAC AGATATTGCA TTAAGACTCA GAGACAACAG TGTAATCGAC
 1741 AATAATACTA GACAAAATGT AAGGACACGA GTCACACAAA TGCAGTATTA TGGATTTCAT
 1801 CTCTCTGTGC GGGACACGTT CAATCCTATT TTAAATGCAG GAAAATTAAC TCAACAGTTT
 1861 ATGTGGATT CATATTCAAA AATGGAGGCC AATCGGATAA ATTTTCATCA AGCAAACCAA
 1921 TCTAAGTTGA GAGTTGAAAA ATATAGTGGT TTGATGGATT ATCTCAAATC TAGATCTGAA
 1981 AATGACAATG TGCCGATTGG TAAAATGATA ATACTTCCAT CATCTTTTGA GGGTAGTCCC
 2041 AGAAATATGC AGCAGCGATA TCAGGATGCT ATGGCAATTG TAACGAAGTA TGGCAAGCCC
 2101 GATTTATTCA TAACCATGAC ATGCAACCCC AAATGGGCAG ATATTACAAA CAATTTACAA
 2161 CGCTGGCAAA AAGTTGAAAA CAGACCTGAC TTGGTAGCCA GAGTTTTTAA TATTAAGCTG
 2221 AATGCTCTTT TAAATGATAT ATGTAAATTC CATTTATTTG GCAAAGTAAT AGCTAAAATT
 2281 CATGTCATTG AATTTAGAA ACGCGACTG CCTCACGCTC ACATATTATT GATATTAGAT
 2341 AGTGAGTCCA AATTACGTTT AGAAGATGAC ATTGACCGTA TAGTTAAGGC AGAAATTCCA
 2401 GATGAAGACC AGTGTCTCG ACTTTTTCAA ATTGTAAAT CAAATATGGT ACATGGACCA
 2461 TGTGAATAC AAAATCCAAA TAGTCCATGT ATGGAATATG GAAATGTTC AAAGGGATAT
 2521 CCAAAAGAAT TTCAAATGC GACCATTGGA AATATTGATG GATATCCCAA ATACAAACGA
 2581 AGATCTGGTA GCACCATGTC TATTGGAAAT AAAGTTGTCG ATAACACTTG GATTGTCCCT
 2641 TATAACCCGT ATTTGTGCCT TAAATATAAC TGTCATATAA ATGTTGAAGT CTGTGCATCA
 2701 ATTAAAGTG TCAAATATTT ATTTAAATAC ATCTATAAAG GGCACGATTG TGCAAATATT
 2761 CAAATTTCTG AAAAAATAT TATCAATCAT GACGAAGTAC AGGACTTCAT TGAATCCAGG
 2821 TATGTAGCG CTCCTGAGGC TGTTTGAGA CTTTTTGCAA TGCGAATGCA TGACCAATCT
 2881 CATGCAATCA CAAGATTAGC TATTCATTG CCAATGATC AGAATTTGTA TTTTCATACC
 2941 GATGATTTTG CTGAAGTTTT AGATAGGGCT AAAAGGCATA ACTCGACTTT GATGGCTTGG
 3001 TTCTTATTGA ATAGAGAAGA TTCTGATGCA CGTAATTATT ATTATTGGGA GATTCCACAG
 3061 CATTATGTGT TTAATAATTC TTTGTGACA AAACGCCGAA AGGGTGGGAA TAAAGTATTA
 3121 GGTAGACTGT TCACTGTGAG CTTTAGAGAA CCAGAACGAT ATTACCTTAG ACTTTTGCTT
 3181 CTGCATGTA AAGGTGCGAT AAGTTTTGAG GATCTGCGAA CTGTAGGAGG TGTAACCTAT
 3241 GATACATTC ATGAAGCTGC TAAACACCGA GGATTATTAC TTGATGACAC TATCTGAAAA
 3301 GATACGATTG ACGATGCAAT CATCCTTAAT ATGCCCAAAC AACTACGGCA ACTTTTGCA
 3361 TATATATGTG TGTGAGATG TCCTTCTGCT GCAGACAAAT TATGGGATGA GAATAAATCT
 3421 CATTTTATTG AAGATTTCTG TTGGAAATTA CACCGAAGAG AAGGTGCCTG TGTGAACGTG
 3481 GAAATGCATG CCCTTAACGA AATTCAGGAG GTATTACAT TGCATGGAAT GAAATGTTCA
 3541 CATTTCAAAC TTCCGGACTA TCCTTTATTA ATGAATGCAA ATACATGTGA TCAATTGTAC

-continued

3601 GAGCAACAAC AGGCAGAGGT TTTGATAAAT TCTCTGAATG ATGAACAGTT GGCAGCCTTT
3661 CAGACTATAA CTTCAGCCAT CGAAGATCAA ACTGTACACC CCAAATGCTT TTTCTTGGAT
3721 GGTCCAGGTG GTAGTGGAAA AACATATCTG TATAAAGTTT TAACACATTA TATTAGAGGT
3781 CGTGGTGGTA CTGTTTTACC CACAGCATCT ACAGGAATTG CTGCAAATTT ACTTCTTGGT
3841 GGAAGAACCT TTCATTCCCA ATATAAATTA CCAATTCCAT TAAATGAAAC TTCAATTTCT
3901 AGACTCGATA TAAAGAGTGA AGTTGCTAAA ACCATTAAAA AGGCCCACT TCTCATTATT
3961 GATGAATGCA CCATGGCATC CAGTCATGCT ATAAACGCCA TAGATAGATT ACTAAGAGAA
4021 ATTATGAATT TGAATGTTGC ATTTGGTGGG AAAGTTCTCC TTCTCGGAGG GGATTTTCGA
4081 CAATGCTCA GTATTGTACC ACATGCTATG CGATCGGCCA TAGTACAAAC GAGTTTAAAG
4141 TACTGTAATG TTTGGGGATG TTTCAGAAAG TTGTCTCTTA AAACAAATAT GAGATCAGAG
4201 GATTCTGCTT ATAGTGAATG GTTAGTAAAA CTTGGAGATG GCAAACCTGA TAGCAGTTTT
4261 CATTTAGGAA TGGATATTAT TGAAATCCCC CATGAAATGA TTTGTAACGG ATCTATTATT
4321 GAAGCTACCT TTGGAAATAG TATATCTATA GATAATATTA AAAATATATC TAAACGTGCA
4381 ATTCTTTGTC CAAAAAATGA GCATGTTCAA AAATTAAATG AAGAAATTTT GGATATACTT
4441 GATGGAGATT TTCACACATA TTTGAGTGAT GATTCCATTG ATTCAACAGA TGATGCTGAA
4501 AAGGAAAATT TTCCCATCGA ATTTCTTAAT AGTATTACTC CTTCGGGAAT GCCGTGTCAT
4561 AAATTAAAT TGAAAGTGGG TGCAATCATC ATGCTATTGA GAAATCTTAA TAGTAAATGG
4621 GGTCTTTGTA ATGGTACTAG ATTTATTATC AAAAGATTAC GACCTAACAT TATCGAAGCT
4681 GAAGTATTAA CAGGATCTGC AGAGGGAGAG GTTGTCTGA TTCCAAGAAT TGATTTGTCC
4741 CCATCTGACA CTGGCCTCCC ATTTAAATTA ATTCGAAGAC AGTTTCCCGT GATGCCAGCA
4801 TTTGCGATGA CTATTAATAA ATCACAAGGA CAACTCTAG ACAGAGTAGG AATATTCCTA
4861 CCTGAACCCG TTTTCGCACA TGGTCAGTTA TATGTTGCTT TCTCTCAGT TCGAAGAGCA
4921 TGTGACGTTA AAGTTAAAGT TGTAATACT TCATCACAAG GGAAATTAGT CAAGCACTCT
4981 GAAAGTGT TTACTCTTAA TGTGGTATAC AGGGAGATAT TAGAATAAGT TTAATCACTT
5041 TATCAGTCAT TGTTCGCATC AATGTTGTTT TTATATCATG TTTTGTGTG TTTTATATCA
5101 TGTCTTTGTT GTTGTATAT CATGTTGTTA TTGTTTATTT ATTAATAAAT TTATGTATTA
5161 TTTTCATATA CATTTTACTC ATTTCTTTC ATCTCTCACA CTTCTATTAT AGAGAAAGGG
5221 CAAATAGCAA TATTAATAA TTTCTCTAA TTAATTCCT TCAATGTGC ACGAATTTCG
5281 TGCACCGGGC CACTAG.

Unlike other transposases, the Helitron transposase does not contain an RNase-H like catalytic domain, but instead comprises a RepHel motif made up of a replication initiator domain (Rep) and a DNA helicase domain. The Rep domain is a nuclease domain of the HUH superfamily of nucleases.

An exemplary Helitron transposase of the disclosure comprises an amino acid sequence comprising:

(SEQ ID NO: 14501)
1 MSKEQLLIQR SSAAERCRRY RQKMSAEQRA 60
SDLERRRRLQ QNVSEEQLLE KRRSEAEKQR
61 RHRQKMSKDQ RAFEVERRRW RRQNSREQS
STSTNTGRN CLLSKNGVHE DAILEHSCGG 65

-continued

121 MTVRCEFCLS LNFSDEKPSD GKPTRCCSKG
KVCNDIHFP DYPAYLKRLM TNEDSDSKNF
181 MENIRSINSS FAFASMGANI ASPSGYGPYC
FRIHQVYHR TGTLHPSDGV SRKFAQLYLIL
241 DTAEATSKRL AMPENQGCSE RLMINNNLM
HEINELTKSY KMLHEVEKEA QSEAAKGIA
301 PTEVIMAIKY DRNSDPGRYN SPRVTEVAVI
PRNEDGEPPF ERDLLIHCKP DPNNPNATKM

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-continued

361 KQISILFPTL DAMTYPILEP HGEKGWGTDI
 ALRLRDNSVI DNNTRQNVRT RVTQMYYGYF
 421 HLSVRDTFNP ILNAGKLTQQ FIVDSYSKME
 ANRINFIKAN QSKLRVEKYS GLMDYLKSRs
 481 ENDNVPIGKM IILPSSFEGS PRNMQQRYQD
 AMAIVTKYKG PDLFITMTCN PKWADITNNL
 541 QRWQKVENRP DLVARVFNIK LNALLNDICK
 FHLFGKVIK IHVIEFQKRG LPHAHILLIL
 601 DSESKLRSED DIDRIVKAEI PDEDQCPRLF
 QIVKSNMVG PCGIQNPNSP CMENGKCSKG
 661 YPKFQNTATI GNIDGYPKYK RRSGSTMSIG
 NKVVDNTWIV PYNPYLCLKY NCHINVEVCA
 721 SIKSVKYLK YIYKGHDCA IQISEKNIIN
 HDEVQDFIDS RYVSAPEAVW RLFAMRMHDQ
 781 SHAITRLAIH LPNDQNLPH TDDFAEVLDR
 AKRHNSLMA WFLNREDS DARNYYYWEIP
 841 QHYVFNNSLW TKRRKGKNV LGRIFTVSFR
 EPERYLLRL LLHVKGAI SF EDLRTVGGVT
 901 YDTFHEAAKH RGLLLDDTIW KDTIDDAIIL
 NMPKQLRQLF AYICVFGCPS AADKLWDENK
 961 SHPIEDFCWK LHRREGACVN CEMHALNEIQ
 EVFTLHGMKC SHFKLPDYPL LMNANTCDQL
 1021 YEQQQAELVI NSLNDEQLAA FQTITSAIED
 QTVHPKCFPL DGPGGSGKTY LYKVLTHYIR
 1081 GRGGTVLPTA STGIAANLLL GGRTFHSQYK
 LPIPLNETSI SRLDIKSEVA KTIKKAQLLI
 1141 IDECTMASSH AINAIDRLR EIMNLNVAFG
 GKVLLGGDF RQCLSIVPHA MRSIVQTSI
 1201 KYCNVWGCFR KLSLKTNMRS EDSAYSEWLV
 KLGDKGLDSS FHLGMDIIEI PHEMICNGSI
 1261 IEATPGNSIS IDNIKNIKR AILCPKNEHV
 QKLNEEILDI LDGDFHTYLS DDSIDSTDDA
 1321 EKENFPIEFL NSITPSGMPK HKLKLKVGAI
 IMLRLNLNSK WGLCNGTRFI IKRLRPNIIE
 1381 AEVLTSAGS EVVLIPRIDL SPSDTGLPFFK
 LIRRQFPVMP AFAMTINKSQ GQTLDRVGIF
 1441 LPEPVFAHQ LYVAFSRVR ACDVKVKVNV
 TSSQGLVKH SESVFTLNVV YREILE.

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transposition generates covalently closed circular intermediates. Furthermore, Helitron transpositions can lack target site duplications. In the Helraiser sequence, the transposase is flanked by left and right terminal sequences termed LTS (5' terminal sequence) and RTS (3' terminal sequence). These sequences terminate with a conserved 5'-TC/CTAG-3' motif. A 19 bp palindromic sequence with the potential to form the hairpin termination structure is located 11 nucleotides upstream of the RTS and consists of the sequence GTGCACGAATTCGTGCACCGGGCCACTAG (SEQ ID NO: 14500).

Tol2 transposons may be isolated or derived from the genome of the medaka fish, and may be similar to transposons of the hAT family. Exemplary Tol2 transposons of the disclosure are encoded by a sequence comprising about 4.7 kilobases and contain a gene encoding the Tol2 transposase, which contains four exons. An exemplary Tol2 transposase of the disclosure comprises an amino acid sequence comprising the following:

(SEQ ID NO: 14502)

1 MEEVCDSSAA ASSTVQNPQ DQEHWPYLR
 25 EFFSLSGVKN DSFKMKCVLC LPLNKEISAF
 61 KSSPSNLRKH IERMHPNYLK NYSKLTAQKR
 30 KIGTSTHASS SKQLKVDSVF PVKHVSPVTV
 121 NKAILRYIIQ GLHPFSTVDL PSFKELISTL
 QPGISVITRP TLRSKIAEAA LIMKQKVTA
 181 MSEVEWIATT TDCWTARRKS FIGVTAHWIN
 35 PGSLEHSA LACKRLMGSH TFEVLASAMN
 241 DIHSEYEIRD KVVCTTDSG SNFMKAPRVF
 40 GVENNDIETE ARRCESDDTD SEGCGEGSDG
 301 VEFQDASRLV DQDDGFEFQL PKHQKACACHL
 LNLVSSVDAQ KALSNEHYKK LYRSVFGKCQ
 45 361 ALWNKSSRSA LAEEAVESES RLQLLRPNTQ
 RWNSTFMAVD RILQICKEAG EGALRNICTS
 421 LEVPMFNP AE MLPLTEWANT MRPVAKVLDI
 50 LQAEINTQLG WLLPSVHQLS LKLQRLHLSL
 481 RYCDPLVDAL QQGIQTRFKH MFEDPEITAA
 55 AILLPKFRTS WTNETIIRK GMDYIRVHLE
 541 PLDHKKELAN SSSDDEFFA SLKPTTHEAS
 KELDGYLACV SDTRESLLTF PAICSLSIKT
 60 601 NTPLPASAAC ERLFSTAGLL FSPKRRLDT
 NNFENQLLLK LNLRFYFNE.

In Helitron transpositions, a hairpin close to the 3' end of the transposon functions as a terminator. However, this hairpin can be bypassed by the transposase, resulting in the transduction of flanking sequences. In addition, Helraiser

An exemplary Tol2 transposon of the disclosure, including inverted repeats, subterminal sequences and the Tol2 transposase, is encoded by a nucleic acid sequence comprising the following:

(SEQ ID NO: 17007)

1 CAGAGGTGTA AAGTACTTGA GTAATTTTAC TTGATTACTG TACTTAAGTA TTATTTTGG
61 GGATTTTAC TTTACTTGAG TACAATTAAA AATCAATACT TTTACTTTTA CTTAATTACA
121 TTTTITTAGA AAAAAAGTA CTTTTTACTC CTTACAATTT TATTACAGT CAAAAAGTAC
181 TTATTTTGTG GAGATCACTT CATTCATTTT TCCCTTGCTA TTACCAAACC AATTGAATTG
241 CGCTGATGCC CAGTTTAATT TAAATGTTAT TTATTCTGCC TATGAAAATC GTTTTCACAT
301 TATATGAAAT TGGTCAGACA TGTTCAATTG TCCTTTGGAA GTGACGTCAT GTCACATCTA
361 TTACCACAAT GCACAGCACC TTGACCTGGA AATTAGGGAA ATTATAACAG TCAATCAGTG
421 GAAGAAAATG GAGGAAGTAT GTGATTTCATC AGCAGCTGCG AGCAGCACAG TCCAAAATCA
481 GCCACAGGAT CAAGAGCACC CGTGGCCGTA TCTTCGCGAA TTCTTTTCTT TAAGTGGTGT
541 AAATAAAGAT TCATTCAAGA TGAATGTGT CCTCTGTCTC CCGCTTAATA AAGAAATATC
601 GGCCTTCAAA AGTTCGCCAT CAAACCTAAG GAAGCATATT GAGGTAAGTA CATTAAGTAT
661 TTTGTTTTAC TGATAGTTTT TTTTTTTTTT **TTTTTTTTTT** TTTTGGGTG TGCATGTTTT
721 GACGTTGATG GCGCGCCTTT TATATGTGTA GTAGGCCTAT TTTCACTAAT GCATGCGATT
781 GACAATATAA GGCTCACGTA ATAAAATGCT AAAATGCATT TGTAAATTGGT AACGTTAGGT
841 CCACGGGAAA TTTGGCGCCT ATTGCAGCTT TGAATAATCA TTATCATTCG GTGCTCTCAT
901 TGTGTTTGAA TTCATGCAAA ACACAAGAAA ACCAAGCGAG AAATTTTTTT CCAAACATGT
961 TGTATTGTCA AAACGGTAAC ACTTTACAAT GAGGTTGATT AGTTCATGTA TTAACATAACA
1021 TTAAATAACC ATGAGCAATA CATTTGTTAC TGTATCTGTT AATCTTTGTT AACGTTAGTT
1081 AATAGAAATA CAGATGTTCA TTGTTTGTTT ATGTTAGTTC ACAGTGCATT AACTAATGTT
1141 AACAGATAT AAAGTATTAG TAAATGTTGA AATTAACATG TATACGTGCA GTTCATTATT
1201 AGTTCATGTT AACTAATGTA GTTAACATAAC GAACCTTATT GTAAAAGTGT TACCATCAAA
1261 ACTAATGTAA TGAAATCAAT TCACCCTGTC ATGTCAGCCT TACAGTCCTG TGTTTTTGTC
1321 AATATAATCA GAAATAAAAT TAATGTTTGA TTGTCACTAA ATGCTACTGT ATTTCTAAAA
1381 TCAACAAGTA TTTAACATTA TAAAGTGTGC AATTGGCTGC AAATGTCAGT TTTATTAAAG
1441 GGTTAGTTCA CCCAAAAATG AAAATAATGT CATTAATGAC TCGCCCTCAT GTCGTTCCAA
1501 GCCCGTAAGA CCTCCGTTCA TCTTCAGAAC ACAGTTTAAG ATATTTTAGA TTTAGTCCGA
1561 GAGCTTCTG TGCCTCCATT GAGAATGTAT GTACGGTATA CTGTCCATGT CCAGAAAGGT
1621 AATAAAACA TCAGAGTAGT CCATGTGACA TCAGTGGGT AGTTAGAATT TTTGAAGCA
1681 TCGAATACAT TTTGGTCCAA AAATAACAAA ACCTACGACT TTATTCGGCA TTGTATTCTC
1741 TTCCGGGTCT GTTGTCATC CGCGTTCACG ACTTCGAGT GACGCTACAA TGCTGAATAA
1801 AGTCGTAGGT TTTGTTATTT TTGGACCAA ATGTATTTTC GATGCTTCAA ATAATTCTAC
1861 CTAACCCACT GATGTCACAT GGACTACTTT GATGTTTTTA TTACCTTTCT GGACATGGAC
1921 AGTATACCGT ACATACATTT TCAGTGGAGG GACAGAAAGC TCTCGGACTA AATCTAAAT
1981 ATCTTAACT GTGTTCCGAA GATGAACGGA GGTGTTACGG GCTTGAACG ACATGAGGGT
2041 GAGTCATTAA TGACATCTTT TCATTTTGG GTGAACATAAC CCTTTAATGC TGTAATCAGA
2101 GAGTGTATGT GTAATTGTTA CATTTATTGC ATACAATATA AATATTTATT TGTGTTTTT
2161 ACAGAGAATG CACCCAAATT ACCTCAAAAA CTACTCTAAA TTGACAGCAC AGAAGAGAAA
2221 GATCGGGACC TCCACCATG CTTCCAGCAG TAAGCAACTG AAAGTTGACT CAGTTTTCCC
2281 AGTCAAACAT GTGCTCCAG TCACTGTGAA CAAAGCTATA TTAAGGTACA TCATTCAAGG
2341 ACTTCATCCT TTCAGCACTG TTGATCTGCC ATCATTTAAA GAGCTGATTA GTACACTGCA

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2401 GCCTGGCATT TCTGTCATTA CAAGGCCTAC TTTACGCTCC AAGATAGCTG AAGCTGCTCT
2461 GATCATGAAA CAGAAAGTGA CTGCTGCCAT GAGTGAAGTT GAATGGATTG CAACCACAAC
2521 GGATTGTTGG ACTGCACGTA GAAAGTCATT CATTGGTGTA ACTGCTCACT GGATCAACCC
2581 TGAAGTCTT GAAAGACATT CCGCTGCACT TGCCTGCAAA AGATTAATGG GCTCTCATAC
2641 TTTTGAGGTA CTGGCCAGTG CCATGAATGA TATCCACTCA GAGTATGAAA TACGTGACAA
2701 GGTGTTTGG ACAACCACAG ACAGTGGTTC CAACTTTATG AAGGCTTTCA GAGTTTTTGG
2761 TGTGAAAAAC AATGATATCG AGACTGAGGC AAGAAGGTGT GAAAGTGATG AACTGATTTC
2821 TGAAGGCTGT GGTGAGGGAA GTGATGGTGT GGAATTCCAA GATGCCTCAC GAGTCTGGAA
2881 CCAAGACGAT GGCTTCGAAT TCCAGCTACC AAAACATCAA AAGTGTGCCT GTCACCTACT
2941 TAACCTAGTC TCAAGCGTTG ATGCCCCAAA AGCTCTCTCA AATGAACACT ACAAGAACT
3001 CTACAGATCT GTCTTTGGCA AATGCCAAGC TTTATGGAAT AAAAGCAGCC GATCGGCTCT
3061 AGCAGCTGAA GCTGTTGAAT CAGAAAGCCG GCTTCAGCTT TTAAGGCCAA ACCAAACGCG
3121 GTGGAATTCA ACTTTTATGG CTGTTGACAG AATTCTTCAA ATTTGCAAAG AAGCAGGAGA
3181 AGGCGCACTT CGGAATATAT GCACCTCTCT TGAGGTTCCA ATGTAAGTGT TTTTCCCTC
3241 TATCGATGTA AACAAATGTG GGTGTTTTT GTTTAATACT CTTTGATTAT GCTGATTCT
3301 CCTGTAGGTT TAATCCAGCA GAAATGCTGT TCTTGACAGA GTGGGCCAAC ACAATGCGTC
3361 CAGTTGCAA AGTACTCGAC ATCTTGCAAG CGGAAACGAA TACACAGCTG GGGTGGCTGC
3421 TGCTAGTGT CCATCAGTTA AGCTTGAAAC TTCAGCGACT CCACCATTCT CTCAGGTACT
3481 GTGACCCACT TGTGGATGCC CTACAACAAG GAATCCAAAC ACGATTCAAG CATATGTTTG
3541 AAGATCCTGA GATCATAGCA GCTGCCATCC TTCTCCCTAA ATTTGCGACC TCTTGACAA
3601 ATGATGAAAC CATCATAAAA CGAGGTAAAT GAATGCAAGC AACATACACT TGACGAATTC
3661 TAATCTGGGC AACCTTTGAG CCATACCAA ATTATTCTTT TATTTATTTA TTTTGCCT
3721 TTTTAGGAAT GTTATATCCC ATCTTTGGCT GTGATCTCAA TATGAATATT GATGTAAAGT
3781 ATTCTTGCA GAGGTTGTAG TTATCCCTCA GTGTTTCTTG AAACCAAAT CATATGTATC
3841 ATATGTGGTT TGGAAATGCA GTTAGATTTT ATGCTAAAAT AAGGGATTG CATGATTTTA
3901 GATGTAGATG ACTGCACGTA AATGTAGTTA ATGACAAAAT CCATAAAAT TGTTCCTAGT
3961 CAGAAGCCCC TCAACCAAAC TTTTCTTGT GTCTGCTCAC TGTGCTTGTA GGCATGGACT
4021 ACATCAGAGT GCATCTGGAG CCTTTGGACC ACAAGAAGGA ATTGCCCAAC AGTTCATCTG
4081 ATGATGAAGA TTTTTTCGCT TCTTTGAAAC CGACAACACA TGAAGCCAGC AAAGAGTTGG
4141 ATGGATATCT GGCCTGTGTT TCAGACACCA GGGAGTCTCT GCTCACGTTT CCTGCTATTT
4201 GCAGCCTCTC TATCAAGACT AATACACCTC TTCCCGCATC GGCTGCCTGT GAGAGGCTTT
4261 TCAGCACTGC AGGATTGCTT TTCAGCCCCA AAAGAGCTAG GCTTGACACT AACAAATTTG
4321 AGAATCAGCT TCTACTGAAG TTAAATCTGA GGTTTTACAA CTTTGAGTAG CGTGTACTGG
4381 CATTAGATTG TCTGTCTTAT AGTTTGATAA TTAAATACAA ACAGTTCTAA AGCAGGATAA
4441 AACCTTGAT GCATTCATT TAATGTTTT TGAGATTAAA AGCTTAAACA AGAATCTCTA
4501 GTTTTCTTTC TTGCTTTTAC TTTTACTTCC TTAATACTCA AGTACAATTT TAATGGAGTA
4561 CTTTTTTACT TTTACTCAAG TAAGATTCTA GCCAGATACT TTTACTTTTA ATTGAGTAAA
4621 ATTTTCCCTA AGTACTTGTA CTTTCACTTG AGTAAATTT TTGAGTACTT TTTACACCTC
4681 TG.

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Exemplary transposon/transposase systems of the disclosure include, but are not limited to, piggyBac and piggyBac-like transposons and transposases.

PiggyBac and piggyBac-like transposases recognizes transposon-specific inverted terminal repeat sequences (ITRs) on the ends of the transposon, and moves the contents between the ITRs into TTAA or TTAT chromosomal sites. The piggyBac or piggyBac-like transposon system has no payload limit for the genes of interest that can be included between the ITRs.

In certain embodiments, and, in particular, those embodiments wherein the transposon is a piggyBac transposon, the transposase is a piggyBac, Super piggyBac (SPB) transposase. In certain embodiments, and, in particular, those embodiments wherein the transposase is a piggyBac, Super piggyBac (SPB), the sequence encoding the transposase is an mRNA sequence.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or a piggyBac-like transposase enzyme. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14487)

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1      MGSSLDDEHI  LSALLQSDDE  LVGEDSDSEI
      SDHVS EDDVQ  SDTEEFIDE  VHEVQPTSSG
61     SEILDEQNVI  EQPGSSLASN  RILTLPQRTI
      RGKNKHCWST  SKSTRSRVS  ALNIVRSQRG
121    PTRMCRNIYD  PLLCFKLFFT  DEIISEIVKW
      TNAEISLKRR  ESMTGATFRD  TNEDEIYAFF
181    GILVMTAVRK  DNHMSTDDL  FDRSLSMVYVS
      VMSRDRFDFL  IRCLRMDDKS  IRPTLRENDV
241    FTPVRKIWDL  FIHQCIQNYT  PGAHLTIDEQ
      LLGFRGRCPF  RMYIPNKPSK  YGIKILMMCD
301    SGTKYMINGM  PYLGRGTQTN  GVPLGEYYVK
      ELSKPVHGSC  RNITCDNWFT  SIPLAKNLLQ
361    EPYKLTIVGT  VRSNKREIPE  VLKNSRSRPV
      GTSMFCFDGP  LTLVSYKPKP  AKMVYLLSSC
421    DEDASINEST  GKPQVMYYN  QTKGGVDTLD
      QMCSVMTCSR  KTNRWPMALL  YGMINIACIN
481    SFIIYSHNVS  SKGEKVQSRK  KFMRLNLYMSL
      TSSFMKRKLE  APTLKRYLRD  NISNILPNEV
541    PGTSDDSTEE  PVMKKRTYCT  YCPSKIRKKA
      NASCKKCKKV  ICREHNIDMC  QSCF.

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In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at one or more of positions 30, 165, 282, or 538 of the sequence:

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(SEQ ID NO: 14487)

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1      MGSSLDDEHI  LSALLQSDDE  LVGEDSDSEI
      SDHVS EDDVQ  SDTEEFIDE  VHEVQPTSSG
61     SEILDEQNVI  EQPGSSLASN  RILTLPQRTI
      RGKNKHCWST  SKSTRSRVS  ALNIVRSQRG
121    PTRMCRNIYD  PLLCFKLFFT  DEIISEIVKW
      TNAEISLKRR  ESMTGATFRD  TNEDEIYAFF
181    GILVMTAVRK  DNHMSTDDL  FDRSLSMVYVS
      VMSRDRFDFL  IRCLRMDDKS  IRPTLRENDV
241    FTPVRKIWDL  FIHQCIQNYT  PGAHLTIDEQ
      LLGFRGRCPF  RMYIPNKPSK  YGIKILMMCD
301    SGTKYMINGM  PYLGRGTQTN  GVPLGEYYVK
      ELSKPVHGSC  RNITCDNWFT  SIPLAKNLLQ
361    EPYKLTIVGT  VRSNKREIPE  VLKNSRSRPV
      GTSMFCFDGP  LTLVSYKPKP  AKMVYLLSSC
421    DEDASINEST  GKPQVMYYN  QTKGGVDTLD
      QMCSVMTCSR  KTNRWPMALL  YGMINIACIN
481    SFIIYSHNVS  SKGEKVQSRK  KFMRLNLYMSL
      TSSFMKRKLE  APTLKRYLRD  NISNILPNEV
541    PGTSDDSTEE  PVMKKRTYCT  YCPSKIRKKA
      NASCKKCKKV  ICREHNIDMC  QSCF.

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In certain embodiments, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at two or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at three or more of positions 30, 165, 282, or 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme that comprises or consists of an amino acid sequence having an amino acid substitution at each of the following positions 30, 165, 282, and 538 of the sequence of SEQ ID NO: 14487. In certain embodiments, the amino acid substitution at position 30 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 165 of the sequence of SEQ ID NO: 14487 is a substitution of a serine (S) for a glycine (G). In certain embodiments, the amino acid substitution at position 282 of the sequence of SEQ ID NO: 14487 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 538 of the sequence of SEQ ID NO: 14487 is a substitution of a lysine (K) for an asparagine (N).

In certain embodiments of the methods of the disclosure, the transposase enzyme is a Super piggyBac (SPB) or piggyBac-like transposase enzyme. In certain embodiments, the Super piggyBac (SPB) or piggyBac-like transposase enzyme of the disclosure may comprise or consist of the amino acid sequence of the sequence of SEQ ID NO: 14487 wherein the amino acid substitution at position 30 is a

substitution of a valine (V) for an isoleucine (I), the amino acid substitution at position 165 is a substitution of a serine (S) for a glycine (G), the amino acid substitution at position 282 is a substitution of a valine (V) for a methionine (M), and the amino acid substitution at position 538 is a substitution of a lysine (K) for an asparagine (N). In certain embodiments, the Super piggyBac (SPB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14484)

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1  MGSSLDDEHI  LSALLQSDDE  LVGEDSDSEV
    SDHVSEDDVQ  SDTEEFIDE  VHEVQPTSSG
61  SEILDEQNV  EQPGSSLASN  RILTLPORTI
    RGKNKHCWST  SKSTRRSRVS  ALNIVRSQRG
121 PTRMCRNIYD  PLLCFKLFFT  DEIISEIVKW
    TNAEISLKRR  EMSATSATFRD  TNEDEIYAFF
181 GILVMTAVRK  DNHMSTDDLF  DRSLSMVYVS
    VMSRDRFDFL  IRCLRMDDKS  IRPTLRENDV
241 FTPVRKIWDL  FIHQCIQNYT  PGAHLTIDEQ
    LLGFRGRCPF  RVYIPNKPSK  YGIKILMMCD
301 SGTKYMINGM  PYLGRGTQTN  GVPLGEYYVK
    ELSKPVHGSC  RNITCDNWFT  SIPLAKNLLQ
361 EPYKLTIVGT  VRSNKREIPE  VLKNSRSRPV
    GTSMFCDFGP  LTLVSYKPKP  AKMVYLLSSC
421 DEDASINEST  GKPQVMYYN  QTKGGVDTLD
    QMCSVMTCSR  KTNRWPMALL  YGMINIACIN
481 SFIIYSHNVS  SKGEKVQSRK  KFMRNLYMSL
    TSSPMKRLE  APTLKRYLRD  NISNILPKEV
541 PGTSDDSTEE  PVMKKRTYCT  YCPSKIRRKA
    NASCKCKCKV  ICREHNIDMC  QSCF.

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In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac, Super piggyBac or piggyBac-like transposase enzyme may further comprise an amino acid substitution at one or more of positions 3, 46, 82, 103, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 258, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 486, 503, 552, 570 and 591 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac, Super piggyBac or piggyBac-like transposase enzyme may further comprise an amino acid substitution at one or more of positions 46, 119, 125, 177, 180, 185, 187, 200, 207, 209, 226, 235, 240, 241, 243, 296, 298, 311, 315, 319, 327, 328, 340, 421, 436, 456, 470, 485, 503, 552 and 570. In certain embodiments, the amino acid substitution at position 3 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for a serine (S). In certain embodi-

ments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an alanine (A). In certain embodiments, the amino acid substitution at position 46 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 82 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for an isoleucine (I). In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 119 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for an arginine (R). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) a cysteine (C). In certain embodiments, the amino acid substitution at position 125 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 177 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 180 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 185 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 187 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for an alanine (A). In certain embodiments, the amino acid substitution at position 200 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 207 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a valine (V). In certain embodiments, the amino acid substitution at position 209 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a valine (V). In certain embodiments, the amino acid substitution at position 226 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a methionine (M). In certain embodiments, the amino acid substitution at position 235 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a leucine (L). In certain embodiments, the amino acid substitution at position 240 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 241 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a phenylalanine (F). In certain embodiments, the amino acid substitution at position 243 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a proline (P). In certain embodiments, the amino acid substitution at position 258 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the amino acid substitution at position 296 of SEQ ID

NO: 14487 or SEQ ID NO: 14484 is a substitution of a tryptophan (W) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a leucine (L). In certain embodiments, the amino acid substitution at position 296 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a methionine (M). In certain embodiments, the amino acid substitution at position 298 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a proline (P). In certain embodiments, the amino acid substitution at position 311 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine for a proline (P). In certain embodiments, the amino acid substitution at position 315 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for an arginine (R). In certain embodiments, the amino acid substitution at position 319 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a threonine (T). In certain embodiments, the amino acid substitution at position 327 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 328 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a tyrosine (Y). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a cysteine (C). In certain embodiments, the amino acid substitution at position 340 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a cysteine (C). In certain embodiments, the amino acid substitution at position 421 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a histidine (H) for the aspartic acid (D). In certain embodiments, the amino acid substitution at position 436 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a valine (V). In certain embodiments, the amino acid substitution at position 456 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a tyrosine (Y) for a methionine (M). In certain embodiments, the amino acid substitution at position 470 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a phenylalanine (F) for a leucine (L). In certain embodiments, the amino acid substitution at position 485 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a serine (S). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a leucine (L) for a methionine (M). In certain embodiments, the amino acid substitution at position 503 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an isoleucine (I) for a methionine (M). In certain embodiments, the amino acid substitution at position 552 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a lysine (K) for a valine (V). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a threonine (T) for an alanine (A). In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a glutamine (Q).

In certain embodiments, the amino acid substitution at position 591 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an arginine (R) for a glutamine (Q).

In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or piggyBac-like transposase enzyme or may comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at one or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments of the methods of the disclosure, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or piggyBac-like transposase enzyme may further comprise the Super piggyBac transposase enzyme may further comprise an amino acid substitution at two, three, four, five, six or more of positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, including those embodiments wherein the transposase comprises the above-described mutations at positions 30, 165, 282 and/or 538, the piggyBac or piggyBac-like transposase enzyme may further comprise or the Super piggyBac transposase enzyme may further comprise an amino acid substitution at positions 103, 194, 372, 375, 450, 509 and 570 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the amino acid substitution at position 103 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a proline (P) for a serine (S). In certain embodiments, the amino acid substitution at position 194 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a valine (V) for a methionine (M). In certain embodiments, the amino acid substitution at position 372 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for an arginine (R). In certain embodiments, the amino acid substitution at position 375 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an alanine (A) for a lysine (K). In certain embodiments, the amino acid substitution at position 450 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of an asparagine (N) for an aspartic acid (D). In certain embodiments, the amino acid substitution at position 509 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a glycine (G) for a serine (S). In certain embodiments, the amino acid substitution at position 570 of SEQ ID NO: 14487 or SEQ ID NO: 14484 is a substitution of a serine (S) for an asparagine (N). In certain embodiments, the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487. In certain embodiments, including those embodiments wherein the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, the piggyBac or piggyBac-like transposase enzyme may further comprise an amino acid substitution at positions 372, 375 and 450 of the sequence of SEQ ID NO: 14487 or SEQ ID NO: 14484. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, and a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme may comprise a substitution of a valine (V) for a methionine (M) at position 194 of SEQ ID NO: 14487, a substitution of an

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alanine (A) for an arginine (R) at position 372 of SEQ ID NO: 14487, a substitution of an alanine (A) for a lysine (K) at position 375 of SEQ ID NO: 14487 and a substitution of an asparagine (N) for an aspartic acid (D) at position 450 of SEQ ID NO: 14487.

In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from an insect. In certain embodiments, the insect is *Trichoplusia ni* (GenBank Accession No. AAA87375; SEQ ID NO: 16796), *Argyrogramma agnata* (GenBank Accession No. GU477713; SEQ ID NO: 14534, SEQ ID NO: 16797), *Anopheles gambiae* (GenBank Accession No. XP_312615 (SEQ ID NO: 16798); GenBank Accession No. XP_320414 (SEQ ID NO: 16799); GenBank Accession No. XP_310729 (SEQ ID NO: 16800)), *Aphis gossypii* (GenBank Accession No. GU329918; SEQ ID NO: 16801, SEQ ID NO: 16802), *Acyrtosiphon pisum* (GenBank Accession No. XP_001948139; SEQ ID NO: 16803), *Agrotis ipsilon* (GenBank Accession No. GU477714; SEQ ID NO: 14537, SEQ ID NO: 16804), *Bombyx mori* (GenBank Accession No. BAD11135; SEQ ID NO: 14505), *Chilo suppressalis* (GenBank Accession No. JX294476; SEQ ID NO: 16805, SEQ ID NO: 16806), *Drosophila melanogaster* (GenBank Accession No. AAL39784; SEQ ID NO: 16807), *Helicoverpa armigera* (GenBank Accession No. ABS18391; SEQ ID NO: 14525), *Heliothis virescens* (GenBank Accession No. ABD76335; SEQ ID NO: 16808), *Macdunnoughia crassignia* (GenBank Accession No. EU287451; SEQ ID NO: 16809, SEQ ID NO: 16810), *Pectinophora gossypiella* (GenBank Accession No. GU270322; SEQ ID NO: 14530, SEQ ID NO: 16811), *Tribolium castaneum* (GenBank Accession No. XP_001814566; SEQ ID NO: 16812), *Ctenoplusia agnata* (also called *Argyrogramma agnata*), *Messour bouvieri*, *Megachile rotundata*, *Bombus impatiens*, *Mamestra brassicae*, *Mayetiola destructor* or *Apis mellifera*.

In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from an insect. In certain embodiments, the insect is *Trichoplusia ni* (AAA87375).

In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from an insect. In certain embodiments, the insect is *Bombyx mori* (BAD11135).

In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from a crustacean. In certain embodiments, the crustacean is *Daphnia pulex* (AAM76342, SEQ ID NO: 16813).

In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from a vertebrate. In certain embodiments, the vertebrate is *Xenopus tropicalis* (GenBank Accession No. BAF82026; SEQ ID NO: 14518), *Homo sapiens* (GenBank Accession No. NP_689808; SEQ ID NO: 16814), *Mus musculus* (GenBank Accession No. NP_741958; SEQ ID NO: 16815), *Macaca fascicularis* (GenBank Accession No. AB179012; SEQ ID NO: 16816, SEQ ID NO: 16817), *Rattus norvegicus* (GenBank Accession No. XP_220453; SEQ ID NO: 16818) or *Myotis lucifugus*.

In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from a urochordate. In certain embodiments, the urochordate is *Ciona intestinalis* (GenBank Accession No. XP_002123602; SEQ ID NO: 16819).

In certain embodiments, the piggyBac or piggyBac-like transposase inserts a transposon at the sequence 5'-TTAT-3' within a chromosomal site (a TTAT target sequence).

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In certain embodiments, the piggyBac or piggyBac-like transposase inserts a transposon at the sequence 5'-TTAA-3' within a chromosomal site (a TTAA target sequence).

In certain embodiments, the target sequence of the piggyBac or piggyBac-like transposon comprises or consists of 5'-CTAA-3', 5'-TTAG-3', 5'-ATAA-3', 5'-TCAA-3', 5'-AGTT-3', 5'-ATTA-3', 5'-GTTA-3', 5'-TTGA-3', 5'-TTTA-3', 5'-TTAC-3', 5'-ACTA-3', 5'-AGGG-3', 5'-CTAG-3', 5'-TGAA-3', 5'-AGGT-3', 5'-ATCA-3', 5'-CTCC-3', 5'-TAAA-3', 5'-TCTC-3', 5'-TGAA-3', 5'-AAAT-3', 5'-AATC-3', 5'-ACAA-3', 5'-ACAT-3', 5'-ACTC-3', 5'-AGTG-3', 5'-ATAG-3', 5'-CAAA-3', 5'-CACA-3', 5'-CATA-3', 5'-CCAG-3', 5'-CCCA-3', 5'-CGTA-3', 5'-GTCC-3', 5'-TAAG-3', 5'-TCTA-3', 5'-TGAG-3', 5'-TGTT-3', 5'-TTCA-3', 5'-TTCT-3' and 5'-TTTT-3'.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Bombyx mori*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14504)

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1 MDIERQEERI RAMLEEELSD YSESSSEDE
    TDHCSEHEVN YDTEERIDS VDVPSNSRQE
61 EANAIIANES DSDPDDDLPL SLVRQRASAS
    RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE
121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD
    MLQEILTHTN SSIRHRQTKT AAENSSAETS
181 FYMQETTLCE LKALIALLYL AGLIKSNRQS
    LKDLWRDGT GVDIFRTTMS LQRFQFLQNN
241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ
    CCQNAYSPSE FLTIDEMLLS FRGRCLFRVY
301 IPNKPAGYGI KILALVDAKN FDVVNLEVYA
    GKQPSGPYAV SNRPFEVVER LIQPVARSHR
361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR
    KNKRQIPESF IRTDRQPNSS VFGFQKDITL
421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK
    QKPEMITFYN STKAGVDVVD ELSANYNVSF
481 NSKRWPMTLF YGVLNMAAIN ACIIYRANKN
    VTIKRTFIR SLGLSMIYEH LHSRNNKKNI
541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRQ
    DCPYKDRKT KHSNCAKAP ICMEHAKFLC
601 ENCAELDSSL
  
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The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

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(SEQ ID NO: 14505)

1	MDIERQEERI	RAMLEEELSD	YSESSSEDE	
	TDHCSEHEVN	YDTEERIDS	VDVPSNSRQE	5
61	EANAIIANES	DSDPDDDLPL	SLVRQRASAS	
	RQVSGPFYTS	KDGTKWYKNC	QRPNVRLRSE	
121	NIVTEQAQVK	NIARDASTEY	ECWNIFVTSD	10
	MLQEILTHTN	SSIRHRQTKT	AAENSSAETS	
181	FYMQETTLCE	LKALIALLYL	AGLIKSNRQS	
	LKDLWRDGT	GVDIFRTTMS	LQRFQFLQNN	
241	IRFDDKSTRD	ERKQTDNMAA	FRSIFDQFVQ	15
	CCQNAYSPE	FLTIDEMLLS	FRGRCLFRVY	
301	IPNKPAKYGI	KILALVDKN	FYVNVLEVYA	
	GKQPSGPYAV	SNRPFEVVER	LIQPVARSHR	20
361	NVTFDNWFTG	YELMLHLLNE	YRLTSVGTVR	
	KNKRQIPESF	IRTDROPNSS	VFGFQKDITL	
421	VSYAPKKNKV	VVMSTMHHD	NSIDESTGEK	25
	QKPEMITFYN	STKAGVDVVD	ELCANYNVS	
481	NSKRWPMTLF	YGVNMAAIN	ACIIYRTNKN	
	VTIKRTEFIR	SLGLSMIYEH	LHSRNKKKNI	30
541	PTYLRQRIEK	QLGEPSPRHV	NVPGRYVRCQ	
	DCPYKKDRKT	KRSCNACAKP	ICMEHAKFLC	
601	ENCAELDSSL			35

In certain embodiments, the piggyBac or piggyBac-like transposase is fused to a nuclear localization signal. In certain embodiments, the amino acid sequence of the piggyBac or piggyBac-like transposase fused to a nuclear localization signal is encoded by a polynucleotide sequence comprising:

(SEQ ID NO: 14629)

1	atggcaccca	aaaagaaacg	taaagtgatg	45
	gacattgaaa	gacaggaaga	agaatcagg	
61	gcgatgctcg	aagaagaact	gagcgactac	
	tccgacgaat	cgtcatcaga	ggatgaaacc	50
121	gaccactgta	gcgagcatga	ggttaactac	
	gacaccgagg	aggagagaat	cgactctgtg	
181	gatgtgcct	ccaactcacg	ccaagaagag	55
	gccaatgcaa	ttatcgcaaa	cgaatcggac	
241	agcgatccag	acgatgatct	gccactgtcc	
	ctcgtgcgcc	agcgggccag	cgttcgaga	60
301	caagtgtcag	gtccattcta	cacttcgaag	
	gacggcacta	agtggtacaa	gaattgccag	
361	cgacctaacg	tcagactccg	ctccgagaat	65
	atcgtgaccg	aacaggctca	ggtcaagaat	

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-continued

421	atcgcccgcg	acgcctcgac	tgagtacgag
	tggtggaata	tcttcgtgac	ttcggacatg
481	ctgcaagaaa	ttctgacgca	caccaacagc
	tcgattagcg	atcgccagac	caagactgca
541	gcggagaact	catcggccga	aacctccttc
	tatatgcaag	agactactct	gtgcgaactg
601	aaggcgctga	ttgcactgct	gtacttggcc
	ggcctcatca	aatcaaatag	gcagagcctc
661	aaagatctct	ggagaacgga	tggaaactgga
	gtggatatct	ttcggacgac	tatgagcttg
721	cagcggttcc	agtttctgca	aaacaatatc
	agattcgacg	acaagttcac	cggggacgaa
781	aggaaacaga	ctgacaacat	ggctgcgttc
	cggtcfaat	tcgatcagtt	tgtgcagtgc
841	tgccaaaacg	cttatagccc	atcggaattc
	ctgaccatcg	acgaaatgct	tctctccttc
901	cgggggctct	gcctgttccg	agtgtagatc
	cgaacaacg	cggctaata	cgaatcaaaa
961	atcctggccc	tggtggacgc	caagaatttc
	tacgtcgtga	atctcgaagt	gtacgcagga
1021	aagcaaccgt	cgggaccgta	cgtgtttctg
	aaccgcccgt	ttgaagtctg	cgagcggctt
1081	attcagccgg	tggccagatc	ccaccgcaat
	gttaccttcg	acaattgggt	caccggctac
1141	gagctgatgc	ttcaccttct	gaacgagtac
	cggctcacta	gcgtggggac	tgtagcgaag
1201	aacaagcggc	agatcccaga	atccttcac
	cgcaccgacc	gccagcctaa	ctcgtccgtg
1261	ttcggatttc	aaaaggatat	cacgcttgct
	tcgtacgccc	ccaagaaaaa	caaggtcgtg
1321	gtcgtgatga	gcaccatgca	tcacgacaac
	agcatcgacg	agtcaaccgg	agaaaagcaa
1381	aagcccagga	tgatcacctt	ctacaattca
	actaaggccg	gcgtcgacgt	cgtggatgaa
1441	ctgtgcgcga	actataacgt	gtcccggaa
	tctaagcggg	ggcctatgac	tctcttctac
1501	ggagtgtgta	atatggccgc	aatcaacgcg
	tgcatcatct	accgcaccaa	caagaacgtg
1561	accatcaagc	gcaccgaggt	catcagatcg
	ctgggtttga	gcgatgatcta	cgagcacctc

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-continued

1621 cattcacgga acaagaagaa gaatatccct
 acttacctga ggcagcgat cgagaagcag
 1681 ttgggagaac caagcccgcg ccacgtgaac
 gtgccggggc gctacgtgcg gtgccaagat
 1741 tgcccggtaca aaaaggaccg caaaacaaaa
 agatcgtgta acgcgtgctg caaacctatc
 1801 tgcacggagc atgccaaatt tctgtgtgaa
 aattgtgctg aactcgattc ctcctcg.

In certain embodiments, the piggyBac or piggyBac-like transposase is hyperactive. A hyperactive piggyBac or piggyBac-like transposase is a transposase that is more active than the naturally occurring variant from which it is derived. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Bombyx mori*. In certain embodiments, the piggyBac or piggyBac-like transposase is a hyperactive variant of SEQ ID NO: 14505. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to:

(SEQ ID NO: 14576)

1 MDIERQEERI RAMLEEELSD YDESSEDE
 TDHCSEHEVN YDTEERIDS VDVPSNSRQE
 61 EANAIIANES DSDPDDLPL SLVRQRASAS
 RQMSGPHYTS KDGTKWYKNC QRPNVRLRSE
 121 NIVTEQAQVK NIARDASTEY ECWNIFVTS
 MLQEILHTN SSIRWRQTKT AAENSSASTS
 181 FYMQETTLCE LKALIGLLYI AGLIKSNRQS
 LKDLWRTDGT GVDIFRTTMS LQRFQFLQNN
 241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ
 SCQNAISPSE FLTIDEMLLS FRGRCLFRVY
 301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR
 361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR
 KNRQIPESF IRTDRQPNSS VFGFQKDITL
 421 VSYAPKKNKV VVMSTMHHD NSIDESTGEK
 QKPEMITFYN STKAGVDVVD ELCANYNVSR
 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN
 VTIKRTFIR SLGLSMIYEH LHSRNKKKNI
 541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRCQ
 DCPYKKDRKT KRSCNACAKP ICMEHAKFLC
 601 ENCAELDSHL.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14576. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

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(SEQ ID NO: 14630)

1 MDIERQEERI RAMLEEELSD YDESSEDE
 TDHCSEHEVN YDTEERIDS VDVPSNSRQE
 61 EANAIIANES DSDPDDLPL SLVRQRASAS
 RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE
 121 NIVTEQAQVK NIARDASTEY ECWNIFVTS
 MLQEILHTN SSIRWRQTKT AAENSSASTS
 181 FYMQETTLCE LKALIGLLYI AGLIKSNRQS
 LKDLWRTDGT GVDIFRTTMS LQRFQFLNN
 241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ
 SCQNAISPSE FLTIDEMLLS FRGRCLFRVY
 301 IPNKPAKYGI KILALVDAKN FYVHNLEVYA
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR
 361 NVTFDNWFTG YEVMLHLLNE YRLTSVGTVR
 KNRQIPESF IRTDRQPNSS VFGFQKDITL
 421 VSYAPKKNKV VVMSTMHHD NSIDESTGEK
 QKPEMITFYN STKAGVDVVD ELCANYNVSR
 481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN
 VTIKRTFIR SLGLSMIYEH LHSRNKKKNI
 541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRCQ
 DCPYKKDRKT KRSCNACAKP ICMEHAKFLC
 601 ENCAHLDS.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14631)

1 MDIERQEERI RAMLEEELSD YDESSEDE
 TDHCSEHEVN YDTEERIDS VDVPSNSRQE
 61 EANAIIANES DSDPDDLPL SLVRQRASAS
 RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE
 121 NIVTEQAQVK NIARDASTEY ECWNIFVTS
 MLQEILHTN SSIRWRQTKT AAENSSASTS
 181 FYMQETTLCE LKALIGLLYI AGLIKSNRQS
 LKDLWRTDGT GVDIFRTTMS LQRFQFLNN
 241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ
 SCQNAISPSE FLTIDEMLLS FRGRCLFRVY
 301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA
 GKQPSGPYAV SNRPFEVVER LIQPVARSHR
 361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR
 KNRQIPESF IRTDRQPNSS VFGFQKDITL
 421 VSYAPKKNKV VVMSTMHHD NSIDESTGEK
 QKPEMITFYN STKAGVDVVD ELCANYNVSR

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-continued

481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN

VTIKRTEFIR SLGLSMIYEH LHSRNNKKNI

541 PTYLRQRIAM QLGEPSPRHV NVPGRYVRCQ

DCPYKKDRKT KRSCNACAKP ICMEHAKFLC

601 ENCAELDSSL.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14632)

1 MDIERQEERI RAMLEEELSD YSDESSSEDE

TDHCSEHEVN YDTEERIDS VDVPSNSRQE

61 EANAIIANES DSDPDDLPL SLVRQRASAS

RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE

121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD

MLQEILTHTN SSIRWRQTKT AAENSSAETS

181 FYMQETTLCE LKALIGLLYI AGLIKSNRQS

LKDLWRTDGT GVDIFRTTMS LQRFQFLNN

241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ

SCQNAYSPSE FLTIDEMLLS FRGRCLFRVY

301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA

GKQPSGPYAV SNRPFEVVER LIQPVARSHR

361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR

KNKTQIPENF IRTDRQPNSS VFGFQKDITL

421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK

QKPEMITFYN STKAGVDVVD ELQANYNVS

481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN

VTIKRTEFIR SLGLSMIYEH LHSRNNKKNI

541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRCQ

DCPYKKDRKT KRSCNACAKP ICMEHAKFLC

601 ENCAELDSSL.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14633)

1 MDIERQEERI RAMLEEELSD YSDESSSEDE

TDHCSEHEVN YDTEERIDS VDVPSNSRQE

61 EANAIIANES DSDPDDLPL SLVRQRASAS

RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE

121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD

MLQEILTHTN SSIRWRQTKT AAENSSAETS

181 FYMQETTLCE LKALIGLLYI AGLIKSNRQS

LKDLWRTDGT GVDIFRTTMS LQRFQFLQNN

94

-continued

241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ

SCQNAYSPSE FLTIDEMLLS FRGRCLFRVY

301 IPNKPAKYGI KILALVDAKN FYVKNLEVYA

GKQPSGPYAV SNRPFEVVER LIQPVARSHR

361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR

KNKRQIPESF IRTDRQPNSS VFGFQKDITL

421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK

QKPEMITFYN STKAGVDVVD ELCANYNVS

481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN

VTIKRTEFIR SLGLSMIYEH LHSRNNKKNI

541 PTYLRQRIEK QLGEPSPRHV NVPGRYVRCQ

DCPYKKDRKT KRSCNACAKP ICMEHAKFLC

601 ENCAELDSSL.

25 In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14634)

1 MDIERQEERI RAMLEEELSD YSDESSSEDE

TDHCSEHEVN YDTEERIDS VDVPSNSRQE

61 EANAIIANES DSDPDDLPL SLVRQRASAS

RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE

121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD

MLQEILTHTN SSIRHRQTKT AAENSSAETS

181 FYMQETTLCE LKALIALLYL AGLIKSNRQS

LKDLWRTDGT GVDIFRTTMS LQRFQFLQNN

241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ

CCQNAYSPSE FLTIDEMLLS FRGRCLFRVY

301 IPNKPAKYGI KILALVDAKN DYVNVLEVYA

GKQPSGPYAV SNRPFEVVER LIQPVARSHR

361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR

KNKRQIPESF IRTDRQPNSS VFGFQKDITL

421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK

QKPEMITFYN STKAGVDVVD ELCANYNVS

481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN

VTIKRTEFIR SLGLSMIYEH LHSRNNKKNI

541 PTYLRQRIEK QLGEPSRSHV NVKGRYVRCQ

DCPYKKDRKT KRSCNACAKP ICMEHAKFLC

601 ENCAELDSSL.

65 In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase is more active than the transposase of SEQ ID NO: 14505. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase is at

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least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or any percentage in between identical to SEQ ID NO: 14505.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution at a position selected from 92, 93, 96, 97, 165, 178, 189, 196, 200, 201, 211, 215, 235, 238, 246, 253, 258, 261, 263, 271, 303, 321, 324, 330, 373, 389, 399, 402, 403, 404, 448, 473, 484, 507, 523, 527, 528, 543, 549, 550, 557, 601, 605, 607, 609, 610 or a combination thereof (relative to SEQ ID NO: 14505). In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution of Q92A, V93L, V93M, P96G, F97H, F97C, H165E, H165W, E178S, E178H, C189P, A196G, L200I, A201Q, L211A, W215Y, G219S, Q235Y, Q235G, Q238L, K246I, K253V, M258V, F261L, S263K, C271S, N303R, F321W, F321D, V324K, V324H, A330V, L373C, L373V, V389L, S399N, R402K, T403L, D404Q, D404S, D404M, N441R, G448W, E449A, V469T, C473Q, R484K, T507C, G523A, I527M, Y528K, Y543I, E549A, K550M, P557S, E601V, E605H, E605W, D607H, S609H, L610I or any combination thereof. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution of Q92A, V93L, V93M, P96G, F97H, F97C, H165E, H165W, E178S, E178H, C189P, A196G, L200I, A201Q, L211A, W215Y, G219S, Q235Y, Q235G, Q238L, K246I, K253V, M258V, F261L, S263K, C271S, N303R, F321W, F321D, V324K, V324H, A330V, L373C, L373V, V389L, S399N, R402K, T403L, D404Q, D404S, D404M, N441R, G448W, E449A, V469T, C473Q, R484K, T507C, G523A, I527M, Y528K, Y543I, E549A, K550M, P557S, E601V, E605H, E605W, D607H, S609H and L610I.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions for a wild type amino acid comprises a substitution of E4X, A12X, M13X, L14X, E15X, D20X, E24X, S25X, S26X, S27X, D32X, H33X, E36X, E44X, E45X, E46X, I48X, D49X, R58X, A62X, N63X, A64X, I65X, I66X, N68X, E69X, D71X, S72X, D76X, P79X, R84X, Q85X, A87X, S88X, Q92X, V93X, S94X, G95X, P96X, F97X, Y98X, T99X, I145X, S149X, D150X, L152X, E154X, T157X, N160X, S161X, S162X, H165X, R166X, T168X, K169X, T170X, A171X, E173X, S175X, S176X, E178X, T179X, M183X, Q184X, T186X, T187X, L188X, C189X, L194X, I195X, A196X, L198X, L200X, A201X, L203X, I204X, K205X, A206X, N207X, Q209X, S210X, L211X, K212X, D213X, L214X, W215X, R216X, T217X, G219X, V222X, D223X, I224X, T227X, M229X, Q235X, L237X, Q238X, N239X, N240X, P302X, N303X, P305X, A306X, K307X, Y308X, I310X, K311X, I312X, L313X, A314X, L315X, V316X, D317X, A318X, K319X, N320X, F321X, Y322X, V323X, V324X, L326X, E327X, V328X, A330X, Q333X, P334X, S335X, G336X, P337X, A339X, V340X, S341X, N342X, R343X, P344X, F345X, E346X, V347X, E349X, I352X, Q353X, V355X, A356X, S357X, N361X, D365X, W367X, T369X, G370X, L373X, M374X, L375X, H376X, N379X, E380X, R382X, V386X, V389X, N392X, R394X, Q395X, S399X, F400X, I401X, R402XT403X, D404X, R405X, Q406X, P407X, N408X, S409X, S410X, V411X, F412X, F414X, Q415X, I418X, T419X, L420X, N428XV432X, M434X, D440X, N441X, S442X, I443X, D444X, E445X, G448X, E449X, Q451X, K452X, M455X, I456X, T457X, F458X, S461X, A464X, V466X, Q468X, V469X, E471X, L472X, C473X, A474X, K483X, W485X, T488X, L489X, Y491X, G492X, V493X,

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M496X, I499X, C502X, I503X, T507X, K509X, N510X, V511X, T512X, I513X, R515X, E517X, S521X, G523X, L524X, S525X, I527X, Y528X, E529X, H532X, S533X, N535X, K536X, K537X, N539X, I540X, T542X, Y543X, Q546X, E549X, K550X, Q551X, G553X, E554X, P555X, S556X, P557X, R558X, H559X, V560X, N561X, V562X, P563X, G564X, R565X, Y566X, V567X, Q570X, D571X, P573X, Y574X, K576X, K581X, S583X, A586X, A588X, E594X, F598X, L599X, E601X, N602X, C603X, A604X, E605X, L606X, D607X, S608X, S609X or L610X (relative to SEQ ID NO: 14505). A list of hyperactive amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated herein by reference in their entirety.

In certain embodiments, the piggyBac or piggyBac-like transposase is integration deficient. In certain embodiments, an integration deficient piggyBac or piggyBac-like transposase is a transposase that can excise its corresponding transposon, but that integrates the excised transposon at a lower frequency than a corresponding wild type transposase. In certain embodiments, the piggyBac or piggyBac-like transposase is an integration deficient variant of SEQ ID NO: 14505.

In certain embodiments, the excision competent, integration deficient piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions for a wild type amino acid comprises a substitution of R9X, A12X, M13X, D20X, Y21K, D23X, E24X, S25X, S26X, S27X, E28X, E30X, D32X, H33X, E36X, H37X, A39X, Y41X, D42X, T43X, E44X, E45X, E46X, R47X, D49X, S50X, S55X, A62X, N63X, A64X, I66X, A67X, N68X, E69X, D70X, D71X, S72X, D73X, P74X, D75X, D76X, D77X, I78X, S81X, V83X, R84X, Q85X, A87X, S88X, A89X, S90X, R91X, Q92X, V93X, S94X, G95X, P96X, F97X, Y98X, T99X, W012X, G103X, Y107X, K108X, L117X, I122X, Q128X, I312X, D135X, S137X, E139X, Y140X, I145X, S149X, D150X, Q153X, E154X, T157X, S161X, S162X, R164X, H165X, R166X, Q167X, T168X, K169X, T170X, A171X, A172X, E173X, R174X, S175X, S176X, A177X, E178X, T179X, S180X, Y182X, Q184X, E185X, T187X, L188X, C189X, L194X, I195X, A196X, L198X, L200X, A201X, L203X, I204X, K205X, N207X, Q209X, L211X, D213X, L214X, W215X, R216X, T217X, G219X, T220X, V222X, D223X, I224X, T227X, T228X, F234X, Q235X, L237X, Q238X, N239X, N240X, N303X, K304X, I310X, I312X, L313X, A314X, L315X, V316X, D317X, A318X, K319X, N320X, F321X, Y322X, V323X, V324X, N325X, L326X, E327X, V328X, A330X, G331X, K332X, Q333X, S335X, P337X, P344X, F345X, E349X, H359X, N361X, V362X, D365X, F368X, Y371X, E372X, L373X, H376X, E380X, R382X, R382X, V386X, G387X, T388X, V389X, K391X, N392X, R394X, Q395X, E398X, S399X, F400X, I401X, R402XT403X, D404X, R405X, Q406X, P407X, N408X, S409X, S410X, Q415X, K416X, A424X, K426X, N428X, V430X, V432X, V433X, M434X, D436X, D440X, N441X, S442X, I443X, D444X, E445X, S446X, T447X, G448X, E449X, K450X, Q451X, E454X, M455X, I456X, T457X, F458X, S461X, A464X, V466X, Q468X, V469X, C473X, A474X, N475X, N477X, K483X, R484X, P486X, T488X, L489X, G492X, V493X, M496X, I499X, I503X, Y505X, T507X, N510X, V511X, T512X, I513X, K514X, T516X, E517X, S521X, G523X, L524X, S525X, I527X, Y528X, L531X, H532X, S533X, N535X, I540X, T542X, Y543X, R545X, Q546X, E549X, L552X, G553X, E554X, P555X, S556X, P557X, R558X, H559X, V560X, N561X, V562X, P563X, G564X, V567X, Q570X, D571X,

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P573X, Y574X, K575X, K576X, N585X, A586X, M593X, K596X, E601X, N602X, A604X, E605X, L606X, D607X, S608X, S609X or L610X (relative to SEQ ID NO: 14505). A list of integration deficient amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated by reference in their entirety.

In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14606)

```

1 MDIERQEERI RAMLEEELSD YSDESSSEDE
    TDHCSEHEVN YDTEERIDS VDVPSNSRQE
61 EANAIIANES DSDPDDDLPL SLVRQRASAS
    RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE
121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD
    MLQEILTHTN SSIRHRQTKT AAENSSAETS
181 FYMQETTLCE LKALIALLYL AGLIKSNRQS
    LKDLWRKDG T GVDIFRTTMS LQRFQFLN
241 IRFDDISTRD ERKQTDNMAA FRSIFDQFVQ
    CCQNAYSPSE FLTIDEMLLS FRGRCLFRVY
301 IPNKPAKYGI KILALVDAKN FYVNVLEVYA
    GKQPSGPYAV SNRPFEVVER LIQPVARSHR
361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR
    KNKRQIPESF IRTDRQPNSS VFGFQKDITL
421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK
    QKPEMITFYN STKAGVDVVD ELCANYNVSR
481 NSKKWPMTLF YGVLNMAAIN ACIIYRTNKN
    VTIKRTEFIR SLGLSMYEH LHSRNNKKNI
541 PTYLRQRIEK QLGEPVPRHV NYPGRYVRCQ
    DCPYKKDRKT KRSCNACAKP ICMEHAKFLC
601 ENCAELDSSL.

```

In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14607)

```

1 MDIERQEERI RAMLEEELSD YSDESSSEDE
    TDHCSEHEVN YDTEERIDS VDVPSNSRQE
61 EANAIIANES DSDPDDDLPL SLVRQRASAS
    RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE
121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD
    MLQEILTHTN SSIRHRQTKT AAENSSAETS
181 FYMQETTLCE LKALIGLLYL AGLIKSNRQS
    LKDLWRKDG T GVDIFRTTMS LQRFYFLQNN
241 IRFDDKSTLD ERKQTDNMAA FRSIFDQFVQ
    SCQNAYSPSE FLTIDEMLLS FRGRCLFRVY

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301 IPNKPAKYGI KILALVDAKN FYVNVLEVYA
    GKQPSGPYAV SNRPFEVVER LIQPVARSHR
361 NVTFDNWFTG YELMLHLLNE YRLTSVGTVR
    KNKRQIPESF IRTDRQPNSS VFGFQKDITL
421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK
    QKPEMITFYN STKAGVDVVD ELCANYNVSR
481 NSKRWPMTLF YGVLNMAAIN ACIIYRTNKN
    VTIKRTEFIR SLGLSMIYEH LHSRNNKKNI
541 PTYLRQRIEK QLGEPSPRHV NYPGRYVRCQ
    DCPYKKDRKT KRSCNACAKP ICMEHAKFLC
601 VNCAELDSSL.

```

In certain embodiments, the piggyBac or piggyBac-like transposase that is is integration deficient comprises a sequence of:

(SEQ ID NO: 14608)

```

1 MDIERQEERI RAMLEEELSD YSDESSSEDE
    TDHCSEHEVN YDTEERIDS VDVPSNSRQE
61 EANAIIANES DSDPDDDLPL SLVRQRASAS
    RQVSGPFYTS KDGTKWYKNC QRPNVRLRSE
121 NIVTEQAQVK NIARDASTEY ECWNIFVTSD
    MLQEILTHTN SSIRHRQTKT AAENSSAETS
181 FYMQETTLCE LKALIALLYL AGLIKSNRQS
    LKDLWRKDG T GVDIFRTTMS LQRFQFLN
241 IRFDDKSTRD ERKQTDNMAA FRSIFDQFVQ
    CCQNAYSPSE FLTIDEMLLS FRGRCLFRVY
301 IPNKPAKYGI KILALVDAKN DYVNVLEVYA
    GKQPSGPYAV SNRPFEVVER LIQPVARSHR
361 NVTFDNWFTG YECMLHLLNE YRLTSVGTVR
    KNKRQIPESF IRTDRQPNSS VFGFQKDITL
421 VSYAPKKNKV VVVMSTMHHD NSIDESTGEK
    QKPEMITFYN STKAGVDVVD ELCANYNVSR
481 NSKKWPMTLF YGVLNMAAIN ACIIYRTNKN
    VTIKRTEFIR SLGLSMIYEH LHSRNNKKNI
541 PTYLRQRIEK QLGEPSPRHV NYPGRYVRCQ
    DCPYKKDRKT KRSCNACAKP ICMEHAKFLC
601 ENCAELDSSL.

```

In certain embodiments, the integration deficient transposase comprises a sequence that is at least 90% identical to SEQ ID NO: 14608.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Bombyx mori*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14506)

```

1 ttatcccggc gagcatgagg cagggtatct
   cataccctgg taaaatttta aagttgtgta
61 ttttataaaa ttttcgtctg acaacactag
   cgcgctcagt agctggaggc aggagcgtgc
121 gggaggggat agtggcgtga tcgcagtgtg
   gcacgggaca ccggcgagat attcgtgtgc
181 aaacctgttt cgggtatggt ataccctgcc
   tcattgttga cgtatttttt ttatgtaatt
241 tttccgatta ttaatttcaa ctgttttatt
   ggtattttta tggtaacctg tgtttttttt
301 ttatgattta ctgtatcggt tgtctttcgt
   tccttttagtt gagttttttt ttattatttt
361 cagtttttga tcaaa .

```

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14507)

```

1 tcatattttt agtttaaaaa aataattata
   tgttttataa tgaaaagaat ctcatatctt
61 ttcagtatta ggttgattta tattccaaaag
   aataataatt ttgttaaatt gttgattttt
121 gtaaacctct aaatgtttgt tgctaaaatt
   actgtgttta agaaaaagat taataataaa
181 taataatttc ataattaaaa acttctttca
   ttgaatgcca ttaataaaac cattatttta
241 caaaataaga tcaacataat tgagtaaata
   ataataagaa caatattata gtacaacaaa
301 atatgggtat gtcataacct gccacattct
   tgatgtaact ttttttcacc tcatgctcgc
361 cgggttat .

```

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14508)

```

1 ttatcccggc gagcatgagg cagggtatct
   cataccctgg taaaatttta aagttgtgta
61 ttttataaaa ttttcgtctg acaacactag
   cgcgctcagt agctggaggc aggagcgtgc
121 gggaggggat agtggcgtga tcgcagtgtg
   gcacgggaca ccggcgagat attcgtgtgc
181 aaacctgttt cgggtatggt ataccctgcc
   tcat .

```

In certain embodiments, the piggyBac (PB) or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14509)

```

1 taaataataa taatttcata attaaaaact
   tctttcattg aatgccatta aataaacctt
61 tattttacaa aataagatca acataattga
   gtaaaataata ataagaacaa tattatagta
121 caacaaaata tgggtatgtc ataccctgcc
   acattcttga tgtaactttt tttcacctca
181 tgctcgccgg gttat .

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In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' sequence corresponding to SEQ ID NO: 14506 and a 3' sequence corresponding to SEQ ID NO: 14507. In certain embodiments, one piggyBac or piggyBac-like transposon end is at least 85%, at least 90%, at least 95%, at least 98%, at least 99% identical or any percentage in between identical to SEQ ID NO: 14506 and the other piggyBac or piggyBac-like transposon end is at least 85%, at least 90%, at least 95%, at least 98%, at least 99% or any percentage in between identical to SEQ ID NO: 14507. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14506 and SEQ ID NO: 14507 or SEQ ID NO: 14509. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14508 and SEQ ID NO: 14507 or SEQ ID NO: 14509. In certain embodiments, the 5' and 3' transposon ends share a 16 bp repeat sequence at their ends of CCCGGCGAGCATGAGG (SEQ ID NO: 14510) immediately adjacent to the 5'-TTAT-3 target insertion site, which is inverted in the orientation in the two ends. In certain embodiments, 5' transposon end begins with a sequence comprising 5'-TTATCCCGGCGAGCATGAGG-3 (SEQ ID NO: 14511), and the 3' transposon ends with a sequence comprising the reverse complement of this sequence: 5'-CCTCATGCTCGCCGGGTTAT-3' (SEQ ID NO: 14512).

In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 14, 16, 18, 20, 30 or 40 contiguous nucleotides of SEQ ID NO: 14506 or SEQ ID NO: 14508. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 14, 16, 18, 20, 30 or 40 contiguous nucleotides of SEQ ID NO: 14507 or SEQ ID NO: 14509. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end with at least 90% identity to SEQ ID NO: 14506 or SEQ ID NO: 14508. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end with at least 90% identity to SEQ ID NO: 14507 or SEQ ID NO: 14509.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14515)

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1 ttaaccggc gagcatgagg cagggtatct
   cataccctgg taaaatttta aagttgtgta
61 ttttataaaa ttttcgtctg acaacactag
   cgcgctcagt agctggaggc aggagcgtgc
121 gggaggggat agtggcgtga tcgcagtgtg
   gcacgggaca ccggcgagat attcgtgtgc

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181 aaacctgttt cgggtatgtt atacctgcc
 tcaatgttga cgtatttttt ttatgtaatt
 241 ttcccgatta ttaatttcaa ctgtttttatt
 ggtattttta tggtatccat tgttcttttt
 301 ttatgattta ctgtatcggt tgtctttcgt
 tccttttagtt gagttttttt ttattatttt
 361 cagtttttga tcaaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14516)
 1 tcatattttt agtttaaaaa aataattata
 tgttttataa tgaaaagaat ctcattatct
 61 ttcagtatta ggttgattta tattccaaag
 aataataatt ttgttaaatt gttgattttt
 121 gtaaacctct aaatgtttgt tgctaaaatt
 actgtgttta agaaaagat taataataaa
 181 taataatttc ataattaaaa acttctttca
 tgaatgccca ttaataaatt cattatttta
 241 caaaataaga tcaacataat tgagtaaata
 ataataagaa caatattata gtacaacaaa
 301 atatgggtat gtcataccct tttttttttt
 tttttttttt ttttttcggg tagagggccg
 361 aacctctac gaggtccccc cgcaaaaggg
 gcgcgcgggg tatgtgagac tcaacgatct
 421 gcatgggtgt gtgagcagac cgcgggccca
 aggatttttag agcccaccca ctaaacgact
 481 cctctgcact cttacacccg acgtccgac
 ccctccgagg tcagaacccg gatgaggtag
 541 gggggctacc gcggtcaaca ctacaaccag
 acggcgcggc tcacccaag gacgcccagc
 601 cgacggagcc ttcgaggcga atcgaaggct
 ctgaaacgct gccgctctcg gtacggcagc
 661 ccgtcgggcc gccacagcg tgccgctggt
 gtcccgaat accccgctgg accagaacca
 721 gcctgcccgg tcgggacgcg atacaccgct
 gaccggtcgc tetaatcact ccacggcagc
 781 gcgctagagt gctggta.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of CCCGCGCAGCATGAGG (SEQ ID NO: 14510). In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of SEQ ID NO: 14510. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTATCCCGCGCAGCATGAGG (SEQ ID NO:

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14511). In certain embodiments, the piggyBac or piggyBac-like transposon comprises at least 16 contiguous nucleotides from SEQ ID NO: 14511. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of CCTCATGCTCGCCGGGTAT (SEQ ID NO: 14512). In certain embodiments, the piggyBac or piggyBac-like transposon comprises at least 16 contiguous nucleotides from SEQ ID NO: 14512. In certain embodiments, the piggyBac or piggyBac-like transposon comprises one end comprising at least 16 contiguous nucleotides from SEQ ID NO: 14511 and one end comprising at least 16 contiguous nucleotides from SEQ ID NO: 14512. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14511 and SEQ ID NO: 14512. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of TTAACCCGCGCAGCATGAGG (SEQ ID NO: 14513). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of

(SEQ ID NO: 14514)
 CCTCATGCTCGCCGGGTAA.

In certain embodiments, the piggyBac or piggyBac-like transposon may have ends comprising SEQ ID NO: 14506 and SEQ ID NO: 14507, or a variant of either or both of these having at least 90% sequence identity to SEQ ID NO: 14506 or SEQ ID NO: 14507, and the piggyBac or piggyBac-like transposase has the sequence of SEQ ID NO: 14504 or SEQ ID NO: 14505, or a sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identity to SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a heterologous polynucleotide inserted between a pair of inverted repeats, where the transposon is capable of transposition by a piggyBac or piggyBac-like transposase having at least 5% 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identity to SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the transposon comprises two transposon ends, each of which comprises SEQ ID NO: 14510 in inverted orientations in the two transposon ends. In certain embodiments, each inverted terminal repeat (ITR) is at least 90% identical to SEQ ID NO: 14510.

In certain embodiments, the piggyBac or piggyBac-like transposon is capable of insertion by a piggyBac or piggyBac-like transposase at the sequence 5'-TTAT-3' within a target nucleic acid. In certain embodiments, one end of the piggyBac or piggyBac-like transposon comprises at least 16 contiguous nucleotides from SEQ ID NO: 14506 and the other transposon end comprises at least 16 contiguous nucleotides from SEQ ID NO: 14507. In certain embodiments, one end of the piggyBac or piggyBac-like transposon comprises at least 17, at least 18, at least 19, at least 20, at least 22, at least 25, at least 30 contiguous nucleotides from SEQ ID NO: 14506 and the other transposon end comprises at least 17, at least 18, at least 19, at least 20, at least 22, at least 25, at least 30 contiguous nucleotides from SEQ ID NO: 14507.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises transposon ends (each end comprising an ITR) corresponding to SEQ ID NO: 14506 and SEQ ID NO: 14507, and has a target sequence corresponding to 5'-TTAT3'. In certain embodiments, the piggyBac or piggyBac-like transposon also comprises a sequence encoding a

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transposase (e.g. SEQ ID NO: 14505). In certain embodiments, the piggyBac or piggyBac-like transposon comprises one transposon end corresponding to SEQ ID NO: 14506 and a second transposon end corresponding to SEQ ID NO: 14516. SEQ ID NO: 14516 is very similar to SEQ ID NO: 14507, but has a large insertion shortly before the ITR. Although the ITR sequences for the two transposon ends are identical (they are both identical to SEQ ID NO: 14510), they have different target sequences: the second transposon has a target sequence corresponding to 5'-TTAA-3', providing evidence that no change in ITR sequence is necessary to modify the target sequence specificity. The piggyBac or piggyBac-like transposase (SEQ ID NO: 14504), which is associated with the 5'-TTAA-3' target site differs from the 5'-TTAT-3'-associated transposase (SEQ ID NO: 14505) by only 4 amino acid changes (D322Y, S473C, A507T, H582R). In certain embodiments, the piggyBac or piggyBac-like transposase (SEQ ID NO: 14504), which is associated with the 5'-TTAA-3' target site is less active than the 5'-TTAT-3'-associated piggyBac or piggyBac-like transposase (SEQ ID NO: 14505) on the transposon with 5'-TTAT-3' ends. In certain embodiments, piggyBac or piggyBac-like transposons with 5'-TTAA-3' target sites can be converted to piggyBac or piggyBac-like transposases with 5'-TTAT-3' target sites by replacing 5'-TTAA-3' target sites with 5'-TTAT-3'. Such transposons can be used either with a

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piggyBac or piggyBac-like transposase such as SEQ ID NO: 14504 which recognizes the 5'-TTAT-3' target sequence, or with a variant of a transposase originally associated with the 5'-TTAA-3' transposon. In certain embodiments, the high similarity between the 5'-TTAA-3' and 5'-TTAT-3' piggyBac or piggyBac-like transposases demonstrates that very few changes to the amino acid sequence of a piggyBac or piggyBac-like transposase alter target sequence specificity. In certain embodiments, modification of any piggyBac or piggyBac-like transposon-transposase gene transfer system, in which 5'-TTAA-3' target sequences are replaced with 5'-TTAT-3'-target sequences, the ITRs remain the same, and the transposase is the original piggyBac or piggyBac-like transposase or a variant thereof resulting from using a low-level mutagenesis to introduce mutations into the transposase. In certain embodiments, piggyBac or piggyBac-like transposon transposase transfer systems can be formed by the modification of a 5'-TTAT-3'-active piggyBac or piggyBac-like transposon-transposase gene transfer systems in which 5'-TTAT-3' target sequences are replaced with 5'-TTAA-3'-target sequences, the ITRs remain the same, and the piggyBac or piggyBac-like transposase is the original transposase or a variant thereof.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Bombyx mori*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14577)

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1  cccggcgagc atgaggcagg gtatctcata ccctggtaaa attttaagt tgtgtatttt
61  ataaaatttt cgtctgacaa cactagcgcg ctcagtagct ggaggcagga gcgtgcggga
121  ggggatagtg gcgtgatcgc agtgtggcac gggacaccgg cgagatatcc gtgtgcaaac
181  ctgtttcggg tatgtttatac cctgcctcat tgttgacgta t.

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In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14578)

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1  tttaagaaaa agattaataa ataataataa tttcataatt aaaaacttct ttcattgaat
61  gccattaaat aaaccattat ttacaaaat aagatcaaca taattgagta aataataata
121  agaacaatat tatagtacaa caaaatatgg gtatgtcata ccctgccaca ttcttgatgt
181  aacttttttt cacctcatgc tcgccggg.

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In certain embodiments, the transposon comprises at least 16 contiguous bases from SEQ ID NO: 14577 and at least 16 contiguous bases from SEQ ID NO: 14578, and inverted terminal repeats that are at least 87% identical to CCCGGCGAGCATGAGG (SEQ ID NO: 14510). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14595)

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1  cccggcgagc atgaggcagg gtatctcata ccctggtaaa attttaagt tgtgtatttt
61  ataaaatttt cgtctgacaa cactagcgcg ctcagtagct ggaggcagga gcgtgcggga
121  ggggatagtg gcgtgatcgc agtgtggcac gggacaccgg cgagatatcc gtgtgcaaac
181  ctgtttcggg tatgtttatac cctgcctcat tgttgacgta ttttttttat gtaatttttt
241  cgattattaa tttcaactgt tttattgga tttttatgtt atccattgtt ctttttttat

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301 gatttactgt atcggttgtc ttctgttctt ttagttgagt tttttttat tattttcagt
 361 ttttgatcaa a.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14596)

1 tcatattttt agtttaaaaa aataattata tgttttataa tgaaaagaat ctcatatct
 61 ttcagtatta ggttgattta tattccaaag aataatattt ttgttaaatt gttgattttt
 121 gtaaacctct aaatgtttgt tgctaaaatt actgtgttta agaaaaagat taataataa
 181 taataatttc ataattaaaa acttctttca ttgaatgcca ttaataaac cattatttta
 241 caaaataaga tcaacataat tgagttaaata ataataagaa caatattata gtacaacaaa
 301 atatgggtat gtcataccct gccacattct tgatgtaact ttttttcacc tcatgctcgc
 361 cggg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14595 and SEQ ID NO: 14596, and is transposed by the piggyBac or piggyBac-like transposase of SEQ ID NO: 14505. In certain embodiments, the ITRs of SEQ ID NO: 14595 and SEQ ID: 14596 are not flanked by a 5'-TTAA-3' sequence. In certain embodiments, the ITRs of SEQ ID NO: 14595 and SEQ ID: 14596 are flanked by a 5'-TTAT-3' sequence.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14597)

1 cccggcgagc atgaggcagg gtatctcata ccttggtaaa attttaaagt tgtgtatttt
 61 ataaaaattt cgtctgacaa cactagcgcg cttagtagct ggaggcagga gcgtgcggga
 121 ggggtagtg gcgtgatcgc agtgtggcac gggacaccgg cgagatatct gtgtgcaaac
 181 ctgtttcggg tatgttatac cctgcctcat tgttgacgta tttttttat gtaatttttc
 241 cgattattaa tttcaactgt tttattgga tttttatggt atccattggt ctttttttat
 301 g.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14598)

1 cagggtatct cataccctgg taaaatttta aagttgtga ttttataaaa ttttcgtctg
 61 acaacactag cgcgctcagt agctggaggc aggagcgtgc gggaggggat agtggcgtga
 121 tcgcagtgtg gcacgggaca ccggcgagat attcgtgtgc aaacctgttt cgggtatggt
 181 ataccctgcc tcattgttga cgtatttttt ttatgtaatt ttccgatta ttaatttcaa
 241 ctgttttatt ggtattttta tgttatccat tgttcttttt ttatg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14599)

1 cagggtatct cataccctgg taaaatttta aagttgtga ttttataaaa ttttcgtctg
 61 acaacactag cgcgctcagt agctggaggc aggagcgtgc gggaggggat agtggcgtga

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121 tcgcagtggtg gcacgggaca ccggcgagat attcgtgtgc aaacctgttt cgggtatggt
 181 ataccctgcc tcattgttga cgtat.

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In certain embodiments, the 5' end of the piggyBac or piggyBac-like transposon comprises a sequence of SEQ ID NO: 14577, SEQ ID NO: 14595, or SEQ ID NOs: 14597-14599. In certain embodiments, the 5' end of the piggyBac or piggyBac-like transposon is preceded by a 5' target

sequence.
 In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

3' transposon end ends with the reverse complement of SEQ ID NO: 14510, 5'-CCTCATGCTCGCCGGG-3' (SEQ ID NO: 14603). In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR with at least 93%, at least 87%, or at least 81% or any percentage in between identity to SEQ ID NO: 14510 or SEQ ID NO: 14603. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a target sequence followed by a 5'

(SEQ ID NO: 14600)

1 tcatatTTTT agtttaaaaa aataattata tgttttataa tgaaaagaat ctcattatct
 61 ttcagtatta ggttgattta tattccaaag aataatattt ttgttaaatt gttgattttt
 121 gtaaacctct aaatgtttgt tgctaaaatt actgtgttta agaaaaagat taataaataa
 181 taataatttc ataattaaaa acttctttca ttgaatgcca ttaaataaac cattatttta
 241 caaataaga tcaacataat tgagtaaata ataataagaa caatattata gtacaacaaa
 301 atatgggtat gtcataacct gccacattct tgatgtaact ttttttcacc tcatgctcgc
 361 cggg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

transposon end comprising a sequence selected from SEQ ID NOs: 88, 105 or 107 and a 3' transposon end comprising

(SEQ ID NO: 14601)

1 tttaagaaaa agattaataa ataataataa ttccataatt aaaaacttct ttcattgaat
 61 gccattaaat aaaccattat ttacaaaaat aagatcaaca taattgagta aataataata
 121 agaacaatat tatagtacaa caaaatatgg gtatgtcata ccctgccaca ttcttgatgt
 181 aacttttttt ca.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

SEQ ID NO: 14578 or 106 followed by a target sequence. in certain embodiments, the piggyBac or piggyBac like trans-

(SEQ ID NO: 14602)

1 cccggcgagc atgaggcagg gtatctcata ccctggtaaa attttaaagt tgtgtatttt
 61 ataaaatttt cgtctgacaa cactagcgcg ctcagtagct ggaggcagga gcgtgcggga
 121 ggggtagtg gcgtgatcgc agtgtggcac gggacaccgg cgagatatcc gtgtgcaaac
 181 ctgtttcggg tatgttatcc cctgcctcat tgttgacgta ttttttttat gtaatttttc
 241 cgattattaa tttcaactgt ttatttggtt tttttatggt atccattggt ctttttttat
 301 gatttactgt atcggttgtc tttcgttcc ttagttgagt ttttttttat tattttcagt
 361 ttttgatcaa a.

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In certain embodiments, the 3' end of the piggyBac or piggyBac-like transposon comprises a sequence of SEQ ID NO: 14578, SEQ ID NO: 14596, or SEQ ID NOs: 14600-14601. In certain embodiments, the 3' end of the piggyBac or piggyBac-like transposon is followed by a 3' target sequence. In certain embodiments, the transposon is transposed by the transposase of SEQ ID NO: 14505. In certain embodiments, the 5' and 3' ends of the piggyBac or piggyBac-like transposon share a 16 bp repeat sequence of SEQ ID NO: 14510 in inverted orientation and immediately adjacent to the target sequence. In certain embodiments, the 5' transposon end begins with SEQ ID NO: 14510, and the

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poson comprises one end that comprises a sequence that is at least 90%, at least 95% or at least 99% or any percentage in between identical to SEQ ID NO: 14577 and one end that comprises a sequence that is at least 90%, at least 95% or at least 99% or any percentage in between identical to SEQ ID NO: 14578. In certain embodiments, one transposon end comprises at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14577 and one transposon end comprises at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14578.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises two transposon ends wherein each

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3transposon ends comprises a sequence that is at least 81% identical, at least 87% identical or at least 93% identical or any percentage in between identical to SEQ ID NO: 14510 in inverted orientation in the two transposon ends. One end may further comprise at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14599, and the other end may further comprise at least 14, at least 16, at least 18 or at least 20 contiguous bases from SEQ ID NO: 14601. The piggyBac or piggyBac-like transposon may be transposed by the transposase of SEQ ID NO: 14505, and the transposase may optionally be fused to a nuclear localization signal.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14595 and SEQ ID NO: 14596 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14597 and SEQ ID NO: 14596 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14595 and SEQ ID NO: 14578 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505. In certain embodiments,

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the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14602 and SEQ ID NO: 14600 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14504 or SEQ ID NO: 14505.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' end comprising 1, 2, 3, 4, 5, 6, or 7 sequences selected from ATGAGGCAGGGTAT (SEQ ID NO: 14614), ATACCCTGCCTCAT (SEQ ID NO: 14615), GGCAGGGTAT (SEQ ID NO: 14616), ATACCCTGCC (SEQ ID NO: 14617), TAAAATTTTA (SEQ ID NO: 14618), ATTTTATAAAAT (SEQ ID NO: 14619), TCAT-ACCCTG (SEQ ID NO: 14620) and TAAATAATAATAA (SEQ ID NO: 14621). In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 3' end comprising 1, 2 or 3 sequences selected from SEQ ID NO: 14617, SEQ ID NO: 14620 and SEQ ID NO: 14621.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Xenopus tropicalis*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14517)

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1  MAKRFYSAEE AAAHCMASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
61  DEDVDDLEDQ EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
121 FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID
241 SLSEFPAAVY TPCQNICIDE SLLLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMILT SIHDESVIRE
421 QRVGRPPKKN PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKGVIY LIQMALRNSY
481 IVYKAAVPGP KLSYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPPT
541 GKQRPQKGCK VCRKRGIRRD TRYCPKCPR NPGLCFKPCF EIYHTQLHY.

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In some embodiments, the piggyBac or piggyBac-like transposase is a hyperactive variant of SEQ ID NO: 14517. In certain embodiments, the piggyBac or piggyBac-like transposase is an integration defective variant of SEQ ID NO: 14517. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14518)

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1  MAKRFYSAEE AAAHCMAPSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
61  DEDVDDLEDQ EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
121 FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
181 SLESYWNNTT VLSIPVFSAT MSRNRYQLLL RFLHFNNNAT AVPPDQPDHD RLHKLRPLID
241 SLSEFPAAVY TPCQNICIDE SLLLFKGRLR FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
361 PACGTINRTR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMILT SIHDESVIRE

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421 QRVGRPPKKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT SAWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMLP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR NPGLCFKPCF EIYHTQLHY.

In certain embodiments, the piggyBac or piggyBac-like transposase is isolated or derived from *Xenopus tropicalis*. In certain embodiments, the piggyBac or piggyBac-like transposase is a hyperactive piggyBac or piggyBac-like transposase. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence at least 90% identical to:

(SEQ ID NO: 14572)

1 MAKRFYSAEE AAAHCSASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLTRG ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SIESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID
 241 SLSERFANVY TPCQNICIDE SLMLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSTGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMLT SIHDESVIRE
 421 QRVGRPPKKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPD SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR NPGLCRKPCF EIYHTQLHY.

In certain embodiments, piggyBac or piggyBac-like transposase is a hyperactive piggyBac or piggyBac-like transposase. A hyperactive piggyBac or piggyBac-like transposase is a transposase that is more active than the naturally occurring variant from which it is derived. In certain

embodiments, a hyperactive piggyBac or piggyBac-like transposase is more active than the transposase of SEQ ID NO: 14517. In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14572)

1 MAKRFYSAEE AAAHCSASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLTRG ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SIESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID
 241 SLSERFANVY TPCQNICIDE SLMLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSTGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMLT SIHDESVIRE
 421 QRVGRPPKKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPD SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYYPCKCPR NPGLCRKPCF EIYHTQLHY.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14624)

1 MAKRFYSAEE AAAHCSASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLTRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN

-continued

181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID
 241 SLSEFFAAVY TPCQNICIDE SLLLFKGRLO FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMLT SIHDESVIRE
 421 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYICPKCPR NPGLCRKPCF EIYHTQLHY.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14625)
 1 MAKRFYSAEE AAAHCMASST EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTN NFEPIINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SLESYWDTTT VLKIPVFSAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID
 241 SLSEFFAAVY TPCQNICIDE SLLLFKGRLO FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMLT SIHDESVIRE
 421 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYICPKCPR NPGLCFKPCF EIYHTQLHY.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14627)
 1 MAKRFYSAEE AAAHCMASST EQTSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTN NFEPIINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLTRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SIESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID
 241 SLSEFANVY TPCQNICIDE SLLLFKGRLO FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMLT SIHDESVIRE
 421 QRVGRKPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYICPKCPR NPGLCRKPCF EIYHTQLHY.

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In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

(SEQ ID NO: 14628)
 1 MAKRFYSAEE AAAHCSASS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTN NFEPIINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLTRG ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SLESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRPLID

-continued

241 SLSERFANVY TPCQNICIDE SLMLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSTGYTSYP
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMILT SIHDESIVIRE
 421 QRVGRPPKKN PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYYPKCPR NPGLCRKPCF EIYHTQLHY.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises a sequence of:

V51X, S52X, A53X, L54X, E55X, E56X, P57X, M58X, E59X, E62X, D63X, V64X, D65X, D66X, L67X, E68X,

(SEQ ID NO: 16820)

1 MAKRFYSAEE AAAHMASSS EEFGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTN NFEPINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQNPLTRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SLESYWDTTT VLSIPVFGAT MSRNRYQLLL RFLHFNNNAT AVPPDQPGHD RLHKLRLPID
 241 SLSERFANVY TPCQNICIDE SLLLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYP
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLNT
 361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMILT SIHDESIVIRE
 421 QRVGRPPKKN PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RHWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRYYPKCPR NPGLCRKPCF EIYHTQLHY.

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution at a position selected from amino acid 6, 7, 16, 19, 20, 21, 22, 23, 24, 26, 28, 31, 34, 67, 73, 76, 77, 88, 91, 141, 145, 146, 148, 150, 157, 162, 179, 182, 189, 192, 193, 196, 198, 200, 210, 212, 218, 248, 263, 270, 294, 297, 308, 310, 333, 336, 354, 357, 358, 359, 377, 423, 426, 428, 438, 447, 450, 462, 469, 472, 498, 502, 517, 520, 523, 533, 534, 576, 577, 582, 583 or 587 (relative to SEQ ID NO: 14517). In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises an amino acid substitution of Y6C, S7G, M16S, S19G, S20Q, S20G, S20D, E21D, E22Q, F23T, F23P, S24Y, S26V, S28Q, V31K, A34E, L67A, G73H, A76V, D77N, P88A, N91D, Y141Q, Y141A, N145E, N145V, P146T, P146V, P146K, P148T, P148H, Y150G, Y150S, Y150C, H157Y, A162C, A179K, L182I, L182V, T189G, L192H, S193N, S193K, V196I, S198G, T200W, L210H, F212N, N218E, A248N, L263M, Q270L, S294T, T297M, S308R, L310R, L333M, Q336M, A354H, C357V, L358F, D359N, L377I, V 423H, P426K, K428R, S438A, T447G, T447A, L450V, A462H, A462Q, I469V, I472L, Q498M, L502V, E517I, P520D, P520G, N523S, I533E, D534A, F576R, F576E, K577, I582R, Y583F, L587Y or L587W, or any combination thereof including at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or all of these mutations (relative to SEQ ID NO: 14517).

In certain embodiments, the hyperactive piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions are for wild type amino acid comprises a substitution of A2X, K3X, R4X, F5X, Y6X, S7X, A11X, A13X, C15X, M16X, A17X, S18X, S19X, S20X, E21X, E22X, F23X, S24X, G25X, 26X, D27X, S28X, E29X, E42X, E43X, S44X, C46X, S47X, S48X, S49X, T50X,

D69X, Q70X, E71X, A72X, G73X, D74X, R75X, A76X, D77X, A78X, A79X, A80X, G81X, G82X, E83X, P84X, A85X, W86X, G87X, P88X, P89X, C90X, N91X, F92X, P93X, E95X, I96X, P97X, P98X, F99X, T100X, T101X, P103X, G104X, V105X, K106X, V107X, D108X, T109X, N111X, P114X, I115X, N116X, F117X, F118X, Q119X, M122X, T123X, E124X, A125X, I126X, L127X, Q128X, D129X, M130X, L132X, Y133X, V126X, Y127X, A138X, E139X, Q140X, Y141X, L142X, Q144X, N145X, P146X, L147X, P148X, Y150X, A151X, A155X, H157X, P158X, I161X, A162X, V168X, T171X, L172X, A173X, M174X, I177X, A179X, L182X, D187X, T188X, T189X, T190X, L192X, S193X, I194X, P195X, V196X, S198X, A199X, T200X, S202X, L208X, L209X, L210X, R211X, F212X, F215X, N217X, N218X, A219X, T220X, A221X, V222X, P224X, D225X, Q226X, P227X, H229X, R231X, H233X, L235X, P237X, I239X, D240X, L242X, S243X, E244X, R244X, F246X, A247X, A248X, V249X, Y250X, T251X, P252X, C253X, Q254X, I256X, C257X, I258X, D259X, E260X, S261X, L262X, L263X, L264X, F265X, K266X, G267X, R268X, L269X, Q270X, F271X, R272X, Q273X, Y274X, I275X, P276X, S277X, K278X, R279X, A280X, R281X, Y282X, G283X, I284X, K285X, F286X, Y287X, K288X, L289X, C290X, E291X, S292X, S293XS294X, G295X, Y296X, T297X, S298X, Y299X, F300X, E304X, L310X, P313X, G314X, P316X, P317X, D318X, L319X, T320X, V321X, K324X, E328X, I330X, S331X, P332X, L333X, L334X, G335X, Q336X, F338X, L340X, D343X, N344X, F345X, Y346X, S347X, L351X, F352X, A354X, L355X, Y356X, C357X, L358X, D359X, T360X, R422X, Y423X, G424X, P426X, K428X, N429X, K430X, P431X, L432X, S434X, K435X, E436X, S438X, K439X, Y440X, G443X, R446X, T447X, L450X, Q451X, N455X, T460X, R461X, A462X, K465X, V467X, G468X, I469X, Y470X,

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L471X, I472X, M474X, A475X, L476X, R477X, S479X, Y480X, V482XY483X, K484X, A485X, A486X, V487X, P488X, P490X, K491X, S493X, Y494X, Y495X, K496X, Y497T, Q498X, L499X, Q500X, I501X, L502X, P503X, A504X, L505X, L506X, F507X, G508X, G509X, V510X, E511X, E512X, Q513X, T514X, V515X, E517X, M518X, P519X, P520X, S521X, D522X, N523X, V524X, A525X, L527X, I528X, K530X, H531X, F532X, I533X, D534X, T535X, L536X, T539X, P540X, Q546X, K550X, R553X, K554X, R555X, G556X, I557X, R558X, R559X, D560X, T561X, Y564X, P566X, K567X, P569X, R570X, N571X, L574X, C575X, F576X, K577X, P578X, F580X, E581X, I582X, Y583X, T585X, Q586X, L587X, H588X or Y589X (relative to SEQ ID NO: 14517). A list of hyperactive amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated by reference in their entirety.

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In certain embodiments, the piggyBac or piggyBac-like transposase is integration deficient. In certain embodiments, an integration deficient piggyBac or piggyBac-like transposase is a transposase that can excise its corresponding transposon, but that integrates the excised transposon at a lower frequency than a corresponding naturally occurring transposase. In certain embodiments, the piggyBac or piggyBac-like transposase is an integration deficient variant of SEQ ID NO: 14517. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase is deficient relative to SEQ ID NO: 14517.

In certain embodiments, the piggyBac or piggyBac-like transposase is active for excision but deficient in integration. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of

(SEQ ID NO: 14605)

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1   MAKRFYSAEE AAAHCMASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
61  DEDVDLLEQD EAGDRVDAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
121 FMTEAILQDM VLYTNVYAEQ YLTQNPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL KFLHFNNEAT AVPPDQPGHD RLHKLRPLID
241 SLSEFPAAYV TPCQNICIDE SLLLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMILT SIHDESIVRE
421 QRVGRPPKNK PLCSKEYSKY MGGVDRDQDL QHYYNATRKT RAWYKKVGIY LIQMALRNSY
481 IVYKAAVPGP KLSYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPPT
541 GKQRPQKGCK VCRKRGIRRD TRYCPKCPR NPGLCFKPCF EIYHTQLHYG RR.

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In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

(SEQ ID NO: 14604)

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1   MAKRFYSAEE AAAHCMASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
61  DEDVDLLEQD EAGDRADAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
121 FMTEAILQDM VLYTNVYAEQ YLTQVPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
181 SLESYWDTTT VLNIPVFSAT MSRNRYQLLL RFLEFNNEAT AVPPDQPGHD RLHKLRPLID
241 SLSEFPAAYV TPCQNICIDE SLLLFKGRLQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
361 PACGTINRNR KGLPRALLDK KLNRGETYAL RKNELLAIKF FDKKNVFMILT SIHDESIVRE
421 QRVGRPPKNK PLCSKEYSKY MGGVDRDQDL QHYYNATRKT RAWYKKVGIY LIQMALRNSY
481 IVYKAAVPGP KLSYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPPT
541 GKQRPQKGCK VCRKRGIRRD TRYCPKCPR NPGLCFKPCF EIYHTQLHY.

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In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

(SEQ ID NO: 14611)

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1   MAKRFYSAEE AAAHCMASSS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
61  DEDVDLLEQD EAGDRADAAA GGEPAGWPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL

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121 FMTEAILQDM VLYTNVYAEQ YLTQNVLPY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNN DAT AVPPDQPGHD RLHKLRPLID
 241 SLTERFAAVY TPCQNICIDE SLLLFKGR LQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
 361 PACGTINRNR KGLPRALLDK KLN RGETYAL RKNELLAIKF FDKKNVFM LT SIHDES VIRE
 421 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRY YCPKCPR NPGLCFKPCF EIYHTQLHYG RR.

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In certain embodiments, the integration deficient piggy-Bac or piggyBac-like transposase comprises SEQ ID NO: 14611. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

piggyBac or piggyBac-like transposase comprises an amino acid substitution wherein the Asn at position 218 is replaced by a Glu or an Asp (N218D or N218E) (relative to SEQ ID NO: 14517).

(SEQ ID NO: 14612)

1 MAKRFYSAEE AAAHCMASS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAP GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQVPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SLESYWDTTT VLSIPVFSAT MSRNRYQLLL RFLHFNN EAT AVPPDQPGHD RLHKLRPLID
 241 SLSEFFAAVY TPCQNICIDE SLLLFKGR LQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
 361 PACGTINRNR KGLPRALLDK KLN RGETYAL RKNELLAIKF FDKKNVFM LT SIHDES VIRE
 421 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRY YCPKCPR NPGLCFKPCF EIYHTQLHYG RR.

In certain embodiments, the integration deficient piggy-Bac or piggyBac-like transposase comprises SEQ ID NO: 14612. In certain embodiments, the integration deficient piggyBac or piggyBac-like transposase comprises a sequence that is at least 90% identical to a sequence of:

⁴⁰ In certain embodiments, the excision competent, integration deficient piggyBac or piggyBac-like transposase comprises one or more substitutions of an amino acid that is not wild type, wherein the one or more substitutions a for wild type amino acid comprises a substitution of A2X, K3X,

(SEQ ID NO: 14613)

1 MAKRFYSAEE AAAHCMASS EEFSGSDSEY VPPASESDSS TEESWCSSST VSALEEPMEV
 61 DEDVDDLEDQ EAGDRADAAA GGEPAWGPPC NFPPEIPPFT TVPGVKVDTS NFEPINFFQL
 121 FMTEAILQDM VLYTNVYAEQ YLTQVPLPRY ARAHAWHPTD IAEMKRFVGL TLAMGLIKAN
 181 SLESYWDTTT VLNIPVFSAT MSRNRYQLLL RFLEFN NNAT AVPPDQPGHD RLHKLRPLID
 241 SLSEFFAAVY TPCQNICIDE SLLLFKGR LQ FRQYIPSKRA RYGIKFYKLC ESSSGYTSYF
 301 LIYEGKDSKL DPPGCPPDLT VSGKIVWELI SPLLGQGFHL YVDNFYSSIP LFTALYCLDT
 361 PACGTINRNR KGLPRALLDK KLN RGETYAL RKNELLAIKF FDKKNVFM LT SIHDES VIRE
 421 QRVGRPPKNK PLCSKEYSKY MGGVDRTDQL QHYYNATRKT RAWYKKVGIY LIQMALRNSY
 481 IVYKAAVPGP KLSYYKYQLQ ILPALLFGGV EEQTVPEMPP SDNVARLIGK HFIDTLPPTP
 541 GKQRPQKGCK VCRKRGIRRD TRY YCPKCPR NPGLCFKPCF EIYHTQLHYG RR.

In certain embodiments, the integration deficient piggy-Bac or piggyBac-like transposase comprises SEQ ID NO: 14613. In certain embodiments, the integration deficient

⁶⁵ R4X, F5X, Y6X, S7X, A8X, E9X, E10X, A11X, A12X, A13X, H14X, C15X, M16X, A17X, S18X, S19X, S20X, E21X, E22X, F23X, S24X, G25X, 26X, D27X, S28X,

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E29X, V31X, P32X, P33X, A34X, S35X, E36X, S37X, D38X, S39X, S40X, T41X, E42X, E43X, S44X, W45X, C46X, S47X, S48X, S49X, T50X, V51X, S52X, A53X, L54X, E55X, E56X, P57X, M58X, E59X, V60X, M122X, T123X, E124X, A125X, L127X, Q128X, D129X, L132X, Y133X, V126X, Y127X, E139X, Q140X, Y141X, L142X, T143X, Q144X, N145X, P146X, L147X, P148X, R149X, Y150X, A151X, H154X, H157X, P158X, T159X, D160X, I161X, A162X, E163X, M164X, K165X, R166X, F167X, V168X, G169X, L170X, T171X, L172X, A173X, M174X, G175X, L176X, I177X, K178X, A179X, N180X, S181X, L182X, S184X, Y185X, D187X, T188X, T189X, T190X, V191X, L192X, S193X, I194X, P195X, V196X, F197X, S198X, A199X, T200X, M201X, S202X, R203X, N204X, R205X, Y206X, Q207X, L208X, L209X, L210X, R211X, F212X, L213X, H241X, F215X, N216X, N217X, N218X, A219X, T220X, A221X, V222X, P223X, P224X, D225X, Q226X, P227X, G228X, H229X, D230X, R231X, H233X, K234X, L235X, R236X, L238X, I239X, D240X, L242X, S243X, E244X, R244X, F246X, A247X, A248X, V249X, Y250X, T251X, P252X, C253X, Q254X, N255X, I256X, C257X, I258X, D259X, E260X, S261X, L262X, L263X, L264X, F265X, K266X, G267X, R268X, L269X, Q270X, F271X, R272X, Q273X, Y274X, I275X, P276X, S277X, K278X, R279X, A280X, R281X, Y282X, G283X, I284X, K285X, F286X, Y287X, K288X, L289X, C290X, E291X, S292X, S293X, S294X, G295X, Y296X, T297X, S298X, Y299X, F300X, I302X, E304X, G305X, K306X, D307X, S308X, K309X, L310X, D311X, P312X, P313X, G314X, C315X, P316X, P317X, D318X, L319X, T320X, V321X, S322X, G323X, K324X, I325X, V326X, W327X, E328X, L329X, I330X, S331X, P332X, L333X, L334X, G335X, Q336X, F338X, H339X, L340X, V342X, N344X, F345X, Y346X, S347X, S348X, I349X, L351X, T353X, A354X, Y356X, C357X, L358X, D359X, T360X, P361X, A362X, C363X, G364X, I366X, N367X, R368X, D369X, K371X,

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G372X, L373X, R375X, A376X, L377X, L378X, D379X, K380X, K381X, L382X, N383X, R384XG385X, T387X, Y388X, A389X, L390X, K392X, N393X, E394X, A397X, K399X, F400X, F401X, D402X, N405X, L406X, L409X, R422X, Y423X, G424X, E425X, P426X, K428X, N429X, K430X, P431X, L432X, S434X, K435X, E436X, S438X, K439X, Y440X, G442X, G443X, V444X, R446X, T447X, L450X, Q451X, H452X, N455X, T457X, R458X, T460X, R461X, A462X, Y464X, K465X, V467X, G468X, I469X, L471X, I472X, Q473X, M474X, L476X, R477X, N478X, S479X, Y480X, V482XY483X, K484X, A485X, A486X, V487X, P488X, G489X, P490X, K491X, L492X, S493X, Y494X, Y495X, K496X, Q498X, L499X, Q500X, I501X, L502X, P503X, A504X, L505X, L506X, F507X, G508X, G509X, V510X, E511X, E512X, Q513X, T514X, V515X, E517X, M518X, P519X, P520X, S521X, D522X, N523X, V524X, A525X, L527X, I528X, G529X, K530X, F532X, I533X, D534X, T535X, L536X, P537X, P538X, T539X, P540X, G541X, F542X, Q543X, R544X, P545X, Q546X, K547X, G548X, C549X, K550X, V551X, C552X, R553X, K554X, R555X, G556X, I557X, R558X, R559X, D560X, T561X, R562X, Y563X, Y564X, C565X, P566X, K567X, C568X, P569X, R570X, N571X, P572X, G573X, L574X, C575X, F576X, K577X, P578X, C579X, F580X, E581X, I582X, Y583X, H584X, T585X, Q586X, L587X, H588X or Y589X (relative to SEQ ID NO: 14517). A list of excision competent, integration deficient amino acid substitutions can be found in U.S. Pat. No. 10,041,077, the contents of which are incorporated by reference in their entirety.

In certain embodiments, the piggyBac or piggyBac-like transposase is fused to a nuclear localization signal. In certain embodiments, SEQ ID NO: 14517 or SEQ ID NO: 14518 is fused to a nuclear localization signal. In certain embodiments, the amino acid sequence of the piggyBac or piggyBac like transposase fused to a nuclear localization signal is encoded by a polynucleotide sequence comprising:

(SEQ ID NO: 14626)

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1  atggcacccca aaaagaaacg taaagtgatg gccaaaagat ttacagcgc cgaagaagca
61  gcagcacatt gcattggcatc gtcattccgaa gaattctcgg ggagcgattc cgaatatgtc
121  ccaccggcct cggaagcgca ttcgagcact gaggagtctg ggtgttcttc ctcaactgtc
181  tcggctcttg aggagccgat ggaagtggat gaggatgtgg acgacttgga ggaccaggaa
241  gccggagaca gggccgacgc tgccgcggga ggggagccgg cgtggggacc tccatgcaat
301  ttctctcccg aaatccacc gtctactact gtgccgggag tgaaggtega cacgtccaac
361  ttcgaaacga tcaatttctt tcaactcttc atgactgaag cgactctgca agatatgggt
421  ctctacacta atgtgtacgc cgagcagtag ctgactcaaa acccgctgcc tcgctacgcg
481  agagcgcatg cgtggcaccg gaccgatatc gcggagatga agcggttcgt gggactgacc
541  ctcgcaatgg gectgatcaa ggccaacagc ctcgagtcac actgggatac cagcactgtg
601  cttagcattc cggtgttctc cgctaccatg tcccgtaacc gctaccaact cctgctcgcg
661  ttctctcact tcaacaacaa tgcgaccgct gtgccacctg accagccagg acacgacaga
721  ctccacaagc tgcggccatt gatcgactcg ctgagcgagc gattcgccgc ggtgtacacc
781  ccttgccaaa acatttgcat cgacgagtcg cttctgctgt ttaaaggccg gcttcagttc
841  cgccagtaca tcccatcgaa gcgcgctcgc tatggtatca aattctacaa actctgcgag
901  tcgtccagcg gtacacgctc atactctctg atctacgagg ggaaggactc taagctggac
961  ccaccggggt gtccaccgga tcttactgtc tccgaaaaa tcgtgtggga actcatctca

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1021 cctctcctcg gacaaggctt tcctctctac gtcgacaatt tctactcadc gateccctctg
 1081 ttcaccgccc tctactgect ggatactcca gcctgtggga ccattaacag aaaccggaag
 1141 ggtctgcca gagcactgct ggataagaag ttgaacaggg gagagactta cgcgctgaga
 1201 aagaacgaac tctcgccat caaatcttc gacaagaaaa atgtgtttat gctcacctcc
 1261 atccacgacg aatccgtcat ccgggagcag cgcgtgggca ggccgcccga aaacaagccg
 1321 ctgtgctcta aggaatactc caagtacatg ggggggtgctg accggaccga tcagctgcag
 1381 cattactaca acgccactag aaagaccgga gcctggtaca agaaagtcgg catctacctg
 1441 atccaaatgg cactgaggaa ttcttatatt gtctacaagg ctgccgttcc gggcccgaaa
 1501 ctgtcactact acaagtacca gcttcaaatc ctgccggcgc tgctgttcgg tggagtggaa
 1561 gaacagactg tgcccagat gccgccatcc gacaacgtgg ccggttgat cggaagcac
 1621 ttcattgata cctcgctcc gacgcctgga aagcagcggc cacagaaggg atgcaaagtt
 1681 tgccgcaagc gcggaatacg gcgcgatacc cgctactatt gcccgaaagt ccccgcaat
 1741 cccggactgt gtttcaagcc ctgttttgaa atctaccaca ccagttgca ttac.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Xenopus tropicalis*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14519)

1 ttaacccttt tactgccaat gacgcatggg atacgtcgtg gcagtaaaag ggcttaaatg
 61 ccaacgacgc gtcccatag ttgttgcat ttaagtctt ctctctgcag cggcagcatg
 121 tgccgccgct gcagagagtt tctagcgtg acagcccctc tgggcaacga gccggggggg
 181 ctgtc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14520)

1 tttgcatttt tagacattta gaagcctata tcttgttaca gaattggaat tacacaaaaa
 61 ttctaccata ttttgaaagc ttaggttggt ctgaaaaaaa caatatattg ttttctggg
 121 taaactaaaa gtcccctcga ggaaaggccc ctaaagtga acagtgcata acgttcaaaa
 181 actgtctggc aatacaagtt ccactttgac caaacggct ggagtaaaa gggttaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14519 and SEQ ID NO:

14520. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14521)

1 ttaacccttt gctgccaat cagcctggg atacgtcgtg gcagtaaaag ggcttaaatg
 61 ccaacgacgc gtcccatag ttgttgcat ttaagtctt ctctctgcag cggcagcatg
 121 tgccgccgct gcagagagtt tctagcgtg acagcccctc tgggcaacga gccggggggg
 181 ctgtc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14522)

1 tttgcatttt tagacattta gaagcctata tcttgttaca gaattggaat tacacaaaaa
 61 ttctaccata ttttgaaagc ttaggttggt ctgaaaaaaa caatatattg ttttctggg

121 taaactaaaa gtccctcga ggaaaggccc ctaaagtga acagtgc aaa acgttcaaaa
181 actgtctggc aatacaagtt ccactttggg acaaatcggc tggcagtgaa aggggttaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14523)
1 ttaacctttt tactgccaat gacgcattggg
 atacgtcgtg gcagtaaaag ggcttaaatg
61 ccaacgcagc gtcccatagc ttgttggcat
 tttaattctt ctctctgcag cggcagcatg
121 tgccgcgcgt gcagagagtt tctagcgatg
 acagcccctc tgggcaacga gccggggggg
181 ctgtc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14520 and SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO: 14523. In certain
embodiments, the piggyBac or piggyBac-like transposon
comprises SEQ ID NO: 14522 and SEQ ID NO: 14519, SEQ
ID NO: 14521 or SEQ ID NO: 14523. In certain embodi-
ments, the piggyBac or piggyBac-like transposon comprises
one end comprising at least 14, 16, 18, 20, 30 or 40
contiguous nucleotides from SEQ ID NO: 14519, SEQ ID
NO: 14521 or SEQ ID NO: 14523. In certain embodiments,
the piggyBac or piggyBac-like transposon comprises one
end comprising at least 14, 16, 18, 20, 30 or 40 contiguous
nucleotides from SEQ ID NO: 14520 or SEQ ID NO: 14522.
In certain embodiments, the piggyBac or piggyBac-like
transposon comprises one end with at least 90% identity to
SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ ID NO:
14523. In certain embodiments, the piggyBac or piggyBac-
like transposon comprises one end with at least 90% identity
to SEQ ID NO: 14520 or SEQ ID NO: 14522. In one
embodiment, one transposon end is at least 90% identical to
SEQ ID NO: 14519 and the other transposon end is at least
90% identical to SEQ ID NO: 14520.

In certain embodiments, the piggyBac or piggyBac-like
transposon comprises a sequence of TTAACCTTTT-
TACTGCCA (SEQ ID NO: 14524). In certain embodiments,
the piggyBac or piggyBac-like transposon comprises a
sequence of TTAACCTTTGCTGCCA (SEQ ID NO:
14526). In certain embodiments, the piggyBac or piggyBac-
like transposon comprises a sequence of TTAACCTTTT-
TACTGCCA (SEQ ID NO: 14527). In certain embodiments,
the piggyBac or piggyBac-like transposon comprises a
sequence of TGGCAGTAAAGGGTTAA (SEQ ID NO:
14529). In certain embodiments, the piggyBac or piggyBac-
like transposon comprises a sequence of TGGCAGT-
GAAAGGGTTAA (SEQ ID NO: 14531). In certain embodi-
ments, the piggyBac or piggyBac-like transposon comprises
a sequence of TTAACCTTTKMCTGCCA (SEQ ID NO:
14533). In certain embodiments, one end of the piggyBac or
piggyBac-like transposon comprises a sequence selected
from SEQ ID NO: 14524, SEQ ID NO: 14526 and SEQ ID
NO: 14527. In certain embodiments, one end of the piggy-
Bac (PB) or piggyBac-like transposon comprises a sequence
selected from SEQ ID NO: 14529 and SEQ ID NO: 14531.
In certain embodiments, each inverted terminal repeat of the
piggyBac or piggyBac-like transposon comprises a

sequence of ITR sequence of CCYTTTKMCTGCCA (SEQ
ID NO: 14563). In certain embodiments, each end of the
piggyBac (PB) or piggyBac-like transposon comprises SEQ
ID NO: 14563 in inverted orientations. In certain embodi-
ments, one ITR of the piggyBac or piggyBac-like transpo-
son comprises a sequence selected from SEQ ID NO: 14524,
SEQ ID NO: 14526 and SEQ ID NO: 14527. In certain
embodiments, one ITR of the piggyBac or piggyBac-like
transposon comprises a sequence selected from SEQ ID NO:
14529 and SEQ ID NO: 14531. In certain embodiments, the
piggyBac or piggyBac like transposon comprises SEQ ID
NO: 14533 in inverted orientation in the two transposon
ends.

In certain embodiments, The piggyBac or piggyBac-like
transposon may have ends comprising SEQ ID NO: 14519
and SEQ ID NO: 14520 or a variant of either or both of these
having at least 90% sequence identity to SEQ ID NO: 14519
or SEQ ID NO: 14520, and the piggyBac or piggyBac-like
transposase has the sequence of SEQ ID NO: 14517 or a
variant showing at least %, 10%, 15%, 20%, 25%, 30%,
35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%,
85%, 90%, 95%, 99% or any percentage in between
sequence identity to SEQ ID NO: 14517 or SEQ ID NO:
14518. In certain embodiments, one piggyBac or piggyBac-
like transposon end comprises at least 14 contiguous nucleo-
tides from SEQ ID NO: 14519, SEQ ID NO: 14521 or SEQ
ID NO: 14523, and the other transposon end comprises at
least 14 contiguous nucleotides from SEQ ID NO: 14520 or
SEQ ID NO: 14522. In certain embodiments, one transposon
end comprises at least 15, at least 16, at least 17, at least 18,
at least 19, at least 20, at least 22, at least 25, at least 30
contiguous nucleotides from SEQ ID NO: 14519, SEQ ID
NO: 14521 or SEQ ID NO: 14523, and the other transposon
end comprises at least 15, at least 16, at least 17, at least 18,
at least 19, at least 20, at least 22, at least 25 or at least 30
contiguous nucleotides from SEQ ID NO: 14520 or SEQ ID
NO: 14522.

In certain embodiments, the piggyBac or piggyBac-like
transposase recognizes a transposon end with a 5' sequence
corresponding to SEQ ID NO: 14519, and a 3' sequence
corresponding to SEQ ID NO: 14520. It will excise the
transposon from one DNA molecule by cutting the DNA at
the 5'-TTAA-3' sequence at the 5' end of one transposon end
to the 5'-TTAA-3' at the 3' end of the second transposon end,
including any heterologous DNA that is placed between
them, and insert the excised sequence into a second DNA
molecule. In certain embodiments, truncated and modified
versions of the 5' and 3' transposon ends will also function
as part of a transposon that can be transposed by the
piggyBac or piggyBac-like transposase. For example, the 5'
transposon end can be replaced by a sequence corresponding
to SEQ ID NO: 14521 or SEQ ID NO: 14523, the 3'
transposon end can be replaced by a shorter sequence
corresponding to SEQ ID NO: 14522. In certain embodi-
ments, the 5' and 3' transposon ends share an 18 bp almost
perfectly repeated sequence at their ends (5'-
TTAACCTTTKMCTGCCA: SEQ ID NO: 14533) that
includes the 5'-TTAA-3' insertion site, which sequence is
inverted in the orientation in the two ends. That is in (SEQ
ID NO: 14519) and SEQ ID NO: 14523 the 5' transposon
end begins with the sequence 5'-TTAACCTTTT-

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TACTGCCA-3' (SEQ ID NO: 14524), or in (SEQ ID NO: 14521) the 5' transposon end begins with the sequence 5'-TTAACCCCTTGCCTGCCA-3' (SEQ ID NO: 14526); the 3' transposon ends with approximately the reverse complement of this sequence: in SEQ ID NO: 14520 it ends 5' TGGCAGTAAAAGGGTTAA-3' (SEQ ID NO: 14529), in (SEQ ID NO: 14522) it ends 5'-TGGCAGT-GAAAGGGTTAA-3' (SEQ ID NO: 14531.) One embodiment of the invention is a transposon that comprises a heterologous polynucleotide inserted between two transposon ends each comprising SEQ ID NO: 14533 in inverted orientations in the two transposon ends. In certain embodiments, one transposon end comprises a sequence selected from SEQ ID NOS: 14524, SEQ ID NO: 14526 and SEQ ID NO: 14527. In some embodiments, one transposon end comprises a sequence selected from SEQ ID NO: 14529 and SEQ ID NO: 14531.

In certain embodiments, the piggyBac (PB) or piggyBac-like transposon is isolated or derived from *Xenopus tropicalis*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14573)
 1 ccctttgctt gccaatcacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa
 61 cgacgcgtcc catacgtt.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14574)
 1 cctgggtaaa ctaaaagtcc cctcgaggaa
 aggcccttaa agtgaaacag tgcaaacagt
 61 tcaaaaactg tctggcaata caagtccac
 tttgggacaa atcggtggc agtgaaagg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at least 16 contiguous bases from SEQ ID NO: 14573 or SEQ ID NO: 14574, and inverted terminal repeat of CCYTTTBMCTGCCA (SEQ ID NO: 14575).

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14579)
 1 ccctttgctt gccaatcacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa
 61 cgacgcgtcc catacgttgt tggcatttta
 agtcttctct ctgcagcggc agcatgtgcc
 121 gccgtgcag agagtttcta gcgatgacag
 cccctctggg caacgagccg ggggggctgt
 181 c.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14580)
 1 cctttttact gccaatgacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa

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-continued

61 cgacgcgtcc catacgttgt tggcatttta
 attcttctct ctgcagcggc agcatgtgcc
 121 gccgtgcag agagtttcta gcgatgacag
 cccctctggg caacgagccg ggggggctgt
 181 c.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14581)
 1 cctttttact gccaatgacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa
 61 cgacgcgtcc catacgttgt tggcatttta
 agtcttctct ctgcagcggc agcatgtgcc
 121 gccgtgcag agagtttcta gcgatgacag
 cccctctggg caacgagccg ggggggctgt
 181 c.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14582)
 1 cctttttact gccaatgacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa
 61 cgacgcgtcc catacgttgt tggcatttta
 agtcttctct ctgcagcggc agcatgtgcc
 121 gccgtgcag agag.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14583)
 1 cctttttact gccaatgacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa
 61 cgacgcgtcc catacgttgt tggcatttta
 agtctt.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14584)
 1 ccctttgctt gccaatcacg catgggatac
 gtcgtggcag taaaagggt taaatgccaa
 61 cgacgcgtcc catacgttgt tggcatttta
 agtctt.

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In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14585) 5
 1 ttatcctttt tactgccaat gacgcatggg
 atacgtcgtg gcagtaaaag ggcttaaatg
 61 ccaacgacgc gtcccatatcg ttgttggcat
 ttttaagtctt ctctctgcag cggcagcatg
 121 tgccgcgcgt gcagagagtt tctagcgatg
 acagccctc tgggcaacga gccggggggg
 181 ctgtc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14586)
 1 tttgcatttt tagacattta gaagcctata
 tcttggtaca gaattggaat tacacaaaaa
 61 ttctaccata ttttgaaagc ttaggttggt
 ctgaaaaaaa caatatattg ttttctggg
 121 taaactaaaa gtccctcga ggaaaggccc
 ctaaagtga acagtgcacaa acgttcaaaa
 181 actgtctggc aatacaagtt ccactttggg
 acaaatcggc tggcagtga aggg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' transposon end sequence selected from SEQ ID NO: 14573 and SEQ ID NOs: 14579-14585. In certain embodiments, the 5' transposon end sequence is preceded by a 5' target sequence. In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14587)
 1 tttgcatttt tagacattta gaagcctata
 tcttggtaca gaattggaat tacacaaaaa
 61 ttctaccata ttttgaaagc ttaggttggt
 ctgaaaaaaa caatatattg ttttctggg
 121 taaactaaaa gtccctcga ggaaaggccc
 ctaaagtga acagtgcacaa acgttcaaaa
 181 actgtctggc aatacaagtt ccactttgac
 caaacggct ggcagtaaaa ggg .

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14588)
 1 ttgttctgaa aaaaacaata tattgttttc
 ctgggtaaac taaaagtccc ctcgaggaaa

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-continued

61 ggcccctaaa gtgaacagc gcaaacagtt
 caaaaactgt ctggcaatac aagttccact
 121 ttgacaaaaa cggtggcag taaaaggg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14589)
 1 tttgcatttt tagacattta gaagcctata
 tcttggtaca gaattggaat tacacaaaaa
 61 ttctaccata ttttgaaagc ttaggttggt
 ctgaaaaaaa caatatattg ttttctggg
 121 taaactaaaa gtccctcga ggaaaggccc
 ctaaagtga acagtgcacaa acgttcaaaa
 181 actgtctggc aatacaagtt ccactttgac
 caaacggct ggcagtaaaa ggggttat.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises at a sequence of:

(SEQ ID NO: 14590)
 1 ttgttctgaa aaaaacaata tattgttttc
 ctgggtaaac taaaagtccc ctcgaggaaa
 61 ggcccctaaa gtgaacagc gcaaacagtt
 caaaaactgt ctggcaatac aagttccact
 121 ttgggacaaa tcggctggca gtgaagggg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 3' transposon end sequence selected from SEQ ID NO: 14574 and SEQ ID NOs: 14587-14590. In certain embodiments, the 3' transposon end sequence is followed by a 3' target sequence. In certain embodiments, the 5' and 3' transposon ends share a 14 repeated sequence inverted in orientation in the two ends (SEQ ID NO: 14575) adjacent to the target sequence. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a 5' transposon end comprising a target sequence and a sequence that is selected from SEQ ID NOs: 14582-14584 and 14573, and a 3' transposon end comprising a sequence selected from SEQ ID NOs: 14588-14590 and 14574 followed by a 3' target sequence.

In certain embodiments, the 5' transposon end of the piggyBac or piggyBac-like transposon comprises

1 atcacgcatg ggatacgtcg tggcagtaaa
 agggcttaaa tgccaacgac gcgtcccata
 61 cggt

(SEQ ID NO: 14591), and an ITR. In certain embodiments, the 5' transposon end comprises

1 atcacgcatg ggatacgtcg tggcagtaaa
 agggcttaaa tgccaacgac gcgtcccata
 61 cggtgttgcc attttaagtc tt

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(SEQ ID NO: 14592) and an ITR. In certain embodiments, the 3' transposon end of the piggyBac or piggyBac-like transposon comprises

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1 cctgggtaaa ctaaaagtcc cctcgaggaa
    aggccccctaa agtgaaacag tgcaaacgt
61 tcaaaaaactg tctggcaata caagttccac
    ttggggacaa atcggc

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(SEQ ID NO: 14593) and an ITR. In certain embodiments, the 3' transposon end comprises

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1 ttgttctgaa aaaaacaata tattgttttc
    ctgggtaaac taaaagtccc ctcgaggaaa
61 ggcccccctaaa gtgaaacagt gcaaacggt
    caaaaactgt ctggcaatac aagttccact
121 ttgacaaaa cggc

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(SEQ ID NO: 14594) and an ITR.

In certain embodiments, one transposon end comprises a sequence that is at least 90%, at least 95%, at least 99% or any percentage in between identical to SEQ ID NO: 14573 and the other transposon end comprises a sequence that is at least 90%, at least 95%, at least 99% or any percentage in between identical to SEQ ID NO: 14574. In certain embodiments, one transposon end comprises at least 14, at least 16, at least 18, at least 20 or at least 25 contiguous nucleotides from SEQ ID NO: 14573 and one transposon end comprises at least 14, at least 16, at least 18, at least 20 or at least 25 contiguous nucleotides from SEQ ID NO: 14574. In certain embodiments, one transposon end comprises at least 14, at least 16, at least 18, at least 20 from SEQ ID NO: 14591, and the other end comprises at least 14, at least 16, at least 18, at least 20 from SEQ ID NO: 14593. In certain embodiments, each transposon end comprises SEQ ID NO: 14575 in inverted orientations.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence selected from of SEQ ID NO: 14573, SEQ ID NO: 14579, SEQ ID NO: 14581, SEQ ID NO: 14582, SEQ ID NO: 14583, and SEQ ID NO: 14588, and a sequence selected from SEQ ID NO: 14587, SEQ ID NO: 14588, SEQ ID NO: 14589 and SEQ ID NO: 14586 and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14517 or SEQ ID NO: 14518.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises ITRs of CCCTTTGCCTGCCA (SEQ ID NO: 14622) (5' ITR) and TGGCAGTAAAAGGG (SEQ ID NO: 14623) (3' ITR) adjacent to the target sequences.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Helicoverpa armigera*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

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(MSEQ ID NO: 14525)
1 MASRQRLNHD EIATILENDD DYSPLDSESE
    KEDCVVEDDV WSDNEDAIVD FVEDTSAQED

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61 PDNNIASRES PNLEVTSLTS HRIITLPQRS
    IRGKNNHVWS TTKGRTTGRT SAINIIRTNR
121 GPTRMCRNIV DPLLFCQLFI TDEIHEIVK
    WTNVEIIVKR QNLKDISASY RDINTMEIWA
181 LVGILTLTAV MKDNHLSTDE LFDATFSGTR
    YVSVMRERF EPLIRCIRMD DKTLRPTLRS
241 DDAFLPVRKI WEIFINQCRQ NHVPGSNLTV
    DEQLLGFRGR CPFRMYIPNK PDKYGIKPPM
301 MCAAATKYMI DAIPYLGKST KTNGLPLGEF
    YVKDLTKTVH GTNRNITCDN WFTSIPLAKN
361 MLQAPYNLTI VGTIRSNKRE MPPEIKNSRS
    RPVGSSMFCF DGPLTLVSYK PKPSKMVPLL
421 SSCDENAVIN ESNKPDML FYNQTKGGVD
    SFDQMCKSMS ANRKTNRWPM AVFYGMLNMA
481 FVNSYIIYCH NKINKQEKPI SRKEFMKKLS
    IQLTTPWMQE RLQAPTLKRT LRDNITNVLK
541 NVVPASSENI SNEPEPKRR YCGVCSYKKR
    RMTKAQCCKC KKAICGEHNI DVCQDCI.

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In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Helicoverpa armigera*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

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(MSEQ ID NO: 14570)
1 ttaaccctag aagcccaatc tacgtaaatt
    tgacgtatac cgcggcgaaa tatctctgtc
61 tctttcatgt ttaccgtcgg atcgccgcta
    acttctgaac caactcagta gccattggga
121 cctgcagga cacagttcgg tcatctcggt
    aagtgcgcgc attttgttgt actctctatt
181 acaacacacg tcacgtcacg tcgttcgacg
    tcattttgac gtataattgg gctttgtgta
241 acttttgaat ttgtttcaaa ttttttatgt
    ttgtgattta ttgagtttaa tcgtattggt
301 tcgttacatt tttcatataa taataatatt
    ttcaggttga gtacaaa.

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In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

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(MSEQ ID NO: 14528)
1 agactgtttt tttctaagag acttctaataa
    tattattacg agttgattta attttatgaa

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61 aacattttaa actagttgat tttttttata
 attacataat ttttaagaaaa agtgtagag
 121 gcttgatttt tttgttgatt ttttctaaga
 tttgattaaa gtgcataat agtattaata
 181 aagagtattt ttttaacttaa aatgtatttt
 atttattaat taaaacttca attatgataa
 241 ctcatgcaaa aatatagttc attaacagaa
 aaaaatagga aaactttgaa gttttgtttt
 301 tacacgtcat ttttacgtat gattgggctt
 tatagctagt taaatatgat tgggcttcta
 361 ggggttaa.

in certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Pectinophora gossypiella*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14530)

1 MDLRKQDEKI RQWLEQDIEE DSKGESDNSS
 SETEDIVEME VHKNTSSESE VSSES DYEPV
 61 CPSKRQRTQI IESEESDNSE SIRPSRRQTS
 RVIDSDETDE DVMSSTPQNI PRNPVNIQPS
 121 SRFLYGKNKH KWSSAAKPSS VRTSRRNIH
 FIPGPKERAR EVSEPIDIFS LFISEDMLQQ
 181 VVTFTNAEML IRKNKYKTET FTVSPTNLEE
 IRALLGLLFN AAAMKSNHLP TRMLFNTHRS
 241 GTIFKACMSA ERLNFLIKCL RFDDKLTRNV
 RQRDDRFAPF RDLWQALISN FQKWYTPGSY
 301 ITVDEQLVGF RGRCSFRMYI PNKPNKYGIK
 LVMAADVNSK YIVNAIPYLG KGTDPOQNQPL
 361 ATFFIKEITS TLHGHTNRNIT MDNWFSTVPL
 ANELLMAPYN LTLVGTLRSN KREIPEKLN
 421 SKSRAIGTSM FCYDGDKTLV SYKAKSNKV
 FILSTIHDQP DINQETGKPE MIHFYNSTKG
 481 AVDTVDQMCS SISTNRKTQR WPLCVFYNML
 NLSIINAYVV YVYNNVRNKK KPMSRRDFVI
 541 KLGDQLMEPV LRQLQTVTL RRDIKVMIQD
 ILGESSDLEA PVPSVSNVRK IYYLCPSEAR
 601 RMTKHRCIKC KQAICGPHNI DICSRDIE.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Pectinophora gos-*

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sypiella. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14532)

1 ttaaccctag ataactaaac attcgccgc
 tcgacgacgc gctatgccgc gaaattgaag
 61 tttacctatt attccgcgtc ccccgcccc
 gccgcttttt ctagcttcct gatttgcaaa
 121 atagtgcacg gcgtgacacg ctcgaggtca
 cacgacaatt aggtcgaaag ttacaggaat
 181 ttcgctcgcc gctcgacgaa agtttagtaa
 ttacgtaagt ttggcaagg taagtgaatg
 241 aagtattttt ttataattat tttttaattc
 tttatagtga taacgtaagg tttatttaa
 301 tttattactt ttatagttat ttagccaatt
 gttataaatt ccttggtatt gctgaaaaat
 361 ttgcctgttt tagtcaaaat ttattaactt
 ttcgacgtt ttttag.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14571)

1 tttcactaag taattttgtt cctatttagt
 agataagtaa cacataatta ttgtgatatt
 61 caaaacttaa gaggtttaat aaataataat
 aaaaaaaaaa tgggttttat ttcgtagtct
 121 gctcgacgaa tgtttagtta ttacgtaacc
 gtgaatatag tttagtagtc taggggttaa.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Ctenoplusia agnata*. The piggyBac or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14534)

1 MASRQHLYQD EIAAILENED DYSPHDTDSE
 MEDCVTQDDV RSDVEDEMVD NIGNGTSPAS
 61 RHEDPETPDP SSEASNLEVT LSSHRIIILP
 QRSIREKNNH IWSTTKGQSS GRTAAINIVR
 121 TNRGPTRMCR NIVDPLLCFQ LFIKEEIVVE
 IVKWTNVEMV QKRVNLKDIS ASYRDTNEME
 181 IWAIISMLTL SAVMKDNHLS TDELNFVSYG
 TRYVSVMSRE RFEFLRLRLR MGDKLLRPNL

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241 RQEDAFTPVR KIWEIFINQC RLNYVPGTNL
TVDEQLLGFR GRCPPRMYP NPKDKYGIKF
301 PMVCDAAATKY MVDALPYLGK STKTQGLPLG
EFYVKELTQT VHGTNRNVTC DNWFTSVPLA
361 KSLLNSPYNL TLVGTIRSNK REIPEEVKNS
RSRQVGSSMF CFDGPLTLVS YKPKPSKMVF
421 LLSSCNEDAV VNQSNKPDPM ILFYNQTKGG
VDSFDQMCS MSTNRKTNRW PMAVFGMLN
481 MAFVNSYIIY CHNMLAKKEK PLSRKDFMKK
LSTDLTTPSM QKRLEAPTLK RSLRDNITNV
541 LKIVPQAAID TSPDEPEPKK RRYCGFCYSK
KKRMTKTQCF KCKKPVCGEH NIDVCQDCI.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Ctenoplia agnata*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14535)
1 ttaaccctag aagcccaatc tacgtcattc
tgacgtgtat gtcgcccgaat atactctgtc
61 tctttctcct gcacgatcgg attgcgcgga
acgtctcgatt caaccagtt ggccgcgaga
121 tctattggag gactgcggcg ttgattcggg
aagtcccgcc attttgtcat agtaacagta
181 ttgcacgtca gcttgacgta tatttgggct
ttgtgttatt ttgttaaatt ttcaacgtta
241 gtttattatt gcatcttttt gttacattac
tggttttatt gcatgtatta ctcaaatatt
301 atttttattt tagcgtagaa aataca.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14536)
1 agactgtttt ttttgtattt gcattatata
ttatattcta aagttgattt aattctaaga
61 aaaacattaa aataagtttc tttttgtaaa
atttaattaa ttataagaaa aagtttaagt
121 tgatctcatt ttttataaaa atttgcaatg
tttccaaagt tattattgta aaagaataaa
181 taaaagtaaa ctgagtttta attgatgttt
tattatatca ttatactata tattacttaa
241 ataaaacaat aactgaatgt atttctaaaa
ggaatcacta gaaaatatag tgatcaaaaa

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-continued

301 tttacacgtc atttttgcgt atgattgggc
tttatagggt ctaaaaatat gattgggcct
361 ctagggttaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTAGAAGCCCAATC (SEQ ID NO: 14564).

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Agrotis ipsilon*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14537)
1 MESRQLNQD EIATILEND DYSPLDSDE
AEDRVVEDDV WSDNEDAMID YVEDTSRQED
61 PDNNIASQES ANLEVTSLTS HRIISLPQRS
ICGKNNHVWS TTKGRTTGRT SAINIIRTNR
121 GPTRMCRNIV DPLLCFQLFI TDEIIHEIVK
WTNVEMIVKR QNLIDISASY RDTNTMEMWA
181 LVGILTLTAV MKDNHLSTDE LFDATFSGTR
YVSVMSRERF EFLIRCMRMD DKTLRPTLRS
241 DDAFIPVRKL WEIFINQCRL NYVPGGNLTV
DEQLLGFRGR CPFRMYIPNK PDKYGIRFPM
301 MCDAATKYMI DAIPYLKST KTNGLPLGEF
YVKELTKTVH GTNRNVTCN WFTSIPLAKN
361 MLQAPYNLTI VGTIRSNKRE IPEEIKNSRS
RPVGSSMFCF DGPLTLVSYK PKPSRMVFL
421 SSCDENAVIN ESNKPDML FYNQTKGGVD
SFDQMCKSMS ANRKTNRWPM AVFYGMLNMA
481 FVNSYIIYCH NKINKQKKPI NRKEFMKNLS
TDLTTPWMQE RLKAPTLKRT LRDNITNVLK
541 NVVPPSPANN SEEPGPKKRS YCGFCYSKKR
RMTKTQFYKC KKAICGEHNI DVCQDCV.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Agrotis ipsilon*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14538)
1 ttaaccctag aagcccaatc tacgttaaatt
tgacgtatac cgccgcccgaat tatactctgtc
61 tctttcacgt ttaccgtcgg attcccgtata
acttcggaac caactcagta gccattgaga

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-continued

121 actcccagga cacagttgcg tcattctcggt
aagtgcgcgc attttgttgt aatagacagg
181 ttgcacgtca ttttgacgta taattgggct
ttgtgtaact ttgaaatta ttataaattt
241 ttattgatgt gatttatttg agttaatcgt
attgtttcgt tacatttttc atatgatatt
301 aatattttca gattgaatat aaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14539)
1 agactgtttt ttttaaaagg cttataaagt attactattg
cgtgatttaa ttttataaaa
61 atatttataaa ccagttgatt tttttaataa ttacctaatt
ttaagaaaaa atgttagaag
121 cttgatattt ttgttgattt ttttctaaga ttgattaaaa
aggccataat tgtattataa
181 aagagtattt ttaacttcaa atttatttta tttattaatt
aaaacttcaa ttatgataat
241 acatgcaaaa atatagttca tcaacagaaa aatataggaa
aacttaata gttttatttt
301 tacacgtcat ttttacgtat gattgggctt tatagctagt
caaataatgat tgggcttcta
361 ggggttaa.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Megachile rotundata*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14540)
1 MNGKDSLGEF YLDDLSDCLD CRSASSTDDE SDSSNIAIRK
RCPIPLIYSD SEDEDMNNV
61 EDNNHFVKES NRYHYQIVEK YKITSKTKKW KDVTVTEMKK
FLGLIILMGQ VKKDVLYDYW
121 STDPSIETPF FSKVMSRNF LQIMQSWHFI NNNDISPNSH
RLVKIQFVID YFKEKFNNVY
181 KSDQQLSLDE CLIPWRGRLS IKTYNPAKIT KYGILVRVLS
EARTGYVSNF CUYAADGKKI
241 EETVLSVIGP YKNMWHVYQ DNYYSVNIA KIFLKNKLRV
CGTIRKNRSL PQILQTVKLS
301 RGQHQLRNG HTLLEWVWNG KRNVMISTI HSAQMAESRN
RSRTSDCPIQ KPISIIDYNK
361 YMKGVDRADQ YLSYYSIFRK TKKWTKRVVM FFINCALFNS
FKVYTTLNGQ KITYKNFLHK

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-continued

421 AALSLIEDCG TEEQGTDLPN SEPTTTRTTS RVDHPGRLEN
FGKHKLNVIV TSGQCKKPLR

481 QCRVCASKKK LSRTGFACKY CNVPLHKGDC FERYHSLKKY.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Megachile rotundata*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14541)
1 ttaaataatg cccactctag atgaacttaa cactttaccg
accggcgctc gattattcga
61 cgtttgctcc ccagcgctta ccgaccggcc atcgattatt
cgacgtttgc ttcccagcgc
121 ttaccgaccg gtcacgact tttgatcttt ccgtagatt
tggttaggtc agattgacaa
181 gtagcaagca tttcgcatc tttattcaaa taatcggtgc
ttttttctaa gcttagtccc
241 ttagaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14542)
1 acaacttctt ttttcaacaa atattgttat atggattatt
tatttattta tttatttatg
61 gtatatttta tgtttattta tttatgggta ttatgggata
ttttatgtaa ataataaact
121 gaaaacgatt gtaatagatg aaataaatat tgttttaaca
ctaataataat taaagtaaaa
181 gatttttaata aatttcgtta cctacaata acacgaagcg
tacaatttta ccagagtta
241 ttaa.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Bombus impatiens*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14543)
1 MNEKNGIGEF YLDDLSDCPD SYSRSNSGDE SDGSDTIIRK
RGSVLPFRYS DSEDEINNV
61 EDNANNVENN DDIWSTNDEA IILEPFEFGSP GLKIMPSSAE
SVTDNVNLFV GDDFFEHLVR
121 ESNRYHYQVM EKYKIPSKAK KWTDTITVPEM KKFLGLIVLM
GQIKKDVLYD YWSTDPSIET
181 PFFSQVMSRN RFVQIMQSWH FCNNDNIPHD SHRLAKIQPV
IDYFRKFNND VYKPCQQLSL
241 DESIIPWRGR LSIKTYNPAK ITKYGILVRV LSEAVTGYVC
NFDVYAADGK KLEDVAVIEP
301 YKNIWHQIYQ DNYYSVKMA RILLKNKVRV CGTIRKNRGL
PRSLKTIQLS RGQYEFRRNH

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361 QILLEVWNNG RRNVNMISTI HSAQLMESRS KSKRSDVPIQ
KPNIIIDYNK YMKGVDRADQ

421 YLAYYSIFRK TKKWKRVVM FFINCALFNS FRVYTILNGK
NITYKNFLHK VAVSWIEDGE

481 TNCTEQDDNL PNSEPTRRAP RLDHPGRLSN YGKHKLINIV
TSGRSLKPQR QCRVCAVQKK

541 RSRTCFCVKF CNVPLHKGDC FERYHTLKKY.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Bombus impatiens*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14544)

1 ttaatttttt aacattttac cgacgcatag ccgattaatc
gggtttttgc cgctgacgct

61 taccgaccga taacctatta atcggtcttt tgcgtcgaa
gcttaccacac ctatagccta

121 cctatagtta atcggttgcc atggcgataa acaatctttc
tcattatag agcagtaatt

181 tgttatttag tactaaggta ccttgcctag ttgcgtcagt
tgcggtgctt tgtaagctcc

241 cacagtttta taccaattcg aaaaacttac cgttcgcg.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14545)

1 actatttcac atttgaacta aaaaccgttg taatagataa
aataaatata atttagtatt

61 aatattatgg aaacaaaaga ttttattcaa ttttaattatc
ctatagtaac aaaaagcggc

121 caattttatc tgagcatatc aaaagcacag atactccgc
ccgacagtct aaaccgaaac

181 agagccggcg ccaggagaaa tctgcgcctg agcagccggt
cggacgtgctg ttgctggtg

241 aaccgctagt ggtcagtaaa ccagaaccag tcagtaagcc
agtaactgat cagttaacta

301 gattgtatag ttcaaatga acttaacta gtttttaagc
gttggaatgt tgtctaactt

361 cgttatatat tatattcttt ttaa.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Mamestra brassicae*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14546)

1 MFSFVPNKEQ TRTVLIFCFH LKTTAAESHR PLVEAFGEQV
PTVKTCERWF QRFKSGDFDV

61 DDKEHGKPPK RYEDAELQAL LDEDDAQTQK QLAEQLEVSQ
QAVSNRLREG GKIQKVGWV

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121 PHELNERQRE RRKNTCEILL SRYKRKSFLH RIVTGEEKWI
FFVNPKRKKS YVDPGQPATS

181 TARPNRFGKK TRLCVWWDQS GVIYYELLKP GETVNTARYQ
QQLINLNRAL QKRPEYQKR

241 QHRVIFLHDN APSHTARAVR DTLETNLWEV LPHAAYSDDL
APSDYHLFAS MGHALAEQRF

301 DSYESVEEWL DEWFAAKDDE FYWIRGIHKL ERWDCVASD
GKYFE.

15 In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Mamestra brassicae*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14547)

1 ttattgggtt gcccaaaaag taattgcgga tttttcatat
acctgtcttt taaacgtaca

61 tagggatcga actcagtaaa actttgacct tgtgaaataa
caaacttgac tgtccaacca

121 ccatagtttg gcgcgaattg agcgtcataa ttgttttgac
tttttgacgt caac.

30 In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14548)

1 atgatttttt ctttttaaac caattttaat tagttaattg
atataaaaat ccgcaattac

61 tttttgggca acccaataa.

40 In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Mayetiola destructor*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14549)

1 MENFENWRKR RHLREVLLGH FFAKKTAAES HRLLEVEYGE
HALAKTQCFE WFQRFKSGDF

61 DTEDKERPGQ PKKFEDEELE ALLDEDCQQT QEELAKSLGV
TQQAISKRLK AAGYIQKQGN

121 WVPHELKPRD VERRFCMSEM LLQRHKKKSF LSRIITGDEK
WIHYDNSKRK KSYVKRGGRA

181 KSTPKSNLHG AKVMLCIWWD QRGVLYYELL EPGQITGDL
YRTQLIRLQK ALAEKRPEYA

241 KRHGAVIFHH DNARPHVALP VKNYLENSGW EVLPHPPYSP
DLAPSDYHLF RSMQNDLAGK

301 RFTSEQGIRK WLDSPFLAAKP AKFFEKGIHE LSERWEKVIA
SDGQYFE.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Mayetiola destructor*.

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In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14550) 5
 1 taagacttcc aaaattttcca cccgaacttt accttccccg
 cgcattatgt ctctcttttc
 61 accctctgat ccttggtatt gttgtcgagc acgatttata
 ttgggtgtac aacttaaaaa 10
 121 ccggaattgg acgctagatg tccacactaa cgaatagtgt
 aaaagcacia atttcata
 181 tacgtcattt tgaaggtaga ttgacagct atcaaaatca
 gtcaataaaa ctattctatc 15
 241 tgtgtgcac atattttttt attaaact.
 20

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14551) 25
 1 tgcattcatt cattttgtta tcgaaataaa gcattaattt
 tcactaaaaa attccggttt
 61 ttaagttgta caccacaatat catccttagt gacaattttc
 aatggcttt cccattgagc
 121 tgaacccgtg gctctagtaa gaaaaacgcc caaccgtca
 tcatatgcct tttttttctc 30
 181 aacatccg.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Apis mellifera*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%,

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40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14552)
 1 MENQKEHYRH ILLFYFRKGK NASQAHKKLC AVYGDEALKE
 RQCQNWFDFK RSGDFSLKDE
 61 KRSGRPVEVD DDLIKAIIDS DRHSTTREIA EKLHVSHTCI
 ENHLKQLGYV QKLDTWVPHE 10
 121 LKEKHLTQRI NSCDLLKKRN ENDPFLKRLI TGDEKWVVYN
 NIKRKRSWSR PREPAQTTSK
 181 AGIHRKKVLL SVWWDYKGIV YFELLPPNRT INSVVYIEQL
 TKLNNAVEEK RPELTNRKGV 15
 241 VFHHDNARPH TSLVTRQKLL ELGWDVLPHP PYSPLAPSD
 YFLFRSLQNS LNGKNFNND 20
 301 DIKSYLIQFF ANKNQKFYER GIMMLPERWQ KVIDQNGQHI TE.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Apis mellifera*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14553)
 1 ttgggttgge aactaagtaa ttgcggattt cactcataga
 tggcttcagt tgaattttta
 61 gggttgctgg cgtagtccaa atgtaaaaca cattttgtta
 ttgatagtt ggcaattcag
 121 ctgtcaatca gtaaaaaaag ttttttgatc ggttgcgtag
 ttttcggttg gcgttcggtg
 181 aaaa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14554)
 1 agttatttag ttccatgaaa aaattgtcct tgattttcta aaaaaaatcc gcaattactt
 61 agttgccaat ccaa.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Messor bouvieri*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14555)
 1 MSSFVPENVH LRHALLFLFH QKKRAESHR LLVETYGEHA PTIRCTETWF RQFKCGDFNV
 61 QDKERPGRPK TFEDAELQEL LDEDSTQTQK QLAEKLNVSR VAICERLQAM GKIQKMGRWV
 121 PHELNDRQME NRKIVSEMLL QRYERKSFLH RIVTGDEKWI YFENPKRKKS WLSPEAGPS
 181 TARPNRFGRK TMLCVWWDQI GVVYELLKP GETVNTDRYR QQMINLNCAL IEKRPQYAQR

-continued

241 HDKVILQHDN APSHTAKPVK EMLKSLGWEV LSHPPYSPDL APSDYHLFAS MGHALAEQHF
 301 ADFEVVKWL DEWFSSKEKL FFWNGIHKLS ERWTKCIESN GQYFE.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Messor bouvieri*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14556)
 1 agtcagaaat gacacctcga tcgacgacta atcgacgtct aatcgacgtc gattttatgt
 61 caacatgtta ccagggtgtgt cggtaatcc tttccggttt ttccggcaga tgtcactagc
 121 cataagtatg aaatgttatg atttgatata tatgtcattt tattctactg acattaacct
 181 taaaactaca caagttacgt tccgccaaaa taacagcgtt atagatttat aattttttga
 241 aa.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14557)
 1 ataaatttga actatccatt ctaagtaacg tgttttcttt aacgaaaaaa ccggaaga
 61 attaccgaca ctctcgttat gtcaacatgt tattttcgac attgaatcgc gtcgattcga
 121 agtcgatcga ggtgtcattt ctgact.

In certain embodiments of the methods of the disclosure, the transposase enzyme is a piggyBac or piggyBac-like transposase enzyme. In certain embodiments, the piggyBac or piggyBac-like transposase enzyme is isolated or derived from *Trichoplusia ni*. The piggyBac (PB) or piggyBac-like transposase enzyme may comprise or consist of an amino acid sequence at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or any percentage in between identical to:

(SEQ ID NO: 14558)
 1 MGSSLDEHI LSALLQSDDE LVGEDSDSEV SDHVEDDVQ SDTEEFIDE VHEVQPTSSG
 61 SEILDEQNV I EQPGSSLASN RILTLPQRTI RGKNKHCWST SKSTRSRVS ALNIVRSQRG
 121 PTRMCRNIYD PLLCPKLFPT DEIISEIVKW TNAEISLKRR ESMTSATFRD TNEDEIYAFF
 181 GILVMTAVRK DNHMSTDDL FDRSLSMVYVS VMSRDRPDL IRCLRMDDKS IRPTLRENDV
 241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF RVYIPNKPSK YGIKILMMCD
 301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK ELSKPVHGSC RNITCDNWFT SIPLAKNLLQ
 361 EPYKLTIVGT VRSNKREIPE VLKNSRSRPV GTSMFCDGP LTLVSYKPKP AKMVYLLSSC
 421 DEDASINEST GKPQVMYYN QTKGGVDTLD QMCSVMTCSR KTNRWPMALL YGMINIACIN
 481 SFIYSHNVS SKGEKVQSRK KFMRNLYMSL TSSFMRKRL APTLKRYLRD NISNILPKEV
 541 PGTSDDSTEE PVMKKRTYCT YCPSKIRKA NASCKKCKV ICREHNIDMC QSCF.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Trichoplusia ni*. In

certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14559)
 1 ttaaccctag aaagatagtc tgcgtaaaat tgacgcatgc attcttgaaa tattgtctc
 61 tctttctaaa tagcgcaat ccgtcgctgt gcatttagga catctcagtc gccgcttgga

-continued

121 gctcccgtag ggcgtgcttg tcaatgcggt aagtgtcact gattttgaac tataacgacc
 181 gcgtgagtag aaatgacgca tgattatctt ttacgtgact ttaagattt aactcatatc
 241 ataattatat tgttatttca tgttctactt acgtgataac ttattatata tatattttct
 301 tgttatagat atc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14560)
 1 tttgttactt tatagaagaa attttgagtt tttgtttttt tttaataaat aaataaacat
 61 aaataaattg tttgttgaat ttattattag tatgtaagt taaatataat aaaacttaat
 121 atctattcaa attaataaat aaacctgat atacagaccg ataaacaca tgcgtcaatt
 181 ttacgcatga ttatctttaa cgtacgtcac aatatgatta tctttctagg gttaa.
 20

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14561)
 1 ccctagaaag atagtctgag taaaattgac gcatgcattc ttgaaatatt gctctctctt
 61 tctaaatagc gcgaatccgt cgctgtgcat ttaggacatc tcagtcgccc cttggagctc
 121 ccgtgaggcg tgcttgtcaa tgcggtaagt gtcactgatt ttgaactata acgaccgcgt
 181 gagtcaaaat gacgcatgat tatcttttac gtgactttta agatttaact catatcgataa
 241 ttatattggt atttcatggt ctacttacgt gataacttat tatatatata ttttcttggt
 301 atagatatc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14562)
 1 tttgttactt tatagaagaa attttgagtt tttgtttttt tttaataaat aaataaacat
 61 aaataaattg tttgttgaat ttattattag tatgtaagt taaatataat aaaacttaat
 121 atctattcaa attaataaat aaacctgat atacagaccg ataaacaca tgcgtcaatt
 181 ttacgcatga ttatctttaa cgtacgtcac aatatgatta tctttctagg g.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14609)
 1 tctaaatagc gcgaatccgt cgctgtgcat ttaggacatc tcagtcgccc cttggagctc
 61 ccgtgaggcg tgcttgtcaa tgcggtaagt gtcactgatt ttgaactata acgaccgcgt
 121 gagtcaaaat gacgcatgat tatcttttac gtgactttta agatttaact catatcgataa
 181 ttatattggt atttcatggt ctacttacgt gataacttat tatatatata ttttcttggt
 241 atagatatc.

In certain embodiments, the piggyBac or piggyBac-like transposon comprises a sequence of:

(SEQ ID NO: 14610)

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1  tttgttactt tatagaagaa attttgagtt ttgtttttt ttaataaat aaataaacat
61  aaataaattg ttgttgaat ttattattag tatgtaagt taaatataat aaaacttaat
121 atctattcaa attaataaat aaacctgat atacagaccg ataaacaca tgcgtcaatt
181 ttacgcatga ttatctttaa cgtacgtcac aatatgatta tctttctagg g.

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In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14561 and SEQ ID NO: 14562, and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14558. In certain embodiments, the piggyBac or piggyBac-like transposon comprises SEQ ID NO: 14609 and SEQ ID NO: 14610, and the piggyBac or piggyBac-like transposase comprises SEQ ID NO: 14558.

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Aphis gossypii*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCTTCCAGCGGGCGCGC (SEQ ID NO: 14565).

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Chilo suppressalis*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCAGATTAGCCT (SEQ ID NO: 14566).

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Heliothis virescens*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTTAAT-TACTCGCG (SEQ ID NO: 14567).

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Pectinophora gossypiella*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTAGA-TAACTAAAC (SEQ ID NO: 14568).

In certain embodiments, the piggyBac or piggyBac-like transposon is isolated or derived from *Anopheles stephensi*. In certain embodiments, the piggyBac or piggyBac-like transposon comprises an ITR sequence of CCCTAGAAA-GATA (SEQ ID NO: 14569).

Non-Transposition Based Methods of Genetic Modification

In some embodiments of the methods of the disclosure, a modified HSC or modified HSC descendent cell of the disclosure may be produced by introducing a transgene into an HSC or an HSC descendent cell of the disclosure. The introducing step may comprise delivery of a nucleic acid sequence and/or a genomic editing construct via a non-transposition delivery system.

In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises one or more of topical delivery, adsorption, absorption, electroporation, spin-fec-tion, co-culture, transfection, mechanical delivery, sonic delivery, vibrational delivery, magnetofection or by nanoparticle-mediated delivery. In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises liposomal transfection, calcium phosphate transfection, fugene transfection, and dendrimer-mediated transfection. In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic

editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ by mechanical transfection

comprises cell squeezing, cell bombardment, or gene gun techniques. In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ by nanoparticle-mediated transfection comprises liposomal delivery, delivery by micelles, and delivery by polymerosomes.

In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises a non-viral vector. In some embodiments, the non-viral vector comprises a nucleic acid. In some embodiments, the non-viral vector comprises plasmid DNA, linear double-stranded DNA (dsDNA), linear single-stranded DNA (ssDNA), DoggyBone™ DNA, nanoplas-mids, minicircle DNA, single-stranded oligodeoxy-nucleotides (ssODN), DDNA oligonucleotides, single-stranded mRNA (ssRNA), and double-stranded mRNA (dsRNA). In some embodiments, the non-viral vector comprises a transposon of the disclosure.

In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises a viral vector. In some embodiments, the viral vector is a non-integrating non-chromosomal vector. Exemplary non-integrating non-chromosomal vectors include, but are not limited to, adeno-associated virus (AAV), adenovirus, and herpes viruses. In some embodiments, the viral vector is an integrating chromosomal vector. Integrating chromosomal vectors include, but are not limited to, adeno-associated vectors (AAV), Lentiviruses, and gamma-retroviruses.

In some embodiments of the methods of the disclosure, introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ comprises a combination of vectors. Exemplary, non-limiting vector combinations include: viral and non-viral vectors, a plurality of non-viral vectors, or a plurality of viral vectors. Exemplary but non-limiting vectors combinations include: a combination of a DNA-derived and an RNA-derived vector, a combination of an RNA and a reverse transcriptase, a combination of a transposon and a transposase, a combination of a non-viral vector and an endonuclease, and a combination of a viral vector and an endonuclease.

In some embodiments of the methods of the disclosure, genome modification comprising introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell ex vivo, in vivo, in vitro or in situ stably integrates a nucleic acid sequence, transiently integrates a nucleic acid sequence, produces site-specific integration a nucleic acid sequence, or produces a biased integration of a nucleic acid sequence. In some embodiments, the nucleic acid sequence is a transgene.

In some embodiments of the methods of the disclosure, genome modification comprising introducing a nucleic acid sequence and/or a genomic editing construct into an HSC or HSC descendent cell *ex vivo*, *in vivo*, *in vitro* or *in situ* stably integrates a nucleic acid sequence. In some embodiments, the stable chromosomal integration can be a random integration, a site-specific integration, or a biased integration. In some embodiments, the site-specific integration can be non-assisted or assisted. In some embodiments, the assisted site-specific integration is co-delivered with a site-directed nuclease. In some embodiments, the site-directed nuclease comprises a transgene with 5' and 3' nucleotide sequence extensions that contain a percentage homology to upstream and downstream regions of the site of genomic integration. In some embodiments, the transgene with homologous nucleotide extensions enable genomic integration by homologous recombination, microhomology-mediated end joining, or nonhomologous end-joining. In some embodiments the site-specific integration occurs at a safe harbor site. Genomic safe harbor sites are able to accommodate the integration of new genetic material in a manner that ensures that the newly inserted genetic elements function reliably (for example, are expressed at a therapeutically effective level of expression) and do not cause deleterious alterations to the host genome that cause a risk to the host organism. Potential genomic safe harbors include, but are not limited to, intronic sequences of the human albumin gene, the adeno-associated virus site 1 (AAVS1), a naturally occurring site of integration of AAV virus on chromosome 19, the site of the chemokine (C—C motif) receptor 5 (CCR5) gene and the site of the human ortholog of the mouse Rosa26 locus.

In some embodiments, the site-specific transgene integration occurs at a site that disrupts expression of a target gene. In some embodiments, disruption of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements. In some embodiments, exemplary target genes targeted by site-specific integration include but are not limited to TRAC, TRAB, PDI, any immunosuppressive gene, and genes involved in allo-rejection.

In some embodiments, the site-specific transgene integration occurs at a site that results in enhanced expression of a target gene. In some embodiments, enhancement of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements.

In some embodiments of the methods of the disclosure, enzymes may be used to create strand breaks in the host genome to facilitate delivery or integration of the transgene. In some embodiments, enzymes create single-strand breaks. In some embodiments, enzymes create double-strand breaks. In some embodiments, examples of break-inducing enzymes include but are not limited to: transposases, integrases, endonucleases, CRISPR-Cas9, transcription activator-like effector nucleases (TALEN), zinc finger nucleases (ZFN), Cas-CLOVER™, and Cpf1. In some embodiments, break-inducing enzymes can be delivered to the cell encoded in DNA, encoded in mRNA, as a protein, as a nucleoprotein complex with a guide RNA (gRNA).

In some embodiments of the methods of the disclosure, the site-specific transgene integration is controlled by a vector-mediated integration site bias. In some embodiments vector-mediated integration site bias is controlled by the

chosen lentiviral vector. In some embodiments vector-mediated integration site bias is controlled by the chosen gamma-retroviral vector.

In some embodiments of the methods of the disclosure, the site-specific transgene integration site is a non-stable chromosomal insertion. In some embodiments, the integrated transgene may become silenced, removed, excised, or further modified.

In some embodiments of the methods of the disclosure, the genome modification is a non-stable integration of a transgene. In some embodiments, the non-stable integration can be a transient non-chromosomal integration, a semi-stable non chromosomal integration, a semi-persistent non-chromosomal insertion, or a non-stable chromosomal insertion. In some embodiments, the transient non-chromosomal insertion can be epi-chromosomal or cytoplasmic.

In some embodiments, the transient non-chromosomal insertion of a transgene does not integrate into a chromosome and the modified genetic material is not replicated during cell division.

In some embodiments of the methods of the disclosure, the genome modification is a semi-stable or persistent non-chromosomal integration of a transgene. In some embodiments, a DNA vector encodes a Scaffold/matrix attachment region (S-MAR) module that binds to nuclear matrix proteins for episomal retention of a non-viral vector allowing for autonomous replication in the nucleus of dividing cells.

In some embodiments of the methods of the disclosure, the genome modification is a non-stable chromosomal integration of a transgene. In some embodiments, the integrated transgene may become silenced, removed, excised, or further modified.

In some embodiments of the methods of the disclosure, the modification to the genome by transgene insertion can occur via host cell-directed double-strand breakage repair (homology-directed repair) by homologous recombination (HR), microhomology-mediated end joining (MMEJ), non-homologous end joining (NHEJ), transposase enzyme-mediated modification, integrase enzyme-mediated modification, endonuclease enzyme-mediated modification, or recombinant enzyme-mediated modification. In some embodiments, the modification to the genome by transgene insertion can occur via CRISPR-Cas9, TALEN, ZFNs, Cas-CLOVER, and Cpf1.

In gene editing systems that involve inserting new or existing nucleotides/nucleic acids, insertion tools (e.g., DNA template vectors, transposable elements (transposons or retrotransposons) must be delivered to the cell in addition to the cutting enzyme (e.g., a nuclease, recombinase, integrase or transposase). Examples of such insertion tools for a recombinase may include a DNA vector. Other gene editing systems require the delivery of an integrase along with an insertion vector, a transposase along with a transposon/retrotransposon, etc. In some embodiments, an example recombinase that may be used as a cutting enzyme is the CRE recombinase. In various embodiments, example integrases that may be used in insertion tools include viral based enzymes taken from any of a number of viruses including, but not limited to, AAV, gamma retrovirus, and lentivirus. Example transposons/retrotransposons that may be used in insertion tools include, but are not limited to, the piggyBac transposon, Sleeping Beauty transposon, and the L1 retrotransposon.

In certain embodiments of the methods of the disclosure, the transgene is delivered *in vivo*. In certain embodiments of the methods of the disclosure, *in vivo* transgene delivery can occur by: topical delivery, adsorption, absorption, electropo-

ration, spin-fecton, co-culture, transfection, mechanical delivery, sonic delivery, vibrational delivery, magnetofection or by nanoparticle-mediated delivery. In certain embodiments of the methods of the disclosure, in vivo transgene delivery by transfection can occur by liposomal transfection, calcium phosphate transfection, fugene transfection, and dendrimer-mediated transfection. In certain embodiments of the methods of the disclosure, in vivo mechanical transgene delivery can occur by cell squeezing, bombardment, and gene gun. In certain embodiments of the methods of the disclosure, in vivo nanoparticle-mediated transgene delivery can occur by liposomal delivery, delivery by micelles, and delivery by polymerosomes. In various embodiments, nucleases that may be used as cutting enzymes include, but are not limited to, Cas9, transcription activator-like effector nucleases (TALENs) and zinc finger nucleases.

In certain embodiments of the methods of the disclosure, non-viral vectors are used for transgene delivery. In certain embodiments, the non-viral vector is a nucleic acid. In certain embodiments, the nucleic acid non-viral vector is plasmid DNA, linear double-stranded DNA (dsDNA), linear single-stranded DNA (ssDNA), DoggyBone™ DNA, nanoplastids, minicircle DNA, single-stranded oligodeoxynucleotides (ssODN), DDNA oligonucleotides, single-stranded mRNA (ssRNA), and double-stranded mRNA (dsRNA). In certain embodiments, the non-viral vector is a transposon. In certain embodiments, the transposon is piggyBac.

In certain embodiments of the methods of the disclosure, transgene delivery can occur via viral vector. In certain embodiments, the viral vector is a non-integrating non-chromosomal vectors. Non-integrating non-chromosomal vectors can include adeno-associated virus (AAV), adenovirus, and herpes viruses. In certain embodiments, the viral vector is an integrating chromosomal vectors. Integrating chromosomal vectors can include adeno-associated vectors (AAV), Lentiviruses, and gamma-retroviruses.

In certain embodiments of the methods of the disclosure, transgene delivery can occur by a combination of vectors. Exemplary but non-limiting vector combinations can include: viral plus non-viral vectors, more than one non-viral vector, or more than one viral vector. Exemplary but non-limiting vectors combinations can include: DNA-derived plus RNA-derived vectors, RNA plus reverse transcriptase, a transposon and a transposase, a non-viral vectors plus an endonuclease, and a viral vector plus an endonuclease.

In certain embodiments of the methods of the disclosure, the genome modification can be a stable integration of a transgene, a transient integration of a transgene, a site-specific integration of a transgene, or a biased integration of a transgene.

In certain embodiments of the methods of the disclosure, the genome modification can be a stable chromosomal integration of a transgene. In certain embodiments, the stable chromosomal integration can be a random integration, a site-specific integration, or a biased integration. In certain embodiments, the site-specific integration can be non-assisted or assisted. In certain embodiments, the assisted site-specific integration is co-delivered with a site-directed nuclease. In certain embodiments, the site-directed nuclease comprises a transgene with 5' and 3' nucleotide sequence extensions that contain homology to upstream and downstream regions of the site of genomic integration. In certain embodiments, the transgene with homologous nucleotide extensions enable genomic integration by homologous

recombination, microhomology-mediated end joining, or nonhomologous end-joining. In certain embodiments the site-specific integration occurs at a safe harbor site. Genomic safe harbor sites are able to accommodate the integration of new genetic material in a manner that ensures that the newly inserted genetic elements function reliably (for example, are expressed at a therapeutically effective level of expression) and do not cause deleterious alterations to the host genome that cause a risk to the host organism. Potential genomic safe harbors include, but are not limited to, intronic sequences of the human albumin gene, the adeno-associated virus site 1 (AAVS1), a naturally occurring site of integration of AAV virus on chromosome 19, the site of the chemokine (C—C motif) receptor 5 (CCR5) gene and the site of the human ortholog of the mouse Rosa26 locus.

In certain embodiments, the site-specific transgene integration occurs at a site that disrupts expression of a target gene. In certain embodiments, disruption of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements. In certain embodiments, exemplary target genes targeted by site-specific integration include but are not limited to TRAC, TRAB, PDI, any immunosuppressive gene, and genes involved in allo-rejection.

In certain embodiments, the site-specific transgene integration occurs at a site that results in enhanced expression of a target gene. In certain embodiments, enhancement of target gene expression occurs by site-specific integration at introns, exons, promoters, genetic elements, enhancers, suppressors, start codons, stop codons, and response elements.

In certain embodiments of the methods of the disclosure, enzymes may be used to create strand breaks in the host genome to facilitate delivery or integration of the transgene. In certain embodiments, enzymes create single-strand breaks. In certain embodiments, enzymes create double-strand breaks. In certain embodiments, examples of break-inducing enzymes include but are not limited to: transposases, integrases, endonucleases, CRISPR-Cas9, transcription activator-like effector nucleases (TALEN), zinc finger nucleases (ZFN), Cas-CLOVER™, and Cpf1. In certain embodiments, break-inducing enzymes can be delivered to the cell encoded in DNA, encoded in mRNA, as a protein, as a nucleoprotein complex with a guide RNA (gRNA).

In certain embodiments of the methods of the disclosure, the site-specific transgene integration is controlled by a vector-mediated integration site bias. In certain embodiments vector-mediated integration site bias is controlled by the chosen lentiviral vector. In certain embodiments vector-mediated integration site bias is controlled by the chosen gamma-retroviral vector.

In certain embodiments of the methods of the disclosure, the site-specific transgene integration site is a non-stable chromosomal insertion. In certain embodiments, the integrated transgene may become silenced, removed, excised, or further modified. In certain embodiments of the methods of the disclosure, the genome modification is a non-stable integration of a transgene. In certain embodiments, the non-stable integration can be a transient non-chromosomal integration, a semi-stable non chromosomal integration, a semi-persistent non-chromosomal insertion, or a non-stable chromosomal insertion. In certain embodiments, the transient non-chromosomal insertion can be epi-chromosomal or cytoplasmic. In certain embodiments, the transient non-chromosomal insertion of a transgene does not integrate into

a chromosome and the modified genetic material is not replicated during cell division.

In certain embodiments of the methods of the disclosure, the genome modification is a semi-stable or persistent non-chromosomal integration of a transgene. In certain embodiments, a DNA vector encodes a Scaffold/matrix attachment region (S-MAR) module that binds to nuclear matrix proteins for episomal retention of a non-viral vector allowing for autonomous replication in the nucleus of dividing cells.

In certain embodiments of the methods of the disclosure, the genome modification is a non-stable chromosomal integration of a transgene. In certain embodiments, the integrated transgene may become silenced, removed, excised, or further modified.

In certain embodiments of the methods of the disclosure, the modification to the genome by transgene insertion can occur via host cell-directed double-strand breakage repair (homology-directed repair) by homologous recombination (HR), microhomology-mediated end joining (MMEJ), non-homologous end joining (NHEJ), transposase enzyme-mediated modification, integrase enzyme-mediated modification, endonuclease enzyme-mediated modification, or recombinant enzyme-mediated modification. In certain embodiments, the modification to the genome by transgene insertion can occur via CRISPR-Cas9, TALEN, ZFNs, Cas-CLOVER, and Cpf1.

In certain embodiments of the methods of the disclosure, a cell with an in vivo or ex vivo genomic modification can be a germline cell or a somatic cell. In certain embodiments the modified cell can be a human, non-human, mammalian, rat, mouse, or dog cell. In certain embodiments, the modified cell can be differentiated, undifferentiated, or immortalized. In certain embodiments, the modified undifferentiated cell can be a stem cell. In certain embodiments, the modified cell can be differentiated, undifferentiated, or immortalized. In certain embodiments, the modified undifferentiated cell can be an induced pluripotent stem cell. In certain embodiments, the modified cell can be a T cell, a hematopoietic stem cell, a natural killer cell, a macrophage, a dendritic cell, a monocyte, a megakaryocyte, or an osteoclast. In certain embodiments, the modified cell can be modified while the cell is quiescent, in an activated state, resting, in interphase, in prophase, in metaphase, in anaphase, or in telophase. In certain embodiments, the modified cell can be fresh, cryo-preserved, bulk, sorted into sub-populations, from whole blood, from leukapheresis, or from an immortalized cell line.

Production and Generation of VCARS

At least one VHH protein or VCAR of the disclosure can be optionally produced by a cell line, a mixed cell line, an immortalized cell or clonal population of immortalized cells, as well known in the art. See, e.g., Ausubel, et al., ed., *Current Protocols in Molecular Biology*, John Wiley & Sons, Inc., NY, N.Y. (1987-2001); Sambrook, et al., *Molecular Cloning: A Laboratory Manual*, 2nd Edition, Cold Spring Harbor, N.Y. (1989); Harlow and Lane, *Antibodies, a Laboratory Manual*, Cold Spring Harbor, N.Y. (1989); Colligan, et al., eds., *Current Protocols in Immunology*, John Wiley & Sons, Inc., NY (1994-2001); Colligan et al., *Current Protocols in Protein Science*, John Wiley & Sons, NY, N.Y., (1997-2001).

Amino acids from a VHH protein can be altered, added and/or deleted to reduce immunogenicity or reduce, enhance or modify binding, affinity, on-rate, off-rate, avidity, specificity, half-life, stability, solubility or any other suitable characteristic, as known in the art.

Optionally, VHH proteins can be engineered with retention of high affinity for the antigen and other favorable biological properties. To achieve this goal, the VHH proteins can be optionally prepared by a process of analysis of the parental sequences and various conceptual engineered products using three-dimensional models of the parental and engineered sequences. Three-dimensional models are commonly available and are familiar to those skilled in the art. Computer programs are available which illustrate and display probable three-dimensional conformational structures of selected candidate sequences and can measure possible immunogenicity (e.g., Immunofilter program of Xencor, Inc. of Monrovia, Calif.). Inspection of these displays permits analysis of the likely role of the residues in the functioning of the candidate sequence, i.e., the analysis of residues that influence the ability of the candidate VHH protein to bind its antigen. In this way, residues can be selected and combined from the parent and reference sequences so that the desired characteristic, such as affinity for the target antigen(s), is achieved. Alternatively, or in addition to, the above procedures, other suitable methods of engineering can be used. Screening of VHH Proteins

Screening VHH for specific binding to similar proteins or fragments can be conveniently achieved using nucleotide (DNA or RNA display) or peptide display libraries, for example, in vitro display. This method involves the screening of large collections of peptides for individual members having the desired function or structure. The displayed nucleotide or peptide sequences can be from 3 to 5000 or more nucleotides or amino acids in length, frequently from 5-100 amino acids long, and often from about 8 to 25 amino acids long. In addition to direct chemical synthetic methods for generating peptide libraries, several recombinant DNA methods have been described. One type involves the display of a peptide sequence on the surface of a bacteriophage or cell. Each bacteriophage or cell contains the nucleotide sequence encoding the particular displayed peptide sequence. The VHH proteins of the disclosure can bind human or other mammalian proteins with a wide range of affinities (KD). In a preferred embodiment, at least one VHH of the present invention can optionally bind to a target protein with high affinity, for example, with a KD equal to or less than about 10^{-7} M, such as but not limited to, 0.1-9.9 (or any range or value therein) $\times 10^{-8}$, 10^{-9} , 10^{-10} , 10^{-11} , 10^{-12} , 10^{-13} , 10^{-14} , 10^{-15} or any range or value therein, as determined by surface plasmon resonance or the Kinexa method, as practiced by those of skill in the art.

The affinity or avidity of a VHH or a VCAR for an antigen can be determined experimentally using any suitable method. (See, for example, Berzofsky, et al., "Antibody-Antigen Interactions," In *Fundamental Immunology*, Paul, W. E., Ed., Raven Press: New York, N.Y. (1984); Kubly, Janis Immunology, W.H. Freeman and Company: New York, N.Y. (1992); and methods described herein). The measured affinity of a particular VHH-antigen or VCAR-antigen interaction can vary if measured under different conditions (e.g., salt concentration, pH). Thus, measurements of affinity and other antigen-binding parameters (e.g., KD, Kon, Koff) are preferably made with standardized solutions of VHH or VCAR and antigen, and a standardized buffer, such as the buffer described herein.

Competitive assays can be performed with the VHH or VCAR of the disclosure in order to determine what proteins, antibodies, and other antagonists compete for binding to a target protein with the VHH or VCAR of the present invention and/or share the epitope region. These assays as readily known to those of ordinary skill in the art evaluate

competition between antagonists or ligands for a limited number of binding sites on a protein. The protein and/or antibody is immobilized or insolubilized before or after the competition and the sample bound to the target protein is separated from the unbound sample, for example, by decanting (where the protein/antibody was preinsolubilized) or by centrifuging (where the protein/antibody was precipitated after the competitive reaction). Also, the competitive binding may be determined by whether function is altered by the binding or lack of binding of the VHH or VCAR to the target protein, e.g., whether the VCAR molecule inhibits or potentiates the enzymatic activity of, for example, a label. ELISA and other functional assays may be used, as well known in the art.

Nucleic Acid Molecules

Nucleic acid molecules of the disclosure encoding VHH proteins or VCARS can be in the form of RNA, such as mRNA, hnRNA, tRNA or any other form, or in the form of DNA, including, but not limited to, cDNA and genomic DNA obtained by cloning or produced synthetically, or any combinations thereof. The DNA can be triple-stranded, double-stranded or single-stranded, or any combination thereof. Any portion of at least one strand of the DNA or RNA can be the coding strand, also known as the sense strand, or it can be the non-coding strand, also referred to as the anti-sense strand.

Isolated nucleic acid molecules of the disclosure can include nucleic acid molecules comprising an open reading frame (ORF), optionally, with one or more introns, e.g., but not limited to, at least one specified portion of at least one VCAR; nucleic acid molecules comprising the coding sequence for a VCAR; and nucleic acid molecules which comprise a nucleotide sequence substantially different from those described above but which, due to the degeneracy of the genetic code, still encode the VCAR as described herein and/or as known in the art. Of course, the genetic code is well known in the art. Thus, it would be routine for one skilled in the art to generate such degenerate nucleic acid variants that code for specific VCARS of the present invention. See, e.g., Ausubel, et al., supra, and such nucleic acid variants are included in the present invention.

As indicated herein, nucleic acid molecules of the disclosure which comprise a nucleic acid encoding a VCAR can include, but are not limited to, those encoding the amino acid sequence of a VHH fragment, by itself, the coding sequence for the entire VCAR or a portion thereof; the coding sequence for a VHH, fragment or portion, as well as additional sequences, such as the coding sequence of at least one signal leader or fusion peptide, with or without the aforementioned additional coding sequences, such as at least one intron, together with additional, non-coding sequences, including but not limited to, non-coding 5' and 3' sequences, such as the transcribed, non-translated sequences that play a role in transcription, mRNA processing, including splicing and polyadenylation signals (for example, ribosome binding and stability of mRNA); an additional coding sequence that codes for additional amino acids, such as those that provide additional functionalities. Thus, the sequence encoding a VCAR can be fused to a marker sequence, such as a sequence encoding a peptide that facilitates purification of the fused VCAR comprising a VHH fragment or portion. Polynucleotides Selectively Hybridizing to a Polynucleotide as Described Herein

The disclosure provides isolated nucleic acids that hybridize under selective hybridization conditions to a polynucleotide disclosed herein. Thus, the polynucleotides of this embodiment can be used for isolating, detecting, and/or

quantifying nucleic acids comprising such polynucleotides. For example, polynucleotides of the present invention can be used to identify, isolate, or amplify partial or full-length clones in a deposited library. In some embodiments, the polynucleotides are genomic or cDNA sequences isolated, or otherwise complementary to, a cDNA from a human or mammalian nucleic acid library.

Preferably, the cDNA library comprises at least 80% full-length sequences, preferably, at least 85% or 90% full-length sequences, and, more preferably, at least 95% full-length sequences. The cDNA libraries can be normalized to increase the representation of rare sequences. Low or moderate stringency hybridization conditions are typically, but not exclusively, employed with sequences having a reduced sequence identity relative to complementary sequences. Moderate and high stringency conditions can optionally be employed for sequences of greater identity. Low stringency conditions allow selective hybridization of sequences having about 70% sequence identity and can be employed to identify orthologous or paralogous sequences.

Optionally, polynucleotides of this invention will encode at least a portion of a VCAR encoded by the polynucleotides described herein. The polynucleotides of this invention embrace nucleic acid sequences that can be employed for selective hybridization to a polynucleotide encoding a VCAR of the present invention. See, e.g., Ausubel, supra; Colligan, supra, each entirely incorporated herein by reference.

Construction of Nucleic Acids

The isolated nucleic acids of the disclosure can be made using (a) recombinant methods, (b) synthetic techniques, (c) purification techniques, and/or (d) combinations thereof, as well-known in the art.

The nucleic acids can conveniently comprise sequences in addition to a polynucleotide of the present invention. For example, a multi-cloning site comprising one or more endonuclease restriction sites can be inserted into the nucleic acid to aid in isolation of the polynucleotide. Also, translatable sequences can be inserted to aid in the isolation of the translated polynucleotide of the disclosure. For example, a hexa-histidine marker sequence provides a convenient means to purify the proteins of the disclosure. The nucleic acid of the disclosure, excluding the coding sequence, is optionally a vector, adapter, or linker for cloning and/or expression of a polynucleotide of the disclosure.

Additional sequences can be added to such cloning and/or expression sequences to optimize their function in cloning and/or expression, to aid in isolation of the polynucleotide, or to improve the introduction of the polynucleotide into a cell. Use of cloning vectors, expression vectors, adapters, and linkers is well known in the art. (See, e.g., Ausubel, supra; or Sambrook, supra).

Recombinant Methods for Constructing Nucleic Acid

The isolated nucleic acid compositions of this disclosure, such as RNA, cDNA, genomic DNA, or any combination thereof, can be obtained from biological sources using any number of cloning methodologies known to those of skill in the art. In some embodiments, oligonucleotide probes that selectively hybridize, under stringent conditions, to the polynucleotides of the present invention are used to identify the desired sequence in a cDNA or genomic DNA library. The isolation of RNA, and construction of cDNA and genomic libraries are well known to those of ordinary skill in the art. (See, e.g., Ausubel, supra; or Sambrook, supra).

Nucleic Acid Screening and Isolation Methods

A cDNA or genomic library can be screened using a probe based upon the sequence of a polynucleotide of the disclo-

sure. Probes can be used to hybridize with genomic DNA or cDNA sequences to isolate homologous genes in the same or different organisms. Those of skill in the art will appreciate that various degrees of stringency of hybridization can be employed in the assay; and either the hybridization or the wash medium can be stringent. As the conditions for hybridization become more stringent, there must be a greater degree of complementarity between the probe and the target for duplex formation to occur. The degree of stringency can be controlled by one or more of temperature, ionic strength, pH and the presence of a partially denaturing solvent, such as formamide. For example, the stringency of hybridization is conveniently varied by changing the polarity of the reactant solution through, for example, manipulation of the concentration of formamide within the range of 0% to 50%. The degree of complementarity (sequence identity) required for detectable binding will vary in accordance with the stringency of the hybridization medium and/or wash medium. The degree of complementarity will optimally be 100%, or 70-100%, or any range or value therein. However, it should be understood that minor sequence variations in the probes and primers can be compensated for by reducing the stringency of the hybridization and/or wash medium.

Methods of amplification of RNA or DNA are well known in the art and can be used according to the disclosure without undue experimentation, based on the teaching and guidance presented herein.

Known methods of DNA or RNA amplification include, but are not limited to, polymerase chain reaction (PCR) and related amplification processes (see, e.g., U.S. Pat. Nos. 4,683,195, 4,683,202, 4,800,159, 4,965,188, to Mullis, et al.; U.S. Pat. Nos. 4,795,699 and 4,921,794 to Tabor, et al.; U.S. Pat. No. 5,142,033 to Innis; U.S. Pat. No. 5,122,464 to Wilson, et al.; U.S. Pat. No. 5,091,310 to Innis; U.S. Pat. No. 5,066,584 to Gyllenstein, et al.; U.S. Pat. No. 4,889,818 to Gelfand, et al.; U.S. Pat. No. 4,994,370 to Silver, et al.; U.S. Pat. No. 4,766,067 to Biswas; U.S. Pat. No. 4,656,134 to Ringold) and RNA mediated amplification that uses antisense RNA to the target sequence as a template for double-stranded DNA synthesis (U.S. Pat. No. 5,130,238 to Malek, et al, with the tradename NASBA), the entire contents of which references are incorporated herein by reference. (See, e.g., Ausubel, *supra*; or Sambrook, *supra*.)

For instance, polymerase chain reaction (PCR) technology can be used to amplify the sequences of polynucleotides of the disclosure and related genes directly from genomic DNA or cDNA libraries. PCR and other in vitro amplification methods can also be useful, for example, to clone nucleic acid sequences that code for proteins to be expressed, to make nucleic acids to use as probes for detecting the presence of the desired mRNA in samples, for nucleic acid sequencing, or for other purposes. Examples of techniques sufficient to direct persons of skill through in vitro amplification methods are found in Berger, *supra*, Sambrook, *supra*, and Ausubel, *supra*, as well as Mullis, et al., U.S. Pat. No. 4,683,202 (1987); and Innis, et al., PCR Protocols A Guide to Methods and Applications, Eds., Academic Press Inc., San Diego, Calif. (1990). Commercially available kits for genomic PCR amplification are known in the art. See, e.g., Advantage-GC Genomic PCR Kit (Clontech). Additionally, e.g., the T4 gene 32 protein (Boehringer Mannheim) can be used to improve yield of long PCR products.

Synthetic Methods for Constructing Nucleic Acids

The isolated nucleic acids of the disclosure can also be prepared by direct chemical synthesis by known methods (see, e.g., Ausubel, et al., *supra*). Chemical synthesis gen-

erally produces a single-stranded oligonucleotide, which can be converted into double-stranded DNA by hybridization with a complementary sequence, or by polymerization with a DNA polymerase using the single strand as a template.

One of skill in the art will recognize that while chemical synthesis of DNA can be limited to sequences of about 100 or more bases, longer sequences can be obtained by the ligation of shorter sequences.

Recombinant Expression Cassettes

The disclosure further provides recombinant expression cassettes comprising a nucleic acid of the disclosure. A nucleic acid sequence of the disclosure, for example, a cDNA or a genomic sequence encoding a VCAR of the disclosure, can be used to construct a recombinant expression cassette that can be introduced into at least one desired host cell. A recombinant expression cassette will typically comprise a polynucleotide of the disclosure operably linked to transcriptional initiation regulatory sequences that will direct the transcription of the polynucleotide in the intended host cell. Both heterologous and non-heterologous (i.e., endogenous) promoters can be employed to direct expression of the nucleic acids of the disclosure.

In some embodiments, isolated nucleic acids that serve as promoter, enhancer, or other elements can be introduced in the appropriate position (upstream, downstream or in the intron) of a non-heterologous form of a polynucleotide of the disclosure so as to up or down regulate expression of a polynucleotide of the disclosure. For example, endogenous promoters can be altered in vivo or in vitro by mutation, deletion and/or substitution.

Nanotransposons

The disclosure provides a nanotransposon comprising: (a) a sequence encoding a transposon insert, comprising a sequence encoding a first inverted terminal repeat (ITR), a sequence encoding a second inverted terminal repeat (ITR), and an intra-ITR sequence; (b) a sequence encoding a backbone, wherein the sequence encoding the backbone comprises a sequence encoding an origin of replication having between 1 and 450 nucleotides, inclusive of the endpoints, and a sequence encoding a selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, and (c) an inter-ITR sequence. In some embodiments, the inter-ITR sequence of (c) comprises the sequence of (b). In some embodiments, the intra-ITR sequence of (a) comprises the sequence of (b).

In some embodiments of the nanotransposons of the disclosure, the sequence encoding the backbone comprises between 1 and 600 nucleotides, inclusive of the endpoints. In some embodiments, the sequence encoding the backbone consists of between 1 and 50 nucleotides, between 50 and 100 nucleotides, between 100 and 150 nucleotides, between 150 and 200 nucleotides, between 200 and 250 nucleotides, between 250 and 300 nucleotides, between 300 and 350 nucleotides, between 350 and 400 nucleotides, between 400 and 450 nucleotides, between 450 and 500 nucleotides, between 500 and 550 nucleotides, between 550 and 600 nucleotides, each range inclusive of the endpoints.

In some embodiments of the nanotransposons of the disclosure, the inter-ITR sequence comprises between 1 and 1000 nucleotides, inclusive of the endpoints. In some embodiments, the inter-ITR sequence consists of between 1 and 50 nucleotides, between 50 and 100 nucleotides, between 100 and 150 nucleotides, between 150 and 200 nucleotides, between 200 and 250 nucleotides, between 250 and 300 nucleotides, between 300 and 350 nucleotides, between 350 and 400 nucleotides, between 400 and 450 nucleotides, between 450 and 500 nucleotides, between 500

and 550 nucleotides, between 550 and 600 nucleotides, between 600 and 650 nucleotides, between 650 and 700 nucleotides, between 700 and 750 nucleotides, between 750 and 800 nucleotides, between 800 and 850 nucleotides, between 850 and 900 nucleotides, between 900 and 950 nucleotides, or between 950 and 1000 nucleotides, each range inclusive of the endpoints.

In some embodiments of the nanotransposons of the disclosure, including the short nanotransposons (SNTs) of the disclosure, the inter-ITR sequence comprises between 1 and 200 nucleotides, inclusive of the endpoints. In some embodiments, the inter-ITR sequence consists of between 1 and 10 nucleotides, between 10 and 20 nucleotides, between 20 and 30 nucleotides, between 30 and 40 nucleotides, between 40 and 50 nucleotides, between 50 and 60 nucleotides, between 60 and 70 nucleotides, between 70 and 80 nucleotides, between 80 and 90 nucleotides, or between 90 and 100 nucleotides, each range inclusive of the endpoints.

In some embodiments of the nanotransposons of the disclosure, the selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, comprises a sequence encoding a sucrose-selectable marker. In some embodiments, the sequence encoding a sucrose-selectable marker comprises a sequence encoding an RNA-OUT sequence. In some embodiments, the sequence encoding an RNA-OUT sequence comprises or consists of 137 base pairs (bp). In some embodiments, the selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, comprises a sequence encoding a fluorescent marker. In some embodiments, the selectable marker having between 1 and 200 nucleotides, inclusive of the endpoints, comprises a sequence encoding a cell surface marker.

In some embodiments of the nanotransposons of the disclosure, the sequence encoding an origin of replication having between 1 and 450 nucleotides, inclusive of the endpoints, comprises a sequence encoding a mini origin of replication. In some embodiments, the sequence encoding an origin of replication having between 1 and 450 nucleotides, inclusive of the endpoints, comprises a sequence encoding an R6K origin of replication. In some embodiments, the R6K origin of replication comprises an R6K gamma origin of replication. In some embodiments, the R6K origin of replication comprises an R6K mini origin of replication. In some embodiments, the R6K origin of replication comprises an R6K gamma mini origin of replication. In some embodiments, the R6K gamma mini origin of replication comprises or consists of 281 base pairs (bp).

In some embodiments of the nanotransposons of the disclosure, the sequence encoding the backbone does not comprise a recombination site, an excision site, a ligation site or a combination thereof. In some embodiments, neither the nanotransposon nor the sequence encoding the backbone comprises a product of a recombination site, an excision site, a ligation site or a combination thereof. In some embodiments, neither the nanotransposon nor the sequence encoding the backbone is derived from a recombination site, an excision site, a ligation site or a combination thereof.

In some embodiments of the nanotransposons of the disclosure, a recombination site comprises a sequence resulting from a recombination event. In some embodiments, a recombination site comprises a sequence that is a product of a recombination event. In some embodiments, the recombination event comprises an activity of a recombinase (e.g., a recombinase site).

In some embodiments of the nanotransposons of the disclosure, the sequence encoding the backbone does not further comprise a sequence encoding foreign DNA.

In some embodiments of the nanotransposons of the disclosure, the inter-ITR sequence does not comprise a recombination site, an excision site, a ligation site or a combination thereof. In some embodiments, the inter-ITR sequence does not comprise a product of a recombination event, an excision event, a ligation event or a combination thereof. In some embodiments, the inter-ITR sequence is not derived from a recombination event, an excision event, a ligation event or a combination thereof.

In some embodiments of the nanotransposons of the disclosure, the inter-ITR sequence comprises a sequence encoding foreign DNA.

In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises at least one sequence encoding an insulator and a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell. In some embodiments, the mammalian cell is a human cell.

In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a first sequence encoding an insulator, a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell and a second sequence encoding an insulator.

In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a first sequence encoding an insulator, a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell, a polyadenosine (polyA) sequence and a second sequence encoding an insulator.

In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a first sequence encoding an insulator, a sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell, at least one exogenous sequence, a polyadenosine (polyA) sequence and a second sequence encoding an insulator.

In some embodiments of the nanotransposons of the disclosure, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell is capable of expressing an exogenous sequence in a human cell. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding a constitutive promoter. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding an inducible promoter. In some embodiments, the intra-ITR sequence comprises a first sequence encoding a first promoter capable of expressing an exogenous sequence in a mammalian cell and a second sequence encoding a second promoter capable of expressing an exogenous sequence in a mammalian cell, wherein the first promoter is a constitutive promoter, wherein the second promoter is an inducible promoter, and wherein the first sequence encoding the first promoter and the second sequence encoding the second promoter are oriented in opposite directions. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding a cell-type or tissue-type specific promoter. In some embodiments, the sequence encoding a promoter capable of expressing an exogenous sequence in a mammalian cell comprises a sequence encoding an EF1a promoter, a sequence encoding a CMV promoter, a sequence encoding an MND promoter, a sequence encoding an SV40 promoter, a sequence encoding a PGK1 promoter, a sequence encoding a Ubc promoter, a sequence encoding a

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CAG promoter, a sequence encoding an H1 promoter, or a sequence encoding a U6 promoter.

In some embodiments of the nanotransposons of the disclosure, the polyadenosine (polyA) sequence is isolated or derived from a viral polyA sequence. In some embodiments, the polyadenosine (polyA) sequence is isolated or derived from an (SV40) polyA sequence.

In some embodiments of the nanotransposons of the disclosure, the at least one exogenous sequence comprises an inducible proapoptotic polypeptide. In some embodiments, the inducible caspase polypeptide comprises (a) a ligand binding region, (b) a linker, and (c) a caspase polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In some embodiments, the inducible caspase polypeptide comprises (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the ligand binding region comprises a FKBP12 polypeptide. In some embodiments, the amino acid sequence of the ligand binding region comprises a FK506 binding protein 12 (FKBP12) polypeptide. In some embodiments, the FK506 binding protein 12 (FKBP12) polypeptide comprises a modification at position 36 of the sequence. In some embodiments, the modification comprises a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In some embodiments, the FKBP12 polypeptide is encoded by an amino acid sequence comprising

(SEQ ID NO: 14635)

GVQVETISPGDGRTPFKRGQTCVVHYTGMLDGGKVDSSRDENKPFK
FMLGKQEVIRGWEEGVAQMSVGQRAKLTI SPDYAYGATGHPGIIPPH
ATLVDFVELLKE.

In some embodiments, the FKBP12 polypeptide is encoded by a nucleic acid sequence comprising

(SEQ ID NO: 14636)

GGGGTCCAGGTCGAGACTATTTCCAGGGGATGGGCGAATTTCC
AAAAAGGGGCCAGACTTGCCTCGTGCATTACACGGGATGCTGGAGG
ACGGGAAGAAAGTGGACAGCTCCAGGGATCGCAACAAGCCCTTCAAG
TTCATGCTGGGAAAGCAGGAAGTATCCGAGGATGGGAGGAAGGCGT
GGCAGAGATGTCAGTCGGCCAGCGGCCAACTGACCATTAGCCCTG
ACTACGCTTATGGAGCAACAGGCCACCCAGGGATCATTCCCCTCAT
GCCACCTCGGTCTTCGATGTGGAAGTGTGAAGCTGGAG.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the linker region is encoded by an amino acid comprising GGGGS (SEQ ID NO: 14637) or a nucleic acid sequence comprising GGAGGAGGAGGATCC (SEQ ID NO: 14638). In some embodiments, the nucleic acid sequence encoding the linker does not comprise a restriction site.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exog-

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enous sequence comprises an inducible proapoptotic polypeptide, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an arginine (R) at position 87 of the sequence. In some embodiments, the truncated caspase 9 polypeptide is encoded by an amino acid sequence that does not comprise an alanine (A) at position 282 the sequence. In some embodiments, the truncated caspase 9 polypeptide is encoded by an amino acid comprising

(SEQ ID NO: 14639)

GGFDVGALES LRGNADLAYILSMEPCGHCLII NNVNFCRESGLRTRT
GSNIDCEKLRRRFSSLFHFMVEVKGDLTAKKMVLALLELAQQDHGALD
CCVVVILSHGCQASHLQFP GAVYGT DGC PVSVEKIVNIFNGTSCPSL
GGKPKLFFIQACGGEQKDHGF EVASTSPEDESPG SNPEPDATPFQEG
LRTFDQLDAISSLP TSDIFVSYSTFP GFVSWRDPKSGSWYVETLDD
IFEQWAHSEDLQSLLLRVANAVSVKGIYKQMPGCFNFLRKKLFKTS.

In some embodiments, the truncated caspase 9 polypeptide is encoded by a nucleic acid sequence comprising

(SEQ ID NO: 14640)

GGATTTGGGGACGTGGGGGCCCTGGAGTCTCTGCGAGGAAATGCCGA
TCTGGCTTACATCCTGAGCATGGAACCTGCGGCCACTGTCTGATCA
TTAACAATGTGAACCTTCTGCAGAGAAAGCGACTGCGAACACGGACT
GGCTCCAATATTGACTGTGAGAAGCTGCGGAGAAGGTTCTCTAGTCT
GCACTTTATGGTCGAAGTAAAGGGGATCTGACCGCCAAGAAAATGG
TGCTGGCCCTGCTGGAGCTGGCTCAGCAGGACCATGGAGCTCTGGAT
TGCTGCGTGGTCTGATCCTGTCCCACGGGTGCCAGGCTTCTCATCT
GCAGTTCCTCCGAGCAGTGACGGAACAGACGGCTGTCTGTACAGC
TGGAGAAGATCGTCAACATCTTCAACGGCACTTCTTGCCCTAGTCTG
GGGGGAAAGCCAAAACCTGTTCTTTATCCAGGCCTGTGGCGGGGAACA
GAAAGATCACGGCTTCGAGGTGGCCAGCACCAGCCCTGAGGACGAAT
CACCAGGAGCAACCTGAACAGATGCAACTCCATTCCAGGAGGGA
CTGAGGACCTTTGACCAGCTGGATGCTATCTCAAGCCTGCCACTCC
TAGTGACATTTTCGTGTCTTACAGTACCTTCCAGGCTTTGTCTCAT
GGCGCGATCCCAAGTCAGGGAGCTGGTACGTGGAGACACTGGACGAC
ATCTTTGAACAGTGGGCCATTTCAGAGGACCTGCAGAGCCTGCTGCT
GCGAGTGGCAAACGCTGTCTCTGTGAAGGGCATCTACAAACAGATGC
CCGGGTGCTTCAATTTCTGAGAAAGAACTGTTCTTTAAGACTTCC.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the inducible proapoptotic polypeptide is encoded by an amino acid sequence comprising

(SEQ ID NO: 14641)

GVQVETISPGDGRTPFKRGQTCVVHYTGMLDGGKVDSSRDENKPFKFML
GKQEVIRGWEEGVAQMSVGQRAKLTI SPDYAYGATGHPGIIPPHATLVDFD

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VELLKLEGGGSGFGDVGALSLRGNADLAYILSMPCGHCLIIINNVC
 RESGLRTRTGSNIDCEKLRRRPFSSLHFMVEVKDILTAKMKVLALLELAQQ
 DHGALDCCVVVILSHGCQASHLQFPFPAVYGTGCPVSVEKIVNIFNGTSC
 PSLGGKPKLFFIQACGGEQKDHGFEVASTSPEDSPGSNPEPDATPFQEG
 LRTFDQLDAISSLPSPDIFVSYSTFPFVSWRDPKSGSWYVETLDDIFE
 QWAHSEDLQSLLLRVANAVSVKGIYKMPGCFNFLRKKLFKTS.

In some embodiments, the inducible proapoptotic polypeptide is encoded by a nucleic acid sequence comprising

(SEQ ID NO: 14642)

gggggtccaggtcgagactatttcaccaggggatggcgcaacatttccaaa
 aagggggcagacttgcgtcgtgcattacaccgggatgctggaggacggga
 agaaagtggacagctccagggatcgcaacaagccctcaagttcgtctg
 ggaaagcaggaagtgatccgaggatgggaggaagcggtggcacagatgtc
 agtcggccagcgggcaaaactgaccattagccctgactacgttattggag
 caacaggccaccagggatcattccccctcatgccaccctggtcttcgat
 gtggaactgctgaagctggaggaggaggaggatccggatttggggacgt
 gggggccctggagtctctcgaggaaatgccgatctggtctacatcctga
 gcatggaaccctgeggccactgtctgatcattaacaatgtgaactctgc
 agagaaagcggactgcgaacacggactggctccaatattgactgtgagaa
 gctgcgagagaaggttctctagtctgcactttatggctgaagtgaagggg
 atctgaccgccaaagaaatgggtgctggccctgctggagctggctcagcag
 gaccatggagctctggattgctgctggctgctgatcctgtcccacgggtg
 ccaggcttctcatctgcagttccccggagcagtgtaacgaacagacggct
 gtctctgacgctggagaagatcgtaacatcttcaacggcactttctgc
 cctagtctgggggaaagccaaaactgttctttatccaggcctgtggcg
 ggaacagaaagatcacggcttcgaggtggccagcaccagccctgaggacg
 aatcaccaggagcaaccctgaaccagatgcaactccattccaggaggga
 ctgaggacctttgaccagctggatgctatctcaagcctgccactectag
 tgacattttcgtgtettacagtaccttcccaggctttgtctcatggcg
 atcccaagtcaggagctggtagctggagacactggagacatctttgaa
 cagtgggcccatcagaggacctgcagagcctgctgctgagtgaggaaa
 cgctgtctctgtgaagggcatctacaaacagatgcccggtgcttcaatt
 ttctgagaagaaactgttctttaagacttcc.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide, the exogenous sequence further comprises a sequence encoding a selectable marker. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a detectable marker. In some embodiments, the detectable marker comprises a fluorescent marker or a cell-surface marker. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a protein that is active in dividing cells and not active in non-dividing cells. In some embodiments, the sequence encoding the selectable marker comprises a

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sequence encoding a metabolic marker. In some embodiments, the sequence encoding the selectable marker comprises a sequence encoding a dihydrofolate reductase (DHFR) mutein enzyme. In some embodiments, the DHFR mutein enzyme comprises or consists of the amino acid sequence of:

(SEQ ID NO: 17012)

MVGSLLNCIVA VSQNMIGIGN GDFWPPLRN
 ESRVFQRMFTT TSSVEGKQNL
 61 VIMGKKTWFS IPEKNRPLKG RINLVLSREL
 KEPPQGAHFL SRSLLDALKL
 121 TEQPELANKV DMVWIVGGSS VYKEAMNHPG
 HLKLFVTRIM QDFESDTFFP
 181 EIDLEKYKLL PEYPGVLSDV QEEKGIKYKF
 EVYEKND.

In some embodiments, the DHFR mutein enzyme is encoded by a the nucleic acid sequence comprising or consisting of

(SEQ ID NO: 17095)

atggtcgggtctctgaattgtatcgctcgccgtgagtcagaacatgggcat
 tgggaagaatggcgatttcccatggccacctctgcgcaacagagtcggat
 30 actttcagcggatgacaactacctcctctgtggaagggaaacagaatctg
 gtcatcatgggaaagaaaacttggttcagcattccagagaagaaccggcc
 cctgaaaggcagaatcaatctggtgctgtcccgagaactgaaggagccac
 35 cacaggagagctcactttctgagccggtccctggacgatgactgaagctg
 acagaacagcctgagctggccaacaaagtcatatggtgtggtatcgctcg
 gggaagttcagtgataaggaggccatgaatcacccggccatctgaaac
 40 tggttcgtcacaggatcatgcaggactttgagagcgatactttcttctct
 gaaattgacctggagaagtacaaactgctgcccgaatatcctggcgtgct
 gtccgatgtccaggaagagaaggcatcaaatacaagttcgaggtctatg
 45 agaagaatgac.

In some embodiments, the amino acid sequence of the DHFR mutein enzyme further comprises a mutation at one or more of positions 80, 113, or 153. In some embodiments, the amino acid sequence of the DHFR mutein enzyme comprises one or more of a substitution of a Phenylalanine (F) or a Leucine (L) at position 80, a substitution of a Leucine (L) or a Valine (V) at position 113, and a substitution of a Valine (V) or an Aspartic Acid (D) at position 153.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide and/or the exogenous sequence comprises a sequence encoding a selectable marker, the exogenous sequence further comprises a sequence encoding a non-naturally occurring antigen receptor, and/or a sequence encoding a therapeutic polypeptide. In some embodiments, the non-naturally occurring antigen receptor comprises a T cell Receptor (TCR). In some embodiments, a sequence encoding the TCR comprises one or more of an insertion, a deletion, a substitution, an inversion, a transposition or a frameshift compared to a corresponding wild type sequence.

In some embodiments, a sequence encoding the TCR comprises a chimeric or recombinant sequence. In some embodiments, the non-naturally occurring antigen receptor comprises a chimeric antigen receptor (CAR). In some embodiments, the CAR comprises: (a) an ectodomain comprising an antigen recognition region, (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the ectodomain of (a) of the CAR further comprises a signal peptide. In some embodiments, the ectodomain of (a) of the CAR further comprises a hinge between the antigen recognition region and the transmembrane domain. In some embodiments, the endodomain comprises a human CD3 ζ endodomain. In some embodiments, the at least one costimulatory domain comprises a human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. In some embodiments, the at least one costimulatory domain comprises a human CD28 and/or a 4-1BB costimulatory domain. In some embodiments, the antigen recognition region comprises one or more of a scFv, a VHH, a VH, and a Centyrin.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises an inducible proapoptotic polypeptide and/or the exogenous sequence comprises a sequence encoding a selectable marker, the exogenous sequence further comprises a sequence encoding a transposase.

In some embodiments of the nanotransposons of the disclosure, the intra-ITR sequence comprises a sequence encoding a selectable marker, an exogenous sequence, a sequence encoding an inducible caspase polypeptide, and at least one sequence encoding a self-cleaving peptide. In some embodiments, the at least one sequence encoding a self-cleaving peptide is positioned between one or more of: (a) the sequence encoding a selectable marker and the exogenous sequence, (b) the sequence encoding a selectable marker and the inducible caspase polypeptide, and (c) the exogenous sequence and the inducible caspase polypeptide. In some embodiments, a first sequence encoding a self-cleaving peptide is positioned between the sequence encoding a selectable marker and the exogenous sequence and a second sequence encoding a self-cleaving peptide is positioned between the exogenous sequence and the inducible caspase polypeptide.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprise a TTAA, a TTAT or a TTAX recognition sequence. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprise a TTAA, a TTAT or a TTAX recognition sequence

and a sequence having at least 50% identity to a sequence isolated or derived from a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprise at least 2 nucleotides (nts), 3 nts, 4 nts, 5 nts, 6 nts, 7 nts, 8 nts, 9 nts, 10 nts, 11 nts, 12 nts, 13 nts, 14 nts, 15 nts, 16 nts, 17 nts, 18 nts, 19 nts, or 20 nts.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAGTCTGCGTAAAT-TGACGCATG (SEQ ID NO: 17096) or a sequence having at least 70% identity to the sequence of CCCTAGAAAGATAGTCTGCGTAAAT-TGACGCATG (SEQ ID NO: 17096). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAATCATAT-TGTGACGTACGTTAAAGATAATCATGCGTAAAT-TGA CGCATG (SEQ ID NO: 17097). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAGTCTGCGTAAAT-TGACGCATG (SEQ ID NO: 17096) and comprises the sequence of CCCTAGAAAGATAATCATAT-TGTGACGTACGTTAAAGATAATCATGCGTAAAT-TGA CGCATG (SEQ ID NO: 17097). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAATCATAT-TGTGACGTACGTTAAAGATAATCATGCGTAAAT-TGA CGCATG (SEQ ID NO: 17098). In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) comprises the sequence of CCCTAGAAAGATAGTCTGCGTAAAT-TGACGCATG (SEQ ID NO: 17096) and comprises the sequence of

(SEQ ID NO: 17099)

TTAACCTAGAAAGATAATCATATTTGTGACGTACGTTAAAGATAATCATG

TGTAAATTTGACGCATGTGTTTATCGGTCTGTATATCGAGGTTTATTTA

TTAATTTGAATAGATATTAAGTTTATTTATATTTACACTTACATACTAAT

AATAAATTCACAAACAATTTATTTATGTTTATTTATTTATTAATAAAAAA

CAAAACTCAAAATTTCTTCTATAAGTAACAAACTTTTA.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized

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by a piggyBac transposase or a piggyBac-like transposase. In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having an amino acid sequence of at least 20% identity to the amino acid sequence of

(SEQ ID NO: 14487)

1 MGSSLDDEHI LSALLQSDDE LVGEDSDSEI SDHVSDDVQ
SDTEEAFIDE VHEVQPTSSG

61 SEILDEQNVI EQPGSSLASN RILTLPORTI RGKNKHCWST
SKSTRRSRVs ALNIVRSQRG

121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR
ESMTGATFRD TNEDEIYAFF

181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRPDDL
IRCLRMDDKS IRPTLRENDV

241 FTPVRKIIDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF
RMYIPNKPSK YGIKILMMCD

301 SGTKYMINGM PYLGRGTQTN GVPLGEYVVK ELSKPVHGSC
RNITCDNWFT SIPLAKNLLQ

361 EPYKLTIVGT VRSNKREIPE VLKNSRSRPV GTSMFCFDGP
LTLVSYKPKP AKMVYLLSSC

421 DEDASINEST GKPQVMVYNN QTKGGVDTL D QMCSVMTC SR
KTNRWPMALL YGMINIACIN

481 SFIIYSHNVS SKGEKVQSRK KPMRNLYMSL TSSFMKRKLE
APTLKRYLRD NISNILPNEV

541 PGTSDSTEE PVMKKRTYCT YCPSKIRKKA NASCKKCKKV
ICREHNIDMC QSCF.

In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having the amino acid sequence of

(SEQ ID NO: 14487)

1 MGSSLDDEHI LSALLQSDDE LVGEDSDSEI SDHVSDDVQ
SDTEEAFIDE VHEVQPTSSG

61 SEILDEQNVI EQPGSSLASN RILTLPORTI RGKNKHCWST
SKSTRRSRVs ALNIVRSQRG

121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR
ESMTGATFRD TNEDEIYAFF

181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRPDDL
IRCLRMDDKS IRPTLRENDV

241 FTPVRKIIDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF
RMYIPNKPSK YGIKILMMCD

301 SGTKYMINGM PYLGRGTQTN GVPLGEYVVK ELSKPVHGSC
RNITCDNWFT SIPLAKNLLQ

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-continued

361 EPYKLTIVGT VRSNKREIPE VLKNSRSRPV GTSMFCFDGP
LTLVSYKPKP AKMVYLLSSC

421 DEDASINEST GKPQVMVYNN QTKGGVDTL D QMCSVMTC SR
KTNRWPMALL YGMINIACIN

481 SFIIYSHNVS SKGEKVQSRK KPMRNLYMSL TSSFMKRKLE
APTLKRYLRD NISNILPNEV

541 PGTSDSTEE PVMKKRTYCT YCPSKIRKKA NASCKKCKKV
ICREHNIDMC QSCF.

In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having an amino acid sequence of at least 20% identity to the amino acid sequence of

(SEQ ID NO: 14484)

1 MGSSLDDEHI LSALLQSDDE LVGEDSDSEV SDHVSDDVQ
SDTEEAFIDE VHEVQPTSSG

61 SEILDEQNVI EQPGSSLASN RILTLPORTI RGKNKHCWST
SKSTRRSRVs ALNIVRSQRG

121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR
ESMTSATFRD TNEDEIYAFF

181 GILVMTAVRK DNHMSTDDL F DRSLSMVYVS VMSRDRPDDL
IRCLRMDDKS IRPTLRENDV

241 FTPVRKIIDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF
RVYIPNKPSK YGIKILMMCD

301 SGTKYMINGM PYLGRGTQTN GVPLGEYVVK ELSKPVHGSC
RNITCDNWFT SIPLAKNLLQ

361 EPYKLTIVGT VRSNKREIPE VLKNSRSRPV GTSMFCFDGP
LTLVSYKPKP AKMVYLLSSC

421 DEDASINEST GKPQVMVYNN QTKGGVDTL D QMCSVMTC SR
KTNRWPMALL YGMINIACIN

481 SFIIYSHNVS SKGEKVQSRK KPMRNLYMSL TSSFMKRKLE
APTLKRYLRD NISNILPKEV

541 PGTSDSTEE PVMKKRTYCT YCPSKIRKKA NASCKKCKKV
ICREHNIDMC QSCF.

In some embodiments, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) is recognized by a piggyBac transposase having the amino acid sequence of

(SEQ ID NO: 14484)

1 MGSSLDDEHI LSALLQSDDE LVGEDSDSEV SDHVSDDVQ
SDTEEAFIDE VHEVQPTSSG

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-continued

61 SEILDEQNV I EQPGSSLASN RILTLPQRTI RGKNKHCWST
SKSTRSRVS ALNIVRSQRG

121 PTRMCRNIYD PLLCFKLFFT DEIISEIVKW TNAEISLKRR
ESMTSATFRD TNEDEIYAFF

181 GILVMTAVRK DNHMSTDDLF DRSLSMVYVS VMSRDRFDFL
IRCLRMDDKS IRPTLRENDV

241 FTPVRKIWDL FIHQCIQNYT PGAHLTIDEQ LLGFRGRCPF
RVYIPNKPSK YGIKILMMCD

301 SGTKYMINGM PYLGRGTQTN GVPLGEYYVK ELSKPVHGSC
RNITCDNWF T SIPLAKNLLQ

361 EPYKLTIVGT VRSNKREIPE VLKNSRSPV GTSMFCFDGP
LTLVSYKPKP AKMVYLLSSC

421 DEDASINEST GKPMVMYYN QTKGGVDTLD QMCSVMTCSR
KTNRPWALL YGMINIACIN

481 SFIIYSHNVS SKGEKVQSRK KFMRLNLYMSL TSSPMRKRLE
APTLKRYLRD NISNILPKEV

541 PGTSDDSTEE PVMKKRTYCT YCPSKIRRA NASCKKCKKV
ICREHNIDMC QSCF.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a Sleeping Beauty transposase. In some embodiments, the Sleeping Beauty transposase is a hyperactive Sleeping Beauty transposase (SB100X).

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a Helitron transposase.

In some embodiments of the nanotransposons of the disclosure, including those wherein the at least one exogenous sequence comprises one or more of an inducible proapoptotic polypeptide, a sequence encoding a selectable marker, and an exogenous sequence, the sequence encoding a first inverted terminal repeat (ITR) or the sequence encoding a second inverted terminal repeat (ITR) are recognized by a Tol2 transposase.

The disclosure provides a cell comprising a nanotransposon of the disclosure. In some embodiments, the cell further comprises a transposase composition. In some embodiments, the transposase composition comprises a transposase or a sequence encoding the transposase that is capable of recognizing the first ITR or the second ITR of the nanotransposon. In some embodiments, the transposase composition comprises a nanotransposon comprising the sequence encoding the transposase. In some embodiments, the cell comprises a first nanotransposon comprising an exogenous

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sequence and a second nanotransposon comprising a sequence encoding a transposase. In some embodiments, the cell is an allogeneic cell.

The disclosure provides a composition comprising the nanotransposon of the disclosure.

The disclosure provides a composition comprising the cell of the disclosure. In some embodiments, the cell comprises a nanotransposon of the disclosure. In some embodiments, the cell is not further modified. In some embodiments, the cell is allogeneic.

The disclosure provides a composition comprising the cell of the disclosure. In some embodiments, the cell comprises a nanotransposon of the disclosure. In some embodiments, the cell is not further modified. In some embodiments, the cell is autologous.

The disclosure provides a composition comprising a plurality of cells of the disclosure. In some embodiments, at least one cell of the plurality of cells comprises a nanotransposon of the disclosure. In some embodiments, a portion of the plurality of cells comprises a nanotransposon of the disclosure. In some embodiments, the portion comprises at least 1%, 2%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 99% or any percentage in between of the plurality of cells. In some embodiments, each cell of the plurality of cells comprises a nanotransposon of the disclosure. In some embodiments, the plurality of cells does not comprise a modified cell of the disclosure. In some embodiments, at least one cell of the plurality of cells is not further modified. In some embodiments, none of the plurality of cells is not further modified. In some embodiments, plurality of cells is allogeneic. In some embodiments, an allogeneic plurality of cells are produced according to the methods of the disclosure. In some embodiments, plurality of cells is autologous. In some embodiments, an autologous plurality of cells are produced according to the methods of the disclosure.

The disclosure provides a modified cell comprising: (a) a nanotransposon of the disclosure; (b) a sequence encoding an inducible proapoptotic polypeptide; and wherein the cell is a T cell, (c) a modification of an endogenous sequence encoding a T cell Receptor (TCR), wherein the modification reduces or eliminates a level of expression or activity of the TCR. In some embodiments, the cell further comprises: (d) a non-naturally occurring sequence comprising an HLA class I histocompatibility antigen, alpha chain E (HLA-E), and (e) a modification of an endogenous sequence encoding Beta-2-Microglobulin (B2M), wherein the modification reduces or eliminates a level of expression or activity of a major histocompatibility complex (MHC) class I (MHC-I).

The disclosure provides a modified cell comprising: (a) a nanotransposon of the disclosure; (b) a sequence encoding an inducible proapoptotic polypeptide; (c) a non-naturally occurring sequence comprising an HLA class I histocompatibility antigen, alpha chain E (HLA-E), and (e) a modification of an endogenous sequence encoding Beta-2-Microglobulin (B2M), wherein the modification reduces or eliminates a level of expression or activity of a major histocompatibility complex (MHC) class I (MHC-I).

In some embodiments of the modified cells of the disclosure, the non-naturally occurring sequence comprising a HLA-E further comprises a sequence encoding a B2M signal peptide. In some embodiments, the non-naturally occurring sequence comprising an HLA-E further comprises a linker, wherein the linker is positioned between the sequence encoding the sequence encoding a B2M polypeptide and the sequence encoding the HLA-E. In some

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embodiments, the non-naturally occurring sequence comprising an HLA-E further comprises a sequence encoding a peptide and a sequence encoding a B2M polypeptide. In some embodiments, the non-naturally occurring sequence comprising an HLA-E further comprises a first linker positioned between the sequence encoding the B2M signal peptide and the sequence encoding the peptide, and a second linker positioned between the sequence encoding the B2M polypeptide and the sequence encoding the HLA-E.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a mammalian cell.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a human cell.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a stem cell.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a differentiated cell.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a somatic cell.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is an immune cell or an immune cell precursor. In some embodiments, the immune cell is a lymphoid progenitor cell, a natural killer (NK) cell, a cytokine induced killer (CIK) cell, a T lymphocyte (T cell), a B lymphocyte (B-cell) or an antigen presenting cell (APC). In some embodiments, the immune cell is a T cell, an early memory T cell, a stem cell-like T cell, a stem memory T cell (Tscm), or a central memory T cell (Tcm). In some embodiments, the immune cell precursor is a hematopoietic stem cell (HSC). In some embodiments, the cell is an antigen presenting cell (APC).

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell further comprises a gene editing composition. In some embodiments, the gene editing composition comprises a sequence encoding a DNA binding domain and a sequence encoding a nuclease protein or a nuclease domain thereof. In some embodiments, the gene editing composition comprises a sequence encoding a nuclease protein or a sequence encoding a nuclease domain thereof. In some embodiments, the sequence encoding a nuclease protein or the sequence encoding a nuclease domain thereof comprises a DNA sequence, an RNA sequence, or a combination thereof. In some embodiments, the nuclease or the nuclease domain thereof comprises one or more of a CRISPR/Cas protein, a Transcription Activator-Like Effector Nuclease (TALEN), a Zinc Finger Nuclease (ZFN), and an endonuclease. In some embodiments, the CRISPR/Cas protein comprises a nuclease-inactivated Cas (dCas) protein.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell further comprises a gene editing composition. In some embodiments, the gene editing composition comprises a sequence encoding a DNA binding domain and a sequence encoding a nuclease protein or a nuclease domain thereof. In some embodiments, the nuclease or the nuclease domain thereof comprises a nuclease-inactivated Cas (dCas) protein and an endonuclease. In some embodiments, the endonuclease comprises a Clo051 nuclease or a nuclease domain thereof. In some embodiments, the gene editing composition comprises a fusion protein. In some embodiments, the fusion protein comprises a nuclease-inactivated Cas9 (dCas9) protein and a Clo051 nuclease or a Clo051 nuclease domain. In some embodiments, the gene editing composition further comprises a guide sequence. In some embodiments, the guide sequence

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comprises an RNA sequence. In some embodiments, the fusion protein comprises or consists of the amino acid sequence:

(SEQ ID NO: 17013)

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5  MAPKKKKRKEGKISNISLLKDELRGQISHISHEYLSLIDLAFDSKQNRLF
  EMKVLELLVNEYGFKGRHLGGSRKPDGIVYSTTLEDNFGIIVDTKAYSEG
10 YSLPISQADEMERYVRENSNRDEEVNPNKWWENFSEEVKKYFVFIGSGF
  KGKFEEQLRRLSMTTGNGSAVNVNLLLGAEKIRSGEMTIEELERAMFN
  NSEFILKYGGGSDKKYSIGLAIGTNSVGWAVITDEYKVPSSKEKVLGNT
15 DRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRNRI CYLQEIFSN
  EMAKVDDSFHRLSEESPLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHL
  RKKLVDSTDKADRLIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFIQQL
20 VQTYNQLFEEPNINASGVDAKAILSARLSKSRRLLENLIAQLPGEKKNGLF
  GNLIALLSLGLTPNEKSNEDLAEDAKLQLSKDITYDDLDNLLAQIGDQYAD
  LFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLLKAL
25 VRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEEFYKFIKPILEKMDGTEE
  LLVKNLREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNRE
  KIEKILTFRIPIYYVGPLARGNSRFAWMTRKSEETITPWNFEVVDKGASA
30 QSFIERMTNEDKNLPNEKVLPHKSLLEYFTVYNELTKVKYVTEGMRKPA
  ELSGEQKKAIVDLLEKTNRKVIVKQLKEDYFKKIECEDSVESIGVEDRPN
  ASLGTYHDLKKI IKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKT
35 YAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTILDFLKSDFG
  ANRNFMLIHDDSLTFKEDIQKAQVSGQDLSLHEHIANLAGSPAIIKKGIL
  QTVKVVDELVKVMGRHKPENIVIEMARENQTTQKQKNSRERMKRIEEGI
40 KELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDV
  DAIVPQSFLKDDSIDNKVLTTRSDKNRGSNDVPSEEVVKMKNYWRQLLN
  AKLITQRKEDNLIKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDNR
45 MNTKYDENDKLIREVKVITLKSCLVSDERKDFQFYKVREINNYHHAHDAY
  LNAVVGITALIKKYPKLESEFVYGDYKVDVRKMIKSEQEI GKATAYFF
  YSNIMNFPKTEITLANGEIRKPLIETNGETGEI VWDKGRDFATVRKVL S
50 MPQVNI VKKTEVQTGGFSKESILPKRNSDKLIARKKWDPKKYGGFDSPT
  VAYSVLVAKVEKGSKKLKSVKELLGITIMERSSEFKNPIDFLEAKGYK
  EVKKDLIIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFYLY
55 ASHYEKLKSGPEDNEQKQLFVEQHKHYLDEII EQISEFSKRVILADANLD
  KVL SAYNKHRDKPIREQAENI IHLFTLTNLGAPAAFKYFDTTIDRKRYTS
  TKEVL DATLIHQSI TGLYETRIDLSQLGGDGSPPKKRKRVS
60

```

or a nucleic acid comprising or consisting of the sequence:

(SEQ ID NO: 17014)

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1  atggcaccacaa agaagaaaag aaaagtggag ggcacaaagt
65  caaacatcag cctgctgaaa

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61 gacgaactgc ggggacagat tagtcacatc agtcacgagt
acctgtcact gattgatctg

121 gccttcgaca gcaagcagaa tagactgttt gagatgaaag
tgctggaact gctggccaac

181 gagtatggct tcaagggcag acatctgggc gggctctagga
aacctgacgg catcgtgtac

241 agtaccacac tggaagacaa ctctcggaatc attgtcgata
ccaaggctta ttccgagggc

301 tactctctgc caattagtca ggcagatgag atggaaaggt
acgtgcgcga aaactcaaat

361 agggacgagg aagtcaaccc caataagtgg tgggagaatt
tcagcgagga agtgaagaaa

421 tactactctg tctttatctc aggcagcttc aaagggaggt
ttgaggaaca gctgcggaga

481 ctgtccatga ctaccgggt gaacggatct gctgtcaacg
tggtcaatct gctgctgggc

541 gcagaaaaga tcaggtcctg ggagatgaca attgaggaac
tggaacgcgc catgttcaac

601 aattctgagt ttatcctgaa gtatggaggc gggggaagcg
ataagaaata ctccatcgga

661 ctggccattg gcaccaattc cgtgggctgg gctgtcatca
cagacgagta caagggtcca

721 agcaagaagt tcaaggtcct ggggaacacc gatcgccaca
gtatcaagaa aaatctgatt

781 ggagccctgc tgttcgactc aggcgagact gctgaagcaa
cccactgaa gcggactgct

841 aggcgcctg ataccggag aaaaaatcgg atctgctacc
tgcaggaaat ttccagcaac

901 gagatggcca aggtggacga tagtttcttt caccgcctgg
aggaatcatt cctgggtggag

961 gaagataaga aacacgagcg gcatcccatc tttggcaaca
ttgtggacga agtcgcttat

1021 cagcagaagt accctactat ctatcatctg aggaagaaac
tggtggactc caccgataag

1081 gcagacctgc gctgatcta tctggccctg gctcacatga
tcaagttccg ggggcatttt

1141 ctgatcgagg gagatctgaa ccctgacaat tctgatgtgg
acaagctgtt catccagctg

1201 gtccagacat acaatcagct gtttgaggaa aacccaatta
atgcctcagg cgtggacgca

174

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1261 aaggccatcc tgagcgccag actgtccaaa tctaggcgcc
tggaaaacct gatcgctcag

5 1321 ctgccaggag agaagaaaaa cggcctgttt gggaatctga
ttgcaactgc cctgggcctg

1381 acaccaact tcaagtctaa tttgatctg gccgaggacg
ctaagctgca gctgtccaaa

10 1441 gacacttatg acgatgacct ggataacctg ctggctcaga
tcggcgatca gtacgcagac

15 1501 ctgttctctg ccgctaagaa tctgagtac gccatcctgc
tgtcagatat tctgcgcgtg

1561 aacacagaga ttactaagc cccactgagt gcttcaatga
tcaaaagata tgacgagcac

20 1621 catcaggatc tgacctgct gaaggctctg gtgaggcagc
agctgccga gaaatacaag

1681 gaaatcttct ttgatcagag caagaatgga tacgccggct
atattgacgg cggggcttcc

25 1741 caggaggagt tctacaagtt catcaagccc attctggaaa
agatggacgg caccgaggaa

30 1801 ctgctggtga agctgaatcg ggaggacctg ctgagaaaac
agaggacatt tgataacgga

1861 agcatccctc accagattca tctggcgaa ctgcacgcca
tctgcgacg gcaggaggac

35 1921 ttctacccat ttctgaagga taaccgcgag aaaatcgaaa
agatcctgac cttcagaatc

40 1981 ccctactatg tggggcctct ggcacgggga aatagtagat
ttgcctggat gacaagaaag

2041 tcagaggaaa ctatcaccoc ctggaacttc gaggaagtgg
tcgataaagg cgctagcgca

2101 cagtccttca ttgaaaggat gacaaatttt gacaagaacc
tgccaaatga gaaggtgctg

50 2161 cccaaacaca gcctgctgta cgaatatttc acagtgtata
acgagctgac taaagtgaag

2221 tacgtcaccg aagggatgag caagcccga ttctgtccg
gagagcagaa gaaagccatc

55 2281 gtggacctgc tgtttaagac aaatcgaaa gtgactgtca
aacagctgaa ggaagactat

60 2341 ttcaagaaaa ttgagtgttt cgattcagtg gaaatcagcg
gcgtcgagga caggtttaac

2401 gcctccctgg ggacctacca cgatctgctg aagatcatca
aggataagga cttcctggac

65

175

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2461 aacgaggaaa atgaggacat cctggaggac attgtgctga
 cactgactct gtttgaggat
 2521 cgcgaaatga tcgaggaacg actgaagact tatgcccatc
 tgttcgatga caaagtgatg
 2581 aagcagctga aaagaaggcg ctacaccgga tggggacgcc
 tgagccgaaa actgatcaat
 2641 gggattagag acaagcagag cggaaaaact atcctggact
 ttctgaagtc cgatggcttc
 2701 gccaacagga acttcatgca gctgattcac gatgactctc
 tgaccttcaa ggaggacatc
 2761 cagaaagcac aggtgtctgg ccagggggac agtctgcacg
 agcatatcgc aaacctggcc
 2821 ggcagccccc ccatcaagaa agggattctg cagaccgtga
 aggtggtgga cgaactggtc
 2881 aaggtcatgg gacgacacaa acctgagaac atcgtgattg
 agatggcccc cgaaaatcag
 2941 acaactcaga agggccagaa aaacagtcga gaacggatga
 agagaatcga ggaaggcatc
 3001 aaggagctgg ggtcacagat cctgaaggag catcctgtgg
 aaaacactca gctgcagaat
 3061 gagaactgt atctgtacta tctgcagaat ggacgggata
 tgtacgtgga ccaggagctg
 3121 gatattaaca gactgagtga ttatgacgtg gatgccatcg
 tccctcagag ctctcgaag
 3181 gatgactcca ttgacaacaa ggtgctgacc aggtccgaca
 agaaccgccc caaatcagat
 3241 aatgtgccaa gcgaggaagt ggtcaagaaa atgaagaact
 actggaggca gctgctgaat
 3301 gccaaactga tcacacagcg gaaatttgat aacctgacta
 aggcagaaaagg agggagcctg
 3361 tctgagctgg acaaggccgg ctatcatcaag cggcagctgg
 tggagacaag acagatcact
 3421 aagcacgtcg ctacagattct ggatagcaga atgaacacaa
 agtacgatga aaacgacaag
 3481 ctgatcaggg aggtgaaagt cattactctg aaatccaagc
 tgggtgtctga ctttagaaaag
 3541 gatttccagt ttataaaagt caggagatc aacaactacc
 accatgctca tgacgcatac
 3601 ctgaacgcag tggtcgggac cggcctgatt aagaaatacc
 ccaagctgga gtccgagttc

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3661 gtgtacggag actataaagt gtacgatgac cggaagatga
 tcgccaatc tgagcaggaa
 5 3721 attggcaagg ccaccgctaa gtatttcttt tacagtaaca
 tcatgaattt ctttaagacc
 3781 gaaatcacac tggcaaatgg ggagatcaga aaaaggcctc
 10 tgattgagac caacggggag
 3841 acaggagaaa tcgtgtggga caagggaagg gattttgcta
 ccgtgcgcaa agtctgtctc
 15 3901 atgccccaaag tgaatattgt caagaaaact gaagtgcaga
 ccgggggatt ctctaaggag
 3961 agtattctgc ctaagcgaag ctctgataaa ctgatcgccc
 20 ggaagaaaga ctgggacccc
 4021 aagaagtatg gcgggttcga ctctccaaca gtggcttaca
 gtgtcctggt ggtcgcaaaag
 25 4081 gtggaaaagg ggaagtccaa gaaactgaag tctgtcaaag
 agctgctggg aatcactatt
 30 4141 atggaacgca gctccttcga gaagaatcct atcgattttc
 tggaaagcaa gggctataaa
 4201 gaggtgaaga aagacctgat cattaagctg ccaaaatact
 35 cactgtttga gctggaaaaac
 4261 ggacgaaagc gaatgctggc aagcgcgga gaactgcaga
 agggcaatga gctggccctg
 40 4321 ccctccaaat acgtgaactt cctgtatctg gctagccact
 acgagaaaact gaaggggtcc
 4381 cctgaggata acgaacagaa gcagctgttt gtggagcagc
 45 acaaacatta tctggacgag
 4441 atcattgaac agatttcaga gttcagcaag agagtgatcc
 tggctgacgc aaatctggat
 50 4501 aaagtcttga gcgcatacaa caagcaccga gacaaaccaa
 tccgggagca ggcgaaaaat
 4561 atcattcatc tgttcacctt gacaaacctg ggcgcccctg
 55 cagccttcaa gtattttgac
 4621 accacaatcg atcggaagag atacacttct accaaagagg
 60 tgctggatgc taccctgatc
 4681 caccagagta ttaccggcct gtatgagaca cgcacgacc
 tgtcacagct gggaggcgat
 65 4741 gggagcccca agaaaaagcg gaaggtgtct agttaa.

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In some embodiments, the fusion protein comprises or consists of the amino acid sequence:

((SEQ ID NO: 17058) 5

1 MPKKKRKVEG IKSNISSLKD ELRGQISHIS HEYLSLIDLA

FDSKQNRLF MKVLELLVNE

61 YGFKGRHLGG SRKPDGIVYS TTLEDNFGII VDTKAYSEGY

SLPISQADEM ERYVRENSNR

10

121 DEEVNPNKWW ENFSEEVKKY YFVFISGSFK GKFEQLRRL

SMTTGVNGSA VNVVNLLGA

181 EKIRSGEMTI EELERAMFNN SEFILKYGGG GSDKKYSIGL

AIGTNSVGWA VITDEYKVPS

241 KKFKVLGNTD RHSIKKNLIG ALLFDSGETA EATRLKRTAR

RRYTRKNRI CYLQEIFSNE

20

301 MAKVDDSPFH RLEESFLVEE DKKHERHPHF GNIVDEVAYH

EKYPTIYHLR KKLVDSTDKA

361 DLRLIYLALA HMIKFRGHFL IEGDLNPDNS DVDKLFQILV

QTYNQLFEEN PINASGVDAK

421 AILSARLSKS RRLLENLIAQL PGEKKNGLFG NLIALSLGLT

PNFKNFDLA EDAKLQLSKD

30

481 TYDDDLNLL AQIGDQYADL FLAAKNLSDA ILLSDILRVN

TEITKAPLSA SMIKRYDEHH

541 QDLTLKALV RQQLPEKYKE IFFDQSKNGY AGYIDGGASQ

EEFYKFIKPI LEKMDGTEEL

601 LVKLNREDLL RKQRTFDNGS IPHQIHLGEL HAILRRQEDF

YPFLKDNREK IEKILTFRIP

661 YYVGPLARGN SRFAWMTRKS EETITPWNFE EVVDKGASAQ

SFIERMTNFD KNLPNEKVL P

721 KHSLLYEFYFT VYNELTKVKY VTEGMRKPAF LSGEQKKAIV

DLLFKTNRKV TVKQLKEDYF

45

781 KKIECFDSVE ISGVEDRFNA SLGTYHDLK IIKDKDFLDN

EENEDIEDI VLTITLTFEDR

841 EMIEERLKTY AHLFDDKVMK QLKRRRYTGW GRLSRKLING

IRDKQSGKTI LDFLKSDGFA

50

901 NRNFMQLIHD DSLTFKEDIQ KAQVSGQGDS LHEHIANLAG

SPAIKKGILQ TVKVVDLVK

55

961 VMGRHKPENI VIEMARENQT TQKGQKNSRE RMKRIEELIK

ELGSQILKEH PVENTQLQNE

1021 KLYLYYLQNG RDMYVDQELD INRLSDYDVD AIVPQSFLKD

DSIDNKVLTR SdkNRGKSDN

1081 VPSEEVVKM KNYWRQLLNA KLITQRKFDN LTKAERGGLS

ELDKAGFIKR QLVETRQITK

60

65

178

-continued

1141 HVAQILDSRM NTKYDENDKL IREVKVITLK SKLVSDFRKD

FQFYKVIN NYHHAHDAYL

1201 NAVVGTALIK KYPKLESEFV YGDYKVYDVR KMIKSEQEI

GKATAKYFFY SNIMNFFKTE

1261 ITLANGEIRK RPLIETNGET GEIVWDKGRD FATVRKVLMS

PQVNIVKTE VQTGGFSKES

1321 ILPKRNSDKL IARKKDWDPK KYGGFDSPTV AYSVLVAVK

EKGKSKKLS VKELLGITIM

15

1381 ERSSFEKNPI DFLEAKGYKE VKKDIIKLP KYSLFELENG

RKRMLASAGE LQKGNELALP

20

1441 SKYVNFLYLA SHYEKLKGP EDNEQKQLFV EQKHLYLDEI

IEQISEFSKR VILADANLDC

1501 VLSAYNKHLD KPIREQAENI IHLFTLINLG APAAFKYFDT

TIDRKRYTST KEVLDTLIH

25

1561 QSITGLYETR IDLSQLGGDG SPKKKRKV

or a nucleic acid comprising or consisting of the sequence:

(SEQ ID NO: 17059)

1 atgcctaaga agaagcggaa ggtggaaggc atcaaaagca

acatctccct cctgaaagac

35

61 gaactccggg gccagattag ccacattagt caccgaatacc

tctccctcat cgacctggct

121 ttcgatagca agcagaacag gctcttttag atgaaagtgc

tggaactgct cgtcaatgag

40

181 tacgggttca agggcgaca cctcgccgga tctaggaaac

cagacggcat cgtgtatagt

45

241 accacactgg aagacaactt tgggatcatt gtggatacca

aggcactatc tgagggttat

301 agtctgcccc tttcacaggc cgacgagatg gaacggtacg

tgccgcgagaa ctcaaataga

361 gatgaggaag tcaaccctaa caagtgggtg gagaaactct

ctgaggaagt gaagaaatac

55

421 tacttcgtct ttatcagcgg gtccttcaag ggtaaatctg

aggaacagct caggagactg

481 agcatgacta ccggcgtgaa tggcagcgcc gtcaactgtg

tcaatctgct cctgggcgct

541 gaaaagattc ggagcgaga gatgaccatc gaagagctgg

agagggcaat gtttaataat

601 agcgagtcta tcctgaaata cggtgccggt ggatccgata

aaaagtattc tattggttta

65

179

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661 gccatcggca ctaattccgt tggatgggct gtcataaccg
atgaatacaa agtaccttca
721 aagaaattta aggtgttggg gaacacagac cgtcattcga
ttaaaaagaa tcttatcggg
781 gccctcctat tcgatagtgg cgaaacggca gagcgactc
gcctgaaacg aaccgctcgg
841 agaaggtata cacgtcgcaa gaaccgaata tggtacttac
aagaaatttt tagcaatgag
901 atggccaaag ttgacgattc tttctttcac cgtttggaag
agtccttcct tgcgaagag
961 gacaagaaac atgaacggca cccatcttt ggaaacatag
tagatgaggt ggcatatcat
1021 gaaaagtacc caacgattta tcacctcaga aaaaagctag
ttgactcaac tgataaagcg
1081 gacctgaggt taatctactt ggctcttgcc catatgataa
agttccgtgg gcactttctc
1141 attgaggggt atctaaatcc ggacaactcg gatgtcgaca
aactgttcat ccagttagta
1201 caaacctata atcagttggt tgaagagaac cctataaatg
caagtggcgt ggatgcgaag
1261 gctattctta gcgcccgct ctctaaatcc cgacggctag
aaaacctgat cgcacaatta
1321 cccggagaga agaaaaatgg gttgttcggg aaccttatag
cgctctcact aggcctgaca
1381 ccaaatttta agtogaactt cgacttagct gaagatgcca
aattgcagct tagtaaggac
1441 acgtacgatg acgatctcga caatctactg gcacaaattg
gagatcagta tgcggactta
1501 tttttggctg ccaaaaaact tagcgatgca atcctcctat
ctgacatact gagagttaat
1561 actgagatta ccaagcgcc gttatccgct tcaatgatca
aaaggtagca tgaacatcac
1621 caagacttga cacttctcaa ggccttagtc cgtcagcaac
tgcttgagaa atataaggaa
1681 atattctttg atcagtcgaa aaacgggtac gcagggtata
ttgacggcgg agcgagtcaa
1741 gaggaattct acaagtttat caaacccata ttagagaaga
tggtggggac ggaagagttg
1801 cttgtaaaac tcaatcgca agatctactg cgaaagcagc
ggactttcga caacggtagc

180

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1861 attccacatc aaatccactt aggcgaattg catgctatac
ttagaaggca ggaggatttt
5 1921 tatccgttcc tcaaagacaa tcgtgaaaag attgagaaaa
tcctaacctt tcgcatacct
1981 tactatgtgg gacccttggc ccgagggaac tctcggttcg
catggatgac aagaaagtcc
10 2041 gaagaaacga ttactccatg gaattttgag gaagttgtcg
ataaagggtgc gtcagctcaa
15 2101 tcgttcacgc agaggatgac caactttgac aagaatttac
cgaacgaaaa agtattgcct
2161 aagcacagtt tactttacga gtatttcaca gtgtacaatg
aactcacgaa agttaagtat
20 2221 gtcactgagg gcatgcgtaa acccgcttt ctaagcggag
aacagaagaa agcaatagta
2281 gatctgttat tcaagaccaa ccgcaaatg acagttaagc
aattgaaaga ggactacttt
2341 aagaaaattg aatgcttcga ttctgtcgag atctccgggg
tagaagatcg atttaatgcg
30 2401 tcacttggtg cgtatcatga cctcctaaag ataattaaag
ataaggactt cctggataac
2461 gaagagaatg aagatatctt agaagatata gtgttgactc
ttacctctt tgaagatcgg
35 2521 gaaatgattg aggaaagact aaaaacatac gtcacctgt
tcgacgataa ggttatgaaa
40 2581 cagttaaaga ggcgtcgcta tacgggctgg ggacgattgt
cgcggaact tatcaacggg
2641 ataagagaca agcaagtgg taaaactatt ctcgattttc
taaagagcga cggtctcgc
45 2701 aataggaact ttatgcagct gatccatgat gactctttaa
cctcaaaga ggatatacaa
50 2761 aaggcacagg tttccggaca aggggactca ttgcacgaac
atattgcgaa tcttgcgtgt
2821 tcgccagcca tcaaaaaggg catactccag acagtcaaag
tagtgatga gctagttaag
55 2881 gtcatgggac gtcacaaacc ggaaaacatt gtaatcgaga
tggcacgcga aaatcaaacg
60 2941 actcagaagg ggcaaaaaa cagtcgagag cggatgaaga
gaatagaaga gggattataa
3001 gaactgggca gccagatctt aaaggagcat cctgtggaaa
65 ataccaatt gcagaacgag

181

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3061 aaactttacc tctattacct acaaatgga agggacatgt
 atgttgatca ggaactggac
 3121 ataaaccgtt tatctgatta cgacgtcgat gccattgtac
 cccaatcctt ttgaaaggac
 3181 gattcaatcg acaataaagt gcttacacgc tcggataaga
 accgagggaa aagtgacaat
 3241 gttccaagcg aggaagtcgt aaagaaaatg aagaactatt
 ggcggcagct cctaaatgcg
 3301 aaactgataa cgcaaagaaa gtctgataac ttaactaaag
 ctgagagggg tggcttgtct
 3361 gaacttgaca aggccggatt tattaacgt cagctcgtgg
 aaacccgcc aatcacaaag
 3421 catgttgac agatactaga ttcccgaatg aatacgaat
 acgacgagaa cgataagctg
 3481 attcgggaag tcaaagtaat cactttaag tcaaatgtg
 tgtcggactt cagaaaggat
 3541 tttcaattct ataaagttag ggagataaat aactaccacc
 atgcgcacga cgcttatctt
 3601 aatgccgtcg tagggaccgc actcattaag aaataccga
 agctagaaa tgagtttgtg
 3661 tatgggtgatt acaaagttta tgacgtccgt aagatgatcg
 cgaaaagcga acaggagata
 3721 ggcaaggcta cagccaaata cttcttttat tctaacatta
 tgaatttctt taagacggaa
 3781 atcactctgg caaacggaga gatacgcaaa cgacctttaa
 ttgaaaccaa tggggagaca
 3841 ggtgaaatcg tatgggataa gggccgggac ttcgcgacgg
 tgagaaaagt tttgtccatg
 3901 cccaagtca acatagtaaa gaaaactgag gtgcagaccg
 gagggtttcc aaaggaatcg
 3961 attcttccaa aaaggaatag tgataagctc atcgctcgta
 aaaaggactg ggacccgaaa
 4021 aagtacgggt gcttcgatag ccctacagtt gcctattctg
 tcctagtagt ggcaaaagtt
 4081 gagaagggaa aatccaagaa actgaagtca gtcaagaat
 tattggggat aacgattatg
 4141 gagcgtcgt cttttgaaaa gaacccatc gacttccttg
 aggcgaaaagg ttacaaggaa
 4201 gtaaaaaagg atctcataat taaactacca aagtatagtc
 tgtttgagtt agaaaatggc

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4261 cgaaaacgga tgttggttag cgccggagag cttcaaaagg
 ggaacgaact cgcactaccg
 5 4321 tctaaatacg tgaatttcct gtatttagcg tccattacg
 agaagttgaa aggttcacct
 4381 gaagataacg aacagaagca actttttgtt gacgagcaca
 10 aacattatct cgacgaaatc
 4441 atagagcaaa ttccggaatt cagtaagaga gtcattctag
 ctgatgccaa tctggacaaa
 15 4501 gtattaagcg catacaacaa gcacagggat aaaccatac
 gtgagcaggc ggaaaatatt
 4561 atccatttgt ttactcttac caacctcgcc gctccagccg
 20 cattcaagta ttttgacaca
 4621 acgatagatc gcaaacgata cacttctacc aaggaggtgc
 tagacgcgac actgattcac
 4681 caatccatca cgggattata tgaaactcgg atagatttgt
 25 cacagcttgg gggtagcggg
 4741 tcccccaaga agaagaggaa agtctga.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, a nanotransposon comprises the gene editing composition comprising a guide sequence and a sequence encoding a fusion protein comprising a sequence encoding an inactivated Cas9 (dCas9) and a sequence encoding a Clo051 nuclease or a nuclease domain thereof.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell expresses the gene editing composition transiently.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the cell is a T cell and the guide RNA comprises a sequence complementary to a target sequence encoding an endogenous TCR. In some embodiments, the guide RNA comprises a sequence complementary to a target sequence encoding a B2M polypeptide.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the guide RNA comprises a sequence complementary to a target sequence within a safe harbor site of a genomic DNA sequence.

In some embodiments of the cells, unmodified cells and modified cells of the disclosure, the Clo051 nuclease or a nuclease domain thereof induces a single or double strand break in a target sequence. In some embodiments, a donor sequence, a donor plasmid, or a donor nanotransposon intra-ITR sequence integrated at a position of single or double strand break and/or at a position of cellular repair within a target sequence.

The disclosure provides a composition comprising a modified cell according to the disclosure. In some embodiments, the composition further comprises a pharmaceutically-acceptable carrier.

The disclosure provides a composition comprising a plurality of modified cells according to the disclosure. In some embodiments, the composition further comprises a pharmaceutically-acceptable carrier.

The disclosure provides a composition of the disclosure for use in the treatment of a disease or disorder.

The disclosure provides the use of a composition of the disclosure for the treatment of a disease or disorder.

The disclosure provides a method of treating a disease or disorder comprising administering to a subject in need thereof a therapeutically-effective amount of a composition of the disclosure. In some embodiments, the subject does not develop graft vs. host (GvH) and/or host vs. graft (HvG) following administration of the composition. In some embodiments, the administration is systemic. In some embodiments, the composition is administered by an intravenous route. In some embodiments, the composition is administered by an intravenous injection or an intravenous infusion.

The disclosure provides a method of treating a disease or disorder comprising administering to a subject in need thereof a therapeutically-effective amount of a composition of the disclosure. In some embodiments, the subject does not develop graft vs. host (GvH) and/or host vs. graft (HvG) following administration of the composition. In some embodiments, the administration is local. In some embodiments, the composition is administered by an intra-tumoral route, an intraspinal route, an intracerebroventricular route, an intraocular route or an intraosseous route. In some embodiments, the composition is administered by an intra-tumoral injection or infusion, an intraspinal injection or infusion, an intracerebroventricular injection or infusion, an intraocular injection or infusion or an intraosseous injection or infusion.

In some embodiments of the methods of treating a disease or disorder of the disclosure, the therapeutically effective dose is a single dose and wherein the allogeneic cells of the composition engraft and/or persist for a sufficient time to treat the disease or disorder. In some embodiments, the single dose is one of at least 2, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100 or any number of doses in between that are manufactured simultaneously.

In some embodiments of the methods of treating a disease or disorder of the disclosure, the therapeutically effective dose is a single dose and wherein the autologous cells of the composition engraft and/or persist for a sufficient time to treat the disease or disorder. In some embodiments, the single dose is one of at least 2, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100 or any number of doses in between that are manufactured simultaneously.

In some embodiments of the composition and methods of the disclosure, allogeneic cells are stem cells. In some embodiments, allogeneic cells are derived from stem cells. Exemplary stem cells include, but are not limited to, embryonic stem cells, adult stem cells, induced pluripotent stem cells (iPSCs), multipotent stem cells, pluripotent stem cells, and hematopoietic stem cells (HSCs).

In some embodiments of the composition and methods of the disclosure, allogeneic cells are differentiated somatic cells.

In some embodiments of the composition and methods of the disclosure, allogeneic cells are immune cells. In some embodiments, allogeneic cells are T lymphocytes (T cells). In some embodiments, allogeneic cells are T cells that do not express one or more components of a naturally-occurring T-cell Receptor (TCR). In some embodiments, allogeneic cells are T cells that express a non-naturally occurring antigen receptor. Alternatively, or in addition, in some embodiments, allogeneic cells are T cells that express a non-naturally occurring Chimeric Stimulatory Receptor (CSR). In some embodiments, the non-naturally occurring CSR comprises or consists of a switch receptor. In some embodiments, the switch receptor comprises an extracellular

domain, a transmembrane domain, and an intracellular domain. In some embodiments, the extracellular domain of the switch receptor binds to a TCR co-stimulatory molecule and transduces a signal to the intracellular space of the allogeneic cell that recapitulates TCR signaling or TCR co-stimulatory signaling.

Chimeric Stimulatory Receptors (CSRs)

Adoptive cell compositions that are “universally” safe for administration to any patient requires a significant reduction or elimination of alloreactivity.

Towards this end, allogeneic cells of the disclosure are modified to interrupt expression or function of a T-cell Receptor (TCR) and/or a class of Major Histocompatibility Complex (MHC). The TCR mediates graft vs host (GvH) reactions whereas the MHC mediates host vs graft (HvG) reactions. In preferred embodiments, any expression and/or function of the TCR is eliminated in allogeneic cells of the disclosure to prevent T-cell mediated GvH that could cause death to the subject. Thus, in particularly preferred embodiments, the disclosure provides a pure TCR-negative allogeneic T-cell composition (e.g. each cell of the composition expresses at a level so low as to either be undetectable or non-existent).

In preferred embodiments, expression and/or function of MHC class I (MHC-I, specifically, HLA-A, HLA-B, and HLA-C) is reduced or eliminated in allogeneic cells of the disclosure to prevent HvG and, consequently, to improve engraftment of allogeneic cells of the disclosure in a subject. Improved engraftment of the allogeneic cells of the disclosure results in longer persistence of the cells, and, therefore, a larger therapeutic window for the subject. Specifically, in the allogeneic cells of the disclosure, expression and/or function of a structural element of MHC-I, Beta-2-Microglobulin (B2M), is reduced or eliminated in allogeneic cells of the disclosure.

The above strategies for generating an allogeneic cell of the disclosure induce further challenges. T Cell Receptor (TCR) knockout (KO) in T cells results in loss of expression of CD3-zeta (CD3z or CD3ζ), which is part of the TCR complex. The loss of CD3ζ in TCR-KO T-cells dramatically reduces the ability of optimally activating and expanding these cells using standard stimulation/activation reagents, including, but not limited to, agonist anti-CD3 mAb. When the expression or function of any one component of the TCR complex is interrupted, all components of the complex are lost, including TCR-alpha (TCRα), TCR-beta (TCRβ), CD3-gamma (CD3γ), CD3-epsilon (CD3ε), CD3-delta (CD3δ), and CD3-zeta (CD3ζ). Both CD3ε and CD3ζ are required for T cell activation and expansion. Agonist anti-CD3 mAbs typically recognize CD3ε and possibly another protein within the complex which, in turn, signals to CD3ζ. CD3ζ provides the primary stimulus for T cell activation (along with a secondary co-stimulatory signal) for optimal activation and expansion. Under normal conditions, full T-cell activation depends on the engagement of the TCR in conjunction with a second signal mediated by one or more co-stimulatory receptors (e.g. CD28, CD2, 4-1BBL, etc. . . .) that boost the immune response. However, when the TCR is not present, T cell expansion is severely reduced when stimulated using standard activation/stimulation reagents, including agonist anti-CD3 mAb. In fact, T cell expansion is reduced to only 20-40% of the normal level of expansion when stimulated using standard activation/stimulation reagents, including agonist anti-CD3 mAb.

The disclosure provides a Chimeric Stimulatory Receptor (CSR) to deliver CD3z primary stimulation to allogeneic T cells in the absence of an endogenous TCR (and, conse-

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quently, an endogenous CD3 ζ) when stimulated using standard activation/stimulation reagents, including agonist anti-CD3 mAb.

In the absence of an endogenous TCR, Chimeric Stimulatory Receptors (CSRs) of the disclosure provide a CD3 ζ stimulus to enhance activation and expansion of allogeneic T cells. In other words, in the absence of an endogenous TCR, Chimeric Stimulatory Receptors (CSRs) of the disclosure rescue the allogeneic cell from an activation-based disadvantage when compared to non-allogeneic T-cells that express an endogenous TCR. In some embodiments, CSRs of the disclosure comprise an agonist mAb epitope extracellularly and a CD3 ζ stimulatory domain intracellularly and, functionally, convert an anti-CD28 or anti-CD2 binding event on the surface into a CD3 ζ signaling event in an allogeneic T cell modified to express the CSR. In some embodiments, a CSR comprises a wild type CD28 or CD2 protein and a CD3 ζ intracellular stimulation domain, to produce CD28 ζ CSR and CD2 ζ CSR, respectively. In preferred embodiments, CD28 ζ CSR and/or CD2 ζ CSR further express a non-naturally occurring antigen receptor and/or a therapeutic protein. In preferred embodiments, the non-naturally occurring antigen receptor comprises a Chimeric Antigen Receptor.

The data provided herein demonstrate that modified allogeneic T cells of the disclosure comprising/expressing a CSR of the disclosure improve or rescue, the expansion of allogeneic T cells that no longer express endogenous TCR when compared to those cells that do not comprise/express a CSR of the disclosure.

Endogenous TCR Knock-Out

Gene editing compositions of the disclosure, including but not limited to, RNA-guided fusion proteins comprising dCas9-Clo051, may be used to target and decrease or eliminate expression of an endogenous T-cell receptor of an allogeneic cell of the disclosure. In preferred embodiments, the gene editing compositions of the disclosure target and delete a gene, a portion of a gene, or a regulatory element of a gene (such as a promoter) encoding an endogenous T-cell receptor of an allogeneic cell of the disclosure.

Nonlimiting examples of primers (including a T7 promoter, genome target sequence, and gRNA scaffold) for the generation of guide RNA (gRNA) templates for targeting and deleting TCR-alpha (TCR-a) are provided in Table 10.

TABLE 10

Target sequences underlined		
Name	Sequence	SEQ ID NO:
TCRa-gRNA-WT1	TAATACGACTCACTATA <u>GCTGGTACACGGCAGGG</u> <u>TCAGTTT</u> TAGAGCTAGA AATAG	16821
TCRa-gRNA-WT2	TAATACGACTCACTATA <u>GAGAATCAAAATCGGTG</u> <u>AAT</u>	16822
TCRa-gRNA-WT4	TAATACGACTCACTATA <u>GTGCTAGACATGAGGTC</u> <u>TA</u>	16823
TCRa-gRNA-WT1-2G	TAATACGACTCACTATA <u>GGCTGGTACACGGCAGG</u> <u>GTCA</u>	16824

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TABLE 10-continued

Target sequences underlined		
Name	Sequence	SEQ ID NO:
TCRa-gRNA-WT2	TAATACGACTCACTATA <u>GAGAATCAAAATCGGTG</u> <u>AATGTTT</u> TAGAGCTAGA AATAG	16825
TCRa-gRNA-WT3	TAATACGACTCACTATA <u>GGATTTAGAGTCTCTCA</u> <u>GCGTTT</u> TAGAGCTAGAA ATAG	16826
TCRa-gRNA-WT4	TAATACGACTCACTATA <u>GTGCTAGACATGAGGTC</u> <u>TAGTTT</u> TAGAGCTAGAA ATAG	16827
TCRa-gRNA-WT5	TAATACGACTCACTATA <u>GACACCTTCTTCCCCAG</u> <u>CCCGTTT</u> TAGAGCTAGA AATAG	16828
TCRa-gRNA-NG1-L	TAATACGACTCACTATA <u>gtggaataatgctgttg</u> <u>ttgaGTTT</u> TAGAGCTAG AAATAG	16829
TCRa-gRNA-NG2-L	TAATACGACTCACTATA <u>gcatcacaggaactttc</u> <u>taaaGTTT</u> TAGAGCTAG AAATAG	16830
TCRa-gRNA-NG3-L	TAATACGACTCACTATA <u>gtaaaaccaagagggcca</u> <u>cagGTTT</u> TAGAGCTAGA AATAG	16831
TCRa-gRNA-NG4-L	TAATACGACTCACTATA <u>gacccggccactttcag</u> <u>gaggGTTT</u> TAGAGCTAG AAATAG	16832
TCRa-gRNA-NGS-L	TAATACGACTCACTATA <u>gattaaacccggccact</u> <u>ttcGTTT</u> TAGAGCTAGA AATAG	16833
TCRa-gRNA-NG1-R	TAATACGACTCACTATA <u>gagcccaggttaaggcca</u> <u>gcttGTTT</u> TAGAGCTAG AAATAG	16834
TCRa-gRNA-NG2-1-R	TAATACGACTCACTATA <u>gagctttgaaacaggta</u> <u>agacGTTT</u> TAGAGCTAG AAATAG	16835
TCRa-gRNA-NG2-2-R	TAATACGACTCACTATA <u>gctttgaaacaggtaag</u> <u>acaGTTT</u> TAGAGCTAGA AATAG	16836
TCRa-gRNA-NG3-R	TAATACGACTCACTATA <u>gtttcaaaacctgtcag</u> <u>tgatGTTT</u> TAGAGCTAG AAATAG	16837
TCRa-gRNA-NG4-R	TAATACGACTCACTATA <u>gctgaggctgtggtcca</u> <u>gctgGTTT</u> TAGAGCTAG AAATAG	16838
TCRa-gRNA-NG5-1-R	TAATACGACTCACTATA <u>gctgtggtccagctgag</u> <u>gtgGTTT</u> TAGAGCTAGA AATAG	16839

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TABLE 10-continued

Target sequences underlined		
Name	Sequence	SEQ ID NO:
TCRa-gRNA-NG5-2-R	TAATACGACTCACTATA <u>gctgtggtccagctgag</u> <u>gtgaGTTTTAGAGCTAG</u> AAATAG	16840
TCRa-gRNA-NG5-3-R	TAATACGACTCACTATA <u>gtgtggtccagctgagg</u> <u>tgagGTTTTAGAGCTAG</u> AAATAG	16841
TCRa-gRNA-NG5-3-Rb	TAATACGACTCACTATA <u>gtgtggtccagctgagg</u> <u>tgagGTTTTAGAGCTAG</u> AAATAG	16842

Nonlimiting examples of primers for the generation of guide RNA (gRNA) templates for targeting and deleting TCR-beta (TCR- β) are provided in Table 11.

TABLE 11

Target sequences underlined		
Name	Sequence	SEQ ID NO:
TCRb-gRNA-WT1	TAATACGACTCACTATAG <u>GCTGCTCCTTGAGGGCT</u> <u>GGTTTTAGAGCTAGAAAT</u> AG	16843
TCRb-gRNA-WT2	TAATACGACTCACTATAG <u>GCAGTATCTGGAGTCATT</u> <u>GGTTTTAGAGCTAGAAAT</u> AG	16844
TCRb-gRNA-WT3	TAATACGACTCACTATAG <u>GCCTCGGCGCTGACGATC</u> T	16845
TCRb-gRNA-WT5	TAATACGACTCACTATAG <u>GCTCTCGGAGAATGACGA</u> G	16846
TCRb-gRNA-WT3	TAATACGACTCACTATAG <u>GCCTCGGCGCTGACGATC</u> <u>TGTTTTAGAGCTAGAAAT</u> AG	16847
TCRb-gRNA-WT4	TAATACGACTCACTATAG <u>GAGAATGACGAGTGGACC</u> <u>CGTTTTAGAGCTAGAAAT</u> AG	16848
TCRb-gRNA-WT5	TAATACGACTCACTATAG <u>GCTCTCGGAGAATGACGA</u> <u>GGTTTTAGAGCTAGAAAT</u> AG	16849
TCRb-gRNA-NG1-L	TAATACGACTCACTATAG <u>CAACACAGCGACCTCGG</u> <u>GTGTTTTAGAGCTAGAAA</u> TAG	16850
TCRb-gRNA-NG2-L	TAATACGACTCACTATAG <u>TGGCTCAAACACAGCGAC</u> <u>CTGTTTTAGAGCTAGAAA</u> TAG	16851

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TABLE 11-continued

Target sequences underlined		
Name	Sequence	SEQ ID NO:
TCRb-gRNA-NG3-L	TAATACGACTCACTATAG <u>AGGGCGGGCTGCTCCTTG</u> <u>AGGTTTTAGAGCTAGAAA</u> TAG	16852
TCRb-gRNA-NG4-L	TAATACGACTCACTATAG <u>TATCTGGAGTCATTGAGG</u> <u>GGTTTTAGAGCTAGAAAT</u> AG	16853
TCRb-gRNA-NGS-L	TAATACGACTCACTATAG <u>ACTGGACTTGACAGCGGA</u> <u>AGGTTTTAGAGCTAGAAA</u> TAG	16854
TCRb-gRNA-NG1-R	TAATACGACTCACTATAG <u>AGAGATCTCCACACCCA</u> <u>AAGTTTTAGAGCTAGAAA</u> TAG	16855
TCRb-gRNA-NG2-R	TAATACGACTCACTATAG <u>CCACACCCAAAAGGCCAC</u> <u>ACGTTTTAGAGCTAGAAA</u> TAG	16856
TCRb-gRNA-NG3-R	TAATACGACTCACTATAG <u>ACTGCCTGAGCAGCCGCC</u> <u>TGGTTTTAGAGCTAGAAA</u> TAG	16857
TCRb-gRNA-NG4-R	TAATACGACTCACTATAG <u>TGAGGGTCTCGGCCACCT</u> <u>TCGTTTTAGAGCTAGAAA</u> TAG	16858
TCRb-gRNA-NGS-R	TAATACGACTCACTATAG <u>ATGACGAGTGGACCCAGG</u> <u>ATGTTTTAGAGCTAGAAA</u> TAG	16859
TCRb-gRNA-NG6-L	TAATACGACTCACTATAG <u>TGGCTCAAACACAGCGAC</u> <u>CTGTTTTAGAGCTAGAAA</u> TAG	16860
TCRb-gRNA-NG6-R	TAATACGACTCACTATAG <u>CCACACCCAAAAGGCCAC</u> <u>ACGTTTTAGAGCTAGAAA</u> TAG	16861

Nonlimiting examples of primers for the generation of guide RNA (gRNA) templates for targeting and deleting beta-2-microglobulin (β 2M) are provided in Table 12.

TABLE 12

Target sequences underlined			
Primer No.	Name	Sequence	SEQ ID NO:
1	B2-Prom-NG1-R	TAATACGACTCACTATAG <u>AGACAGGTGACGGTCCCTGC</u> <u>GTTTTAGAGCTAGAAATAG</u>	16862
2	B2-Prom-NG1-L	TAATACGACTCACTATA <u>GCAGTGCCAGGTTAGAGAGA</u> <u>GTTTTAGAGCTAGAAATAG</u>	16863
3	B2-Ex2-NG-R	TAATACGACTCACTATA <u>GAAGTTGACTTACTGAAGAA</u> <u>GTTTTAGAGCTAGAAATAG</u>	16864

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TABLE 12 -continued

Target sequences underlined			
Primer No.	Name	Sequence	SEQ ID NO:
4	B2-Ex2-NG-L	TAATACGACTCACTATA G ACCCAGACACATAGCAATTC GTTTATAGCTAGAAATAG	16865
5	Ex2-B2-NG2-R	TAATACGACTCACTATA G TCACGTCATCCAGCAGAGAA GTTTATAGCTAGAAATAG	16866
6	B2-Ex2-NG2-L	TAATACGACTCACTATA gatattcctcagGTACTCCA GTTTATAGCTAGAAATAG	16867
7	b2MEx1-NG-left	TAATACGACTCACTATA GGCCACGGAGCGAGACATCT GTTTATAGCTAGAAATAG	16868
8	b2MEx1-NG-right	TAATACGACTCACTATAG ACTCTCTCTTTCTGGCCTGG GTTTATAGCTAGAAATAG	16869
9	b2M-gRNA WT Ex2	TAATACGACTCACTATAG GAGAGAGAATTGAAAAAG GTTTATAGCTAGAAATAG	16870

Endogenous MHC Knock-Out

Gene editing compositions of the disclosure, including but not limited to, RNA-guided fusion proteins comprising dCas9-Clo51, may be used to target and decrease or eliminate expression of an endogenous MHCI, MHCII, or MHC activator of an allogeneic cell of the disclosure. In preferred embodiments, the gene editing compositions of the disclosure target and delete agene, apportion of agene, or a regulatory element of agene (such as a promoter) encoding one or more components of an endogenous MHCI, MHCII, or MHC activator of an allogeneic cell of the disclosure.

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Nonlimiting examples of guide RNAs (gRNAs) for targeting and deleting MHC activators are provided in Tables 13 and 14.

TABLE 13

Gene	Reagent/Left Target Type	Sequence	SEQ		SEQ ID NO
			ID	Right Target Sequence	
C2TA	C2TA exon 4	CATCGCTGTTA	16871	CTACCACTTCTA	16880
	NG	AGAAGCTCC		TGACCAGA	
	C2TA exon6	GGCCCTCCAGC	16872	CAGTAAGTTTGT	16881
	NG	TGGGAGTCC		GGTGGGTG	
RFXANK	RFXANK exon1	GGGTCTGCTGG	16873	GGACCTGAAGA	16882
	NG1	GTCTGGATG		CCCCGGAG	
	RFXANK exon1	GTTCTGAGGCA	16874	CCCGGAGAGGAG	16883
	NG2	GGGGTCTGC		GCTGCAGA	
RFXAP	RFXAP Exon 1	CCCGCCCCAAC	16875	CTGTGCGAAGGG	16884
	NG1	GCTGCCCCC		GCCGGGGA	
	RFXAP Exon 1	CCTTCGCACAG	16876	AGAGGAGGCTGG	16885
	NG2	GTACCTAAC		GGAGGACG	
RFX5	RFX5 exon 1	GTCTTGGGGCT	16877	CCCAGGTGGTGC	16886
	NG1	CTTAGCATC		TGAGGCTG	
	RFX5 exon 2	ACGGCCTTGCT	16878	GGGATCCTGGTA	16887
	NG2	GTGGGAAG		AGTGTGTT	
	RFX5 exon5	TCTGATGATCT	16879	ATCAAAGCTCGA	16888
	NG3	TGCCAAAGT		AGGCTTGG	

TABLE 14

Gene	Reagent/Type	Exon or region	NG-Left Target Sequence	SEQ	SEQ	SEQ ID NO
				ID NO:	NG-Right Target Sequence	
Beta2-MG	B2-Promoter-NG1	promoter	GCAGTGCCAGGTTAG	16889	AGACAGGTGACGGTC	16913
			AGAGA		CCTGC	
	B2-Promoter-NG2	promoter	CAAGCCAGCGACGCA	16890	CCTGCGGCGCTTGTC	16914
			GTGCC		CTGAT	
	B2-Promoter-NG3	promoter	CCAATCAGGACAAGG	16891	TATAAGTTGAGGCGT	16915
			CCCCG		CGCGC	
	B2-Ex2-NG	exon 2	ACCCAGACACATAGC	16892	GAAAGTTGACTTACTG	16916
			AATTC		AAGAA	
	B2-Ex2-NG2	exon 2	gatattcctcagGTA	16893	TCACGTCATCCAGCA	16917
			CTCCA		GAGAA	
Beta2-MG	B2-Ex1-NG	exon 1	GGCCACGGAGCGAGA	16894	ACTCTCTCTTTCTGG	16918
			CATCT		CCTGG	
	WT-B2MG-exon2					GGAGAGAGAATTGAAAAAG 16937
	WT-B2MG-promoter-4	cuts in promoter region Y				GGGCGCTTGTCCTGATTGGC 16938
	WT-B2MG-promoter-5	cuts in promoter region				GGCACTGCGTCGCTGGCT 16939
C2TA	C2TA exon 4	exon 4	CATCGCTGTTAAGAA	16895	CTACCACTTCTATGA	16919
	NG		GCTCC		CCAGA	
	C2TA exon4	exon 4	GGTCCATCTGGTCAT	16896	AGATTGAGCTCTACT	16920
	NG2		AGAAG		CAGGT	
	C2TA exon6	exon6	GGCCCTCCAGCTGGG	16897	CAGTAAGTTTGTGGT	16921
	NG		AGTCC		GGGTG	

TABLE 14 -continued

Gene	Reagent /Type	Exon or region	NG-Left Target Sequence	SEQ ID NO:	NG-Right Target Sequence	SEQ ID NO:	Target sequence (if WT crispr)	SEQ ID NO
	C2TA exon4- WT	exon 4					GGTCCATCTGGTCATAGAAG16940	
	C2TA exon6- WT	exon6					GGAGTCCTGGAAGACATAC 16941	
	C2TA exon6 NG2	exon6	CCTTGCTCAGGCCCT16898		TGTGGTGGGTGGGA16922			
			CCAGC		GGTCT			
RFXANK	RFXANK NG1	exon1	GGGTCTGCTGGGTCT16899		GGACCTGAAGACCC16923			
	RFXANK NG2	exon1	GGATG		CGGAG			
	RFXANK NG1	exon1	GTTCTGAGGCAGGG16900		CCCGAGAGGAGGCT16924			
	RFXANK NG1	exon2	TCTGC		GCAGA			
	RFXANK NG1	exon2	TGAGAGTGGTGGAGT16901		GAACGAGGTGTCAGC16925			
	RFXANK NG2	Exon2	GCTTC		TCTGC			
	RFXANK NG2	Exon2	CTCGTCCCTCGCTG16902		GGCCACCCCTAGACTG16926			
	RFXANK-WT-exon1-3	exon 1	CCGGT		TGAGT		GGTCCCCAAGTTCTGAGGC 16942	
	RFXANK-WT-exon1-4	exon1					GGCAGGGGTCTGCTGGGTC 16943	
RFXAP	RFXAP NG1	Exon 1	CCCCCCCCAACGCTG16903		CTGTGCGAAGGGGCC16927			
	RFXAP NG2	Exon 1	CCCCC		GGGGA			
	RFXAP NG2	Exon1	CCTTCGCACAGGTAC16904		AGAGGAGGCTGGGGA16928			
	RFXAP NG3	Exon1	CTAAC		GGACG			
	RFXAP NG3	Exon1	CAGCCGGGGCTAGGG16905		CTTGGCGCCAGCCTC16929			
	RFXAP NG4	Exon1	CCGCG		GGTGG			
	RFXAP NG4	Exon1	GCCGCGGCCGCCACC16906		CTAGTGATGCAACCC16930			
	RFXAP NG5	Exon1	GAGGC		TGTGC			
	RFXAP NG5	Exon1	GCCGCGCTCTCGCCT16907		GAGGACGAGGAGACT16931			
	WT- RFXAP-ex1-3	exon 1	CCCCC		CACTC		GGCCCCGGGGCAGCGTT 16944	
	WT- RFXAP-ex1-4	exon 1					GGTACCTGTGCGAAGGGGC 16945	
RFX5	RFX5 NG1	exon1	GTCTTGGGGCTCTTA16908		CCCAGGTGGTGTGA16932			
	RFX5 NG2	exon2	GCATC		GGCTG			
	RFX5 NG2	exon2	ACGGCCTTGCTGTGG16909		GGGATCCTGGTAAGT16933			
	RFX5 NG3	exon5	GGAAG		GTGTT			
	RFX5 NG3	exon5	TCTGATGATCTTGCC16910		ATCAAAGCTCGAAGG16934			
	RFX5 NG2	exon1	AAAGT		CTTGG			
	RFX5 NG2	exon1	GTCTTGGGGCTCTTA16911		CCCAGGTGGTGTGCTG16935			
	RFX5 NG3	exon1	GCATC		AGGCT			
	RFX5 NG3	exon1	AGGCTCATCTTCTG16912		ACTGGGGGAAGGGCC16936			
	WT-RFX5-ex1-4	Exon1	CATCC		CCCCC		GGGAAGGGCCCCCAGG 16946	
	WT-RFX5-ex5-5	Exon 5					GCCTTCGAGCTTTGATGTC 16947	

Engineered HLA-E Compositions

MHCI knockout (KO) renders cells resistant to killing by T cells, but also makes them susceptible to natural killer (NK) cell-mediated cytotoxicity (“Missing-self hypothesis”) (see FIG. 23). It is hypothesized that NK rejection would reduce the in vivo efficacy and/or persistence of these KO cells in a therapeutic setting, such as allogeneic (allo) CAR-T therapy. Retention of MHCI on the surface of allo CAR-T cells would render them susceptible to killing by host T cells, as observed in the classic mixed lymphocyte reaction (MLR) experiment. It is estimated that up to 10% of a person’s T cells are specific to foreign MHC, which would mediate the rejection of foreign cells and tissues. A targeted KO of MHCI, specifically HLA-A, B and C, which can be achieved by targeted KO of B2M, results in a loss of additional HLA molecules including HLA-E. Loss of HLA-E, for example, renders the KO cells more susceptible to NK cell-mediated cytotoxicity due to the “Missing-self Hypothesis”. NK-mediated cytotoxicity against missing-self cells is a defense mechanism against pathogens that downregulate

MHC on the surface of infected cells to evade detection and killing by cells of the adaptive immune system.

Two strategies are contemplated by the disclosure for engineering allo (MHCI-neg) T cells (including CAR-T cells) more resistant to NK cell-mediated cytotoxicity. In some embodiments, a sequence encoding a molecule (such as single-chain HLA-E) that reduces or prevents NK killing is introduced or delivered to an allogeneic cell. Alternatively, or in addition, gene editing methods of the disclosure retain certain endogenous HLA molecules (such as endogenous HLA-E). For example, the first approach involves piggyBac (PB) delivery of a single-chain (sc)HLA-E molecule to B2M KO T cells.

The second approach uses a gene editing composition with guide RNAs selective for HLA-A, HLA-B and HLA-C, but not, for example, HLA-E or other molecules that are protective against natural-killer cell mediated cytotoxicity for MHCI KO cells.

Alternative or additional molecules to HLA-E that are protective against NK cell-mediated cytotoxicity include,

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but are not limited to, interferon alpha/beta receptor 1 (IFNAR1), human IFNAR1, interferon alpha/beta receptor 2 (IFNAR2), human IFNAR2, HLA-G1, HLA-G2, HLA-G3, HLA-G4, HLA-G5, HLA-G6, HLA-G7, human carcinoembryonic antigen-related cell adhesion molecule 1 (CEACAM1), viral hemoagglutinins, CD48, LIT1 (also

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referred to as C-type lectin domain family 2 member (CLC2D)), ULBP2, ULBP3, and sMICA or a variant thereof.

An exemplary INFAR1 protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide, Extracellular, TM, Cytoplasmic):

(SEQ ID NO: 17017)

MMVVLGATTLVLVAVAPWVLSAAAGGKNLKSPOKVEVDIIDDNFILRWNRSDSVGNVT
 FSPDYQKTGMNDNWKLSGCQNTSTKCNFSSSLKNVYEEIKLRIRAEKENTSSWYEVDSF
 TPFRKAQIGPPEVHLEAEDKAIVIHISPGTKDSVMWALDGLSFTYSLVIWKNSSGVEERI
 ENIYSRHKIYKLSPETTYCLKVKAALLTSWKIGVYSPVHCIKTTVENELPPPENIEVSVQ
 NQNYVLKWDYTYANMTFQVQWLHAFKRNPGNHLKWKQIPDCENVKTTQCVFPQNVFQK
 GIYLLRVQASDGNNTSFWSEEIKFDTEIQAFLLPPVFNIRSLSDSFHIYIGAPKQSGNTP
 VIQDYPLIYEIIFWENTSNAERKIEKKTDTVTPNLKPLTVYCVKARAHTMDEKLNKSSV
 FSDAVCEKTKPGNTSITWLVIGICIALFALPFVVIYAAKVFLRCINVFPPSLKPSSSIDE
 YFSEQPLKNLLSTSEEQIEKCFIENISTATVEETNQTDEDHKYSSQTSQDSGNYSN
 EDESEKSTSEELQQDFV.

25

An exemplary INFAR2 protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide, Extracellular, TM, Cytoplasmic):

(SEQ ID NO: 17018)

MLLSQNAFIFRSLNLVLMVYISLVFGISYDSPDYTDESCTFKISLRNFRSILSWELKNHS
 IVPHTHYLLYTIMSKPEDLKVVKNCAITIRSFCDLTDEWRSTHEAYVTVLEGFSGNITLF
 SCSHNFWLAIDMSFEPPEFEIVGFTNHINVMVKFPSIVEEELQFDLSLVIEEQSEGIVKK
 HKPEIKGNMSGNTFYIIDKLIPNTNYCVSVYLEHSDEQAVIKSPLKCTLLPPGQESAE
 SAKIGGIITVFLIALVLTSTIVTLKWIGYICLRNSLPKVLNFHNLAWFFPNLPPLEAMD
 MVEVIYINRKKKVDYNYDDESDDTEAAPRTSGGGYTMHGLTVRPLGQASATSTESQLI
 DPESEEEPDLPEDVELPTMPKDSPPQLELLSGPCERRKSPLQDPFPEEDYSSTEGSGGR
 ITFNVDLNSVFLRVLDDESDDLAPMLLSSHLEEMVDPEDPDNVQSNHLLASGEGTOPT
 FSPSPSEGLWSEDAPSDQSDISESDVDLGDGYIMR.

50 An exemplary HLA-G1 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

(SEQ ID NO: 17019)

MVVMAPRTLFLLLSGALTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS
 DSACPRMEPRAPVVEQEGPEYWEETRNTKAHAQTDRMNLQTLRGYYNQSEASSHTLQWMIG
 CDLGSDGRLRLRGYEQYAYDGKDYALNEDLRSWTAADTAAOISKRC EAANVAEQRRAYLEG
 TCVEWLHRYLENGKEMLORADPPKTHVTHHPVFDYEATLRCAWALGFYPAEIIITWQRDGEDQ
 TQDVELVETRPAGDGTFOKWAAVVPSGEEQRYTCHVQHEGLPEPLMLRWKQSSSLPTIPIMG
 IVAGLVVLAAVVTGAAVAVALWRKKSSD.

An exemplary HLA-G2 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

(SEQ ID NO: 17020)
 MVVMAPRTLFLLLSGALTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS
 DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEADPPKTHVTHH
 PVFDYEATLRCWALGFYPAEIIITWQRDGEDQTQDVELVETRPAGDGTQKWAADVVP SGEE
 QRYTCHVQHEGLPEPLMLRWKQSSLPTIPIMGIVAGLVVLAAVVTGAAVA AVLWRKKSSD .

An exemplary HLA-G3 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):
 MVVMAPRTLFLLLSGALTLTETWAGSHSM-
 RYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS
 DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQT-
 DRMNLQTLRGYYNQSEAKQSSLPTIPI MGIVAGLVV-
 LAAVVTGAAVA AVLWRKKSSD (SEQ ID NO: 17021).

An exemplary HLA-G4 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3):

(SEQ ID NO: 17022)
 MVVMAPRTLFLLLSGALTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS
 DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEASSHTLQWMIG
 CDLGSDGRLRLRGYEQYAYDGKDKLALNEDLRSWTAADTAAQISKRKCEANVAEORRAYLEG
 TCVEWLHRYLENGKEMLQRAKQSSLPTIPIMGIVAGLVVLAAVVTGAAVA AVLWRKKSSD .

An exemplary HLA-G5 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3, intron 4):

(SEQ ID NO: 17023)
 MVVMAPRTLFLLLSGALTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS
 DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEASSHTLQWMIG
 CDLGSDGRLRLRGYEQYAYDGKDKLALNEDLRSWTAADTAAQISKRKCEANVAEORRAYLEG
 TCVEWLHRYLENGKEMLQRAADPPKTHVTHHPVFDYEATLRCWALGFYPAEIIITWQRDGEDQ
 TQDVELVETRPAGDGTQKWAADVVP SGEEQRYTCHVQHEGLPEPLMLRWKSGEGGGI:SVR
 ESRSLSEDL .

An exemplary HLA-G5 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3, intron 4):

(SEQ ID NO: 17024)
 MVVMAPRTLFLLLSGALTLTETWAGSHSMRYFSAAVSRPGRGEPRFIAMGYVDDTQFVRFDS
 DSACPRMEPRAPWVEQEGPEYWEEETRNTKAHAQTDRMNLQTLRGYYNQSEADPPKTHVTHH
 PVFDYEATLRCWALGFYPAEIIITWQRDGEDQTQDVELVETRPAGDGTQKWAADVVP SGEE
 QRYTCHVQHEGLPEPLMLRWKSGEGGGI:SVRESRSLSEDL .

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An exemplary HLA-G5 protein of the disclosure comprises or consists of the amino acid sequence of (Alpha chain 1, Alpha chain 2, Alpha chain 3, intron 2):

(SEQ ID NO: 17025)

MVVMAPGALTTLTETWAGSHSMRYFSAAVSRPGRGEPFIAMGYVDDTQFV
 RFDSDSACPRMEPRAPWVEQEGPEYWEETRNTKAHAQTDRMNLQTLRGY
 YNQSEASE.

An exemplary CEACAM1 protein of the disclosure comprises or consists of the amino acid sequence of (Extracellular, TM, Cytoplasmic):

(SEQ ID NO: 17026)

MGHSAPLHRVRVPWQGLLLTASLLTFWNPPTAQLTTESMPFNVAEGKE
 VLLL VHNLPQQLFGYSWYKGERVDGNRQIVGYAIGTQQATPGPANSGRE
 IYPNASLLIQNVTDNDTGFTTLQVIKSDLVNEEATGQFHVPELPKPSIS
 SNNSNPVEDKDAVAFTCEPETQDITYLWWINNQSPLVSPRLQLSNGNRTL
 TLLSVTRNDTPGYECEIQNPVSANRSDPVTNLNVTYGPDTPTTISPSTYYR
 PGANLSLSCYAASNPPAQYSWLINGTFQQSTQELFIPNITVNNSGSYTCH
 ANNSVTGCNRTTVKTIIVTELSPPVAKPQIKASKTTVTGDKDSVNLTCST
 NDGTGISIRWFFKNQSLPSSERMKLSQGNNTLSINPVKREDAGTYWCEVFN
 PISKNQSDPIMLVNINYNALPQENGLSPGAIGIVIGVVALVALIAVALAC
 FLHFGKTGRASQDRDLTEHKPSVSNTQDHSNDPPNKMNEVITYSTLNFEA
 QQPTQPTSASPSLTATEIIYSEVKKQ.

An exemplary viral hemagglutinin protein of the disclosure comprises or consists of the amino acid sequence of (HAforInfluenzaAvirus(A/NewCaledonia/20/1999(H1N1); TM):

(SEQ ID NO: 17027)

MKAKLLVLLCTFTATYADTICIGYHANNSTDVDTVLEKNVTVTHSVNLL
 EDSHNGKLCCLKGIAPLQLGNCVAGWILGNPECELLISKESWSYIVETP
 NPENGTCYPGYFADYEELREQLSSVSSFERFEIFPKESSWPNHVTGVSA
 SCSHNGKSSFYRNLLWLTGKNGLYPNLSKSYVNNKEEVLVLWGWHPPN
 IGQRALYHTENAYSVSVSSHYRRFTPEIAKRPKVRDQEGRINYWILL
 EPGDTIIFEANGNLIAPWYAFALSRGFGSGIITSNAPMDECDKQTPQG
 AINSSLPPQNVHPVTIGCEPKYVRS AKLRMTGLRNIPSIQSRGLFGAIA
 GFIEGGWTGMVDGWYGYHHQNEQSGYAADQKSTQNAINGITNKVNSVIE
 KMNTQFTAVGKEFNKLERRMENLNKKVDDGFLDIWTYNAELLVLLNERT
 LDFHDSNVKNLYEKVKSQKNNAKEIGNGCFEYHKCNNECMESVKNQTY
 DYPKYSEESKLNREKIDGVKLESMGVYQILAIYSTVASSLVLLVSLGAIS
 FWMCSNGLQCRICI.

An exemplary CD48 protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide, Chain, Pro peptide removed in mature form):

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(SEQ ID NO: 17028)

MCSRGWDSCLALELLLLPLSLVTSIQGHLVHMTVVSGSNVTLNISESLP
 ENYKQLTWFFYTEDQKIVEWDSRKSKEYFESKFGRVRLDPQSGALYISKVQ
 KEDNSTYIMRVLKKTGNEQEWKIKLQVLDPVKPKVIKIEKIEDMDNCYL
 KLSCVIPGESVNYTWYGDKRFPFKELQNSVLETTLMPHNYSRCYTCQVSN
 SVSSKNGTVCLSPPTLARSFGVEWIASWLVTVTPTILGLLLT.

An exemplary LLT1 protein of the disclosure comprises or consists of the amino acid sequence of (Cytoplasmic, TM, Extracellular):

(SEQ ID NO: 17029)

MHDSNNVEKDITPSEL PANPGCLHSKEHSIKATLIWRLFFLIMFLTIIVC
 GMVAALSAIRANCHQEPSVCLQAACPESWIGFQRKCFYFSDDTKNWTSSQ
 RFCDSDADLAQVESFQELNLLRYKGPSDHWIGLSREQGQPWKWINGTE
 WTRQFPILGAGECAYLNDKGASSARHYTERKWICKSKSDIHV.

An exemplary ULBP2 protein of the disclosure comprises or consists of the amino acid sequence of (also known as NKG2D ligand; Genbank ACCESSION No. AAQ89028):

(SEQ ID NO: 17030)

1 maaaaatkil lclpllllls gwsragradp hslcyditvi
 pkfrpgprwc avggqvdekt
 61 flhydcgnkt vtpvslpgkk lntttawkaq npvlrevvdi
 lteqlrdiql enytpkeplt
 121 lqarmsceqk aeghsqswq fsfdgqifll fdsekrmwtt
 vhpqarkmke kwendkvvam
 181 sfhyfsmgdc igwledflmg mdstlepszag aplanssgtt
 qlratattli lccilliilpc
 241 filpgi.

An exemplary ULBP3 protein of the disclosure comprises or consists of the amino acid sequence of (also known as NKG2D ligand; Genbank ACCESSION No. NP_078794):

(SEQ ID NO: 17031)

1 maaaaspail prlailpyll fdwsgtgrad ahslywnfti
 ihlprhgqqw cevqsgvdgk
 61 nflsydcgsd kvlsmghee glyatdawgk qlemlrevgq
 rlrleladte ledftpsgpl
 121 tlqvrmscec eadgyirgsw qfsfdgrkfl lfdsnnrkwt
 vvhagarmk ekwekdsglt
 181 tffkmvsmrd ckwirdflm hrkkrelepta pptmapglaq
 pkaiattlsp wsfliilcfl
 241 lpgi.

An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Signal Peptide, Portion of Extracellular domain, TM and cytoplasmic domain) (Genbank Accession No. Q29983):

(SEQ ID NO: 17032)

1 mglgpvfilll agifpfappg aaaephslry nltvlswdgs
 vqsgfltevh ldgqpfllrcd
 61 rqkcrakpgg qwaedvlgkn twdretrdlt gngkdlrmtl
 ahikdqkegl hslqeirvce

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121 ihednstrss qhfydydgelf lsnletkew tmpqssraqt
lamnvrnflk edamktkthy

181 hamhadclqe lrrylksgvv lrrtvppmvn vtrseasegn
itvtcrasgf ypnwntlsr

241 qdgvslshdt qqwgdvlpdg ngtyqtwvat ricqgeeqr
tcymehsgnh sthpvpsgkv

301 lvlqshwqtf hvsavaaaai fviiifyvrc ckkttsaaeg
pelvslqvld qhpvgtsdhr

361 datqlgfqpl msdlgstgst ega.

An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Alpha-1, Alpha-2, Alpha-3):

(SEQ ID NO: 17033)

1 mglgpvfilll agifpfappg aaaphslry nlvlswdgs
vqsgfltevh ldgqpfllrcd

61 rqkcrakpgg qwaedvlgk twdretrdlt gngkdlrmtl
ahikdqkegl hslqeirvce

121 ihednstrss qhfydydgelf lsnletkew tmpassraqt
lamnvrnflk edamktkthy

181 hamhadclqe lrrylksgvv lrrtvppmvn vtrseasegn
itvtcrasgf ypnwntlsr

241 qdgvslshdt qqwgdvlpdg ngtyqtwvat ricqgeeqr
tcymehsgnh sthpvpsgkv

301 lvlqshwqtf hvsavaaaai fviiifyvrc ckkttsaaeg
pelvslqvld qhpvgtsdhr

361 datqlgfqpl msdlgstgst ega.

An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide; Alpha-, Alpha-2, Alpha-3):

(SEQ ID NO: 17034)

MGGVLLTQRTL LSLVLALLFPMSASM ephslry nltvlswdgs
vqsgfltevh ldgqpfllrcd

61 rqkcrakpgg qwaedvlgk twdretrdlt gngkdlrmtl
ahikdqkegl hslqeirvce

121 ihednstrss qhfydydgelf lsnletkew tmpassraqt
1 thy

181 hamhadclqe lrrylksgvv lrrtvppmvn vtrseasegn
itvtcrasgf ypnwntlsr

241 qdgvslshdt qqwgdvlpdg ngtyqtwvat ricqgeeqr
tcymehsgnh sthpvpsgkv

301 lvlqshw.

An exemplary sMICA protein of the disclosure comprises or consists of the amino acid sequence of (Signal peptide):

(SEQ ID NO: 17035)

MGGVLLTQRTLISLVALLFPMSASMEPHSLRYNLTVLSWDGSSVQSGFLT

EVHLDGQPFLLRCRQKRAKPGQWAEDVLGNKTWDRETRDLTGNGKLDL

RMTLAHIKLDQKEGLHSLQEIRVCEIHEDNSTRSSQHFIYNGELFLSQNL

ETKEWTPQSSRAQTLYTHAMHADCLQELRRLKSGVLRRTVPPMVDV

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TRSEASEGNITVTCRASGFYPWNITLSWRQDGVLSHDTQQWGDVLPDGN

5 GTYQTWVATRICQGEEQRFTCYMEHSGNHSTHPVPSGKVLVLQSHW.

An exemplary bGBE Trimer (20G and 484S) protein of the disclosure comprises or consists of the amino acid sequence of:

10 (SEQ ID NO: 16972)

MSRSVALAVLALLSLGLEAVMAPRILILGGGSGGGSGGGSIQRTF

KIQVYSRHPAENGKSNFLNCYVSGFHPSDIEVDLLKNGERIEKVEHSDL

15 SFSKDWSFYLLYYTEFTPTTEKDEYACRVNHVLSQPKIVKWDRDMGGGG

SGGGSGGGSGGGSGSHSLKYFHTSVSRPGRGEPRFISVGYYDDTQF

VRFDNDAA SPRMPRAPWMEQEGSEYWDRETRSARDTAQIFRVNLRTL

20 GYYNQSEAGSHTLQWMHGCELGPDGRFLRGYEQFAYDGKDYLTNLNEDLR

SWTAVDTAAQISEQKSNDAEAEHQRAYLEDTCVEWLHKYLEKGKETLL

HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQQDGEHTQDTE

25 LVETRPAGDGTGQWAAVVVPSGEEQRYTCHVQHEGLPEPVTLRWKPAS

QPTIPIVGIIAGLVLLGSVVSGAVVAIVWRKKS SGGKGSYSKAESWD

SAQGSSEHSL*.

An exemplary bGBE Trimer (270G and 484S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(SEQ ID NO: 16973)

35 atgtctcgcagcgtggccctggcgtgctggccctgctgctcctgctggtg

cctggaggccgtgatggccccggaccctgatcctggaggaggaggca

gcggcgaggaggctccggaggcgccgctctatccagcgcacacctaa

40 atccaggtgtattctcgccacccagccgagaaacggcaagagcaactcct

gaattgctacgtgagcggccttccaccctccgacatcgaggtggatctgc

tgaagaatggcgagagaatcgagaaggtggagcactccgacctgagcttc

tccaaggattggtctttttatctgctgtactataccaggtttaccctac

45 agagaaggacgagtagcctgtgcgtgaaccacgtgacactgtccagc

caaagatcgtgaagtgggacccggatattggcgccggcggtctctggcgcc

ggcgccagcggcgccggcggtccggaggaggcggtcttggcagccactc

50 cctgaagtatttccacacctctgtgagcggccaggcagaggagagccac

ggttcactctctgtgggtacgtggacgatacacagttcgtgaggtttgac

aatgatgccgccagcccaagaatgggtgctaggccccatggatggagca

55 ggagggcagcagtagtattgggacaggagaccggagcgcagagacacag

cacagattttccgggtgaacctgagaacctgaggggctactataatcag

tccgaggccggtctcacacactccagtgatgcacggatgcagagctggg

60 accagatggccgcttctgctggggctacgagcagtttgcctatgacggca

aggattacctgacctgaacgaggacctgagatcctggaccgctggat

acagccgccagatcagcagcagaagccaatgacgcatctgaggcaga

65 gcaccagagggcatatctggaggatacctgctggagtggtgctgcacaagt

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acctggagaagggcaaggagacactgctgcacctggagccccctaagacc
 cactgacacaccaccaatcagcgaccacgagggccacctgaggtgttg
 ggcaactgggcttctatcccgccgagatcacctgacatggcagcaggacg
 gagagggacacaccaggatacagagctggtggagaccaggcccccgccg
 gatggcacatttcagaagtggcgccgctggtggtgcttccggagagga
 gcagagatacacctgtcacgtgcagcacgaggactgccagagccagtga
 ccctgaggtggaagcctgccagccagccacaatccctatcgtgggaatc
 atcgaggcctggtgctgctgggctctgtggtgagcggagcagtggtggc
 cgccgtgatctggcggaagaagagcagcggagggaaggaggtcctact
 ccaaggcagagtggagcgactccgcccagggtctgagagccactccctg
 tga.

An exemplary bGBE Trimer (270R and 484S) protein of the disclosure comprises or consists of the amino acid sequence of:

(SEQ ID NO: 16974)

MSRSVALAVLALLSLSGLEAVMAPRTLILGGGSGGGSGGGSIQRT
 KIQVYSRHPAENGKSNFLNCYVSGFHPSDIEVDLLKNGERIEKVEHSDL
 SFSKDWSEFYLLYYTEFTPEKDEYACRVNHVILSQPKIVKWRDMGGGG
 SGGGSGGGSGGGSGGSHSLKYFHTSVSRPGRGEPRFISVGVDVTQF
 VRFDNDAAAPRMVPRAPWMEQEGSEYWDRETRSARDTAQIFRVNLRTLR
 GYYNQSEAGSHTLQWMHGCELGPDRFLRGYEQFAYDGKDYLTNLNEDLR
 SWTAVDTAAQISEQKSDASEAEHQRAYLEDTCVEWLHKYLEKGKETLL
 HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQQDGEGHTQDTE
 LVETRPAGDGTQKWAAVVPSGEEQRYTCHVQHEGLPEPVTLRWKPAS
 QPTIPIVGIIAGLVLLGSVVSAGVAAVWRKKS SGGKGSYSKAESD
 SAQGSSESHSL*.

An exemplary bGBE Trimer (270R and 484S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(SEQ ID NO: 16975)

atgtctcgacgctggccctggccgtgctggccctgctgctccctgtctgg
 cctggaggccgtgatggcccccgaccctgacctgggaggaggagga
 gcggcgaggaggctccggaggcgccgctctatccagcgacacctaag
 atccaggtgtattctcgccaccagccgagaaaggcaagagcaacttcct
 gaattgctacgtgagcggtttccacctccgacatcgaggtggatctgc
 tgaagaatggcgagagaatcgagaaggtggagcactccgacctgagcttc
 tccaaggattggtctttttatctgctgtactataccgagtttaccctac
 agagaaggacgagtacgcctgtcgctgaaccacgtgacactgtcccagc
 caaagatcgtgaagtgggaccgggatatggcgccgagcggtctgtggcgc
 ggccgagcggcgccgagcgtccggaggaggcggtctgtgcagccactc
 cctgaagtatttccacacctctgtgagccggccagcagaggagagccac
 ggttcactctctgtgggtacgtggagacatacacagttcgtgaggtttgac

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aatgatgccgccagcccaagaatggtgcctaggcccccatggatggagca
 ggagggcagcgagattatgggacagggagaccggagcgccagagacacag
 cacagattttccgggtgaacctgagaacctgaggggctactataatcag
 tccgaggccggtctcacacactccagtggtgacggatgcgagctggg
 accagatggccgcttctgctggggtacgagcagtttgcctatgacggca
 aggtattacctgacctgaacgaggacctgagatcctggaccgctggtat
 acagccgcccagatcagcgagcagaagccaatgacgcacatctgaggcaga
 gcaccagagggcatatctggaggatacctgcgtggagtggtgcacaagt
 acctggagaagggcaaggagacactgctgcacctggagccccctaagacc
 cactgacacaccaccaatcagcgaccacgagggccacctgaggtgttg
 ggcaactgggcttctatcccgccgagatcacctgacatggcagcaggacg
 gagagggacacaccaggatacagagctggtggagaccaggcccccgccg
 gatggcacatttcagaagtggcgccgctggtggtgcttccggagagga
 gcagagatacacctgtcacgtgcagcagcagggactgccagagccagtga
 ccctgaggtggaagcctgccagccagccacaatccctatcgtgggaatc
 atcgaggcctggtgctgctgggctctgtggtgagcggagcagtggtggc
 cgccgtgatctggcggaagaagagcagcggagggaaggaggtcctact
 Ccaaggcagagtggagcgactccgcccagggtctgagagccactccctg
 tga.

An exemplary gBE Dimer (Rand S) protein of the disclosure comprises or consists of the amino acid sequence of:

(SEQ ID NO: 16976)

MSRSVALAVLALLSLSGLEAIQRTPKIQVYSRHPAENGKSNFLNCYVSGF
 HPDSDIEVDLLKNGERIEKVEHSDLSFSKDWSEFYLLYYTEFTPEKDEYAC
 RVNHVTLSPKIVKWRDMGGGSGGGSGGGSGGSHSLKYFHTS
 VSRPGRGEPRFISVGVDVTQFVRFDNDAAAPRMVPRAPWMEQEGSEYWD
 RETRSARDTAQIFRVNLRTLRGYYNQSEAGSHTLQWMHGCELGPDRFLR
 GYEQFAYDGKDYLTNLNEDLRWTAVDTAAQISEQKSDASEAEHQRAYLE
 DTCVEWLHKYLEKGKETLLHLEPPKTHVTHHPISDHEATLRCWALGFYP
 AEITLTWQQDGEGHTQDTELVETRPAGDGTQKWAAVVPSGEEQRYTCHV
 QHEGLPEPVTLRWKPASQPTIPIVGIIAGLVLLGSVVSAGVAAVWRKKS
 SSGKGSYSKAESD SAQGSSESHSL.

An exemplary gBE Dimer (R and S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(SEQ ID NO: 16977)

ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTGTCTGG
 CCTGGAAGCCATCCAGCGGACCCCTAAGATCCAGGTGTACAGCAGACACC
 CCGCCGAGAACGGCAAGAGCAACTTCTGAAGTGTCTGCTGCTGGCTTT
 CACCCAGCGACATTGAGGTGGACCTGCTGAAGAACGGCGAGCGGATCGA
 GAAGGTGGAACACAGCGATCTGAGCTTACGCAAGGACTGCTCTTCTACC

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TGCTGTACTACACCGAGTTCACCCCTACCGAGAAGGACGAGTACGCCTGC
 AGAGTGAACCACTGACACTGAGCCAGCCTAAGATCGTGAAGTGGGACAG
 AGATATGGGCGGAGGCGGATCTGGTGGCGGAGGAAGTGGCGGCGGAGGAT
 CTGGCGGTGGTGGTTCTGGATCTCACAGCCTGAAGTACTTTACACCTCC
 GTGTCCAGACCTGGCAGAGGCGAGCCTAGATTATCAGCGTGGGCTACGT
 GGACGACACCCAGTTCTGTAGATTTCGACAACGACGCCCTCTCTCTCGGA
 TGGTTCCTAGAGACCCCTGGATGGAACAAGAGGGCAGCGAGTACTGGGAT
 CGCGAGACAAGAAGCGCCAGAGACACAGCCAGATCTTCCGCGTGAACCT
 GAGAACCCTGCGGGGCTACTACAATCAGTCTGAGCCGGCTCTCACACCC
 TGCAGTGGATGCATGGATGTGAAGTGGGCCCCGACAGACCGGTTCTCTGAGA
 GGCTATGAGCAGTTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGA
 GGACCTGAGAAGCTGGACCGCGTGGATACAGCCCTCAGATCAGCGAGC
 AGAAGTCTAACGACGCCAGCGAGGCGAACACCAGAGAGCCTATCTGGAA
 GATACCTGCGTGAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGAC
 ACTGCTGCACCTGGAACCTCCAAGACACATGTGACCCACCATCCTATCA
 GCGACCACGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCTGCC
 GAGATCACACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATAC
 AGAGCTGGTGGAAACAAGACCTGCCGGCGCAGGCACCTTCCAGAAATGGG
 CTGCTGTGGTTGTGCCCAGCGCGGAGGAACAGAGATACACCTGTACGTG
 CAGCACGAGGGACTGCCTGAACCTGTGACTCTGAGATGGAAGCCTGCCAG
 CCAGCCAACAATCCCCATCGTGGGAATCATTGCCGGCCTGGTGTCTGTGG
 GATCTGTGGTTTCTGGTGTGTGGTGGCGCGTGAATTGGAGAAAGAAG
 TCCTCTGGCGGCAAAGGGCGGCTCCTACTCTAAGGCCGAGTGGAGCGATTCT
 TGCCACGGCTCTGAAAGCCACAGCCTGTAGATAA.

An exemplary gBE Dimer (G and S) protein of the disclosure comprises or consists of the amino acid sequence of:

(SEQ ID NO: 16978)

DLKNGERIEKVEHSDLSFSKDWISFYLLYYTEFTPEKDEYACRVNHVTL
 SQPKIVKWDRDMGGGSGGGGSGGGGSGSHSLKYFHTSVSRPGRG
 EPRFISVGIVDDTQFVRFDNDAASPRMVPAPWMEQEGSEYWDRETRSAR
 DTAQIFRVNLRLRGYYNQSEAGSHTLQWMHGCELGPDRFLRGYEQFAY
 DGKDYLTNLNEDLRSWTAVDTAAQISEQKSNDAEAEHQRAYLEDTCVEWL
 HKYLEKGKETLLHLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQ
 QDGEHTQDTELVEVTRPAGDGTQKWAAVVPSGEEQRYTCHVQHEGLPE
 PVTLRWKPASQPTIPIVIGIIAGLVLLGSVVSAGVVAIVWRKSSGGKGG
 SYSKAEWSDSAQGSSEHSL

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An exemplary gBE Dimer (G and S) protein of the disclosure comprises or consists of the amino acid sequence of:

(SEQ ID NO: 16979)
 ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG
 CCTGGAAGCCATCCAGCGGACCCCTAAGATCCAGGTGTACAGCAGACACC
 CCGCCGAGAACGGCAAGAGCAACTTCCTGAAGTGTACGTGTCCGGCTTT
 CACCCACGCGACATTGAGGTGGACCTGCTGAAGAACGGCGAGCGGATCGA
 GAAGGTGGAACACAGCGATCTGAGCTTCAGCAAGGACTGGTCTTCTTACC
 TGCTGTACTACACCGAGTTCACCCCTACCGAGAAGGACGAGTACGCCTGC
 AGAGTGAACCACGTGACACTGAGCCAGCCTAAGATCGTGAAGTGGGACAG
 AGATATGGGCGGAGGCGGATCTGGTGGCGGAGGAAGTGGCGGCGGAGGAT
 CTGGCGGTGGTGGTTCTGGATCTCACAGCCTGAAGTACTTTACACCTCC
 GTGTCCAGACCTGGCAGAGGCGAGCCTAGATTATCAGCGTGGGCTACGT
 GGACGACACCCAGTTCTGTAGATTTCGACAACGACGCCCGCTCTCTCTCGGA
 TGGTTCCTAGAGCACCCCTGGATGGAACAAGAGGGCAGCGAGTACTGGGAT
 CGCGAGACAAGAAGCGCCAGAGACACAGCCAGATCTTCCGCGTGAACCT
 GAGAACCCTGCGGGGCTACTACAATCAGTCTGAGGCCGGCTCTCACACCC
 TGCAGTGGATGCATGGATGTGAAGTGGGCCCCGACAGACCAGTTCTCTGAGA
 GGCTATGAGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGA
 GGACCTGAGAAGCTGGACCGCGTGGATACAGCCGCTCAGATCAGCGAGC
 AGAAGTCTAACGACGCCAGCGAGGCGAACACCAGAGAGCCTATCTGGAA
 GATACCTGCGTGGAAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGAC
 ACTGCTGCACCTGGAACCTCCAAGACACATGTGACCCACCATCCTATCA
 GCGACCACGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCTGCC
 GAGATCACACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATAC
 AGAGCTGGTGGAAACAAGACCTGCCGGCGCAGGCACCTTCCAGAAATGGG
 CTGCTGTGGTTGTGCCCAGCGCGGAGGAACAGAGATACACCTGTACAGTG
 CAGCACGAGGGACTGCCTGAACCTGTGACTCTGAGATGGAAGCCTGCCAG
 CCAGCCAACAATCCCCATCGTGGGAATCATTGCCGGCCTGGTGTCTGTGG
 GATCTGTGGTTTCTGGTGTGTGGTGGCGCGGCTGAATTGGAGAAAGAAG
 TCCTCTGGCGGCAAAGGGCGGCTCCTACTCTAGGCCGAGTGGAGCGATTCT
 TGCCACGGCTCTGAAAGCCACAGCCTGTAGATAA

An exemplary WT HLA-E Monomer (Rand S) protein of the disclosure comprises or consists of the amino acid sequence of:

(SEQ ID NO: 16980)

MSRSVALAVLALLSLSGLEAGSHSLKYFHTSVSRPGRGEPRFISVGIVDD
 TQFVRFDNDAASPRMVPAPWMEQEGSEYWDRETRSARDTAQIFRVNLRL
 LRGYYNQSEAGSHTLQWMHGCELGPDRRFLRGYEQFAYDGKDYLTNLNEDL
 RSWTAVDTAAQISEQKSNDAEAEHQRAYLEDTCVEWLHKYLEKGKETLL
 HLEPPKTHVTHHPISDHEATLRCWALGFYPAEITLTWQDGEHTQDTEL

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VETRPAGDGTQKWAADVVPSEGEQRYTCHVQHEGLPEPVTLRWKPASQP
 TIPIVGIIAGLVLLGSVSVGAVVAAVIWRKSSGGKGGSY~~9~~KAEWSDSAQ
 GSESHSL

An exemplary WT HLA-E Monomer (Rand S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(SEQ ID NO: 16981)
 ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG
 ACTGGAAGCCGGCAGCCACAGCCTGAAGTACTTTACACCAGCGTGTCCA
 GACCTGGCAGAGGCGAGCCTAGATTTCATCAGCGTGGCTACGTGGACGAC
 ACCCAGTTCGTTCAGATTTCGACAACGACGCGCCTCTCCTCGGATGGTTCC
 TAGAGCACCTGGATGGAACAAGAGGGCAGCGAGTACTGGGACAGAGAGA
 CAAGAAGCGCCAGAGACACAGCCAGATCTTCAGAGTGAACCTGCGGACC
 CTGCGGGGCTACTACAATCAGTCTGAAGCCGGCTCTCACACCTGCAGTG
 GATGCAACGATGTGAACCTGGGCCCCGAC~~CAGAA~~GATTCTGAGAGGCTACG
 AGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGAGGACCTG
 AGAAGCTGGACCGCGCTGGATACAGCCGCTCAGATCAGCGAGCAGAAGTC
 TAACGACGCCTCTGAGGCCGAACACCAGAGAGCCTACCTGGAAGATACCT
 GCGTGGAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGACACTGCTG
 CACCTGGAACCTCCAAAGACACACGTGACCCACCATCTATCAGCGACCA
 CGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCCCGCCGAGATCA
 CACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATACAGAGCTG
 GTGGAACAAGACCTGCCGCGCAGGCACCTTCCAGAAATGGGCTGCTGT
 GGTGGTTCACGCGCGAGGAACAGAGATACACCTGTACGTGCAGCAGC
 AGGGACTGCCTGAACCTGTGACACTGAGGTGGAAGCCTGCCAGCCAGCCT
 ACAATCCCCATCGTGGGAATCATTGCCCGCTGGTGCTGCTGGGATCTGT
 GGTTTCTGGTGCAGTGGTGGCCCGCTGATCTGGCGGAAAAAAGCTCAG
 GCGGCAAAGCGGCTCCTACT~~CTCC~~AAAGCCGAGTGGAGCGATTCTGCCACG
 GGCTCTGAAAGCCACTCTCTGTAGATAA.

An exemplary WT HLA-E Monomer (G and S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(SEQ ID NO: 16982)
 MSRSVALAVLALLSLSGLEAGSHSLKYFHTSVSRPGRGEPRFISVGYVDD
 TQFVRFDNDAASPRMVPAPWMEQEGSEYWDRETRSARDTAQIFRVNLR
 LRGYYNQSEAGSHTLQWMHGCELGPD~~GR~~FLRGYEQFAYDGKDYLTINEDL
 RSWTAVDTAAQISEQKSNDAEAEHQRAYLEDTCVEWLHKYLEKGKETLL
 HLEPPKTHVTHHPISDHEATLRWCALGFYPAEITLTWQQDGBGHTQDTEL
 VETRPAGDGTQKWAADVVPSEGEQRYTCHVQHEGLPEPVTLRWKPASQP
 TIPIVGIIAGLVLLGSVSVGAVVAAVIWRKSSGGKGGSY~~9~~KAEWSDSAQ
 GSESHSL.

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An exemplary WT HLA-E Monomer (Q and S) protein of the disclosure comprises or consists of the nucleic acid sequence of:

(SEQ ID NO: 16983)
 ATGAGCAGATCTGTGGCCCTGGCTGTTCTGGCTCTGCTGTCTCTGTCTGG
 ACTGGAAGCCGGCAGCCACAGCCTGAAGTACTTTACACCAGCGTGTCCA
 GACCTGGCAGAGGCGAGCCTAGATTTCATCAGCGTGGGCTACGTGGACGAC
 ACCCAGTTCGTTCAGATTTCGACAACGACGCGCCTCTCCTCGGATGGTTCC
 TAGAGCACCTGGATGGAACAAGAGGGCAGCGAGTACTGGGACAGAGAGA
 CAAGAAGCGCCAGAGACACAGCCAGATCTTCAGAGTGAACCTGCGGACC
 CTGCGGGGCTACTACAATCAGTCTGAAGCCGGCTCTCACACCTGCAGTG
 GATGCAACGATGTGAACCTGGGCCCCGAC~~CAGAA~~GATTCTGAGAGGCTACG
 AGCAGTTCGCCTACGACGGCAAGGACTACCTGACACTGAACGAGGACCTG
 AGAAGCTGGACCGCGCTGGATACAGCCGCTCAGATCAGCGAGCAGAAGTC
 TAACGACGCCTCTGAGGCCGAACACCAGAGAGCCTACCTGGAAGATACCT
 GCGTGGAATGGCTGCACAAGTACCTGGAAAAGGGCAAAGAGACACTGCTG
 CACCTGGAACCTCCAAAGACACACGTGACCCACCATCTATCAGCGACCA
 CGAGGCCACACTGAGATGTTGGGCCCTGGGCTTTTACCCCGCCGAGATCA
 CACTGACATGGCAGCAGGATGGCGAGGGCCACACACAGGATACAGAGCTG
 GTGGAACAAGACCTGCCGCGCAGGCACCTTCCAGAAATGGGCTGCTGT
 GGTGGTTCACGCGCGAGGAACAGAGATACACCTGTACGTGCAGCAGC
 AGGGACTGCCTGAACCTGTGACACTGAGGTGGAAGCCTGCCAGCCAGCCT
 ACAATCCCCATCGTGGGAATCATTGCCCGCTGGTGCTGCTGGGATCTGT
 GGTTTCTGGTGCAGTGGTGGCCCGCTGATCTGGCGGAAAAAAGCTCAG
 GCGGCAAAGCGGCTCCTACT~~CTCC~~AAAGCCGAGTGGAGCGATTCTGCCACG
 GGCTCTGAAAGCCACTCTCTGTAGATAA.

Vectors and Host Cells

The disclosure also relates to vectors that include isolated nucleic acid molecules of the disclosure, host cells that are genetically engineered with the recombinant vectors, and the production of at least one VHH or VCAR by recombinant techniques, as is well known in the art. See, e.g., Sambrook, et al., supra; Ausubel, et al., supra, each entirely incorporated herein by reference.

For example, the PB-EF1a vector may be used. The vector comprises the following nucleotide sequence:

(SEQ ID NO: 17502)
 tgtacatagattaaccctagaagataatcatattgtgacgtacgttaaa
 gataatcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcga
 ggtttatttattaatttgaatagatattaagttttatttatatttacactt
 acataactaataataaattcaacaacaattttatttatgtttattttatta
 ttaaaaaaaaaaaaaaactcaaaatttcttctataaagtaacaaaacttt
 tatcgaatacctgcagcccggggatgcagaggagacagccccccccaaa
 gccccaggatgtaattacgtccctcccccgtagggggcagcagcgag
 ccgccccgggctccgctccggctccggcgtcccccgcatccccgagccg

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gcagcgtgccccgacagccccgggcacgggaaggtggcacgggatcgctt
 tcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggga
 tacggggaaaagttagctgtgcctttcgatcgaacctggacagttagct
 ttgcaaagatggataaagttttaaacagagaggaatctttgcagctaag
 gaccttctaggtcttgaaaggagtgggaattggctccggtgcccgtagt
 gggcagagcgcacatcgccacagtcgccgagaagtggggggaggggtc
 ggcaattgaaccgggtgcctagagaagtgggcggggtaaactgggaaag
 tgatgtcgctgactggctcgctcttttcccgaggtgggggagaacgt
 atataagtgcagtagtcgctgaaagcttcttttgcgaacgggtttgcc
 gccagaacacaggtaagtgcctgtgtggttcccgggcctggcctctt
 tacgggttatggcccttgcgtgccttgaaattacttccacctggctgcagt
 acgtgattcttgatcccgagcttcgggttggaagtgggtgggagagtctg
 aggccttgctgcttaaggagccccctcgctctgcttgagttgaggcctg
 gcctggcgctggggccgcgcgtgcgaatctgggtggcaccttcgcgcct
 gtctcgctgctttcgataagtctctagccatttaaaatttttgatgacct
 gctgcgacgcttttttctggcaagatagtcttgtaaatgcgggccaaga
 tctgcacactggtatttctgggttttggggccgcggggcgacggggccc
 gtgcctccagcgcacatgttcggcgaggcggggcctgcgagcgcggcca
 ccgagaatcggaacggggtagtctcaagctggccgctgctctggtgcc
 tggcctcgcgccgcgtgtatcgccccccctggcgccaaggctggccc
 ggtcggcaccagttagctgagcggaaagatggccgcttcccgccctgct
 gcagggagctcaaaatggaggacgcggcgtcgggagagcggcggggtga
 gtcacccacacaaaggaaaaggccttccgctcctcagccgctcgcttcat
 gtgactccacggagtaccggcgccgctccaggcacctcgattagttctcg
 agcttttgagtagctcgctctttaggttggggggaggggttttatgcgat
 ggagtttccccacactgagtggggtggagactgaagttaggccagcttggc
 acttgatgtaatttctccttggaatttgccctttttgagtttggatcttgg
 ttcatctcaagcctcagacagtgggtcaaaagttttttcttccatttca
 ggtgtcgtagaatttetaatacgactcactatagggtgtgctgtctcatc
 attttggcaagattgccaccaagcttgtcctgcaggaggggtcgacgcc
 tctagacggggcgccgctccggatccacgggtaccgatcacatagcctt
 taattaaacactagttctatagtgacctaataatcccttttagtgagggt
 taatggccgtaggccgcgaatagggtccagacatgataagatacattg
 atgagtttggacaaaccacaactagaatgcagtgaaaaaaatgctttat
 tgtgaaatttgtgatgctattgctttatgttaaccattataagctgcaa
 taaacaagttaacaacaacaattgcattcattttatgtttcaggttcagg
 gggaggtgtgggaggttttttccgactctaggacctgcgcagctgcttgg
 cgtaatcatggtcatagctgtttcctgttttcccgctatccccccaggtg
 tctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccg
 gccaccttcccggtgccccgggtgtccccgcacgctgcgggtcggggat

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gcggggggagcgccggaccggagcggagcggcggcggtcgctgctgcc
 ccctagcggggggaggacgtaattacatccctgggggcttggggggggg
 5 ctgtccctctcaccgcggtggagctccagctttgttcgaattggggccc
 cccctcgagggtatcgatgatctataacaagaaaatatatatataata
 agttatcacgtaagtagaacatgaataacaatataattatcgtagagt
 10 taaatctttaaagtcacgtaaaagataatcatgcgtcattttgactcacg
 cggtcggttagttcaaaatcagtgacacttaccgcattgacaagcacgc
 ctacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacg
 15 gattcgcgctatttagaaagagagagcaatatccaagaatgcagcgctc
 aattttacgcagactatctttctagggttaacttagctagccttaagggc
 gctattgctgtgcgctcactgcccgctttccagtcgggaaacctgtcgt
 gccagctgcattaatgaatcgccaacgcgggggagaggcggtttgcgt
 20 atggggcgctcttcgcttctcgctcactgactcgctgcgctcggtcgt
 tcggctgcggcgagcggtatcagctcactcaaaggcggttaacggttat
 ccacagaatcaggggataaacgcaggaaagaacatgacaaaatcccttaa
 25 cgtgagtttctgctccactgagcgtcagaccccgtagaaaagatcaaagg
 atcttcttgagatccttttttctgcgcgtaactgtgctgtgcaaacaa
 aaaaaccaccgctaccagcggtggtttgtttgccggatcaagagctacca
 30 actcttttccgaaggtaactggcttcagcagagcgagataccaaatac
 tgttcttctagtgtagccgtagttaggccaccacttcaagaactctgtag
 caccgcctacatacctcgctctgctaactcctgttaccagtggtgctgcc
 35 agtggcgataaagtcgtgtcttaccgggttggaactcaagacgatagttacc
 ggataaggcgcagcggtcggtgtaacggggggttcgtgcacacagccca
 gcttgagcgaacacgacctacaccgaactgagatacctacagcgtgagcta
 40 tgagaaagcgccacgcttcccgaaaggagaaaggcgacaggtatccggt
 aagcggcgagggtcggaacagagagcgcacagggagcttccagggggaa
 acgctggtatctttatagtcctgtcggtttcgccacctctgacttgag
 45 cgtcgatttttgtgatgctcgtcagggggcgagcctatggaaaaacgc
 cagcaacgcggccttttacgggtcctggccttttgcgtggccttttgcct
 acatgagattatcaaaaaggatcttcacctagatccttttaaaataaaaa
 50 tgaagttttaaatcaatctaaagtatatatagtaaaacttggtctgacag
 tcagaagaactcgtaagaaggcgatagaaggcgatgcgctgcgaatcgg
 gagcggcgataccgtaagcacgaggaaggcggtcagccccatcgccgccca
 55 agctcttcagcaatatcacgggttagccaacgctatgtcctgatagcgggtc
 cgccacaccagcggccacagtcgatgaatccagaaaagcggccatttt
 ccaccatgatattcggaagcagggatcgccatgggtcacgacgagatcc
 tcgccgtcgggcatgctcgcttgagcctggcgaaacagttcggtggcgcc
 60 gagccctgatgctcttcgtccagatcatcctgatcgacaagacggcctt
 ccatccgagtagctgctcgctcgatgcgatgttcgcttggtggtcgaat
 gggcaggtagccggatcaagcgtatgcagccgcgcgcatgcatcagccat
 65 gatggatactttctcggcaggagcaaggtagatgacaggagatcctgcc

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ccggcacttcgcccataagcagccagtccttcccgttcagtgacaacg
 tcgagcacagctgcgcaaggaacgcccgtcgtggccagccagatagccg
 cgctgectcgtcttgagttcattcagggcacggacaggtcggtcttga
 caaaaagaacggcgccctgcgctgacagccggaacacggcgcatca
 gagcagccgattgtctgtgtgccagtcatagcgaatagcctctccac
 ccaagcggcggagaaacctgcgtgcaatccattgttcaatcataatat
 tattgaagcatttatcagggttcgtctcgtccgggtctcctcccaatgca
 tgtcaatatgtgccattagccatattattcattggttatatagcataaat
 caatatgtgctattggccattgcatacgttgatctatatcataata

The polynucleotides can optionally be joined to a vector containing a selectable marker for propagation in a host. Generally, a plasmid vector is introduced in a precipitate, such as a calcium phosphate precipitate, or in a complex with a charged lipid. If the vector is a virus, it can be packaged in vitro using an appropriate packaging cell line and then transduced into host cells.

The DNA insert should be operatively linked to an appropriate promoter. The expression constructs will further contain sites for transcription initiation, termination and, in the transcribed region, a ribosome binding site for translation. The coding portion of the mature transcripts expressed by the constructs will preferably include a translation initiating at the beginning and a termination codon (e.g., UAA, UGA or UAG) appropriately positioned at the end of the mRNA to be translated, with UAA and UAG preferred for mammalian or eukaryotic cell expression.

Expression vectors will preferably but optionally include at least one selectable marker. Such markers include, e.g., but are not limited to, ampicillin, zeocin (Sh bla gene), puromycin (pac gene), hygromycin B (hygB gene), G418/Geneticin (neo gene), mycophenolic acid, or glutamine synthetase (GS, U.S. Pat. Nos. 5,122,464; 5,770,359; 5,827,739), blasticidin (bsd gene), resistance genes for eukaryotic cell culture as well as ampicillin, zeocin (Sh bla gene), puromycin (pac gene), hygromycin B (hygB gene), G418/Geneticin (neo gene), kanamycin, spectinomycin, streptomycin, carbenicillin, bleomycin, erythromycin, polymyxin B, or tetracycline resistance genes for culturing in *E. coli* and other bacteria or prokaryotics (the above patents are entirely incorporated hereby by reference). Appropriate culture mediums and conditions for the above-described host cells are known in the art. Suitable vectors will be readily apparent to the skilled artisan. Introduction of a vector construct into a host cell can be effected by calcium phosphate transfection, DEAE-dextran mediated transfection, cationic lipid-mediated transfection, electroporation, transduction, infection or other known methods. Such methods are described in the art, such as Sambrook, supra, Chapters 1-4 and 16-18; Ausubel, supra, Chapters 1, 9, 13, 15, 16.

Expression vectors will preferably but optionally include at least one selectable cell surface marker for isolation of cells modified by the compositions and methods of the disclosure. Selectable cell surface markers of the disclosure comprise surface proteins, glycoproteins, or group of proteins that distinguish a cell or subset of cells from another defined subset of cells. Preferably the selectable cell surface marker distinguishes those cells modified by a composition or method of the disclosure from those cells that are not modified by a composition or method of the disclosure. Such

cell surface markers include, e.g., but are not limited to, "cluster of designation" or "classification determinant" proteins (often abbreviated as "CD") such as a truncated or full length form of CD19, CD271, CD34, CD22, CD20, CD33, CD52, or any combination thereof. Cell surface markers further include the suicide gene marker RQR8 (Philip B et al. Blood. 2014 Aug. 21; 124(8):1277-87).

Expression vectors will preferably but optionally include at least one selectable drug resistance marker for isolation of cells modified by the compositions and methods of the disclosure. Selectable drug resistance markers of the disclosure may comprise wild-type or mutant Neo, DHFR, TYMS, FRANCF, RAD51C, GCS, MDR1, ALDH1, NKX2.2, or any combination thereof.

At least one VHH or VCAR of the disclosure can be expressed in a modified form, such as a fusion protein, and can include not only secretion signals, but also additional heterologous functional regions. For instance, a region of additional amino acids, particularly charged amino acids, can be added to the N-terminus of a VHH or VCAR to improve stability and persistence in the host cell, during purification, or during subsequent handling and storage. Also, peptide moieties can be added to a VHH or VCAR of the disclosure to facilitate purification. Such regions can be removed prior to final preparation of a VHH or VCAR or at least one fragment thereof. Such methods are described in many standard laboratory manuals, such as Sambrook, supra, Chapters 17.29-17.42 and 18.1-18.74; Ausubel, supra, Chapters 16, 17 and 18.

Those of ordinary skill in the art are knowledgeable in the numerous expression systems available for expression of a nucleic acid encoding a protein of the disclosure. Alternatively, nucleic acids of the disclosure can be expressed in a host cell by turning on (by manipulation) in a host cell that contains endogenous DNA encoding a VHH or VCAR of the disclosure. Such methods are well known in the art, e.g., as described in U.S. Pat. Nos. 5,580,734, 5,641,670, 5,733,746, and 5,733,761, entirely incorporated herein by reference.

Illustrative of cell cultures useful for the production of the VHH or VCARs, specified portions or variants thereof, are bacterial, yeast, and mammalian cells as known in the art. Mammalian cell systems often will be in the form of monolayers of cells although mammalian cell suspensions or bioreactors can also be used. A number of suitable host cell lines capable of expressing intact glycosylated proteins have been developed in the art, and include the COS-1 (e.g., ATCC CRL 1650), COS-7 (e.g., ATCC CRL-1651), HEK293, BHK21 (e.g., ATCC CRL-10), CHO (e.g., ATCC CRL 1610) and BSC-1 (e.g., ATCC CRL-26) cell lines, Cos-7 cells, CHO cells, hep G2 cells, P3X63 μ g8.653, SP2/0-Ag14, 293 cells, HeLa cells and the like, which are readily available from, for example, American Type Culture Collection, Manassas, Va. (www.atcc.org). Preferred host cells include cells of lymphoid origin, such as myeloma and lymphoma cells. Particularly preferred host cells are P3X63Ag8.653 cells (ATCC Accession Number CRL-1580) and SP2/0-Ag14 cells (ATCC Accession Number CRL-1851). In a particularly preferred embodiment, the recombinant cell is a P3X63Ab8.653 or an SP2/0-Ag14 cell.

Expression vectors for these cells can include one or more of the following expression control sequences, such as, but not limited to, an origin of replication; a promoter (e.g., late or early SV40 promoters, the CMV promoter (U.S. Pat. Nos. 5,168,062; 5,385,839), an HSV tk promoter, a pgk (phosphoglycerate kinase) promoter, an EF-1 alpha promoter (U.S. Pat. No. 5,266,491), at least one human promoter; an enhancer, and/or processing information sites, such as ribo-

some binding sites, RNA splice sites, polyadenylation sites (e.g., an SV40 large T Ag poly A addition site), and transcriptional terminator sequences. See, e.g., Ausubel et al., supra; Sambrook, et al., supra. Other cells useful for production of nucleic acids or proteins of the present invention are known and/or available, for instance, from the American Type Culture Collection Catalogue of Cell Lines and Hybridomas (www.atcc.org) or other known or commercial sources.

When eukaryotic host cells are employed, polyadenylation or transcription terminator sequences are typically incorporated into the vector. An example of a terminator sequence is the polyadenylation sequence from the bovine growth hormone gene. Sequences for accurate splicing of the transcript can also be included. An example of a splicing sequence is the VP1 intron from SV40 (Sprague, et al., J. Virol. 45:773-781 (1983)). Additionally, gene sequences to control replication in the host cell can be incorporated into the vector, as known in the art.

Purification of a VH, VHH or VCAR

A VHH protein or VCAR can be recovered and purified from recombinant cell cultures by well-known methods including, but not limited to, protein A purification, ammonium sulfate or ethanol precipitation, acid extraction, anion or cation exchange chromatography, phosphocellulose chromatography, hydrophobic interaction chromatography, affinity chromatography, hydroxylapatite chromatography and lectin chromatography. High performance liquid chromatography ("HPLC") can also be employed for purification. See, e.g., Colligan, Current Protocols in Immunology, or Current Protocols in Protein Science, John Wiley & Sons, NY, N.Y., (1997-2001), e.g., Chapters 1, 4, 6, 8, 9, 10, each entirely incorporated herein by reference.

VHs, VHHs and/or VCARs of the disclosure include purified products, products of chemical synthetic procedures, and products produced by recombinant techniques from a prokaryotic or eukaryotic host, including, for example, *E. coli*, yeast, higher plant, insect and mammalian cells. Depending upon the host employed in a recombinant production procedure, the VCAR of the disclosure can be glycosylated or can be non-glycosylated. Such methods are described in many standard laboratory manuals, such as Sambrook, supra, Sections 17.37-17.42; Ausubel, supra, Chapters 10, 12, 13, 16, 18 and 20, Colligan, Protein Science, supra, Chapters 12-14, all entirely incorporated herein by reference.

Amino Acid Codes

The amino acids that make up VCARs of the disclosure are often abbreviated. The amino acid designations can be indicated by designating the amino acid by its single letter code, its three letter code, name, or three nucleotide codon(s) as is well understood in the art (see Alberts, B., et al., Molecular Biology of The Cell, Third Ed., Garland Publishing, Inc., New York, 1994). A VCAR of the disclosure can include one or more amino acid substitutions, deletions or additions, from spontaneous or mutations and/or human manipulation, as specified herein. Amino acids in a VCAR of the disclosure that are essential for function can be identified by methods known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (e.g., Ausubel, supra, Chapters 8, 15; Cunningham and Wells, Science 244:1081-1085 (1989)). The latter procedure introduces single alanine mutations at every residue in the molecule. The resulting mutant molecules are then tested for biological activity, such as, but not limited to, at least one neutralizing activity. Sites that are critical for VCAR binding can also be identified by structural analysis, such as crys-

tallization, nuclear magnetic resonance or photoaffinity labeling (Smith, et al., J. Mol. Biol. 224:899-904 (1992) and de Vos, et al., Science 255:306-312 (1992)).

As those of skill will appreciate, the invention includes at least one biologically active VCAR of the disclosure. Biologically active VCARs have a specific activity at least 20%, 30%, or 40%, and, preferably, at least 50%, 60%, or 70%, and, most preferably, at least 80%, 90%, or 95%-99% or more of the specific activity of the native (non-synthetic), endogenous or related and known VCAR. Methods of assaying and quantifying measures of enzymatic activity and substrate specificity are well known to those of skill in the art.

In another aspect, the disclosure relates to VHH proteins and fragments, as described herein, which are modified by the covalent attachment of an organic moiety. Such modification can produce a VCAR fragment with improved pharmacokinetic properties (e.g., increased in vivo serum half-life). The organic moiety can be a linear or branched hydrophilic polymeric group, fatty acid group, or fatty acid ester group. In particular embodiments, the hydrophilic polymeric group can have a molecular weight of about 800 to about 120,000 Daltons and can be a polyalkane glycol (e.g., polyethylene glycol (PEG), polypropylene glycol (PPG)), carbohydrate polymer, amino acid polymer or polyvinyl pyrrolidone, and the fatty acid or fatty acid ester group can comprise from about eight to about forty carbon atoms.

The modified VHH proteins and fragments of the disclosure can comprise one or more organic moieties that are covalently bonded, directly or indirectly, to the antibody. Each organic moiety that is bonded to a VHH protein or fragment of the disclosure can independently be a hydrophilic polymeric group, a fatty acid group or a fatty acid ester group. As used herein, the term "fatty acid" encompasses mono-carboxylic acids and dicarboxylic acids. A "hydrophilic polymeric group," as the term is used herein, refers to an organic polymer that is more soluble in water than in octane. For example, polylysine is more soluble in water than in octane. Thus, a VHH protein modified by the covalent attachment of polylysine is encompassed by the disclosure. Hydrophilic polymers suitable for modifying VCARs of the disclosure can be linear or branched and include, for example, polyalkane glycols (e.g., PEG, monomethoxy-polyethylene glycol (mPEG), PPG and the like), carbohydrates (e.g., dextran, cellulose, oligosaccharides, polysaccharides and the like), polymers of hydrophilic amino acids (e.g., polylysine, polyarginine, polyaspartate and the like), polyalkane oxides (e.g., polyethylene oxide, polypropylene oxide and the like) and polyvinyl pyrrolidone. Preferably, the hydrophilic polymer that modifies the VCAR of the disclosure has a molecular weight of about 800 to about 150,000 Daltons as a separate molecular entity. For example, PEG5000 and PEG 20,000, wherein the subscript is the average molecular weight of the polymer in Daltons, can be used. The hydrophilic polymeric group can be substituted with one to about six alkyl, fatty acid or fatty acid ester groups. Hydrophilic polymers that are substituted with a fatty acid or fatty acid ester group can be prepared by employing suitable methods. For example, a polymer comprising an amine group can be coupled to a carboxylate of the fatty acid or fatty acid ester, and an activated carboxylate (e.g., activated with N,N-carbonyl diimidazole) on a fatty acid or fatty acid ester can be coupled to a hydroxyl group on a polymer.

Fatty acids and fatty acid esters suitable for modifying VHH proteins of the disclosure can be saturated or can contain one or more units of unsaturation. Fatty acids that

are suitable for modifying VCARs of the disclosure include, for example, n-dodecanoate (C12, laurate), n-tetradecanoate (C14, myristate), n-octadecanoate (C18, stearate), n-eicosanoate (C20, arachidate), n-docosanoate (C22, behenate), n-triacontanoate (C30), n-tetracontanoate (C40), cis- Δ^9 -octadecanoate (C18, oleate), all cis- $\Delta^{5,8,11,14}$ -eicosatetraenoate (C20, arachidonate), octanedioic acid, tetradecanedioic acid, octadecanedioic acid, docosanedioic acid, and the like. Suitable fatty acid esters include mono-esters of dicarboxylic acids that comprise a linear or branched lower alkyl group. The lower alkyl group can comprise from one to about twelve, preferably, one to about six, carbon atoms.

The modified VHH proteins and fragments can be prepared using suitable methods, such as by reaction with one or more modifying agents. A "modifying agent" as the term is used herein, refers to a suitable organic group (e.g., hydrophilic polymer, a fatty acid, a fatty acid ester) that comprises an activating group. An "activating group" is a chemical moiety or functional group that can, under appropriate conditions, react with a second chemical group thereby forming a covalent bond between the modifying agent and the second chemical group. For example, amine-reactive activating groups include electrophilic groups, such as tosylate, mesylate, halo (chloro, bromo, fluoro, iodo), N-hydroxysuccinimidyl esters (NHS), and the like. Activating groups that can react with thiols include, for example, maleimide, iodoacetyl, acryloyl, pyridyl disulfides, 5-thiol-2-nitrobenzoic acid thiol (TNB-thiol), and the like. An aldehyde functional group can be coupled to amine- or hydrazide-containing molecules, and an azide group can react with a trivalent phosphorous group to form phosphoramidate or phosphorimide linkages. Suitable methods to introduce activating groups into molecules are known in the art (see for example, Hermanson, G. T., *Bioconjugate Techniques*, Academic Press: San Diego, Calif. (1996)). An activating group can be bonded directly to the organic group (e.g., hydrophilic polymer, fatty acid, fatty acid ester), or through a linker moiety, for example, a divalent C1-C12 group wherein one or more carbon atoms can be replaced by a heteroatom, such as oxygen, nitrogen or sulfur. Suitable linker moieties include, for example, tetraethylene glycol, $-(CH_2)_3-$, $-NH-(CH_2)_6-NH-$, $-(CH_2)_2-NH-$ and $-CH_2-O-CH_2-CH_2-O-CH_2-CH_2-O-CH-NH-$.

Modifying agents that comprise a linker moiety can be produced, for example, by reacting a mono-Boc-alkyldiamine (e.g., mono-Boc-ethylenediamine, mono-Boc-diaminohexane) with a fatty acid in the presence of 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC) to form an amide bond between the free amine and the fatty acid carboxylate. The Boc protecting group can be removed from the product by treatment with trifluoroacetic acid (TFA) to expose a primary amine that can be coupled to another carboxylate, as described, or can be reacted with maleic anhydride and the resulting product cyclized to produce an activated maleimido derivative of the fatty acid. (See, for example, Thompson, et al., WO 92/16221, the entire teachings of which are incorporated herein by reference.)

The modified VHH proteins and fragments of the disclosure can be produced by reacting VHH protein or fragment with a modifying agent. For example, the organic moieties can be bonded to the VHH protein in a non-site specific manner by employing an amine-reactive modifying agent, for example, an NHS ester of PEG. Modified VHH proteins and fragments comprising an organic moiety that is bonded to specific sites of a VHH protein of the disclosure can be prepared using suitable methods, such as reverse proteolysis (Fisch et al., *Bioconjugate Chem.*, 3:147-153 (1992); Werlen

et al., *Bioconjugate Chem.*, 5:411-417 (1994); Kumaran et al., *Protein Sci.* 6(10):2233-2241 (1997); Itoh et al., *Bioorg. Chem.*, 24(1): 59-68 (1996); Capellas et al., *Biotechnol. Bioeng.*, 56(4):456-463 (1997)), and the methods described in Hermanson, G. T., *Bioconjugate Techniques*, Academic Press: San Diego, Calif. (1996).

VH or VHH Protein Compositions Comprising Further Therapeutically Active Ingredients

VHH protein or VCAR compounds, compositions or combinations of the present disclosure can further comprise at least one of any suitable auxiliary, such as, but not limited to, diluent, binder, stabilizer, buffers, salts, lipophilic solvents, preservative, adjuvant or the like. Pharmaceutically acceptable auxiliaries are preferred. Non-limiting examples of, and methods of preparing such sterile solutions are well known in the art, such as, but limited to, Gennaro, Ed., *Remington's Pharmaceutical Sciences*, 18th Edition, Mack Publishing Co. (Easton, Pa.) 1990. Pharmaceutically acceptable carriers can be routinely selected that are suitable for the mode of administration, solubility and/or stability of the VHH protein or VCAR, fragment or variant composition as well known in the art or as described herein.

Pharmaceutical excipients and additives useful in the present composition include, but are not limited to, proteins, peptides, amino acids, lipids, and carbohydrates (e.g., sugars, including monosaccharides, di-, tri-, tetra-, and oligosaccharides; derivatized sugars, such as alditols, aldonic acids, esterified sugars and the like; and polysaccharides or sugar polymers), which can be present singly or in combination, comprising alone or in combination 1-99.99% by weight or volume. Exemplary protein excipients include serum albumin, such as human serum albumin (HSA), recombinant human albumin (rHA), gelatin, casein, and the like. Representative amino acid/protein components, which can also function in a buffering capacity, include alanine, glycine, arginine, betaine, histidine, glutamic acid, aspartic acid, cysteine, lysine, leucine, isoleucine, valine, methionine, phenylalanine, aspartame, and the like. One preferred amino acid is glycine.

Carbohydrate excipients suitable for use in the invention include, for example, monosaccharides, such as fructose, maltose, galactose, glucose, D-mannose, sorbose, and the like; disaccharides, such as lactose, sucrose, trehalose, cellobiose, and the like; polysaccharides, such as raffinose, melezitose, maltodextrins, dextrans, starches, and the like; and alditols, such as mannitol, xylitol, maltitol, lactitol, xylitol sorbitol (glucitol), myoinositol and the like. Preferred carbohydrate excipients for use in the present invention are mannitol, trehalose, and raffinose.

VH protein, VHH protein or VCAR compositions can also include a buffer or a pH-adjusting agent; typically, the buffer is a salt prepared from an organic acid or base. Representative buffers include organic acid salts, such as salts of citric acid, ascorbic acid, gluconic acid, carbonic acid, tartaric acid, succinic acid, acetic acid, or phthalic acid; Tris, tromethamine hydrochloride, or phosphate buffers. Preferred buffers for use in the present compositions are organic acid salts, such as citrate.

Additionally, VH protein, VHH protein or VCAR compositions of the disclosure can include polymeric excipients/additives, such as polyvinylpyrrolidones, ficolls (a polymeric sugar), dextrans (e.g., cyclodextrins, such as 2-hydroxypropyl- β -cyclodextrin), polyethylene glycols, flavoring agents, antimicrobial agents, sweeteners, antioxidants, antistatic agents, surfactants (e.g., polysorbates, such

as “TWEEN 20” and “TWEEN 80”), lipids (e.g., phospholipids, fatty acids), steroids (e.g., cholesterol), and chelating agents (e.g., EDTA).

These and additional known pharmaceutical excipients and/or additives suitable for use in the VH protein, VHH protein, VCAR, portion or variant compositions according to the invention are known in the art, e.g., as listed in “Remington: The Science & Practice of Pharmacy”, 19th ed., Williams & Williams, (1995), and in the “Physician’s Desk Reference”, 52nd ed., Medical Economics, Montvale, N.J. (1998), the disclosures of which are entirely incorporated herein by reference. Preferred carrier or excipient materials are carbohydrates (e.g., saccharides and alditols) and buffers (e.g., citrate) or polymeric agents. An exemplary carrier molecule is the mucopolysaccharide, hyaluronic acid, which may be useful for intraarticular delivery.

T Cell Isolation from a Leukapheresis Product

A leukapheresis product or blood may be collected from a subject at clinical site using a closed system and standard methods (e.g., a COBE Spectra Apheresis System). Preferably, the product is collected according to standard hospital or institutional Leukapheresis procedures in standard Leukapheresis collection bags. For example, in preferred embodiments of the methods of the disclosure, no additional anticoagulants or blood additives (heparin, etc.) are included beyond those normally used during leukapheresis.

Alternatively, white blood cells (WBC)/Peripheral Blood Mononuclear Cells (PBMC) (using Biosafe Sepax 2 (Closed/Automated)) or T cells (using CliniMACS® Prodigy (Closed/Automated)) may be isolated directly from whole blood. However, in certain subjects (e.g. those diagnosed and/or treated for cancer), the WBC/PBMC yield may be significantly lower when isolated from whole blood than when isolated by leukapheresis.

Either the leukapheresis procedure and/or the direct cell isolation procedure may be used for any subject of the disclosure.

The leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should be packed in insulated containers and should be kept at controlled room temperature (+19° C. to +25° C.) according to standard hospital or institutional blood collection procedures approved for use with the clinical protocol. The leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should not be refrigerated.

The cell concentration leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should not exceed 0.2×10^9 cells per mL during transportation. Intense mixing of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should be avoided.

If the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition has to be stored, e.g., overnight, it should be kept at controlled room temperature (same as above). During storage, the concentration of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should never exceed 0.2×10^9 cell per mL.

Preferably, cells of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should be stored in autologous plasma. In certain embodiments, if the cell concentration of the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition is higher than 0.2×10^9 cell per mL, the product should be diluted with autologous plasma.

Preferably, the leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition should not be

older than 24 hours when starting the labeling and separation procedure. The leukapheresis product, blood, WBC/PBMC composition and/or T-cell composition may be processed and/or prepared for cell labeling using a closed and/or automated system (e.g., CliniMACS Prodigy).

An automated system may perform additional buffy coat isolation, possibly by ficoliation, and/or washing of the cellular product (e.g., the leukapheresis product, blood, WBC/PBMC composition and/or T cell composition).

A closed and/or automated system may be used to prepare and label cells for T-Cell isolation (from, for example, the leukapheresis product, blood, WBC/PBMC composition and/or T cell composition).

Although WBC/PBMCs may be nucleofected directly (which is easier and saves additional steps), the methods of the disclosure may include first isolating T cells prior to nucleofection. The easier strategy of directly nucleofecting PBMC requires selective expansion of VCAR+ cells that is mediated via VCAR signaling, which by itself is proving to be an inferior expansion method that directly reduces the in vivo efficiency of the product by rendering T cells functionally exhausted. The product may be a heterogeneous composition of VCAR+ cells including T cells, NK cells, NKT cells, monocytes, or any combination thereof, which increases the variability in product from patient to patient and makes dosing and CRS management more difficult. Since T cells are thought to be the primary effectors in tumor suppression and killing, T cell isolation for the manufacture of an autologous product may result in significant benefits over the other more heterogeneous composition.

T cells may be isolated directly, by enrichment of labeled cells or depletion of labeled cells in a one-way labeling procedure or, indirectly, in a two-step labeling procedure. According to certain enrichment strategies of the disclosure, T cells may be collected in a Cell Collection Bag and the non-labeled cells (non-target cells) in a Negative Fraction Bag. In contrast to an enrichment strategy of the disclosure, the non-labeled cells (target cells) are collected in a Cell Collection Bag and the labeled cells (non-target cells) are collected in a Negative Fraction Bag or in the Non-Target Cell Bag, respectively. Selection reagents may include, but are not limited to, antibody-coated beads. Antibody-coated beads may either be removed prior to a modification and/or an expansion step, or, retained on the cells prior to a modification and/or an expansion step. One or more of the following non-limiting examples of cellular markers may be used to isolate T-cells: CD3, CD4, CD8, CD25, anti-biotin, CD1c, CD3/CD19, CD3/CD56, CD14, CD19, CD34, CD45RA, CD56, CD62L, CD133, CD137, CD271, CD304, IFN-gamma, TCR alpha/beta, and/or any combination thereof. Methods for the isolation of T-cells may include one or more reagents that specifically bind and/or detectably-label one or more of the following non-limiting examples of cellular markers may be used to isolate T-cells: CD3, CD4, CD8, CD25, anti-biotin, CD1c, CD3/CD19, CD3/CD56, CD14, CD19, CD34, CD45RA, CD56, CD62L, CD133, CD137, CD271, CD304, IFN-gamma, TCR alpha/beta, and/or any combination thereof. These reagents may or may not be “Good Manufacturing Practices” (“GMP”) grade. Reagents may include, but are not limited to, Thermo DynaBeads and Miltenyi CliniMACS products. Methods of isolating T-cells of the disclosure may include multiple iterations of labeling and/or isolation steps. At any point in the methods of isolating T-cells of the disclosure, unwanted cells and/or unwanted cell types may be depleted from a T cell product composition of the disclosure by positively or negatively selecting for the unwanted cells and/or unwanted

cell types. A T cell product composition of the disclosure may contain additional cell types that may express CD4, CD8, and/or another T cell marker(s).

Methods of the disclosure for nucleofection of T cells may eliminate the step of T cell isolation by, for example, a process for nucleofection of T cells in a population or composition of WBC/PBMCs that, following nucleofection, includes an isolation step or a selective expansion step via TCR signaling.

Certain cell populations may be depleted by positive or negative selection before or after T cell enrichment and/or sorting. Examples of cell compositions that may be depleted from a cell product composition may include myeloid cells, CD25+ regulatory T cells (T Regs), dendritic cells, macrophages, red blood cells, mast cells, gamma-delta T cells, natural killer (NK) cells, a Natural Killer (NK)-like cell (e.g., a Cytokine Induced Killer (CIK) cell), induced natural killer (iNK) T cells, NK T cells, B cells, or any combination thereof.

T cell product compositions of the disclosure may include CD4+ and CD8+ T-Cells. CD4+ and CD8+ T-Cells may be isolated into separate collection bags during an isolation or selection procedure. CD4+ T cells and CD8+ T cells may be further treated separately, or treated after reconstitution (combination into the same composition) at a particular ratio.

The particular ratio at which CD4+ T cells and CD8+ T cells may be reconstituted may depend upon the type and efficacy of expansion technology used, cell medium, and/or growth conditions utilized for expansion of T-cell product compositions. Examples of possible CD4+: CD8+ ratios include, but are not limited to, 50%:50%, 60%:40%, 40%:60% 75%:25% and 25%:75%.

CD8+ T cells exhibit a potent capacity for tumor cell killing, while CD4+ T cells provide many of the cytokines required to support CD8+ T cell proliferative capacity and function. Because T cells isolated from normal donors are predominantly CD4+, the T-cell product compositions are artificially adjusted in vitro with respect to the CD4+:CD8+ ratio to improve upon the ratio of CD4+ T cells to CD8+ T cells that would otherwise be present in vivo. An optimized ratio may also be used for the ex vivo expansion of the autologous T-cell product composition. In view of the artificially adjusted CD4+:CD8+ ratio of the T-cell product composition, it is important to note that the product compositions of the disclosure may be significantly different and provide significantly greater advantage than any endogenous-occurring population of T-cells.

Preferred methods for T cell isolation may include a negative selection strategy for yielding untouched pan T cell, meaning that the resultant T-cell composition includes T-cells that have not been manipulated and that contain an endogenously-occurring variety/ratio of T-cells.

Reagents that may be used for positive or negative selection include, but are not limited to, magnetic cell separation beads. Magnetic cell separation beads may or may not be removed or depleted from selected populations of CD4+ T cells, CD8+ T cells, or a mixed population of both CD4+ and CD8+ T cells before performing the next step in a T-cell isolation method of the disclosure.

T cell compositions and T cell product compositions may be prepared for cryopreservation, storage in standard T Cell Culture Medium, and/or genetic modification.

T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be cryopreserved using a standard cryopreservation method optimized for storing and

recovering human cells with high recovery, viability, phenotype, and/or functional capacity. Commercially-available cryopreservation media and/or protocols may be used. Cryopreservation methods of the disclosure may include a DMSO free cryopreservant (e.g., CryoSOfree™ DMSO-free Cryopreservation Medium) reduce freezing-related toxicity.

T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be stored in a culture medium. T cell culture media of the disclosure may be optimized for cell storage, cell genetic modification, cell phenotype and/or cell expansion. T cell culture media of the disclosure may include one or more antibiotics. Because the inclusion of an antibiotic within a cell culture media may decrease transfection efficiency and/or cell yield following genetic modification via nucleofection, the specific antibiotics (or combinations thereof) and their respective concentration(s) may be altered for optimal transfection efficiency and/or cell yield following genetic modification via nucleofection.

T cell culture media of the disclosure may include serum, and, moreover, the serum composition and concentration may be altered for optimal cell outcomes. Human AB serum is preferred over FBS/FCS for culture of T cells because, although contemplated for use in T cell culture media of the disclosure, FBS/FCS may introduce xeno-proteins. Serum may be isolated from the blood of the subject for whom the T-cell composition in culture is intended for administration, thus, a T cell culture medium of the disclosure may comprise autologous serum. Serum-free media or serum-substitute may also be used in T-cell culture media of the disclosure. In certain embodiments of the T-cell culture media and methods of the disclosure, serum-free media or serum-substitute may provide advantages over supplementing the medium with xeno-serum, including, but not limited to, healthier cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

T cell culture media may include a commercially-available cell growth media. Exemplary commercially-available cell growth media include, but are not limited to, PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium, or any combination thereof.

T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be prepared for genetic modification. Preparation of T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof for genetic modification may include cell washing and/or resuspension in a desired nucleofection buffer. Cryopreserved T-cell compositions may be thawed and prepared for genetic modification by nucleofection. Cryopreserved cells may be thawed according to standard or known protocols. Thawing and preparation of cryopreserved cells may be optimized to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. For example, Grifols Albuten (25% human albumin) may be used in the thawing and/or preparation process.

Genetic Modification of an Autologous T Cell Product Composition

T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be genetically modified using, for example, a nucleofection strategy such as electroporation. The total number of cells to be nucleofected, the total volume of the nucleofection reaction, and the precise timing of the preparation of the sample may be optimized to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

Nucleofection and/or electroporation may be accomplished using, for example, Lonza Amaxa, MaxCyte PulseAgile, Harvard Apparatus BTX, and/or Invitrogen Neon. Non-metal electrode systems, including, but not limited to, plastic polymer electrodes, may be preferred for nucleofection.

Prior to genetic modification by nucleofection, T cell compositions, T cell product compositions, unstimulated T cell compositions, resting T cell compositions or any portion thereof may be resuspended in a nucleofection buffer. Nucleofection buffers of the disclosure include commercially available nucleofection buffers. Nucleofection buffers of the disclosure may be optimized to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Nucleofection buffers of the disclosure may include, but are not limited to, PBS, HBSS, OptiMEM, BTXpress, Amaxa Nucleofector, Human T cell nucleofection buffer and any combination thereof. Nucleofection buffers of the disclosure may comprise one or more supplemental factors to yield cells that have greater viability, nucleofect with higher efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Exemplary supplemental factors include, but are not limited to, recombinant human cytokines, chemokines, interleukins and any combination thereof. Exemplary cytokines, chemokines, and interleukins include, but are not limited to, IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN- γ , IL-1 α /IL-1F1, IL-1 β /IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 β , IL-32 γ , IL-33, LAP (TGF- β 1), Lymphotoxin- α /TNF- β , TGF- β , TNF- α , TRANCE/TNFSF11/RANK L and any combination thereof. Exemplary supplemental factors include, but are not limited to, salts, minerals, metabolites or any combination thereof. Exemplary salts, minerals, and metabolites include, but are not limited to, HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl₂, Na₂HPO₄, NaH₂PO₄, Sodium lactobionate, Manitol, Sodium succinate, Sodium Chloride, CINa, Glucose, Ca(NO₃)₂, Tris/HCl, K₂HPO₄, KH₂PO₄, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, and any combination thereof. Exemplary

supplemental factors include, but are not limited to, media such as PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium and any combination thereof. Exemplary supplemental factors include, but are not limited to, inhibitors of cellular DNA sensing, metabolism, differentiation, signal transduction, the apoptotic pathway and combinations thereof. Exemplary inhibitors include, but are not limited to, inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1 β , PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 β (GSK-3 β) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK and any combination thereof. Exemplary supplemental factors include, but are not limited to, reagents that modify or stabilize one or more nucleic acids in a way to enhance cellular delivery, enhance nuclear delivery or transport, enhance the facilitated transport of nucleic acid into the nucleus, enhance degradation of epi-chromosomal nucleic acid, and/or decrease DNA-mediated toxicity. Exemplary reagents that modify or stabilize one or more nucleic acids include, but are not limited to, pH modifiers, DNA-binding proteins, lipids, phospholipids, CaPO₄, net neutral charge DNA binding peptides with or without NLS sequences, TREX1 enzyme, and any combination thereof.

Transposition reagents, including a transposon and a transposase, may be added to a nucleofection reaction of the disclosure prior to, simultaneously with, or after an addition of cells to a nucleofection buffer (optionally, contained within a nucleofection reaction vial or cuvette). Transposons of the disclosure may comprise plasmid DNA, linearized plasmid DNA, a PCR product, DOGGYBONE™ DNA, an mRNA template, a single or double-stranded DNA, a protein-nucleic acid combination or any combination thereof. Transposons of the disclosure may comprise one or more sequences that encode one or more TTAA site(s), one or more inverted terminal repeat(s) (ITRs), one or more long terminal repeat(s) (LTRs), one or more insulator(s), one or more promoter(s), one or more full-length or truncated gene(s), one or more polyA signal(s), one or more self-cleaving 2A peptide cleavage site(s), one or more internal ribosome entry site(s) (IRES), one or more enhancer(s), one or more regulator(s), one or more replication origin(s), and any combination thereof.

Transposons of the disclosure may comprise one or more sequences that encode one or more full-length or truncated gene(s). Full-length and/or truncated gene(s) introduced by transposons of the disclosure may encode one or more of a signal peptide, a VCAR, a single chain variable fragment (scFv), a hinge, a transmembrane domain, a costimulatory domain, a chimeric antigen receptor (CAR), a VCAR, a chimeric T-cell receptor (CAR-T or VCAR-T), a receptor, a ligand, a cytokine, a drug resistance gene, a tumor antigen, an allo or auto antigen, an enzyme, a protein, a peptide, a poly-peptide, a fluorescent protein, a mutein or any combination thereof.

Transposons of the disclosure may be prepared in water, TAE, TBE, PBS, HBSS, media, a supplemental factor of the disclosure or any combination thereof.

Transposons of the disclosure may be designed to optimize clinical safety and/or improve manufacturability. As a non-limiting example, transposons of the disclosure may be designed to optimize clinical safety and/or improve manu-

facturability by eliminating unnecessary sequences or regions and/or including a non-antibiotic selection marker. Transposons of the disclosure may or may not be GMP grade.

Transposase enzymes of the disclosure may be encoded by one or more sequences of plasmid DNA, mRNA, protein, protein-nucleic acid combination or any combination thereof.

Transposase enzymes of the disclosure may be prepared in water, TAE, TBE, PBS, HBSS, media, a supplemental factor of the disclosure or any combination thereof. Transposase enzymes of the disclosure or the sequences/constructs encoding or delivering them may or may not be GMP grade.

Transposons and transposase enzymes of the disclosure may be delivered to a cell by any means.

Although compositions and methods of the disclosure include delivery of a transposon and/or transposase of the disclosure to a cell by plasmid DNA (pDNA), the use of a plasmid for delivery may allow the transposon and/or transposase to be integrated into the chromosomal DNA of the cell, which may lead to continued transposase expression. Accordingly, transposon and/or transposase enzymes of the disclosure may be delivered to a cell as either mRNA or protein to remove any possibility for chromosomal integration.

Transposons and transposases of the disclosure may be pre-incubated alone or in combination with one another prior to the introduction of the transposon and/or transposase into a nucleofection reaction. The absolute amounts of each of the transposon and the transposase, as well as the relative amounts, e.g., a ratio of transposon to transposase may be optimized.

Following preparation of nucleofection reaction, optionally, in a vial or cuvette, the reaction may be loaded into a nucleofector apparatus and activated for delivery of an electric pulse according to the manufacturer's protocol. Electric pulse conditions used for delivery of a transposon and/or a transposase of the disclosure (or a sequence encoding a transposon and/or a transposase of the disclosure) to a cell may be optimized for yielding cells with enhanced viability, higher nucleofection efficiency, greater viability post-nucleofection, desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. When using Amaxa nucleofector technology, each of the various nucleofection programs for the Amaxa 2B or 4D nucleofector are contemplated.

Following a nucleofection reaction of the disclosure, cells may be gently added to a cell medium. For example, when T cells undergo the nucleofection reaction, the T cells may be added to a T cell medium. Post-nucleofection cell media of the disclosure may comprise any one or more commercially-available media. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be optimized to yield cells with greater viability, higher nucleofection efficiency, exhibit greater viability post-nucleofection, display a more desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may comprise PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium and any combination thereof. Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure)

may comprise one or more supplemental factors of the disclosure to enhance viability, nucleofection efficiency, viability post-nucleofection, cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

Exemplary supplemental factors include, but are not limited to, recombinant human cytokines, chemokines, interleukins and any combination thereof. Exemplary cytokines, chemokines, and interleukins include, but are not limited to, IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L and any combination thereof. Exemplary supplemental factors include, but are not limited to, salts, minerals, metabolites or any combination thereof. Exemplary salts, minerals, and metabolites include, but are not limited to, HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl2, Na2HPO4, NaH2PO4, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, ClNa, Glucose, Ca(NO3)2, Tris/HCl, K2HPO4, KH2PO4, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5, and any combination thereof. Exemplary supplemental factors include, but are not limited to, media such as PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium and any combination thereof. Exemplary supplemental factors include, but are not limited to, inhibitors of cellular DNA sensing, metabolism, differentiation, signal transduction, the apoptotic pathway and combinations thereof. Exemplary inhibitors include, but are not limited to, inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-I, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 β (GSK-3 β) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK and any combination thereof. Exemplary supplemental factors include, but are not limited to, reagents that modify or stabilize one or more nucleic acids in a way to enhance cellular delivery, enhance nuclear delivery or transport, enhance the facilitated transport of nucleic acid into the nucleus, enhance degradation of epi-chromosomal nucleic acid, and/or decrease DNA-mediated toxicity. Exemplary reagents that modify or stabilize one or more nucleic acids include, but are not limited to, pH modifiers, DNA-binding proteins, lipids, phospholipids, CaPO4, net neutral charge DNA binding peptides with or without NLS sequences, TREX1 enzyme, and any combination thereof.

Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be used at room temperature or pre-warmed to, for example to between 32° C. to 37° C., inclusive of the endpoints. Post-nucleofection cell media of the disclosure (including

post-nucleofection T cell media of the disclosure) may be pre-warmed to any temperature that maintains or enhances cell viability and/or expression of a transposon or portion thereof of the disclosure.

Post-nucleofection cell media of the disclosure (including post-nucleofection T cell media of the disclosure) may be contained in tissue culture flasks or dishes, G-Rex flasks, Bioreactor or cell culture bags, or any other standard receptacle. Post-nucleofection cell cultures of the disclosure (including post-nucleofection T cell cultures of the disclosure) may be kept still, or, alternatively, they may be perturbed (e.g., rocked, swirled, or shaken).

Post-nucleofection cell cultures may comprise genetically-modified cells. Post-nucleofection T cell cultures may comprise genetically-modified T cells. Genetically modified cells of the disclosure may be either rested for a defined period of time or stimulated for expansion by, for example, the addition of a T Cell Expander technology. In certain embodiments, genetically modified cells of the disclosure may be either rested for a defined period of time or immediately stimulated for expansion by, for example, the addition of a T Cell Expander technology. Genetically modified cells of the disclosure may be rested to allow them sufficient time to acclimate, time for transposition to occur, and/or time for positive or negative selection, resulting in cells with enhanced viability, higher nucleofection efficiency, greater viability post-nucleofection, desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies. Genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or more hours. In certain embodiments, genetically modified cells of the disclosure may be rested, for example, for an overnight. In certain aspects, an overnight is about 12 hours. Genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more days.

Genetically modified cells of the disclosure may be selected following a nucleofection reaction and prior to addition of an expander technology. For optimal selection of genetically-modified cells, the cells may be allowed to rest in a post-nucleofection cell medium for at least 2-14 days to facilitate identification of modified cells (e.g., differentiation of modified from non-modified cells).

As early as 24-hours post-nucleofection, expression of a CAR/VCAR and selection marker of the disclosure may be detectable in modified T cells upon successful nucleofection of a transposon of the disclosure. Due to epi-chromosomal expression of the transposon, expression of a selection marker alone may not differentiate modified T cells (those cells in which the transposon has been successfully integrated) from unmodified T cells (those cells in which the transposon was not successfully integrated). When epi-chromosomal expression of the transposon obscures the detection of modified cells by the selection marker, the nucleofected cells (both modified and unmodified cells) may be rested for a period of time (e.g., 2-14 days) to allow the cells to cease expression or lose all epi-chromosomal transposon expression. Following this extended resting period, only modified T cells should remain positive for expression of selection marker. The length of this extended resting period may be optimized for each nucleofection reaction and selection process. When epi-chromosomal expression of the transposon obscures the detection of modified cells by the selection marker, selection may be performed without this extended resting period, however, an additional selection step may be included at a later time point (e.g., either during or after the expansion stage).

Selection of genetically modified cells of the disclosure may be performed by any means. In certain embodiments of the methods of the disclosure, selection of genetically modified cells of the disclosure may be performed by isolating cells expressing a specific selection marker. Selection markers of the disclosure may be encoded by one or more sequences in the transposon. Selection markers of the disclosure may be expressed by the modified cell as a result of successful transposition (i.e., not encoded by one or more sequences in the transposon). In certain embodiments, genetically modified cells of the disclosure contain a selection marker that confers resistance to a deleterious compound of the post-nucleofection cell medium. The deleterious compound may comprise, for example, an antibiotic or a drug that, absent the resistance conferred by the selection marker to the modified cells, would result in cell death. Exemplary selection markers include, but are not limited to, wild type (WT) or mutant forms of one or more of the following genes: neo, DHFR, TYMS, ALDH, MDR1, MGMT, FANCE, RAD51C, GCS, and NKX2.2. Exemplary selection markers include, but are not limited to, a surface-expressed selection marker or surface-expressed tag may be targeted by Ab-coated magnetic bead technology or column selection, respectively. A cleavable tag such as those used in protein purification may be added to a selection marker of the disclosure for efficient column selection, washing, and elution. In certain embodiments, selection markers of the disclosure are not expressed by the modified cells (including modified T cells) endogenously and, therefore, may be useful in the physical isolation of modified cells (by, for example, cell sorting techniques). Exemplary selection markers of the disclosure are not expressed by the modified cells (including modified T cells) endogenously include, but are not limited to, full-length, mutated, or truncated forms of CD271, CD19 CD52, CD34, RQR8, CD22, CD20, CD33 and any combination thereof.

Genetically modified cells of the disclosure may be selectively expanded following a nucleofection reaction. In certain embodiments, modified T cells comprising a VCAR may be selectively expanded by VCAR stimulation. Modified T cells comprising a VCAR may be stimulated by contact with a target-covered reagent (e.g., a tumor line or a normal cell line expressing a target or expander beads covered in a target). Alternatively, modified T cells comprising a VCAR may be stimulated by contact with an irradiated tumor cell, an irradiated allogeneic normal cell, an irradiated autologous PBMC. To minimize contamination of cell product compositions of the disclosure with a target-expressing cell used for stimulation, for example, when the cell product composition may be administered directly to a subject, the stimulation may be performed using expander beads coated with VCAR target protein. Selective expansion of modified T cells comprising a VCAR by VCAR stimulation may be optimized to avoid functionally-exhausting the modified T-cells.

Selected genetically-modified cells of the disclosure may be cryopreserved, rested for a defined period of time, or stimulated for expansion by the addition of a Cell Expander technology. Selected genetically-modified cells of the disclosure may be cryopreserved, rested for a defined period of time, or immediately stimulated for expansion by the addition of a Cell Expander technology. When the selected genetically-modified cells are T cells, the T cells may be stimulated for expansion by the addition of a T-Cell Expander technology. Selected genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22,

23, 24, or more hours. In certain embodiments, selected genetically modified cells of the disclosure may be rested, for example, for an overnight. In certain aspects, an overnight is about 12 hours. Selected genetically modified cells of the disclosure may be rested, for example, for 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 or more days. Selected genetically modified cells of the disclosure may be rested for any period of time resulting in cells with enhanced viability, higher nucleofection efficiency, greater viability post-nucleofection, desirable cell phenotype, and/or greater/faster expansion upon addition of expansion technologies.

Selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be cryopreserved using any standard cryopreservation method, which may be optimized for storing and/or recovering human cells with high recovery, viability, phenotype, and/or functional capacity. Cryopreservation methods of the disclosure may include commercially-available cryopreservation media and/or protocols.

A transposition efficiency of selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be assessed by any means. For example, prior to the application of an expander technology, expression of the transposon by selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be measured by fluorescence-activated cell sorting (FACS). Determination of a transposition efficiency of selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may include determining a percentage of selected cells expressing the transposon (e.g., a VCAR). Alternatively, or in addition, a purity of T cells, a Mean Fluorescence Intensity (MFI) of the transposon expression (e.g., CAR expression), an ability of a VCAR (delivered in the transposon) to mediate degranulation and/or killing of a target cell expressing the VCAR ligand, and/or a phenotype of selected genetically-modified cells (including selected genetically-modified T cells of the disclosure) may be assessed by any means.

Cell product compositions of the disclosure may be released for administration to a subject upon meeting certain release criteria. Exemplary release criteria may include, but are not limited to, a particular percentage of modified, selected and/or expanded T cells expressing detectable levels of a VCAR on the cell surface.

Genetic Modification of an Autologous T Cell Product Composition

Genetically-modified cells (including genetically-modified T cells) of the disclosure may be expanded using an expander technology. Expander technologies of the disclosure may comprise a commercially-available expander technology. Exemplary expander technologies of the disclosure include stimulation a genetically-modified T cell of the disclosure via the TCR. While all means for stimulation of a genetically-modified T cell of the disclosure are contemplated, stimulation a genetically-modified T cell of the disclosure via the TCR is a preferred method, yielding a product with a superior level of killing capacity.

To stimulate a genetically-modified T cell of the disclosure via the TCR, Thermo Expander DynaBeads may be used at a 3:1 bead to T cell ratio. If the expander beads are not biodegradable, the beads may be removed from the expander composition. For example, the beads may be removed from the expander composition after about 5 days. To stimulate a genetically-modified T cell of the disclosure via the TCR, a Miltenyi T Cell Activation/Expansion Reagent may be used. To stimulate a genetically-modified T cell of the disclosure via the TCR, StemCell Technologies'

ImmunoCult Human CD3/CD28 or CD3/CD28/CD2 T Cell Activator Reagent may be used. This technology may be preferred since the soluble tetrameric antibody complexes would degrade after a period and would not require removal from the process.

Artificial antigen presenting cells (APCs) may be engineered to co-express the target antigen and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure. Artificial APCs may comprise or may be derived from a tumor cell line (including, for example, the immortalized myelogenous leukemia line K562) and may be engineered to co-express multiple costimulatory molecules or technologies (such as CD28, 4-1BBL, CD64, mBIL-21, mBIL-15, CAR target molecule, etc.). When artificial APCs of the disclosure are combined with costimulatory molecules, conditions may be optimized to prevent the development or emergence of an undesirable phenotype and functional capacity, namely terminally-differentiated effector T cells.

Irradiated PBMCs (auto or allo) may express some target antigens, such as CD19, and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure. Alternatively, or in addition, irradiated tumor cells may express some target antigens and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure.

Plate-bound and/or soluble anti-CD3, anti-CD2 and/or anti-CD28 stimulate may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure.

Antigen-coated beads may display target protein and may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure. Alternatively, or in addition, expander beads coated with a VCAR target protein may be used to stimulate a cell or T-cell of the disclosure through a TCR and/or VCAR of the disclosure.

Expansion methods drawn to stimulation of a cell or T-cell of the disclosure through the TCR or VCAR and via surface-expressed CD2, CD3, CD28, 4-1BB, and/or other markers on genetically-modified T cells.

An expansion technology may be applied to a cell of the disclosure immediately post-nucleofection until approximately 24 hours post-nucleofection. While various cell media may be used during an expansion procedure, a desirable T Cell Expansion Media of the disclosure may yield cells with, for example, greater viability, cell phenotype, total expansion, or greater capacity for in vivo persistence, engraftment, and/or CAR-mediated killing. Cell media of the disclosure may be optimized to improve/enhance expansion, phenotype, and function of genetically-modified cells of the disclosure. A preferred phenotype of expanded T cells may include a mixture of T stem cell memory, T central, and T effector memory cells. Expander Dynabeads may yield mainly central memory T cells which may lead to superior performance in the clinic.

Exemplary T cell expansion media of the disclosure may include, in part or in total, PBS, HBSS, OptiMEM, DMEM, RPMI 1640, AIM-V, X-VIVO 15, CellGro DC Medium, CTS OpTimizer T Cell Expansion SFM, TexMACS Medium, PRIME-XV T Cell Expansion Medium, ImmunoCult-XF T Cell Expansion Medium, or any combination thereof. T cell expansion media of the disclosure may further include one or more supplemental factors. Supplemental factors that may be included in a T cell expansion media of the disclosure enhance viability, cell phenotype, total expansion, or increase capacity for in vivo persistence, engraftment, and/or VCAR-mediated killing. Supplemental factors

that may be included in a T cell expansion media of the disclosure include, but are not limited to, recombinant human cytokines, chemokines, and/or interleukins such as IL2, IL7, IL12, IL15, IL21, IL1, IL3, IL4, IL5, IL6, IL8, CXCL8, IL9, IL10, IL11, IL13, IL14, IL16, IL17, IL18, IL19, IL20, IL22, IL23, IL25, IL26, IL27, IL28, IL29, IL30, IL31, IL32, IL33, IL35, IL36, GM-CSF, IFN-gamma, IL-1 alpha/IL-1F1, IL-1 beta/IL-1F2, IL-12 p70, IL-12/IL-35 p35, IL-13, IL-17/IL-17A, IL-17A/F Heterodimer, IL-17F, IL-18/IL-1F4, IL-23, IL-24, IL-32, IL-32 beta, IL-32 gamma, IL-33, LAP (TGF-beta 1), Lymphotoxin-alpha/TNF-beta, TGF-beta, TNF-alpha, TRANCE/TNFSF11/RANK L, or any combination thereof. Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, salts, minerals, and/or metabolites such as HEPES, Nicotinamide, Heparin, Sodium Pyruvate, L-Glutamine, MEM Non-Essential Amino Acid Solution, Ascorbic Acid, Nucleosides, FBS/FCS, Human serum, serum-substitute, anti-biotics, pH adjusters, Earle's Salts, 2-Mercaptoethanol, Human transferrin, Recombinant human insulin, Human serum albumin, Nucleofector PLUS Supplement, KCL, MgCl2, Na2HPO4, NAH2PO4, Sodium lactobionate, Mannitol, Sodium succinate, Sodium Chloride, C1Na, Glucose, Ca(NO3)2, Tris/HCl, K2HPO4, KH2PO4, Polyethylenimine, Poly-ethylene-glycol, Poloxamer 188, Poloxamer 181, Poloxamer 407, Poly-vinylpyrrolidone, Pop313, Crown-5 or any combination thereof. Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, inhibitors of cellular DNA sensing, metabolism, differentiation, signal transduction, and/or the apoptotic pathway such as inhibitors of TLR9, MyD88, IRAK, TRAF6, TRAF3, IRF-7, NF-KB, Type 1 Interferons, pro-inflammatory cytokines, cGAS, STING, Sec5, TBK1, IRF-3, RNA pol III, RIG-1, IPS-1, FADD, RIP1, TRAF3, AIM2, ASC, Caspase1, Pro-IL1B, PI3K, Akt, Wnt3A, inhibitors of glycogen synthase kinase-3 β (GSK-3 β) (e.g., TWS119), Bafilomycin, Chloroquine, Quinacrine, AC-YVAD-CMK, Z-VAD-FMK, Z-IETD-FMK, or any combination thereof.

Supplemental factors that may be included in a T cell expansion media of the disclosure include, but are not limited to, reagents that modify or stabilize nucleic acids in a way to enhance cellular delivery, enhance nuclear delivery or transport, enhance the facilitated transport of nucleic acid into the nucleus, enhance degradation of epi-chromosomal nucleic acid, and/or decrease DNA-mediated toxicity, such as pH modifiers, DNA-binding proteins, lipids, phospholipids, CaPO4, net neutral charge DNA binding peptides with or without NLS sequences, TREX1 enzyme, or any combination thereof.

Genetically-modified cells of the disclosure may be selected during the expansion process by the use of selectable drugs or compounds. For example, in certain embodiments, when a transposon of the disclosure may encode a selection marker that confers to genetically-modified cells resistance to a drug added to the culture medium, selection may occur during the expansion process and may require approximately 1-14 days of culture for selection to occur. Examples of drug resistance genes that may be used as selection markers encoded by a transposon of the disclosure, include, but are not limited to, wild type (WT) or mutant forms of the genes neo, DHFR, TYMS, ALDH, MDR1, MGMT, FANCF, RAD51C, GCS, NKX2.2, or any combination thereof. Examples of corresponding drugs or compounds that may be added to the culture medium to which a selection marker may confer resistance include, but are not limited to, G418, Puromycin, Ampicillin, Kanamycin,

Methotrexate, Mephalan, Temozolomide, Vincristine, Etoposide, Doxorubicin, Bendamustine, Fludarabine, Are-dia (Pamidronate Disodium), Becenun (Carmustine), BiCNU (Carmustine), Bortezomib, Carfilzomib, Carmubris (Carmustine), Carmustine, Clafen (Cyclophosphamide), Cyclophosphamide, Cytosan (Cyclophosphamide), Daratumumab, Darzalex (Daratumumab), Doxil (Doxorubicin Hydrochloride Liposome), Doxorubicin Hydrochloride Liposome, Dox-SL (Doxorubicin Hydrochloride Liposome), Elotuzumab, Emluciti (Elotuzumab), Evacet (Doxorubicin Hydrochloride Liposome), Farydak (Panobinostat), Ixazomib Citrate, Kyprolis (Carfilzomib), Lenalidomide, LipoDox (Doxorubicin Hydrochloride Liposome), Mozobil (Plerixafor), Neosar (Cyclophosphamide), Ninlaro (Ixazomib Citrate), Pamidronate Disodium, Panobinostat, Plerixafor, Pomalidomide, Pomalyst (Pomalidomide), Revlimid (Lenalidomide), Synovir (Thalidomide), Thalidomide, Thalomid (Thalidomide), Velcade (Bortezomib), Zoledronic Acid, Zometa (Zoledronic Acid), or any combination thereof.

A T-Cell Expansion process of the disclosure may occur in a cell culture bag in a WAVE Bioreactor, a G-Rex flask, or in any other suitable container and/or reactor.

A cell or T-cell culture of the disclosure may be kept steady, rocked, swirled, or shaken.

A cell or T-cell expansion process of the disclosure may optimize certain conditions, including, but not limited to culture duration, cell concentration, schedule for T cell medium addition/removal, cell size, total cell number, cell phenotype, purity of cell population, percentage of genetically-modified cells in growing cell population, use and composition of supplements, the addition/removal of expander technologies, or any combination thereof.

A cell or T-cell expansion process of the disclosure may continue until a predefined endpoint prior to formulation of the resultant expanded cell population. For example, a cell or T-cell expansion process of the disclosure may continue for a predetermined amount of time: at least, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24 hours; at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30 days; at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 weeks; at least 1, 2, 3, 4, 5, 6, months, or at least 1 year. A cell or T-cell expansion process of the disclosure may continue until the resultant culture reaches a predetermined overall cell density: 1, 10, 100, 1000, 104, 105, 106, 107, 108, 109, 1010 cells per volume (μ l, ml, L) or any density in between. A cell or T-cell expansion process of the disclosure may continue until the genetically-modified cells of a resultant culture demonstrate a predetermined level of expression of a transposon of the disclosure: 1%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or 100% or any percentage in between of a threshold level of expression (a minimum, maximum or mean level of expression indicating the resultant genetically-modified cells are clinically-efficacious). A cell or T-cell expansion process of the disclosure may continue until the proportion of genetically-modified cells of a resultant culture to the proportion of unmodified cells reaches a predetermined threshold: at least 1:10, 1:9, 1:8, 1:7, 1:6, 1:5, 1:4, 1:3, 1:2, 1:1, 2:1, 2:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1 10:1 or any ratio in between.

Analysis of Genetically-Modified Autologous T Cells for Release

A percentage of genetically-modified cells may be assessed during or after an expansion process of the disclosure. Cellular expression of a transposon by a genetically-modified cell of the disclosure may be measured by fluorescence-activated cell sorting (FACS). For example, FACS

may be used to determine a percentage of cells or T cells expressing a VCAR of the disclosure. Alternatively, or in addition, a purity of genetically-modified cells or T cells, the Mean Fluorescence Intensity (MFI) of a VCAR expressed by a genetically-modified cell or T cell of the disclosure, an ability of the VCAR to mediate degranulation and/or killing of a target cell expressing the VCAR ligand, and/or a phenotype of VCAR+ T cells may be assessed.

Compositions of the disclosure intended for administration to a subject may be required to meet one or more “release criteria” that indicate that the composition is safe and efficacious for formulation as a pharmaceutical product and/or administration to a subject. Release criteria may include a requirement that a composition of the disclosure (e.g., a T-cell product of the disclosure) comprises a particular percentage of T cells expressing detectable levels of a VCAR of the disclosure on their cell surface.

The expansion process should be continued until a specific criterion has been met (e.g., achieving a certain total number of cells, achieving a particular population of memory cells, achieving a population of a specific size).

Certain criterion signal a point at which the expansion process should end. For example, cells should be formulated, reactivated, or cryopreserved once they reach a cell size of 300 fL (otherwise, cells reaching a size above this threshold may start to die). Cryopreservation immediately once a population of cells reaches an average cell size of less than 300 fL may yield better cell recovery upon thawing and culture because the cells haven’t yet reached a fully quiescent state prior to cryopreservation (a fully quiescent size is approximately 180 fL). Prior to expansion, T cells of the disclosure may have a cell size of about 180 fL, but may more than quadruple their cell size to approximately 900 fL at 3 days post-expansion. Over the next 6-12 days, the population of T-cells will slowly decrease cell size to full quiescence at 180 fL.

A process for preparing a cell population for formulation may include, but is not limited to the steps of, concentrating the cells of the cell population, washing the cells, and/or further selection of the cells via drug resistance or magnetic bead sorting against a particular surface-expressed marker. A process for preparing a cell population for formulation may further include a sorting step to ensure the safety and purity of the final product. For example, if a tumor cell from a patient has been used to stimulate a genetically-modified T-cell of the disclosure or that have been genetically-modified in order to stimulate a genetically-modified T-cell of the disclosure that is being prepared for formulation, it is critical that no tumor cells from the patient are included in the final product.

Cell Product Infusion and/or Cryopreservation for Infusion

A pharmaceutical formulation of the disclosure may be distributed into bags for infusion, cryopreservation, and/or storage.

A pharmaceutical formulation of the disclosure may be cryopreserved using a standard protocol and, optionally, an infusible cryopreservation medium. For example, a DMSO free cryopreservant (e.g., CryoSOfree™ DMSO-free Cryopreservation Medium) may be used to reduce freezing-related toxicity. A cryopreserved pharmaceutical formulation of the disclosure may be stored for infusion to a patient at a later date. An effective treatment may require multiple administrations of a pharmaceutical formulation of the disclosure and, therefore, pharmaceutical formulations may be packaged in pre-aliquoted “doses” that may be stored frozen but separated for thawing of individual doses.

A pharmaceutical formulation of the disclosure may be stored at room temperature. An effective treatment may require multiple administrations of a pharmaceutical formulation of the disclosure and, therefore, pharmaceutical formulations may be packaged in pre-aliquoted “doses” that may be stored together but separated for administration of individual doses.

A pharmaceutical formulation of the disclosure may be archived for subsequent re-expansion and/or selection for generation of additional doses to the same patient in the case of an allogenic therapy who may need an administration at a future date following, for example, a remission and relapse of a condition.

Formulations

As noted above, the disclosure provides for stable formulations, which preferably comprise a phosphate buffer with saline or a chosen salt, as well as preserved solutions and formulations containing a preservative as well as multi-use preserved formulations suitable for pharmaceutical or veterinary use, comprising at least one VCAR in a pharmaceutically acceptable formulation. Preserved formulations contain at least one known preservative or optionally selected from the group consisting of at least one phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, phenylmercuric nitrite, phenoxyethanol, formaldehyde, chlorobutanol, magnesium chloride (e.g., hexahydrate), alkylparaben (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal, polymers, or mixtures thereof in an aqueous diluent. Any suitable concentration or mixture can be used as known in the art, such as about 0.0015%, or any range, value, or fraction therein. Non-limiting examples include, no preservative, about 0.1-2% m-cresol (e.g., 0.2, 0.3, 0.4, 0.5, 0.9, 1.0%), about 0.1-3% benzyl alcohol (e.g., 0.5, 0.9, 1.1, 1.5, 1.9, 2.0, 2.5%), about 0.001-0.5% thimerosal (e.g., 0.005, 0.01), about 0.001-2.0% phenol (e.g., 0.05, 0.25, 0.28, 0.5, 0.9, 1.0%), 0.0005-1.0% alkylparaben(s) (e.g., 0.00075, 0.0009, 0.001, 0.002, 0.005, 0.0075, 0.009, 0.01, 0.02, 0.05, 0.075, 0.09, 0.1, 0.2, 0.3, 0.5, 0.75, 0.9, 1.0%), and the like.

As noted above, the invention provides an article of manufacture, comprising packaging material and at least one vial comprising a solution of at least one VCAR with the prescribed buffers and/or preservatives, optionally in an aqueous diluent, wherein said packaging material comprises a label that indicates that such solution can be held over a period of 1, 2, 3, 4, 5, 6, 9, 12, 18, 20, 24, 30, 36, 40, 48, 54, 60, 66, 72 hours or greater. The invention further comprises an article of manufacture, comprising packaging material, a first vial comprising lyophilized at least one VCAR, and a second vial comprising an aqueous diluent of prescribed buffer or preservative, wherein said packaging material comprises a label that instructs a patient to reconstitute the at least one VCAR in the aqueous diluent to form a solution that can be held over a period of twenty-four hours or greater.

The at least one VCAR used in accordance with the present invention can be produced by recombinant means, including from mammalian cell or transgenic preparations, or can be purified from other biological sources, as described herein or as known in the art.

The range of at least one VCAR in the product of the present invention includes amounts yielding upon reconstitution, if in a wet/dry system, concentrations from about 1.0 pg/ml to about 1000 mg/ml, although lower and higher concentrations are operable and are dependent on the intended delivery vehicle, e.g., solution formulations will

differ from transdermal patch, pulmonary, transmucosal, or osmotic or micro pump methods.

Preferably, the aqueous diluent optionally further comprises a pharmaceutically acceptable preservative. Preferred preservatives include those selected from the group consisting of phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, alkylparaben (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal, or mixtures thereof. The concentration of preservative used in the formulation is a concentration sufficient to yield an antimicrobial effect. Such concentrations are dependent on the preservative selected and are readily determined by the skilled artisan.

Other excipients, e.g., isotonicity agents, buffers, antioxidants, and preservative enhancers, can be optionally and preferably added to the diluent. An isotonicity agent, such as glycerin, is commonly used at known concentrations. A physiologically tolerated buffer is preferably added to provide improved pH control. The formulations can cover a wide range of pHs, such as from about pH 4 to about pH 10, and preferred ranges from about pH 5 to about pH 9, and a most preferred range of about 6.0 to about 8.0. Preferably, the formulations of the present invention have a pH between about 6.8 and about 7.8. Preferred buffers include phosphate buffers, most preferably, sodium phosphate, particularly, phosphate buffered saline (PBS).

Other additives, such as a pharmaceutically acceptable solubilizers like Tween 20 (polyoxyethylene (20) sorbitan monolaurate), Tween 40 (polyoxyethylene (20) sorbitan monopalmitate), Tween 80 (polyoxyethylene (20) sorbitan monooleate), Pluronic F68 (polyoxyethylene polyoxypropylene block copolymers), and PEG (polyethylene glycol) or non-ionic surfactants, such as polysorbate 20 or 80 or poloxamer 184 or 188, Pluronic® polyols, other block copolymers, and chelators, such as EDTA and EGTA, can optionally be added to the formulations or compositions to reduce aggregation. These additives are particularly useful if a pump or plastic container is used to administer the formulation. The presence of pharmaceutically acceptable surfactant mitigates the propensity for the protein to aggregate.

The formulations of the present invention can be prepared by a process which comprises mixing at least one VCAR and a preservative selected from the group consisting of phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, alkylparaben, (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate and thimerosal or mixtures thereof in an aqueous diluent. Mixing the at least one VCAR and preservative in an aqueous diluent is carried out using conventional dissolution and mixing procedures. To prepare a suitable formulation, for example, a measured amount of at least one VCAR in buffered solution is combined with the desired preservative in a buffered solution in quantities sufficient to provide the protein and preservative at the desired concentrations. Variations of this process would be recognized by one of ordinary skill in the art. For example, the order the components are added, whether additional additives are used, the temperature and pH at which the formulation is prepared, are all factors that can be optimized for the concentration and means of administration used.

The claimed formulations can be provided to patients as clear solutions or as dual vials comprising a vial of lyophilized at least one VCAR that is reconstituted with a second vial containing water, a preservative and/or excipients, preferably, a phosphate buffer and/or saline and a chosen salt, in an aqueous diluent. Either a single solution

vial or dual vial requiring reconstitution can be reused multiple times and can suffice for a single or multiple cycles of patient treatment and thus can provide a more convenient treatment regimen than currently available.

The present claimed articles of manufacture are useful for administration over a period ranging from immediate to twenty-four hours or greater. Accordingly, the presently claimed articles of manufacture offer significant advantages to the patient. Formulations of the invention can optionally be safely stored at temperatures of from about 2° C. to about 40° C. and retain the biological activity of the protein for extended periods of time, thus allowing a package label indicating that the solution can be held and/or used over a period of 6, 12, 18, 24, 36, 48, 72, or 96 hours or greater. If preserved diluent is used, such label can include use up to 1-12 months, one-half, one and a half, and/or two years.

The solutions of at least one VCAR of the invention can be prepared by a process that comprises mixing at least one VCAR in an aqueous diluent. Mixing is carried out using conventional dissolution and mixing procedures. To prepare a suitable diluent, for example, a measured amount of at least one VCAR in water or buffer is combined in quantities sufficient to provide the protein and, optionally, a preservative or buffer at the desired concentrations. Variations of this process would be recognized by one of ordinary skill in the art. For example, the order the components are added, whether additional additives are used, the temperature and pH at which the formulation is prepared, are all factors that can be optimized for the concentration and means of administration used.

The claimed products can be provided to patients as clear solutions or as dual vials comprising a vial of lyophilized at least one VCAR that is reconstituted with a second vial containing the aqueous diluent. Either a single solution vial or dual vial requiring reconstitution can be reused multiple times and can suffice for a single or multiple cycles of patient treatment and thus provides a more convenient treatment regimen than currently available.

The claimed products can be provided indirectly to patients by providing to pharmacies, clinics, or other such institutions and facilities, clear solutions or dual vials comprising a vial of lyophilized at least one VCAR that is reconstituted with a second vial containing the aqueous diluent. The clear solution in this case can be up to one liter or even larger in size, providing a large reservoir from which smaller portions of the at least one VCAR solution can be retrieved one or multiple times for transfer into smaller vials and provided by the pharmacy or clinic to their customers and/or patients.

Recognized devices comprising single vial systems include pen-injector devices for delivery of a solution, such as BD Pens, BD Autojector®, Humaject®, NovoPen®, B-D® Pen, AutoPen®, and OptiPen®, GenotropinPen®, GenotroNorm Pen®, Humatro Pen®, Reco-Pen®, Roferon Pen®, Biojector®, Iject®, J-tip Needle-Free Injector®, Intraject®, Medi-Ject®, e.g., as made or developed by Becton Dickinson (Franklin Lakes, N.J., www.bectondickenson.com), Disetronic (Burgdorf, Switzerland, www.disetronic.com), Bioject, Portland, Oreg. (www.bioject.com); National Medical Products, Weston Medical (Peterborough, UK, www.weston-medical.com), Medi-Ject Corp (Minneapolis, Minn., www.mediject.com), and similarly suitable devices. Recognized devices comprising a dual vial system include those pen-injector systems for reconstituting a lyophilized drug in a cartridge for delivery of the reconstituted solution, such as the HumatroPen®. Examples of other devices suit-

able include pre-filled syringes, auto-injectors, needle free injectors and needle free IV infusion sets.

The products presently claimed include packaging material. The packaging material provides, in addition to the information required by the regulatory agencies, the conditions under which the product can be used. The packaging material of the present invention provides instructions to the patient to reconstitute at least one VCAR in the aqueous diluent to form a solution and to use the solution over a period of 2-24 hours or greater for the two vial, wet/dry, product. For the single vial, solution product, the label indicates that such solution can be used over a period of 2-24 hours or greater. The presently claimed products are useful for human pharmaceutical product use.

The formulations of the present invention can be prepared by a process that comprises mixing at least one VCAR and a selected buffer, preferably, a phosphate buffer containing saline or a chosen salt. Mixing at least one VCAR and buffer in an aqueous diluent is carried out using conventional dissolution and mixing procedures. To prepare a suitable formulation, for example, a measured amount of at least one VCAR in water or buffer is combined with the desired buffering agent in water in quantities sufficient to provide the protein and buffer at the desired concentrations. Variations of this process would be recognized by one of ordinary skill in the art. For example, the order the components are added, whether additional additives are used, the temperature and pH at which the formulation is prepared, are all factors that can be optimized for the concentration and means of administration used.

The claimed stable or preserved formulations can be provided to patients as clear solutions or as dual vials comprising a vial of lyophilized VCAR that is reconstituted with a second vial containing a preservative or buffer and excipients in an aqueous diluent. Either a single solution vial or dual vial requiring reconstitution can be reused multiple times and can suffice for a single or multiple cycles of patient treatment and thus provides a more convenient treatment regimen than currently available.

Other formulations or methods of stabilizing the VCAR may result in other than a clear solution of lyophilized powder comprising the VCAR. Among non-clear solutions are formulations comprising particulate suspensions, said particulates being a composition containing the VCAR in a structure of variable dimension and known variously as a microsphere, microparticle, nanoparticle, nanosphere, or liposome. Such relatively homogenous, essentially spherical, particulate formulations containing an active agent can be formed by contacting an aqueous phase containing the active agent and a polymer and a nonaqueous phase followed by evaporation of the nonaqueous phase to cause the coalescence of particles from the aqueous phase as taught in U.S. Pat. No. 4,589,330. Porous microparticles can be prepared using a first phase containing active agent and a polymer dispersed in a continuous solvent and removing said solvent from the suspension by freeze-drying or dilution-extraction-precipitation as taught in U.S. Pat. No. 4,818,542. Preferred polymers for such preparations are natural or synthetic copolymers or polymers selected from the group consisting of gelatin agar, starch, arabinogalactan, albumin, collagen, polyglycolic acid, polylactic acid, glycolide-L(-) lactide poly(epsilon-caprolactone), poly(epsilon-caprolactone-CO-lactic acid), poly(epsilon-caprolactone-CO-glycolic acid), poly(beta-hydroxy butyric acid), polyethylene oxide, polyethylene, poly(alkyl-2-cyanoacrylate), poly(hydroxyethyl methacrylate), polyamides, poly(amino acids), poly(2-hydroxyethyl DL-aspartamide), poly

(ester urea), poly(L-phenylalanine/ethylene glycol/1,6-diisocyanatohexane) and poly(methyl methacrylate). Particularly preferred polymers are polyesters, such as polyglycolic acid, polylactic acid, glycolide-L(-) lactide poly(epsilon-caprolactone), poly(epsilon-caprolactone-CO-lactic acid), and poly(epsilon-caprolactone-CO-glycolic acid). Solvents useful for dissolving the polymer and/or the active include: water, hexafluoroisopropanol, methylenechloride, tetrahydrofuran, hexane, benzene, or hexafluoroacetone sesquihydrate. The process of dispersing the active containing phase with a second phase may include pressure forcing said first phase through an orifice in a nozzle to affect droplet formation.

Dry powder formulations may result from processes other than lyophilization, such as by spray drying or solvent extraction by evaporation or by precipitation of a crystalline composition followed by one or more steps to remove aqueous or nonaqueous solvent. Preparation of a spray-dried VCAR preparation is taught in U.S. Pat. No. 6,019,968. The VCAR-based dry powder compositions may be produced by spray drying solutions or slurries of the VCAR and, optionally, excipients, in a solvent under conditions to provide a respirable dry powder. Solvents may include polar compounds, such as water and ethanol, which may be readily dried. VCAR stability may be enhanced by performing the spray drying procedures in the absence of oxygen, such as under a nitrogen blanket or by using nitrogen as the drying gas. Another relatively dry formulation is a dispersion of a plurality of perforated microstructures dispersed in a suspension medium that typically comprises a hydrofluoroalkane propellant as taught in WO 9916419. The stabilized dispersions may be administered to the lung of a patient using a metered dose inhaler. Equipment useful in the commercial manufacture of spray dried medicaments are manufactured by Buchi Ltd. or Niro Corp.

At least one VCAR in either the stable or preserved formulations or solutions described herein, can be administered to a patient in accordance with the present invention via a variety of delivery methods including SC or IM injection; transdermal, pulmonary, transmucosal, implant, osmotic pump, cartridge, micro pump, or other means appreciated by the skilled artisan, as well-known in the art.

Therapeutic Applications

The present invention also provides a method for modulating or treating a disease, in a cell, tissue, organ, animal, or patient, as known in the art or as described herein, using at least one VCAR of the present invention, e.g., administering or contacting the cell, tissue, organ, animal, or patient with a therapeutic effective amount of VCAR. The present invention also provides a method for modulating or treating a disease, in a cell, tissue, organ, animal, or patient including, but not limited to, a malignant disease.

The present invention also provides a method for modulating or treating at least one malignant disease in a cell, tissue, organ, animal or patient, including, but not limited to, at least one of leukemia, acute leukemia, acute lymphoblastic leukemia (ALL), acute lymphocytic leukemia, B-cell, T-cell or FAB ALL, acute myeloid leukemia (AML), acute myelogenous leukemia, chronic myelocytic leukemia (CML), chronic lymphocytic leukemia (CLL), hairy cell leukemia, myelodysplastic syndrome (MDS), a lymphoma, Hodgkin's disease, a malignant lymphoma, non-Hodgkin's lymphoma, Burkitt's lymphoma, multiple myeloma, Kaposi's sarcoma, colorectal carcinoma, pancreatic carcinoma, nasopharyngeal carcinoma, malignant histiocytosis, paraneoplastic syndrome/hypercalcemia of malignancy, solid tumors, bladder cancer, breast cancer, colorectal cancer,

endometrial cancer, head cancer, neck cancer, hereditary nonpolyposis cancer, Hodgkin's lymphoma, liver cancer, lung cancer, non-small cell lung cancer, ovarian cancer, pancreatic cancer, prostate cancer, renal cell carcinoma, testicular cancer, adenocarcinomas, sarcomas, malignant melanoma, hemangioma, metastatic disease, cancer related bone resorption, cancer related bone pain, and the like.

Any method of the present invention can comprise administering an effective amount of a composition or pharmaceutical composition comprising at least one VCAR to a cell, tissue, organ, animal or patient in need of such modulation, treatment or therapy. Such a method can optionally further comprise co-administration or combination therapy for treating such diseases or disorders, wherein the administering of said at least one VCAR, specified portion or variant thereof, further comprises administering, before concurrently, and/or after, at least one selected from at least one of an alkylating agent, an a mitotic inhibitor, and a radiopharmaceutical. Suitable dosages are well known in the art. See, e.g., Wells et al., eds., *Pharmacotherapy Handbook*, 2nd Edition, Appleton and Lange, Stamford, Conn. (2000); *PDR Pharmacopoeia*, Tarascon Pocket Pharmacopoeia 2000, Deluxe Edition, Tarascon Publishing, Loma Linda, Calif. (2000); *Nursing 2001 Handbook of Drugs*, 21st edition, Springhouse Corp., Springhouse, Pa., 2001; *Health Professional's Drug Guide 2001*, ed., Shannon, Wilson, Stang, Prentice-Hall, Inc., Upper Saddle River, N.J. each of which references are entirely incorporated herein by reference.

Preferred doses can optionally include about 0.1-99 and/or 100-500 mg/kg/administration, or any range, value or fraction thereof, or to achieve a serum concentration of about 0.1-5000 µg/ml serum concentration per single or multiple administration, or any range, value or fraction thereof. A preferred dosage range for the VCAR of the present invention is from about 1 mg/kg, up to about 3, about 6 or about 12 mg/kg of body weight of the patient.

Alternatively, the dosage administered can vary depending upon known factors, such as the pharmacodynamic characteristics of the particular agent, and its mode and route of administration; age, health, and weight of the recipient; nature and extent of symptoms, kind of concurrent treatment, frequency of treatment, and the effect desired. Usually a dosage of active ingredient can be about 0.1 to 100 milligrams per kilogram of body weight. Ordinarily 0.1 to 50, and preferably, 0.1 to 10 milligrams per kilogram per administration or in sustained release form is effective to obtain desired results.

As a non-limiting example, treatment of humans or animals can be provided as a one-time or periodic dosage of at least one VCAR of the present invention about 0.1 to 100 mg/kg or any range, value or fraction thereof per day, on at least one of day 1-40, or, alternatively or additionally, at least one of week 1-52, or, alternatively or additionally, at least one of 1-20 years, or any combination thereof, using single, infusion or repeated doses.

Dosage forms (composition) suitable for internal administration generally contain from about 0.001 milligram to about 500 milligrams of active ingredient per unit or container. In these pharmaceutical compositions the active ingredient will ordinarily be present in an amount of about 0.5-99.999% by weight based on the total weight of the composition.

For parenteral administration, the VCAR can be formulated as a solution, suspension, emulsion, particle, powder, or lyophilized powder in association, or separately provided, with a pharmaceutically acceptable parenteral vehicle. Examples of such vehicles are water, saline, Ringer's solu-

tion, dextrose solution, and about 1-10% human serum albumin. Liposomes and nonaqueous vehicles, such as fixed oils, can also be used. The vehicle or lyophilized powder can contain additives that maintain isotonicity (e.g., sodium chloride, mannitol) and chemical stability (e.g., buffers and preservatives). The formulation is sterilized by known or suitable techniques.

Suitable pharmaceutical carriers are described in the most recent edition of Remington's *Pharmaceutical Sciences*, A. Osol, a standard reference text in this field. *Alternative Administration*

Many known and developed modes can be used according to the present invention for administering pharmaceutically effective amounts of at least one VCAR according to the present invention. While pulmonary administration is used in the following description, other modes of administration can be used according to the present invention with suitable results. VCARS of the present invention can be delivered in a carrier, as a solution, emulsion, colloid, or suspension, or as a dry powder, using any of a variety of devices and methods suitable for administration by inhalation or other modes described here within or known in the art.

Parenteral Formulations and Administration

Formulations for parenteral administration can contain as common excipients sterile water or saline, polyalkylene glycols, such as polyethylene glycol, oils of vegetable origin, hydrogenated naphthalenes and the like. Aqueous or oily suspensions for injection can be prepared by using an appropriate emulsifier or humidifier and a suspending agent, according to known methods. Agents for injection can be a non-toxic, non-orally administrable diluting agent, such as aqueous solution, a sterile injectable solution or suspension in a solvent. As the usable vehicle or solvent, water, Ringer's solution, isotonic saline, etc. are allowed; as an ordinary solvent or suspending solvent, sterile involatile oil can be used. For these purposes, any kind of involatile oil and fatty acid can be used, including natural or synthetic or semisynthetic fatty oils or fatty acids; natural or synthetic or semi-synthetic mono- or di- or tri-glycerides. Parental administration is known in the art and includes, but is not limited to, conventional means of injections, a gas pressured needleless injection device as described in U.S. Pat. No. 5,851,198, and a laser perforator device as described in U.S. Pat. No. 5,839,446 entirely incorporated herein by reference.

Alternative Delivery

The invention further relates to the administration of at least one VCAR by parenteral, subcutaneous, intramuscular, intravenous, intraarticular, intrabronchial, intraabdominal, intracapsular, intracartilaginous, intracavitary, intracelial, intracerebellar, intracerebroventricular, intracolic, intracervical, intragastric, intrahepatic, intramyocardial, intraosteal, intrapelvic, intrapericardiac, intraperitoneal, intrapleural, intraprostatic, intrapulmonary, intrarectal, intrarenal, intraretinal, intraspinal, intrasynovial, intrathoracic, intrauterine, intravesical, intralesional, bolus, vaginal, rectal, buccal, sublingual, intranasal, or transdermal means. At least one VCAR composition can be prepared for use for parenteral (subcutaneous, intramuscular or intravenous) or any other administration particularly in the form of liquid solutions or suspensions; for use in vaginal or rectal administration particularly in semisolid forms, such as, but not limited to, creams and suppositories; for buccal, or sublingual administration, such as, but not limited to, in the form of tablets or capsules; or intranasally, such as, but not limited to, the form of powders, nasal drops or aerosols or certain agents; or transdermally, such as not limited to a gel, ointment, lotion, suspension or patch delivery system with

chemical enhancers such as dimethyl sulfoxide to either modify the skin structure or to increase the drug concentration in the transdermal patch (Junginger, et al. In "Drug Permeation Enhancement;" Hsieh, D. S., Eds., pp. 59-90 (Marcel Dekker, Inc. New York 1994, entirely incorporated herein by reference), or with oxidizing agents that enable the application of formulations containing proteins and peptides onto the skin (WO 98/53847), or applications of electric fields to create transient transport pathways, such as electroporation, or to increase the mobility of charged drugs through the skin, such as iontophoresis, or application of ultrasound, such as sonophoresis (U.S. Pat. Nos. 4,309,989 and 4,767,402) (the above publications and patents being entirely incorporated herein by reference).

Pulmonary/Nasal Administration

For pulmonary administration, preferably, at least one VCAR composition is delivered in a particle size effective for reaching the lower airways of the lung or sinuses. According to the invention, at least one VCAR can be delivered by any of a variety of inhalation or nasal devices known in the art for administration of a therapeutic agent by inhalation. These devices capable of depositing aerosolized formulations in the sinus cavity or alveoli of a patient include metered dose inhalers, nebulizers, dry powder generators, sprayers, and the like. Other devices suitable for directing the pulmonary or nasal administration of VCARS are also known in the art. All such devices can use formulations suitable for the administration for the dispensing of VCAR in an aerosol. Such aerosols can be comprised of either solutions (both aqueous and nonaqueous) or solid particles.

Metered dose inhalers like the Ventolin metered dose inhaler, typically use a propellant gas and require actuation during inspiration (See, e.g., WO 94/16970, WO 98/35888). Dry powder inhalers like Turbuhaler™ (Astra), Rotahaler® (Glaxo), Diskus® (Glaxo), Spiros™ inhaler (Dura), devices marketed by Inhale Therapeutics, and the Spinhaler® powder inhaler (Fisons), use breath-actuation of a mixed powder (U.S. Pat. No. 4,668,218 Astra, EP 237507 Astra, WO 97/25086 Glaxo, WO 94/08552 Dura, U.S. Pat. No. 5,458,135 Inhale, WO 94/06498 Fisons, entirely incorporated herein by reference). Nebulizers like AERx™ Aradigm, the Ultravent® nebulizer (Mallinckrodt), and the Acorn II® nebulizer (Marquest Medical Products) (U.S. Pat. No. 5,404,871 Aradigm, WO 97/22376), the above references entirely incorporated herein by reference, produce aerosols from solutions, while metered dose inhalers, dry powder inhalers, etc. generate small particle aerosols. These specific examples of commercially available inhalation devices are intended to be a representative of specific devices suitable for the practice of this invention, and are not intended as limiting the scope of the invention.

Preferably, a composition comprising at least one VCAR is delivered by a dry powder inhaler or a sprayer. There are several desirable features of an inhalation device for administering at least one VCAR of the present invention. For example, delivery by the inhalation device is advantageously reliable, reproducible, and accurate. The inhalation device can optionally deliver small dry particles, e.g., less than about 10 μm , preferably about 1-5 μm , for good respirability.

Administration of VCAR Compositions as a Spray

A spray including VCAR composition can be produced by forcing a suspension or solution of at least one VCAR through a nozzle under pressure. The nozzle size and configuration, the applied pressure, and the liquid feed rate can be chosen to achieve the desired output and particle size. An

electrospray can be produced, for example, by an electric field in connection with a capillary or nozzle feed. Advantageously, particles of at least one VCAR composition delivered by a sprayer have a particle size less than about 10 μm , preferably, in the range of about 1 μm to about 5 μm , and, most preferably, about 2 μm to about 3 μm .

Formulations of at least one VCAR composition suitable for use with a sprayer typically include VCAR composition in an aqueous solution at a concentration of about 0.1 mg to about 100 mg of at least one VCAR composition per ml of solution or mg/gm, or any range, value, or fraction therein. The formulation can include agents, such as an excipient, a buffer, an isotonicity agent, a preservative, a surfactant, and, preferably, zinc. The formulation can also include an excipient or agent for stabilization of the VCAR composition, such as a buffer, a reducing agent, a bulk protein, or a carbohydrate. Bulk proteins useful in formulating VCAR compositions include albumin, protamine, or the like. Typical carbohydrates useful in formulating VCAR compositions include sucrose, mannitol, lactose, trehalose, glucose, or the like. The VCAR composition formulation can also include a surfactant, which can reduce or prevent surface-induced aggregation of the VCAR composition caused by atomization of the solution in forming an aerosol. Various conventional surfactants can be employed, such as polyoxyethylene fatty acid esters and alcohols, and polyoxyethylene sorbitol fatty acid esters. Amounts will generally range between 0.001 and 14% by weight of the formulation. Especially preferred surfactants for purposes of this invention are polyoxyethylene sorbitan monooleate, polysorbate 80, polysorbate 20, or the like. Additional agents known in the art for formulation of a protein, such as VCARS, or specified portions or variants, can also be included in the formulation. Administration of VCAR Compositions by a Nebulizer

VCAR compositions of the invention can be administered by a nebulizer, such as jet nebulizer or an ultrasonic nebulizer. Typically, in a jet nebulizer, a compressed air source is used to create a high-velocity air jet through an orifice. As the gas expands beyond the nozzle, a low-pressure region is created, which draws a solution of VCAR composition through a capillary tube connected to a liquid reservoir. The liquid stream from the capillary tube is sheared into unstable filaments and droplets as it exits the tube, creating the aerosol. A range of configurations, flow rates, and baffle types can be employed to achieve the desired performance characteristics from a given jet nebulizer. In an ultrasonic nebulizer, high-frequency electrical energy is used to create vibrational, mechanical energy, typically employing a piezoelectric transducer. This energy is transmitted to the formulation of VCAR composition either directly or through a coupling fluid, creating an aerosol including the VCAR composition. Advantageously, particles of VCAR composition delivered by a nebulizer have a particle size less than about 10 μm , preferably, in the range of about 1 μm to about 5 μm , and, most preferably, about 2 μm to about 3 μm .

Formulations of at least one VCAR suitable for use with a nebulizer, either jet or ultrasonic, typically include a concentration of about 0.1 mg to about 100 mg of at least one VCAR per ml of solution. The formulation can include agents, such as an excipient, a buffer, an isotonicity agent, a preservative, a surfactant, and, preferably, zinc. The formulation can also include an excipient or agent for stabilization of the at least one VCAR composition, such as a buffer, a reducing agent, a bulk protein, or a carbohydrate. Bulk proteins useful in formulating at least one VCAR compositions include albumin, protamine, or the like. Typical carbohydrates useful in formulating at least one VCAR include

sucrose, mannitol, lactose, trehalose, glucose, or the like. The at least one VCAR formulation can also include a surfactant, which can reduce or prevent surface-induced aggregation of the at least one VCAR caused by atomization of the solution in forming an aerosol. Various conventional surfactants can be employed, such as polyoxyethylene fatty acid esters and alcohols, and polyoxyethylene sorbital fatty acid esters. Amounts will generally range between about 0.001 and 4% by weight of the formulation. Especially preferred surfactants for purposes of this invention are polyoxyethylene sorbitan mono-oleate, polysorbate 80, polysorbate 20, or the like. Additional agents known in the art for formulation of a protein, such as VCAR, can also be included in the formulation.

Administration of VCAR Compositions by a Metered Dose Inhaler

In a metered dose inhaler (MDI), a propellant, at least one VCAR, and any excipients or other additives are contained in a canister as a mixture including a liquefied compressed gas. Actuation of the metering valve releases the mixture as an aerosol, preferably containing particles in the size range of less than about 10 μm , preferably, about 1 μm to about 5 μm , and, most preferably, about 2 μm to about 3 μm . The desired aerosol particle size can be obtained by employing a formulation of VCAR composition produced by various methods known to those of skill in the art, including jet-milling, spray drying, critical point condensation, or the like. Preferred metered dose inhalers include those manufactured by 3M or Glaxo and employing a hydrofluorocarbon propellant. Formulations of at least one VCAR for use with a metered-dose inhaler device will generally include a finely divided powder containing at least one VCAR as a suspension in a non-aqueous medium, for example, suspended in a propellant with the aid of a surfactant. The propellant can be any conventional material employed for this purpose, such as chlorofluorocarbon, a hydrochlorofluorocarbon, a hydrofluorocarbon, or a hydrocarbon, including trichlorofluoromethane, dichlorodifluoromethane, dichlorotetrafluoroethanol and 1,1,1,2-tetrafluoroethane, HFA-134a (hydrofluoroalkane-134a), HFA-227 (hydrofluoroalkane-227), or the like. Preferably, the propellant is a hydrofluorocarbon. The surfactant can be chosen to stabilize the at least one VCAR as a suspension in the propellant, to protect the active agent against chemical degradation, and the like. Suitable surfactants include sorbitan trioleate, soya lecithin, oleic acid, or the like. In some cases, solution aerosols are preferred using solvents, such as ethanol. Additional agents known in the art for formulation of a protein can also be included in the formulation. One of ordinary skill in the art will recognize that the methods of the current invention can be achieved by pulmonary administration of at least one VCAR composition via devices not described herein.

Oral Formulations and Administration

Formulations for oral administration rely on the co-administration of adjuvants (e.g., resorcinols and nonionic surfactants, such as polyoxyethylene oleyl ether and n-hexadecylpolyethylene ether) to increase artificially the permeability of the intestinal walls, as well as the co-administration of enzymatic inhibitors (e.g., pancreatic trypsin inhibitors, diisopropylfluorophosphate (DFF) and trasylol) to inhibit enzymatic degradation. Formulations for delivery of hydrophilic agents including proteins and VCARs and a combination of at least two surfactants intended for oral, buccal, mucosal, nasal, pulmonary, vaginal transmembrane, or rectal administration are taught in U.S. Pat. No. 6,309,663. The active constituent compound of the solid-type dosage form for oral administration can be mixed with at

least one additive, including sucrose, lactose, cellulose, mannitol, trehalose, raffinose, maltitol, dextran, starches, agar, arginates, chitins, chitosans, pectins, gum tragacanth, gum arabic, gelatin, collagen, casein, albumin, synthetic or semisynthetic polymer, and glyceride. These dosage forms can also contain other type(s) of additives, e.g., inactive diluting agent, lubricant, such as magnesium stearate, paraben, preserving agent, such as sorbic acid, ascorbic acid, α -tocopherol, antioxidant such as cysteine, disintegrator, binder, thickener, buffering agent, sweetening agent, flavoring agent, perfuming agent, etc.

Tablets and pills can be further processed into enteric-coated preparations. The liquid preparations for oral administration include emulsion, syrup, elixir, suspension and solution preparations allowable for medical use. These preparations can contain inactive diluting agents ordinarily used in said field, e.g., water. Liposomes have also been described as drug delivery systems for insulin and heparin (U.S. Pat. No. 4,239,754). More recently, microspheres of artificial polymers of mixed amino acids (proteinoids) have been used to deliver pharmaceuticals (U.S. Pat. No. 4,925,673). Furthermore, carrier compounds described in U.S. Pat. Nos. 5,879,681 and 5,871,753 and used to deliver biologically active agents orally are known in the art.

Mucosal Formulations and Administration

A formulation for orally administering a bioactive agent encapsulated in one or more biocompatible polymer or copolymer excipients, preferably, a biodegradable polymer or copolymer, affording microcapsules which due to the proper size of the resultant microcapsules results in the agent reaching and being taken up by the folliculi lymphatic aggregati, otherwise known as the "Peyer's patch," or "GALT" of the animal without loss of effectiveness due to the agent having passed through the gastrointestinal tract. Similar folliculi lymphatic aggregati can be found in the bronchi tubes (BALT) and the large intestine. The above-described tissues are referred to in general as mucosally associated lymphoreticular tissues (MALT). For absorption through mucosal surfaces, compositions and methods of administering at least one VCAR include an emulsion comprising a plurality of submicron particles, a mucoadhesive macromolecule, a bioactive peptide, and an aqueous continuous phase, which promotes absorption through mucosal surfaces by achieving mucoadhesion of the emulsion particles (U.S. Pat. No. 5,514,670). Mucous surfaces suitable for application of the emulsions of the present invention can include corneal, conjunctival, buccal, sublingual, nasal, vaginal, pulmonary, stomachic, intestinal, and rectal routes of administration. Formulations for vaginal or rectal administration, e.g., suppositories, can contain as excipients, for example, polyalkyleneglycols, vaseline, cocoa butter, and the like. Formulations for intranasal administration can be solid and contain as excipients, for example, lactose or can be aqueous or oily solutions of nasal drops. For buccal administration, excipients include sugars, calcium stearate, magnesium stearate, pregelatinized starch, and the like (U.S. Pat. No. 5,849,695).

Transdermal Formulations and Administration

For transdermal administration, the at least one VCAR is encapsulated in a delivery device, such as a liposome or polymeric nanoparticles, microparticle, microcapsule, or microspheres (referred to collectively as microparticles unless otherwise stated). A number of suitable devices are known, including microparticles made of synthetic polymers, such as polyhydroxy acids, such as polylactic acid, polyglycolic acid and copolymers thereof, polyorthoesters, polyanhydrides, and polyphosphazenes, and natural poly-

mers, such as collagen, polyamino acids, albumin and other proteins, alginate and other polysaccharides, and combinations thereof (U.S. Pat. No. 5,814,599).

Prolonged Administration and Formulations

It can be desirable to deliver the compounds of the present invention to the subject over prolonged periods of time, for example, for periods of one week to one year from a single administration. Various slow release, depot or implant dosage forms can be utilized. For example, a dosage form can contain a pharmaceutically acceptable non-toxic salt of the compounds that has a low degree of solubility in body fluids, for example, (a) an acid addition salt with a polybasic acid, such as phosphoric acid, sulfuric acid, citric acid, tartaric acid, tannic acid, pamoic acid, alginic acid, polyglutamic acid, naphthalene mono- or di-sulfonic acids, polygalacturonic acid, and the like; (b) a salt with a polyvalent metal cation, such as zinc, calcium, bismuth, barium, magnesium, aluminum, copper, cobalt, nickel, cadmium and the like, or with an organic cation formed from e.g., N,N'-dibenzylethylenediamine or ethylenediamine; or (c) combinations of (a) and (b), e.g., a zinc tannate salt. Additionally, the compounds of the present invention or, preferably, a relatively insoluble salt, such as those just described, can be formulated in a gel, for example, an aluminum monostearate gel with, e.g., sesame oil, suitable for injection. Particularly preferred salts are zinc salts, zinc tannate salts, pamoate salts, and the like. Another type of slow release depot formulation for injection would contain the compound or salt dispersed for encapsulation in a slow degrading, non-toxic, non-antigenic polymer, such as a polylactic acid/polyglycolic acid polymer for example as described in U.S. Pat. No. 3,773,919. The compounds or, preferably, relatively insoluble salts, such as those described above, can also be formulated in cholesterol matrix silastic pellets, particularly for use in animals. Additional slow release, depot or implant formulations, e.g., gas or liquid liposomes, are known in the literature (U.S. Pat. No. 5,770,222 and "Sustained and Controlled Release Drug Delivery Systems", J. R. Robinson ed., Marcel Dekker, Inc., N.Y., 1978).

Infusion of Modified Cells as Adoptive Cell Therapy

The disclosure provides modified cells that express one or more CARs and/or VCARs of the disclosure that have been selected and/or expanded for administration to a subject in need thereof. Modified cells of the disclosure may be formulated for storage at any temperature including room temperature and body temperature. Modified cells of the disclosure may be formulated for cryopreservation and subsequent thawing. Modified cells of the disclosure may be formulated in a pharmaceutically acceptable carrier for direct administration to a subject from sterile packaging. Modified cells of the disclosure may be formulated in a pharmaceutically acceptable carrier with an indicator of cell viability and/or CAR/VCAR expression level to ensure a minimal level of cell function and CAR/VCAR expression. Modified cells of the disclosure may be formulated in a pharmaceutically acceptable carrier at a prescribed density with one or more reagents to inhibit further expansion and/or prevent cell death.

Inducible Proapoptotic Polypeptides

Inducible proapoptotic polypeptides of the disclosure are superior to existing inducible polypeptides because the inducible proapoptotic polypeptides of the disclosure are far less immunogenic. While inducible proapoptotic polypeptides of the disclosure are recombinant polypeptides, and, therefore, non-naturally occurring, the sequences that are recombined to produce the inducible proapoptotic polypeptides of the disclosure do not comprise non-human

sequences that the host human immune system could recognize as "non-self" and, consequently, induce an immune response in the subject receiving an inducible proapoptotic polypeptide of the disclosure, a cell comprising the inducible proapoptotic polypeptide or a composition comprising the inducible proapoptotic polypeptide or the cell comprising the inducible proapoptotic polypeptide.

The disclosure provides inducible proapoptotic polypeptides comprising a ligand binding region, a linker, and a proapoptotic peptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. In certain embodiments, the proapoptotic peptide is a caspase polypeptide. In certain embodiments, the caspase polypeptide is a caspase 9 polypeptide. In certain embodiments, the caspase 9 polypeptide is a truncated caspase 9 polypeptide. Inducible proapoptotic polypeptides of the disclosure may be non-naturally occurring.

Caspase polypeptides of the disclosure include, but are not limited to, caspase 1, caspase 2, caspase 3, caspase 4, caspase 5, caspase 6, caspase 7, caspase 8, caspase 9, caspase 10, caspase 11, caspase 12, and caspase 14. Caspase polypeptides of the disclosure include, but are not limited to, those caspase polypeptides associated with apoptosis including caspase 2, caspase 3, caspase 6, caspase 7, caspase 8, caspase 9, and caspase 10. Caspase polypeptides of the disclosure include, but are not limited to, those caspase polypeptides that initiate apoptosis, including caspase 2, caspase 8, caspase 9, and caspase 10. Caspase polypeptides of the disclosure include, but are not limited to, those caspase polypeptides that execute apoptosis, including caspase 3, caspase 6, and caspase 7.

Caspase polypeptides of the disclosure may be encoded by an amino acid or a nucleic acid sequence having one or more modifications compared to a wild type amino acid or a nucleic acid sequence. The nucleic acid sequence encoding a caspase polypeptide of the disclosure may be codon optimized. The one or more modifications to an amino acid and/or nucleic acid sequence of a caspase polypeptide of the disclosure may increase an interaction, a cross-linking, a cross-activation, or an activation of the caspase polypeptide of the disclosure compared to a wild type amino acid or a nucleic acid sequence. Alternatively, or in addition, the one or more modifications to an amino acid and/or nucleic acid sequence of a caspase polypeptide of the disclosure may decrease the immunogenicity of the caspase polypeptide of the disclosure compared to a wild type amino acid or a nucleic acid sequence.

Caspase polypeptides of the disclosure may be truncated compared to a wild type caspase polypeptide. For example, a caspase polypeptide may be truncated to eliminate a sequence encoding a Caspase Activation and Recruitment Domain (CARD) to eliminate or minimize the possibility of activating a local inflammatory response in addition to initiating apoptosis in the cell comprising an inducible caspase polypeptide of the disclosure. The nucleic acid sequence encoding a caspase polypeptide of the disclosure may be spliced to form a variant amino acid sequence of the caspase polypeptide of the disclosure compared to a wild type caspase polypeptide. Caspase polypeptides of the disclosure may be encoded by recombinant and/or chimeric sequences. Recombinant and/or chimeric caspase polypeptides of the disclosure may include sequences from one or more different caspase polypeptides. Alternatively, or in addition, recombinant and/or chimeric caspase polypeptides of the disclosure may include sequences from one or more

species (e.g. a human sequence and a non-human sequence). Caspase polypeptides of the disclosure may be non-naturally occurring.

The ligand binding region of an inducible proapoptotic polypeptide of the disclosure may include any polypeptide sequence that facilitates or promotes the dimerization of a first inducible proapoptotic polypeptide of the disclosure with a second inducible proapoptotic polypeptide of the disclosure, the dimerization of which activates or induces cross-linking of the proapoptotic polypeptides and initiation of apoptosis in the cell.

The ligand-binding ("dimerization") region may comprise any polypeptide or functional domain thereof that will allow for induction using an endogenous or non-naturally-occurring ligand (i.e. and induction agent), for example, a non-naturally-occurring synthetic ligand. The ligand-binding region may be internal or external to the cellular membrane, depending upon the nature of the inducible proapoptotic polypeptide and the choice of ligand (i.e. induction agent). A wide variety of ligand-binding polypeptides and functional domains thereof, including receptors, are known. Ligand-binding regions of the disclosure may include one or more sequences from a receptor. Of particular interest are ligand-binding regions for which ligands (for example, small organic ligands) are known or may be readily produced. These ligand-binding regions or receptors may include, but are not limited to, the FKBP and cyclophilin receptors, the steroid receptors, the tetracycline receptor, and the like, as well as "non-naturally-occurring" receptors, which can be obtained from antibodies, particularly the heavy or light chain subunit, mutated sequences thereof, random amino acid sequences obtained by stochastic procedures, combinatorial syntheses, and the like. In certain embodiments, the ligand-binding region is selected from the group consisting of a FKBP ligand-binding region, a cyclophilin receptor ligand-binding region, a steroid receptor ligand-binding region, a cyclophilin receptors ligand-binding region, and a tetracycline receptor ligand-binding region.

The ligand-binding regions comprising one or more receptor domain(s) may be at least about 50 amino acids, and fewer than about 350 amino acids, usually fewer than 200 amino acids, either as the endogenous domain or truncated active portion thereof. The binding region may, for example, be small (<25 kDa, to allow efficient transfection in viral vectors), monomeric, nonimmunogenic, have synthetically accessible, cell permeable, nontoxic ligands that can be configured for dimerization.

The ligand-binding regions comprising one or more receptor domain(s) may be intracellular or extracellular depending upon the design of the inducible proapoptotic polypeptide and the availability of an appropriate ligand (i.e. induction agent). For hydrophobic ligands, the binding region can be on either side of the membrane, but for hydrophilic ligands, particularly protein ligands, the binding region will usually be external to the cell membrane, unless there is a transport system for internalizing the ligand in a form in which it is available for binding. For an intracellular receptor, the inducible proapoptotic polypeptide or a transposon or vector comprising the inducible proapoptotic polypeptide may encode a signal peptide and transmembrane domain 5' or 3' of the receptor domain sequence or may have a lipid attachment signal sequence 5' of the receptor domain sequence. Where the receptor domain is between the signal peptide and the transmembrane domain, the receptor domain will be extracellular.

Antibodies and antibody subunits, e.g., heavy or light chain, particularly fragments, more particularly all or part of

the variable region, or fusions of heavy and light chain to create high-affinity binding, can be used as a ligand binding region of the disclosure. Antibodies that are contemplated include ones that are an ectopically expressed human product, such as an extracellular domain that would not trigger an immune response and generally not expressed in the periphery (i.e., outside the CNS/brain area). Such examples, include, but are not limited to low affinity nerve growth factor receptor (LNGFR), and embryonic surface proteins (i.e., carcinoembryonic antigen). Yet further, antibodies can be prepared against haptenic molecules, which are physiologically acceptable, and the individual antibody subunits screened for binding affinity. The cDNA encoding the subunits can be isolated and modified by deletion of the constant region, portions of the variable region, mutagenesis of the variable region, or the like, to obtain a binding protein domain that has the appropriate affinity for the ligand. In this way, almost any physiologically acceptable haptenic compound can be employed as the ligand or to provide an epitope for the ligand. Instead of antibody units, endogenous receptors can be employed, where the binding region or domain is known and there is a useful or known ligand for binding.

For multimerizing the receptor, the ligand for the ligand-binding region/receptor domains of the inducible proapoptotic polypeptides may be multimeric in the sense that the ligand can have at least two binding sites, with each of the binding sites capable of binding to a ligand receptor region (i.e. a ligand having a first binding site capable of binding the ligand-binding region of a first inducible proapoptotic polypeptide and a second binding site capable of binding the ligand-binding region of a second inducible proapoptotic polypeptide, wherein the ligand-binding regions of the first and the second inducible proapoptotic polypeptides are either identical or distinct). Thus, as used herein, the term "multimeric ligand binding region" refers to a ligand-binding region of an inducible proapoptotic polypeptide of the disclosure that binds to a multimeric ligand. Multimeric ligands of the disclosure include dimeric ligands. A dimeric ligand of the disclosure may have two binding sites capable of binding to the ligand receptor domain. In certain embodiments, multimeric ligands of the disclosure are a dimer or higher order oligomer, usually not greater than about tetrameric, of small synthetic organic molecules, the individual molecules typically being at least about 150 Da and less than about 5 kDa, usually less than about 3 kDa. A variety of pairs of synthetic ligands and receptors can be employed. For example, in embodiments involving endogenous receptors, dimeric FK506 can be used with an FKBP12 receptor, dimerized cyclosporin A can be used with the cyclophilin receptor, dimerized estrogen with an estrogen receptor, dimerized glucocorticoids with a glucocorticoid receptor, dimerized tetracycline with the tetracycline receptor, dimerized vitamin D with the vitamin D receptor, and the like. Alternatively higher orders of the ligands, e.g., trimeric can be used. For embodiments involving non-naturally occurring receptors, e.g., antibody subunits, modified antibody subunits, single chain antibodies comprised of heavy and light chain variable regions in tandem, separated by a flexible linker, or modified receptors, and mutated sequences thereof, and the like, any of a large variety of compounds can be used. A significant characteristic of the units comprising a multimeric ligand of the disclosure is that each binding site is able to bind the receptor with high affinity, and preferably, that they are able to be dimerized chemically. Also, methods are available to balance the hydrophobicity/hydrophilicity of

the ligands so that they are able to dissolve in serum at functional levels, yet diffuse across plasma membranes for most applications.

Activation of inducible proapoptotic polypeptides of the disclosure may be accomplished through, for example, chemically induced dimerization (CID) mediated by an induction agent to produce a conditionally controlled protein or polypeptide. Proapoptotic polypeptides of the disclosure not only inducible, but the induction of these polypeptides is also reversible, due to the degradation of the labile dimerizing agent or administration of a monomeric competitive inhibitor.

In certain embodiments, the ligand-binding region comprises a FKBP12 binding protein 12 (FKBP12) polypeptide. In certain embodiments, the ligand-binding region comprises a FKBP12 polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In certain embodiments, in which the ligand binding region comprises a FKBP12 polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V), the induction agent may comprise AP1903, a synthetic drug (CAS Index Name: 2-Piperidinecarboxylic acid, 1-[(2S)-1-oxo-2-(3,4,5-trimethoxyphenyl)butyl]-, 1,2-ethanediylbis[imino(2-oxo-2,1-ethanediyl)oxy-3,1-phenylene[(1R)-3-(3,4-dimethoxyphenyl)propylidene]]ester, [2S-[1(R*),2R*[S*[S*[1(R*),2R*]]]]-(9C1) CAS Registry Number: 195514-63-7; Molecular Formula: C₇₈H₉₈N₄O₂₀; Molecular Weight: 1411.65)). In certain embodiments, in which the ligand-binding region comprises a FKBP12 polypeptide having a substitution of valine (V) for phenylalanine (F) at position 36 (F36V), the induction agent may comprise AP20187 (CAS Registry Number: 195514-80-8 and Molecular Formula: C₈₂H₁₀₇N₅O₂₀). In certain embodiments, the induction agent is an AP20187 analog, such as, for example, AP1510. As used herein, the induction agents AP20187, AP1903 and AP1510 may be used interchangeably.

AP1903 API is manufactured by Alphora Research Inc. and AP1903 Drug Product for Injection is made by Formattech Inc. It is formulated as a 5 mg/mL solution of AP1903 in a 25% solution of the non-ionic solubilizer Solutol HS 15 (250 mg/mL, BASF). At room temperature, this formulation is a clear, slightly yellow solution. Upon refrigeration, this formulation undergoes a reversible phase transition, resulting in a milky solution. This phase transition is reversed upon re-warming to room temperature. The fill is 2.33 mL in a 3 mL glass vial (approximately 10 mg AP1903 for Injection total per vial). Upon determining a need to administer AP1903, patients may be, for example, administered a single fixed dose of AP1903 for Injection (0.4 mg/kg) via IV infusion over 2 hours, using a non-DEHP, non-ethylene oxide sterilized infusion set. The dose of AP1903 is calculated individually for all patients, and is not be recalculated unless body weight fluctuates by $\geq 10\%$. The calculated dose is diluted in 100 mL in 0.9% normal saline before infusion. In a previous Phase I study of AP1903, 24 healthy volunteers were treated with single doses of AP1903 for Injection at dose levels of 0.01, 0.05, 0.1, 0.5 and 1.0 mg/kg infused IV over 2 hours. AP1903 plasma levels were directly proportional to dose, with mean C_{max} values ranging from approximately 10-1275 ng/mL over the 0.01-1.0 mg/kg dose range. Following the initial infusion period, blood concentrations demonstrated a rapid distribution phase, with plasma levels reduced to approximately 18, 7, and 1% of maximal concentration at 0.5, 2 and 10 hours post-dose, respectively. AP1903 for Injection was shown to be safe and

well tolerated at all dose levels and demonstrated a favorable pharmacokinetic profile. Iuliucci J D, et al., J Clin Pharmacol. 41: 870-9, 2001.

The fixed dose of AP1903 for injection used, for example, may be 0.4 mg/kg intravenously infused over 2 hours. The amount of AP1903 needed in vitro for effective signaling of cells is 10-100 nM (1600 Da MW). This equates to 16-160 pg/L or 0.016-1.6 pg/kg (1.6-160 pg/kg). Doses up to 1 mg/kg were well tolerated in the Phase I study of AP1903 described above. Therefore, 0.4 mg/kg may be a safe and effective dose of AP1903 for this Phase I study in combination with the therapeutic cells.

The amino acid and/or nucleic acid sequence encoding ligand binding of the disclosure may contain sequence one or more modifications compared to a wild type amino acid or nucleic acid sequence. For example, the amino acid and/or nucleic acid sequence encoding ligand-binding region of the disclosure may be a codon-optimized sequence. The one or more modifications may increase the binding affinity of a ligand (e.g., an induction agent) for the ligand-binding region of the disclosure compared to a wild type polypeptide. Alternatively, or in addition, the one or more modifications may decrease the immunogenicity of the ligand-binding region of the disclosure compared to a wild type polypeptide. Ligand binding regions of the disclosure and/or induction agents of the disclosure may be non-naturally occurring.

Inducible proapoptotic polypeptides of the disclosure comprise a ligand-binding region, a linker and a proapoptotic peptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In certain embodiments, the non-human sequence comprises a restriction site. The linker may comprise any organic or inorganic material that permits, upon dimerization of the ligand-binding region, interaction, cross-linking, cross-activation, or activation of the proapoptotic polypeptides such that the interaction or activation of the proapoptotic polypeptides initiates apoptosis in the cell. In certain embodiments, the linker is a polypeptide. In certain embodiments, the linker is a polypeptide comprising a G/S rich amino acid sequence (a "GS" linker). In certain embodiments, the linker is a polypeptide comprising the amino acid sequence GGGGS (SEQ ID NO: 18024). In preferred embodiments, the linker is a polypeptide and the nucleic acid encoding the polypeptide does not contain a restriction site for a restriction endonuclease. Linkers of the disclosure may be non-naturally occurring.

Inducible proapoptotic polypeptides of the disclosure may be expressed in a cell under the transcriptional regulation of any promoter capable of initiating and/or regulating the expression of an inducible proapoptotic polypeptide of the disclosure in that cell. The term "promoter" as used herein refers to a promoter that acts as the initial binding site for RNA polymerase to transcribe a gene. For example, inducible proapoptotic polypeptides of the disclosure may be expressed in a mammalian cell under the transcriptional regulation of any promoter capable of initiating and/or regulating the expression of an inducible proapoptotic polypeptide of the disclosure in a mammalian cell, including, but not limited to native, endogenous, exogenous, and heterologous promoters. Preferred mammalian cells include human cells. Thus, inducible proapoptotic polypeptides of the disclosure may be expressed in a human cell under the transcriptional regulation of any promoter capable of initiating and/or regulating the expression of an inducible proapoptotic polypeptide of the disclosure in a human cell, including, but not limited to, a human promoter or a viral promoter.

Exemplary promoters for expression in human cells include, but are not limited to, a human cytomegalovirus (CMV) immediate early gene promoter, a SV40 early promoter, a Rous sarcoma virus long terminal repeat, j-actin promoter, a rat insulin promoter and a glyceraldehyde-3-phosphate dehydrogenase promoter, each of which may be used to obtain high-level expression of an inducible proapoptotic polypeptide of the disclosure. The use of other viral or mammalian cellular or bacterial phage promoters that are well known in the art to achieve expression of an inducible proapoptotic polypeptide of the disclosure is contemplated as well, provided that the levels of expression are sufficient for initiating apoptosis in a cell. By employing a promoter with well-known properties, the level and pattern of expression of the protein of interest following transfection or transformation can be optimized.

Selection of a promoter that is regulated in response to specific physiologic or synthetic signals can permit inducible expression of the inducible proapoptotic polypeptide of the disclosure. The ecdysone system (Invitrogen, Carlsbad, Calif.) is one such system. This system is designed to allow regulated expression of a gene of interest in mammalian cells. It consists of a tightly regulated expression mechanism that allows virtually no basal level expression of a transgene, but over 200-fold inducibility. The system is based on the heterodimeric ecdysone receptor of *Drosophila*, and when ecdysone or an analog such as muristerone A binds to the receptor, the receptor activates a promoter to turn on expression of the downstream transgene high levels of mRNA transcripts are attained. In this system, both monomers of the heterodimeric receptor are constitutively expressed from one vector, whereas the ecdysone-responsive promoter, which drives expression of the gene of interest, is on another plasmid. Engineering of this type of system into a vector of interest may therefore be useful. Another inducible system that may be useful is the Tet-Off™ or Tet-On™ system (Clontech, Palo Alto, Calif.) originally developed by Gossen and Bujard (Gossen and Bujard, Proc. Natl. Acad. Sci. USA, 89:5547-5551, 1992; Gossen et al., Science, 268:1766-1769, 1995). This system also allows high levels of gene expression to be regulated in response to tetracycline or tetracycline derivatives such as doxycycline. In the Tet-On™ system, gene expression is turned on in the presence of doxycycline, whereas in the Tet-Off™ system, gene expression is turned on in the absence of doxycycline. These systems are based on two regulatory elements derived from the tetracycline resistance operon of *E. coli*: the tetracycline operator sequence (to which the tetracycline repressor binds) and the tetracycline repressor protein. The gene of interest is cloned into a plasmid behind a promoter that has tetracycline-responsive elements present in it. A second plasmid contains a regulatory element called the tetracycline-controlled transactivator, which is composed, in the Tet-Off™ system, of the VP16 domain from the herpes simplex virus and the wild-type tetracycline repressor. Thus in the absence of doxycycline, transcription is constitutively on. In the Tet-On™ system, the tetracycline repressor is not wild type and in the presence of doxycycline activates transcription. For gene therapy vector production, the Tet-Off™ system may be used so that the producer cells could be grown in the presence of tetracycline or doxycycline and prevent expression of a potentially toxic transgene, but when the vector is introduced to the patient, the gene expression would be constitutively on.

In some circumstances, it is desirable to regulate expression of a transgene in a gene therapy vector. For example, different viral promoters with varying strengths of activity

are utilized depending on the level of expression desired. In mammalian cells, the CMV immediate early promoter is often used to provide strong transcriptional activation. The CMV promoter is reviewed in Donnelly, J. J., et al., 1997. Annu. Rev. Immunol. 15:617-48. Modified versions of the CMV promoter that are less potent have also been used when reduced levels of expression of the transgene are desired. When expression of a transgene in hematopoietic cells is desired, retroviral promoters such as the LTRs from MLV or MMTV are often used. Other viral promoters that are used depending on the desired effect include SV40, RSV LTR, HIV-1 and HIV-2 LTR, adenovirus promoters such as from the E1A, E2A, or MLP region, AAV LTR, HSV-TK, and avian sarcoma virus.

In other examples, promoters may be selected that are developmentally regulated and are active in particular differentiated cells. Thus, for example, a promoter may not be active in a pluripotent stem cell, but, for example, where the pluripotent stem cell differentiates into a more mature cell, the promoter may then be activated.

Similarly, tissue specific promoters are used to effect transcription in specific tissues or cells so as to reduce potential toxicity or undesirable effects to non-targeted tissues. These promoters may result in reduced expression compared to a stronger promoter such as the CMV promoter, but may also result in more limited expression, and immunogenicity (Bojak, A., et al., 2002. Vaccine. 20:1975-79; Cazeaux, N., et al., 2002. Vaccine 20:3322-31). For example, tissue specific promoters such as the PSA associated promoter or prostate-specific glandular kallikrein, or the muscle creatine kinase gene may be used where appropriate.

Examples of tissue specific or differentiation specific promoters include, but are not limited to, the following: B29 (B cells); CD14 (monocytic cells); CD43 (leukocytes and platelets); CD45 (hematopoietic cells); CD68 (macrophages); desmin (muscle); elastase-1 (pancreatic acinar cells); endoglin (endothelial cells); fibronectin (differentiating cells, healing tissues); and Flt-1 (endothelial cells); GFAP (astrocytes).

In certain indications, it is desirable to activate transcription at specific times after administration of the gene therapy vector. This is done with such promoters as those that are hormone or cytokine regulatable. Cytokine and inflammatory protein responsive promoters that can be used include K and T kininogen (Kageyama et al., (1987) J. Biol. Chem., 262, 2345-2351), c-fos, TNF-alpha, C-reactive protein (Arcone, et al., (1988) Nucl. Acids Res., 16(8), 3195-3207), haptoglobin (Oliviero et al., (1987) EMBO J., 6, 1905-1912), serum amyloid A2, C/EBP alpha, IL-1, IL-6 (Poli and Cortese, (1989) Proc. Nat'l Acad. Sci. USA, 86, 8202-8206), Complement C3 (Wilson et al., (1990) Mol. Cell. Biol., 10:1611-1619), IL-8, alpha-1 acid glycoprotein (Prowse and Baumann, (1988) Mol Cell Biol, 8, 42-51), alpha-1 antitrypsin, lipoprotein lipase (Zechner et al., Mol. Cell. Biol., 2394-2401, 1988), angiotensinogen (Ron, et al., (1991) Mol. Cell. Biol., 2887-2895), fibrinogen, c-jun (inducible by phorbol esters, TNF-alpha, UV radiation, retinoic acid, and hydrogen peroxide), collagenase (induced by phorbol esters and retinoic acid), metallothionein (heavy metal and glucocorticoid inducible), Stromelysin (inducible by phorbol ester, interleukin-1 and EGF), alpha-2 macroglobulin and alpha-1 anti-chymotrypsin. Other promoters include, for example, SV40, MMTV, Human Immunodeficiency Virus (MV), Moloney virus, ALV, Epstein Barr virus, Rous Sarcoma virus, human actin, myosin, hemoglobin, and creatine.

It is envisioned that any of the above promoters alone or in combination with another can be useful depending on the action desired. Promoters, and other regulatory elements, are selected such that they are functional in the desired cells or tissue. In addition, this list of promoters should not be construed to be exhaustive or limiting; other promoters that are used in conjunction with the promoters and methods disclosed herein.

Armored T-Cells “Knock Down” Strategy

T-cells of the disclosure may be genetically modified to enhance their therapeutic potential. Alternatively, or in addition, T-cells of the disclosure may be modified to render them less sensitive to immunologic and/or metabolic checkpoints. Modifications of this type “armor” the T cells of the disclosure, which, following the modification, may be referred to here as “armored” T cells. Armored T cells of the disclosure may be produced by, for example, blocking and/or diluting specific checkpoint signals delivered to the T-cells (i.e. checkpoint inhibition) naturally, within the tumor immunosuppressive microenvironment, for example.

In some embodiments, an armored T-cell of the disclosure is derived from a T cell, a NK cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell (including a T cell isolated or derived from G-CSF-mobilized peripheral blood), or an umbilical cord blood (UCB) derived T cell. In some embodiments, an armored T-cell of the disclosure comprises one or more of a chimeric ligand receptor (CLR comprising a single domain antibody,)/chimeric antigen receptor (CAR comprising a protein scaffold, an antibody, an ScFv, or an antibody mimetic), a CARTyrin (a CAR comprising a Centyrin), and/or a VCAR (a CAR comprising a camelid VHH or a single domain VH) of the disclosure. In some embodiments, an armored T-cell of the disclosure comprises an inducible proapoptotic polypeptide comprising (a) a ligand binding region, (b) a linker, and (c) a truncated caspase 9 polypeptide, wherein the inducible proapoptotic polypeptide does not comprise a non-human sequence. In some embodiments, the non-human sequence is a restriction site. In some embodiments, the ligand binding region inducible caspase polypeptide comprises a FK506 binding protein 12 (FKBP12) polypeptide. In some embodiments, the amino acid sequence of the FK506 binding

protein 12 (FKBP12) polypeptide comprises a modification at position 36 of the sequence. In some embodiments, the modification is a substitution of valine (V) for phenylalanine (F) at position 36 (F36V). In some embodiments, an armored T-cell of the disclosure comprises an exogenous sequence. In some embodiments, the exogenous sequence comprises a sequence encoding a therapeutic protein. Exemplary therapeutic proteins may be nuclear, cytoplasmic, intracellular, transmembrane, cell-surface bound, or secreted proteins. Exemplary therapeutic proteins expressed by the armored T cell may modify an activity of the armored T cell or may modify an activity of a second cell. In some embodiments, an armored T-cell of the disclosure comprises a selection gene or a selection marker. In some embodiments, an armored T-cell of the disclosure comprises a synthetic gene expression cassette (also referred to herein as an inducible transgene construct).

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression one or more gene(s) encoding receptor(s) of inhibitory checkpoint signals to produce an armored T-cell of the disclosure. Examples of inhibitory checkpoint signals include, but are not limited to, a PD-L1 ligand binding to a PD-1 receptor on a CAR-T cell of the disclosure or a TGF β cytokine binding to a TGF β RII receptor on a CAR-T cell. Receptors of inhibitory checkpoint signals are expressed on the cell surface or within the cytoplasm of a T-cell. Silencing or reducing expressing of the gene encoding the receptor of the inhibitory checkpoint signal results a loss of protein expression of the inhibitory checkpoint receptors on the surface or within the cytoplasm of an armored T-cell of the disclosure. Thus, armored T cells of the disclosure having silenced or reduced expression of one or more genes encoding an inhibitory checkpoint receptor is resistant, non-receptive or insensitive to checkpoint signals. The armored T cell's resistance or decreased sensitivity to inhibitory checkpoint signals enhances the armored T cell's therapeutic potential in the presence of these inhibitory checkpoint signals. Inhibitory checkpoint signals include but are not limited to the examples listed in Table 1. Exemplary inhibitory checkpoint signals that may be silenced in an armored T cell of the disclosure include, but are not limited to, PD-1 and TGF β RII.

TABLE 1

Exemplary Inhibitory Checkpoint Signals (and proteins that induce immunosuppression). A CSR of the disclosure may comprise an endodomain of any one of the proteins of this table.		
Full Name	Abbreviation	SEQ ID NO:
Programmed cell death protein 1	PD1	14643-14644
transforming growth factor β Receptor 1	TGF β R1	14645
transforming growth factor β Receptor 2	TGF β R2	14646
T-cell immunoglobulin and mucin-domain containing-3	TIM3	14647
Lymphocyte-activation gene 3	LAG3	14648
Cytotoxic T-lymphocyte protein 4	CTLA4	14649
B- and T-lymphocyte attenuator	BTLA	14650
Killer cell immunoglobulin-like receptor	KIR	14651
Alpha-2A adrenergic receptor	A2aR	14652
V-type immunoglobulin domain-containing suppressor of T-cell activation	VISTA	14653
T-cell immunoreceptor with Ig and ITIM domains	TIGIT	14654
Programmed cell death 1 ligand 1	B7H1 or PD-L1	14655
Programmed cell death 1 ligand 2	B7DC or PD-L2	14656
T-lymphocyte activation antigen CD80	B7-1 or CD80	14657
T-lymphocyte activation antigen CD86	B7-2 or CD86	14658
CD160 antigen	CD160	14659
Leukocyte-associated immunoglobulin-like receptor 1	LAIR1	14660

TABLE 1-continued

Exemplary Inhibitory Checkpoint Signals (and proteins that induce immunosuppression). A CSR of the disclosure may comprise an endodomain of any one of the proteins of this table.		
Full Name	Abbreviation	SEQ ID NO:
T-cell immunoglobulin and mucin domain-containing protein 4	TIM4 or TIMD4	14661
Natural killer cell receptor 2B4	2B4 or CD244	14662
Major Histocompatibility Complex type I	MHC I	14663
Major Histocompatibility Complex type II	MHC II	
Putative 2-methylcitrate dehydratase receptor	PDH1R	
T-cell immunoglobulin and mucin domain 1 receptor	TIM1R	
T-cell immunoglobulin and mucin domain 4 receptor	TIM4R	
B7-H3 receptor	B7H3R or CD176 Receptor	
B7-H4 receptor	B7H4R	
Immunoglobulin-like transcript (ILT) 3 receptor	ILT3R	
phosphoinositide 3-kinase, subunit alpha	PI3K alpha	14664
phosphoinositide 3-kinase, subunit gamma	PI3K gamma	14665
Tyrosine-protein phosphatase non-receptor type 11	SHP2 or PTPN11	14666
Protein phosphatase 2, subunit gamma	PP2A gamma	14667
Protein phosphatase 2, subunit beta	PP2A beta	14668
Protein phosphatase 2, subunit delta	PP2A delta	14669
Protein phosphatase 2, subunit epsilon	PP2A epsilon	14670
Protein phosphatase 2, subunit alpha	PP2A alpha	14671
T-cell Receptor, subunit alpha	TCR alpha	14672
T-cell Receptor, subunit beta	TCR beta	14673
T-cell Receptor, subunit zeta	TCR zeta	14674
T-cell Receptor, subunit CD3 epsilon	TCR CD3 epsilon	14675
T-cell Receptor, subunit CD3 gamma	TCR CD3 gamma	14676
T-cell Receptor, subunit CD3 delta	TCR CD3 delta	14677
Cluster of Differentiation 28	CD28	14678
Galectins	Galectins	
Galectin 9	Galectin 9	14679
High Mobility Group Box 1	HMGB1	14680
Arginase 1	ARG1	14681
Prostaglandin-Endoperoxide Synthase 1	PTGS1	14682
Prostaglandin-Endoperoxide Synthase 2	PTGS2	14683
Mucin 1, Cell Surface Associated	MUC1	14684
Mucin 2, Oligomeric Mucus/Gel-Forming	MUC2	14685
Mucin 3A, Cell Surface Associated	MUC3A	14686
Mucin 3B, Cell Surface Associated	MUC3B	14687
Mucin 4, Cell Surface Associated	MUC4	14688
Mucin 5AC, Oligomeric Mucus/Gel-Forming	MUC5AC	14689
Mucin 5B, Oligomeric Mucus/Gel-Forming	MUC5B	14690
Mucin 6, Oligomeric Mucus/Gel-Forming	MUC6	14691
Mucin 7, Secreted	MUC7	14692
Mucin 8	MUC8	
Mucin 12, Cell Surface Associated	MUC12	14693
Mucin 13, Cell Surface Associated	MUC13	14694
Mucin 15, Cell Surface Associated	MUC15	14695
Mucin 16, Cell Surface Associated	MUC16	14696
Mucin 17, Cell Surface Associated	MUC17	14697
Mucin 19, Oligomeric	MUC19	14698
Mucin 20, Cell Surface Associated	MUC20	14699
Mucin 21, Cell Surface Associated	MUC21	14700
Mucin 22	MUC22	14701
Indoleamine 2,3-Dioxygenase 1	IDO1	14702
Indoleamine 2,3-Dioxygenase 2	IDO2	14703
Inducible T Cell Costimulator Ligand	ICOSLG	14704
ROS Proto-Oncogene 1, Receptor Tyrosine Kinase	ROS1	14705
Tumor Necrosis Factor Receptor Superfamily Member 9	4-1BB, CD137, ILA or TNFRSF9	14706
4-1BB Ligand	4-1BB-L	14707
Glucocorticoid-induced TNFR family related gene	GITR	14708
Glucocorticoid-induced TNFR family related gene ligand	GITRL	14709

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding intracellular proteins involved in checkpoint signaling to produce an armored T-cell of the disclosure. The

activity of a T-cell of the disclosure may be enhanced by targeting any intracellular signaling protein involved in a checkpoint-signaling pathway, thereby achieving checkpoint inhibition or interference to one or more checkpoint

pathways. Intracellular signaling proteins involved in checkpoint signaling include, but are not limited to, exemplary intracellular signaling proteins listed in Table 2.

TABLE 2

Exemplary Intracellular Signaling Proteins.		
Full Name	Abbreviation	SEQ ID NO:
phosphoinositide 3-kinase, subunit alpha	PI3K alpha	14710
phosphoinositide 3-kinase, subunit gamma	PI3K gamma	14711
Tyrosine-protein phosphatase non-receptor type 11	SHP2 or PTPN11	14712
Protein phosphatase 2, subunit gamma	PP2A gamma	14713
Protein phosphatase 2, subunit beta	PP2A beta	14714
Protein phosphatase 2, subunit delta	PP2A delta	14715
Protein phosphatase 2, subunit epsilon	PP2A epsilon	14716
Protein phosphatase 2, subunit alpha	PP2A alpha	14717
RAC-alpha serine/threonine-protein kinase	AKT or PKB	14718
Tyrosine-protein kinase ZAP-70	ZAP70	14719
Amino acid sequence (KIEELE)-containing domain protein	KIEELE-domain containing proteins	
BCL2 associated athanogene 6	Bat3, Bag6 or Scythe	14720
B-cell lymphoma-extra large	Bcl-xL	14721
Bcl-2-related protein A1	Bfl-1 or BCL2A1	14722

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a transcription factor that hinders the efficacy of a therapy to produce an armored T-cell of the disclosure. The activity of armored T-cells may be enhanced or modulated by silencing or reducing expression (or repressing a function) of a transcription factor that hinders the efficacy of a therapy. Exemplary transcription factors that may be modified to silence or reduce expression or to repress a function thereof include, but are not limited to, the exemplary transcription factors listed in Table 3. For example, expression of a FOXP3 gene may be silenced or reduced in an armored T cell of the disclosure to prevent or reduce the formation of T regulatory CAR-T-cells (CAR-Treg cells), the expression or activity of which may reduce efficacy of a therapy.

TABLE 3

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
activity-dependent neuroprotector homeobox	ADNP	14723
ADNP homeobox 2	ADNP2	14724
AE binding protein 1	AEBP1	14725
AE binding protein 2	AEBP2	14726
AF4/FMR2 family member 1	AFF1	14727
AF4/FMR2 family member 2	AFF2	14728
AF4/FMR2 family member 3	AFF3	14729
AF4/FMR2 family member 4	AFF4	14730
AT-hook containing transcription factor 1	AHCTF1	14731
aryl hydrocarbon receptor	AHR	14732
aryl-hydrocarbon receptor repressor	AHRR	14733
autoimmune regulator	AIRE	14734
AT-hook transcription factor	AKNA	14735
ALX homeobox 1	ALX1	14736
ALX homeobox 3	ALX3	14737
ALX homeobox 4	ALX4	14738
ankyrin repeat and zinc finger domain containing 1	ANKZF1	14739
adaptor related protein complex 5 zeta 1 subunit	AP5Z1	14740
androgen receptor	AR	14741
arginine-fifty homeobox	ARGFX	14742
Rho GTPase activating protein 35	ARHGAP35	14743
AT-rich interaction domain 1A	ARID1A	14744
AT-rich interaction domain 1B	ARID1B	14745
AT-rich interaction domain 2	ARID2	14746
AT-rich interaction domain 3A	ARID3A	14747
AT-rich interaction domain 3B	ARID3B	14748
AT-rich interaction domain 3C	ARID3C	14749
AT-rich interaction domain 4A	ARID4A	14750
AT-rich interaction domain 4B	ARID4B	14751
AT-rich interaction domain 5A	ARID5A	14752
AT-rich interaction domain 5B	ARID5B	14753
aryl hydrocarbon receptor nuclear translocator	ARNT	14754
aryl hydrocarbon receptor nuclear translocator 2	ARNT2	14755
aryl hydrocarbon receptor nuclear translocator like	ARNTL	14756
aryl hydrocarbon receptor nuclear translocator like 2	ARNTL2	14757
aristaless related homeobox	ARX	14758

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
achaete-scute family bHLH transcription factor 1	ASCL1	14759
achaete-scute family bHLH transcription factor 2	ASCL2	14760
achaete-scute family bHLH transcription factor 3	ASCL3	14761
achaete-scute family bHLH transcription factor 4	ASCL4	14762
achaete-scute family bHLH transcription factor 5	ASCL5	14763
ash1 (absent, small, or homeotic)-like (<i>Drosophila</i>)	ASH1L	14764
ash2 (absent, small, or homeotic)-like (<i>Drosophila</i>)	ASH2L	14765
activating transcription factor 1	ATF1	14766
activating transcription factor 2	ATF2	14767
activating transcription factor 3	ATF3	14768
activating transcription factor 4	ATF4	14769
activating transcription factor 5	ATF5	14770
activating transcription factor 6	ATF6	14771
activating transcription factor 6 beta	ATF6B	14772
activating transcription factor 7	ATF7	14773
atonal bHLH transcription factor 1	ATOH1	14774
atonal bHLH transcription factor 7	ATOH7	14775
atonal bHLH transcription factor 8	ATOH8	14776
alpha thalassemia/mental retardation syndrome X-linked	ATRX	14777
ataxin 7	ATXN7	14778
BTB and CNC homology 1, basic leucine zipper transcription factor1	BACH1	14779-14780
BTB domain and CNC homolog 2	BACH2	14781
BarH like homeobox 1	BARHL1	14782
BarH like homeobox 2	BARHL2	14783
BARX homeobox 1	BARX1	14784
BARX homeobox 2	BARX2	14785
Basic Leucine Zipper ATF-Like Transcription Factor,	Batf	14786
basic leucine zipper transcription factor, ATF-like	BATF	14786
basic leucine zipper transcription factor, ATF-like 2	BATF2	14787
basic leucine zipper transcription factor, ATF-like 3	BATF3	14788
bobby sox homolog (<i>Drosophila</i>)	BBX	14789
B-cell CLL/lymphoma 11A	BCL11A	14790
B-cell CLL/lymphoma 11B	BCL11B	14791
B-cell CLL/lymphoma 3	BCL3	14792
B-cell CLL/lymphoma 6	BCL6	14793
B-cell CLL/lymphoma 6, member B	BCL6B	14794
BCL2 associated transcription factor 1	BCLAF1	14795
basic helix-loop-helix family member a15	BHLHA15	14796
basic helix-loop-helix family member a9	BHLHA9	14797
basic helix-loop-helix domain containing, class B, 9	BHLHB9	14798
basic helix-loop-helix family member e22	BHLHE22	14799
basic helix-loop-helix family member e23	BHLHE23	14800
basic helix-loop-helix family member e40	BHLHE40	14801
basic helix-loop-helix family member e41	BHLHE41	14802
Beta-Interferon Gene Positive-Regulatory Domain I Binding Factor	Blimp-1	14803
bone morphogenetic protein 2	BMP2	14804
basonuclin 1	BNC1	14805
basonuclin 2	BNC2	14806
bolA family member 1	BOLA1	14807
bolA family member 2	BOLA2	14808
bolA family member 3	BOLA3	14809
bromodomain PHD finger transcription factor	BPTF	14810
breast cancer 1	BRCA1	14811
brain specific homeobox	BSX	14812
chromosome 20 open reading frame 194	C20orf194	14813
calmodulin binding transcription activator 1	CAMTA1	14814
calmodulin binding transcription activator 2	CAMTA2	14815
calcium regulated heat stable protein 1	CARHSP1	14816
castor zinc finger 1	CASZ1	14817
core-binding factor, beta subunit	CBFB	14818
coiled-coil domain containing 79	CCDC79	14819
cell division cycle 5 like	CDC5L	14820
caudal type homeobox 1	CDX1	14821
caudal type homeobox 2	CDX2	14822
caudal type homeobox 4	CDX4	14823
CCAAT/enhancer binding protein alpha	CEBPA	14824
CCAAT/enhancer binding protein beta	CEBPB	14825
CCAAT/enhancer binding protein delta	CEBPD	14826
CCAAT/enhancer binding protein epsilon	CEBPE	14827
CCAAT/enhancer binding protein gamma	CEBPG	14828
CCAAT/enhancer binding protein zeta	CEBPZ	14829
centromere protein T	CENPT	14830
ceramide synthase 3	CERS3	14831

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
ceramide synthase 6	CERS6	14832
chromosome alignment maintaining phosphoprotein 1	CHAMP1	14833
capicua transcriptional repressor	CIC	14834
CDKN1A interacting zinc finger protein 1	CIZ1	14835
clock circadian regulator	CLOCK	14836
CCR4-NOT transcription complex subunit 4	CNOT4	14837
CPX chromosome region, candidate 1	CPXCR1	14838
cramped chromatin regulator homolog 1	CRAMP1	14839
cAMP responsive element binding protein 1	CREB1	14840
cAMP responsive element binding protein 3	CREB3	14841
cAMP responsive element binding protein 3-like 1	CREB3L1	14842
cAMP responsive element binding protein 3-like 2	CREB3L2	14843
cAMP responsive element binding protein 3-like 3	CREB3L3	14844
cAMP responsive element binding protein 3-like 4	CREB3L4	14845
cAMP responsive element binding protein 5	CREB5	14846
CREB binding protein	CREBBP	14847
cAMP responsive element binding protein-like 2	CREBL2	14848
CREB3 regulatory factor	CREBRF	14849
CREB/ATF bZIP transcription factor	CREBZF	14850
cAMP responsive element modulator	CREM	14851
cone-rod homeobox	CRX	14852
cysteine-serine-rich nuclear protein 1	CSRNP1	14853
cysteine-serine-rich nuclear protein 2	CSRNP2	14854
cysteine-serine-rich nuclear protein 3	CSRNP3	14855
CCCTC-binding factor (zinc finger protein)	CTCF	14856
CCCTC-binding factor like	CTCFL	14857
cut-like homeobox 1	CUX1	14858-14859
cut-like homeobox 2	CUX2	14860
CXXC finger protein 1	CXXC1	14861
dachshund family transcription factor 1	DACH1	14862
dachshund family transcription factor 2	DACH2	14863
D site of albumin promoter (albumin D-box) binding protein	DBP	14864
developing brain homeobox 1	DBX1	14865
developing brain homeobox 2	DBX2	14866
damage specific DNA binding protein 2	DDB2	14867
DNA damage inducible transcript 3	DDIT3	14868
DEAF1, transcription factor	DEAF1	14869
distal-less homeobox 1	DLX1	14870
distal-less homeobox 2	DLX2	14871
distal-less homeobox 3	DLX3	14872
distal-less homeobox 4	DLX4	14873
distal-less homeobox 5	DLX5	14874
distal-less homeobox 6	DLX6	14875
DNA methyltransferase 1 associated protein 1	DMAP1	14876
diencephalon/mesencephalon homeobox 1	DMBX1	14877
doublesex and mab-3 related transcription factor 1	DMRT1	14878
doublesex and mab-3 related transcription factor 2	DMRT2	14879
doublesex and mab-3 related transcription factor 3	DMRT3	14880
DMRT like family A1	DMRTA1	14881
DMRT like family A2	DMRTA2	14882
DMRT like family B with proline rich C-terminal 1	DMRTB1	14883
DMRT like family C1	DMRTC1	14884
DMRT like family C1B	DMRTC1B	14884
DMRT like family C2	DMRTC2	14885
cyclin D binding myb like transcription factor 1	DMTF1	14886
DnaJ heat shock protein family (Hsp40) member C1	DNAJC1	14887
DnaJ heat shock protein family (Hsp40) member C2	DNAJC2	14888
DnaJ heat shock protein family (Hsp40) member C21	DNAJC21	14889
DNA (cytosine-5-)-methyltransferase 1	DNMT1	14890
DNA (cytosine-5-)-methyltransferase 3 alpha	DNMT3A	14891
DNA (cytosine-5-)-methyltransferase 3 beta	DNMT3B	14892
DNA (cytosine-5-)-methyltransferase 3-like	DNMT3L	14893
double PHD fingers 1	DPF1	14894
double PHD fingers 2	DPF2	14895
double PHD fingers 3	DPF3	14896
divergent-paired related homeobox	DPRX	14897
down-regulator of transcription 1	DR1	14898
DR1 associated protein 1	DRAP1	14899
dorsal root ganglia homeobox	DRGX	14900
double homeobox 4	DUX4	14901
double homeobox 4 like 9	DUX4L9	14902
double homeobox A	DUXA	14903
E2F transcription factor 1	E2F1	14904
E2F transcription factor 2	E2F2	14905
E2F transcription factor 3	E2F3	14906

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
E2F transcription factor 4	E2F4	14907
E2F transcription factor 5	E2F5	14908
E2F transcription factor 6	E2F6	14909
E2F transcription factor 7	E2F7	14910
E2F transcription factor 8	E2F8	14911
E4F transcription factor 1	E4F1	14912
early B-cell factor 1	EBF1	14913
early B-cell factor 2	EBF2	14914
early B-cell factor 3	EBF3	14915
early B-cell factor 4	EBF4	14916
early growth response 1	EGR1	14917
early growth response 2	EGR2	14918
early growth response 3	EGR3	14919
early growth response 4	EGR4	14920
ets homologous factor	EHF	14921
E74-like factor 1 (ets domain transcription factor)	ELF1	14922
E74-like factor 2 (ets domain transcription factor)	ELF2	14923
E74-like factor 3 (ets domain transcription factor, epithelial-specific)	ELF3	14924
E74-like factor 4 (ets domain transcription factor)	ELF4	14925
E74-like factor 5 (ets domain transcription factor)	ELF5	14926
ELK1, member of ETS oncogene family	ELK1	14927
ELK3, ETS-domain protein (SRF accessory protein 2)	ELK3	14928
ELK4, ETS-domain protein (SRF accessory protein 1)	ELKA	14929
ELM2 and Myb/SANT-like domain containing 1	ELMSAN1	14930
empty spiracles homeobox 1	EMX1	14931
empty spiracles homeobox 2	EMX2	14932
engrailed homeobox 1	EN1	14933
engrailed homeobox 2	EN2	14934
enolase 1, (alpha)	ENO1	14935
eomesodermin	EOMES	14936
endothelial PAS domain protein 1	EPAS1	14937
Ets2 repressor factor	ERF	14938
v-ets avian erythroblastosis virus E26 oncogene homolog	ERG	14939-14940
estrogen receptor 1	ESR1	14941
estrogen receptor 2 (ER beta)	ESR2	14942
estrogen related receptor alpha	ESRRA	14943
estrogen related receptor beta	ESRRB	14944
estrogen related receptor gamma	ESRRG	14945
ESX homeobox 1	ESX1	14946
v-ets avian erythroblastosis virus E26 oncogene homolog 1	ETS1	14947
v-ets avian erythroblastosis virus E26 oncogene homolog 2	ETS2	14948
ets variant 1	ETV1	14949
ets variant 2	ETV2	14950
ets variant 3	ETV3	14951
ets variant 3-like	ETV3L	14952
ets variant 4	ETV4	14953
ets variant 5	ETV5	14954
ets variant 6	ETV6	14955
ets variant 7	ETV7	14956
even-skipped homeobox 1	EVX1	14957
even-skipped homeobox 2	EVX2	14958
enhancer of zeste 1 polycomb repressive complex 2 subunit	EZH1	14959
enhancer of zeste 2 polycomb repressive complex 2 subunit	EZH2	14960
family with sequence similarity 170 member A	FAM170A	14961
Fer3-like bHLH transcription factor	FERD3L	14962
FEV (ETS oncogene family)	FEV	14963
FEZ family zinc finger 1	FEZF1	14964
FEZ family zinc finger 2	FEZF2	14965
folliculogenesis specific bHLH transcription factor	FIGLA	14966
FLT3-interacting zinc finger 1	FIZ1	14967
Fli-1 proto-oncogene, ETS transcription factor	FLI1	14968
FBJ murine osteosarcoma viral oncogene homolog	FOS	14969
FBJ murine osteosarcoma viral oncogene homolog B	FOSB	14970
FOS like antigen 1	FOSL1	14971
FOS like antigen 2	FOSL2	14972
forkhead box A1	FOXA1	14973
forkhead box A2	FOXA2	14974
forkhead box A3	FOXA3	14975
forkhead box B1	FOXB1	14976
forkhead box B2	FOXB2	14977

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
forkhead box C1	FOXC1	14978
forkhead box C2	FOXC2	14979
forkhead box D1	FOXD1	14980
forkhead box D2	FOXD2	14981
forkhead box D3	FOXD3	14982
forkhead box D4	FOXD4	14983
forkhead box D4-like 1	FOXD4L1	14984
forkhead box D4-like 3	FOXD4L3	14985
forkhead box D4-like 4	FOXD4L4	14986
forkhead box D4-like 5	FOXD4L5	14987
forkhead box D4-like 6	FOXD4L6	14988
forkhead box E1	FOX E1	14989
forkhead box E3	FOX E3	14990
forkhead box F1	FOX F1	14991
forkhead box F2	FOX F2	14992
forkhead box G1	FOX G1	14993
forkhead box H1	FOX H1	14994
forkhead box I1	FOX I1	14995
forkhead box I2	FOX I2	14996
forkhead box I3	FOX I3	14997
forkhead box J1	FOX J1	14998
forkhead box J2	FOX J2	14999
forkhead box J3	FOX J3	15000
forkhead box K1	FOX K1	15001
forkhead box K2	FOX K2	15002
forkhead box L1	FOX L1	15003
forkhead box L2	FOX L2	15004
forkhead box M1	FOX M1	15005
forkhead box N1	FOX N1	15006
forkhead box N2	FOX N2	15007
forkhead box N3	FOX N3	15008
forkhead box N4	FOX N4	15009
forkhead box O1	FOX O1	15010
forkhead box O3	FOX O3	15011
forkhead box O4	FOX O4	15012
forkhead box O6	FOX O6	15013
forkhead box P1	FOX P1	15014
forkhead box P2	FOX P2	15015
forkhead box P3	FOX P3	15016
forkhead box P4	FOX Q1	15017
forkhead box Q1	FOX R1	15018
forkhead box R1	FOX R2	15019
forkhead box R2	FOX S1	15020
forkhead box S1	FOX P3	15021
far upstream element binding protein 1	FUBP1	15022
far upstream element (FUSE) binding protein 3	FUBP3	15023
GA binding protein transcription factor alpha subunit	GABPA	15024
GA binding protein transcription factor, beta subunit 1	GABPB1	15025
GA binding protein transcription factor, beta subunit 2	GABPB2	15026
GATA binding protein 1 (globin transcription factor 1)	GATA1	15027
GATA binding protein 2	GATA2	15028
GATA binding protein 3	GATA3	15029
GATA binding protein 4	GATA4	15030
GATA binding protein 5	GATA5	15031
GATA binding protein 6	GATA6	15032
GATA zinc finger domain containing 1	GATAD1	15033
GATA zinc finger domain containing 2A	GATAD2A	15034
GATA zinc finger domain containing 2B	GATAD2B	15035
gastrulation brain homeobox 1	GBX1	15036
gastrulation brain homeobox 2	GBX2	15037
GC-rich sequence DNA-binding factor 2	GCFC2	15038
glial cells missing homolog 1	GCM1	15039
glial cells missing homolog 2	GCM2	15040
growth factor independent 1 transcription repressor	GFI1	15041
growth factor independent 1B transcription repressor	GFI1B	15042
GLI family zinc finger 1	GLI1	15043
GLI family zinc finger 2	GLI2	15044
GLI family zinc finger 3	GLI3	15045
GLI family zinc finger 4	GLI4	15046
GLIS family zinc finger 1	GLIS1	15047
GLIS family zinc finger 2	GLIS2	15048
GLIS family zinc finger 3	GLIS3	15049
glucocorticoid modulatory element binding protein 1	GMEB1	15050
glucocorticoid modulatory element binding protein 2	GMEB2	15051
gon-4-like (<i>C. elegans</i>)	GON4L	15052
grainyhead like transcription factor 1	GRHL1	15053

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
grainyhead like transcription factor 2	GRHL2	15054
grainyhead like transcription factor 3	GRHL3	15055
goosecoid homeobox	GSC	15056
goosecoid homeobox 2	GSC2	15057
GS homeobox 1	GSX1	15058
GS homeobox 2	GSX2	15059
general transcription factor IIIi	GTF2I	15060
general transcription factor IIIA	GTF3A	15061
GDNF inducible zinc finger protein 1	GZF1	15062
heart and neural crest derivatives expressed 1	HAND1	15063
heart and neural crest derivatives expressed 2	HAND2	15064
HMG-box transcription factor 1	HBP1	15065-15066
highly divergent homeobox	HDX	15067
helt bHLH transcription factor	HELT	15068
hes family bHLH transcription factor 1	HES1	15069-15070
hes family bHLH transcription factor 2	HES2	15071
hes family bHLH transcription factor 3	HES3	15072
hes family bHLH transcription factor 4	HES4	15073
hes family bHLH transcription factor 5	HES5	15074
hes family bHLH transcription factor 6	HES6	15075
hes family bHLH transcription factor 7	HES7	15076
HESX homeobox 1	HESX1	15077
hes-related family bHLH transcription factor with YRPW motif 1	HEY1	15078
hes-related family bHLH transcription factor with YRPW motif 2	HEY2	15079
hes-related family bHLH transcription factor with YRPW motif-like	HEYL	15080
hematopoietically expressed homeobox	HHEX	15081
hypermethylated in cancer 1	HIC1	15082
hypermethylated in cancer 2	HIC2	15083
hypoxia inducible factor 1, alpha subunit (basic helix-loop-helix transcription factor)	HIF1A	15084
hypoxia inducible factor 3, alpha subunit	HIF3A	15085
histone H4 transcription factor	HINFP	15086
human immunodeficiency virus type I enhancer binding protein 1	HIVEP1	15087
human immunodeficiency virus type I enhancer binding protein 2	HIVEP2	15088
human immunodeficiency virus type I enhancer binding protein 3	HIVEP3	15089
HKR1, GLI-Kruppel zinc finger family member	HKR1	15090
hepatic leukemia factor	HLF	15091
helicase-like transcription factor	HLTF	15092
H2.0-like homeobox	HLX	15093
homeobox containing 1	HMBOX1	15094
high mobility group 20A	HMG20A	15095
high mobility group 20B	HMG20B	15096
high mobility group AT-hook 1	HMG1	15097
high mobility group AT-hook 2	HMG2	15098
HMG-box containing 3	HMGXB3	15099
HMG-box containing 4	HMGXB4	15100
H6 family homeobox 1	HMX1	15101
H6 family homeobox 2	HMX2	15102
H6 family homeobox 3	HMX3	15103-15104
HNF1 homeobox A	HNF1A	15105
HNF1 homeobox B	HNF1B	15106
hepatocyte nuclear factor 4 alpha	HNF4A	15107
hepatocyte nuclear factor 4 gamma	HNF4G	15108
heterogeneous nuclear ribonucleoprotein K	HNRNPK	15109
homeobox and leucine zipper encoding	HOMEZ	15110
HOP homeobox	HOPX	15111
homeobox A1	HOXA1	15112
homeobox A10	HOXA10	15113
homeobox A11	HOXA11	15114
homeobox A13	HOXA13	15115
homeobox A2	HOXA2	15116
homeobox A3	HOXA3	15117
homeobox A4	HOXA4	15118
homeobox A5	HOXA5	15119
homeobox A6	HOXA6	15120
homeobox A7	HOXA7	15121
homeobox A9	HOXA9	15122
homeobox B1	HOXB1	15123
homeobox B13	HOXB13	15124
homeobox B2	HOXB2	15125

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
homeobox B3	HOXB3	15126
homeobox B4	HOXB4	15127
homeobox B5	HOXB5	15128
homeobox B6	HOXB6	15129
homeobox B7	HOXB7	15130
homeobox B8	HOXB8	15131
homeobox B9	HOXB9	15132
homeobox C10	HOXC10	15133
homeobox C11	HOXC11	15134
homeobox C12	HOXC12	15135
homeobox C13	HOXC13	15136
homeobox C4	HOXC4	15137
homeobox C5	HOXC5	15138
homeobox C6	HOXC6	15139
homeobox C8	HOXC8	15140
homeobox C9	HOXC9	15141
homeobox D1	HOXD1	15142
homeobox D10	HOXD10	15143
homeobox D11	HOXD11	15144
homeobox D12	HOXD12	15145
homeobox D13	HOXD13	15146
homeobox D3	HOXD3	15147
homeobox D4	HOXD4	15148
homeobox D8	HOXD8	15149
homeobox D9	HOXD9	15150
heat shock transcription factor 1	HSF1	15151
heat shock transcription factor 2	HSF2	15152
heat shock transcription factor 4	HSF4	15153
heat shock transcription factor family member 5	HSF5	15154
heat shock transcription factor family, X-linked 1	HSFX1	15155
heat shock transcription factor, Y-linked 1	HSFY1	15156
heat shock transcription factor, Y-linked 2	HSFY2	15156
inhibitor of DNA binding 1, dominant negative helix-loop-helix protein	ID1	15157
inhibitor of DNA binding 2, dominant negative helix-loop-helix protein	ID2	15158
inhibitor of DNA binding 3, dominant negative helix-loop-helix protein	ID3	15159
inhibitor of DNA binding 4, dominant negative helix-loop-helix protein	ID4	15160
interferon, gamma-inducible protein 16	IFI16	15161
IKAROS family zinc finger 1	IKZF1	15162
IKAROS family zinc finger 2	IKZF2	15163
IKAROS family zinc finger 3	IKZF3	15164
IKAROS family zinc finger 4	IKZF4	15165
IKAROS family zinc finger 5	IKZF5	15166
insulinoma associated 1	INSM1	15167
insulinoma-associated 2	INSM2	15168
interferon regulatory factor 1	IRF1	15169
interferon regulatory factor 2	IRF2	15170
interferon regulatory factor 3	IRF3	15171
interferon regulatory factor 4	IRF4	15172
interferon regulatory factor 5	IRF5	15173
interferon regulatory factor 6	IRF6	15174
interferon regulatory factor 7	IRF7	15175
interferon regulatory factor 8	IRF8	15176
interferon regulatory factor 9	IRF9	15177
iroquois homeobox 1	IRX1	15178
iroquois homeobox 2	IRX2	15179
iroquois homeobox 3	IRX3	15180
iroquois homeobox 4	IRX4	15181
iroquois homeobox 5	IRX5	15182
iroquois homeobox 6	IRX6	15183
ISL LIM homeobox 1	ISL1	15184
ISL LIM homeobox 2	ISL2	15185
intestine specific homeobox	ISX	15186
jumonji and AT-rich interaction domain containing 2	JARID2	15187
JAZF zinc finger 1	JAZF1	15188
Jun dimerization protein 2	JDP2	15189
jun proto-oncogene	JUN	15190
jun B proto-oncogene	JUNB	15191
jun D proto-oncogene	JUND	15192
K(lysine) acetyltransferase 5	KAT5	15193
lysine acetyltransferase 6A	KAT6A	15194
lysine acetyltransferase 6B	KAT6B	15195
lysine acetyltransferase 7	KAT7	15196

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
lysine acetyltransferase 8	KAT8	15197
potassium channel modulatory factor 1	KCMF1	15198
potassium voltage-gated channel interacting protein 3	KCNIP3	15199
lysine demethylase 2A	KDM2A	15200
lysine demethylase 5A	KDM5A	15201
lysine demethylase 5B	KDM5B	15202
lysine demethylase 5C	KDM5C	15203
lysine demethylase 5D	KDM5D	15204
KH-type splicing regulatory protein	KHSRP	15205
KIAA1549	KIAA1549	15206
Kruppel-like factor 1 (erythroid)	KLF1	15207
Kruppel-like factor 10	KLF10	15208
Kruppel-like factor 11	KLF11	15209
Kruppel-like factor 12	KLF12	15210
Kruppel-like factor 13	KLF13	15211
Kruppel-like factor 14	KLF14	15212
Kruppel-like factor 15	KLF15	15213
Kruppel-like factor 16	KLF16	15214
Kruppel-like factor 17	KLF17	15215
Kruppel-like factor 2	KLF2	15216
Kruppel-like factor 3 (basic)	KLF3	15217
Kruppel-like factor 4 (gut)	KLF4	15218
Kruppel-like factor 5 (intestinal)	KLF5	15219
Kruppel-like factor 6	KLF6	15220
Kruppel-like factor 7 (ubiquitous)	KLF7	15221
Kruppel-like factor 8	KLF8	15222
Kruppel-like factor 9	KLF9	15223
lysine methyltransferase 2A	KMT2A	15224
lysine methyltransferase 2B	KMT2B	15225
lysine methyltransferase 2C	KMT2C	15226
lysine methyltransferase 2E	KMT2E	15227
l(3)mbt-like 1 (<i>Drosophila</i>)	L3MBTL1	15228
l(3)mbt-like 2 (<i>Drosophila</i>)	L3MBTL2	15229
l(3)mbt-like 3 (<i>Drosophila</i>)	L3MBTL3	15230
l(3)mbt-like 4 (<i>Drosophila</i>)	L3MBTL4	15231
ladybird homeobox 1	LBX1	15232
ladybird homeobox 2	LBX2	15233
ligand dependent nuclear receptor corepressor	LCOR	15234
ligand dependent nuclear receptor corepressor like	LCORL	15235
lymphoid enhancer binding factor 1	LEF1	15236
leucine twenty homeobox	LEUTX	15237
LIM homeobox 1	LHX1	15238
LIM homeobox 2	LHX2	15239
LIM homeobox 3	LHX3	15240
LIM homeobox 4	LHX4	15241
LIM homeobox 5	LHX5	15242
LIM homeobox 6	LHX6	15243
LIM homeobox 8	LHX8	15244
LIM homeobox 9	LHX9	15245
LIM homeobox transcription factor 1, alpha	LMX1A	15246
LIM homeobox transcription factor 1, beta	LMX1B	15247
LOC730110	LOC730110	
leucine rich repeat (in FLII) interacting protein 1	LRRFIP1	15248
leucine rich repeat (in FLII) interacting protein 2	LRRFIP2	15249
Ly1 antibody reactive	LYAR	15250
lymphoblastic leukemia associated hematopoiesis regulator 1	LYL1	15251
maelstrom spermatogenic transposon silencer	MAEL	15252
v-maf avian musculoaponeurotic fibrosarcoma oncogene homolog	MAF	15253
MAF1 homolog, negative regulator of RNA polymerase III	MAF1	15254
v-maf avian musculoaponeurotic fibrosarcoma oncogene homolog A	MAFA	15255-15256
v-maf avian musculoaponeurotic fibrosarcoma oncogene homolog B	MAFB	15257
v-maf avian musculoaponeurotic fibrosarcoma oncogene homolog F	MAFF	15258
v-maf avian musculoaponeurotic fibrosarcoma oncogene homolog G	MAFG	15259
v-maf avian musculoaponeurotic fibrosarcoma oncogene homolog K	MAFK	15260
matrin 3	MATR3	15261
MYC associated factor X	MAX	15262
MYC associated zinc finger protein	MAZ	15263

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
methyl-CpG binding domain protein 1	MBD1	15264
methyl-CpG binding domain protein 2	MBD2	15265
methyl-CpG binding domain protein 3	MBD3	15266
methyl-CpG binding domain protein 3-like 1	MBD3L1	15267
methyl-CpG binding domain protein 3-like 2	MBD3L2	15268
methyl-CpG binding domain 4 DNA glycosylase	MBD4	15269
methyl-CpG binding domain protein 5	MBD5	15270
methyl-CpG binding domain protein 6	MBD6	15271
muscleblind like splicing regulator 3	MBNL3	15272
MDS1 and EVI1 complex locus	MECOM	15273
methyl-CpG binding protein 2	MECP2	15274
myocyte enhancer factor 2A	MEF2A	15275
myocyte enhancer factor 2B	MEF2B	15276
myocyte enhancer factor 2C	MEF2C	15277
myocyte enhancer factor 2D	MEF2D	15278
Meis homeobox 1	MEIS1	15279
Meis homeobox 2	MEIS2	15280
Meis homeobox 3	MEIS3	15281
Meis homeobox 3 pseudogene 1	MEIS3P1	15282
Meis homeobox 3 pseudogene 2	MEIS3P2	15283
mesenchyme homeobox 1	MEOX1	15284
mesenchyme homeobox 2	MEOX2	15285
mesoderm posterior bHLH transcription factor 1	MESP1	15286
mesoderm posterior bHLH transcription factor 2	MESP2	15287
MGA, MAX dimerization protein	MGA	15288-15289
MIER1 transcriptional regulator	MIER1	15290
MIER family member 2	MIER2	15291
MIER family member 3	MIER3	15292
MIS18 binding protein 1	MIS18BP1	15293
microphthalmia-associated transcription factor	MITF	15294
Mix paired-like homeobox	MIXL1	15295
mohawk homeobox	MKX	15296
myeloid/lymphoid or mixed-lineage leukemia; translocated to, 1	MLLT1	15297
myeloid/lymphoid or mixed-lineage leukemia; translocated to, 10	MLLT10	15298
myeloid/lymphoid or mixed-lineage leukemia; translocated to, 11	MLLT11	15299
myeloid/lymphoid or mixed-lineage leukemia; translocated to, 3	MLLT3	15300
myeloid/lymphoid or mixed-lineage leukemia; translocated to, 4	MLLT4	15301
myeloid/lymphoid or mixed-lineage leukemia; translocated to, 6	MLLT6	15302
MLX, MAX dimerization protein	MLX	15303
MLX interacting protein	MLXIP	15304
MLX interacting protein-like	MLXIPL	15305
MAX network transcriptional repressor	MNT	15306
motor neuron and pancreas homeobox 1	MNX1	15307
musculin	MSC	15308
mesogenin 1	MSGN1	15309
msh homeobox 1	MSX1	15310
msh homeobox 2	MSX2	15311
metastasis associated 1	MTA1	15312
metastasis associated 1 family member 2	MTA2	15313
metastasis associated 1 family member 3	MTA3	15314
metal-regulatory transcription factor 1	MTF1	15315
metal response element binding transcription factor 2	MTF2	15316
MAX dimerization protein 1	MXD1	15317
MAX dimerization protein 3	MXD3	15318
MAX dimerization protein 4	MXD4	15319
MAX interactor 1, dimerization protein	MXI1	15320
v-myb avian myeloblastosis viral oncogene homolog	MYB	15321
v-myb avian myeloblastosis viral oncogene homolog- like 1	MYBL1	15322
v-myb avian myeloblastosis viral oncogene homolog- like 2	MYBL2	15323
v-myc avian myelocytomatosis viral oncogene homolog	MYC	15324
v-myc avian myelocytomatosis viral oncogene lung carcinoma derived homolog	MYCL	15325
MYCL pseudogene 1	MYCLP1	15326
v-myc avian myelocytomatosis viral oncogene neuroblastoma derived homolog	MYCN	15327
myogenic factor 5	MYF5	15328
myogenic factor 6	MYF6	15329

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
myoneurin	MYNN	15330
myogenic differentiation 1	MYOD1	15331
myogenin (myogenic factor 4)	MYOG	15332
myelin regulatory factor	MYRF	15333
Myb-like, SWIRM and MPN domains 1	MYSM1	15334
myelin transcription factor 1	MYT1	15335-15336
myelin transcription factor 1 like	MYT1L	15337
myeloid zinc finger 1	MZF1	15338
Nanog homeobox	NANOG	15339
NANOG neighbor homeobox	NANOGNB	15340
Nanog homeobox pseudogene 1	NANOGP1	15341
Nanog homeobox pseudogene 8	NANOGP8	15342
nuclear receptor coactivator 1	NCOA1	15343
nuclear receptor coactivator 2	NCOA2	15344
nuclear receptor coactivator 3	NCOA3	15345
nuclear receptor coactivator 4	NCOA4	15346
nuclear receptor coactivator 5	NCOA5	15347
nuclear receptor coactivator 6	NCOA6	15348
nuclear receptor coactivator 7	NCOA7	15349
nuclear receptor corepressor 1	NCOR1	15350
nuclear receptor corepressor 2	NCOR2	15351
neuronal differentiation 1	NEUROD1	15352
neuronal differentiation 2	NEUROD2	15353
neuronal differentiation 4	NEUROD4	15354
neuronal differentiation 6	NEUROD6	15355
neurogenin 1	NEUROG1	15356
neurogenin 2	NEUROG2	15357
neurogenin 3	NEUROG3	15358
nuclear factor of activated T-cells 5, tonic- responsive	NFAT5	15359
nuclear factor of activated T-cells, cytoplasmic, calcineurin-dependent 1	NFATC1	15360
nuclear factor of activated T-cells, cytoplasmic, calcineurin-dependent 2	NFATC2	15361
nuclear factor of activated T-cells, cytoplasmic, calcineurin-dependent 3	NFATC3	15362
nuclear factor of activated T-cells, cytoplasmic, calcineurin-dependent 4	NFATC4	15363
nuclear factor, erythroid 2	NFE2	15364
nuclear factor, erythroid 2 like 1	NFE2L1	15365
nuclear factor, erythroid 2 like 2	NFE2L2	15366
nuclear factor, erythroid 2 like 3	NFE2L3	15367
nuclear factor I/A	NFIA	15368
nuclear factor I/B	NFIB	15369
nuclear factor I/C (CCAAT-binding transcription factor)	NFIC	15370
nuclear factor, interleukin 3 regulated	NFIL3	15371
nuclear factor I/X (CCAAT-binding transcription factor)	NFIX	15372
nuclear factor of kappa light polypeptide gene enhancer in B-cells 1	NFKB1	15373
nuclear factor of kappa light polypeptide gene enhancer in B-cells 2 (p49/p100)	NFKB2	15374
nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor, alpha	NFKBIA	15375
nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor, beta	NFKBIB	15376
nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor, delta	NFKBID	15377
nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor, epsilon	NFKBIE	15378
nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor-like 1	NFKBIL1	15379
nuclear factor of kappa light polypeptide gene enhancer in B-cells inhibitor, zeta	NFKBIZ	15380
nuclear factor related to kappaB binding protein	NFRKB	15381
nuclear transcription factor, X-box binding 1	NFX1	15382
nuclear transcription factor, X-box binding-like 1	NFXL1	15383
nuclear transcription factor Y subunit alpha	NFYA	15384
nuclear transcription factor Y subunit beta	NFYB	15385
nuclear transcription factor Y subunit gamma	NFYC	15386
nescient helix-loop-helix 1	NHLH1	15387
nescient helix-loop-helix 2	NHLH2	15388
NFKB repressing factor	NKRF	15389
NK1 homeobox 1	NKX1-1	15390
NK1 homeobox 2	NKX1-2	15391

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
NK2 homeobox 1	NKX2-1	15392
NK2 homeobox 2	NKX2-2	15393
NK2 homeobox 3	NKX2-3	15394
NK2 homeobox 4	NKX2-4	15395
NK2 homeobox 5	NKX2-5	15396
NK2 homeobox 6	NKX2-6	15397
NK2 homeobox 8	NKX2-8	15398
NK3 homeobox 1	NKX3-1	15399
NK3 homeobox 2	NKX3-2	15400
NK6 homeobox 1	NKX6-1	15401
NK6 homeobox 2	NKX6-2	15402
NK6 homeobox 3	NKX6-3	15403
NOBOX oogenesis homeobox	NOBOX	15404
NOC3 like DNA replication regulator	NOC3L	15405
nucleolar complex associated 4 homolog	NOC4L	15406
non-POU domain containing, octamer-binding	NONO	15407
notochord homeobox	NOTO	15408
neuronal PAS domain protein 1	NPAS1	15409
neuronal PAS domain protein 2	NPAS2	15410
neuronal PAS domain protein 3	NPAS3	15411
neuronal PAS domain protein 4	NPAS4	15412
nuclear receptor subfamily 0 group B member 1	NR0B1	15413
nuclear receptor subfamily 0 group B member 2	NR0B2	15414
nuclear receptor subfamily 1 group D member 1	NR1D1	15415
nuclear receptor subfamily 1 group D member 2	NR1D2	15416
nuclear receptor subfamily 1 group H member 2	NR1H2	15417
nuclear receptor subfamily 1 group H member 3	NR1H3	15418
nuclear receptor subfamily 1 group H member 4	NR1H4	15419
nuclear receptor subfamily 1 group I member 2	NR1I2	15420
nuclear receptor subfamily 1 group I member 3	NR1I3	15421
nuclear receptor subfamily 2 group C member 1	NR2C1	15422
nuclear receptor subfamily 2 group C member 2	NR2C2	15423
nuclear receptor subfamily 2 group E member 1	NR2E1	15424
nuclear receptor subfamily 2 group E member 3	NR2E3	15425
nuclear receptor subfamily 2 group F member 1	NR2F1	15426
nuclear receptor subfamily 2 group F member 2	NR2F2	15427
nuclear receptor subfamily 2 group F member 6	NR2F6	15428
nuclear receptor subfamily 3 group C member 1	NR3C1	15429
nuclear receptor subfamily 3 group C member 2	NR3C2	15430
nuclear receptor subfamily 4 group A member 1	NR4A1	15431
nuclear receptor subfamily 4 group A member 2	NR4A2	15432
nuclear receptor subfamily 4 group A member 3	NR4A3	15433
nuclear receptor subfamily 5 group A member 1	NR5A1	15434
nuclear receptor subfamily 5 group A member 2	NR5A2	15435
nuclear receptor subfamily 6 group A member 1	NR6A1	15436
nuclear respiratory factor 1	NRF1	15437-15438
neural retina leucine zipper	NRL	15439
oligodendrocyte transcription factor 1	OLIG1	15440
oligodendrocyte lineage transcription factor 2	OLIG2	15441
oligodendrocyte transcription factor 3	OLIG3	15442
one cut homeobox 1	ONECUT1	15443
one cut homeobox 2	ONECUT2	15444
one cut homeobox 3	ONECUT3	15445
odd-skipped related transcription factor 1	OSR1	15446
odd-skipped related transcription factor 2	OSR2	15447
orthopedia homeobox	OTP	15448
orthodenticle homeobox 1	OTX1	15449
orthodenticle homeobox 2	OTX2	15450
ovo like zinc finger 1	OVOL1	15451
ovo like zinc finger 2	OVOL2	15452
ovo like zinc finger 3	OVOL3	15453
poly(ADP-ribose) polymerase 1	PARP1	15454
poly(ADP-ribose) polymerase family member 12	PARP12	15455
POZ/BTB and AT hook containing zinc finger 1	PATZ1	15456
PRKC, apoptosis, WT1, regulator	PAWR	15457
paired box 1	PAX1	15458
paired box 2	PAX2	15459
paired box 3	PAX3	15460
paired box 4	PAX4	15461
paired box 5	PAX5	15462
paired box 6	PAX6	15463
paired box 7	PAX7	15464
paired box 8	PAX8	15465
paired box 9	PAX9	15466
PAX3 and PAX7 binding protein 1	PAXBP1	15467
polybromo 1	PBRM1	15468

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
pre-B-cell leukemia homeobox 1	PBX1	15469
pre-B-cell leukemia homeobox 2	PBX2	15470
pre-B-cell leukemia homeobox 3	PBX3	15471
pre-B-cell leukemia homeobox 4	PBX4	15472
poly(rC) binding protein 1	PCBP1	15473
poly(rC) binding protein 2	PCBP2	15474
poly(rC) binding protein 3	PCBP3	15475
poly(rC) binding protein 4	PCBP4	15476
poly comb group ring finger 6	PCGF6	15477
pancreatic and duodenal homeobox 1	PDX1	15478-15479
paternally expressed 3	PEG3	15480
progesterone receptor	PGR	15481
prohibitin	PHB	15482
prohibitin 2	PHB2	15483
PHD finger protein 20	PHF20	15484
PHD finger protein 5A	PHF5A	15485
paired like homeobox 2a	PHOX2A	15486
paired like homeobox 2b	PHOX2B	15487
putative homeodomain transcription factor 1	PHTF1	15488
putative homeodomain transcription factor 2	PHTF2	15489
paired like homeodomain 1	PITX1	15490
paired like homeodomain 2	PITX2	15491
paired like homeodomain 3	PITX3	15492
PBX/knotted 1 homeobox 1	PKNOX1	15493
PBX/knotted 1 homeobox 2	PKNOX2	15494
PLAG1 zinc finger	PLAG1	15495
PLAG1 like zinc finger 1	PLAGL1	15496
PLAG1 like zinc finger 2	PLAGL2	15497
pleckstrin	PLEK	15498
promyelocytic leukaemia zinc finger	PLZF	15499
pogo transposable element with ZNF domain	POGZ	15500
POU class 1 homeobox 1	POU1F1	15501
POU class 2 associating factor 1	POU2AF1	15502
POU class 2 homeobox 1	POU2F1	15503
POU class 2 homeobox 2	POU2F2	15504
POU class 2 homeobox 3	POU2F3	15505
POU class 3 homeobox 1	POU3F1	15506
POU class 3 homeobox 2	POU3F2	15507
POU class 3 homeobox 3	POU3F3	15508
POU class 3 homeobox 4	POU3F4	15509
POU class 4 homeobox 1	POU4F1	15510
POU class 4 homeobox 2	POU4F2	15511
POU class 4 homeobox 3	POU4F3	15512
POU class 5 homeobox 1	POU5F1	15513
POU class 5 homeobox 1B	POU5F1B	15514
POU domain class 5, transcription factor 2	POU5F2	15515
POU class 6 homeobox 1	POU6F1	15516
POU class 6 homeobox 2	POU6F2	15517
peroxisome proliferator activated receptor alpha	PPARA	15518
peroxisome proliferator activated receptor delta	PPARD	15519
peroxisome proliferator activated receptor gamma	PPARG	15520
protein phosphatase 1 regulatory subunit 13 like	PPP1R13L	15521
PR domain 1	PRDM1	15522
PR domain 10	PRDM10	15523
PR domain 11	PRDM11	15524
PR domain 12	PRDM12	15525
PR domain 13	PRDM13	15526
PR domain 14	PRDM14	15527
PR domain 15	PRDM15	15528
PR domain 16	PRDM16	15529
PR domain 2	PRDM2	15530
PR domain 4	PRDM4	15531
PR domain 5	PRDM5	15532
PR domain 6	PRDM6	15533
PR domain 7	PRDM7	15534
PR domain 8	PRDM8	15535
PR domain 9	PRDM9	15536
prolactin regulatory element binding	PREB	15537
PROP paired-like homeobox 1	PROP1	15538
prospero homeobox 1	PROX1	15539
prospero homeobox 2	PROX2	15540
paired related homeobox 1	PRRX1	15541
paired related homeobox 2	PRRX2	15542
paraspeckle component 1	PSPC1	15543
pancreas specific transcription factor, 1a	PTF1A	15544
purine-rich element binding protein A	PURA	15545

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
purine-rich element binding protein B	PURB	15546
purine-rich element binding protein G	PURG	15547
retinoic acid receptor alpha	RARA	15548
retinoic acid receptor beta	RARB	15549
retinoic acid receptor gamma	RARG	15550
retina and anterior neural fold homeobox	RAX	15551-15552
retina and anterior neural fold homeobox 2	RAX2	15553
RB associated KRAB zinc finger	RBAK	15554
RNA binding motif protein 22	RBM22	15555
recombination signal binding protein for immunoglobulin kappa J region	RBPJ	15556
recombination signal binding protein for immunoglobulin kappa J region-like	RBPJL	15557
ring finger and CCCH-type domains 1	RC3H1	15558
ring finger and CCCH-type domains 2	RC3H2	15559
REST corepressor 1	RCOR1	15560
REST corepressor 2	RCOR2	15561
REST corepressor 3	RCOR3	15562
v-rel avian reticuloendotheliosis viral oncogene homolog	REL	15563
v-rel avian reticuloendotheliosis viral oncogene homolog A	RELA	15564
v-rel avian reticuloendotheliosis viral oncogene homolog B	RELB	15565
arginine-glutamic acid dipeptide (RE) repeats	RERE	15566
RE1-silencing transcription factor	REST	15567
regulatory factor X1	RFX1	15568
regulatory factor X2	RFX2	15569
regulatory factor X3	RFX3	15570
regulatory factor X4	RFX4	15571
regulatory factor X5	RFX5	15572
regulatory factor X6	RFX6	15573
regulatory factor X7	RFX7	15574
RFX family member 8, lacking RFX DNA binding domain	RFX8	15575
regulatory factor X associated ankyrin containing protein	RFXANK	15576
regulatory factor X associated protein	RFXAP	15577
Rhox homeobox family member 1	RHOXF1	15578
Rhox homeobox family member 2	RHOXF2	15579
Rhox homeobox family member 2B	RHOXF2B	15580
rearranged L-myc fusion	RLF	15581-15582
RAR related orphan receptor A	RORA	15583
RAR related orphan receptor B	RORB	15584
RAR related orphan receptor C	RORC	15585
retinoic acid receptor-related orphan nuclear receptor gamma	RORgT	15586
ras responsive element binding protein 1	RREB1	15587
runt related transcription factor 1	RUNX1	15588
runt related transcription factor 1; translocated to, 1 (cyclin D related)	RUNX1T1	15589
runt related transcription factor 2	RUNX2	15590
runt related transcription factor 3	RUNX3	15591
retinoid X receptor alpha	RXRA	15592
retinoid X receptor beta	RXRB	15593
retinoid X receptor gamma	RXRG	15594
spalt-like transcription factor 1	SALL1	15595
spalt-like transcription factor 2	SALL2	15596
spalt-like transcription factor 3	SALL3	15597
spalt-like transcription factor 4	SALL4	15598
SATB homeobox 1	SATB1	15599
SATB homeobox 2	SATB2	15600
S-phase cyclin A-associated protein in the ER	SCAPER	15601
scratch family zinc finger 1	SCRT1	15602
scratch family zinc finger 2	SCRT2	15603
scleraxis bHLH transcription factor	SCX	15604
SEBOX homeobox	SEBOX	15605
SET binding protein 1	SETBP1	15606
splicing factor proline/glutamine-rich	SFPQ	15607
short stature homeobox	SHOX	15608
short stature homeobox 2	SHOX2	15609
single-minded family bHLH transcription factor 1	SIM1	15610
single-minded family bHLH transcription factor 2	SIM2	15611
SIX homeobox 1	SIX1	15612
SIX homeobox 2	SIX2	15613
SIX homeobox 3	SIX3	15614

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
SIX homeobox 4	SIX4	15615
SIX homeobox 5	SIX5	15616
SIX homeobox 6	SIX6	15617
SKI proto-oncogene	SKI	15618
SKI-like proto-oncogene	SKIL	15619
SKI family transcriptional corepressor 1	SKOR1	15620
SKI family transcriptional corepressor 2	SKOR2	15621
solute carrier family 30 (zinc transporter), member 9	SLC30A9	15622
SMAD family member 1	SMAD1	15623
SMAD family member 2	SMAD2	15624
SMAD family member 3	SMAD3	15625
SMAD family member 4	SMAD4	15626
SMAD family member 5	SMAD5	15627
SMAD family member 6	SMAD6	15628
SMAD family member 7	SMAD7	15629
SMAD family member 9	SMAD9	15630
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 1	SMARCA1	15631
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 2	SMARCA2	15632
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 4	SMARCA4	15633
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 5	SMARCA5	15634
SWI/SNF-related, matrix-associated actin-dependent regulator of chromatin, subfamily a, containing DEAD/H box 1	SMARCAD1	15635
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a-like 1	SMARCAL1	15636
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily b, member 1	SMARCB1	15637
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily c, member 1	SMARCC1	15638
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily c, member 2	SMARCC2	15639
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily d, member 1	SMARCD1	15640
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily d, member 2	SMARCD2	15641
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily d, member 3	SMARCD3	15642
SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily e, member 1	SMARCE1	15643
snail family zinc finger 1	SNAI1	15644
snail family zinc finger 2	SNAI2	15645
snail family zinc finger 3	SNAI3	15646
small nuclear RNA activating complex polypeptide 4	SNAPC4	15647
spermatogenesis and oogenesis specific basic helix-loop-helix 1	SOHLH1	15648
spermatogenesis and oogenesis specific basic helix-loop-helix 2	SOHLH2	15649
SRY-box 1	SOX1	15650
SRY-box 10	SOX10	15651
SRY-box 11	SOX11	15652
SRY-box 12	SOX12	15653
SRY-box 13	SOX13	15654
SRY-box 14	SOX14	15655
SRY-box 15	SOX15	15656
SRY-box 17	SOX17	15657
SRY-box 18	SOX18	15658
SRY-box 2	SOX2	15659
SRY-box 21	SOX21	15660
SRY-box 3	SOX3	15661
SRY-box 30	SOX30	15662
SRY-box 4	SOX4	15663
SRY-box 5	SOX5	15664
SRY-box 6	SOX6	15665
SRY-box 7	SOX7	15666
SRY-box 8	SOX8	15667
SRY-box 9	SOX9	15668
Sp1 transcription factor	SP1	15669-15670
SP100 nuclear antigen	SP100	15671
SP110 nuclear body protein	SP110	15672
SP140 nuclear body protein	SP140	15673
SP140 nuclear body protein like	SP140L	15674
Sp2 transcription factor	SP2	15675

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
Sp3 transcription factor	SP3	15676
Sp4 transcription factor	SP4	15677
Sp5 transcription factor	SP5	15678
Sp6 transcription factor	SP6	15679
Sp7 transcription factor	SP7	15680
Sp8 transcription factor	SP8	15681
Sp9 transcription factor	SP9	15682
SAM pointed domain containing ETS transcription factor	SPDEF	15683
Spi-1 proto-oncogene	SPI1	15684
Spi-B transcription factor (Spi-1/PU.1 related)	SPIB	15685
Spi-C transcription factor (Spi-1/PU.1 related)	SPIC	15686
spermatogenic leucine zipper 1	SPZ1	15687
sterol regulatory element binding transcription factor 1	SREBF1	15688
sterol regulatory element binding transcription factor 2	SREBF2	15689
serum response factor	SRF	15690
sex determining region Y	SRY	15691
structure specific recognition protein 1	SSRP1	15692
suppression of tumorigenicity 18, zinc finger	ST18	15693
signal transducer and activator of transcription 1	STAT1	15694
signal transducer and activator of transcription 2	STAT2	15695
signal transducer and activator of transcription 3 (acute-phase response factor)	STAT3	15696
signal transducer and activator of transcription 4	STAT4	15697
signal transducer and activator of transcription 5	STAT5	15698
signal transducer and activator of transcription 5A	STAT5A	15699
signal transducer and activator of transcription 5B	STAT5B	15700
signal transducer and activator of transcription 6, interleukin-4 induced	STAT6	15701
transcriptional adaptor 2A	TADA2A	15702
transcriptional adaptor 2B	TADA2B	15703
TATA-box binding protein associated factor 1	TAF1	15704
T-cell acute lymphocytic leukemia 1	TAL1	15705
T-cell acute lymphocytic leukemia 2	TAL2	15706
Tax1 (human T-cell leukemia virus type I) binding protein 1	TAX1BP1	15707
Tax1 (human T-cell leukemia virus type I) binding protein 3	TAX1BP3	15708
T-box transcription factor T-bet	Tbet	15709
TATA-box binding protein	TBP	15710
TATA-box binding protein like 1	TBPL1	15711
TATA-box binding protein like 2	TBPL2	15712
T-box, brain 1	TBR1	15713
T-box 1	TBX1	15714
T-box 10	TBX10	15715
T-box 15	TBX15	15716
T-box 18	TBX18	15717
T-box 19	TBX19	15718
T-box 2	TBX2	15719
T-box 20	TBX20	15720
T-box 21	TBX21	15721
T-box 22	TBX22	15722
T-box 3	TBX3	15723
T-box 4	TBX4	15724
T-box 5	TBX5	15725
T-box 6	TBX6	15726
transcription factor 12	TCF12	15727
transcription factor 15 (basic helix-loop-helix)	TCF15	15728
transcription factor 19	TCF19	15729
transcription factor 20 (AR1)	TCF20	15730
transcription factor 21	TCF21	15731
transcription factor 23	TCF23	15732
transcription factor 24	TCF24	15733
transcription factor 25 (basic helix-loop-helix)	TCF25	15734
transcription factor 3	TCF3	15735
transcription factor 4	TCF4	15736
transcription factor 7 (T-cell specific, HMG-box, TCF1)	TCF7	15737
transcription factor 7 like 1	TCF7L1	15738
transcription factor 7 like 2	TCF7L2	15739
transcription factor-like 5 (basic helix-loop-helix)	TCFL5	15740
TEA domain transcription factor 1	TEAD1	15741
TEA domain transcription factor 2	TEAD2	15742
TEA domain transcription factor 3	TEAD3	15743
TEA domain transcription factor 4	TEAD4	15744
thyrotrophic embryonic factor	TEF	15745

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
telomeric repeat binding factor (NIMA-interacting) 1	TERF1	15746
telomeric repeat binding factor 2	TERF2	15747
tet methylcytosine dioxygenase 1	TET1	15748
tet methylcytosine dioxygenase 2	TET2	15749
tet methylcytosine dioxygenase 3	TET3	15750
transcription factor A, mitochondrial	TFAM	15751
transcription factor AP-2 alpha (activating enhancer binding protein 2 alpha)	TFAP2A	15752
transcription factor AP-2 beta (activating enhancer binding protein 2 beta)	TFAP2B	15753
transcription factor AP-2 gamma (activating enhancer binding protein 2 gamma)	TFAP2C	15754
transcription factor AP-2 delta (activating enhancer binding protein 2 delta)	TFAP2D	15755
transcription factor AP-2 epsilon (activating enhancer binding protein 2 epsilon)	TFAP2E	15756
transcription factor AP-4 (activating enhancer binding protein 4)	TFAP4	15757
transcription factor B1, mitochondrial	TFB1M	15758
transcription factor B2, mitochondrial	TFB2M	15759
transcription factor CP2	TFCP2	15760
transcription factor CP2-like 1	TFCP2L1	15761
transcription factor Dp-1	TFDP1	15762
transcription factor Dp-2 (E2F dimerization partner 2)	TFDP2	15763
transcription factor Dp family member 3	TFDP3	15764
transcription factor binding to IGHM enhancer 3	TFE3	15765
transcription factor EB	TFEB	15766
transcription factor EC	TFEC	15767
TGFB induced factor homeobox 1	TGIF1	15768
TGFB induced factor homeobox 2	TGIF2	15769
TGFB induced factor homeobox 2 like, X-linked	TGIF2LX	15770
TGFB induced factor homeobox 2 like, Y-linked	TGIF2LY	15771
THAP domain containing, apoptosis associated protein 1	THAP1	15772
THAP domain containing 10	THAP10	15773
THAP domain containing 11	THAP11	15774
THAP domain containing 12	THAP12	15775
THAP domain containing, apoptosis associated protein 2	THAP2	15776
THAP domain containing, apoptosis associated protein 3	THAP3	15777
THAP domain containing 4	THAP4	15778
THAP domain containing 5	THAP5	15779
THAP domain containing 6	THAP6	15780
THAP domain containing 7	THAP7	15781
THAP domain containing 8	THAP8	15782
THAP domain containing 9	THAP9	15783
Th inducing POZ-Kruppel Factor	ThPOK	15784
thyroid hormone receptor, alpha	THRA	15785
thyroid hormone receptor, beta	THRB	15786
T-cell leukemia homeobox 1	TLX1	15787
T-cell leukemia homeobox 2	TLX2	15788
T-cell leukemia homeobox 3	TLX3	15789
target of EGR1, member 1 (nuclear)	TOE1	15790
tonsoku-like, DNA repair protein	TONSL	15791
topoisomerase I binding, arginine/serine-rich, E3	TOPORS	15792
ubiquitin protein ligase		
thymocyte selection associated high mobility group box	TOX	15793
TOX high mobility group box family member 2	TOX2	15794
TOX high mobility group box family member 3	TOX3	15795
TOX high mobility group box family member 4	TOX4	15796
tumor protein p53	TP53	15797
tumor protein p63	TP63	15798
tumor protein p73	TP73	15799
tetra-peptide repeat homeobox 1	TPRX1	15800
tetra-peptide repeat homeobox-like	TPRXL	15801
transcriptional regulating factor 1	TRERF1	15802
trichorhinophalangeal syndrome 1	TRPS1	15803
TSC22 domain family member 1	TSC22D1	15804
TSC22 domain family member 2	TSC22D2	15805
TSC22 domain family member 3	TSC22D3	15806
TSC22 domain family member 4	TSC22D4	15807
teashirt zinc finger homeobox 1	TSHZ1	15808
teashirt zinc finger homeobox 2	TSHZ2	15809
teashirt zinc finger homeobox 3	TSHZ3	15810

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
transcription termination factor, RNA polymerase I	TTF1	15811-15812
transcription termination factor, RNA polymerase II	TTF2	15813-15814
tubby bipartite transcription factor	TUB	15815
twist family bHLH transcription factor 1	TWIST1	15816
twist family bHLH transcription factor 2	TWIST2	15817
upstream binding protein 1 (LBP-1a)	UBP1	15818
upstream binding transcription factor, RNA polymerase I	UBTF	15819
upstream binding transcription factor, RNA polymerase I-like 1	UBTFL1	15820
upstream binding transcription factor, RNA polymerase I-like 6 (pseudogene)	UBTFL6	15821
UNC homeobox	UNCX	15822
unkempt family zinc finger	UNK	15823
unkempt family like zinc finger	UNKL	15824
upstream transcription factor 1	USF1	15825
upstream transcription factor 2, c-fos interacting	USF2	15826
upstream transcription factor family member 3	USF3	15827
undifferentiated embryonic cell transcription factor 1	UTF1	15828
ventral anterior homeobox 1	VAX1	15829
ventral anterior homeobox 2	VAX2	15830
vitamin D (1,25-dihydroxyvitamin D3) receptor	VDR	15831
VENT homeobox	VENTX	15832
vascular endothelial zinc finger 1	VEZF1	15833
visual system homeobox 1	VSX1	15834
visual system homeobox 2	VSX2	15835
WD repeat and HMG-box DNA binding protein 1	WDHD1	15836
Wolf-Hirschhorn syndrome candidate 1	WHSC1	15837
widely interspaced zinc finger motifs	WIZ	15838
Wilms tumor 1	WT1	15839
X-box binding protein 1	XBP1	15840
Y-box binding protein 1	YBX1	15841
Y-box binding protein 2	YBX2	15842
Y-box binding protein 3	YBX3	15843
YEATS domain containing 2	YEATS2	15844
YEATS domain containing 4	YEATS4	15845
YY1 transcription factor	YY1	15846
YY2 transcription factor	YY2	15847
zinc finger BED-type containing 1	ZBED1	15848
zinc finger BED-type containing 2	ZBED2	15849
zinc finger BED-type containing 3	ZBED3	15850
zinc finger BED-type containing 4	ZBED4	15851
zinc finger BED-type containing 5	ZBED5	15852
zinc finger, BED-type containing 6	ZBED6	15853
Z-DNA binding protein 1	ZBP1	15854-15855
zinc finger and BTB domain containing 1	ZBTB1	15856
zinc finger and BTB domain containing 10	ZBTB10	15857
zinc finger and BTB domain containing 11	ZBTB11	15858
zinc finger and BTB domain containing 12	ZBTB12	15859
zinc finger and BTB domain containing 14	ZBTB14	15860
zinc finger and BTB domain containing 16	ZBTB16	15861
zinc finger and BTB domain containing 17	ZBTB17	15862
zinc finger and BTB domain containing 18	ZBTB18	15863
zinc finger and BTB domain containing 2	ZBTB2	15864
zinc finger and BTB domain containing 20	ZBTB20	15865
zinc finger and BTB domain containing 21	ZBTB21	15866
zinc finger and BTB domain containing 22	ZBTB22	15867
zinc finger and BTB domain containing 24	ZBTB24	15868
zinc finger and BTB domain containing 25	ZBTB25	15869
zinc finger and BTB domain containing 26	ZBTB26	15870
zinc finger and BTB domain containing 3	ZBTB3	15871
zinc finger and BTB domain containing 32	ZBTB32	15872
zinc finger and BTB domain containing 33	ZBTB33	15873
zinc finger and BTB domain containing 34	ZBTB34	15874
zinc finger and BTB domain containing 37	ZBTB37	15875
zinc finger and BTB domain containing 38	ZBTB38	15876
zinc finger and BTB domain containing 39	ZBTB39	15877
zinc finger and BTB domain containing 4	ZBTB4	15878
zinc finger and BTB domain containing 40	ZBTB40	15879
zinc finger and BTB domain containing 41	ZBTB41	15880
zinc finger and BTB domain containing 42	ZBTB42	15881
zinc finger and BTB domain containing 43	ZBTB43	15882
zinc finger and BTB domain containing 44	ZBTB44	15883
zinc finger and BTB domain containing 45	ZBTB45	15884
zinc finger and BTB domain containing 46	ZBTB46	15885
zinc finger and BTB domain containing 47	ZBTB47	15886

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger and BTB domain containing 48	ZBTB48	15887
zinc finger and BTB domain containing 49	ZBTB49	15888
zinc finger and BTB domain containing 5	ZBTB5	15889
zinc finger and BTB domain containing 6	ZBTB6	15890
zinc finger and BTB domain containing 7A	ZBTB7A	15891
zinc finger and BTB domain containing 7B	ZBTB7B	15892
zinc finger and BTB domain containing 7C	ZBTB7C	15893
zinc finger and BTB domain containing 8A	ZBTB8A	15894
zinc finger and BTB domain containing 9	ZBTB9	15895
zinc finger CCCH-type containing 10	ZC3H10	15896
zinc finger CCCH-type containing 11A	ZC3H11A	15897
zinc finger CCCH-type containing 12A	ZC3H12A	15898
zinc finger CCCH-type containing 12B	ZC3H12B	15899
zinc finger CCCH-type containing 13	ZC3H13	15900
zinc finger CCCH-type containing 14	ZC3H14	15901
zinc finger CCCH-type containing 15	ZC3H15	15902
zinc finger CCCH-type containing 18	ZC3H18	15903
zinc finger CCCH-type containing 3	ZC3H3	15904
zinc finger CCCH-type containing 4	ZC3H4	15905
zinc finger CCCH-type containing 6	ZC3H6	15906
zinc finger CCCH-type containing 7A	ZC3H7A	15907
zinc finger CCCH-type containing 7B	ZC3H7B	15908
zinc finger CCCH-type containing 8	ZC3H8	15909
zinc finger CCHC-type containing 11	ZCCHC11	15910
zinc finger CCHC-type containing 6	ZCCHC6	15911
zinc finger E-box binding homeobox 1	ZEB1	15912
zinc finger E-box binding homeobox 2	ZEB2	15913
zinc finger and AT-hook domain containing	ZFAT	15914
zinc finger homeobox 2	ZFH2	15915
zinc finger homeobox 3	ZFH3	15916
zinc finger homeobox 4	ZFH4	15917
ZFP1 zinc finger protein	ZFP1	15918
ZFP14 zinc finger protein	ZFP14	15919
ZFP2 zinc finger protein	ZFP2	15920
ZFP28 zinc finger protein	ZFP28	15921
ZFP3 zinc finger protein	ZFP3	15922
ZFP30 zinc finger protein	ZFP30	15923
ZFP36 ring finger protein-like 1	ZFP36L1	15924
ZFP36 ring finger protein-like 2	ZFP36L2	15925
ZFP37 zinc finger protein	ZFP37	15926
ZFP41 zinc finger protein	ZFP41	15927
ZFP42 zinc finger protein	ZFP42	15928
ZFP57 zinc finger protein	ZFP57	15929
ZFP62 zinc finger protein	ZFP62	15930
ZFP64 zinc finger protein	ZFP64	15931
ZFP69 zinc finger protein	ZFP69	15932-15933
ZFP69 zinc finger protein B	ZFP69B	15934
ZFP82 zinc finger protein	ZFP82	15935
ZFP90 zinc finger protein	ZFP90	15936
ZFP91 zinc finger protein	ZFP91	15937
ZFP92 zinc finger protein	ZFP92	15938
zinc finger protein, FOG family member 1	ZFPM1	15939
zinc finger protein, FOG family member 2	ZFPM2	15940
zinc finger protein, X-linked	ZFX	15941
zinc finger protein, Y-linked	ZFY	15942
zinc finger, FYVE domain containing 26	ZFYVE26	15943
zinc finger, GATA-like protein 1	ZGLP1	15944
zinc finger CCCH-type and G-patch domain containing	ZGPAT	15945
zinc fingers and homeoboxes 1	ZHX1	15946
zinc fingers and homeoboxes 2	ZHX2	15947
zinc fingers and homeoboxes 3	ZHX3	15948
Zic family member 1	ZIC1	15949
Zic family member 2	ZIC2	15950
Zic family member 3	ZIC3	15951
Zic family member 4	ZIC4	15952
Zic family member 5	ZIC5	15953
zinc finger protein interacting with K protein 1	ZIK1	15954
zinc finger, imprinted 2	ZIM2	15955
zinc finger, imprinted 3	ZIM3	15956
zinc finger with KRAB and SCAN domains 1	ZKSCAN1	15957
zinc finger with KRAB and SCAN domains 2	ZKSCAN2	15958
zinc finger with KRAB and SCAN domains 3	ZKSCAN3	15959
zinc finger with KRAB and SCAN domains 4	ZKSCAN4	15960
zinc finger with KRAB and SCAN domains 5	ZKSCAN5	15961
zinc finger with KRAB and SCAN domains 7	ZKSCAN7	15962

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger with KRAB and SCAN domains 8	ZKSCAN8	15963
zinc finger matrin-type 1	ZMAT1	15964
zinc finger matrin-type 2	ZMAT2	15965
zinc finger matrin-type 3	ZMAT3	15966
zinc finger matrin-type 4	ZMAT4	15967
zinc finger matrin-type 5	ZMAT5	15968
zinc finger protein 10	ZNF10	15969
zinc finger protein 100	ZNF100	15970
zinc finger protein 101	ZNF101	15971
zinc finger protein 106	ZNF106	15972
zinc finger protein 107	ZNF107	15973
zinc finger protein 112	ZNF112	15974
zinc finger protein 114	ZNF114	15975
zinc finger protein 117	ZNF117	15976
zinc finger protein 12	ZNF12	15977
zinc finger protein 121	ZNF121	15978
zinc finger protein 124	ZNF124	15979
zinc finger protein 131	ZNF131	15980
zinc finger protein 132	ZNF132	15981
zinc finger protein 133	ZNF133	15982
zinc finger protein 134	ZNF134	15983
zinc finger protein 135	ZNF135	15984
zinc finger protein 136	ZNF136	15985
zinc finger protein 137, pseudogene	ZNF137P	15986
zinc finger protein 138	ZNF138	15987
zinc finger protein 14	ZNF14	15988
zinc finger protein 140	ZNF140	15989
zinc finger protein 141	ZNF141	15990
zinc finger protein 142	ZNF142	15991
zinc finger protein 143	ZNF143	15992
zinc finger protein 146	ZNF146	15993
zinc finger protein 148	ZNF148	15994
zinc finger protein 154	ZNF154	15995
zinc finger protein 155	ZNF155	15996
zinc finger protein 157	ZNF157	15997
zinc finger protein 16	ZNF16	15998
zinc finger protein 160	ZNF160	15999
zinc finger protein 165	ZNF165	16000
zinc finger protein 169	ZNF169	16001
zinc finger protein 17	ZNF17	16002
zinc finger protein 174	ZNF174	16003
zinc finger protein 175	ZNF175	16004
zinc finger protein 18	ZNF18	16005
zinc finger protein 180	ZNF180	16006
zinc finger protein 181	ZNF181	16007
zinc finger protein 182	ZNF182	16008
zinc finger protein 184	ZNF184	16009
zinc finger protein 189	ZNF189	16010
zinc finger protein 19	ZNF19	16011
zinc finger protein 195	ZNF195	16012
zinc finger protein 197	ZNF197	16013
zinc finger protein 2	ZNF2	16014
zinc finger protein 20	ZNF20	16015-16016
zinc finger protein 200	ZNF200	16017
zinc finger protein 202	ZNF202	16018
zinc finger protein 205	ZNF205	16019
zinc finger protein 207	ZNF207	16020
zinc finger protein 208	ZNF208	16021
zinc finger protein 211	ZNF211	16022
zinc finger protein 212	ZNF212	16023
zinc finger protein 213	ZNF213	16024
zinc finger protein 214	ZNF214	16025
zinc finger protein 215	ZNF215	16026
zinc finger protein 217	ZNF217	16027
zinc finger protein 219	ZNF219	16028
zinc finger protein 22	ZNF22	16029
zinc finger protein 221	ZNF221	16030
zinc finger protein 223	ZNF223	16031
zinc finger protein 224	ZNF224	16032
zinc finger protein 225	ZNF225	16033-16034
zinc finger protein 226	ZNF226	16035
zinc finger protein 227	ZNF227	16036
zinc finger protein 229	ZNF229	16037
zinc finger protein 23	ZNF23	16038
zinc finger protein 230	ZNF230	16039-16040
zinc finger protein 232	ZNF232	16041

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger protein 233	ZNF233	16042-16043
zinc finger protein 234	ZNF234	16044
zinc finger protein 235	ZNF235	16045
zinc finger protein 236	ZNF236	16046
zinc finger protein 239	ZNF239	16047
zinc finger protein 24	ZNF24	16048
zinc finger protein 248	ZNF248	16049
zinc finger protein 25	ZNF25	16050
zinc finger protein 250	ZNF250	16051
zinc finger protein 251	ZNF251	16052
zinc finger protein 252, pseudogene	ZNF252P	16053
zinc finger protein 253	ZNF253	16054
zinc finger protein 254	ZNF254	16055
zinc finger protein 256	ZNF256	16056
zinc finger protein 257	ZNF257	16057
zinc finger protein 26	ZNF26	16058
zinc finger protein 260	ZNF260	16059
zinc finger protein 263	ZNF263	16060
zinc finger protein 264	ZNF264	16061
zinc finger protein 266	ZNF266	16062
zinc finger protein 267	ZNF267	16063
zinc finger protein 268	ZNF268	16064
zinc finger protein 273	ZNF273	16065
zinc finger protein 274	ZNF274	16066
zinc finger protein 275	ZNF275	16067
zinc finger protein 276	ZNF276	16068
zinc finger protein 277	ZNF277	16069
zinc finger protein 28	ZNF28	16070
zinc finger protein 280A	ZNF280A	16071
zinc finger protein 280B	ZNF280B	16072
zinc finger protein 280C	ZNF280C	16073
zinc finger protein 280D	ZNF280D	16074
zinc finger protein 281	ZNF281	16075
zinc finger protein 282	ZNF282	16076
zinc finger protein 283	ZNF283	16077
zinc finger protein 284	ZNF284	16078
zinc finger protein 285	ZNF285	16079
zinc finger protein 286A	ZNF286A	16080
zinc finger protein 286B	ZNF286B	16081
zinc finger protein 287	ZNF287	16082
zinc finger protein 292	ZNF292	16083
zinc finger protein 296	ZNF296	16084
zinc finger protein 3	ZNF3	16085
zinc finger protein 30	ZNF30	16086
zinc finger protein 300	ZNF300	16087
zinc finger protein 302	ZNF302	16088
zinc finger protein 304	ZNF304	16089
zinc finger protein 311	ZNF311	16090
zinc finger protein 316	ZNF316	16091
zinc finger protein 317	ZNF317	16092
zinc finger protein 318	ZNF318	16093
zinc finger protein 319	ZNF319	16094
zinc finger protein 32	ZNF32	16095
zinc finger protein 320	ZNF320	16096
zinc finger protein 322	ZNF322	16097
zinc finger protein 324	ZNF324	16098
zinc finger protein 324B	ZNF324B	16099
zinc finger protein 326	ZNF326	16100
zinc finger protein 329	ZNF329	16101
zinc finger protein 331	ZNF331	16102
zinc finger protein 333	ZNF333	16103
zinc finger protein 334	ZNF334	16104
zinc finger protein 335	ZNF335	16105
zinc finger protein 337	ZNF337	16106
zinc finger protein 33A	ZNF33A	16107
zinc finger protein 33B	ZNF33B	16108
zinc finger protein 34	ZNF34	16109
zinc finger protein 341	ZNF341	16110
zinc finger protein 343	ZNF343	16111
zinc finger protein 345	ZNF345	16112
zinc finger protein 346	ZNF346	16113
zinc finger protein 347	ZNF347	16114
zinc finger protein 35	ZNF35	16115
zinc finger protein 350	ZNF350	16116
zinc finger protein 354A	ZNF354A	16117

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Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger protein 354B	ZNF354B	16118
zinc finger protein 354C	ZNF354C	16119
zinc finger protein 355, pseudogene	ZNF355P	16120
zinc finger protein 358	ZNF358	16121
zinc finger protein 362	ZNF362	16122
zinc finger protein 365	ZNF365	16123-16124
zinc finger protein 366	ZNF366	16125
zinc finger protein 367	ZNF367	16126
zinc finger protein 37A	ZNF37A	16127
zinc finger protein 382	ZNF382	16128
zinc finger protein 383	ZNF383	16129
zinc finger protein 384	ZNF384	16130
zinc finger protein 385A	ZNF385A	16131
zinc finger protein 385B	ZNF385B	16132
zinc finger protein 385C	ZNF385C	16133
zinc finger protein 385D	ZNF385D	16134
zinc finger protein 391	ZNF391	16135
zinc finger protein 394	ZNF394	16136
zinc finger protein 395	ZNF395	16137
zinc finger protein 396	ZNF396	16138
zinc finger protein 397	ZNF397	16139
zinc finger protein 398	ZNF398	16140
zinc finger protein 404	ZNF404	16141
zinc finger protein 407	ZNF407	16142
zinc finger protein 408	ZNF408	16143
zinc finger protein 41	ZNF41	16144
zinc finger protein 410	ZNF410	16145
zinc finger protein 414	ZNF414	16146
zinc finger protein 415	ZNF415	16147
zinc finger protein 416	ZNF416	16148
zinc finger protein 417	ZNF417	16149
zinc finger protein 418	ZNF418	16150
zinc finger protein 419	ZNF419	16151
zinc finger protein 420	ZNF420	16152
zinc finger protein 423	ZNF423	16153
zinc finger protein 425	ZNF425	16154
zinc finger protein 426	ZNF426	16155
zinc finger protein 428	ZNF428	16156
zinc finger protein 429	ZNF429	16157
zinc finger protein 43	ZNF43	16158
zinc finger protein 430	ZNF430	16159
zinc finger protein 431	ZNF431	16160
zinc finger protein 432	ZNF432	16161
zinc finger protein 433	ZNF433	16162
zinc finger protein 436	ZNF436	16163
zinc finger protein 438	ZNF438	16164
zinc finger protein 439	ZNF439	16165
zinc finger protein 44	ZNF44	16166
zinc finger protein 440	ZNF440	16167
zinc finger protein 441	ZNF441	16168
zinc finger protein 442	ZNF442	16169
zinc finger protein 443	ZNF443	16170
zinc finger protein 444	ZNF444	16171
zinc finger protein 445	ZNF445	16172
zinc finger protein 446	ZNF446	16173
zinc finger protein 449	ZNF449	16174
zinc finger protein 45	ZNF45	16175
zinc finger protein 451	ZNF451	16176
zinc finger protein 454	ZNF454	16177
zinc finger protein 460	ZNF460	16178
zinc finger protein 461	ZNF461	16179
zinc finger protein 462	ZNF462	16180
zinc finger protein 467	ZNF467	16181
zinc finger protein 468	ZNF468	16182
zinc finger protein 469	ZNF469	16183
zinc finger protein 470	ZNF470	16184
zinc finger protein 471	ZNF471	16185
zinc finger protein 473	ZNF473	16186
zinc finger protein 474	ZNF474	16187-16188
zinc finger protein 479	ZNF479	16189
zinc finger protein 48	ZNF48	16190
zinc finger protein 480	ZNF480	16191
zinc finger protein 483	ZNF483	16192
zinc finger protein 484	ZNF484	16193
zinc finger protein 485	ZNF485	16194
zinc finger protein 486	ZNF486	16195

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger protein 487	ZNF487	16196
zinc finger protein 488	ZNF488	16197
zinc finger protein 490	ZNF490	16198
zinc finger protein 491	ZNF491	16199
zinc finger protein 492	ZNF492	16200
zinc finger protein 493	ZNF493	16201
zinc finger protein 496	ZNF496	16202
zinc finger protein 497	ZNF497	16203
zinc finger protein 500	ZNF500	16204
zinc finger protein 501	ZNF501	16205
zinc finger protein 502	ZNF502	16206
zinc finger protein 503	ZNF503	16207
zinc finger protein 506	ZNF506	16208
zinc finger protein 507	ZNF507	16209
zinc finger protein 510	ZNF510	16210
zinc finger protein 511	ZNF511	16211
zinc finger protein 512	ZNF512	16212
zinc finger protein 512B	ZNF512B	16213
zinc finger protein 513	ZNF513	16214
zinc finger protein 514	ZNF514	16215
zinc finger protein 516	ZNF516	16216
zinc finger protein 517	ZNF517	16217
zinc finger protein 518A	ZNF518A	16218
zinc finger protein 518B	ZNF518B	16219
zinc finger protein 519	ZNF519	16220
zinc finger protein 521	ZNF521	16221
zinc finger protein 524	ZNF524	16222
zinc finger protein 526	ZNF526	16223
zinc finger protein 527	ZNF527	16224
zinc finger protein 528	ZNF528	16225
zinc finger protein 529	ZNF529	16226
zinc finger protein 530	ZNF530	16227
zinc finger protein 532	ZNF532	16228
zinc finger protein 534	ZNF534	16229
zinc finger protein 536	ZNF536	16230
zinc finger protein 540	ZNF540	16231
zinc finger protein 541	ZNF541	16232
zinc finger protein 542, pseudogene	ZNF542P	16233
zinc finger protein 543	ZNF543	16234
zinc finger protein 544	ZNF544	16235
zinc finger protein 546	ZNF546	16236
zinc finger protein 547	ZNF547	16237
zinc finger protein 548	ZNF548	16238
zinc finger protein 549	ZNF549	16239
zinc finger protein 550	ZNF550	16240
zinc finger protein 552	ZNF552	16241
zinc finger protein 554	ZNF554	16242
zinc finger protein 555	ZNF555	16243
zinc finger protein 556	ZNF556	16244
zinc finger protein 557	ZNF557	16245
zinc finger protein 558	ZNF558	16246
zinc finger protein 559	ZNF559	16247
zinc finger protein 56	ZNF56	16248
zinc finger protein 560	ZNF560	16249
zinc finger protein 561	ZNF561	16250
zinc finger protein 562	ZNF562	16251
zinc finger protein 563	ZNF563	16252
zinc finger protein 564	ZNF564	16253
zinc finger protein 565	ZNF565	16254
zinc finger protein 566	ZNF566	16255
zinc finger protein 567	ZNF567	16256
zinc finger protein 568	ZNF568	16257
zinc finger protein 569	ZNF569	16258
zinc finger protein 57	ZNF57	16259
zinc finger protein 570	ZNF570	16260
zinc finger protein 571	ZNF571	16261
zinc finger protein 572	ZNF572	16262
zinc finger protein 573	ZNF573	16263
zinc finger protein 574	ZNF574	16264
zinc finger protein 575	ZNF575	16265
zinc finger protein 576	ZNF576	16266-16267
zinc finger protein 577	ZNF577	16268
zinc finger protein 578	ZNF578	16269
zinc finger protein 579	ZNF579	16270
zinc finger protein 580	ZNF580	16271
zinc finger protein 581	ZNF581	16272

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger protein 582	ZNF582	16273
zinc finger protein 583	ZNF583	16274
zinc finger protein 584	ZNF584	16275
zinc finger protein 585A	ZNF585A	16276
zinc finger protein 585B	ZNF585B	16277
zinc finger protein 586	ZNF586	16278
zinc finger protein 587	ZNF587	16279
zinc finger protein 589	ZNF589	16280
zinc finger protein 592	ZNF592	16281
zinc finger protein 593	ZNF593	16282
zinc finger protein 594	ZNF594	16283
zinc finger protein 595	ZNF595	16284
zinc finger protein 596	ZNF596	16285
zinc finger protein 597	ZNF597	16286
zinc finger protein 598	ZNF598	16287
zinc finger protein 599	ZNF599	16288
zinc finger protein 600	ZNF600	16289
zinc finger protein 605	ZNF605	16290
zinc finger protein 606	ZNF606	16291
zinc finger protein 607	ZNF607	16292
zinc finger protein 608	ZNF608	16293
zinc finger protein 609	ZNF609	16294
zinc finger protein 610	ZNF610	16295
zinc finger protein 611	ZNF611	16296
zinc finger protein 613	ZNF613	16297
zinc finger protein 614	ZNF614	16298
zinc finger protein 615	ZNF615	16299
zinc finger protein 616	ZNF616	16300
zinc finger protein 618	ZNF618	16301
zinc finger protein 619	ZNF619	16302
zinc finger protein 620	ZNF620	16303
zinc finger protein 621	ZNF621	16304
zinc finger protein 622	ZNF622	16305
zinc finger protein 623	ZNF623	16306
zinc finger protein 624	ZNF624	16307
zinc finger protein 625	ZNF625	16308
zinc finger protein 626	ZNF626	16309
zinc finger protein 627	ZNF627	16310
zinc finger protein 628	ZNF628	16311
zinc finger protein 629	ZNF629	16312
zinc finger protein 639	ZNF639	16313
zinc finger protein 641	ZNF641	16314
zinc finger protein 644	ZNF644	16315
zinc finger protein 645	ZNF645	16316
zinc finger protein 646	ZNF646	16317
zinc finger protein 648	ZNF648	16318
zinc finger protein 649	ZNF649	16319
zinc finger protein 652	ZNF652	16320
zinc finger protein 653	ZNF653	16321
zinc finger protein 654	ZNF654	16322
zinc finger protein 655	ZNF655	16323
zinc finger protein 658	ZNF658	16324
zinc finger protein 658B (pseudogene)	ZNF658B	16325
zinc finger protein 66	ZNF66	16326
zinc finger protein 660	ZNF660	16327
zinc finger protein 662	ZNF662	16328
zinc finger protein 664	ZNF664	16329
zinc finger protein 665	ZNF665	16330
zinc finger protein 667	ZNF667	16331
zinc finger protein 668	ZNF668	16332
zinc finger protein 669	ZNF669	16333
zinc finger protein 670	ZNF670	16334
zinc finger protein 671	ZNF671	16335
zinc finger protein 672	ZNF672	16336
zinc finger protein 674	ZNF674	16337
zinc finger protein 675	ZNF675	16338
zinc finger protein 676	ZNF676	16339
zinc finger protein 677	ZNF677	16340
zinc finger protein 678	ZNF678	16341
zinc finger protein 679	ZNF679	16342
zinc finger protein 680	ZNF680	16343
zinc finger protein 681	ZNF681	16344
zinc finger protein 682	ZNF682	16345
zinc finger protein 683	ZNF683	16346
zinc finger protein 684	ZNF684	16347
zinc finger protein 687	ZNF687	16348

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger protein 688	ZNF688	16349
zinc finger protein 689	ZNF689	16350
zinc finger protein 69	ZNF69	16351
zinc finger protein 691	ZNF691	16352
zinc finger protein 692	ZNF692	16353
zinc finger protein 695	ZNF695	16354
zinc finger protein 696	ZNF696	16355
zinc finger protein 697	ZNF697	16356
zinc finger protein 699	ZNF699	16357
zinc finger protein 7	ZNF7	16358
zinc finger protein 70	ZNF70	16359
zinc finger protein 701	ZNF701	16360
zinc finger protein 702, pseudogene	ZNF702P	16361
zinc finger protein 703	ZNF703	16362
zinc finger protein 704	ZNF704	16363
zinc finger protein 705A	ZNF705A	16364
zinc finger protein 705D	ZNF705D	16365
zinc finger protein 705E	ZNF705E	16366
zinc finger protein 705G	ZNF705G	16367
zinc finger protein 706	ZNF706	16368
zinc finger protein 707	ZNF707	16369
zinc finger protein 708	ZNF708	16370
zinc finger protein 709	ZNF709	16371
zinc finger protein 71	ZNF71	16372
zinc finger protein 710	ZNF710	16373
zinc finger protein 711	ZNF711	16374
zinc finger protein 713	ZNF713	16375
zinc finger protein 714	ZNF714	16376
zinc finger protein 716	ZNF716	16377
zinc finger protein 717	ZNF717	16378
zinc finger protein 718	ZNF718	16379
zinc finger protein 720	ZNF720	16380
zinc finger protein 721	ZNF721	16381
zinc finger protein 724, pseudogene	ZNF724P	16382
zinc finger protein 726	ZNF726	16383
zinc finger protein 727	ZNF727	16384
zinc finger protein 729	ZNF729	16385
zinc finger protein 730	ZNF730	16386
zinc finger protein 732	ZNF732	16387
zinc finger protein 735	ZNF735	16388
zinc finger protein 737	ZNF737	16389
zinc finger protein 74	ZNF74	16390
zinc finger protein 740	ZNF740	16391
zinc finger protein 746	ZNF746	16392
zinc finger protein 747	ZNF747	16393
zinc finger protein 749	ZNF749	16394
zinc finger protein 750	ZNF750	16395
zinc finger protein 75a	ZNF75A	16396
zinc finger protein 75D	ZNF75D	16397
zinc finger protein 76	ZNF76	16398
zinc finger protein 761	ZNF761	16399
zinc finger protein 763	ZNF763	16400
zinc finger protein 764	ZNF764	16401
zinc finger protein 765	ZNF765	16402
zinc finger protein 766	ZNF766	16403
zinc finger protein 768	ZNF768	16404
zinc finger protein 77	ZNF77	16405
zinc finger protein 770	ZNF770	16406
zinc finger protein 771	ZNF771	16407
zinc finger protein 772	ZNF772	16408
zinc finger protein 773	ZNF773	16409
zinc finger protein 774	ZNF774	16410
zinc finger protein 775	ZNF775	16411
zinc finger protein 776	ZNF776	16412
zinc finger protein 777	ZNF777	16413
zinc finger protein 778	ZNF778	16414
zinc finger protein 780A	ZNF780A	16415
zinc finger protein 780B	ZNF780B	16416
zinc finger protein 781	ZNF781	16417
zinc finger protein 782	ZNF782	16418
zinc finger family member 783	ZNF783	16419
zinc finger protein 784	ZNF784	16420
zinc finger protein 785	ZNF785	16421
zinc finger protein 786	ZNF786	16422
zinc finger protein 787	ZNF787	16423
zinc finger family member 788	ZNF788	16424

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger protein 789	ZNF789	16425
zinc finger protein 79	ZNF79	16426
zinc finger protein 790	ZNF790	16427
zinc finger protein 791	ZNF791	16428
zinc finger protein 792	ZNF792	16429
zinc finger protein 793	ZNF793	16430
zinc finger protein 799	ZNF799	16431
zinc finger protein 8	ZNF8	16432
zinc finger protein 80	ZNF80	16433
zinc finger protein 800	ZNF800	16434
zinc finger protein 804A	ZNF804A	16435
zinc finger protein 804B	ZNF804B	16436
zinc finger protein 805	ZNF805	16437
zinc finger protein 806	ZNF806	16438
zinc finger protein 808	ZNF808	16439
zinc finger protein 81	ZNF81	16440
zinc finger protein 813	ZNF813	16441
zinc finger protein 814	ZNF814	16442
zinc finger protein 816	ZNF816	16443
zinc finger protein 821	ZNF821	16444
zinc finger protein 823	ZNF823	16445
zinc finger protein 827	ZNF827	16446
zinc finger protein 829	ZNF829	16447
zinc finger protein 83	ZNF83	16448
zinc finger protein 830	ZNF830	16449
zinc finger protein 831	ZNF831	16450
zinc finger protein 833, pseudogene	ZNF833P	16451
zinc finger protein 835	ZNF835	16452
zinc finger protein 836	ZNF836	16453
zinc finger protein 837	ZNF837	16454
zinc finger protein 839	ZNF839	16455
zinc finger protein 84	ZNF84	16456
zinc finger protein 840, pseudogene	ZNF840P	16457
zinc finger protein 841	ZNF841	16458
zinc finger protein 843	ZNF843	16459
zinc finger protein 844	ZNF844	16460
zinc finger protein 845	ZNF845	16461
zinc finger protein 846	ZNF846	16462
zinc finger protein 85	ZNF85	16463
zinc finger protein 853	ZNF853	16464
zinc finger protein 860	ZNF860	16465
zinc finger protein 876, pseudogene	ZNF876P	16466
zinc finger protein 878	ZNF878	16467
zinc finger protein 879	ZNF879	16468
zinc finger protein 880	ZNF880	16469
zinc finger protein 891	ZNF891	16470
zinc finger protein 90	ZNF90	16471
zinc finger protein 91	ZNF91	16472
zinc finger protein 92	ZNF92	16473
zinc finger protein 93	ZNF93	16474
zinc finger protein 98	ZNF98	16475
zinc finger protein 99	ZNF99	16476
zinc finger, NFX1-type containing 1	ZNFX1	16477
zinc finger and SCAN domain containing 1	ZSCAN1	16478
zinc finger and SCAN domain containing 10	ZSCAN10	16479
zinc finger and SCAN domain containing 12	ZSCAN12	16480
zinc finger and SCAN domain containing 16	ZSCAN16	16481
zinc finger and SCAN domain containing 18	ZSCAN18	16482
zinc finger and SCAN domain containing 2	ZSCAN2	16483
zinc finger and SCAN domain containing 20	ZSCAN20	16484
zinc finger and SCAN domain containing 21	ZSCAN21	16485
zinc finger and SCAN domain containing 22	ZSCAN22	16486
zinc finger and SCAN domain containing 23	ZSCAN23	16487
zinc finger and SCAN domain containing 25	ZSCAN25	16488
zinc finger and SCAN domain containing 26	ZSCAN26	16489
zinc finger and SCAN domain containing 29	ZSCAN29	16490
zinc finger and SCAN domain containing 30	ZSCAN30	16491
zinc finger and SCAN domain containing 31	ZSCAN31	16492
zinc finger and SCAN domain containing 32	ZSCAN32	16493
zinc finger and SCAN domain containing 4	ZSCAN4	16494
zinc finger and SCAN domain containing 5A	ZSCAN5A	16495
zinc finger and SCAN domain containing 5B	ZSCAN5B	16496
zinc finger and SCAN domain containing 5C, pseudogene	ZSCAN5CP	16497
zinc finger and SCAN domain containing 9	ZSCAN9	16498
zinc finger with UFM1-specific peptidase domain	ZUFSP	16499

TABLE 3-continued

Exemplary Transcription Factors.		
Full Name	Abbreviation	SEQ ID NO:
zinc finger, X-linked, duplicated A	ZXDA	16500
zinc finger, X-linked, duplicated B	ZXDB	16501
ZXD family zinc finger C	ZXDC	16502
zinc finger ZZ-type containing 3	ZZZ3	16503

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a cell death or cell apoptosis receptor to produce an armored T-cell of the disclosure. Interaction of a death receptor and its endogenous ligand results in the initiation of apoptosis. Disruption of an expression, an activity, or an interaction of a cell death and/or cell apoptosis receptor and/or ligand render an armored T-cell of the disclosure less receptive to death signals, consequently, making the armored T cell of the disclosure more efficacious in a tumor environment. An exemplary cell death receptor which may be modified in an armored T cell of the disclosure is Fas (CD95). Exemplary cell death and/or cell apoptosis receptors and ligands of the disclosure include, but are not limited to, the exemplary receptors and ligands provided in Table 4.

TABLE 4

Exemplary Cell Death and/or Cell Apoptosis Receptors and Ligands.		
Full Name	Abbreviation	SEQ ID NO:
Cluster of Differentiation 120	CD120a	16504-16505
Death receptor 3	DR3	16506
Death receptor 6	DR6	16507
first apoptosis signal (Fas) receptor	Fas	16508-16509
	(CD95/APO-1)	
Fas Ligand	FasL	16510
cellular tumor antigen p53	p53	16511
Tumor necrosis factor receptor 1	TNF-R1	16512
Tumor necrosis factor receptor 2	TNF-R2	16513
Tumor necrosis factor-related apoptosis-inducing ligand receptor 1	TRAIL-R1	16514
	(DR4)	
Tumor necrosis factor-related apoptosis-inducing ligand receptor 2	TRAIL-R2	16515
	(DR5)	
Fas-associated protein with death domain	FADD	16516
Tumor necrosis factor receptor type 1-associated DEATH domain protein	TRADD	16517
Bcl-2-associated X protein	Bax	16518
Bcl-2 homologous killer	BAK	16519
14-3-3 protein	14-3-3	16520
B-cell lymphoma 2	Bcl-2	16521
Cytochrome C	CytC	16522

TABLE 4-continued

Exemplary Cell Death and/or Cell Apoptosis Receptors and Ligands.		
Full Name	Abbreviation	SEQ ID NO:
Second mitochondria-derived activator of caspase	Smac/Diablo	16523
High temperature requirement protein A2	HTRA2/Omi	16524
Apoptosis inducing factor	AIF	16525
Endonuclease G	EXO G	16526
Caspase 9	Cas9	16527
Caspase 2	Cas2	16528
Caspase 8	Cas8	16529
Caspase 10	Cas10	16530
Caspase 3	Cas3	16531
Caspase 6	Cas6	16532
Caspase 7	Cas7	16533
Tumor Necrosis Factor alpha	TNF-alpha	16534
TNF-related weak inducer of apoptosis	TWEAK	16535
TNF-related weak inducer of apoptosis receptor	TWEAK -R	16536
Tumor necrosis factor-related apoptosis-inducing ligand	TRAIL	16537
TNF ligand-related molecule 1	TL1A	16538
Receptor-interacting serine/threonine-protein kinase 1	RIP1	16539
Cellular inhibitor of apoptosis 1	cIAP-1	16540
TNF receptor-associated factor 2	TRAF-2	16541

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a metabolic sensing protein to produce an armored T-cell of the disclosure. Disruption to the metabolic sensing of the immunosuppressive tumor microenvironment (characterized by low levels of oxygen, pH, glucose and other molecules) by an armored T-cell of the disclosure leads to extended retention of T-cell function and, consequently, more tumor cell skilled per armored T cell. For example, HIF1a and VHL play a role in T-cell function while in a hypoxic environment. An armored T-cell of the disclosure may have silenced or reduced expression of one or more genes encoding HIF1a or VHL. Genes and proteins involved in metabolic sensing include, but are not limited to, the exemplary genes and proteins provided in Table 5.

TABLE 5

Exemplary Metabolic Sensing Genes (and encoded Proteins).			
Full Name	Metabolite	Abbreviation	SEQ ID NO:
hypoxia-inducible factor 1a	Low oxygen	HIF-1a	16542
von Hippel-Lindau tumor suppressor	Low oxygen	VHL	16543
Prolyl-hydroxylase domain proteins	High oxygen	PHD proteins	
Glucose transporter 1	glucose	GLUT1	16544
Linker of Activated T cells	Amino acid (leucine)	LAT	16545
CD98 glycoprotein	Amino acid (leucine)	CD98	16546
Alanine, serine, cysteine-preferring transporter 2	Cationic Amino acid (glutamine)	ASCT2/Slc1a5	16547
Solute carrier family 7 member 1	Cationic Amino acids	Slc7a1	16548
Solute carrier family 7 member 2	Cationic Amino acids	Slc7a2	16549
Solute carrier family 7 member 3	Cationic Amino acids	Slc7a3	16550
Solute carrier family 7 member 4	Cationic Amino acids	Slc7a4	16551

TABLE 5-continued

Exemplary Metabolic Sensing Genes (and encoded Proteins).			
Full Name	Metabolite	Abbreviation	SEQ ID NO:
Solute carrier family 7 member 5	Glycoprotein associated Amino acids	Slc7a5	16552
Solute carrier family 7 member 6	Glycoprotein associated Amino acids	Slc7a6	16553
Solute carrier family 7 member 7	Glycoprotein associated Amino acids	Slc7a7	16554
Solute carrier family 7 member 8	Glycoprotein associated Amino acids	Slc7a8	16555
Solute carrier family 7 member 9	Glycoprotein associated Amino acids	Slc7a9	16556
Solute carrier family 7 member 10	Glycoprotein associated Amino acids	Slc7a10	16557
Solute carrier family 7 member 11	Glycoprotein associated Amino acids	Slc7a11	16558
Solute carrier family 7 member 13	Glycoprotein associated Amino acids	Slc7a13	16559
Solute carrier family 7 member 14	Cationic Amino acids	Slc7a14	16560
Solute carrier family 3 member 2	Amino acid	Slc3a2	16561
Calcium transport protein 2	Cationic Amino acid (arginine)	CAT2	16562
Calcium transport protein 3	Cationic Amino acid (arginine)	CAT3	16563
Calcium transport protein 4	Cationic Amino acid (arginine)	CAT4	16564
Bromodomain adjacent to zinc finger domain protein 1B	Amino acid (arginine)	BAZ1B	16565
PC4 and SFRS1-interacting protein	Amino acid (arginine)	PSIP1	16566
Translin	Amino acid (arginine)	TSN	16567
G-protein-coupled receptors	Fatty Acid and Cholesterol	GPCRs	
T-cell Receptor, subunit alpha	Fatty Acid and Cholesterol	TCR alpha	16568
T-cell Receptor, subunit beta	Fatty Acid and Cholesterol	TCR beta	16569
T-cell Receptor, subunit zeta	Fatty Acid and Cholesterol	TCR zeta	16570
T-cell Receptor, subunit CD3 epsilon	Fatty Acid and Cholesterol	TCR CD3 epsilon	16571
T-cell Receptor, subunit CD3 gamma	Fatty Acid and Cholesterol	TCR CD3 gamma	16572
T-cell Receptor, subunit CD3 delta	Fatty Acid and Cholesterol	TCR CD3 delta	16573
peroxisome proliferator-activated receptors	Fatty Acid and Cholesterol	PPARs	
AMP-activated protein kinase	Energy homeostasis (intracellular AMP to ATP ratio)	AMPK	16574-16575
P2X purinoceptor 7	Redox homeostasis	P2X7	16576

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding proteins that confer sensitivity to a cancer therapy, including a monoclonal antibody, to produce an armored T-cell of the disclosure. Thus, an armored T-cell of the disclosure can function and may demonstrate superior func-

tion or efficacy whilst in the presence of a cancer therapy (e.g., a chemotherapy, a monoclonal antibody therapy, or another anti-tumor treatment). Proteins involved in conferring sensitivity to a cancer therapy include, but are not limited to, the exemplary proteins provided in Table 6.

TABLE 6

Exemplary Proteins that Confer Sensitivity to a Cancer Therapeutic.		
Full Name	Abbreviation	SEQ ID NO:
Copper-transporting ATPase 2	ATP7B	16577
Breakpoint cluster region protein	BCR	16578

TABLE 6-continued

Exemplary Proteins that Confer Sensitivity to a Cancer Therapeutic.		
Full Name	Abbreviation	SEQ ID NO:
Abelson tyrosine-protein kinase 1	ABL	16579
Breast cancer resistance protein	BCRP	16580
Breast cancer type 1 susceptibility protein	BRCA1	16581
Breast cancer type 2 susceptibility protein	BRCA2	16582
CAMPATH-1 antigen	CD52	16583
Cytochrome P450 2D6	CYP2D6	16584
Deoxycytidine kinase	dCK	16585
Dihydrofolate reductase	DHFR	16586
Dihydropyrimidine dehydrogenase [NADP(+)]	DPYD	16587
Epidermal growth factor receptor	EGFR	16588
DNA excision repair protein ERCC-1	ERCC1	16589
Estrogen Receptor	ESR	16590
Low affinity immunoglobulin gamma Fc region receptor III-A	FCGR3A	16591
Receptor tyrosine-protein kinase erbB-2	HER2 or ERBB2	16592
Insulin-like growth factor 1 receptor	IGF1R	16593
GTPase KRas	KRAS	16594
Multidrug resistance protein 1	MDR1 or ABCB1	16595
Methylated-DNA--protein-cysteine methyltransferase	MGMT	16596
Multidrug resistance-associated protein 1	MRP1 or ABCC1	16597
Progesterone Receptor	PGR	16598
Regulator of G-protein signaling 10	RGS10	16599
Suppressor of cytokine signaling 3	SOCS-3	16600
Thymidylate synthase	TYMS	16601
UDP-glucuronosyltransferase 1-1	UGT1A1	16602

In some embodiments, a T-cell of the disclosure is modified to silence or reduce expression of one or more gene(s) encoding a growth advantage factor to produce an armored T-cell. Silencing or reducing expression of an oncogene can confer a growth advantage for an armored T-cell of the disclosure. For example, silencing or reducing expression (e.g., disrupting expression) of a TET2 gene during a CAR-T manufacturing process results in the generation of an armored CAR-T with a significant capacity for expansion and subsequent eradication of a tumor when compared to a non-armored CAR-T lacking this capacity for expansion. This strategy may be coupled to a safety switch (e.g., an iC9 safety switch of the disclosure), which allows for the targeted disruption of an armored CAR-T-cell in the event of an adverse reaction from a subject or uncontrolled growth of the armored CAR-T. Exemplary growth advantage factors include, but are not limited to, the factors provided in Table 7.

TABLE 7

Exemplary Growth Advantage Factors.		
Full Name	Abbreviation	SEQ ID NO:
Ten Eleven Translocation 2	TET2	16603
DNA (cytosine-5)-methyltransferase 3A	DNMT3A	16604
Transforming protein RhoA	RHOA	16605
Proto-oncogene vav	VAV1	16606
Rhomotin-2	LMO2	16607
T-cell acute lymphocytic leukemia protein 1	TAL1	16608
Suppressor of cytokine signaling 1	SOCS1	16609
herpes virus entry mediator	HVEM	16610
T cell death-associated gene 8	TDAG8	16611
BCL6 corepressor	BCOR	16612
B and T cell attenuator	BTLA	16613
SPARC-like protein 1	SPARCL1	16614
Msh homeobox 1-like protein	MSX1	16615

Armored T-Cells "Null or Switch Receptor" Strategy

In some embodiments, a T-cell of the disclosure is modified to express a modified/chimeric checkpoint receptor to produce an armored T-cell of the disclosure.

In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor. A null receptor, decoy receptor or dominant negative receptor of the disclosure may be modified/chimeric receptor/protein. A null receptor, decoy receptor or dominant negative receptor of the disclosure may be truncated for expression of the intracellular signaling domain. Alternatively, or in addition, a null receptor, decoy receptor or dominant negative receptor of the disclosure may be mutated within an intracellular signaling domain at one or more amino acid positions that are determinative or required for effective signaling. Truncation or mutation of null receptor, decoy receptor or dominant negative receptor of the disclosure may result in loss of the receptor's capacity to convey or transduce a checkpoint signal to the cell or within the cell.

For example, a dilution or a blockage of an immunosuppressive checkpoint signal from a PD-L1 receptor expressed on the surface of a tumor cell may be achieved by expressing a modified/chimeric PD-1 null receptor on the surface of an armored T-cell of the disclosure, which effectively competes with the endogenous (non-modified) PD-1 receptors also expressed on the surface of the armored T-cell to reduce or inhibit the transduction of the immunosuppressive checkpoint signal through endogenous PD-1 receptors of the armored T cell. In this exemplary embodiment, competition between the two different receptors for binding to PD-L1 expressed on the tumor cell reduces or diminishes a level of effective checkpoint signaling, thereby enhancing a therapeutic potential of the armored T-cell expressing the PD-1 null receptor.

In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is a transmembrane receptor.

In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is a membrane-associated or membrane-linked receptor/protein.

In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is an intracellular receptor/protein.

In some embodiments, the modified/chimeric checkpoint receptor comprises a null receptor, decoy receptor or dominant negative receptor that is an intracellular receptor/protein. Exemplary null, decoy, or dominant negative intracellular receptors/proteins of the disclosure include, but are not limited to, signaling components downstream of an inhibitory checkpoint signal (as provided, for example, in Tables 1 and 2), a transcription factor (as provided, for

example, in Table 3), a cytokine or a cytokine receptor, a chemokine or a chemokine receptor, a cell death or apoptosis receptor/ligand (as provided, for example, in Table 4), a metabolic sensing molecule (as provided, for example, in Table 5), a protein conferring sensitivity to a cancer therapy (as provided, for example, in Table 6), and an oncogene or a tumor suppressor gene (as provided, for example, in Table 7). Exemplary cytokines, cytokine receptors, chemokines and chemokine receptors of the disclosure include, but are not limited to, the cytokines and cytokine receptors as well as chemokines and chemokine receptors provided in Table 8.

TABLE 8

Exemplary Cytokines, Cytokine receptors, Chemokines and Chemokine Receptors.		
Full Name	Abbreviation	SEQ ID NO:
4-1BB Ligand	4-1BBL	16616
Tumor necrosis factor receptor superfamily member 25	Apo3 or TNFRSF25	16617
Tumor necrosis factor receptor superfamily member 13	APRIL or TNFRSF13	16618
Bcl2-associated agonist of cell death	Bcl-xL or BAD	16619
Tumor necrosis factor receptor superfamily member 17	BCMA or TNFRSF17	16620
C-C motif chemokine 1	CCL1	16621
C-C motif chemokine 11	CCL11	16622
C-C motif chemokine 13	CCL13	16623
C-C motif chemokine 14	CCL14	16624
C-C motif chemokine 15	CCL15	16625
C-C motif chemokine 16	CCL16	16626
C-C motif chemokine 17	CCL17	16627
C-C motif chemokine 18	CCL18	16628
C-C motif chemokine 19	CCL19	16629
C-C motif chemokine 2	CCL2	16630
C-C motif chemokine 20	CCL20	16631
C-C motif chemokine 21	CCL21	16632
C-C motif chemokine 22	CCL22	16633
C-C motif chemokine 23	CCL23	16634
C-C motif chemokine 24	CCL24	16635
C-C motif chemokine 25	CCL25	16636
C-C motif chemokine 26	CCL26	16637
C-C motif chemokine 27	CCL27	16638
C-C motif chemokine 28	CCL28	16639
C-C motif chemokine 3	CCL3	16640
C-C motif chemokine 4	CCL4	16641
C-C motif chemokine 5	CCL5	16642
C-C motif chemokine 7	CCL7	16643
C-C motif chemokine 8	CCL8	16644
C-C chemokine receptor type 1	CCR1	16645
C-C chemokine receptor type 10	CCR10	16646
C-C chemokine receptor type 11	CCR11	16647
C-C chemokine receptor type 2	CCR2	16648
C-C chemokine receptor type 3	CCR3	16649
C-C chemokine receptor type 4	CCR4	16650
C-C chemokine receptor type 5	CCR5	16651
C-C chemokine receptor type 6	CCR6	16652
C-C chemokine receptor type 7	CCR7	16653
C-C chemokine receptor type 8	CCR8	16654
C-C chemokine receptor type 9	CCR9	16655
Granulocyte colony-stimulating factor receptor	CD114 or CSF3R	16656
Macrophage colony-stimulating factor 1 receptor	CD115 or CSF1R	16657
Granulocyte-macrophage colony-stimulating factor receptor subunit alpha	CD116 or CSF2RA	16658
Mast/stem cell growth factor receptor Kit	CD117 or KIT	16659
Leukemia inhibitory factor receptor	CD118 or LIFR	16660
Tumor necrosis factor receptor superfamily member 1A	CD120a or TNFRSF1A	16661
Tumor necrosis factor receptor superfamily member 1B	CD120b or TNFRSF1B	16662
Interleukin-1 receptor type 1	CD121a or IL1R1	16663
Interleukin-2 receptor subunit beta	CD122 or IL2RB	16664
Interleukin-3 receptor subunit alpha	CD123 or IL3RA	16665

TABLE 8-continued

Exemplary Cytokines, Cytokine receptors, Chemokines and Chemokine Receptors.		
Full Name	Abbreviation	SEQ ID NO:
Interleukin-4 receptor subunit alpha	CD124 or IL4R	16666
Interleukin-6 receptor subunit alpha	CD126 or IL6R	16667
Interleukin-7 receptor subunit alpha	CD127 or IL7R	16668
Interleukin-6 receptor subunit beta	CD130 or IL6ST	16669
Cytokine receptor common subunit gamma	CD132 or IL2RG	16670
Tumor necrosis factor ligand superfamily member 8	CD153 or TNFSF8	16671
CD40 ligand	CD154 or CD40L	16672
Tumor necrosis factor ligand superfamily member 6	CD178 or FASLG	16673
Interleukin-12 receptor subunit beta-1	CD212 or IL12RB1	16674
Interleukin-13 receptor subunit alpha-1	CD213a1 or IL13RA1	16675
Interleukin-13 receptor subunit alpha-2	CD213a2 or IL13RA2	16676
Interleukin-2 receptor subunit alpha	CD25 or IL2RA	16677
CD27 antigen	CD27	16678
Tumor necrosis factor receptor superfamily member 8	CD30 or TNFRSF8	16679
T-cell surface glycoprotein CD4	CD4	16680
Tumor necrosis factor receptor superfamily member 5	CD40 or TNFRSF5	16681
CD70 antigen	CD70	16682
Tumor necrosis factor receptor superfamily member 6	CD95 or FAS or TNFRSF6	16683
Granulocyte-macrophage colony-stimulating factor receptor subunit alpha	CDw116 or CSF2RA	16684
Interferon gamma receptor 1	CDw119 or IFNGR1	16685
Interleukin-1 receptor type 2	CDw121b or IL1R2	16686
Interleukin-5 receptor subunit alpha	CDw125 or IL5RA	16687
Cytokine receptor common subunit beta	CDw131 or CSF2RB	16688
Tumor necrosis factor receptor superfamily member 9	CDw137 or TNFRSF9	16689
Interleukin-10 receptor	CDw210 or IL10R	16690
Interleukin-17 receptor A	CDw217 or IL17RA	16691
C-X3-C motif chemokine 1	CX3CL1	16692
CX3C chemokine receptor 1	CX3CR1	16693
C-X-C motif chemokine 1	CXCL1	16694
C-X-C motif chemokine 10	CXCL10	16695
C-X-C motif chemokine 11	CXCL11	16696
C-X-C motif chemokine 12	CXCL12	16697
C-X-C motif chemokine 13	CXCL13	16698
C-X-C motif chemokine 14	CXCL14	16699
C-X-C motif chemokine 16	CXCL16	16700
C-X-C motif chemokine 2	CXCL2	16701
C-X-C motif chemokine 3	CXCL3	16702
C-X-C motif chemokine 4	CXCL4	16703
C-X-C motif chemokine 5	CXCL5	16704
C-X-C motif chemokine 6	CXCL6	16705
C-X-C motif chemokine 7	CXCL7	16706
C-X-C motif chemokine 8	CXCL8	16707
C-X-C motif chemokine 9	CXCL9	16708
C-X-C chemokine receptor type 1	CXCR1	16709
C-X-C chemokine receptor type 2	CXCR2	16710
C-X-C chemokine receptor type 3	CXCR3	16711
C-X-C chemokine receptor type 4	CXCR4	16712
C-X-C chemokine receptor type 5	CXCR5	16713
C-X-C chemokine receptor type 6	CXCR6	16714
C-X-C chemokine receptor type 7	CXCR7	16715
Atypical chemokine receptor 1	DARC or ACKR1	16716
Erythropoietin	Epo	16717
Erythropoietin receptor	EpoR	16718
Receptor-type tyrosine-protein kinase FLT3	Flt-3	16719
FLT3 Ligand	Flt-3L	16720
Granulocyte colony-stimulating factor receptor	G-CSF or GSF3R	16721
Tumor necrosis factor receptor superfamily member 18	GITR or TNFRSF18	16722
GITR Ligand	GITRL	16723
Cytokine receptor common subunit beta	GM-CSF or CSF2RB	16724
Interleukin-6 receptor subunit beta	gp130 or IL6ST	16725
Tumor necrosis factor receptor superfamily member 14	HVEM or TNFRSF14	16726
Interferon gamma	IFN γ	16727
Interferon gamma receptor 2	IFNGR2	16728

TABLE 8-continued

Exemplary Cytokines, Cytokine receptors, Chemokines and Chemokine Receptors.		
Full Name	Abbreviation	SEQ ID NO:
Interferon-alpha	IFN- α	16729
Interferon-beta	IFN- β	16730
Interleukin-1 alpha	IL1	16731
Interleukin-10	IL10	16732
Interleukin-10 receptor	IL10R	16733
Interleukin-11	IL-11	16734
Interleukin-11 receptor alpha	IL-11Ra	16735
Interleukin-12	IL12	16736
Interleukin-13	IL13	16737
Interleukin-13 receptor	IL13R	16738
Interleukin-14	IL-14	16739
Interleukin-15	IL15	16740
Interleukin-15 receptor alpha	IL-15Ra	16741
Interleukin-16	IL-16	16742
Interleukin-17	IL17	16743
Interleukin-17 receptor	IL17R	16744
Interleukin-18	IL18	16745
Interleukin-1 receptor alpha	IL-1RA	16746
Interleukin-1 alpha	IL-1 α	16747
Interleukin-1beta	IL-1 β	16748
Interleukin-2	IL2	16749
Interleukin-20	IL-20	16750
Interleukin-20 receptor alpha	IL-20R α	16751
Interleukin-20 receptor beta	IL-20R β	16752
Interleukin-21	IL21	16753
Interleukin-3	IL-3	16754
Interleukin-35	IL35	16755
Interleukin-4	IL4	16756
Interleukin-4 receptor	IL4R	16757
Interleukin-5	IL5	16758
Interleukin-5 receptor	IL5R	16759
Interleukin-6	IL6	16760
Interleukin-6 receptor	IL6R	16761
Interleukin-7	IL7	16762
Interleukin-9 receptor	IL-9R	16763
Leukemia inhibitory factor	LIF	16764
Leukemia inhibitory factor receptor	LIFR	16765
tumor necrosis factor superfamily member 14	LIGHT or TNFSF14	16766
Tumor necrosis factor receptor superfamily member 3	LT β R or TNFRSF3	16767
Lymphotoxin-beta	LT- β	16768
Macrophage colony-stimulating factor 1	M-CSF	16769
Tumor necrosis factor receptor superfamily member 11B	OPG or TNFRSF11B	16770
Oncostatin-M	OSM	16771
Oncostatin-M receptor	OSMR	16772
Tumor necrosis factor receptor superfamily member 4	OX40 or TNFRSF4	16773
Tumor necrosis factor ligand superfamily member 4	OX40L or TNFSF4	16774
Tumor necrosis factor receptor superfamily member 11A	RANK or TNFRSF11A	16775
Kit Ligand	SCF or KITLG	16776
Tumor necrosis factor receptor superfamily member 13B	TACI or TNFRSF13B	16777
Tumor necrosis factor ligand superfamily member 13B	TALL-1 or TNFSF13B	16778
TGF-beta receptor type-1	TGF- β R1	16779
TGF-beta receptor type-2	TGF- β R2	16780
TGF-beta receptor type-3	TGF- β R3	16781
Transforming growth factor beta-1	TGF- β 1	16782
Transforming growth factor beta-2	TGF- β 2	16783
Transforming growth factor beta-3	TGF- β 3	16784
Tumor necrosis factor alpha	TNF or TNF- α	16785
Tumor necrosis factor beta	TNF- β	16786
Thyroid peroxidase	Tpo	16787
Thyroid peroxidase receptor	TpoR	16788
Tumor necrosis factor ligand superfamily member 10	TRAIL or TNFSF10	16789
Tumor necrosis factor receptor superfamily member 10A	TRAILR1 or TNFRSF10A	16790
Tumor necrosis factor receptor superfamily member 10B	TRAILR2 or TNFRSF10B	16791

TABLE 8-continued

Exemplary Cytokines, Cytokine receptors, Chemokines and Chemokine Receptors.		
Full Name	Abbreviation	SEQ ID NO:
Tumor necrosis factor ligand superfamily member 11	TRANCE or TNFSF11	16792
Tumor necrosis factor ligand superfamily member 12	TWEAK or TNFSF11	16793
Lymphotoxin	XCL1	16794
Cytokine SCM-1 beta	XCL2	16795

In some embodiments, the modified/chimeric checkpoint receptor comprises a switch receptor. Exemplary switch receptors may comprise a modified/chimeric receptor/protein of the disclosure wherein a native or wild type intracellular signaling domain is switched or replaced with a different intracellular signaling domain that is either non-native to the protein and/or not a wild-type domain. For example, replacement of an inhibitory signaling domain with a stimulatory signaling domain would switch an immunosuppressive signal into an immunostimulatory signal. Alternatively, replacement of an inhibitory signaling domain with a different inhibitory domain can reduce or enhance the level of inhibitory signaling. Expression or overexpression, of a switch receptor can result in the dilution and/or blockage of a cognate checkpoint signal via competition with an endogenous wildtype checkpoint receptor (not a switch receptor) for binding to the cognate checkpoint receptor expressed within the immunosuppressive tumor microenvironment. Armored T cells of the disclosure may comprise a sequence encoding switch receptors of the disclosure, leading to the expression of one or more switch receptors of the disclosure, and consequently, altering an activity of an armored T-cell of the disclosure. Armored T cells of the disclosure may express a switch receptor of the disclosure that targets an intracellularly expressed protein downstream of a checkpoint receptor, a transcription factor, a cytokine receptor, a death receptor, a metabolic sensing molecule, a cancer therapy, an oncogene, and/or a tumor suppressor protein or gene of the disclosure.

Exemplary switch receptors of the disclosure may comprise or may be derived from a protein including, but are not limited to, the signaling components downstream of an inhibitory checkpoint signal (as provided, for example, in Tables 1 and 2), a transcription factor (as provided, for example, in Table 3), a cytokine or a cytokine receptor, a chemokine or a chemokine receptor, a cell death or apoptosis receptor/ligand (as provided, for example, in Table 4), a metabolic sensing molecule (as provided, for example, in Table 5), a protein conferring sensitivity to a cancer therapy (as provided, for example, in Table 6), and an oncogene or a tumor suppressor gene (as provided, for example, in Table 7). Exemplary cytokines, cytokine receptors, chemokines and chemokine receptors of the disclosure include, but are not limited to, the cytokines and cytokine receptors as well as chemokines and chemokine receptors provided in Table 8. Armored T-Cells—Conditional Gene Expression System

In some embodiments, a T-cell of the disclosure is modified to express chimeric ligand receptor (CLR) or a chimeric antigen receptor (CAR) that mediates conditional gene expression to produce an armored T-cell of the disclosure. The combination of the CLR/CAR and the conditional gene expression system in the nucleus of the armored T cell constitutes a synthetic gene expression system that is conditionally activated upon binding of cognate ligand(s) with

CLR or cognate antigen(s) with CAR. This system may help to ‘armor’ or enhance therapeutic potential of modified T cells by reducing or limiting synthetic gene expression at the site of ligand or antigen binding, at or within the tumor environment for example.

Exogenous Receptors

In some embodiments, the armored T-cell comprises a composition comprising (a) an inducible transgene construct, comprising a sequence encoding an inducible promoter and a sequence encoding a transgene, and (b) a receptor construct, comprising a sequence encoding a constitutive promoter and a sequence encoding an exogenous receptor, such as a CLR or CAR, wherein, upon integration of the construct of (a) and the construct of (b) into a genomic sequence of a cell, the exogenous receptor is expressed, and wherein the exogenous receptor, upon binding a ligand or antigen, transduces an intracellular signal that targets directly or indirectly the inducible promoter regulating the expression of the inducible transgene (a) to modify gene expression.

In some embodiments of a synthetic gene expression system of the disclosure, the composition modifies gene expression by decreasing gene expression. In some embodiments, the composition modifies gene expression by transiently modifying gene expression (e.g., for the duration of binding of the ligand to the exogenous receptor). In some embodiments, the composition modifies gene expression acutely (e.g., the ligand reversibly binds to the exogenous receptor). In some embodiments, the composition modifies gene expression chronically (e.g., the ligand irreversibly binds to the exogenous receptor).

In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises an endogenous receptor with respect to the genomic sequence of the cell. Exemplary receptors include, but are not limited to, intracellular receptors, cell-surface receptors, transmembrane receptors, ligand-gated ion channels, and G-protein coupled receptors.

In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the non-naturally occurring receptor is a synthetic, modified, recombinant, mutant or chimeric receptor. In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a T-cell receptor (TCR). In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a scaffold protein. In some embodiments, including those wherein the non-naturally occurring receptor does not comprise a transmembrane domain, the non-naturally occurring receptor interacts with a second transmembrane, membrane-bound and/or an intracellular receptor that, following contact with the non-naturally occurring receptor, transduces an intracellular signal.

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In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the non-naturally occurring receptor is a synthetic, modified, recombinant, mutant or chimeric receptor. In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a T-cell receptor (TCR). In some embodiments, the non-naturally occurring receptor comprises one or more sequences isolated or derived from a scaffold protein. In some embodiments, the non-naturally occurring receptor comprises a transmembrane domain. In some embodiments, the non-naturally occurring receptor interacts with an intracellular receptor that transduces an intracellular signal. In some embodiments, the non-naturally occurring receptor comprises an intracellular signaling domain. In some embodiments, the non-naturally occurring receptor is a chimeric ligand receptor (CLR). In some embodiments, the CLR is a chimeric antigen receptor (CAR).

In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the CLR is a chimeric antigen receptor (CAR). In some embodiments, the chimeric ligand receptor comprises (a) an ectodomain comprising a ligand recognition region, wherein the ligand recognition region comprises at least scaffold protein; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the ectodomain of (a) further comprises a signal peptide. In some embodiments, the ectodomain of (a) further comprises a hinge between the ligand recognition region and the transmembrane domain.

In some embodiments of the CLR/CARs of the disclosure, the signal peptide comprises a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR signal peptide. In some embodiments, the signal peptide comprises a sequence encoding a human CD8α signal peptide. In some embodiments, the signal peptide comprises an amino acid sequence comprising MALPVTALLPLALLHAARP (SEQ ID NO: 17503). In some embodiments, the signal peptide is encoded by a nucleic acid sequence comprising atggcactgccagt-caccgccctgctgctgctgctgctgctgcacgcagctagacca (SEQ ID NO: 17504).

In some embodiments of the CLR/CARs of the disclosure, the transmembrane domain comprises a sequence encoding a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain. In some embodiments, the transmembrane domain comprises a sequence encoding a human CD8α transmembrane domain. In some embodiments, the transmembrane domain comprises an amino acid sequence comprising IYI-WAPLAGTCGVLLLSLVITLYC (SEQ ID NO: 17505). In some embodiments, the transmembrane domain is encoded by a nucleic acid sequence comprising

(SEQ ID NO: 17506)
atctacatttgggcaccactggccgggacctgtggagtgtgctgctgag
cctgtgcatcacactgtactgc.

In some embodiments of the CLR/CARs of the disclosure, the endodomain comprises a human CD3ζ endodomain. In some embodiments, the at least one costimulatory domain comprises a human 4-1BB, CD28, CD3ζ, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof. In some embodiments, the at least one

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costimulatory domain comprises a human CD3ζ and/or a 4-1BB costimulatory domain. In some embodiments, the CD3ζ costimulatory domain comprises an amino acid sequence comprising RVKFSRSADAPAYKQGQNQLY-NELNLGRREEYDVLDKRRGRDPEMGGKPRRKNPQ
5 EGLYNELQKDKMAEAYSEIGMKGERRRGKGHDG-LYQGLSTATKDTYDALHMQALP PR (SEQ ID NO: 17507). In some embodiments, the CD3ζ costimulatory domain is encoded by a nucleic acid sequence comprising
10 cgcgtgaagttagtcgatcagcagatgccccagcttacaacagggacagaac-cagctgtataacgagctgaatctggccgccga gaggaatatgagtgctgga-taagcggagaggacgcagccccgaaatggaggcaagcccaggcgca-
aaaccctcagggaagg cctgtataacgagctgcagaaggacaaatggcag-aagcctattctgagatcgccatgaaggggagcgcagcgagaggcaaaagg
15 gcacgatgggctgtaccagggactgagcaccgccacaaaggacacctat-gatgctctgcatatgcaggcactgcctccaagg (SEQ ID NO: 17508). In some embodiments, the 4-1BB costimulatory domain comprises an amino acid sequence comprising KRGRKKLLY-IFKQPFMRPVQTTQEEDGCSCRFPEEEEGGCEL (SEQ ID NO: 17509). In some embodiments, the 4-1BB costimulatory domain is encoded by a nucleic acid sequence comprising
20 aagagaggcaggaagaaactgctgtatatatttcaaacacccctt-catgcgccccgtgcagactaccaggaggaagacgggtgctcc tgtcgatt-cctgaggaagaggaaggcgggtgtgagctg (SEQ ID NO: 17510). In some embodiments, the 4-1BB costimulatory domain is located between the transmembrane domain and the CD3ζ costimulatory domain.

In some embodiments of the CLR/CARs of the disclosure, the hinge comprises a sequence derived from a human CD8α, IgG4, and/or CD4 sequence. In some embodiments, the hinge comprises a sequence derived from a human CD8α sequence. In some embodiments, the hinge comprises an amino acid sequence comprising

(SEQ ID NO: 17511)
TTTPAPRPPTPAPTIASQPLSLRPEACRPAGGAVHTRGLDFACD.

In some embodiments, the hinge is encoded by a nucleic acid sequence comprising actaccacaccagcactagaccac-caactccagctccaaccatgcgagtcagccctgagctgagacctgagg-cctgcaggcc agctgcaggaggagctgtgcacaccaggggcctgactt-cgctgcgac (SEQ ID NO: 17512). In some embodiments, the hinge is encoded by

(SEQ ID NO: 17513)
ACCACAACCCCTGCCCCAGACCTCCACACCCGCCCTACCATCGCGAG
50 TCAGCCCTGAGTCTGAGACCTGAGGCCTGCAGGCCAGCTGCAGGAGGAG
CTGTGCACACAGGGGCCTGGACTTCGCCTGCGAC.

In some embodiments, the at least one protein scaffold specifically binds the ligand.

In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the CLR is a chimeric antigen receptor (CAR). In some embodiments, the chimeric ligand receptor comprises (a) an ectodomain comprising a ligand recognition region, wherein the ligand recognition region comprises at least scaffold protein; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the at least one protein scaffold comprises an antibody, an antibody fragment, a single domain antibody, a single chain antibody, an antibody mimetic, or a Centyrin (referred to herein as a CARTyrin). In some embodiments, the ligand

recognition region comprises one or more of an antibody, an antibody fragment, a single domain antibody, a single chain antibody, an antibody mimetic, and a Centyrin. In some embodiments, the single domain antibody comprises or consists of a VHH or a VH (referred to herein as a VCAR). In some embodiments, the single domain antibody comprises or consists of a VHH or a VH comprising human complementarity determining regions (CDRs). In some embodiments, the VH is a recombinant or chimeric protein. In some embodiments, the VH is a recombinant or chimeric human protein. In some embodiments, the antibody mimetic comprises or consists of an affibody, an affilin, an affimer, an affitin, an alphabody, an anticalin, an avimer, a DARPin, a Fynomer, a Kunitz domain peptide or a monobody. In some embodiments, the Centyrin comprises or consists of a consensus sequence of at least one fibronectin type III (FN3) domain.

In some embodiments of the compositions of the disclosure, the exogenous receptor of (b) comprises a non-naturally occurring receptor. In some embodiments, the CLR is a chimeric antigen receptor (CAR). In some embodiments, the chimeric ligand receptor comprises (a) an ectodomain comprising a ligand recognition region, wherein the ligand recognition region comprises at least scaffold protein; (b) a transmembrane domain, and (c) an endodomain comprising at least one costimulatory domain. In some embodiments, the Centyrin comprises or consists of a consensus sequence of at least one fibronectin type III (FN3) domain. In some embodiments, the at least one fibronectin type III (FN3) domain is derived from a human protein. In some embodiments, the human protein is Tenascin-C. In some embodiments, the consensus sequence comprises LPAPKNLVVSEVTEDSLRLSWTAPDAAFDLSFLIQYQESEKVGGEAINLTVPGSERSYDL TGLKPGTEYTVSIYGVKGGHRSNPLSAEFTT (SEQ ID NO: 17514). In some embodiments, the consensus sequence comprises MLPAPKNLVVSEVTEDSLRLSWTAPDAAFDLSFLIQYQESEKVGGEAINLTVPGSERSYD LTGLKPGTEYTVSIYGVKGGHRSNPLSAEFTT (SEQ ID NO: 17515). In some embodiments, the consensus sequence is modified at one or more positions within (a) a A-B loop comprising or consisting of the amino acid residues TEDS (SEQ ID NO: 17516) at positions 13-16 of the consensus sequence; (b) a B-C loop comprising or consisting of the amino acid residues TAPDAAF (SEQ ID NO: 17517) at positions 22-28 of the consensus sequence; (c) a C-D loop comprising or consisting of the amino acid residues SEKVGE (SEQ ID NO: 17518) at positions 38-43 of the consensus sequence; (d) a D-E loop comprising or consisting of the amino acid residues GSER (SEQ ID NO: 17519) at positions 51-54 of the consensus sequence; (e) a E-F loop comprising or consisting of the amino acid residues GLKPG (SEQ ID NO: 17520) at positions 60-64 of the consensus sequence; (f) a F-G loop comprising or consisting of the amino acid residues KGGHRSN (SEQ ID NO: 17521) at positions 75-81 of the consensus sequence; or (g) any combination of (a)-(f). In some embodiments, the Centyrin

comprises a consensus sequence of at least 5 fibronectin type III (FN3) domains. In some embodiments, the Centyrin comprises a consensus sequence of at least 10 fibronectin type III (FN3) domains. In some embodiments, the Centyrin comprises a consensus sequence of at least 15 fibronectin type III (FN3) domains. In some embodiments, the scaffold binds an antigen with at least one affinity selected from a K_D of less than or equal to 10^{-9} M, less than or equal to 10^{-10} M, less than or equal to 10^{-11} M, less than or equal to 10^{-12} M, less than or equal to 10^{-13} M, less than or equal to 10^{-14} M, and less than or equal to 10^{-15} M. In some embodiments, the K_D is determined by surface plasmon resonance.

15 Inducible Promoters

In some embodiments of the compositions of the disclosure, the sequence encoding the inducible promoter of (a) comprises a sequence encoding an NF κ B promoter. In some embodiments of the compositions of the disclosure, the sequence encoding the inducible promoter of (a) comprises a sequence encoding an interferon (IFN) promoter or a sequence encoding an interleukin-2 promoter. In some embodiments, the interferon (IFN) promoter is an IFN γ promoter. In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of a cytokine or a chemokine. In some embodiments, the cytokine or chemokine comprises IL2, IL3, IL4, IL5, IL6, IL10, IL12, IL13, IL17A/F, IL21, IL22, IL23, transforming growth factor beta (TGF β), colony stimulating factor 2 (GM-CSF), interferon gamma (IFN γ), Tumor necrosis factor (TNF α), LT α , perform, Granzyme C (Gzmc), Granzyme B (Gzmb), C—C motif chemokine ligand 5 (CCL5), C—C motif chemokine ligand 4 (Ccl4), C—C motif chemokine ligand 3 (Ccl3), X-C motif chemokine ligand 1 (Xcl1) and LIF interleukin 6 family cytokine (Lif).

In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of a gene comprising a surface protein involved in cell differentiation, activation, exhaustion and function. In some embodiments, the gene comprises CD69, CD71, CTLA4, PD-1, TIGIT, LAG3, TIM-3, GITR, MHCII, COX-2, FASL and 4-1BB.

In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of a gene involved in CD metabolism and differentiation. In some embodiments of the compositions of the disclosure, the inducible promoter is isolated or derived from the promoter of Nr4a1, Nr4a3, Tnfrsf9 (4-1BB), Sema7a, Zfp3612, Gadd45b, Dusp5, Dusp6 and Neto2.

50 Inducible Transgene

In some embodiments, the inducible transgene construct comprises or drives expression of a signaling component downstream of an inhibitory checkpoint signal (as provided, for example, in Tables 1 and 2), a transcription factor (as provided, for example, in Table 3), a cytokine or a cytokine receptor, a chemokine or a chemokine receptor, a cell death or apoptosis receptor/ligand (as provided, for example, in Table 4), a metabolic sensing molecule (as provided, for example, in Table 5), a protein conferring sensitivity to a cancer therapy (as provided, for example, in Table 6 or 9), and an oncogene or a tumor suppressor gene (as provided, for example, in Table 7). Exemplary cytokines, cytokine receptors, chemokines and chemokine receptors of the disclosure include, but are not limited to, the cytokines and cytokine receptors as well as chemokines and chemokine receptors provided in Table 8.

TABLE 9

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
A1BG	Alpha-1-B glycoprotein	SEQ ID NOS: 1-2
A2M	Alpha-2-macroglobulin	SEQ ID NOS: 3-6
A2ML1	Alpha-2-macroglobulin-like 1	SEQ ID NOS: 7-12
A4GNT	Alpha-1,4-N-acetylglucosaminyltransferase	SEQ ID NO: 13
AADACL2	Arylacetamide deacetylase-like 2	SEQ ID NOS: 14-15
AANAT	Aralkylamine N-acetyltransferase	SEQ ID NOS: 16-19
ABCG1	ATP-binding cassette, sub-family G (WHITE), member 1	SEQ ID NOS: 20-26
ABHD1	Abhydrolase domain containing 1	SEQ ID NOS: 27-31
ABHD10	Abhydrolase domain containing 10	SEQ ID NOS: 32-35
ABHD14A	Abhydrolase domain containing 14A	SEQ ID NOS: 36-40
ABHD15	Abhydrolase domain containing 15	SEQ ID NO: 41
ABI3BP	ABI family, member 3 (NESH) binding protein	SEQ ID NOS: 42-63
AC008641.1		SEQ ID NO: 73
AC009133.22		SEQ ID NO: 76
AC009491.2		SEQ ID NO: 77
AC011513.3		SEQ ID NOS: 92-93
AC136352.5		SEQ ID NO: 88
AC145212.4	MaFF-interacting protein	SEQ ID NO: 90
AC233755.1		SEQ ID NO: 91
ACACB	Acetyl-CoA carboxylase beta	SEQ ID NOS: 94-100
ACAN	Aggrecan	SEQ ID NOS: 101-108
ACE	Angiotensin I converting enzyme	SEQ ID NOS: 109-121
ACHE	Acetylcholinesterase (Yt blood group)	SEQ ID NOS: 122-134
ACP2	Acid phosphatase 2, lysosomal	SEQ ID NOS: 135-142
ACP5	Acid phosphatase 5, tartrate resistant	SEQ ID NOS: 143-151
ACP6	Acid phosphatase 6, lysophosphatidic	SEQ ID NOS: 152-158
ACPP	Acid phosphatase, prostate	SEQ ID NOS: 163-167
ACR	Acrosin	SEQ ID NOS: 168-169
ACRBP	Acrosin binding protein	SEQ ID NOS: 170-174
ACRV1	Acrosomal vesicle protein 1	SEQ ID NOS: 175-178
ACSF2	Acyl-CoA synthetase family member 2	SEQ ID NOS: 179-187
ACTL10	Actin-like 10	SEQ ID NO: 188
ACVR1	Activin A receptor, type I	SEQ ID NOS: 189-197
ACVR1C	Activin A receptor, type IC	SEQ ID NOS: 198-201
ACVRL1	Activin A receptor type II-like 1	SEQ ID NOS: 202-207
ACYP1	Acylphosphatase 1, erythrocyte (common) type	SEQ ID NOS: 208-213
ACYP2	Acylphosphatase 2, muscle type	SEQ ID NOS: 214-221
ADAM10	ADAM metalloproteinase domain 10	SEQ ID NOS: 230-237
ADAM12	ADAM metalloproteinase domain 12	SEQ ID NOS: 238-240
ADAM15	ADAM metalloproteinase domain 15	SEQ ID NOS: 241-252
ADAM17	ADAM metalloproteinase domain 17	SEQ ID NOS: 253-255
ADAM18	ADAM metalloproteinase domain 18	SEQ ID NOS: 256-260
ADAM22	ADAM metalloproteinase domain 22	SEQ ID NOS: 261-269
ADAM28	ADAM metalloproteinase domain 28	SEQ ID NOS: 270-275
ADAM29	ADAM metalloproteinase domain 29	SEQ ID NOS: 276-284
ADAM32	ADAM metalloproteinase domain 32	SEQ ID NOS: 285-291
ADAM33	ADAM metalloproteinase domain 33	SEQ ID NOS: 292-296
ADAM7	ADAM metalloproteinase domain 7	SEQ ID NOS: 297-300
ADAM8	ADAM metalloproteinase domain 8	SEQ ID NOS: 301-305
ADAM9	ADAM metalloproteinase domain 9	SEQ ID NOS: 306-311
ADAMDEC1	ADAM-like, decysin 1	SEQ ID NOS: 312-314
ADAMTS1	ADAM metalloproteinase with thrombospondin type 1 motif, 1	SEQ ID NOS: 315-318
ADAMTS10	ADAM metalloproteinase with thrombospondin type 1 motif, 10	SEQ ID NOS: 319-324
ADAMTS12	ADAM metalloproteinase with thrombospondin type 1 motif, 12	SEQ ID NOS: 325-327
ADAMTS13	ADAM metalloproteinase with thrombospondin type 1 motif, 13	SEQ ID NOS: 328-335
ADAMTS14	ADAM metalloproteinase with thrombospondin type 1 motif, 14	SEQ ID NOS: 336-337
ADAMTS15	ADAM metalloproteinase with thrombospondin type 1 motif, 15	SEQ ID NO: 338
ADAMTS16	ADAM metalloproteinase with thrombospondin type 1 motif, 16	SEQ ID NOS: 339-340
ADAMTS17	ADAM metalloproteinase with thrombospondin type 1 motif, 17	SEQ ID NOS: 341-344
ADAMTS18	ADAM metalloproteinase with thrombospondin type 1 motif, 18	SEQ ID NOS: 345-348
ADAMTS19	ADAM metalloproteinase with thrombospondin type 1 motif, 19	SEQ ID NOS: 349-352
ADAMTS2	ADAM metalloproteinase with thrombospondin type 1 motif, 2	SEQ ID NOS: 353-355

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
ADAMTS20	ADAM metalloproteinase with thrombospondin type 1 motif, 20	SEQ ID NOS: 356-359
ADAMTS3	ADAM metalloproteinase with thrombospondin type 1 motif, 3	SEQ ID NOS: 360-361
ADAMTS5	ADAM metalloproteinase with thrombospondin type 1 motif, 5	SEQ ID NO: 362
ADAMTS6	ADAM metalloproteinase with thrombospondin type 1 motif, 6	SEQ ID NOS: 363-364
ADAMTS7	ADAM metalloproteinase with thrombospondin type 1 motif, 7	SEQ ID NO: 365
ADAMTS8	ADAM metalloproteinase with thrombospondin type 1 motif, 8	SEQ ID NO: 366
ADAMTS9	ADAM metalloproteinase with thrombospondin type 1 motif, 9	SEQ ID NOS: 367-371
ADAMTSL1	ADAMTS-like 1	SEQ ID NOS: 372-382
ADAMTSL2	ADAMTS-like 2	SEQ ID NOS: 383-385
ADAMTSL3	ADAMTS-like 3	SEQ ID NOS: 386-387
ADAMTSL4	ADAMTS-like 4	SEQ ID NOS: 388-391
ADAMTSL5	ADAMTS-like 5	SEQ ID NOS: 392-397
ADCK1	AarF domain containing kinase 1	SEQ ID NOS: 398-402
ADCYAP1	Adenylate cyclase activating polypeptide 1 (pituitary)	SEQ ID NOS: 403-404
ADCYAP1R1	Adenylate cyclase activating polypeptide 1 (pituitary) receptor type I	SEQ ID NOS: 405-411
ADGRA3	Adhesion G protein-coupled receptor A3	SEQ ID NOS: 412-416
ADGRB2	Adhesion G protein-coupled receptor B2	SEQ ID NOS: 417-425
ADGRD1	Adhesion G protein-coupled receptor D1	SEQ ID NOS: 426-431
ADGRE3	Adhesion G protein-coupled receptor E3	SEQ ID NOS: 432-436
ADGRE5	Adhesion G protein-coupled receptor E5	SEQ ID NOS: 437-442
ADGRF1	Adhesion G protein-coupled receptor F1	SEQ ID NOS: 443-447
ADGRG1	Adhesion G protein-coupled receptor G1	SEQ ID NOS: 448-512
ADGRG5	Adhesion G protein-coupled receptor G5	SEQ ID NOS: 513-515
ADGRG6	Adhesion G protein-coupled receptor G6	SEQ ID NOS: 516-523
ADGRV1	Adhesion G protein-coupled receptor V1	SEQ ID NOS: 524-540
ADI1	Acireductone dioxygenase 1	SEQ ID NOS: 541-543
ADIG	Adipogenin	SEQ ID NOS: 544-547
ADIPOQ	Adiponectin, C1Q and collagen domain containing	SEQ ID NOS: 548-549
ADM	Adrenomedullin	SEQ ID NOS: 550-557
ADM2	Adrenomedullin 2	SEQ ID NOS: 558-559
ADM5	Adrenomedullin 5 (putative)	SEQ ID NO: 560
ADPGK	ADP-dependent glucokinase	SEQ ID NOS: 561-570
ADPRHL2	ADP-ribosylhydrolase like 2	SEQ ID NO: 571
AEBP1	AE binding protein 1	SEQ ID NOS: 572-579
AFM	Afamin	SEQ ID NO: 584
AFP	Alpha-fetoprotein	SEQ ID NOS: 585-586
AGA	Aspartylglucosaminidase	SEQ ID NOS: 587-589
AGER	Advanced glycosylation end product-specific receptor	SEQ ID NOS: 590-600
AGK	Acylglycerol kinase	SEQ ID NOS: 601-606
AGPS	Alkylglycerone phosphate synthase	SEQ ID NOS: 607-610
AGR2	Anterior gradient 2, protein disulphide isomerase family member	SEQ ID NOS: 611-614
AGR3	Anterior gradient 3, protein disulphide isomerase family member	SEQ ID NOS: 615-617
AGRN	Agrin	SEQ ID NOS: 618-621
AGRP	Agouti related neuropeptide	SEQ ID NO: 622
AGT	Angiotensinogen (serpin peptidase inhibitor, clade A, member 8)	SEQ ID NO: 623
AGTPBP1	ATP/GTP binding protein 1	SEQ ID NOS: 624-627
AGTRAP	Angiotensin II receptor-associated protein	SEQ ID NOS: 628-635
AHCYL2	Adenosylhomocysteinase-like 2	SEQ ID NOS: 636-642
AHSG	Alpha-2-HS-glycoprotein	SEQ ID NOS: 643-644
AI1	Androgen-induced 1	SEQ ID NOS: 645-653
AK4	Adenylate kinase 4	SEQ ID NOS: 654-657
AKAP10	A kinase (PRKA) anchor protein 10	SEQ ID NOS: 658-666
AKR1C1	Aldo-keto reductase family 1, member C1	SEQ ID NOS: 667-669
AL356289.1		SEQ ID NO: 677
AL589743.1		SEQ ID NO: 678
ALAS2	5'-aminolevulinate synthase 2	SEQ ID NOS: 684-691
ALB	Albumin	SEQ ID NOS: 692-701
ALDH9A1	Aldehyde dehydrogenase 9 family, member A1	SEQ ID NO: 702
ALDOA	Aldolase A, fructose-bisphosphate	SEQ ID NOS: 703-717
ALG1	ALG1, chitobiosylidiphosphodolichol beta-mannosyltransferase	SEQ ID NOS: 718-723

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
ALG5	ALG5, dolichyl-phosphate beta-glucosyltransferase	SEQ ID NOS: 724-725
ALG9	ALG9, alpha-1,2-mannosyltransferase	SEQ ID NOS: 726-736
ALKBH1	AlkB homolog 1, histone H2A dioxygenase	SEQ ID NOS: 746-748
ALKBH5	AlkB homolog 5, RNA demethylase	SEQ ID NOS: 749-750
ALPI	Alkaline phosphatase, intestinal	SEQ ID NOS: 751-752
ALPL	Alkaline phosphatase, liver/bone/kidney	SEQ ID NOS: 753-757
ALPP	Alkaline phosphatase, placental	SEQ ID NO: 758
ALPPL2	Alkaline phosphatase, placental-like 2	SEQ ID NO: 759
AMBN	Ameloblastin (enamel matrix protein)	SEQ ID NOS: 760-762
AMBP	Alpha-1-microglobulin/bikunin precursor	SEQ ID NOS: 763-765
AMELX	Amelogenin, X-linked	SEQ ID NOS: 766-768
AMELY	Amelogenin, Y-linked	SEQ ID NOS: 769-770
AMH	Anti-Mullerian hormone	SEQ ID NO: 771
AMICA1	Adhesion molecule, interacts with CXADR antigen 1	SEQ ID NOS: 7348-7356
AMPD1	Adenosine monophosphate deaminase 1	SEQ ID NOS: 772-774
AMTN	Amelotin	SEQ ID NOS: 775-776
AMY1A	Amylase, alpha 1A (salivary)	SEQ ID NOS: 777-779
AMY1B	Amylase, alpha 1B (salivary)	SEQ ID NOS: 780-783
AMY1C	Amylase, alpha 1C (salivary)	SEQ ID NO: 784
AMY2A	Amylase, alpha 2A (pancreatic)	SEQ ID NOS: 785-787
AMY2B	Amylase, alpha 2B (pancreatic)	SEQ ID NOS: 788-792
ANG	Angiogenin, ribonuclease, RNase A family, 5	SEQ ID NOS: 793-794
ANGEL1	Angel homolog 1 (<i>Drosophila</i>)	SEQ ID NOS: 795-798
ANGPT1	Angiopoietin 1	SEQ ID NOS: 799-803
ANGPT2	Angiopoietin 2	SEQ ID NOS: 804-807
ANGPT4	Angiopoietin 4	SEQ ID NO: 808
ANGPTL1	Angiopoietin-like 1	SEQ ID NOS: 809-811
ANGPTL2	Angiopoietin-like 2	SEQ ID NOS: 812-813
ANGPTL3	Angiopoietin-like 3	SEQ ID NO: 814
ANGPTL4	Angiopoietin-like 4	SEQ ID NOS: 815-822
ANGPTL5	Angiopoietin-like 5	SEQ ID NOS: 823-824
ANGPTL6	Angiopoietin-like 6	SEQ ID NOS: 825-827
ANGPTL7	Angiopoietin-like 7	SEQ ID NO: 828
ANK1	Ankyrin 1, erythrocytic	SEQ ID NOS: 833-843
ANKDD1A	Ankyrin repeat and death domain containing 1A	SEQ ID NOS: 844-850
ANKRD54	Ankyrin repeat domain 54	SEQ ID NOS: 851-859
ANKRD60	Ankyrin repeat domain 60	SEQ ID NO: 860
ANO7	Anoctamin 7	SEQ ID NOS: 861-864
ANO1	#N/A	SEQ ID NO: 865
ANTXR1	Anthrax toxin receptor 1	SEQ ID NOS: 866-869
AOAH	Acyloxyacyl hydrolase (neutrophil)	SEQ ID NOS: 870-874
AOC1	Amine oxidase, copper containing 1	SEQ ID NOS: 875-880
AOC2	Amine oxidase, copper containing 2 (retina-specific)	SEQ ID NOS: 881-882
AOC3	Amine oxidase, copper containing 3	SEQ ID NOS: 883-889
AP000721.4		SEQ ID NO: 890
APBB1	Amyloid beta (A4) precursor protein-binding, family B, member 1 (Fe65)	SEQ ID NOS: 891-907
APCDD1	Adenomatosis polyposis coli down-regulated 1	SEQ ID NOS: 908-913
APCS	Amyloid P component, serum	SEQ ID NO: 914
APELA	Apelin receptor early endogenous ligand	SEQ ID NOS: 915-917
APLN	Apelin	SEQ ID NO: 918
APLP2	Amyloid beta (A4) precursor-like protein 2	SEQ ID NOS: 919-928
APOA1	Apolipoprotein A-I	SEQ ID NOS: 929-933
APOA1BP	Apolipoprotein A-I binding protein	SEQ ID NOS: 9177-9179
APOA2	Apolipoprotein A-II	SEQ ID NOS: 934-942
APOA4	Apolipoprotein A-IV	SEQ ID NO: 943
APOA5	Apolipoprotein A-V	SEQ ID NOS: 944-946
APOB	Apolipoprotein B	SEQ ID NOS: 947-948
APOC1	Apolipoprotein C-I	SEQ ID NOS: 949-957
APOC2	Apolipoprotein C-II	SEQ ID NOS: 958-962
APOC3	Apolipoprotein C-III	SEQ ID NOS: 963-966
APOC4	Apolipoprotein C-IV	SEQ ID NOS: 967-968
APOC4-	APOC4-APOC2 readthrough (NMD candidate)	SEQ ID NOS: 969-970
APOC2		
APOD	Apolipoprotein D	SEQ ID NOS: 971-974
APOE	Apolipoprotein E	SEQ ID NOS: 975-978
APOF	Apolipoprotein F	SEQ ID NO: 979
APOH	Apolipoprotein H (beta-2-glycoprotein I)	SEQ ID NOS: 980-983
APOL1	Apolipoprotein L, 1	SEQ ID NOS: 984-994

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
APOL3	Apolipoprotein L, 3	SEQ ID NOS: 995-1009
APOM	Apolipoprotein M	SEQ ID NOS: 1010-1012
APOOL	Apolipoprotein O-like	SEQ ID NOS: 1013-1015
ARCN1	Archain 1	SEQ ID NOS: 1016-1020
ARFIP2	ADP-ribosylation factor interacting protein 2	SEQ ID NOS: 1021-1027
ARHGAP36	Rho GTPase activating protein 36	SEQ ID NOS: 1028-1033
ARHGAP6	Rho GTPase activating protein 6	SEQ ID NOS: 1043-1048
ARHGEF4	Rho guanine nucleotide exchange factor (GEF) 4	SEQ ID NOS: 1049-1059
ARL16	ADP-ribosylation factor-like 16	SEQ ID NOS: 1060-1068
ARMC5	Armadillo repeat containing 5	SEQ ID NOS: 1069-1075
ARNTL	Aryl hydrocarbon receptor nuclear translocator-like	SEQ ID NOS: 1076-1090
ARSA	Arylsulfatase A	SEQ ID NOS: 1091-1096
ARSB	Arylsulfatase B	SEQ ID NOS: 1097-1100
ARSE	Arylsulfatase E (chondrodysplasia punctata 1)	SEQ ID NOS: 1101-1104
ARSG	Arylsulfatase G	SEQ ID NOS: 1105-1108
ARSI	Arylsulfatase family, member I	SEQ ID NOS: 1109-1111
ARSK	Arylsulfatase family, member K	SEQ ID NOS: 1112-1116
ART3	ADP-ribosyltransferase 3	SEQ ID NOS: 1117-1124
ART4	ADP-ribosyltransferase 4 (Dombrock blood group)	SEQ ID NOS: 1125-1128
ART5	ADP-ribosyltransferase 5	SEQ ID NOS: 1129-1133
ARTN	Artemin	SEQ ID NOS: 1134-1144
ASAH1	N-acylsphingosine amidohydrolase (acid ceramidase) 1	SEQ ID NOS: 1145-1195
ASAH2	N-acylsphingosine amidohydrolase (non-lysosomal ceramidase) 2	SEQ ID NOS: 1196-1201
ASCL1	Achaete-scute family bHLH transcription factor 1	SEQ ID NO: 1202
ASIP	Agouti signaling protein	SEQ ID NOS: 1203-1204
ASPN	Asporin	SEQ ID NOS: 1205-1206
ASTL	Astacin-like metallo-endorpeptidase (M12 family)	SEQ ID NO: 1207
ATAD5	ATPase family, AAA domain containing 5	SEQ ID NOS: 1208-1209
ATAT1	Alpha tubulin acetyltransferase 1	SEQ ID NOS: 1210-1215
ATG2A	Autophagy related 2A	SEQ ID NOS: 1216-1218
ATG5	Autophagy related 5	SEQ ID NOS: 1219-1227
ATMIN	ATM interactor	SEQ ID NOS: 1228-1231
ATP13A1	ATPase type 13A1	SEQ ID NOS: 1232-1234
ATP5F1	ATP synthase, H ⁺ transporting, mitochondrial Fo complex, subunit B1	SEQ ID NOS: 1235-1236
ATP6AP1	ATPase, H ⁺ transporting, lysosomal accessory protein 1	SEQ ID NOS: 1237-1244
ATP6AP2	ATPase, H ⁺ transporting, lysosomal accessory protein 2	SEQ ID NOS: 1245-1267
ATPAF1	ATP synthase mitochondrial F1 complex assembly factor 1	SEQ ID NOS: 1268-1278
AUH	AU RNA binding protein/enoyl-CoA hydratase	SEQ ID NOS: 1279-1280
AVP	Arginine vasopressin	SEQ ID NO: 1281

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
AXIN2	Axin 2	SEQ ID NOS: 1282-1289
AZGP1	Alpha-2-glycoprotein 1, zinc-binding	SEQ ID NOS: 1290-1292
AZU1	Azurocidin 1	SEQ ID NOS: 1293-1294
B2M	Beta-2-microglobulin	SEQ ID NOS: 1295-1301
B3GALNT1	Beta-1,3-N-acetylgalactosaminyltransferase 1 (globoside blood group)	SEQ ID NOS: 1302-1314
B3GALNT2	Beta-1,3-N-acetylgalactosaminyltransferase 2	SEQ ID NOS: 1315-1317
B3GALT1	UDP-Gal:betaGlcNAc beta 1,3-galactosyltransferase, polypeptide 1	SEQ ID NO: 1318
B3GALT4	UDP-Gal:betaGlcNAc beta 1,3-galactosyltransferase, polypeptide 4	SEQ ID NO: 1319
B3GALT5	UDP-Gal:betaGlcNAc beta 1,3-galactosyltransferase, polypeptide 5	SEQ ID NOS: 1320-1324
B3GALT6	UDP-Gal:betaGal beta 1,3-galactosyltransferase polypeptide 6	SEQ ID NO: 1325
B3GAT3	Beta-1,3-glucuronyltransferase 3	SEQ ID NOS: 1326-1330
B3GLCT	Beta 3-glucosyltransferase	SEQ ID NO: 1331
B3GNT3	UDP-GlcNAc:betaGal beta-1,3-N-acetylglucosaminyltransferase 3	SEQ ID NOS: 1332-1335
B3GNT4	UDP-GlcNAc:betaGal beta-1,3-N-acetylglucosaminyltransferase 4	SEQ ID NOS: 1336-1339
B3GNT6	UDP-GlcNAc:betaGal beta-1,3-N-acetylglucosaminyltransferase 6	SEQ ID NOS: 1340-1341
B3GNT7	UDP-GlcNAc:betaGal beta-1,3-N-acetylglucosaminyltransferase 7	SEQ ID NO: 1342
B3GNT8	UDP-GlcNAc:betaGal beta-1,3-N-acetylglucosaminyltransferase 8	SEQ ID NO: 1343
B3GNT9	UDP-GlcNAc:betaGal beta-1,3-N-acetylglucosaminyltransferase 9	SEQ ID NO: 1344
B4GALNT1	Beta-1,4-N-acetyl-galactosaminyl transferase 1	SEQ ID NOS: 1345-1356
B4GALNT3	Beta-1,4-N-acetyl-galactosaminyl transferase 3	SEQ ID NOS: 1357-1358
B4GALNT4	Beta-1,4-N-acetyl-galactosaminyl transferase 4	SEQ ID NOS: 1359-1361
B4GALT4	UDP-Gal:betaGlcNAc beta 1,4-galactosyltransferase, polypeptide 4	SEQ ID NOS: 1362-1374
B4GALT5	UDP-Gal:betaGlcNAc beta 1,4-galactosyltransferase, polypeptide 5	SEQ ID NO: 1375
B4GALT6	UDP-Gal:betaGlcNAc beta 1,4-galactosyltransferase, polypeptide 6	SEQ ID NOS: 1376-1379
B4GAT1	Beta-1,4-glucuronyltransferase 1	SEQ ID NO: 1380
B9D1	B9 protein domain 1	SEQ ID NOS: 1381-1397
BACE2	Beta-site APP-cleaving enzyme 2	SEQ ID NOS: 1398-1400
BAGE5	B melanoma antigen family, member 5	SEQ ID NO: 1401
BCAM	Basal cell adhesion molecule (Lutheran blood group)	SEQ ID NOS: 1402-1405
BCAN	Brevican	SEQ ID NOS: 1406-1412
BCAP29	B-cell receptor-associated protein 29	SEQ ID NOS: 1413-1425
BCAR1	Breast cancer anti-estrogen resistance 1	SEQ ID NOS: 1426-1443
BCHE	Butyrylcholinesterase	SEQ ID NOS: 1444-1448
BCKDHB	Branched chain keto acid dehydrogenase E1, beta polypeptide	SEQ ID NOS: 1449-1451
BDNF	Brain-derived neurotrophic factor	SEQ ID NOS: 1452-1469
BGLAP	Bone gamma-carboxyglutamate (gla) protein	SEQ ID NO: 1470
BGN	Biglycan	SEQ ID NOS: 1471-1472
BLVRB	Biliverdin reductase B	SEQ ID NOS: 1473-1477
BMP1	Bone morphogenetic protein 1	SEQ ID NOS: 1478-1489
BMP10	Bone morphogenetic protein 10	SEQ ID NO: 1490

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
BMP15	Bone morphogenetic protein 15	SEQ ID NO: 1491
BMP2	Bone morphogenetic protein 2	SEQ ID NO: 1492
BMP3	Bone morphogenetic protein 3	SEQ ID NO: 1493
BMP4	Bone morphogenetic protein 4	SEQ ID NOS: 1494-1501
BMP6	Bone morphogenetic protein 6	SEQ ID NO: 1502
BMP7	Bone morphogenetic protein 7	SEQ ID NOS: 1503-1506
BMP8A	Bone morphogenetic protein 8a	SEQ ID NO: 1507
BMP8B	Bone morphogenetic protein 8b	SEQ ID NO: 1508
BMPER	BMP binding endothelial regulator	SEQ ID NOS: 1509-1512
BNC1	Basenuclin 1	SEQ ID NOS: 1513-1514
BOC	BOC cell adhesion associated, oncogene regulated	SEQ ID NOS: 1515-1525
BOD1	Biorientation of chromosomes in cell division 1	SEQ ID NOS: 1526-1530
BOLA1	BolA family member 1	SEQ ID NOS: 1531-1533
BPI	Bactericidal/permeability-increasing protein	SEQ ID NOS: 1534-1537
BPIFA1	BPI fold containing family A, member 1	SEQ ID NOS: 1538-1541
BPIFA2	BPI fold containing family A, member 2	SEQ ID NOS: 1542-1543
BPIFA3	BPI fold containing family A, member 3	SEQ ID NOS: 1544-1545
BPIFB1	BPI fold containing family B, member 1	SEQ ID NOS: 1546-1547
BPIFB2	BPI fold containing family B, member 2	SEQ ID NO: 1548
BPIFB3	BPI fold containing family B, member 3	SEQ ID NO: 1549
BPIFB4	BPI fold containing family B, member 4	SEQ ID NOS: 1550-1551
BPIFB6	BPI fold containing family B, member 6	SEQ ID NOS: 1552-1553
BPIFC	BPI fold containing family C	SEQ ID NOS: 1554-1557
BRF1	BRF1, RNA polymerase III transcription initiation factor 90 kDa subunit	SEQ ID NOS: 1558-1573
BRINP1	Bone morphogenetic protein/retinoic acid inducible neural-specific 1	SEQ ID NOS: 1574-1575
BRINP2	Bone morphogenetic protein/retinoic acid inducible neural-specific 2	SEQ ID NO: 1576
BRINP3	Bone morphogenetic protein/retinoic acid inducible neural-specific 3	SEQ ID NOS: 1577-1579
BSG	Basigin (Ok blood group)	SEQ ID NOS: 1580-1590
BSPH1	Binder of sperm protein homolog 1	SEQ ID NO: 1591
BST1	Bone marrow stromal cell antigen 1	SEQ ID NOS: 1592-1596
BTBD17	BTB (POZ) domain containing 17	SEQ ID NO: 1597
BTB	Biotinidase	SEQ ID NOS: 1598-1607
BTN2A2	Butyrophilin, subfamily 2, member A2	SEQ ID NOS: 1608-1621
BTN3A1	Butyrophilin, subfamily 3, member A1	SEQ ID NOS: 1622-1628
BTN3A2	Butyrophilin, subfamily 3, member A2	SEQ ID NOS: 1629-1639
BTN3A3	Butyrophilin, subfamily 3, member A3	SEQ ID NOS: 1640-1648
C10orf10	Chromosome 10 open reading frame 10	SEQ ID NOS: 1649-4170
C10orf99	Chromosome 10 open reading frame 99	SEQ ID NO: 1650
C11orf1	Chromosome 11 open reading frame 1	SEQ ID NOS: 1651-1655
C11orf24	Chromosome 11 open reading frame 24	SEQ ID NOS: 1656-1658
C11orf45	Chromosome 11 open reading frame 45	SEQ ID NOS: 1659-1660
C11orf94	Chromosome 11 open reading frame 94	SEQ ID NO: 1661
C12orf10	Chromosome 12 open reading frame 10	SEQ ID NOS: 1662-1665
C12orf49	Chromosome 12 open reading frame 49	SEQ ID NOS: 1666-1669

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
C12orf73	Chromosome 12 open reading frame 73	SEQ ID NOS: 1670-1679
C12orf76	Chromosome 12 open reading frame 76	SEQ ID NOS: 1680-1687
C14orf80	Chromosome 14 open reading frame 80	SEQ ID NOS: 13083-13096
C14orf93	Chromosome 14 open reading frame 93	SEQ ID NOS: 1688-1703
C16orf89	Chromosome 16 open reading frame 89	SEQ ID NOS: 1704-1706
C16orf90	Chromosome 16 open reading frame 90	SEQ ID NOS: 1707-1708
C17orf67	Chromosome 17 open reading frame 67	SEQ ID NO: 1709
C17orf75	Chromosome 17 open reading frame 75	SEQ ID NOS: 1710-1718
C17orf99	Chromosome 17 open reading frame 99	SEQ ID NOS: 1719-1721
C18orf54	Chromosome 18 open reading frame 54	SEQ ID NOS: 1722-1726
C19orf47	Chromosome 19 open reading frame 47	SEQ ID NOS: 1727-1734
C19orf70	Chromosome 19 open reading frame 70	SEQ ID NOS: 1735-1738
C19orf80	Chromosome 19 open reading frame 80	SEQ ID NOS: 829-832
C1GALT1	Core 1 synthase, glycoprotein-N-acetylgalactosamine 3-beta-galactosyltransferase 1	SEQ ID NOS: 1739-1743
C1orf127	Chromosome 1 open reading frame 127	SEQ ID NOS: 1744-1747
C1orf159	Chromosome 1 open reading frame 159	SEQ ID NOS: 1748-1760
C1orf198	Chromosome 1 open reading frame 198	SEQ ID NOS: 1761-1765
C1orf234	Chromosome 1 open reading frame 234	SEQ ID NOS: 13118-13120
C1orf54	Chromosome 1 open reading frame 54	SEQ ID NOS: 1766-1768
C1orf56	Chromosome 1 open reading frame 56	SEQ ID NO: 1769
C1QA	Complement component 1, q subcomponent, A chain	SEQ ID NOS: 1770-1772
C1QB	Complement component 1, q subcomponent, B chain	SEQ ID NOS: 1773-1776
C1QC	Complement component 1, q subcomponent, C chain	SEQ ID NOS: 1777-1779
C1QL1	Complement component 1, q subcomponent-like 1	SEQ ID NO: 1780
C1QL2	Complement component 1, q subcomponent-like 2	SEQ ID NO: 1781
C1QL3	Complement component 1, q subcomponent-like 3	SEQ ID NOS: 1782-1783
C1QL4	Complement component 1, q subcomponent-like 4	SEQ ID NO: 1784
C1QTNF1	C1q and tumor necrosis factor related protein 1	SEQ ID NOS: 1785-1794
C1QTNF2	C1q and tumor necrosis factor related protein 2	SEQ ID NO: 1796
C1QTNF3	C1q and tumor necrosis factor related protein 3	SEQ ID NOS: 1797-1798
C1QTNF4	C1q and tumor necrosis factor related protein 4	SEQ ID NOS: 1799-1800
C1QTNF5	C1q and tumor necrosis factor related protein 5	SEQ ID NOS: 1801-1803
C1QTNF7	C1q and tumor necrosis factor related protein 7	SEQ ID NOS: 1804-1808
C1QTNF8	C1q and tumor necrosis factor related protein 8	SEQ ID NOS: 1809-1810
C1QTNF9	C1q and tumor necrosis factor related protein 9	SEQ ID NOS: 1811-1812
C1QTNF9B	C1q and tumor necrosis factor related protein 9B	SEQ ID NOS: 1813-1815
C1R	Complement component 1, r subcomponent	SEQ ID NOS: 1816-1824
C1RL	Complement component 1, r subcomponent-like	SEQ ID NOS: 1825-1833
C1S	Complement component 1, s subcomponent	SEQ ID NOS: 1834-1843

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
C2	Complement component 2	SEQ ID NOS: 1844-1858
C21orf33	Chromosome 21 open reading frame 33	SEQ ID NOS: 1859-1867
C21orf62	Chromosome 21 open reading frame 62	SEQ ID NOS: 1868-1871
C22orf15	Chromosome 22 open reading frame 15	SEQ ID NOS: 1872-1874
C22orf46	Chromosome 22 open reading frame 46	SEQ ID NO: 1875
C2CD2	C2 calcium-dependent domain containing 2	SEQ ID NOS: 1876-1878
C2orf40	Chromosome 2 open reading frame 40	SEQ ID NOS: 1879-1881
C2orf66	Chromosome 2 open reading frame 66	SEQ ID NO: 1882
C2orf69	Chromosome 2 open reading frame 69	SEQ ID NO: 1883
C2orf78	Chromosome 2 open reading frame 78	SEQ ID NO: 1884
C3	Complement component 3	SEQ ID NOS: 1885-1889
C3orf33	Chromosome 3 open reading frame 33	SEQ ID NOS: 1890-1894
C3orf58	Chromosome 3 open reading frame 58	SEQ ID NOS: 1895-1898
C4A	Complement component 4A (Rodgers blood group)	SEQ ID NOS: 1899-1900
C4B	Complement component 4B (Chido blood group)	SEQ ID NOS: 1901-1902
C4BPA	Complement component 4 binding protein, alpha	SEQ ID NOS: 1903-1905
C4BPB	Complement component 4 binding protein, beta	SEQ ID NOS: 1906-1910
C4orf26	Chromosome 4 open reading frame 26	SEQ ID NOS: 1975-1974
C4orf48	Chromosome 4 open reading frame 48	SEQ ID NOS: 1911-1912
C5	Complement component 5	SEQ ID NO: 1913
C5orf46	Chromosome 5 open reading frame 46	SEQ ID NOS: 1914-1915
C6	Complement component 6	SEQ ID NOS: 1916-1919
C6orf120	Chromosome 6 open reading frame 120	SEQ ID NO: 1920
C6orf15	Chromosome 6 open reading frame 15	SEQ ID NO: 1921
C6orf25	Chromosome 6 open reading frame 25	SEQ ID NOS: 1832-8839
C6orf58	Chromosome 6 open reading frame 58	SEQ ID NO: 1922
C7	Complement component 7	SEQ ID NO: 1923
C7orf57	Chromosome 7 open reading frame 57	SEQ ID NOS: 1924-1928
C7orf73	Chromosome 7 open reading frame 73	SEQ ID NOS: 12924-12925
C8A	Complement component 8, alpha polypeptide	SEQ ID NO: 1929
C8B	Complement component 8, beta polypeptide	SEQ ID NOS: 1930-1932
C8G	Complement component 8, gamma polypeptide	SEQ ID NOS: 1933-1934
C9	Complement component 9	SEQ ID NO: 1935
C9orf47	Chromosome 9 open reading frame 47	SEQ ID NOS: 1936-1938
CA10	Carbonic anhydrase X	SEQ ID NOS: 1939-1945
CA11	Carbonic anhydrase XI	SEQ ID NOS: 1946-1947
CA6	Carbonic anhydrase VI	SEQ ID NOS: 1948-1952
CA9	Carbonic anhydrase IX	SEQ ID NOS: 1953-1954
CABLES1	Cdk5 and Abl enzyme substrate 1	SEQ ID NOS: 1955-1960
CABP1	Calcium binding protein 1	SEQ ID NOS: 1961-1964
CACNA2D1	Calcium channel, voltage-dependent, alpha 2/delta subunit 1	SEQ ID NOS: 1965-1968
CACNA2D4	Calcium channel, voltage-dependent, alpha 2/delta subunit 4	SEQ ID NOS: 1969-1982
CADM3	Cell adhesion molecule 3	SEQ ID NOS: 1983-1985

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CALCA	Calcitonin-related polypeptide alpha	SEQ ID NOS: 1986-1990
CALCB	Calcitonin-related polypeptide beta	SEQ ID NOS: 1991-1993
CALCR	Calcitonin receptor	SEQ ID NOS: 1994-2000
CALCRL	Calcitonin receptor-like	SEQ ID NOS: 2001-2005
CALR	Calreticulin	SEQ ID NOS: 2011-2014
CCDC88B	Coiled-coil domain containing 88B	SEQ ID NOS: 2112-2114
CCER2	Coiled-coil glutamate-rich protein 2	SEQ ID NOS: 2115-2116
CKK	Cholecystokinin	SEQ ID NOS: 2117-2119
CCL1	Chemokine (C-C motif) ligand 1	SEQ ID NO: 2120
CCL11	Chemokine (C-C motif) ligand 11	SEQ ID NO: 2121
CCL13	Chemokine (C-C motif) ligand 13	SEQ ID NOS: 2122-2123
CCL14	Chemokine (C-C motif) ligand 14	SEQ ID NOS: 2124-2127
CCL15	Chemokine (C-C motif) ligand 15	SEQ ID NOS: 2128-2129
CCL16	Chemokine (C-C motif) ligand 16	SEQ ID NOS: 2130-2132
CCL17	Chemokine (C-C motif) ligand 17	SEQ ID NOS: 2133-2134
CCL18	Chemokine (C-C motif) ligand 18 (pulmonary and activation-regulated)	SEQ ID NO: 2135
CCL19	Chemokine (C-C motif) ligand 19	SEQ ID NOS: 2136-2137
CCL2	Chemokine (C-C motif) ligand 2	SEQ ID NOS: 2138-2139
CCL20	Chemokine (C-C motif) ligand 20	SEQ ID NOS: 2140-2142
CCL21	Chemokine (C-C motif) ligand 21	SEQ ID NOS: 2143-2144
CCL22	Chemokine (C-C motif) ligand 22	SEQ ID NO: 2145
CCL23	Chemokine (C-C motif) ligand 23	SEQ ID NOS: 2146-2148
CCL24	Chemokine (C-C motif) ligand 24	SEQ ID NOS: 2149-2150
CCL25	Chemokine (C-C motif) ligand 25	SEQ ID NOS: 2151-2154
CCL26	Chemokine (C-C motif) ligand 26	SEQ ID NOS: 2155-2156
CCL27	Chemokine (C-C motif) ligand 27	SEQ ID NO: 2157
CCL28	Chemokine (C-C motif) ligand 28	SEQ ID NOS: 2158-2160
CCL3	Chemokine (C-C motif) ligand 3	SEQ ID NO: 2161
CCL3L3	Chemokine (C-C motif) ligand 3-like 3	SEQ ID NO: 2162
CCL4	Chemokine (C-C motif) ligand 4	SEQ ID NOS: 2163-2164
CCL4L2	Chemokine (C-C motif) ligand 4-like 2	SEQ ID NOS: 2165-2174
CCL5	Chemokine (C-C motif) ligand 5	SEQ ID NOS: 2175-2177
CCL7	Chemokine (C-C motif) ligand 7	SEQ ID NOS: 2178-2180
CCL8	Chemokine (C-C motif) ligand 8	SEQ ID NO: 2181
CCNB1IP1	Cyclin B1 interacting protein 1, E3 ubiquitin protein ligase	SEQ ID NOS: 2182-2193
CCNL1	Cyclin L1	SEQ ID NOS: 2194-2202
CCNL2	Cyclin L2	SEQ ID NOS: 2203-2210
CD14	CD14 molecule	SEQ ID NOS: 2211-2215
CD160	CD160 molecule	SEQ ID NOS: 2216-2220
CD164	CD164 molecule, sialomucin	SEQ ID NOS: 2221-2226
CD177	CD177 molecule	SEQ ID NOS: 2227-2229

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CD1E	CD1e molecule	SEQ ID NOS: 2230-2243
CD2	CD2 molecule	SEQ ID NOS: 2244-2245
CD200	CD200 molecule	SEQ ID NOS: 2246-2252
CD200R1	CD200 receptor 1	SEQ ID NOS: 2253-2257
CD22	CD22 molecule	SEQ ID NOS: 2258-2275
CD226	CD226 molecule	SEQ ID NOS: 2276-2283
CD24	CD24 molecule	SEQ ID NOS: 2284-2290
CD276	CD276 molecule	SEQ ID NOS: 2291-2306
CD300A	CD300a molecule	SEQ ID NOS: 2307-2311
CD300LB	CD300 molecule-like family member b	SEQ ID NOS: 2312-2313
CD300LF	CD300 molecule-like family member f	SEQ ID NOS: 2314-2322
CD300LG	CD300 molecule-like family member g	SEQ ID NOS: 2323-2328
CD3D	CD3d molecule, delta (CD3-TCR complex)	SEQ ID NOS: 2329-2332
CD4	CD4 molecule	SEQ ID NOS: 2333-2335
CD40	CD40 molecule, TNF receptor superfamily member 5	SEQ ID NOS: 2336-2339
CD44	CD44 molecule (Indian blood group)	SEQ ID NOS: 2340-2366
CD48	CD48 molecule	SEQ ID NOS: 2367-2369
CD5	CD5 molecule	SEQ ID NOS: 2370-2371
CD55	CD55 molecule, decay accelerating factor for complement (Cromer blood group)	SEQ ID NOS: 2372-2382
CD59	CD59 molecule, complement regulatory protein	SEQ ID NOS: 2383-2393
CD5L	CD5 molecule-like	SEQ ID NO: 2394
CD6	CD6 molecule	SEQ ID NOS: 2395-2402
CD68	CD68 molecule	SEQ ID NOS: 2403-2406
CD7	CD7 molecule	SEQ ID NOS: 2407-2412
CD79A	CD79a molecule, immunoglobulin-associated alpha	SEQ ID NOS: 2413-2415
CD80	CD80 molecule	SEQ ID NOS: 2416-2418
CD86	CD86 molecule	SEQ ID NOS: 2419-2425
CD8A	CD8a molecule	SEQ ID NOS: 2426-2429
CD8B	CD8b molecule	SEQ ID NOS: 2430-2435
CD99	CD99 molecule	SEQ ID NOS: 2436-2444
CDC23	Cell division cycle 23	SEQ ID NOS: 2445-2449
CDC40	Cell division cycle 40	SEQ ID NOS: 2450-2452
CDC45	Cell division cycle 45	SEQ ID NOS: 2453-2459
CDCP1	CUB domain containing protein 1	SEQ ID NOS: 2460-2461
CDCP2	CUB domain containing protein 2	SEQ ID NOS: 2462-2463
CDH1	Cadherin 1, type 1	SEQ ID NOS: 2464-2471
CDH11	Cadherin 11, type 2, OB-cadherin (osteoblast)	SEQ ID NOS: 2472-2481
CDH13	Cadherin 13	SEQ ID NOS: 2482-2491

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CDH17	Cadherin 17, LI cadherin (liver-intestine)	SEQ ID NOS: 2492-2496
CDH18	Cadherin 18, type 2	SEQ ID NOS: 2497-2503
CDH19	Cadherin 19, type 2	SEQ ID NOS: 2504-2508
CDH23	Cadherin-related 23	SEQ ID NOS: 2509-2524
CDH5	Cadherin 5, type 2 (vascular endothelium)	SEQ ID NOS: 2525-2532
CDHR1	Cadherin-related family member 1	SEQ ID NOS: 2533-2538
CDHR4	Cadherin-related family member 4	SEQ ID NOS: 2539-2543
CDHR5	Cadherin-related family member 5	SEQ ID NOS: 2544-2550
CDKN2A	Cyclin-dependent kinase inhibitor 2A	SEQ ID NOS: 2551-2561
CDNF	Cerebral dopamine neurotrophic factor	SEQ ID NOS: 2562-2563
CDON	Cell adhesion associated, oncogene regulated	SEQ ID NOS: 2564-2571
CDSN	Corneodesmosin	SEQ ID NO: 2572
CEACAM16	Carcinoembryonic antigen-related cell adhesion molecule 16	SEQ ID NOS: 2573-2574
CEACAM18	Carcinoembryonic antigen-related cell adhesion molecule 18	SEQ ID NO: 2575
CEACAM19	Carcinoembryonic antigen-related cell adhesion molecule 19	SEQ ID NOS: 2576-2582
CEACAM5	Carcinoembryonic antigen-related cell adhesion molecule 5	SEQ ID NOS: 2583-2590
CEACAM7	Carcinoembryonic antigen-related cell adhesion molecule 7	SEQ ID NOS: 2591-2593
CEACAM8	Carcinoembryonic antigen-related cell adhesion molecule 8	SEQ ID NOS: 2594-2595
CECR1	Cat eye syndrome chromosome region, candidate 1	SEQ ID NOS: 222-229
CECR5	Cat eye syndrome chromosome region, candidate 5	SEQ ID NOS: 6411-6413
CEL	Carboxyl ester lipase	SEQ ID NO: 2596
CELA2A	Chymotrypsin-like elastase family, member 2A	SEQ ID NO: 2597
CELA2B	Chymotrypsin-like elastase family, member 2B	SEQ ID NOS: 2598-2599
CELA3A	Chymotrypsin-like elastase family, member 3A	SEQ ID NOS: 2600-2602
CELA3B	Chymotrypsin-like elastase family, member 3B	SEQ ID NOS: 2603-2605
CEMIP	Cell migration inducing protein, hyaluronan binding	SEQ ID NOS: 2606-2610
CEP89	Centrosomal protein 89 kDa	SEQ ID NOS: 2611-2616
CER1	Cerberus 1, DAN family BMP antagonist	SEQ ID NO: 2617
CERCAM	Cerebral endothelial cell adhesion molecule	SEQ ID NOS: 2618-2625
CERS1	Ceramide synthase 1	SEQ ID NOS: 2626-2630
CES1	Carboxylesterase 1	SEQ ID NOS: 2631-2636
CES3	Carboxylesterase 3	SEQ ID NOS: 2637-2641
CES4A	Carboxylesterase 4A	SEQ ID NOS: 2642-2647
CES5A	Carboxylesterase 5A	SEQ ID NOS: 2648-2655
CETP	Cholesteryl ester transfer protein, plasma	SEQ ID NOS: 2656-2658
CFB	Complement factor B	SEQ ID NOS: 2669-2673
CFC1	Cripto, FRL-1, cryptic family 1	SEQ ID NOS: 2674-2676
CFC1B	Cripto, FRL-1, cryptic family 1B	SEQ ID NOS: 2677-2679
CFD	Complement factor D (adipsin)	SEQ ID NOS: 2680-2681

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CFDP1	Craniofacial development protein 1	SEQ ID NOS: 2682-2685
CFH	Complement factor H	SEQ ID NOS: 2686-2688
CFHR1	Complement factor H-related 1	SEQ ID NOS: 2689-2690
CFHR2	Complement factor H-related 2	SEQ ID NOS: 2691-2692
CFHR3	Complement factor H-related 3	SEQ ID NOS: 2693-2697
CFHR4	Complement factor H-related 4	SEQ ID NOS: 2698-2701
CFHR5	Complement factor H-related 5	SEQ ID NO: 2702
CFI	Complement factor I	SEQ ID NOS: 2703-2707
CFP	Complement factor properdin	SEQ ID NOS: 2708-2711
CGA	Glycoprotein hormones, alpha polypeptide	SEQ ID NOS: 2712-2716
CGB	Chorionic gonadotropin, beta polypeptide	SEQ ID NO: 2721
CGB1	Chorionic gonadotropin, beta polypeptide 1	SEQ ID NOS: 2717-2718
CGB2	Chorionic gonadotropin, beta polypeptide 2	SEQ ID NOS: 2719-2720
CGB5	Chorionic gonadotropin, beta polypeptide 5	SEQ ID NO: 2722
CGB7	Chorionic gonadotropin, beta polypeptide 7	SEQ ID NOS: 2723-2725
CGB8	Chorionic gonadotropin, beta polypeptide 8	SEQ ID NO: 2726
CGREF1	Cell growth regulator with EF-hand domain 1	SEQ ID NOS: 2727-2734
CH507-9B2.3		SEQ ID NOS: 5532-5538
CHAD	Chondroadherin	SEQ ID NOS: 2735-2737
CHADL	Chondroadherin-like	SEQ ID NOS: 2738-2740
CHEK2	Checkpoint kinase 2	SEQ ID NOS: 2741-2762
CHGA	Chromogranin A	SEQ ID NOS: 2763-2765
CHGB	Chromogranin B	SEQ ID NOS: 2766-2767
CHI3L1	Chitinase 3-like 1 (cartilage glycoprotein-39)	SEQ ID NOS: 2768-2769
CHI3L2	Chitinase 3-like 2	SEQ ID NOS: 2770-2783
CHIA	Chitinase, acidic	SEQ ID NOS: 2784-2792
CHID1	Chitinase domain containing 1	SEQ ID NOS: 2793-2811
CHIT1	Chitinase 1 (chitotriosidase)	SEQ ID NOS: 2812-2815
CHL1	Cell adhesion molecule L1-like	SEQ ID NOS: 2816-2824
CHN1	Chimerin 1	SEQ ID NOS: 2825-2835
CHPF	Chondroitin polymerizing factor	SEQ ID NOS: 2836-2838
CHPF2	Chondroitin polymerizing factor 2	SEQ ID NOS: 2839-2842
CHRD	Chordin	SEQ ID NOS: 2843-2848
CHRDL1	Chordin-like 1	SEQ ID NOS: 2849-2853
CHRDL2	Chordin-like 2	SEQ ID NOS: 2854-2862
CHRNA2	Cholinergic receptor, nicotinic, alpha 2 (neuronal)	SEQ ID NOS: 2863-2871
CHRNA5	Cholinergic receptor, nicotinic, alpha 5 (neuronal)	SEQ ID NOS: 2872-2875
CHRNB1	Cholinergic receptor, nicotinic, beta 1 (muscle)	SEQ ID NOS: 2876-2881
CHRND	Cholinergic receptor, nicotinic, delta (muscle)	SEQ ID NOS: 2882-2887
CHST1	Carbohydrate (keratan sulfate Gal-6) sulfotransferase 1	SEQ ID NO: 2888

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CHST10	Carbohydrate sulfotransferase 10	SEQ ID NOS: 2889-2896
CHST11	Carbohydrate (chondroitin 4) sulfotransferase 11	SEQ ID NOS: 2897-2901
CHST13	Carbohydrate (chondroitin 4) sulfotransferase 13	SEQ ID NOS: 2902-2903
CHST4	Carbohydrate (N-acetylglucosamine 6-O) sulfotransferase 4	SEQ ID NOS: 2904-2905
CHST5	Carbohydrate (N-acetylglucosamine 6-O) sulfotransferase 5	SEQ ID NOS: 2906-2907
CHST6	Carbohydrate (N-acetylglucosamine 6-O) sulfotransferase 6	SEQ ID NOS: 2908-2909
CHST7	Carbohydrate (N-acetylglucosamine 6-O) sulfotransferase 7	SEQ ID NO: 2910
CHST8	Carbohydrate (N-acetylgalactosamine 4-O) sulfotransferase 8	SEQ ID NOS: 2911-2914
CHSY1	Chondroitin sulfate synthase 1	SEQ ID NOS: 2915-2916
CHSY3	Chondroitin sulfate synthase 3	SEQ ID NO: 2917
CHTF8	Chromosome transmission fidelity factor 8	SEQ ID NOS: 2918-2928
CILP	Cartilage intermediate layer protein, nucleotide pyrophosphohydrolase	SEQ ID NO: 2929
CILP2	Cartilage intermediate layer protein 2	SEQ ID NOS: 2930-2931
CIRH1A	Cirrrosis, autosomal recessive 1A (cirhin)	SEQ ID NOS: 13974-13983
CKLF	Chemokine-like factor	SEQ ID NOS: 2932-2937
CKMT1A	Creatine kinase, mitochondrial 1A	SEQ ID NOS: 2938-2943
CKMT1B	Creatine kinase, mitochondrial 1B	SEQ ID NOS: 2944-2953
CLCA1	Chloride channel accessory 1	SEQ ID NOS: 2954-2955
CLCF1	Cardiotrophin-like cytokine factor 1	SEQ ID NOS: 2956-2957
CLDN15	Claudin 15	SEQ ID NOS: 2958-2963
CLDN7	Claudin 7	SEQ ID NOS: 2964-2970
CLDND1	Claudin domain containing 1	SEQ ID NOS: 2971-2996
CLEC11A	C-type lectin domain family 11, member A	SEQ ID NOS: 2997-2999
CLEC16A	C-type lectin domain family 16, member A	SEQ ID NOS: 3000-3005
CLEC18A	C-type lectin domain family 18, member A	SEQ ID NOS: 3006-3011
CLEC18B	C-type lectin domain family 18, member B	SEQ ID NOS: 3012-3015
CLEC18C	C-type lectin domain family 18, member C	SEQ ID NOS: 3016-3022
CLEC19A	C-type lectin domain family 19, member A	SEQ ID NOS: 3023-3026
CLEC2B	C-type lectin domain family 2, member B	SEQ ID NOS: 3027-3028
CLEC3A	C-type lectin domain family 3, member A	SEQ ID NOS: 3029-3030
CLEC3B	C-type lectin domain family 3, member B	SEQ ID NOS: 3031-3032
CLGN	Calmegin	SEQ ID NOS: 3033-3035
CLN5	Ceroid-lipofuscinosis, neuronal 5	SEQ ID NOS: 3036-3047
CLPS	Colipase, pancreatic	SEQ ID NOS: 3048-3050
CLPSL1	Colipase-like 1	SEQ ID NOS: 3051-3052
CLPSL2	Colipase-like 2	SEQ ID NOS: 3053-3054
CLPX	Caseinolytic mitochondrial matrix peptidase chaperone subunit	SEQ ID NOS: 3055-3057
CLSTN3	Calsyntenin 3	SEQ ID NOS: 3058-3064

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CLU	Clusterin	SEQ ID NOS: 3065-3078
CLUL1	Clusterin-like 1 (retinal)	SEQ ID NOS: 3079-3086
CMA1	Chymase 1, mast cell	SEQ ID NOS: 3087-3088
CMPK1	Cytidine monophosphate (UMP-CMP) kinase 1, cytosolic	SEQ ID NOS: 3089-3092
CNBD1	Cyclic nucleotide binding domain containing 1	SEQ ID NOS: 3093-3096
CNDP1	Carnosine dipeptidase 1 (metallopeptidase M20 family)	SEQ ID NOS: 3097-3099
CNPY2	Canopy FGF signaling regulator 2	SEQ ID NOS: 3107-3111
CNPY3	Canopy FGF signaling regulator 3	SEQ ID NOS: 3112-3113
CNPY4	Canopy FGF signaling regulator 4	SEQ ID NOS: 3114-3116
CNTFR	Ciliary neurotrophic factor receptor	SEQ ID NOS: 3117-3120
CNTN1	Contactin 1	SEQ ID NOS: 3121-3130
CNTN2	Contactin 2 (axonal)	SEQ ID NOS: 3131-3142
CNTN3	Contactin 3 (plasmacytoma associated)	SEQ ID NO: 3143
CNTN4	Contactin 4	SEQ ID NOS: 3144-3152
CNTN5	Contactin 5	SEQ ID NOS: 3153-3158
CNTNAP2	Contactin associated protein-like 2	SEQ ID NOS: 3159-3162
CNTNAP3	Contactin associated protein-like 3	SEQ ID NOS: 3163-3167
CNTNAP3B	Contactin associated protein-like 3B	SEQ ID NOS: 3168-3176
COASY	CoA synthase	SEQ ID NOS: 3177-3186
COCH	Cochlin	SEQ ID NOS: 3187-3198
COG3	Component of oligomeric golgi complex 3	SEQ ID NOS: 3199-3202
COL10A1	Collagen, type X, alpha 1	SEQ ID NOS: 3203-3206
COL11A1	Collagen, type XI, alpha 1	SEQ ID NOS: 3207-3217
COL11A2	Collagen, type XI, alpha 2	SEQ ID NOS: 3218-3222
COL12A1	Collagen, type XII, alpha 1	SEQ ID NOS: 3223-3230
COL14A1	Collagen, type XIV, alpha 1	SEQ ID NOS: 3231-3238
COL15A1	Collagen, type XV, alpha 1	SEQ ID NOS: 3239-3240
COL16A1	Collagen, type XVI, alpha 1	SEQ ID NOS: 3241-3245
COL18A1	Collagen, type XVIII, alpha 1	SEQ ID NOS: 3246-3250
COL19A1	Collagen, type XIX, alpha 1	SEQ ID NOS: 3251-3253
COL1A1	Collagen, type I, alpha 1	SEQ ID NOS: 3254-3255
COL1A2	Collagen, type I, alpha 2	SEQ ID NOS: 3256-3257
COL20A1	Collagen, type XX, alpha 1	SEQ ID NOS: 3258-3261
COL21A1	Collagen, type XXI, alpha 1	SEQ ID NOS: 3262-3267
COL22A1	Collagen, type XXII, alpha 1	SEQ ID NOS: 3268-3270
COL24A1	Collagen, type XXIV, alpha 1	SEQ ID NOS: 3271-3274
COL26A1	Collagen, type XXVI, alpha 1	SEQ ID NOS: 3275-3276

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
COL27A1	Collagen, type XXVII, alpha 1	SEQ ID NOS: 3277-3279
COL28A1	Collagen, type XXVIII, alpha 1	SEQ ID NOS: 3280-3284
COL2A1	Collagen, type II, alpha 1	SEQ ID NOS: 3285-3286
COL3A1	Collagen, type III, alpha 1	SEQ ID NOS: 3287-3289
COL4A1	Collagen, type IV, alpha 1	SEQ ID NOS: 3290-3292
COL4A2	Collagen, type IV, alpha 2	SEQ ID NOS: 3293-3295
COL4A3	Collagen, type IV, alpha 3 (Goodpasture antigen)	SEQ ID NOS: 3296-3299
COL4A4	Collagen, type IV, alpha 4	SEQ ID NOS: 3300-3301
COL4A5	Collagen, type IV, alpha 5	SEQ ID NOS: 3302-3308
COL4A6	Collagen, type IV, alpha 6	SEQ ID NOS: 3309-3314
COL5A1	Collagen, type V, alpha 1	SEQ ID NOS: 3315-3317
COL5A2	Collagen, type V, alpha 2	SEQ ID NOS: 3318-3319
COL5A3	Collagen, type V, alpha 3	SEQ ID NO: 3320
COL6A1	Collagen, type VI, alpha 1	SEQ ID NOS: 3321-3322
COL6A2	Collagen, type VI, alpha 2	SEQ ID NOS: 3323-3328
COL6A3	Collagen, type VI, alpha 3	SEQ ID NOS: 3329-3337
COL6A5	Collagen, type VI, alpha 5	SEQ ID NOS: 3338-3342
COL6A6	Collagen, type VI, alpha 6	SEQ ID NOS: 3343-3345
COL7A1	Collagen, type VII, alpha 1	SEQ ID NOS: 3346-3347
COL8A1	Collagen, type VIII, alpha 1	SEQ ID NOS: 3348-3351
COL8A2	Collagen, type VIII, alpha 2	SEQ ID NOS: 3352-3354
COL9A1	Collagen, type IX, alpha 1	SEQ ID NOS: 3355-3358
COL9A2	Collagen, type IX, alpha 2	SEQ ID NOS: 3359-3362
COL9A3	Collagen, type IX, alpha 3	SEQ ID NOS: 3363-3364
COLEC10	Collectin sub-family member 10 (C-type lectin)	SEQ ID NO: 3365
COLEC11	Collectin sub-family member 11	SEQ ID NOS: 3366-3375
COLGALT1	Collagen beta(1-O)galactosyltransferase 1	SEQ ID NOS: 3376-3378
COLGALT2	Collagen beta(1-O)galactosyltransferase 2	SEQ ID NOS: 3379-3381
COLQ	Collagen-like tail subunit (single strand of homotrimer) of asymmetric acetylcholinesterase	SEQ ID NOS: 3382-3386
COMP	Cartilage oligomeric matrix protein	SEQ ID NOS: 3387-3389
COPS6	COP9 signalosome subunit 6	SEQ ID NOS: 3390-3393
COQ6	Coenzyme Q6 monooxygenase	SEQ ID NOS: 3394-3401
CORT	Cortistatin	SEQ ID NO: 3402
CP	Ceruloplasmin (ferroxidase)	SEQ ID NOS: 3403-3407
CPA1	Carboxypeptidase A1 (pancreatic)	SEQ ID NOS: 3408-3412
CPA2	Carboxypeptidase A2 (pancreatic)	SEQ ID NOS: 3413-3414
CPA3	Carboxypeptidase A3 (mast cell)	SEQ ID NO: 3415
CPA4	Carboxypeptidase A4	SEQ ID NOS: 3416-3421
CPA6	Carboxypeptidase A6	SEQ ID NOS: 3422-3424

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CPAMD8	C3 and PZP-like, alpha-2-macroglobulin domain containing 8	SEQ ID NOS: 3425-3430
CPB1	Carboxypeptidase B1 (tissue)	SEQ ID NOS: 3431-3435
CPB2	Carboxypeptidase B2 (plasma)	SEQ ID NOS: 3436-3438
CPE	Carboxypeptidase E	SEQ ID NOS: 3439-3443
CPM	Carboxypeptidase M	SEQ ID NOS: 3444-3453
CPN1	Carboxypeptidase N, polypeptide 1	SEQ ID NOS: 3454-3455
CPN2	Carboxypeptidase N, polypeptide 2	SEQ ID NOS: 3456-3457
CPO	Carboxypeptidase O	SEQ ID NO: 3458
CPQ	Carboxypeptidase Q	SEQ ID NOS: 3459-3464
CPVL	Carboxypeptidase, vitellogenic-like	SEQ ID NOS: 3465-3475
CPXM1	Carboxypeptidase X (M14 family), member 1	SEQ ID NO: 3476
CPXM2	Carboxypeptidase X (M14 family), member 2	SEQ ID NOS: 3477-3478
CPZ	Carboxypeptidase Z	SEQ ID NOS: 3479-3482
CR1L	Complement component (3b/4b) receptor 1-like	SEQ ID NOS: 3483-3484
CRB2	Crumbs family member 2	SEQ ID NOS: 3485-3487
CREG1	Cellular repressor of E1A-stimulated genes 1	SEQ ID NO: 3488
CREG2	Cellular repressor of E1A-stimulated genes 2	SEQ ID NO: 3489
CRELD1	Cysteine-rich with EGF-like domains 1	SEQ ID NOS: 3490-3495
CRELD2	Cysteine-rich with EGF-like domains 2	SEQ ID NOS: 3496-3500
CRH	Corticotropin releasing hormone	SEQ ID NO: 3501
CRHBP	Corticotropin releasing hormone binding protein	SEQ ID NOS: 3502-3503
CRHR1	Corticotropin releasing hormone receptor 1	SEQ ID NOS: 3504-3515
CRHR2	Corticotropin releasing hormone receptor 2	SEQ ID NOS: 3516-3522
CRISP1	Cysteine-rich secretory protein 1	SEQ ID NOS: 3523-3526
CRISP2	Cysteine-rich secretory protein 2	SEQ ID NOS: 3527-3529
CRISP3	Cysteine-rich secretory protein 3	SEQ ID NOS: 3530-3533
CRISPLD2	Cysteine-rich secretory protein LCCL domain containing 2	SEQ ID NOS: 3534-3541
CRLF1	Cytokine receptor-like factor 1	SEQ ID NOS: 3542-3543
CRP	C-reactive protein, pentraxin-related	SEQ ID NOS: 3544-3548
CRTAC1	Cartilage acidic protein 1	SEQ ID NOS: 3549-3553
CRTAP	Cartilage associated protein	SEQ ID NOS: 3554-3555
CRY2	Cryptochrome circadian clock 2	SEQ ID NOS: 3556-3559
CSAD	Cysteine sulfinic acid decarboxylase	SEQ ID NOS: 3560-3572
CSF1	Colony stimulating factor 1 (macrophage)	SEQ ID NOS: 3573-3580
CSF1R	Colony stimulating factor 1 receptor	SEQ ID NOS: 3581-3585
CSF2	Colony stimulating factor 2 (granulocyte-macrophage)	SEQ ID NO: 3586
CSF2RA	Colony stimulating factor 2 receptor, alpha, low-affinity (granulocyte-macrophage)	SEQ ID NOS: 3587-3598
CSF3	Colony stimulating factor 3 (granulocyte)	SEQ ID NOS: 3599-3605
CSGALNACT1	Chondroitin sulfate N-acetylgalactosaminyltransferase 1	SEQ ID NOS: 3606-3614

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CSH1	Chorionic somatomammotropin hormone 1 (placental lactogen)	SEQ ID NOS: 3615-3618
CSH2	Chorionic somatomammotropin hormone 2	SEQ ID NOS: 3619-3623
CSHL1	Chorionic somatomammotropin hormone-like 1	SEQ ID NOS: 3624-3630
CSN1S1	Casein alpha s1	SEQ ID NOS: 3631-3636
CSN2	Casein beta	SEQ ID NO: 3637
CSN3	Casein kappa	SEQ ID NO: 3638
CST1	Cystatin SN	SEQ ID NOS: 3639-3640
CST11	Cystatin 11	SEQ ID NOS: 3641-3642
CST2	Cystatin SA	SEQ ID NO: 3643
CST3	Cystatin C	SEQ ID NOS: 3644-3646
CST4	Cystatin S	SEQ ID NO: 3647
CST5	Cystatin D	SEQ ID NO: 3648
CST6	Cystatin E/M	SEQ ID NO: 3649
CST7	Cystatin F (leukocystatin)	SEQ ID NO: 3650
CST8	Cystatin 8 (cystatin-related epididymal specific)	SEQ ID NOS: 3651-3652
CST9	Cystatin 9 (testatin)	SEQ ID NO: 3653
CST9L	Cystatin 9-like	SEQ ID NO: 3654
CSTL1	Cy statin-like 1	SEQ ID NOS: 3655-3657
CT55	Cancer/testis antigen 55	SEQ ID NOS: 3658-3659
CTB-60B18.6		SEQ ID NOS: 74-75
CTBS	Chitinase, di-N-acetyl-	SEQ ID NOS: 3660-3662
CTD-2313N18.7		SEQ ID NO: 4160
CTD-2370N5.3		SEQ ID NOS: 81-84
CTGF	Connective tissue growth factor	SEQ ID NO: 3663
CTHRC1	Collagen triple helix repeat containing 1	SEQ ID NOS: 3664-3667
CTLA4	Cytotoxic T-lymphocyte-associated protein 4	SEQ ID NOS: 3668-3671
CTNS	Cystinosis, lysosomal cystine transporter	SEQ ID NOS: 3672-3679
CTRB1	Chymotrypsinogen B1	SEQ ID NOS: 3680-3682
CTRB2	Chymotrypsinogen B2	SEQ ID NOS: 3683-3686
CTRC	Chymotrypsin C (caldecrin)	SEQ ID NOS: 3687-3688
CTRL	Chymotrypsin-like	SEQ ID NOS: 3689-3691
CTSA	Cathepsin A	SEQ ID NOS: 3692-3700
CTSB	Cathepsin B	SEQ ID NOS: 3701-3725
CTSC	Cathepsin C	SEQ ID NOS: 3726-3730
CTSD	Cathepsin D	SEQ ID NOS: 3731-3741
CTSE	Cathepsin E	SEQ ID NOS: 3742-3743
CTSF	Cathepsin F	SEQ ID NOS: 3744-3747
CTSG	Cathepsin G	SEQ ID NO: 3748
CTSH	Cathepsin H	SEQ ID NOS: 3749-3754
CTSK	Cathepsin K	SEQ ID NOS: 3755-3756
CTSL	Cathepsin L	SEQ ID NOS: 3757-3759
CTSO	Cathepsin O	SEQ ID NO: 3760
CTSS	Cathepsin S	SEQ ID NOS: 3761-3765

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CTSV	Cathepsin V	SEQ ID NOS: 3766-3767
CTSW	Cathepsin W	SEQ ID NOS: 3768-3770
CTSZ	Cathepsin Z	SEQ ID NO: 3771
CUBN	Cubilin (intrinsic factor-cobalamin receptor)	SEQ ID NOS: 3772-3775
CUTA	CutA divalent cation tolerance homolog (<i>E. coli</i>)	SEQ ID NOS: 3776-3785
CX3CL1	Chemokine (C-X3-C motif) ligand 1	SEQ ID NOS: 3786-3789
CXADR	Coxsackie virus and adenovirus receptor	SEQ ID NOS: 3790-3794
CXCL1	Chemokine (C-X-C motif) ligand 1 (melanoma growth stimulating activity, alpha)	SEQ ID NO: 3795
CXCL10	Chemokine (C-X-C motif) ligand 10	SEQ ID NO: 3796
CXCL11	Chemokine (C-X-C motif) ligand 11	SEQ ID NOS: 3797-3798
CXCL12	Chemokine (C-X-C motif) ligand 12	SEQ ID NOS: 3799-3804
CXCL13	Chemokine (C-X-C motif) ligand 13	SEQ ID NO: 3805
CXCL14	Chemokine (C-X-C motif) ligand 14	SEQ ID NOS: 3806-3807
CXCL17	Chemokine (C-X-C motif) ligand 17	SEQ ID NOS: 3808-3809
CXCL2	Chemokine (C-X-C motif) ligand 2	SEQ ID NO: 3810
CXCL3	Chemokine (C-X-C motif) ligand 3	SEQ ID NO: 3811
CXCL5	Chemokine (C-X-C motif) ligand 5	SEQ ID NO: 3812
CXCL6	Chemokine (C-X-C motif) ligand 6	SEQ ID NOS: 3813-3814
CXCL8	Chemokine (C-X-C motif) ligand 8	SEQ ID NOS: 3815-3816
CXCL9	Chemokine (C-X-C motif) ligand 9	SEQ ID NO: 3817
CXorf36	Chromosome X open reading frame 36	SEQ ID NOS: 3818-3819
CYB5D2	Cytochrome b5 domain containing 2	SEQ ID NOS: 3820-3823
CYHR1	Cysteine/histidine-rich 1	SEQ ID NOS: 3824-3831
CYP17A1	Cytochrome P450, family 17, subfamily A, polypeptide 1	SEQ ID NOS: 3832-3836
CYP20A1	Cytochrome P450, family 20, subfamily A, polypeptide 1	SEQ ID NOS: 3837-3843
CYP21A2	Cytochrome P450, family 21, subfamily A, polypeptide 2	SEQ ID NOS: 3844-3851
CYP26B1	Cytochrome P450, family 26, subfamily B, polypeptide 1	SEQ ID NOS: 3852-3856
CYP2A6	Cytochrome P450, family 2, subfamily A, polypeptide 6	SEQ ID NOS: 3857-3858
CYP2A7	Cytochrome P450, family 2, subfamily A, polypeptide 7	SEQ ID NOS: 3859-3861
CYP2B6	Cytochrome P450, family 2, subfamily B, polypeptide 6	SEQ ID NOS: 3862-3865
CYP2C18	Cytochrome P450, family 2, subfamily C, polypeptide 18	SEQ ID NOS: 3866-3867
CYP2C19	Cytochrome P450, family 2, subfamily C, polypeptide 19	SEQ ID NOS: 3868-3869
CYP2C8	Cytochrome P450, family 2, subfamily C, polypeptide 8	SEQ ID NOS: 3870-3877
CYP2C9	Cytochrome P450, family 2, subfamily C, polypeptide 9	SEQ ID NOS: 3878-3880
CYP2E1	Cytochrome P450, family 2, subfamily E, polypeptide 1	SEQ ID NOS: 3881-3886
CYP2F1	Cytochrome P450, family 2, subfamily F, polypeptide 1	SEQ ID NOS: 3887-3890
CYP2J2	Cytochrome P450, family 2, subfamily J, polypeptide 2	SEQ ID NO: 3891
CYP2R1	Cytochrome P450, family 2, subfamily R, polypeptide 1	SEQ ID NOS: 3892-3897
CYP2S1	Cytochrome P450, family 2, subfamily S, polypeptide 1	SEQ ID NOS: 3898-3903
CYP2W1	Cytochrome P450, family 2, subfamily W, polypeptide 1	SEQ ID NOS: 3904-3906
CYP46A1	Cytochrome P450, family 46, subfamily A, polypeptide 1	SEQ ID NOS: 3907-3911

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
CYP4F11	Cytochrome P450, family 4, subfamily F, polypeptide 11	SEQ ID NOS: 3912-3916
CYP4F2	Cytochrome P450, family 4, subfamily F, polypeptide 2	SEQ ID NOS: 3917-3921
CYR61	Cysteine-rich, angiogenic inducer, 61	SEQ ID NO: 3922
CYTL1	Cytokine-like 1	SEQ ID NOS: 3923-3925
D2HGDH	D-2-hydroxyglutarate dehydrogenase	SEQ ID NOS: 3926-3934
DAG1	Dystroglycan 1 (dystrophin-associated glycoprotein 1)	SEQ ID NOS: 3935-3949
DAND5	DAN domain family member 5, BMP antagonist	SEQ ID NOS: 3950-3951
DAO	D-amino-acid oxidase	SEQ ID NOS: 3952-3957
DAZAP2	DAZ associated protein 2	SEQ ID NOS: 3958-3966
DBH	Dopamine beta-hydroxylase (dopamine beta-monoxygenase)	SEQ ID NOS: 3967-3968
DBNL	Drebrin-like	SEQ ID NOS: 3969-3986
DCD	Dermcidin	SEQ ID NOS: 3987-3989
DCN	Decorin	SEQ ID NOS: 3990-4008
DDIAS	DNA damage-induced apoptosis suppressor	SEQ ID NOS: 4009-4018
DDOST	Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit (non-catalytic)	SEQ ID NOS: 4019-4022
DDR1	Discoidin domain receptor tyrosine kinase 1	SEQ ID NOS: 4023-4068
DDR2	Discoidin domain receptor tyrosine kinase 2	SEQ ID NOS: 4069-4074
DDT	D-dopachrome tautomerase	SEQ ID NOS: 4075-4080
DDX17	DEAD (Asp-Glu-Ala-Asp) box helicase 17	SEQ ID NOS: 4081-4085
DDX20	DEAD (Asp-Glu-Ala-Asp) box polypeptide 20	SEQ ID NOS: 4086-4088
DDX25	DEAD (Asp-Glu-Ala-Asp) box helicase 25	SEQ ID NOS: 4089-4095
DDX28	DEAD (Asp-Glu-Ala-Asp) box polypeptide 28	SEQ ID NO: 4096
DEAF1	DEAF1 transcription factor	SEQ ID NOS: 4097-4099
DEF8	Differentially expressed in FDCP 8 homolog (mouse)	SEQ ID NOS: 4100-4119
DEFA1	Defensin, alpha 1	SEQ ID NOS: 4120-4121
DEFA1B	Defensin, alpha 1B	SEQ ID NO: 4122
DEFA3	Defensin, alpha 3, neutrophil-specific	SEQ ID NO: 4123
DEFA4	Defensin, alpha 4, corticostatin	SEQ ID NO: 4124
DEFA5	Defensin, alpha 5, Paneth cell-specific	SEQ ID NO: 4125
DEFA6	Defensin, alpha 6, Paneth cell-specific	SEQ ID NO: 4126
DEFB1	Defensin, beta 1	SEQ ID NO: 4127
DEFB103A	Defensin, beta 103A	SEQ ID NO: 4128
DEFB103B	Defensin, beta 103B	SEQ ID NO: 4129
DEFB104A	Defensin, beta 104A	SEQ ID NO: 4130
DEFB104B	Defensin, beta 104B	SEQ ID NO: 4131
DEFB105A	Defensin, beta 105A	SEQ ID NO: 4132
DEFB105B	Defensin, beta 105B	SEQ ID NO: 4133
DEFB106A	Defensin, beta 106A	SEQ ID NO: 4134
DEFB106B	Defensin, beta 106B	SEQ ID NO: 4135
DEFB107A	Defensin, beta 107A	SEQ ID NO: 4136
DEFB107B	Defensin, beta 107B	SEQ ID NO: 4137
DEFB108B	Defensin, beta 108B	SEQ ID NO: 4138
DEFB110	Defensin, beta 110	SEQ ID NOS: 4139-4140
DEFB113	Defensin, beta 113	SEQ ID NO: 4141
DEFB114	Defensin, beta 114	SEQ ID NO: 4142
DEFB115	Defensin, beta 115	SEQ ID NO: 4143
DEFB116	Defensin, beta 116	SEQ ID NO: 4144
DEFB118	Defensin, beta 118	SEQ ID NO: 4145
DEFB119	Defensin, beta 119	SEQ ID NOS: 4146-4148
DEFB121	Defensin, beta 121	SEQ ID NO: 4149

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
DEFB123	Defensin, beta 123	SEQ ID NO: 4150
DEFB124	Defensin, beta 124	SEQ ID NO: 4151
DEFB125	Defensin, beta 125	SEQ ID NO: 4152
DEFB126	Defensin, beta 126	SEQ ID NO: 4153
DEFB127	Defensin, beta 127	SEQ ID NO: 4154
DEFB128	Defensin, beta 128	SEQ ID NO: 4155
DEFB129	Defensin, beta 129	SEQ ID NO: 4156
DEFB130	Defensin, beta 130	SEQ ID NO: 4157
DEFB131	Defensin, beta 131	SEQ ID NO: 4159
DEFB132	Defensin, beta 132	SEQ ID NO: 4161
DEFB133	Defensin, beta 133	SEQ ID NO: 4162
DEFB134	Defensin, beta 134	SEQ ID NOS: 4163-4164
DEFB135	Defensin, beta 135	SEQ ID NO: 4165
DEFB136	Defensin, beta 136	SEQ ID NO: 4166
DEFB4A	Defensin, beta 4A	SEQ ID NO: 4167
DEFB4B	Defensin, beta 4B	SEQ ID NO: 4168
DFNA5	Deafness, autosomal dominant 5	SEQ ID NOS: 6271-6279
DFNB31	Deafness, autosomal recessive 31	SEQ ID NOS: 14251-14254
DGCR2	DiGeorge syndrome critical region gene 2	SEQ ID NOS: 4171-4174
DHH	Desert hedgehog	SEQ ID NO: 4175
DHRS4	Dehydrogenase/reductase (SDR family) member 4	SEQ ID NOS: 4176-4183
DHRS4L2	Dehydrogenase/reductase (SDR family) member 4 like 2	SEQ ID NOS: 4184-4193
DHRS7	Dehydrogenase/reductase (SDR family) member 7	SEQ ID NOS: 4194-4201
DHRS7C	Dehydrogenase/reductase (SDR family) member 7C	SEQ ID NOS: 4202-4204
DHRS9	Dehydrogenase/reductase (SDR family) member 9	SEQ ID NOS: 4205-4212
DHRSX	Dehydrogenase/reductase (SDR family) X-linked	SEQ ID NOS: 4213-4217
DHX29	DEAH (Asp-Glu-Ala-His) box polypeptide 29	SEQ ID NOS: 4218-4220
DHX30	DEAH (Asp-Glu-Ala-His) box helicase 30	SEQ ID NOS: 4221-4228
DHX8	DEAH (Asp-Glu-Ala-His) box polypeptide 8	SEQ ID NOS: 4229-4233
DIO2	Deiodinase, iodothyronine, type II	SEQ ID NOS: 4234-4243
DIXDC1	DIX domain containing 1	SEQ ID NOS: 4244-4247
DKK1	Dickkopf WNT signaling pathway inhibitor 1	SEQ ID NO: 4248
DKK2	Dickkopf WNT signaling pathway inhibitor 2	SEQ ID NOS: 4249-4251
DKK3	Dickkopf WNT signaling pathway inhibitor 3	SEQ ID NOS: 4252-4257
DKK4	Dickkopf WNT signaling pathway inhibitor 4	SEQ ID NO: 4258
DKKL1	Dickkopf-like 1	SEQ ID NOS: 4259-4264
DLG4	Discs, large homolog 4 (<i>Drosophila</i>)	SEQ ID NOS: 4265-4273
DLK1	Delta-like 1 homolog (<i>Drosophila</i>)	SEQ ID NOS: 4274-4277
DLL1	Delta-like 1 (<i>Drosophila</i>)	SEQ ID NOS: 4278-4279
DLL3	Delta-like 3 (<i>Drosophila</i>)	SEQ ID NOS: 4280-4282
DMBT1	Deleted in malignant brain tumors 1	SEQ ID NOS: 4283-4289
DMKN	Dermokine	SEQ ID NOS: 4290-4336
DMP1	Dentin matrix acidic phosphoprotein 1	SEQ ID NOS: 4337-4338
DMRTA2	DMRT-like family A2	SEQ ID NOS: 4339-4340
DNAAF5	Dynein, axonemal, assembly factor 5	SEQ ID NOS: 4341-4344
DNAH14	Dynein, axonemal, heavy chain 14	SEQ ID NOS: 4345-4359

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
DNAJB11	DnaJ (Hsp40) homolog, subfamily B, member 11	SEQ ID NOS: 4360-4361
DNAJB9	DnaJ (Hsp40) homolog, subfamily B, member 9	SEQ ID NO: 4362
DNAJC25-GNG10	DNAJC25-GNG10 readthrough	SEQ ID NO: 4363
DNAJC3	DnaJ (Hsp40) homolog, subfamily C, member 3	SEQ ID NOS: 4364-4365
DNASE1	Deoxyribonuclease I	SEQ ID NOS: 4366-4376
DNASE1L1	Deoxyribonuclease I-like 1	SEQ ID NOS: 4377-4387
DNASE1L2	Deoxyribonuclease I-like 2	SEQ ID NOS: 4388-4393
DNASE1L3	Deoxyribonuclease I-like 3	SEQ ID NOS: 4394-4399
DNASE2	Deoxyribonuclease II, lysosomal	SEQ ID NOS: 4400-4401
DNASE2B	Deoxyribonuclease II beta	SEQ ID NOS: 4402-4403
DPEP1	Dipeptidase 1 (renal)	SEQ ID NOS: 4404-4408
DPEP2	Dipeptidase 2	SEQ ID NOS: 4409-4415
DPEP3	Dipeptidase 3	SEQ ID NO: 4416
DPF3	D4, zinc and double PHD fingers, family 3	SEQ ID NOS: 4417-4423
DPP4	Dipeptidyl-peptidase 4	SEQ ID NOS: 4424-4428
DPP7	Dipeptidyl-peptidase 7	SEQ ID NOS: 4429-4434
DPT	Dermatopontin	SEQ ID NO: 4435
DRAXIN	Dorsal inhibitory axon guidance protein	SEQ ID NO: 4436
DSE	Dermatan sulfate epimerase	SEQ ID NOS: 4437-4445
DSG2	Desmoglein 2	SEQ ID NOS: 4446-4447
DSPP	Dentin sialophosphoprotein	SEQ ID NOS: 4448-4449
DST	Dystonin	SEQ ID NOS: 4450-4468
DUOX1	Dual oxidase 1	SEQ ID NOS: 4469-4473
DYNLT3	Dynein, light chain, Tetex-type 3	SEQ ID NOS: 4474-4476
E2F5	E2F transcription factor 5, p130-binding	SEQ ID NOS: 4477-4483
EBAG9	Estrogen receptor binding site associated, antigen, 9	SEQ ID NOS: 4484-4492
EBI3	Epstein-Barr virus induced 3	SEQ ID NO: 4493
ECHDC1	Ethylmalonyl-CoA decarboxylase 1	SEQ ID NOS: 4494-4512
ECM1	Extracellular matrix protein 1	SEQ ID NOS: 4513-4515
ECM2	Extracellular matrix protein 2, female organ and adipocyte specific	SEQ ID NOS: 4516-4519
ECSIT	ECSIT signalling integrator	SEQ ID NOS: 4520-4531
EDDM3A	Epididymal protein 3A	SEQ ID NO: 4532
EDDM3B	Epididymal protein 3B	SEQ ID NO: 4533
EDEM2	ER degradation enhancer, mannosidase alpha-like 2	SEQ ID NOS: 4534-4535
EDEM3	ER degradation enhancer, mannosidase alpha-like 3	SEQ ID NOS: 4536-4538
EDIL3	EGF-like repeats and discoidin I-like domains 3	SEQ ID NOS: 4539-4540
EDN1	Endothelin 1	SEQ ID NO: 4541
EDN2	Endothelin 2	SEQ ID NO: 4542
EDN3	Endothelin 3	SEQ ID NOS: 4543-4548
EDNRB	Endothelin receptor type B	SEQ ID NOS: 4549-4557
EFEMP1	EGF containing fibulin-like extracellular matrix protein 1	SEQ ID NOS: 4558-4568
EFEMP2	EGF containing fibulin-like extracellular matrix protein 2	SEQ ID NOS: 4569-4580

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
EFNA1	Ephrin-A1	SEQ ID NOS: 4581-4582
EFNA2	Ephrin-A2	SEQ ID NO: 4583
EFNA4	Ephrin-A4	SEQ ID NOS: 4584-4586
EGFL6	EGF-like-domain, multiple 6	SEQ ID NOS: 4587-4588
EGFL7	EGF-like-domain, multiple 7	SEQ ID NOS: 4589-4593
EGFL8	EGF-like-domain, multiple 8	SEQ ID NOS: 4594-4596
EGFLAM	EGF-like, fibronectin type III and laminin G domains	SEQ ID NOS: 4597-4605
EGFR	Epidermal growth factor receptor	SEQ ID NOS: 4606-4613
EHBP1	EH domain binding protein 1	SEQ ID NOS: 4614-4625
EHF	Ets homologous factor	SEQ ID NOS: 4626-4635
EHMT1	Euchromatic histone-lysine N-methyltransferase 1	SEQ ID NOS: 4636-4661
EHMT2	Euchromatic histone-lysine N-methyltransferase 2	SEQ ID NOS: 4662-4666
EIF2AK1	Eukaryotic translation initiation factor 2-alpha kinase 1	SEQ ID NOS: 4667-4670
ELANE	Elastase, neutrophil expressed	SEQ ID NOS: 4671-4672
ELN	Elastin	SEQ ID NOS: 4673-4695
ELP2	Elongator acetyltransferase complex subunit 2	SEQ ID NOS: 4696-4708
ELSPBP1	Epididymal sperm binding protein 1	SEQ ID NOS: 4709-4714
EMC1	ER membrane protein complex subunit 1	SEQ ID NOS: 4715-4721
EMC10	ER membrane protein complex subunit 10	SEQ ID NOS: 4722-4728
EMC9	ER membrane protein complex subunit 9	SEQ ID NOS: 4729-4732
EMCN	Endomucin	SEQ ID NOS: 4733-4737
EMID1	EMI domain containing 1	SEQ ID NOS: 4738-4744
EMILIN1	Elastin microfibril interfacer 1	SEQ ID NOS: 4745-4746
EMILIN2	Elastin microfibril interfacer 2	SEQ ID NO: 4747
EMILIN3	Elastin microfibril interfacer 3	SEQ ID NO: 4748
ENAM	Enamelin	SEQ ID NO: 4749
ENDOG	Endonuclease G	SEQ ID NO: 4750
ENDOU	Endonuclease, polyU-specific	SEQ ID NOS: 4751-4753
ENHO	Energy homeostasis associated	SEQ ID NO: 4754
ENO4	Enolase family member 4	SEQ ID NOS: 4755-4759
ENPP6	Ectonucleotide pyrophosphatase/phosphodiesterase 6	SEQ ID NOS: 4760-4761
ENPP7	Ectonucleotide pyrophosphatase/phosphodiesterase 7	SEQ ID NOS: 4762-4763
ENTPD5	Ectonucleoside triphosphate diphosphohydrolase 5	SEQ ID NOS: 4764-4768
ENTPD8	Ectonucleoside triphosphate diphosphohydrolase 8	SEQ ID NOS: 4769-4772
EOGT	EGF domain-specific O-linked N-acetylglucosamine (GlcNAc) transferase	SEQ ID NOS: 4773-4780
EPCAM	Epithelial cell adhesion molecule	SEQ ID NOS: 4781-4784
EPDR1	Ependymin related 1	SEQ ID NOS: 4785-4788
EPGN	Epithelial mitogen	SEQ ID NOS: 4789-4797
EPHA10	EPH receptor A10	SEQ ID NOS: 4798-4805
EPHA3	EPH receptor A3	SEQ ID NOS: 4806-4808
EPHA4	EPH receptor A4	SEQ ID NOS: 4809-4818

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
EPHA7	EPH receptor A7	SEQ ID NOS: 4819-4820
EPHA8	EPH receptor A8	SEQ ID NOS: 4821-4822
EPHB2	EPH receptor B2	SEQ ID NOS: 4823-4827
EPHB4	EPH receptor B4	SEQ ID NOS: 4828-4830
EPHX3	Epoxide hydrolase 3	SEQ ID NOS: 4831-4834
EPO	Erythropoietin	SEQ ID NO: 4835
EPPIN	Epididymal peptidase inhibitor	SEQ ID NOS: 4836-4838
EPPIN-WFDC6	EPPIN-WFDC6 readthrough	SEQ ID NO: 4839
EPS15	Epidermal growth factor receptor pathway substrate 15	SEQ ID NOS: 4840-4842
EPS8L1	EPS8-like 1	SEQ ID NOS: 4843-4848
EPX	Eosinophil peroxidase	SEQ ID NO: 4849
EPYC	Epiphycan	SEQ ID NOS: 4850-4851
EQTN	Equatorin, sperm acrosome associated	SEQ ID NOS: 4852-4854
ERAP1	Endoplasmic reticulum aminopeptidase 1	SEQ ID NOS: 4855-4859
ERAP2	Endoplasmic reticulum aminopeptidase 2	SEQ ID NOS: 4860-4867
ERBB3	Erb-b2 receptor tyrosine kinase 3	SEQ ID NOS: 4868-4881
ERLIN1	ER lipid raft associated 1	SEQ ID NOS: 4885-4887
ERLIN2	ER lipid raft associated 2	SEQ ID NOS: 4888-4896
ERN1	Endoplasmic reticulum to nucleus signaling 1	SEQ ID NOS: 4897-4898
ERN2	Endoplasmic reticulum to nucleus signaling 2	SEQ ID NOS: 4899-4903
ERO1A	Endoplasmic reticulum oxidoreductase alpha	SEQ ID NOS: 4904-4910
ERO1B	Endoplasmic reticulum oxidoreductase beta	SEQ ID NOS: 4911-4913
ERP27	Endoplasmic reticulum protein 27	SEQ ID NOS: 4914-4915
ERP29	Endoplasmic reticulum protein 29	SEQ ID NOS: 4916-4919
ERP44	Endoplasmic reticulum protein 44	SEQ ID NO: 4920
ERV3-1	Endogenous retrovirus group 3, member 1	SEQ ID NO: 4921
ESM1	Endothelial cell-specific molecule 1	SEQ ID NOS: 4922-4924
ESRP1	Epithelial splicing regulatory protein 1	SEQ ID NOS: 4925-4933
EXOG	Endo/exonuclease (5'-3'), endonuclease G-like	SEQ ID NOS: 4934-4947
EXTL1	Exostosin-like glycosyltransferase 1	SEQ ID NO: 4948
EXTL2	Exostosin-like glycosyltransferase 2	SEQ ID NOS: 4949-4953
F10	Coagulation factor X	SEQ ID NOS: 4954-4957
F11	Coagulation factor XI	SEQ ID NOS: 4958-4962
F12	Coagulation factor XII (Hageman factor)	SEQ ID NO: 4963
F13B	Coagulation factor XIII, B polypeptide	SEQ ID NO: 4964
F2	Coagulation factor II (thrombin)	SEQ ID NOS: 4965-4967
F2R	Coagulation factor II (thrombin) receptor	SEQ ID NOS: 4968-4969
F2RL3	Coagulation factor II (thrombin) receptor-like 3	SEQ ID NOS: 4970-4971
F5	Coagulation factor V (proaccelerin, labile factor)	SEQ ID NOS: 4972-4973
F7	Coagulation factor VII (serum prothrombin conversion accelerator)	SEQ ID NOS: 4974-4977
F8	Coagulation factor VIII, procoagulant component	SEQ ID NOS: 4978-4983

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
F9	Coagulation factor IX	SEQ ID NOS: 4984-4985
FABP6	Fatty acid binding protein 6, ileal	SEQ ID NOS: 4986-4988
FAM107B	Family with sequence similarity 107, member B	SEQ ID NOS: 4989-5010
FAM131A	Family with sequence similarity 131, member A	SEQ ID NOS: 5011-5019
FAM132A	Family with sequence similarity 132, member A	SEQ ID NO: 1795
FAM132B	Family with sequence similarity 132, member B	SEQ ID NOS: 4882-4884
FAM150A	Family with sequence similarity 150, member A	SEQ ID NOS: 737-738
FAM150B	Family with sequence similarity 150, member B	SEQ ID NOS: 739-745
FAM171A1	Family with sequence similarity 171, member A1	SEQ ID NOS: 5020-5021
FAM171B	Family with sequence similarity 171, member B	SEQ ID NOS: 5022-5023
FAM172A	Family with sequence similarity 172, member A	SEQ ID NOS: 5024-5028
FAM175A	Family with sequence similarity 175, member A	SEQ ID NOS: 64-71
FAM177A1	Family with sequence similarity 177, member A1	SEQ ID NOS: 5029-5038
FAM179B	Family with sequence similarity 179, member B	SEQ ID NOS: 13628-13633
FAM180A	Family with sequence similarity 180, member A	SEQ ID NOS: 5039-5041
FAM189A1	Family with sequence similarity 189, member A1	SEQ ID NOS: 5042-5043
FAM198A	Family with sequence similarity 198, member A	SEQ ID NOS: 5044-5046
FAM19A1	Family with sequence similarity 19 (chemokine (C-C motif)-like), member A1	SEQ ID NOS: 5047-5049
FAM19A2	Family with sequence similarity 19 (chemokine (C-C motif)-like), member A2	SEQ ID NOS: 5050-5057
FAM19A3	Family with sequence similarity 19 (chemokine (C-C motif)-like), member A3	SEQ ID NOS: 5058-5059
FAM19A4	Family with sequence similarity 19 (chemokine (C-C motif)-like), member A4	SEQ ID NOS: 5060-5062
FAM19A5	Family with sequence similarity 19 (chemokine (C-C motif)-like), member A5	SEQ ID NOS: 5063-5066
FAM20A	Family with sequence similarity 20, member A	SEQ ID NOS: 5067-5070
FAM20C	Family with sequence similarity 20, member C	SEQ ID NO: 5071
FAM213A	Family with sequence similarity 213, member A	SEQ ID NOS: 5072-5077
FAM26D	Family with sequence similarity 26, member D	SEQ ID NOS: 2006-2010
FAM46B	Family with sequence similarity 46, member B	SEQ ID NO: 5078
FAM57A	Family with sequence similarity 57, member A	SEQ ID NOS: 5079-5084
FAM78A	Family with sequence similarity 78, member A	SEQ ID NOS: 5085-5087
FAM96A	Family with sequence similarity 96, member A	SEQ ID NOS: 5088-5092
FAM9B	Family with sequence similarity 9, member B	SEQ ID NOS: 5093-5096
FAP	Fibroblast activation protein, alpha	SEQ ID NOS: 5097-5103
FAS	Fas cell surface death receptor	SEQ ID NOS: 5104-5113
FAT1	FAT atypical cadherin 1	SEQ ID NOS: 5114-5120
FBLN1	Fibulin 1	SEQ ID NOS: 5121-5133
FBLN2	Fibulin 2	SEQ ID NOS: 5134-5139
FBLN5	Fibulin 5	SEQ ID NOS: 5140-5145
FBLN7	Fibulin 7	SEQ ID NOS: 5146-5151

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
FBN1	Fibrillin 1	SEQ ID NOS: 5152-5155
FBN2	Fibrillin 2	SEQ ID NOS: 5156-5161
FBN3	Fibrillin 3	SEQ ID NOS: 5162-5166
FBXW7	F-box and WD repeat domain containing 7, E3 ubiquitin protein ligase	SEQ ID NOS: 5167-5177
FCAR	Fc fragment of IgA receptor	SEQ ID NOS: 5178-5187
FCGBP	Fc fragment of IgG binding protein	SEQ ID NOS: 5188-5190
FCGR1B	Fc fragment of IgG, high affinity Ib, receptor (CD64)	SEQ ID NOS: 5191-5196
FCGR3A	Fc fragment of IgG, low affinity IIIa, receptor (CD16a)	SEQ ID NOS: 5197-5203
FCGRT	Fc fragment of IgG, receptor, transporter, alpha	SEQ ID NOS: 5204-5214
FCMR	Fc fragment of IgM receptor	SEQ ID NOS: 5215-5221
FCN1	Ficolin (collagen/fibrinogen domain containing) 1	SEQ ID NOS: 5222-5223
FCN2	Ficolin (collagen/fibrinogen domain containing lectin) 2	SEQ ID NOS: 5224-5225
FCN3	Ficolin (collagen/fibrinogen domain containing) 3	SEQ ID NOS: 5226-5227
FCRL1	Fc receptor-like 1	SEQ ID NOS: 5228-5230
FCRL3	Fc receptor-like 3	SEQ ID NOS: 5231-5236
FCRL5	Fc receptor-like 5	SEQ ID NOS: 5237-5239
FCRLA	Fc receptor-like A	SEQ ID NOS: 5240-5251
FCRLB	Fc receptor-like B	SEQ ID NOS: 5252-5256
FDCSP	Follicular dendritic cell secreted protein	SEQ ID NO: 5257
FETUB	Fetuin B	SEQ ID NOS: 5258-5264
FGA	Fibrinogen alpha chain	SEQ ID NOS: 5265-5267
FGB	Fibrinogen beta chain	SEQ ID NOS: 5268-5270
FGF10	Fibroblast growth factor 10	SEQ ID NOS: 5271-5272
FGF17	Fibroblast growth factor 17	SEQ ID NOS: 5273-5274
FGF18	Fibroblast growth factor 18	SEQ ID NO: 5275
FGF19	Fibroblast growth factor 19	SEQ ID NO: 5276
FGF21	Fibroblast growth factor 21	SEQ ID NOS: 5277-5278
FGF22	Fibroblast growth factor 22	SEQ ID NOS: 5279-5280
FGF23	Fibroblast growth factor 23	SEQ ID NO: 5281
FGF3	Fibroblast growth factor 3	SEQ ID NO: 5282
FGF4	Fibroblast growth factor 4	SEQ ID NO: 5283
FGF5	Fibroblast growth factor 5	SEQ ID NOS: 5284-5286
FGF7	Fibroblast growth factor 7	SEQ ID NOS: 5287-5291
FGF8	Fibroblast growth factor 8 (androgen-induced)	SEQ ID NOS: 5292-5297
FGFBP1	Fibroblast growth factor binding protein 1	SEQ ID NO: 5298
FGFBP2	Fibroblast growth factor binding protein 2	SEQ ID NO: 5299
FGFBP3	Fibroblast growth factor binding protein 3	SEQ ID NO: 5300
FGFR1	Fibroblast growth factor receptor 1	SEQ ID NOS: 5301-5322
FGFR2	Fibroblast growth factor receptor 2	SEQ ID NOS: 5323-5344
FGFR3	Fibroblast growth factor receptor 3	SEQ ID NOS: 5345-5352
FGFR4	Fibroblast growth factor receptor 4	SEQ ID NOS: 5353-5362
FGFRL1	Fibroblast growth factor receptor-like 1	SEQ ID NOS: 5363-5368

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
FGG	Fibrinogen gamma chain	SEQ ID NOS: 5369-5374
FGL1	Fibrinogen-like 1	SEQ ID NOS: 5375-5381
FGL2	Fibrinogen-like 2	SEQ ID NOS: 5382-5383
FHL1	Four and a half LIM domains 1	SEQ ID NOS: 5384-5411
FHOD3	Formin homology 2 domain containing 3	SEQ ID NOS: 5412-5418
FIBIN	Fin bud initiation factor homolog (zebrafish)	SEQ ID NO: 5419
FICD	FIC domain containing	SEQ ID NOS: 5420-5423
FIGF	C-fos induced growth factor (vascular endothelial growth factor D)	SEQ ID NO: 14054
FJX1	Four jointed box 1	SEQ ID NO: 5424
FKBP10	FK506 binding protein 10, 65 kDa	SEQ ID NOS: 5425-5430
FKBP11	FK506 binding protein 11, 19 kDa	SEQ ID NOS: 5431-5437
FKBP14	FK506 binding protein 14, 22 kDa	SEQ ID NOS: 5438-5440
FKBP2	FK506 binding protein 2, 13 kDa	SEQ ID NOS: 5441-5444
FKBP7	FK506 binding protein 7	SEQ ID NOS: 5445-5450
FKBP9	FK506 binding protein 9, 63 kDa	SEQ ID NOS: 5451-5454
FLT1	Fms-related tyrosine kinase 1	SEQ ID NOS: 5455-5463
FLT4	Fms-related tyrosine kinase 4	SEQ ID NOS: 5464-5468
FMO1	Flavin containing monooxygenase 1	SEQ ID NOS: 5469-5473
FMO2	Flavin containing monooxygenase 2 (non-functional)	SEQ ID NOS: 5474-5476
FMO3	Flavin containing monooxygenase 3	SEQ ID NOS: 5477-5479
FMO5	Flavin containing monooxygenase 5	SEQ ID NOS: 5480-5486
FMOD	Fibromodulin	SEQ ID NO: 5487
FN1	Fibronectin 1	SEQ ID NOS: 5488-5500
FNDC1	Fibronectin type III domain containing 1	SEQ ID NOS: 5501-5502
FNDC7	Fibronectin type III domain containing 7	SEQ ID NOS: 5503-5504
FOCAD	Focadhesin	SEQ ID NOS: 5505-5511
FOLR2	Folate receptor 2 (fetal)	SEQ ID NOS: 5512-5521
FOLR3	Folate receptor 3 (gamma)	SEQ ID NOS: 5522-5526
FOXRED2	FAD-dependent oxidoreductase domain containing 2	SEQ ID NOS: 5527-5530
FP325331.1	Uncharacterized protein UNQ6126/PRO20091	SEQ ID NO: 5531
FPGS	Folypolyglutamate synthase	SEQ ID NOS: 5539-5545
FRAS1	Fraser extracellular matrix complex subunit 1	SEQ ID NOS: 5546-5551
FREM1	FRAS1 related extracellular matrix 1	SEQ ID NOS: 5552-5556
FREM3	FRAS1 related extracellular matrix 3	SEQ ID NO: 5557
FRMPD2	FERM and PDZ domain containing 2	SEQ ID NOS: 5558-5561
FRZB	Frizzled-related protein	SEQ ID NO: 5562
FSHB	Follicle stimulating hormone, beta polypeptide	SEQ ID NOS: 5563-5565
FSHR	Follicle stimulating hormone receptor	SEQ ID NOS: 5566-5569
FST	Follistatin	SEQ ID NOS: 5570-5573
FSTL1	Follistatin-like 1	SEQ ID NOS: 5574-5577

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
FSTL3	Follistatin-like 3 (secreted glycoprotein)	SEQ ID NOS: 5578-5583
FSTL4	Follistatin-like 4	SEQ ID NOS: 5584-5586
FSTL5	Follistatin-like 5	SEQ ID NOS: 5587-5589
FTCDNL1	Forniminotransferase cyclodeaminase N-terminal like	SEQ ID NOS: 5590-5593
FUCA1	Fucosidase, alpha-L- 1, tissue	SEQ ID NO: 5594
FUCA2	Fucosidase, alpha-L- 2, plasma	SEQ ID NOS: 5595-5596
FURIN	Furin (paired basic amino acid cleaving enzyme)	SEQ ID NOS: 5597-5603
FUT10	Fucosyltransferase 10 (alpha (1,3) fucosyltransferase)	SEQ ID NOS: 5604-5606
FUT11	Fucosyltransferase 11 (alpha (1,3) fucosyltransferase)	SEQ ID NOS: 5607-5608
FXN	Frataxin	SEQ ID NOS: 5609-5616
FXR1	Fragile X mental retardation, autosomal homolog 1	SEQ ID NOS: 5617-5629
FXD3	FXD domain containing ion transport regulator 3	SEQ ID NOS: 5630-5642
GABBR1	Gamma-aminobutyric acid (GABA) B receptor, 1	SEQ ID NOS: 5643-5654
GABRA1	Gamma-aminobutyric acid (GABA) A receptor, alpha 1	SEQ ID NOS: 5655-5670
GABRA2	Gamma-aminobutyric acid (GABA) A receptor, alpha 2	SEQ ID NOS: 5671-5685
GABRA5	Gamma-aminobutyric acid (GABA) A receptor, alpha 5	SEQ ID NOS: 5686-5694
GABRG3	Gamma-aminobutyric acid (GABA) A receptor, gamma 3	SEQ ID NOS: 5695-5700
GABRP	Gamma-aminobutyric acid (GABA) A receptor, pi	SEQ ID NOS: 5701-5709
GAL	Galanin/GMAP prepropeptide	SEQ ID NO: 5710
GAL3ST1	Galactose-3-O-sulfotransferase 1	SEQ ID NOS: 5711-5732
GAL3ST2	Galactose-3-O-sulfotransferase 2	SEQ ID NO: 5733
GAL3ST3	Galactose-3-O-sulfotransferase 3	SEQ ID NOS: 5734-5735
GALC	Galactosylceramidase	SEQ ID NOS: 5736-5745
GALNS	Galactosamine (N-acetyl)-6-sulfatase	SEQ ID NOS: 5746-5751
GALNT10	Polypeptide N-acetylgalactosaminyltransferase 10	SEQ ID NOS: 5752-5755
GALNT12	Polypeptide N-acetylgalactosaminyltransferase 12	SEQ ID NOS: 5756-5757
GALNT15	Polypeptide N-acetylgalactosaminyltransferase 15	SEQ ID NOS: 5758-5761
GALNT2	Polypeptide N-acetylgalactosaminyltransferase 2	SEQ ID NO: 5762
GALNT6	Polypeptide N-acetylgalactosaminyltransferase 6	SEQ ID NOS: 5763-5774
GALNT8	Polypeptide N-acetylgalactosaminyltransferase 8	SEQ ID NOS: 5775-5778
GALNTL6	Polypeptide N-acetylgalactosaminyltransferase-like 6	SEQ ID NOS: 5779-5782
GALP	Galanin-like peptide	SEQ ID NOS: 5783-5785
GANAB	Glucosidase, alpha; neutral AB	SEQ ID NOS: 5786-5794
GARS	Glycyl-tRNA synthetase	SEQ ID NOS: 5795-5798
GAS1	Growth arrest-specific 1	SEQ ID NO: 5799
GAS6	Growth arrest-specific 6	SEQ ID NO: 5800
GAST	Gastrin	SEQ ID NO: 5801
GBA	Glucosidase, beta, acid	SEQ ID NOS: 5811-5814
GBGT1	Globoside alpha-1,3-N-acetylgalactosaminyltransferase 1	SEQ ID NOS: 5815-5823
GC	Group-specific component (vitamin D binding protein)	SEQ ID NOS: 5824-5828
GCG	Glucagon	SEQ ID NOS: 5829-5830

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
GCGR	Glucagon receptor	SEQ ID NOS: 5831-5833
GCNT7	Glucosaminyl (N-acetyl) transferase family member 7	SEQ ID NOS: 5834-5835
GCSH	Glycine cleavage system protein H (aminomethyl carrier)	SEQ ID NOS: 5836-5844
GDF1	Growth differentiation factor 1	SEQ ID NO: 5845
GDF10	Growth differentiation factor 10	SEQ ID NO: 5846
GDF11	Growth differentiation factor 11	SEQ ID NOS: 5847-5848
GDF15	Growth differentiation factor 15	SEQ ID NOS: 5849-5851
GDF2	Growth differentiation factor 2	SEQ ID NO: 5852
GDF3	Growth differentiation factor 3	SEQ ID NO: 5853
GDF5	Growth differentiation factor 5	SEQ ID NOS: 5854-5855
GDF6	Growth differentiation factor 6	SEQ ID NOS: 5856-5858
GDF7	Growth differentiation factor 7	SEQ ID NO: 5859
GDF9	Growth differentiation factor 9	SEQ ID NOS: 5860-5864
GDNF	Glial cell derived neurotrophic factor	SEQ ID NOS: 5865-5872
GFOD2	Glucose-fructose oxidoreductase domain containing 2	SEQ ID NOS: 5873-5878
GFPT2	Glutamine-fructose-6-phosphate transaminase 2	SEQ ID NOS: 5879-5881
GFRA2	GDNF family receptor alpha 2	SEQ ID NOS: 5882-5888
GFRA4	GDNF family receptor alpha 4	SEQ ID NOS: 5889-5891
GGA2	Golgi-associated, gamma adaptin ear containing, ARF binding protein 2	SEQ ID NOS: 5892-5900
GGH	Gamma-glutamyl hydrolase (conjugase, folylpolyglutamyldihydrolase)	SEQ ID NO: 5901
GGT1	Gamma-glutamyltransferase 1	SEQ ID NOS: 5902-5924
GGT5	Gamma-glutamyltransferase 5	SEQ ID NOS: 5925-5929
GH1	Growth hormone 1	SEQ ID NOS: 5930-5934
GH2	Growth hormone 2	SEQ ID NOS: 5935-5939
GHDC	GH3 domain containing	SEQ ID NOS: 5940-5947
GHRH	Growth hormone releasing hormone	SEQ ID NOS: 5948-5950
GHRHR	Growth hormone releasing hormone receptor	SEQ ID NOS: 5951-5956
GHRL	Ghrelin/obestatin prepropeptide	SEQ ID NOS: 5957-5967
GIF	Gastric intrinsic factor (vitamin B synthesis)	SEQ ID NOS: 5968-5969
GIP	Gastric inhibitory polypeptide	SEQ ID NO: 5970
GKN1	Gastrophilin 1	SEQ ID NO: 5971
GKN2	Gastrophilin 2	SEQ ID NOS: 5972-5973
GLA	Galactosidase, alpha	SEQ ID NOS: 5974-5975
GLB1	Galactosidase, beta 1	SEQ ID NOS: 5976-5984
GLB1L	Galactosidase, beta 1-like	SEQ ID NOS: 5985-5992
GLB1L2	Galactosidase, beta 1-like 2	SEQ ID NOS: 5993-5994
GLCE	Glucuronic acid epimerase	SEQ ID NOS: 5995-5996
GLG1	Golgi glycoprotein 1	SEQ ID NOS: 5997-6004
GLIPR1	GLI pathogenesis-related 1	SEQ ID NOS: 6005-6008
GLIPR1L1	GLI pathogenesis-related 1 like 1	SEQ ID NOS: 6009-6012
GLIS3	GLIS family zinc finger 3	SEQ ID NOS: 6013-6021

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
GLMP	Glycosylated lysosomal membrane protein	SEQ ID NOS: 6022-6030
GLRB	Glycine receptor, beta	SEQ ID NOS: 6031-6036
GLS	Glutaminase	SEQ ID NOS: 6037-6044
GLT6D1	Glycosyltransferase 6 domain containing 1	SEQ ID NOS: 6045-6046
GLTPD2	Glycolipid transfer protein domain containing 2	SEQ ID NO: 6047
GLUD1	Glutamate dehydrogenase 1	SEQ ID NO: 6048
GM2A	GM2 ganglioside activator	SEQ ID NOS: 6049-6051
GML	Glycosylphosphatidylinositol anchored molecule like	SEQ ID NOS: 6052-6053
GNAS	GNAS complex locus	SEQ ID NOS: 6054-6075
GNLY	Granulysin	SEQ ID NOS: 6076-6079
GNPTG	N-acetylglucosamine-1-phosphate transferase, gamma subunit	SEQ ID NOS: 6080-6084
GNRH1	Gonadotropin-releasing hormone 1 (luteinizing-releasing hormone)	SEQ ID NOS: 6085-6086
GNRH2	Gonadotropin-releasing hormone 2	SEQ ID NOS: 6087-6090
GNS	Glucosamine (N-acetyl)-6-sulfatase	SEQ ID NOS: 6091-6096
GOLM1	Golgi membrane protein 1	SEQ ID NOS: 6097-6101
GORAB	Golgin, RAB6-interacting	SEQ ID NOS: 6102-6104
GOT2	Glutamic-oxaloacetic transaminase 2, mitochondrial	SEQ ID NOS: 6105-6107
GP2	Glycoprotein 2 (zymogen granule membrane)	SEQ ID NOS: 6108-6116
GP6	Glycoprotein VI (platelet)	SEQ ID NOS: 6117-6120
GPC2	Glypican 2	SEQ ID NOS: 6121-6122
GPC5	Glypican 5	SEQ ID NOS: 6123-6125
GPC6	Glypican 6	SEQ ID NOS: 6126-6127
GPD2	Glycerol-3-phosphate dehydrogenase 2 (mitochondrial)	SEQ ID NOS: 6128-6136
GPED1	G protein-coupled estrogen receptor 1	SEQ ID NOS: 6137-6143
GPHA2	Glycoprotein hormone alpha 2	SEQ ID NOS: 6144-6146
GPHB5	Glycoprotein hormone beta 5	SEQ ID NOS: 6147-6148
GPIHBP1	Glycosylphosphatidylinositol anchored high density lipoprotein binding protein 1	SEQ ID NO: 6149
GPLD1	Glycosylphosphatidylinositol specific phospholipase D1	SEQ ID NO: 6150
GPNMB	Glycoprotein (transmembrane) nmb	SEQ ID NOS: 6151-6153
GPR162	G protein-coupled receptor 162	SEQ ID NOS: 6154-6157
GPX3	Glutathione peroxidase 3	SEQ ID NOS: 6158-6165
GPX4	Glutathione peroxidase 4	SEQ ID NOS: 6166-6176
GPX5	Glutathione peroxidase 5	SEQ ID NOS: 6177-6178
GPX6	Glutathione peroxidase 6	SEQ ID NOS: 6179-6181
GPX7	Glutathione peroxidase 7	SEQ ID NO: 6182
GREM1	Gremlin 1, DAN family BMP antagonist	SEQ ID NOS: 6183-6185
GREM2	Gremlin 2, DAN family BMP antagonist	SEQ ID NO: 6186
GRHL3	Grainyhead-like transcription factor 3	SEQ ID NOS: 6187-6192
GRIA2	Glutamate receptor, ionotropic, AMPA 2	SEQ ID NOS: 6193-6204

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
GRIA3	Glutamate receptor, ionotropic, AMPA 3	SEQ ID NOS: 6205-6210
GRIA4	Glutamate receptor, ionotropic, AMPA 4	SEQ ID NOS: 6211-6222
GRIK2	Glutamate receptor, ionotropic, kainate 2	SEQ ID NOS: 6223-6231
GRIN2B	Glutamate receptor, ionotropic, N-methyl-D-aspartate 2B	SEQ ID NOS: 6232-6235
GRM2	Glutamate receptor, metabotropic 2	SEQ ID NOS: 6236-6239
GRM3	Glutamate receptor, metabotropic 3	SEQ ID NOS: 6240-6244
GRM5	Glutamate receptor, metabotropic 5	SEQ ID NOS: 6245-6249
GRN	Granulin	SEQ ID NOS: 6250-6265
GRP	Gastrin-releasing peptide	SEQ ID NOS: 6266-6270
GSG1	Germ cell associated 1	SEQ ID NOS: 6280-6288
GSN	Gelsolin	SEQ ID NOS: 6289-6297
GTDC1	Glycosyltransferase-like domain containing 1	SEQ ID NOS: 6298-6311
GTPBP10	GTP-binding protein 10 (putative)	SEQ ID NOS: 6312-6320
GUCA2A	Guanylate cyclase activator 2A (guanylin)	SEQ ID NO: 6321
GUCA2B	Guanylate cyclase activator 2B (uroguanylin)	SEQ ID NO: 6322
GUSB	Glucuronidase, beta	SEQ ID NOS: 6323-6327
GVQW1	GVQW motif containing 1	SEQ ID NO: 6328
GXYLT1	Glucoside xylosyltransferase 1	SEQ ID NOS: 6329-6330
GXYLT2	Glucoside xylosyltransferase 2	SEQ ID NOS: 6331-6333
GYLTL1B	Glycosyltransferase-like 1B	SEQ ID NOS: 7702-7707
GYPB	Glycophorin B (MNS blood group)	SEQ ID NOS: 6334-6342
GZMA	Granzyme A (granzyme 1, cytotoxic T-lymphocyte-associated serine esterase 3)	SEQ ID NO: 6343
GZMB	Granzyme B (granzyme 2, cytotoxic T-lymphocyte-associated serine esterase 1)	SEQ ID NOS: 6344-6352
GZMH	Granzyme H (cathepsin G-like 2, protein h-CCPX)	SEQ ID NOS: 6353-6355
GZMK	Granzyme K (granzyme 3; tryptase II)	SEQ ID NO: 6356
GZMM	Granzyme M (lymphocyte met-ase 1)	SEQ ID NOS: 6357-6358
H6PD	Hexose-6-phosphate dehydrogenase (glucose 1-dehydrogenase)	SEQ ID NOS: 6359-6360
HABP2	Hyaluronan binding protein 2	SEQ ID NOS: 6361-6362
HADHB	Hydroxyacyl-CoA dehydrogenase/3-ketoacyl-CoA thiolase/enoyl-CoA hydratase (trifunctional protein), beta subunit	SEQ ID NOS: 6363-6369
HAMP	Hepcidin antimicrobial peptide	SEQ ID NOS: 6370-6371
HAPLN1	Hyaluronan and proteoglycan link protein 1	SEQ ID NOS: 6372-6378
HAPLN2	Hyaluronan and proteoglycan link protein 2	SEQ ID NOS: 6379-6380
HAPLN3	Hyaluronan and proteoglycan link protein 3	SEQ ID NOS: 6381-6384
HAPLN4	Hyaluronan and proteoglycan link protein 4	SEQ ID NO: 6385
HARS2	Histidyl-tRNA synthetase 2, mitochondrial	SEQ ID NOS: 6386-6401
HAVCR1	Hepatitis A virus cellular receptor 1	SEQ ID NOS: 6402-6406
HCCS	Holocytochrome c synthase	SEQ ID NOS: 6407-6409
HCRT	Hypocretin (orexin) neuropeptide precursor	SEQ ID NO: 6410
HEATR5A	HEAT repeat containing 5A	SEQ ID NOS: 6414-6420
HEPH	Hephaestin	SEQ ID NOS: 6421-6428

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
HEXA	Hexosaminidase A (alpha polypeptide)	SEQ ID NOS: 6429-6438
HEXB	Hexosaminidase B (beta polypeptide)	SEQ ID NOS: 6439-6444
HFE2	Hemochromatosis type 2 (juvenile)	SEQ ID NOS: 6445-6451
HGF	Hepatocyte growth factor (hepapoietin A; scatter factor)	SEQ ID NOS: 6452-6462
HGFAC	HGF activator	SEQ ID NOS: 6463-6464
HHIP	Hedgehog interacting protein	SEQ ID NOS: 6465-6466
HHIPL1	HHIP-like 1	SEQ ID NOS: 6467-6468
HHIPL2	HHIP-like 2	SEQ ID NO: 6469
HHLA1	HERV-H LTR-associating 1	SEQ ID NOS: 6470-6471
HHLA2	HERV-H LTR-associating 2	SEQ ID NOS: 6472-6482
HIBADH	3-hydroxyisobutyrate dehydrogenase	SEQ ID NOS: 6483-6485
HINT2	Histidine triad nucleotide binding protein 2	SEQ ID NO: 6486
HLA-A	Major histocompatibility complex, class I, A	SEQ ID NOS: 6487-6491
HLA-C	Major histocompatibility complex, class I, C	SEQ ID NOS: 6492-6496
HLA-DOA	Major histocompatibility complex, class II, DO alpha	SEQ ID NOS: 6497-6498
HLA-DPA1	Major histocompatibility complex, class II, DP alpha 1	SEQ ID NOS: 6499-6502
HLA-DQA1	Major histocompatibility complex, class II, DQ alpha 1	SEQ ID NOS: 6503-6508
HLA-DQB1	Major histocompatibility complex, class II, DQ beta 1	SEQ ID NOS: 6509-6514
HLA-DQB2	Major histocompatibility complex, class II, DQ beta 2	SEQ ID NOS: 6515-6518
HMCN1	Hemicentin 1	SEQ ID NOS: 6519-6520
HMCN2	Hemicentin 2	SEQ ID NOS: 6521-6524
HMGCL	3-hydroxymethyl-3-methylglutaryl-CoA lyase	SEQ ID NOS: 6525-6528
HMHA1	Histocompatibility (minor) HA-1	SEQ ID NOS: 1034-1042
HMSD	Histocompatibility (minor) serpin domain containing	SEQ ID NOS: 6529-6530
HP	Haptoglobin	SEQ ID NOS: 6531-6544
HPR	Haptoglobin-related protein	SEQ ID NOS: 6545-6547
HPSE	Heparanase	SEQ ID NOS: 6548-6554
HPSE2	Heparanase 2 (inactive)	SEQ ID NOS: 6555-6560
HPX	Hemopexin	SEQ ID NOS: 6561-6562
HRC	Histidine rich calcium binding protein	SEQ ID NOS: 6563-6565
HRG	Histidine-rich glycoprotein	SEQ ID NO: 6566
HRSP12	Heat-responsive protein 12	SEQ ID NOS: 11389-11392
HS2ST1	Heparan sulfate 2-O-sulfotransferase 1	SEQ ID NOS: 6567-6569
HS3ST1	Heparan sulfate (glucosamine) 3-O-sulfotransferase 1	SEQ ID NOS: 6570-6572
HS6ST1	Heparan sulfate 6-O-sulfotransferase 1	SEQ ID NO: 6573
HS6ST3	Heparan sulfate 6-O-sulfotransferase 3	SEQ ID NOS: 6574-6575
HSD11B1L	Hydroxysteroid (11-beta) dehydrogenase 1-like	SEQ ID NOS: 6576-6594
HSD17B11	Hydroxysteroid (17-beta) dehydrogenase 11	SEQ ID NOS: 6595-6596
HSD17B7	Hydroxysteroid (17-beta) dehydrogenase 7	SEQ ID NOS: 6597-6601
HSP90B1	Heat shock protein 90 kDa beta (Grp94), member 1	SEQ ID NOS: 6602-6607

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
HSPA13	Heat shock protein 70 kDa family, member 13	SEQ ID NO: 6608
HSPA5	Heat shock 70 kDa protein 5 (glucose-regulated protein, 78 kDa)	SEQ ID NO: 6609
HSPG2	Heparan sulfate proteoglycan 2	SEQ ID NOS: 6610-6614
HTATIP2	HIV-1 Tat interactive protein 2, 30 kDa	SEQ ID NOS: 6615-6622
HTN1	Histatin 1	SEQ ID NOS: 6623-6625
HTN3	Histatin 3	SEQ ID NOS: 6626-6628
HTRA1	HtrA serine peptidase 1	SEQ ID NOS: 6629-6630
HTRA3	HtrA serine peptidase 3	SEQ ID NOS: 6631-6632
HTRA4	HtrA serine peptidase 4	SEQ ID NO: 6633
HYAL1	Hyaluronoglucosaminidase 1	SEQ ID NOS: 6634-6642
HYAL2	Hyaluronoglucosaminidase 2	SEQ ID NOS: 6643-6651
HYAL3	Hyaluronoglucosaminidase 3	SEQ ID NOS: 6652-6658
HYOU1	Hypoxia up-regulated 1	SEQ ID NOS: 6659-6673
IAPP	Islet amyloid polypeptide	SEQ ID NOS: 6674-6678
IBSP	Integrin-binding sialoprotein	SEQ ID NO: 6679
ICAM1	Intercellular adhesion molecule 1	SEQ ID NOS: 6680-6682
ICAM2	Intercellular adhesion molecule 2	SEQ ID NOS: 6683-6693
ICAM4	Intercellular adhesion molecule 4 (Landsteiner-Wiener blood group)	SEQ ID NOS: 6694-6696
ID1	Inhibitor of DNA binding 1, dominant negative helix-loop-helix protein	SEQ ID NOS: 6697-6698
IDE	Insulin-degrading enzyme	SEQ ID NOS: 6699-6702
IDNK	IdnK, gluconokinase homolog (<i>E. coli</i>)	SEQ ID NOS: 6703-6708
IDS	Iduronate 2-sulfatase	SEQ ID NOS: 6709-6714
IDUA	Iduronidase, alpha-L-	SEQ ID NOS: 6715-6720
IFI27L2	Interferon, alpha-inducible protein 27-like 2	SEQ ID NOS: 6721-6722
IFI30	Interferon, gamma-inducible protein 30	SEQ ID NOS: 6723-6724
IFNA1	Interferon, alpha 1	SEQ ID NO: 6725
IFNA10	Interferon, alpha 10	SEQ ID NO: 6726
IFNA13	Interferon, alpha 13	SEQ ID NOS: 6727-6728
IFNA14	Interferon, alpha 14	SEQ ID NO: 6729
IFNA16	Interferon, alpha 16	SEQ ID NO: 6730
IFNA17	Interferon, alpha 17	SEQ ID NO: 6731
IFNA2	Interferon, alpha 2	SEQ ID NO: 6732
IFNA21	Interferon, alpha 21	SEQ ID NO: 6733
IFNA4	Interferon, alpha 4	SEQ ID NO: 6734
IFNA5	Interferon, alpha 5	SEQ ID NO: 6735
IFNA6	Interferon, alpha 6	SEQ ID NOS: 6736-6737
IFNA7	Interferon, alpha 7	SEQ ID NO: 6738
IFNA8	Interferon, alpha 8	SEQ ID NO: 6739
IFNAR1	Interferon (alpha, beta and omega) receptor 1	SEQ ID NOS: 6740-6741
IFNB1	Interferon, beta 1, fibroblast	SEQ ID NO: 6742
IFNE	Interferon, epsilon	SEQ ID NO: 6743
IFNG	Interferon, gamma	SEQ ID NO: 6744
IFNGR1	Interferon gamma receptor 1	SEQ ID NOS: 6745-6755
IFNL1	Interferon, lambda 1	SEQ ID NO: 6756
IFNL2	Interferon, lambda 2	SEQ ID NO: 6757
IFNL3	Interferon, lambda 3	SEQ ID NOS: 6758-6759
IFNLR1	Interferon, lambda receptor 1	SEQ ID NOS: 6760-6764

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
IFNW1	Interferon, omega 1	SEQ ID NO: 6765
IGF1	Insulin-like growth factor 1 (somatomedin C)	SEQ ID NOS: 6766-6771
IGF2	Insulin-like growth factor 2	SEQ ID NOS: 6772-6779
IGFALS	Insulin-like growth factor binding protein, acid labile subunit	SEQ ID NOS: 6780-6782
IGFBP1	Insulin-like growth factor binding protein 1	SEQ ID NOS: 6783-6785
IGFBP2	Insulin-like growth factor binding protein 2, 36 kDa	SEQ ID NOS: 6786-6789
IGFBP3	Insulin-like growth factor binding protein 3	SEQ ID NOS: 6790-6797
IGFBP4	Insulin-like growth factor binding protein 4	SEQ ID NO: 6798
IGFBP5	Insulin-like growth factor binding protein 5	SEQ ID NOS: 6799-6800
IGFBP6	Insulin-like growth factor binding protein 6	SEQ ID NOS: 6801-6803
IGFBP7	Insulin-like growth factor binding protein 7	SEQ ID NOS: 6804-6805
IGFBPL1	Insulin-like growth factor binding protein-like 1	SEQ ID NO: 6806
IGFL1	IGF-like family member 1	SEQ ID NO: 6807
IGFL2	IGF-like family member 2	SEQ ID NOS: 6808-6810
IGFL3	IGF-like family member 3	SEQ ID NO: 6811
IGFLR1	IGF-like family receptor 1	SEQ ID NOS: 6812-6820
IGIP	IgA-inducing protein	SEQ ID NO: 6821
IGLON5	IgLON family member 5	SEQ ID NO: 6822
IGSF1	Immunoglobulin superfamily, member 1	SEQ ID NOS: 6823-6828
IGSF10	Immunoglobulin superfamily, member 10	SEQ ID NOS: 6829-6830
IGSF11	Immunoglobulin superfamily, member 11	SEQ ID NOS: 6831-6838
IGSF21	Immunoglobulin superfamily, member 21	SEQ ID NO: 6839
IGSF8	Immunoglobulin superfamily, member 8	SEQ ID NOS: 6840-6843
IGSF9	Immunoglobulin superfamily, member 9	SEQ ID NOS: 6844-6846
IHH	Indian hedgehog	SEQ ID NO: 6847
IL10	Interleukin 10	SEQ ID NOS: 6848-6849
IL11	Interleukin 11	SEQ ID NOS: 6850-6853
IL11RA	Interleukin 11 receptor, alpha	SEQ ID NOS: 6854-6864
IL12B	Interleukin 12B	SEQ ID NO: 6865
IL12RB1	Interleukin 12 receptor, beta 1	SEQ ID NOS: 6866-6871
IL12RB2	Interleukin 12 receptor, beta 2	SEQ ID NOS: 6872-6876
IL13	Interleukin 13	SEQ ID NOS: 6877-6878
IL13RA1	Interleukin 13 receptor, alpha 1	SEQ ID NOS: 6879-6880
IL15RA	Interleukin 15 receptor, alpha	SEQ ID NOS: 6881-6898
IL17A	Interleukin 17A	SEQ ID NO: 6899
IL17B	Interleukin 17B	SEQ ID NO: 6900
IL17C	Interleukin 17C	SEQ ID NO: 6901
IL17D	Interleukin 17D	SEQ ID NOS: 6902-6904
IL17F	Interleukin 17F	SEQ ID NO: 6905
IL17RA	Interleukin 17 receptor A	SEQ ID NOS: 6906-6907
IL17RC	Interleukin 17 receptor C	SEQ ID NOS: 6908-6923
IL17RE	Interleukin 17 receptor E	SEQ ID NOS: 6924-6930
IL18BP	Interleukin 18 binding protein	SEQ ID NOS: 6931-6941
IL18R1	Interleukin 18 receptor 1	SEQ ID NOS: 6942-6945

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
IL18RAP	Interleukin 18 receptor accessory protein	SEQ ID NOS: 6946-6948
IL19	Interleukin 19	SEQ ID NOS: 6949-6951
IL1R1	Interleukin 1 receptor, type I	SEQ ID NOS: 6952-6964
IL1R2	Interleukin 1 receptor, type II	SEQ ID NOS: 6965-6968
IL1RAP	Interleukin 1 receptor accessory protein	SEQ ID NOS: 6969-6982
IL1RL1	Interleukin 1 receptor-like 1	SEQ ID NOS: 6983-6988
IL1RL2	Interleukin 1 receptor-like 2	SEQ ID NOS: 6989-6991
IL1RN	Interleukin 1 receptor antagonist	SEQ ID NOS: 6992-6996
IL2	Interleukin 2	SEQ ID NO: 6997
IL20	Interleukin 20	SEQ ID NOS: 6998-7000
IL20RA	Interleukin 20 receptor, alpha	SEQ ID NOS: 7001-7007
IL21	Interleukin 21	SEQ ID NOS: 7008-7009
IL22	Interleukin 22	SEQ ID NOS: 7010-7011
IL22RA2	Interleukin 22 receptor, alpha 2	SEQ ID NOS: 7012-7014
IL23A	Interleukin 23, alpha subunit p19	SEQ ID NO: 7015
IL24	Interleukin 24	SEQ ID NOS: 7016-7021
IL25	Interleukin 25	SEQ ID NOS: 7022-7023
IL26	Interleukin 26	SEQ ID NO: 7024
IL27	Interleukin 27	SEQ ID NOS: 7025-7026
IL2RB	Interleukin 2 receptor, beta	SEQ ID NOS: 7027-7031
IL3	Interleukin 3	SEQ ID NO: 7032
IL31	Interleukin 31	SEQ ID NO: 7033
IL31RA	Interleukin 31 receptor A	SEQ ID NOS: 7034-7041
IL32	Interleukin 32	SEQ ID NOS: 7042-7071
IL34	Interleukin 34	SEQ ID NOS: 7072-7075
IL3RA	Interleukin 3 receptor, alpha (low affinity)	SEQ ID NOS: 7076-7078
IL4	Interleukin 4	SEQ ID NOS: 7079-7081
IL4I1	Interleukin 4 induced 1	SEQ ID NOS: 7082-7089
IL4R	Interleukin 4 receptor	SEQ ID NOS: 7090-7103
IL5	Interleukin 5	SEQ ID NOS: 7104-7105
IL5RA	Interleukin 5 receptor, alpha	SEQ ID NOS: 7106-7115
IL6	Interleukin 6	SEQ ID NOS: 7116-7122
IL6R	Interleukin 6 receptor	SEQ ID NOS: 7123-7128
IL6ST	Interleukin 6 signal transducer	SEQ ID NOS: 7129-7138
IL7	Interleukin 7	SEQ ID NOS: 7139-7146
IL7R	Interleukin 7 receptor	SEQ ID NOS: 7147-7153
IL9	Interleukin 9	SEQ ID NO: 7154
ILDR1	Immunoglobulin-like domain containing receptor 1	SEQ ID NOS: 7155-7159
ILDR2	Immunoglobulin-like domain containing receptor 2	SEQ ID NOS: 7160-7166
IMP4	IMP4, U3 small nucleolar ribonucleoprotein	SEQ ID NOS: 7167-7172
IMPG1	Interphotoreceptor matrix proteoglycan 1	SEQ ID NOS: 7173-7176

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
INHA	Inhibin, alpha	SEQ ID NO: 7177
INHBA	Inhibin, beta A	SEQ ID NOS: 7178-7180
INHBB	Inhibin, beta B	SEQ ID NO: 7181
INHBC	Inhibin, beta C	SEQ ID NO: 7182
INHBE	Inhibin, beta E	SEQ ID NOS: 7183-7184
INPP5A	Inositol polyphosphate-5-phosphatase A	SEQ ID NOS: 7185-7189
INS	Insulin	SEQ ID NOS: 7190-7194
INS-IGF2	INS-IGF2 readthrough	SEQ ID NOS: 7195-7196
INSL3	Insulin-like 3 (Leydig cell)	SEQ ID NOS: 7197-7199
INSL4	Insulin-like 4 (placenta)	SEQ ID NO: 7200
INSL5	Insulin-like 5	SEQ ID NO: 7201
INSL6	Insulin-like 6	SEQ ID NO: 7202
INTS3	Integrator complex subunit 3	SEQ ID NOS: 7203-7208
IPO11	Importin 11	SEQ ID NOS: 7209-7217
IPO9	Importin 9	SEQ ID NOS: 7218-7219
IQCF6	IQ motif containing F6	SEQ ID NOS: 7220-7221
IRAK3	Interleukin-1 receptor-associated kinase 3	SEQ ID NOS: 7222-7224
IRS4	Insulin receptor substrate 4	SEQ ID NO: 7225
ISLR	Immunoglobulin superfamily containing leucine-rich repeat	SEQ ID NOS: 7226-7229
ISLR2	Immunoglobulin superfamily containing leucine-rich repeat 2	SEQ ID NOS: 7230-7239
ISM1	Isthmin 1, angiogenesis inhibitor	SEQ ID NO: 7240
ISM2	Isthmin 2	SEQ ID NOS: 7241-7246
ITGA4	Integrin, alpha 4 (antigen CD49D, alpha 4 subunit of VLA-4 receptor)	SEQ ID NOS: 7247-7249
ITGA9	Integrin, alpha 9	SEQ ID NOS: 7250-7252
ITGAL	Integrin, alpha L (antigen CD11A (p180), lymphocyte function-associated antigen 1; alpha polypeptide)	SEQ ID NOS: 7253-7262
ITGAX	Integrin, alpha X (complement component 3 receptor 4 subunit)	SEQ ID NOS: 7263-7265
ITGB1	Integrin, beta 1 (fibronectin receptor, beta polypeptide, antigen CD29 includes MDF2, MSK12)	SEQ ID NOS: 7266-7281
ITGB2	Integrin, beta 2 (complement component 3 receptor 3 and 4 subunit)	SEQ ID NOS: 7282-7298
ITGB3	Integrin, beta 3 (platelet glycoprotein IIIa, antigen CD61)	SEQ ID NOS: 7299-7301
ITGB7	Integrin, beta 7	SEQ ID NOS: 7302-7309
ITGBL1	Integrin, beta-like 1 (with EGF-like repeat domains)	SEQ ID NOS: 7310-7315
ITIH1	Inter-alpha-trypsin inhibitor heavy chain 1	SEQ ID NOS: 7316-7321
ITIH2	Inter-alpha-trypsin inhibitor heavy chain 2	SEQ ID NOS: 7322-7324
ITIH3	Inter-alpha-trypsin inhibitor heavy chain 3	SEQ ID NOS: 7325-7327
ITIH4	Inter-alpha-trypsin inhibitor heavy chain family, member 4	SEQ ID NOS: 7328-7331
ITIH5	Inter-alpha-trypsin inhibitor heavy chain family, member 5	SEQ ID NOS: 7332-7335
ITIH6	Inter-alpha-trypsin inhibitor heavy chain family, member 6	SEQ ID NO: 7336
ITLN1	Intelectin 1 (galactofuranose binding)	SEQ ID NO: 7337
ITLN2	Intelectin 2	SEQ ID NO: 7338
IZUMO1R	IZUMO1 receptor, JUNO	SEQ ID NOS: 7339-7340
IZUMO4	IZUMO family member 4	SEQ ID NOS: 7341-7347
JCHAIN	Joining chain of multimeric IgA and IgM	SEQ ID NOS: 7357-7362

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
JMJD8	Jumonji domain containing 8	SEQ ID NOS: 7363-7367
JSRP1	Junctional sarcoplasmic reticulum protein 1	SEQ ID NO: 7368
KANSL2	KAT8 regulatory NSL complex subunit 2	SEQ ID NOS: 7369-7379
KAZALD1	Kazal-type serine peptidase inhibitor domain 1	SEQ ID NO: 7380
KCNIP3	Kv channel interacting protein 3, calseinilin	SEQ ID NOS: 7381-7383
KCNK7	Potassium channel, two pore domain subfamily K, member 7	SEQ ID NOS: 7384-7389
KCNN4	Potassium channel, calcium activated intermediate/small conductance subfamily N alpha, member 4	SEQ ID NOS: 7390-7395
KCNU1	Potassium channel, subfamily U, member 1	SEQ ID NOS: 7396-7400
KCP	Kielin/chordin-like protein	SEQ ID NOS: 7401-7404
KDELC1	KDEL (Lys-Asp-Glu-Leu) containing 1	SEQ ID NO: 7405
KDELC2	KDEL (Lys-Asp-Glu-Leu) containing 2	SEQ ID NOS: 7406-7409
KDM1A	Lysine (K)-specific demethylase 1A	SEQ ID NOS: 7410-7413
KDM3B	Lysine (K)-specific demethylase 3B	SEQ ID NOS: 7414-7417
KDM6A	Lysine (K)-specific demethylase 6A	SEQ ID NOS: 7418-7427
KDM7A	Lysine (K)-specific demethylase 7A	SEQ ID NOS: 7428-7429
KDSR	3-ketodihydrosphingosine reductase	SEQ ID NOS: 7430-7436
KERA	Keratocan	SEQ ID NO: 7437
KIAA0100	KIAA0100	SEQ ID NOS: 7438-7443
KIAA0319	KIAA0319	SEQ ID NOS: 7444-7449
KIAA1324	KIAA1324	SEQ ID NOS: 7450-7458
KIFC2	Kinesin family member C2	SEQ ID NOS: 7459-7461
KIR2DL4	Killer cell immunoglobulin-like receptor, two domains, long cytoplasmic tail, 4	SEQ ID NOS: 7462-7468
KIR3DX1	Killer cell immunoglobulin-like receptor, three domains, X1	SEQ ID NOS: 7469-7473
KIRREL2	Kin of IRRE like 2 (<i>Drosophila</i>)	SEQ ID NOS: 7474-7478
KISS1	KiSS-1 metastasis-suppressor	SEQ ID NOS: 7479-7480
KLHL11	Kelch-like family member 11	SEQ ID NO: 7481
KLHL22	Kelch-like family member 22	SEQ ID NOS: 7482-7488
KLK1	Kallikrein 1	SEQ ID NOS: 7489-7490
KLK10	Kallikrein-related peptidase 10	SEQ ID NOS: 7491-7495
KLK11	Kallikrein-related peptidase 11	SEQ ID NOS: 7496-7504
KLK12	Kallikrein-related peptidase 12	SEQ ID NOS: 7505-7511
KLK13	Kallikrein-related peptidase 13	SEQ ID NOS: 7512-7520
KLK14	Kallikrein-related peptidase 14	SEQ ID NOS: 7521-7522
KLK15	Kallikrein-related peptidase 15	SEQ ID NOS: 7523-7527
KLK2	Kallikrein-related peptidase 2	SEQ ID NOS: 7528-7540
KLK3	Kallikrein-related peptidase 3	SEQ ID NOS: 7541-7552
KLK4	Kallikrein-related peptidase 4	SEQ ID NOS: 7553-7557
KLK5	Kallikrein-related peptidase 5	SEQ ID NOS: 7558-7561
KLK6	Kallikrein-related peptidase 6	SEQ ID NOS: 7562-7568

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
KLK7	Kallikrein-related peptidase 7	SEQ ID NOS: 7569-7573
KLK8	Kallikrein-related peptidase 8	SEQ ID NOS: 7574-7581
KLK9	Kallikrein-related peptidase 9	SEQ ID NOS: 7582-7583
KLKB1	Kallikrein B, plasma (Fletcher factor) 1	SEQ ID NOS: 7584-7588
KNDC1	Kinase non-catalytic C-lobe domain (KIND) containing 1	SEQ ID NOS: 7593-7594
KN1G	Kininogen 1	SEQ ID NOS: 7595-7599
KRBA2	KRAB-A domain containing 2	SEQ ID NOS: 7600-7603
KREMEN2	Kringle containing transmembrane protein 2	SEQ ID NOS: 7604-7609
KRTDAP	Keratinocyte differentiation-associated protein	SEQ ID NOS: 7610-7611
L1CAM	L1 cell adhesion molecule	SEQ ID NOS: 7612-7621
L3MBTL2	L(3)mbt-like 2 (<i>Drosophila</i>)	SEQ ID NOS: 7622-7626
LA16c-380H5.3		SEQ ID NO: 72
LACE1	Lactation elevated 1	SEQ ID NOS: 580-583
LACRT	Lacritin	SEQ ID NOS: 7627-7629
LACTB	Lactamase, beta	SEQ ID NOS: 7630-7632
LAG3	Lymphocyte-activation gene 3	SEQ ID NOS: 7633-7634
LAIR2	Leukocyte-associated immunoglobulin-like receptor 2	SEQ ID NOS: 7635-7638
LALBA	Lactalbumin, alpha-	SEQ ID NOS: 7639-7640
LAMA1	Laminin, alpha 1	SEQ ID NOS: 7641-7642
LAMA2	Laminin, alpha 2	SEQ ID NOS: 7643-7646
LAMA3	Laminin, alpha 3	SEQ ID NOS: 7647-7656
LAMA4	Laminin, alpha 4	SEQ ID NOS: 7657-7671
LAMA5	Laminin, alpha 5	SEQ ID NOS: 7672-7674
LAMB1	Laminin, beta 1	SEQ ID NOS: 7675-7679
LAMB2	Laminin, beta 2 (laminin S)	SEQ ID NOS: 7680-7682
LAMB3	Laminin, beta 3	SEQ ID NOS: 7683-7687
LAMB4	Laminin, beta 4	SEQ ID NOS: 7688-7691
LAMC1	Laminin, gamma 1 (formerly LAMB2)	SEQ ID NOS: 7692-7693
LAMC2	Laminin, gamma 2	SEQ ID NOS: 7694-7695
LAMC3	Laminin, gamma 3	SEQ ID NOS: 7696-7697
LAMP3	Lysosomal-associated membrane protein 3	SEQ ID NOS: 7698-7701
LAT	Linker for activation of T cells	SEQ ID NOS: 7708-7717
LAT2	Linker for activation of T cells family, member 2	SEQ ID NOS: 7718-7726
LBP	Lipopolysaccharide binding protein	SEQ ID NO: 7727
LCAT	Lecithin-cholesterol acyltransferase	SEQ ID NOS: 7728-7734
LCN1	Lipocalin 1	SEQ ID NOS: 7735-7736
LCN10	Lipocalin 10	SEQ ID NOS: 7737-7742
LCN12	Lipocalin 12	SEQ ID NOS: 7743-7745
LCN15	Lipocalin 15	SEQ ID NO: 7746

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
LCN2	Lipocalin 2	SEQ ID NOS: 7747-7749
LCN6	Lipocalin 6	SEQ ID NOS: 7750-7751
LCN8	Lipocalin 8	SEQ ID NOS: 7752-7753
LCN9	Lipocalin 9	SEQ ID NOS: 7754-7755
LCORL	Ligand dependent nuclear receptor corepressor-like	SEQ ID NOS: 7756-7761
LDLR	Low density lipoprotein receptor	SEQ ID NOS: 7762-7770
LDLRAD2	Low density lipoprotein receptor class A domain containing 2	SEQ ID NOS: 7771-7772
LEAP2	Liver expressed antimicrobial peptide 2	SEQ ID NO: 7773
LECT2	Leukocyte cell-derived chemotaxin 2	SEQ ID NOS: 7774-7777
LEFTY1	Left-right determination factor 1	SEQ ID NOS: 7778-7779
LEFTY2	Left-right determination factor 2	SEQ ID NOS: 7780-7781
LEP	Leptin	SEQ ID NO: 7782
LFNG	LFNG O-fucosylpeptide 3-beta-N-acetylglucosaminyltransferase	SEQ ID NOS: 7783-7788
LGALS3BP	Lectin, galactoside-binding, soluble, 3 binding protein	SEQ ID NOS: 7789-7803
LGI1	Leucine-rich, glioma inactivated 1	SEQ ID NOS: 7804-7822
LGI2	Leucine-rich repeat LGI family, member 2	SEQ ID NOS: 7823-7824
LGI3	Leucine-rich repeat LGI family, member 3	SEQ ID NOS: 7825-7828
LGI4	Leucine-rich repeat LGI family, member 4	SEQ ID NOS: 7829-7832
LGMN	Legumain	SEQ ID NOS: 7833-7846
LGR4	Leucine-rich repeat containing G protein-coupled receptor 4	SEQ ID NOS: 7847-7849
LHB	Luteinizing hormone beta polypeptide	SEQ ID NO: 7850
LHCGR	Luteinizing hormone/choriogonadotropin receptor	SEQ ID NOS: 7851-7855
LIF	Leukemia inhibitory factor	SEQ ID NOS: 7856-7857
LIFR	Leukemia inhibitory factor receptor alpha	SEQ ID NOS: 7858-7862
LILRA1	Leukocyte immunoglobulin-like receptor, subfamily A (with TM domain), member 1	SEQ ID NOS: 7863-7864
LILRA2	Leukocyte immunoglobulin-like receptor, subfamily A (with TM domain), member 2	SEQ ID NOS: 7865-7871
LILRB3	Leukocyte immunoglobulin-like receptor, subfamily B (with TM and ITIM domains), member 3	SEQ ID NOS: 7872-7876
LIME1	Lck interacting transmembrane adaptor 1	SEQ ID NOS: 7877-7882
LINGO1	Leucine rich repeat and Ig domain containing 1	SEQ ID NOS: 7883-7893
LIPA	Lipase A, lysosomal acid, cholesterol esterase	SEQ ID NOS: 7894-7898
LIPC	Lipase, hepatic	SEQ ID NOS: 7899-7902
LIPF	Lipase, gastric	SEQ ID NOS: 7903-7906
LIPG	Lipase, endothelial	SEQ ID NOS: 7907-7912
LIPH	Lipase, member H	SEQ ID NOS: 7913-7917
LIPK	Lipase, family member K	SEQ ID NO: 7918
LIPM	Lipase, family member M	SEQ ID NOS: 7919-7920
LIPN	Lipase, family member N	SEQ ID NO: 7921
LMAN2	Lectin, mannose-binding 2	SEQ ID NOS: 7922-7926
LMNTD1	Lamin tail domain containing 1	SEQ ID NOS: 7927-7937
LNK1	Ligand of numb-protein X 1, E3 ubiquitin protein ligase	SEQ ID NOS: 7938-7944

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
LOX	Lysyl oxidase	SEQ ID NOS: 7945-7947
LOXL1	Lysyl oxidase-like 1	SEQ ID NOS: 7948-7949
LOXL2	Lysyl oxidase-like 2	SEQ ID NOS: 7950-7958
LOXL3	Lysyl oxidase-like 3	SEQ ID NOS: 7959-7965
LOXL4	Lysyl oxidase-like 4	SEQ ID NO: 7966
LPA	Lipoprotein, Lp(a)	SEQ ID NOS: 7967-7969
LPL	Lipoprotein lipase	SEQ ID NOS: 7970-7974
LPO	Lactoperoxidase	SEQ ID NOS: 7975-7981
LRAT	Lecithin retinol acyltransferase (phosphatidylcholine--retinol O-acyltransferase)	SEQ ID NOS: 7982-7984
LRCH3	Leucine-rich repeats and calponin homology (CH) domain containing 3	SEQ ID NOS: 7985-7993
LRCOL1	Leucine rich colipase-like 1	SEQ ID NOS: 7994-7997
LRFN4	Leucine rich repeat and fibronectin type III domain containing 4	SEQ ID NOS: 7998-7999
LRFN5	Leucine rich repeat and fibronectin type III domain containing 5	SEQ ID NOS: 8000-8002
LRG1	Leucine-rich alpha-2-glycoprotein 1	SEQ ID NO: 8003
LRP1	Low density lipoprotein receptor-related protein 1	SEQ ID NOS: 8004-8009
LRP11	Low density lipoprotein receptor-related protein 11	SEQ ID NOS: 8010-8011
LRP1B	Low density lipoprotein receptor-related protein 1B	SEQ ID NOS: 8012-8015
LRP2	Low density lipoprotein receptor-related protein 2	SEQ ID NOS: 8016-8017
LRP4	Low density lipoprotein receptor-related protein 4	SEQ ID NOS: 8018-8019
LRPAP1	Low density lipoprotein receptor-related protein associated protein 1	SEQ ID NOS: 8020-8021
LRRC17	Leucine rich repeat containing 17	SEQ ID NOS: 8022-8024
LRRC32	Leucine rich repeat containing 32	SEQ ID NOS: 8025-8028
LRRC3B	Leucine rich repeat containing 3B	SEQ ID NOS: 8029-8033
LRRC4B	Leucine rich repeat containing 4B	SEQ ID NOS: 8034-8036
LRRC70	Leucine rich repeat containing 70	SEQ ID NOS: 8037-8038
LRRN3	Leucine rich repeat neuronal 3	SEQ ID NOS: 8039-8042
LRRTM1	Leucine rich repeat transmembrane neuronal 1	SEQ ID NOS: 8043-8049
LRRTM2	Leucine rich repeat transmembrane neuronal 2	SEQ ID NOS: 8050-8052
LRRTM4	Leucine rich repeat transmembrane neuronal 4	SEQ ID NOS: 8053-8058
LRTM2	Leucine-rich repeats and transmembrane domains 2	SEQ ID NOS: 8059-8063
LSR	Lipolysis stimulated lipoprotein receptor	SEQ ID NOS: 8064-8074
LST1	Leukocyte specific transcript 1	SEQ ID NOS: 8075-8092
LTA	Lymphotoxin alpha	SEQ ID NOS: 8093-8094
LTBP1	Latent transforming growth factor beta binding protein 1	SEQ ID NOS: 8095-8104
LTBP2	Latent transforming growth factor beta binding protein 2	SEQ ID NOS: 8105-8108
LTBP3	Latent transforming growth factor beta binding protein 3	SEQ ID NOS: 8109-8121
LTBP4	Latent transforming growth factor beta binding protein 4	SEQ ID NOS: 8122-8137
LTBR	Lymphotoxin beta receptor (TNFR superfamily, member 3)	SEQ ID NOS: 8138-8143

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
LTF	Lactotransferrin	SEQ ID NOS: 8144-8148
LYTK	Leukocyte receptor tyrosine kinase	SEQ ID NOS: 8149-8152
LUM	Lumican	SEQ ID NO: 8153
LUZP2	Leucine zipper protein 2	SEQ ID NOS: 8154-8157
LVRN	Laeverin	SEQ ID NOS: 8158-8163
LY6E	Lymphocyte antigen 6 complex, locus E	SEQ ID NOS: 8164-8177
LY6G5B	Lymphocyte antigen 6 complex, locus G5B	SEQ ID NOS: 8178-8179
LY6G6D	Lymphocyte antigen 6 complex, locus G6D	SEQ ID NOS: 8180-8181
LY6G6E	Lymphocyte antigen 6 complex, locus G6E (pseudogene)	SEQ ID NOS: 8182-8185
LY6H	Lymphocyte antigen 6 complex, locus H	SEQ ID NOS: 8186-8189
LY6K	Lymphocyte antigen 6 complex, locus K	SEQ ID NOS: 8190-8193
LY86	Lymphocyte antigen 86	SEQ ID NOS: 8195-8196
LY96	Lymphocyte antigen 96	SEQ ID NOS: 8197-8198
LYG1	Lysozyme G-like 1	SEQ ID NOS: 8199-8200
LYG2	Lysozyme G-like 2	SEQ ID NOS: 8201-8206
LYNX1	Ly6/neurotoxin 1	SEQ ID NOS: 8207-8211
LYPD1	LY6/PLAUR domain containing 1	SEQ ID NOS: 8212-8214
LYPD2	LY6/PLAUR domain containing 2	SEQ ID NO: 8215
LYPD4	LY6/PLAUR domain containing 4	SEQ ID NOS: 8216-8218
LYPD6	LY6/PLAUR domain containing 6	SEQ ID NOS: 8219-8223
LYPD6B	LY6/PLAUR domain containing 6B	SEQ ID NOS: 8224-8230
LYPD8	LY6/PLAUR domain containing 8	SEQ ID NOS: 8231-8232
LYZ	Lysozyme	SEQ ID NOS: 8233-8235
LYZL4	Lysozyme-like 4	SEQ ID NOS: 8236-8237
LYZL6	Lysozyme-like 6	SEQ ID NOS: 8238-8240
M6PR	Mannose-6-phosphate receptor (cation dependent)	SEQ ID NOS: 8241-8251
MAD1L1	MAD1 mitotic arrest deficient-like 1 (yeast)	SEQ ID NOS: 8252-8264
MAG	Myelin associated glycoprotein	SEQ ID NOS: 8265-8270
MAGT1	Magnesium transporter 1	SEQ ID NOS: 8271-8274
MALSU1	Mitochondrial assembly of ribosomal large subunit 1	SEQ ID NO: 8275
MAMDC2	MAM domain containing 2	SEQ ID NO: 8276
MAN2B1	Mannosidase, alpha, class 2B, member 1	SEQ ID NOS: 8277-8282
MAN2B2	Mannosidase, alpha, class 2B, member 2	SEQ ID NOS: 8283-8285
MANBA	Mannosidase, beta A, lysosomal	SEQ ID NOS: 8286-8299
MANEAL	Mannosidase, endo-alpha-like	SEQ ID NOS: 8300-8304
MANF	Mesencephalic astrocyte-derived neurotrophic factor	SEQ ID NOS: 8305-8306
MANSC1	MANSC domain containing 1	SEQ ID NOS: 8307-8310
MAP3K9	Mitogen-activated protein kinase 9	SEQ ID NOS: 8311-8316
MASP1	Mannan-binding lectin serine peptidase 1 (C4/C2 activating component of Ra-reactive factor)	SEQ ID NOS: 8317-8324

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
MASP2	Mannan-binding lectin serine peptidase 2	SEQ ID NOS: 8325-8326
MATN1	Matrilin 1, cartilage matrix protein	SEQ ID NO: 8327
MATN2	Matrilin 2	SEQ ID NOS: 8328-8340
MATN3	Matrilin 3	SEQ ID NOS: 8341-8342
MATN4	Matrilin 4	SEQ ID NOS: 8343-8347
MATR3	Matrin 3	SEQ ID NOS: 8348-8375
MAU2	MAU2 sister chromatid cohesion factor	SEQ ID NOS: 8376-8378
MAZ	MYC-associated zinc finger protein (purine-binding transcription factor)	SEQ ID NOS: 8379-8393
MBD6	Methyl-CpG binding domain protein 6	SEQ ID NOS: 8394-8405
MBL2	Mannose-binding lectin (protein C) 2, soluble	SEQ ID NO: 8406
MBNL1	Muscleblind-like splicing regulator 1	SEQ ID NOS: 8407-8425
MCCC1	Methylcrotonoyl-CoA carboxylase 1 (alpha)	SEQ ID NOS: 8426-8437
MCCD1	Mitochondrial coiled-coil domain 1	SEQ ID NO: 8438
MCEE	Methylmalonyl CoA epimerase	SEQ ID NOS: 8439-8442
MCF2L	MCF.2 cell line derived transforming sequence-like	SEQ ID NOS: 8443-8464
MCFD2	Multiple coagulation factor deficiency 2	SEQ ID NOS: 8465-8476
MDFIC	MyoD family inhibitor domain containing	SEQ ID NOS: 8477-8484
MDGA1	MAM domain containing glycosylphosphatidylinositol anchor 1	SEQ ID NOS: 8485-8490
MDK	Midkine (neurite growth-promoting factor 2)	SEQ ID NOS: 8491-8500
MED20	Mediator complex subunit 20	SEQ ID NOS: 8501-8505
MEGF10	Multiple EGF-like-domains 10	SEQ ID NOS: 8506-8509
MEGF6	Multiple EGF-like-domains 6	SEQ ID NOS: 8510-8513
MEI1	Meiotic double-stranded break formation protein 1	SEQ ID NOS: 8514-8517
MEI4	Meiotic double-stranded break formation protein 4	SEQ ID NO: 8518
MEIS1	Meis homeobox 1	SEQ ID NOS: 8519-8524
MEIS3	Meis homeobox 3	SEQ ID NOS: 8525-8534
MEPE	Matrix extracellular phosphoglycoprotein	SEQ ID NOS: 8538-8544
MESDC2	Mesoderm development candidate 2	SEQ ID NOS: 8545-8549
MEST	Mesoderm specific transcript	SEQ ID NOS: 8550-8563
MET	MET proto-oncogene, receptor tyrosine kinase	SEQ ID NOS: 8564-8569
METRN	Meteorin, glial cell differentiation regulator	SEQ ID NOS: 8570-8574
METRNL	Meteorin, glial cell differentiation regulator-like	SEQ ID NOS: 8575-8578
METTL17	Methyltransferase like 17	SEQ ID NOS: 8579-8589
METTL24	Methyltransferase like 24	SEQ ID NO: 8590
METTL7B	Methyltransferase like 7B	SEQ ID NOS: 8591-8592
METTL9	Methyltransferase like 9	SEQ ID NOS: 8593-8601
MEX3C	Mex-3 RNA binding family member C	SEQ ID NOS: 8602-8604
MFAP2	Microfibrillar-associated protein 2	SEQ ID NOS: 8605-8606
MFAP3	Microfibrillar-associated protein 3	SEQ ID NOS: 8607-8611

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
MFAP3L	Microfibrillar-associated protein 3-like	SEQ ID NOS: 8612-8621
MFAP4	Microfibrillar-associated protein 4	SEQ ID NOS: 8622-8624
MFAP5	Microfibrillar associated protein 5	SEQ ID NOS: 8625-8635
MFGE8	Milk fat globule-EGF factor 8 protein	SEQ ID NOS: 8636-8642
MFI2	Antigen p97 (melanoma associated) identified by monoclonal antibodies 133.2 and 96.5	SEQ ID NOS: 8535-8537
MFNG	MFNG O-fucosylpeptide 3-beta-N-acetylglucosaminyltransferase	SEQ ID NOS: 8643-8650
MGA	MGA, MAX dimerization protein	SEQ ID NOS: 8651-8659
MGAT2	Mannosyl (alpha-1,6-)-glycoprotein beta-1,2-N-acetylglucosaminyltransferase	SEQ ID NO: 8660
MGAT3	Mannosyl (beta-1,4-)-glycoprotein beta-1,4-N-acetylglucosaminyltransferase	SEQ ID NOS: 8661-8663
MGAT4A	Mannosyl (alpha-1,3-)-glycoprotein beta-1,4-N-acetylglucosaminyltransferase, isozyme A	SEQ ID NOS: 8664-8668
MGAT4B	Mannosyl (alpha-1,3-)-glycoprotein beta-1,4-N-acetylglucosaminyltransferase, isozyme B	SEQ ID NOS: 8669-8679
MGAT4D	MGAT4 family, member D	SEQ ID NOS: 8680-8685
MGLL	Monoglyceride lipase	SEQ ID NOS: 8686-8695
MGP	Matrix Gla protein	SEQ ID NOS: 8696-8698
MGST2	Microsomal glutathione S-transferase 2	SEQ ID NOS: 8699-8702
MIA	Melanoma inhibitory activity	SEQ ID NOS: 8703-8708
MIA2	Melanoma inhibitory activity 2	SEQ ID NO: 8709
MIA3	Melanoma inhibitory activity family, member 3	SEQ ID NOS: 8710-8714
MICU1	Mitochondrial calcium uptake 1	SEQ ID NOS: 8715-8724
MIER1	Mesoderm induction early response 1, transcriptional regulator	SEQ ID NOS: 8725-8733
MINOS1-NBL1	MINOS1-NBL1 readthrough	SEQ ID NOS: 8734-8736
MINPP1	Multiple inositol-polyphosphate phosphatase 1	SEQ ID NOS: 8737-8739
MLEC	Malectin	SEQ ID NOS: 8740-8743
MLN	Motilin	SEQ ID NOS: 8744-8746
MLXIP	MLX interacting protein	SEQ ID NOS: 8747-8752
MLXIPL	MLX interacting protein-like	SEQ ID NOS: 8753-8760
MMP1	Matrix metalloproteinase 1	SEQ ID NO: 8761
MMP10	Matrix metalloproteinase 10	SEQ ID NOS: 8762-8763
MMP11	Matrix metalloproteinase 11	SEQ ID NOS: 8764-8767
MMP12	Matrix metalloproteinase 12	SEQ ID NO: 8768
MMP13	Matrix metalloproteinase 13	SEQ ID NOS: 8769-8771
MMP14	Matrix metalloproteinase 14 (membrane-inserted)	SEQ ID NOS: 8772-8774
MMP17	Matrix metalloproteinase 17 (membrane-inserted)	SEQ ID NOS: 8775-8782
MMP19	Matrix metalloproteinase 19	SEQ ID NOS: 8783-8788
MMP2	Matrix metalloproteinase 2	SEQ ID NOS: 8789-8796
MMP20	Matrix metalloproteinase 20	SEQ ID NO: 8797
MMP21	Matrix metalloproteinase 21	SEQ ID NO: 8798
MMP25	Matrix metalloproteinase 25	SEQ ID NOS: 8799-8800
MMP26	Matrix metalloproteinase 26	SEQ ID NOS: 8801-8802

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
MMP27	Matrix metallopeptidase 27	SEQ ID NO: 8803
MMP28	Matrix metallopeptidase 28	SEQ ID NOS: 8804-8809
MMP3	Matrix metallopeptidase 3	SEQ ID NOS: 8810-8812
MMP7	Matrix metallopeptidase 7	SEQ ID NO: 8813
MMP8	Matrix metallopeptidase 8	SEQ ID NOS: 8814-8819
MMP9	Matrix metallopeptidase 9	SEQ ID NO: 8820
MMRN1	Multimerin 1	SEQ ID NOS: 8821-8823
MMRN2	Multimerin 2	SEQ ID NOS: 8824-8828
MOXD1	Monooxygenase, DBH-like 1	SEQ ID NOS: 8829-8831
MPO	Myeloperoxidase	SEQ ID NOS: 8840-8841
MPPED1	Metallophosphoesterase domain containing 1	SEQ ID NOS: 8842-8845
MPZL1	Myelin protein zero-like 1	SEQ ID NOS: 8846-8850
MR1	Major histocompatibility complex, class I-related	SEQ ID NOS: 8851-8856
MRPL2	Mitochondrial ribosomal protein L2	SEQ ID NOS: 8857-8861
MRPL21	Mitochondrial ribosomal protein L21	SEQ ID NOS: 8862-8868
MRPL22	Mitochondrial ribosomal protein L22	SEQ ID NOS: 8869-8873
MRPL24	Mitochondrial ribosomal protein L24	SEQ ID NOS: 8874-8878
MRPL27	Mitochondrial ribosomal protein L27	SEQ ID NOS: 8879-8884
MRPL32	Mitochondrial ribosomal protein L32	SEQ ID NOS: 8885-8887
MRPL34	Mitochondrial ribosomal protein L34	SEQ ID NOS: 8888-8892
MRPL35	Mitochondrial ribosomal protein L35	SEQ ID NOS: 8893-8896
MRPL52	Mitochondrial ribosomal protein L52	SEQ ID NOS: 8897-8907
MRPL55	Mitochondrial ribosomal protein L55	SEQ ID NOS: 8908-8933
MRPS14	Mitochondrial ribosomal protein S14	SEQ ID NOS: 8934-8935
MRPS22	Mitochondrial ribosomal protein S22	SEQ ID NOS: 8936-8944
MRPS28	Mitochondrial ribosomal protein S28	SEQ ID NOS: 8945-8952
MS4A14	Membrane-spanning 4-domains, subfamily A, member 14	SEQ ID NOS: 8953-8963
MS4A3	Membrane-spanning 4-domains, subfamily A, member 3 (hematopoietic cell-specific)	SEQ ID NOS: 8964-8968
MSH3	MutS homolog 3	SEQ ID NO: 8969
MSH5	MutS homolog 5	SEQ ID NOS: 8970-8981
MSLN	Mesothelin	SEQ ID NOS: 8982-8989
MSMB	Microseminoprotein, beta-	SEQ ID NOS: 8990-8991
MSRA	Methionine sulfoxide reductase A	SEQ ID NOS: 8992-8999
MSRB2	Methionine sulfoxide reductase B2	SEQ ID NOS: 9000-9001
MSRB3	Methionine sulfoxide reductase B3	SEQ ID NOS: 9002-9015
MST1	Macrophage stimulating 1	SEQ ID NOS: 9016-9017
MSTN	Myostatin	SEQ ID NO: 9018
MT1G	Metallothionein 1G	SEQ ID NOS: 9019-9022
MTHFD2	Methylenetetrahydrofolate dehydrogenase (NADP+ dependent) 2, methenyltetrahydrofolate cyclohydrolase	SEQ ID NOS: 9023-9027
MTMR14	Myotubularin related protein 14	SEQ ID NOS: 9028-9038

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
MTRNR2L11	MT-RNR2-like 11 (pseudogene)	SEQ ID NO: 9039
MTRR	5-methyltetrahydrofolate-homocysteine methyltransferase reductase	SEQ ID NOS: 9040-9052
MTTP	Microsomal triglyceride transfer protein	SEQ ID NOS: 9053-9063
MTX2	Metaxin 2	SEQ ID NOS: 9064-9068
MUC1	Mucin 1, cell surface associated	SEQ ID NOS: 9069-9094
MUC13	Mucin 13, cell surface associated	SEQ ID NOS: 9095-9096
MUC20	Mucin 20, cell surface associated	SEQ ID NOS: 9097-9101
MUC3A	Mucin 3A, cell surface associated	SEQ ID NOS: 9102-9104
MUC5AC	Mucin 5AC, oligomeric mucus/gel-forming	SEQ ID NO: 9105
MUC5B	Mucin 5B, oligomeric mucus/gel-forming	SEQ ID NOS: 9106-9107
MUC6	Mucin 6, oligomeric mucus/gel-forming	SEQ ID NOS: 9108-9111
MUC7	Mucin 7, secreted	SEQ ID NOS: 9112-9115
MUCL1	Mucin-like 1	SEQ ID NOS: 9116-9118
MXRA5	Matrix-remodelling associated 5	SEQ ID NO: 9119
MXRA7	Matrix-remodelling associated 7	SEQ ID NOS: 9120-9126
MYDGF	Myeloid-derived growth factor	SEQ ID NOS: 9127-9129
MYL1	Myosin, light chain 1, alkali; skeletal, fast	SEQ ID NOS: 9130-9131
MYOC	Myocilin, trabecular meshwork inducible glucocorticoid response	SEQ ID NOS: 9132-9133
MYRFL	Myelin regulatory factor-like	SEQ ID NOS: 9134-9138
MZB1	Marginal zone B and B1 cell-specific protein	SEQ ID NOS: 9139-9143
N4BP2L2	NEDD4 binding protein 2-like 2	SEQ ID NOS: 9144-9149
NAA38	N(alpha)-acetyltransferase 38, NatC auxiliary subunit	SEQ ID NOS: 9150-9155
NAAA	N-acyl ethanolamine acid amidase	SEQ ID NOS: 9156-9161
NAGA	N-acetyl galactosaminidase, alpha-	SEQ ID NOS: 9162-9164
NAGLU	N-acetyl glucosaminidase, alpha	SEQ ID NOS: 9165-9169
NAGS	N-acetyl glutamate synthase	SEQ ID NOS: 9170-9171
NAPSA	Napsin A aspartic peptidase	SEQ ID NOS: 9172-9174
NBL1	Neuroblastoma 1, DAN family BMP antagonist	SEQ ID NOS: 9180-9193
NCAM1	Neural cell adhesion molecule 1	SEQ ID NOS: 9194-9213
NCAN	Neurocan	SEQ ID NOS: 9214-9215
NCBP2-AS2	NCBP2 antisense RNA 2 (head to head)	SEQ ID NO: 9216
NCSTN	Nicastrin	SEQ ID NOS: 9217-9226
NDNF	Neuron-derived neurotrophic factor	SEQ ID NOS: 9227-9229
NDP	Norrie disease (pseudoglioma)	SEQ ID NOS: 9230-9232
NDUFA10	NADH dehydrogenase (ubiquinone) 1 alpha subcomplex, 10, 42 kDa	SEQ ID NOS: 9233-9242
NDUFB5	NADH dehydrogenase (ubiquinone) 1 beta subcomplex, 5, 16 kDa	SEQ ID NOS: 9243-9251
NDUFS8	NADH dehydrogenase (ubiquinone) Fe—S protein 8, 23 kDa (NADH-coenzyme Q reductase)	SEQ ID NOS: 9252-9261
NDUFV1	NADH dehydrogenase (ubiquinone) flavoprotein 1, 51 kDa	SEQ ID NOS: 9262-9275
NECAB3	N-terminal EF-hand calcium binding protein 3	SEQ ID NOS: 9276-9285

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
NELL1	Neural EGFL like 1	SEQ ID NOS: 9289-9292
NELL2	Neural EGFL like 2	SEQ ID NOS: 9293-9307
NENF	Neudesin neurotrophic factor	SEQ ID NO: 9308
NETO1	Neuropilin (NRP) and tolloid (TLL)-like 1	SEQ ID NOS: 9309-9312
NFASC	Neurofascin	SEQ ID NOS: 9313-9327
NFE2L1	Nuclear factor, erythroid 2-like 1	SEQ ID NOS: 9328-9346
NFE2L3	Nuclear factor, erythroid 2-like 3	SEQ ID NOS: 9347-9348
NGEF	Neuronal guanine nucleotide exchange factor	SEQ ID NOS: 9349-9354
NGF	Nerve growth factor (beta polypeptide)	SEQ ID NO: 9355
NGLY1	N-glycanase 1	SEQ ID NOS: 9356-9362
NGRN	Neugrin, neurite outgrowth associated	SEQ ID NOS: 9363-9364
NHLRC3	NHL repeat containing 3	SEQ ID NOS: 9365-9367
NID1	Nidogen 1	SEQ ID NOS: 9368-9369
NID2	Nidogen 2 (osteonidogen)	SEQ ID NOS: 9370-9372
NKG7	Natural killer cell granule protein 7	SEQ ID NOS: 9373-9377
NLGN3	Neurologin 3	SEQ ID NOS: 9378-9382
NLGN4Y	Neurologin 4, Y-linked	SEQ ID NOS: 9383-9389
NLRP5	NLR family, pyrin domain containing 5	SEQ ID NOS: 9390-9392
NMB	Neuromedin B	SEQ ID NOS: 9393-9394
NME1	NME/NM23 nucleoside diphosphate kinase 1	SEQ ID NOS: 9395-9401
NME1-NME2	NME1-NME2 readthrough	SEQ ID NOS: 9402-9404
NME3	NME/NM23 nucleoside diphosphate kinase 3	SEQ ID NOS: 9405-9409
NMS	Neuromedin S	SEQ ID NO: 9410
NMU	Neuromedin U	SEQ ID NOS: 9411-9414
NOA1	Nitric oxide associated 1	SEQ ID NO: 9415
NODAL	Nodal growth differentiation factor	SEQ ID NOS: 9416-9417
NOG	Noggin	SEQ ID NO: 9418
NOMO3	NODAL modulator 3	SEQ ID NOS: 9419-9425
NOS1AP	Nitric oxide synthase 1 (neuronal) adaptor protein	SEQ ID NOS: 9426-9430
NOTCH3	Notch 3	SEQ ID NOS: 9431-9434
NOTUM	Notum pectinacetyltransferase homolog (<i>Drosophila</i>)	SEQ ID NOS: 9435-9437
NOV	Nephroblastoma overexpressed	SEQ ID NO: 9438
NPB	Neuropeptide B	SEQ ID NOS: 9439-9440
NPC2	Niemann-Pick disease, type C2	SEQ ID NOS: 9441-9449
NPFF	Neuropeptide FF-amide peptide precursor	SEQ ID NO: 9450
NPFFR2	Neuropeptide FF receptor 2	SEQ ID NOS: 9451-9454
NPHS1	Nephrosis 1, congenital, Finnish type (nephrin)	SEQ ID NOS: 9455-9456
NPNT	Nephronectin	SEQ ID NOS: 9457-9467
NPPA	Natriuretic peptide A	SEQ ID NOS: 9468-9470
NPPB	Natriuretic peptide B	SEQ ID NO: 9471
NPPC	Natriuretic peptide C	SEQ ID NOS: 9472-9473
NPS	Neuropeptide S	SEQ ID NO: 9474
NPTX1	Neuronal pentraxin I	SEQ ID NO: 9475

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
NPTX2	Neuronal pentraxin II	SEQ ID NO: 9476
NPTXR	Neuronal pentraxin receptor	SEQ ID NOS: 9477-9478
NPVF	Neuropeptide VF precursor	SEQ ID NO: 9479
NPW	Neuropeptide W	SEQ ID NOS: 9480-9482
NPY	Neuropeptide Y	SEQ ID NOS: 9483-9485
NQO2	NAD(P)H dehydrogenase, quinone 2	SEQ ID NOS: 9486-9494
NRCAM	Neuronal cell adhesion molecule	SEQ ID NOS: 9495-9507
NRG1	Neuregulin 1	SEQ ID NOS: 9508-9525
NRN1L	Neuritin 1-like	SEQ ID NOS: 9526-9528
NRP1	Neuropilin 1	SEQ ID NOS: 9529-9542
NRP2	Neuropilin 2	SEQ ID NOS: 9543-9549
NRTN	Neurturin	SEQ ID NO: 9550
NRXN1	Neurexin 1	SEQ ID NOS: 9551-9581
NRXN2	Neurexin 2	SEQ ID NOS: 9582-9590
NT5C3A	5'-nucleotidase, cytosolic IIIA	SEQ ID NOS: 9591-9601
NT5DC3	5'-nucleotidase domain containing 3	SEQ ID NOS: 9602-9604
NT5E	5'-nucleotidase, ecto (CD73)	SEQ ID NOS: 9605-9609
NTF3	Neurotrophin 3	SEQ ID NOS: 9610-9611
NTF4	Neurotrophin 4	SEQ ID NOS: 9612-9613
NTM	Neurotrimin	SEQ ID NOS: 9614-9623
NTN1	Netrin 1	SEQ ID NOS: 9624-9625
NTN3	Netrin 3	SEQ ID NO: 9626
NTN4	Netrin 4	SEQ ID NOS: 9627-9631
NTN5	Netrin 5	SEQ ID NOS: 9632-9633
NTNG1	Netrin G1	SEQ ID NOS: 9634-9640
NTNG2	Netrin G2	SEQ ID NOS: 9641-9642
NTS	Neurotensin	SEQ ID NOS: 9643-9644
NUBPL	Nucleotide binding proteindike	SEQ ID NOS: 9645-9651
NUCB1	Nucleobindin 1	SEQ ID NOS: 9652-9658
NUCB2	Nucleobindin 2	SEQ ID NOS: 9659-9674
NUDT19	Nudix (nucleoside diphosphate linked moiety X)-type motif 19	SEQ ID NO: 9675
NUDT9	Nudix (nucleoside diphosphate linked moiety X)-type motif 9	SEQ ID NOS: 9676-9680
NUP155	Nucleoporin 155 kDa	SEQ ID NOS: 9681-9684
NUP214	Nucleoporin 214 kDa	SEQ ID NOS: 9685-9696
NUP85	Nucleoporin 85 kDa	SEQ ID NOS: 9697-9711
NXPE3	Neurexophilin and PC-esterase domain family, member 3	SEQ ID NOS: 9712-9716
NXPE4	Neurexophilin and PC-esterase domain family, member 4	SEQ ID NOS: 9717-9718
NXPH1	Neurexophilin 1	SEQ ID NOS: 9719-9722
NXPH2	Neurexophilin 2	SEQ ID NO: 9723
NXPH3	Neurexophilin 3	SEQ ID NOS: 9724-9725

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
NXPH4	Neurexophilin 4	SEQ ID NOS: 9726-9727
NYX	Nyctalopin	SEQ ID NOS: 9728-9729
OAF	Out at first homolog	SEQ ID NOS: 9730-9731
OBP2A	Odorant binding protein 2A	SEQ ID NOS: 9732-9738
OBP2B	Odorant binding protein 2B	SEQ ID NOS: 9739-9742
OC90	Otoconin 90	SEQ ID NO: 9743
OCLN	Occludin	SEQ ID NOS: 9744-9746
ODAM	Odontogenic, ameloblast associated	SEQ ID NOS: 9747-9750
OGG1	8-oxoguanine DNA glycosylase	SEQ ID NOS: 9755-9768
OGN	Osteoglycin	SEQ ID NOS: 9769-9771
OIT3	Oncoprotein induced transcript 3	SEQ ID NOS: 9772-9773
OLFM1	Olfactomedin 1	SEQ ID NOS: 9774-9784
OLFM2	Olfactomedin 2	SEQ ID NOS: 9785-9788
OLFM3	Olfactomedin 3	SEQ ID NOS: 9789-9791
OLFM4	Olfactomedin 4	SEQ ID NO: 9792
OLFML1	Olfactomedin-like 1	SEQ ID NOS: 9793-9796
OLFML2A	Olfactomedin-like 2A	SEQ ID NOS: 9797-9799
OLFML2B	Olfactomedin-like 2B	SEQ ID NOS: 9800-9804
OLFML3	Olfactomedin-like 3	SEQ ID NOS: 9805-9807
OMD	Osteomodulin	SEQ ID NO: 9808
OMG	Oligodendrocyte myelin glycoprotein	SEQ ID NO: 9809
OOSP2	Oocyte secreted protein 2	SEQ ID NOS: 9810-9811
OPCML	Opioid binding protein/cell adhesion molecule-like	SEQ ID NOS: 9812-9816
OPTC	Opticin	SEQ ID NOS: 9818-9819
ORAI1	ORAI calcium release-activated calcium modulator 1	SEQ ID NO: 9820
ORM1	Orosomucoid 1	SEQ ID NO: 9821
ORM2	Orosomucoid 2	SEQ ID NO: 9822
ORMDL2	ORMDL sphingolipid biosynthesis regulator 2	SEQ ID NOS: 9823-9826
OS9	Osteosarcoma amplified 9, endoplasmic reticulum lectin	SEQ ID NOS: 9827-9841
OSCAR	Osteoclast associated, immunoglobulin-like receptor	SEQ ID NOS: 9842-9852
OSM	Oncostatin M	SEQ ID NOS: 9853-9855
OSMR	Oncostatin M receptor	SEQ ID NOS: 9856-9860
OSTN	Osteocrin	SEQ ID NOS: 9861-9862
OTOA	Otoancorin	SEQ ID NOS: 9863-9868
OTOG	Otogelin	SEQ ID NOS: 9869-9871
OTOGL	Otogelin-like	SEQ ID NOS: 9872-9878
OTOL1	Otolin 1	SEQ ID NO: 9879
OTOR	Otoraplin	SEQ ID NO: 9880
OTOS	Otospiralin	SEQ ID NOS: 9881-9882
OVCH1	Ovochymase 1	SEQ ID NOS: 9883-9885
OVCH2	Ovochymase 2 (gene/pseudogene)	SEQ ID NOS: 9886-9887
OVGP1	Oviductal glycoprotein 1, 120 kDa	SEQ ID NO: 9888

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
OXCT1	3-oxoacid CoA transferase 1	SEQ ID NOS: 9889-9892
OXCT2	3-oxoacid CoA transferase 2	SEQ ID NO: 9893
OXNAD1	Oxidoreductase NAD-binding domain containing 1	SEQ ID NOS: 9894-9900
OXT	Oxytocin/neurophysin I prepropeptide	SEQ ID NO: 9901
P3H1	Prolyl 3-hydroxylase 1	SEQ ID NOS: 9902-9906
P3H2	Prolyl 3-hydroxylase 2	SEQ ID NOS: 9907-9910
P3H3	Prolyl 3-hydroxylase 3	SEQ ID NO: 9911
P3H4	Prolyl 3-hydroxylase family member 4 (non-enzymatic)	SEQ ID NOS: 9912-9916
P4HA1	Prolyl 4-hydroxylase, alpha polypeptide I	SEQ ID NOS: 9917-9921
P4HA2	Prolyl 4-hydroxylase, alpha polypeptide II	SEQ ID NOS: 9922-9936
P4HA3	Prolyl 4-hydroxylase, alpha polypeptide III	SEQ ID NOS: 9937-9941
P4HB	Prolyl 4-hydroxylase, beta polypeptide	SEQ ID NOS: 9942-9953
PAEP	Progestagen-associated endometrial protein	SEQ ID NOS: 9954-9962
PAM	Peptidylglycine alpha-amidating monooxygenase	SEQ ID NOS: 9963-9976
PAMR1	Peptidase domain containing associated with muscle regeneration 1	SEQ ID NOS: 9977-9983
PAPL	Iron/zinc purple acid phosphatase-like protein	SEQ ID NOS: 159-162
PAPLN	Papilin, proteoglycan-like sulfated glycoprotein	SEQ ID NOS: 9984-9991
PAPPA	Pregnancy-associated plasma protein A, pappalysin 1	SEQ ID NO: 9992
PAPPA2	Pappalysin 2	SEQ ID NOS: 9993-9994
PARP15	Poly (ADP-ribose) polymerase family, member 15	SEQ ID NOS: 9995-9998
PARVB	Parvin, beta	SEQ ID NOS: 9999-10003
PATE1	Prostate and testis expressed 1	SEQ ID NOS: 10004-10005
PATE2	Prostate and testis expressed 2	SEQ ID NOS: 10006-10007
PATE3	Prostate and testis expressed 3	SEQ ID NO: 10008
PATE4	Prostate and testis expressed 4	SEQ ID NOS: 10009-10010
PATL2	Protein associated with topoisomerase II homolog 2 (yeast)	SEQ ID NOS: 10011-10016
PAX2	Paired box 2	SEQ ID NOS: 10017-10022
PAX4	Paired box 4	SEQ ID NOS: 10023-10029
PCCB	Propionyl CoA carboxylase, beta polypeptide	SEQ ID NOS: 10030-10044
PCDH1	Protocadherin 1	SEQ ID NOS: 10045-10050
PCDH12	Protocadherin 12	SEQ ID NOS: 10051-10052
PCDH15	Protocadherin-related 15	SEQ ID NOS: 10053-10086
PCDHA1	Protocadherin alpha 1	SEQ ID NOS: 10087-10089
PCDHA10	Protocadherin alpha 10	SEQ ID NOS: 10090-10092
PCDHA11	Protocadherin alpha 11	SEQ ID NOS: 10093-10095
PCDHA6	Protocadherin alpha 6	SEQ ID NOS: 10096-10098
PCDHB12	Protocadherin beta 12	SEQ ID NOS: 10099-10101
PCDHGA11	Protocadherin gamma subfamily A, 11	SEQ ID NOS: 10102-10104
PCF11	PCF11 cleavage and polyadenylation factor subunit	SEQ ID NOS: 10105-10109
PCOLCE	Procollagen C-endopeptidase enhancer	SEQ ID NO: 10110

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
PCOLCE2	Procollagen C-endopeptidase enhancer 2	SEQ ID NOS: 10111-10114
PCSK1	Proprotein convertase subtilisin/kexin type 1	SEQ ID NOS: 10115-10117
PCSK1N	Proprotein convertase subtilisin/kexin type 1 inhibitor	SEQ ID NO: 10118
PCSK2	Proprotein convertase subtilisin/kexin type 2	SEQ ID NOS: 10119-10121
PCSK4	Proprotein convertase subtilisin/kexin type 4	SEQ ID NOS: 10122-10124
PCSK5	Proprotein convertase subtilisin/kexin type 5	SEQ ID NOS: 10125-10129
PCSK9	Proprotein convertase subtilisin/kexin type 9	SEQ ID NO: 10130
PCYOX1	Prenylcysteine oxidase 1	SEQ ID NOS: 10131-10135
PCYOX1L	Prenylcysteine oxidase 1 like	SEQ ID NOS: 10136-10140
PDDC1	Parkinson disease 7 domain containing 1	SEQ ID NOS: 5802-5810
PDE11A	Phosphodiesterase 11A	SEQ ID NOS: 10141-10146
PDE2A	Phosphodiesterase 2A, cGMP-stimulated	SEQ ID NOS: 10147-10168
PDE7A	Phosphodiesterase 7A	SEQ ID NOS: 10169-10172
PDF	Peptide deformylase (mitochondrial)	SEQ ID NO: 10173
PDGFA	Platelet-derived growth factor alpha polypeptide	SEQ ID NOS: 10174-10177
PDGFB	Platelet-derived growth factor beta polypeptide	SEQ ID NOS: 10178-10181
PDGFC	Platelet derived growth factor C	SEQ ID NOS: 10182-10185
PDGFD	Platelet derived growth factor D	SEQ ID NOS: 10186-10188
PDGFRA	Platelet-derived growth factor receptor, alpha polypeptide	SEQ ID NOS: 10189-10195
PDGFRB	Platelet-derived growth factor receptor, beta polypeptide	SEQ ID NOS: 10196-10199
PDGFRL	Platelet-derived growth factor receptor-like	SEQ ID NOS: 10200-10201
PDHA1	Pyruvate dehydrogenase (lipoamide) alpha 1	SEQ ID NOS: 10202-10210
PDIA2	Protein disulfide isomerase family A, member 2	SEQ ID NOS: 10211-10214
PDIA3	Protein disulfide isomerase family A, member 3	SEQ ID NOS: 10215-10218
PDIA4	Protein disulfide isomerase family A, member 4	SEQ ID NOS: 10219-10220
PDIA5	Protein disulfide isomerase family A, member 5	SEQ ID NOS: 10221-10224
PDIA6	Protein disulfide isomerase family A, member 6	SEQ ID NOS: 10225-10231
PDILT	Protein disulfide isomerase-like, testis expressed	SEQ ID NOS: 10232-10233
PDYN	Prodynorphin	SEQ ID NOS: 10234-10236
PDZD8	PDZ domain containing 8	SEQ ID NO: 10237
PDZRN4	PDZ domain containing ring finger 4	SEQ ID NOS: 10238-10240
PEAR1	Platelet endothelial aggregation receptor 1	SEQ ID NOS: 10241-10244
PEBP4	Phosphatidylethanolamine-binding protein 4	SEQ ID NOS: 10245-10246
PECAM1	Platelet/endothelial cell adhesion molecule 1	SEQ ID NOS: 10247-10250
PENK	Proenkephalin	SEQ ID NOS: 10251-10256
PET117	PET117 homolog	SEQ ID NO: 10257
PF4	Platelet factor 4	SEQ ID NO: 10258
PF4V1	Platelet factor 4 variant 1	SEQ ID NO: 10259
PFKP	Phosphofructokinase, platelet	SEQ ID NOS: 10260-10268
PFN1	Profilin 1	SEQ ID NOS: 10269-10271

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
PGA3	Pepsinogen 3, group I (pepsinogen A)	SEQ ID NOS: 10272-10275
PGA4	Pepsinogen 4, group I (pepsinogen A)	SEQ ID NOS: 10276-10278
PGA5	Pepsinogen 5, group I (pepsinogen A)	SEQ ID NOS: 10279-10281
PGAM5	PGAM family member 5, serine/threonine protein phosphatase, mitochondrial	SEQ ID NOS: 10282-10285
PGAP3	Post-GPI attachment to proteins 3	SEQ ID NOS: 10286-10293
PGC	Progastricsin (pepsinogen C)	SEQ ID NOS: 10294-10297
PGF	Placental growth factor	SEQ ID NOS: 10298-10301
PGLYRP1	Peptidoglycan recognition protein 1	SEQ ID NO: 10302
PGLYRP2	Peptidoglycan recognition protein 2	SEQ ID NOS: 10303-10306
PGLYRP3	Peptidoglycan recognition protein 3	SEQ ID NO: 10307
PGLYRP4	Peptidoglycan recognition protein 4	SEQ ID NOS: 10308-10309
PHACTR1	Phosphatase and actin regulator 1	SEQ ID NOS: 10310-10316
PHB	Prohibitin	SEQ ID NOS: 10317-10325
PI15	Peptidase inhibitor 15	SEQ ID NOS: 10326-10327
PI3	Peptidase inhibitor 3, skin-derived	SEQ ID NO: 10328
PIANP	PILR alpha associated neural protein	SEQ ID NOS: 10329-10334
PIGK	Phosphatidylinositol glycan anchor biosynthesis, class K	SEQ ID NOS: 10335-10338
PIGL	Phosphatidylinositol glycan anchor biosynthesis, class L	SEQ ID NOS: 10339-10346
PIGT	Phosphatidylinositol glycan anchor biosynthesis, class T	SEQ ID NOS: 10347-10400
PIGZ	Phosphatidylinositol glycan anchor biosynthesis, class Z	SEQ ID NOS: 10401-10403
PIK3AP1	Phosphoinositide-3-kinase adaptor protein 1	SEQ ID NOS: 10404-10406
PIK3IP1	Phosphoinositide-3-kinase interacting protein 1	SEQ ID NOS: 10407-10410
PILRA	Paired immunoglobulin-like type 2 receptor alpha	SEQ ID NOS: 10411-10415
PILRB	Paired immunoglobulin-like type 2 receptor beta	SEQ ID NOS: 10416-10427
PINLYP	Phospholipase A2 inhibitor and LY6/PLAUR domain containing	SEQ ID NOS: 10428-10432
PIP	Prolactin-induced protein	SEQ ID NO: 10433
PIWIL4	Piwi-like RNA-mediated gene silencing 4	SEQ ID NOS: 10434-10438
PKDCC	Protein kinase domain containing, cytoplasmic	SEQ ID NOS: 10439-10440
PKHD1	Polycystic kidney and hepatic disease 1 (autosomal recessive)	SEQ ID NOS: 10441-10442
PLA1A	Phospholipase A1 member A	SEQ ID NOS: 10443-10447
PLA2G10	Phospholipase A2, group X	SEQ ID NOS: 10448-10449
PLA2G12A	Phospholipase A2, group XIIA	SEQ ID NOS: 10450-10452
PLA2G12B	Phospholipase A2, group XIIB	SEQ ID NO: 10453
PLA2G15	Phospholipase A2, group XV	SEQ ID NOS: 10454-10461
PLA2G1B	Phospholipase A2, group IB (pancreas)	SEQ ID NOS: 10462-10464
PLA2G2A	Phospholipase A2, group IIA (platelets, synovial fluid)	SEQ ID NOS: 10465-10466
PLA2G2C	Phospholipase A2, group IIC	SEQ ID NOS: 10467-10468
PLA2G2D	Phospholipase A2, group IID	SEQ ID NOS: 10469-10470
PLA2G2E	Phospholipase A2, group IIE	SEQ ID NO: 10471
PLA2G3	Phospholipase A2, group III	SEQ ID NO: 10472
PLA2G5	Phospholipase A2, group V	SEQ ID NO: 10473
PLA2G7	Phospholipase A2, group VII (platelet-activating factor acetylhydrolase, plasma)	SEQ ID NOS: 10474-10475

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
PLA2R1	Phospholipase A2 receptor 1, 180 kDa	SEQ ID NOS: 10476-10477
PLAC1	Placenta-specific 1	SEQ ID NO: 10478
PLAC9	Placenta-specific 9	SEQ ID NOS: 10479-10481
PLAT	Plasminogen activator, tissue	SEQ ID NOS: 10482-10490
PLAU	Plasminogen activator, urokinase	SEQ ID NOS: 10491-10493
PLAUR	Plasminogen activator, urokinase receptor	SEQ ID NOS: 10494-10505
PLBD1	Phospholipase B domain containing 1	SEQ ID NOS: 10506-10508
PLBD2	Phospholipase B domain containing 2	SEQ ID NOS: 10509-10511
PLG	Plasminogen	SEQ ID NOS: 10512-10514
PLGLB1	Plasminogen-like B1	SEQ ID NOS: 10515-10518
PLGLB2	Plasminogen-like B2	SEQ ID NOS: 10519-10520
PLOD1	Procollagen-lysine, 2-oxoglutarate 5-dioxygenase 1	SEQ ID NOS: 10521-10523
PLOD2	Procollagen-lysine, 2-oxoglutarate 5-dioxygenase 2	SEQ ID NOS: 10524-10529
PLOD3	Procollagen-lysine, 2-oxoglutarate 5-dioxygenase 3	SEQ ID NOS: 10530-10536
PLTP	Phospholipid transfer protein	SEQ ID NOS: 10537-10541
PLXNA4	Plexin A4	SEQ ID NOS: 10542-10545
PLXNB2	Plexin B2	SEQ ID NOS: 10546-10554
PM20D1	Peptidase M20 domain containing 1	SEQ ID NO: 10555
PMCH	Pro-melanin-concentrating hormone	SEQ ID NO: 10556
PMEL	Premelanosome protein	SEQ ID NOS: 10557-10568
PMEPA1	Prostate transmembrane protein, androgen induced 1	SEQ ID NOS: 10569-10575
PNLIP	Pancreatic lipase	SEQ ID NO: 10576
PNLIPRP1	Pancreatic lipase-related protein 1	SEQ ID NOS: 10577-10585
PNLIPRP3	Pancreatic lipase-related protein 3	SEQ ID NO: 10586
PNOC	Prepronociceptin	SEQ ID NOS: 10587-10589
PNP	Purine nucleoside phosphorylase	SEQ ID NOS: 10590-10593
PNPLA4	Patatin-like phospholipase domain containing 4	SEQ ID NOS: 10594-10597
PODNL1	Podocan-like 1	SEQ ID NOS: 10598-10609
POFUT1	Protein O-fucosyltransferase 1	SEQ ID NOS: 10610-10611
POFUT2	Protein O-fucosyltransferase 2	SEQ ID NOS: 10612-10617
POGLUT1	Protein O-glucosyltransferase 1	SEQ ID NOS: 10618-10622
POLL	Polymerase (DNA directed), lambda	SEQ ID NOS: 10623-10635
POMC	Proopiomelanocortin	SEQ ID NOS: 10636-10640
POMGNT2	Protein O-linked mannose N-acetylglucosaminyltransferase 2 (beta 1,4-)	SEQ ID NOS: 10641-10642
PON1	Paraoxonase 1	SEQ ID NOS: 10643-10644
PON2	Paraoxonase 2	SEQ ID NOS: 10645-10657
PON3	Paraoxonase 3	SEQ ID NOS: 10658-10663
POSTN	Periostin, osteoblast specific factor	SEQ ID NOS: 10664-10669
PPBP	Pro-platelet basic protein (chemokine (C-X-C motif) ligand 7)	SEQ ID NO: 10670
PPIB	Peptidylprolyl isomerase B (cyclophilin B)	SEQ ID NO: 10671
PPIC	Peptidylprolyl isomerase C (cyclophilin C)	SEQ ID NO: 10672

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
PPOX	Protoporphyrinogen oxidase	SEQ ID NOS: 10673-10683
PPP1CA	Protein phosphatase 1, catalytic subunit, alpha isozyme	SEQ ID NOS: 10684-10689
PPT1	Palmitoyl-protein thioesterase 1	SEQ ID NOS: 10690-10706
PPT2	Palmitoyl-protein thioesterase 2	SEQ ID NOS: 10707-10714
PPY	Pancreatic polypeptide	SEQ ID NOS: 10715-10719
PRAC2	Prostate cancer susceptibility candidate 2	SEQ ID NOS: 10720-10721
PRADC1	Protease-associated domain containing 1	SEQ ID NO: 10722
PRAP1	Proline-rich acidic protein 1	SEQ ID NOS: 10723-10724
PRB1	Proline-rich protein BstNI subfamily 1	SEQ ID NOS: 10725-10728
PRB2	Proline-rich protein BstNI subfamily 2	SEQ ID NOS: 10729-10730
PRB3	Proline-rich protein BstNI subfamily 3	SEQ ID NOS: 10731-10732
PRB4	Proline-rich protein BstNI subfamily 4	SEQ ID NOS: 10733-10736
PRCD	Progressive rod-cone degeneration	SEQ ID NOS: 10737-10738
PRCP	Prolylcarboxypeptidase (angiotensinase C)	SEQ ID NOS: 10739-10750
PRDM12	PR domain containing 12	SEQ ID NO: 10751
PRDX4	Peroxiredoxin 4	SEQ ID NOS: 10752-10755
PRELP	Proline/arginine-rich end leucine-rich repeat protein	SEQ ID NO: 10756
PRF1	Perforin 1 (pore forming protein)	SEQ ID NOS: 10757-10759
PRG2	Proteoglycan 2, bone marrow (natural killer cell activator, eosinophil granule major basic protein)	SEQ ID NOS: 10760-10762
PRG3	Proteoglycan 3	SEQ ID NO: 10763
PRG4	Proteoglycan 4	SEQ ID NOS: 10764-10769
PRH1	Proline-rich protein HaeIII subfamily 1	SEQ ID NOS: 10770-10772
PRH2	Proline-rich protein HaeIII subfamily 2	SEQ ID NOS: 10773-10774
PRKAG1	Protein kinase, AMP-activated, gamma 1 non-catalytic subunit	SEQ ID NOS: 10775-10789
PRKCSH	Protein kinase C substrate 80K-H	SEQ ID NOS: 10790-10799
PRKD1	Protein kinase D1	SEQ ID NOS: 10800-10805
PRL	Prolactin	SEQ ID NOS: 10806-10808
PRLH	Prolactin releasing hormone	SEQ ID NO: 10809
PRLR	Prolactin receptor	SEQ ID NOS: 10810-10828
PRNP	Prion protein	SEQ ID NOS: 10829-10832
PRNT	Prion protein (testis specific)	SEQ ID NO: 10833
PROC	Protein C (inactivator of coagulation factors Va and VIIIa)	SEQ ID NOS: 10834-10841
PROK1	Prokineticin 1	SEQ ID NO: 10842
PROK2	Prokineticin 2	SEQ ID NOS: 10843-10844
PROL1	Proline rich, lacrimal 1	SEQ ID NO: 9817
PROM1	Prominin 1	SEQ ID NOS: 10845-10856
PROS1	Protein S (alpha)	SEQ ID NOS: 10857-10860
PROZ	Protein Z, vitamin K-dependent plasma glycoprotein	SEQ ID NOS: 10861-10862
PRR27	Proline rich 27	SEQ ID NOS: 10863-10866
PRR4	Proline rich 4 (lacrimal)	SEQ ID NOS: 10867-10869
PRRG2	Proline rich Gla (G-carboxyglutamic acid) 2	SEQ ID NOS: 10870-10872

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
PRRT3	Proline-rich transmembrane protein 3	SEQ ID NOS: 10873-10875
PRRT4	Proline-rich transmembrane protein 4	SEQ ID NOS: 10876-10882
PRSS1	Protease, serine, 1 (trypsin 1)	SEQ ID NOS: 10883-10886
PRSS12	Protease, serine, 12 (neotrypsin, motopsin)	SEQ ID NO: 10887
PRSS16	Protease, serine, 16 (thymus)	SEQ ID NOS: 10888-10895
PRSS2	Protease, serine, 2 (trypsin 2)	SEQ ID NOS: 10896-10899
PRSS21	Protease, serine, 21 (testisin)	SEQ ID NOS: 10900-10905
PRSS22	Protease, serine, 22	SEQ ID NOS: 10906-10908
PRSS23	Protease, serine, 23	SEQ ID NOS: 10909-10912
PRSS27	Protease, serine 27	SEQ ID NOS: 10913-10915
PRSS3	Protease, serine, 3	SEQ ID NOS: 10916-10920
PRSS33	Protease, serine, 33	SEQ ID NOS: 10921-10924
PRSS35	Protease, serine, 35	SEQ ID NO: 10925
PRSS36	Protease, serine, 36	SEQ ID NOS: 10926-10929
PRSS37	Protease, serine, 37	SEQ ID NOS: 10930-10933
PRSS38	Protease, serine, 38	SEQ ID NO: 10934
PRSS42	Protease, serine, 42	SEQ ID NOS: 10935-10936
PRSS48	Protease, serine, 48	SEQ ID NOS: 10937-10938
PRSS50	Protease, serine, 50	SEQ ID NO: 10939
PRSS53	Protease, serine, 53	SEQ ID NO: 10940
PRSS54	Protease, serine, 54	SEQ ID NOS: 10941-10945
PRSS55	Protease, serine, 55	SEQ ID NOS: 10946-10948
PRSS56	Protease, serine, 56	SEQ ID NOS: 10949-10950
PRSS57	Protease, serine, 57	SEQ ID NOS: 10951-10952
PRSS58	Protease, serine, 58	SEQ ID NOS: 10953-10954
PRSS8	Protease, serine, 8	SEQ ID NOS: 10955-10958
PRTG	Protogenin	SEQ ID NOS: 10959-10962
PRTN3	Proteinase 3	SEQ ID NOS: 10963-10964
PSAP	Prosaposin	SEQ ID NOS: 10965-10968
PSAPL1	Prosaposin-like 1 (gene/pseudogene)	SEQ ID NO: 10969
PSG1	Pregnancy specific beta-1-glycoprotein 1	SEQ ID NOS: 10970-10977
PSG11	Pregnancy specific beta-1-glycoprotein 11	SEQ ID NOS: 10978-10982
PSG2	Pregnancy specific beta-1-glycoprotein 2	SEQ ID NOS: 10983-10984
PSG3	Pregnancy specific beta-1-glycoprotein 3	SEQ ID NOS: 10985-10988
PSG4	Pregnancy specific beta-1-glycoprotein 4	SEQ ID NOS: 10989-11000
PSG5	Pregnancy specific beta-1-glycoprotein 5	SEQ ID NOS: 11001-11006
PSG6	Pregnancy specific beta-1-glycoprotein 6	SEQ ID NOS: 11007-11012
PSG7	Pregnancy specific beta-1-glycoprotein 7 (gene/pseudogene)	SEQ ID NOS: 11013-11015
PSG8	Pregnancy specific beta-1-glycoprotein 8	SEQ ID NOS: 11016-11020
PSG9	Pregnancy specific beta-1-glycoprotein 9	SEQ ID NOS: 11021-11028

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
PSMD1	Proteasome 26S subunit, non-ATPase 1	SEQ ID NOS: 11029-11036
PSORS1C2	Psoriasis susceptibility 1 candidate 2	SEQ ID NO: 11037
PSPN	Persephin	SEQ ID NOS: 11038-11039
PTGDS	Prostaglandin D2 synthase 21 kDa (brain)	SEQ ID NOS: 11040-11044
PTGIR	Prostaglandin I2 (prostacyclin) receptor (IP)	SEQ ID NOS: 11045-11049
PTGS1	Prostaglandin-endoperoxide synthase 1 (prostaglandin G/H synthase and cyclooxygenase)	SEQ ID NOS: 11050-11058
PTGS2	Prostaglandin-endoperoxide synthase 2 (prostaglandin G/H synthase and cyclooxygenase)	SEQ ID NOS: 11059-11060
PTH	Parathyroid hormone	SEQ ID NOS: 11061-11062
PTH2	Parathyroid hormone 2	SEQ ID NO: 11063
PTH1H	Parathyroid hormone-like hormone	SEQ ID NOS: 11064-11072
PTK7	Protein tyrosine kinase 7 (inactive)	SEQ ID NOS: 11073-11088
PTN	Pleiotrophin	SEQ ID NOS: 11089-11090
PTPRA	Protein tyrosine phosphatase, receptor type, A	SEQ ID NOS: 11091-11098
PTPRB	Protein tyrosine phosphatase, receptor type, B	SEQ ID NOS: 11099-11106
PTPRC	Protein tyrosine phosphatase, receptor type, C	SEQ ID NOS: 11107-11117
PTPRCAP	Protein tyrosine phosphatase, receptor type, C-associated protein	SEQ ID NO: 11118
PTPRD	Protein tyrosine phosphatase, receptor type, D	SEQ ID NOS: 11119-11130
PTPRF	Protein tyrosine phosphatase, receptor type, F	SEQ ID NOS: 11131-11138
PTPRJ	Protein tyrosine phosphatase, receptor type, J	SEQ ID NOS: 11139-11144
PTPRO	Protein tyrosine phosphatase, receptor type, O	SEQ ID NOS: 11145-11153
PTPRS	Protein tyrosine phosphatase, receptor type, S	SEQ ID NOS: 11154-11161
PTTG1IP	Pituitary tumor-transforming 1 interacting protein	SEQ ID NOS: 11162-11165
PTX3	Pentraxin 3, long	SEQ ID NO: 11166
PTX4	Pentraxin 4, long	SEQ ID NOS: 11167-11169
PVR	Poliovirus receptor	SEQ ID NOS: 11170-11175
PVRL1	Poliovirus receptor-related 1 (herpesvirus entry mediator C)	SEQ ID NOS: 9286-9288
PXDN	Peroxidasin	SEQ ID NOS: 11176-11180
PXDNL	Peroxidasin-like	SEQ ID NOS: 11181-11183
PXYLP1	2-phosphoxylose phosphatase 1	SEQ ID NOS: 11184-11196
PYY	Peptide YY	SEQ ID NOS: 11197-11198
PZP	Pregnancy-zone protein	SEQ ID NOS: 11199-11200
QPCT	Glutaminyl-peptide cyclotransferase	SEQ ID NOS: 11201-11203
QPRT	Quinolinate phosphoribosyltransferase	SEQ ID NOS: 11204-11205
QRF	Pyroglutamylated RFamide peptide	SEQ ID NOS: 11206-11207
QSOX1	Quiescin Q6 sulfhydryl oxidase 1	SEQ ID NOS: 11208-11211
R3HDM1	R3H domain containing-like	SEQ ID NO: 11212
RAB26	RAB26, member RAS oncogene family	SEQ ID NOS: 11213-11216
RAB36	RAB36, member RAS oncogene family	SEQ ID NOS: 11217-11219
RAB9B	RAB9B, member RAS oncogene family	SEQ ID NO: 11220

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
RAET1E	Retinoic acid early transcript 1E	SEQ ID NOS: 11221-11226
RAET1G	Retinoic acid early transcript 1G	SEQ ID NOS: 11227-11229
RAMP2	Receptor (G protein-coupled) activity modifying protein 2	SEQ ID NOS: 11230-11234
RAPGEF5	Rap guanine nucleotide exchange factor (GEF) 5	SEQ ID NOS: 11235-11241
RARRES1	Retinoic acid receptor responder (tazarotene induced) 1	SEQ ID NOS: 11242-11243
RARRES2	Retinoic acid receptor responder (tazarotene induced) 2	SEQ ID NOS: 11244-11247
RASA2	RAS p21 protein activator 2	SEQ ID NOS: 11248-11250
RBM3	RNA binding motif (RNP1, RRM) protein 3	SEQ ID NOS: 11251-11253
RBP3	Retinol binding protein 3, interstitial	SEQ ID NO: 11254
RBP4	Retinol binding protein 4, plasma	SEQ ID NOS: 11255-11258
RCN1	Reticulocalbin 1, EF-hand calcium binding domain	SEQ ID NOS: 11259-11262
RCN2	Reticulocalbin 2, EF-hand calcium binding domain	SEQ ID NOS: 11263-11266
RCN3	Reticulocalbin 3, EF-hand calcium binding domain	SEQ ID NOS: 11267-11270
RCOR1	REST corepressor 1	SEQ ID NOS: 11271-11272
RDH11	Retinol dehydrogenase 11 (all-trans/9-cis/11-cis)	SEQ ID NOS: 11273-11280
RDH12	Retinol dehydrogenase 12 (all-trans/9-cis/11-cis)	SEQ ID NOS: 11281-11282
RDH13	Retinol dehydrogenase 13 (all-trans/9-cis)	SEQ ID NOS: 11283-11291
RDH5	Retinol dehydrogenase 5 (11-cis/9-cis)	SEQ ID NOS: 11292-11296
RDH8	Retinol dehydrogenase 8 (all-trans)	SEQ ID NOS: 11297-11298
REG1A	Regenerating islet-derived 1 alpha	SEQ ID NO: 11299
REG1B	Regenerating islet-derived 1 beta	SEQ ID NOS: 11300-11301
REG3A	Regenerating islet-derived 3 alpha	SEQ ID NOS: 11302-11304
REG3G	Regenerating islet-derived 3 gamma	SEQ ID NOS: 11305-11307
REG4	Regenerating islet-derived family, member 4	SEQ ID NOS: 11308-11311
RELN	Reelin	SEQ ID NOS: 11312-11315
RELT	RELT tumor necrosis factor receptor	SEQ ID NOS: 11316-11319
REN	Renin	SEQ ID NOS: 11320-11321
REPIN1	Replication initiator 1	SEQ ID NOS: 11322-11335
REPS2	RALBP1 associated Eps domain containing 2	SEQ ID NOS: 11336-11337
RET	Ret proto-oncogene	SEQ ID NOS: 11338-11343
RETN	Resistin	SEQ ID NOS: 11344-11346
RETNLB	Resistin like beta	SEQ ID NO: 11347
RETSAT	Retinol saturase (all-trans-retinol 13,14-reductase)	SEQ ID NOS: 11348-11352
RFNG	RFNG O-fucosylpeptide 3-beta-N-acetylglucosaminyltransferase	SEQ ID NOS: 11353-11355
RGCC	Regulator of cell cycle	SEQ ID NO: 11356
RGL4	Ral guanine nucleotide dissociation stimulator-like 4	SEQ ID NOS: 11357-11363
RGMA	Repulsive guidance molecule family member a	SEQ ID NOS: 11364-11373
RGMB	Repulsive guidance molecule family member b	SEQ ID NOS: 11374-11375
RHOQ	Ras homolog family member Q	SEQ ID NOS: 11376-11380
RIC3	RIC3 acetylcholine receptor chaperone	SEQ ID NOS: 11381-11388

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
RIMS1	Regulating synaptic membrane exocytosis 1	SEQ ID NOS: 11393-11408
RIPPLY1	Ripply transcriptional repressor 1	SEQ ID NOS: 11409-11410
RLN1	Relaxin 1	SEQ ID NO: 11411
RLN2	Relaxin 2	SEQ ID NOS: 11412-11413
RLN3	Relaxin 3	SEQ ID NOS: 11414-11415
RMDN1	Regulator of microtubule dynamics 1	SEQ ID NOS: 11416-11429
RNASE1	Ribonuclease, RNase A family, 1 (pancreatic)	SEQ ID NOS: 11430-11434
RNASE10	Ribonuclease, RNase A family, 10 (non-active)	SEQ ID NOS: 11435-11436
RNASE11	Ribonuclease, RNase A family, 11 (non-active)	SEQ ID NOS: 11437-11447
RNASE12	Ribonuclease, RNase A family, 12 (non-active)	SEQ ID NO: 11448
RNASE13	Ribonuclease, RNase A family, 13 (non-active)	SEQ ID NO: 11449
RNASE2	Ribonuclease, RNase A family, 2 (liver, eosinophil-derived neurotoxin)	SEQ ID NO: 11450
RNASE3	Ribonuclease, RNase A family, 3	SEQ ID NO: 11451
RNASE4	Ribonuclease, RNase A family, 4	SEQ ID NOS: 11452-11454
RNASE6	Ribonuclease, RNase A family, k6	SEQ ID NO: 11455
RNASE7	Ribonuclease, RNase A family, 7	SEQ ID NOS: 11456-11457
RNASE8	Ribonuclease, RNase A family, 8	SEQ ID NO: 11458
RNASE9	Ribonuclease, RNase A family, 9 (non-active)	SEQ ID NOS: 11459-11469
RNASEH1	Ribonuclease H1	SEQ ID NOS: 11470-11472
RNASET2	Ribonuclease T2	SEQ ID NOS: 11473-11480
RNF146	Ring finger protein 146	SEQ ID NOS: 11481-11492
RNF148	Ring finger protein 148	SEQ ID NOS: 11493-11494
RNF150	Ring finger protein 150	SEQ ID NOS: 11495-11499
RNF167	Ring finger protein 167	SEQ ID NOS: 11500-11510
RNF220	Ring finger protein 220	SEQ ID NOS: 11511-11517
RNF34	Ring finger protein 34, E3 ubiquitin protein ligase	SEQ ID NOS: 11518-11525
RNLS	Renalase, FAD-dependent amine oxidase	SEQ ID NOS: 11526-11528
RNPEP	Arginyl aminopeptidase (aminopeptidase B)	SEQ ID NOS: 11529-11534
ROR1	Receptor tyrosine kinase-like orphan receptor 1	SEQ ID NOS: 11535-11537
RP11-1236K1.1		SEQ ID NO: 4158
RP11-14J7.7		SEQ ID NOS: 674-675
RP11-196G11.1		SEQ ID NOS: 85-87
RP11-350O14.18		SEQ ID NO: 683
RP11-520P18.5		SEQ ID NO: 8194
RP11-812E19.9		SEQ ID NO: 89
RP11-903H12.5		SEQ ID NO: 676
RP11-977G19.10		SEQ ID NOS: 78-80
RP4-576H24.4		SEQ ID NOS: 670-672
RP4-608O15.3	Complement factor H-related protein 2	SEQ ID NO: 1649
RPL3	Ribosomal protein L3	SEQ ID NOS: 11538-11543
RPLP2	Ribosomal protein, large, P2	SEQ ID NOS: 11544-11546

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
RPN2	Ribophorin II	SEQ ID NOS: 11547-11553
RPS27L	Ribosomal protein S27-like	SEQ ID NOS: 11554-11559
RQCD1	RCD1 required for cell differentiation1 homolog (<i>S. pombe</i>)	SEQ ID NOS: 3100-3106
RS1	Retinoschisin 1	SEQ ID NO: 11560
RSF1	Remodeling and spacing factor 1	SEQ ID NOS: 11561-11567
RSPO1	R-spondin 1	SEQ ID NOS: 11568-11571
RSPO2	R-spondin 2	SEQ ID NOS: 11572-11579
RSPO3	R-spondin 3	SEQ ID NOS: 11580-11581
RSPO4	R-spondin 4	SEQ ID NOS: 11582-11583
RSPRY1	Ring finger and SPRY domain containing 1	SEQ ID NOS: 11584-11590
RTBDN	Retbindin	SEQ ID NOS: 11591-11603
RTN4RL1	Reticulon 4 receptor-like 1	SEQ ID NO: 11604
RTN4RL2	Reticulon 4 receptor-like 2	SEQ ID NOS: 11605-11607
SAA1	Serum amyloid A1	SEQ ID NOS: 11608-11610
SAA2	Serum amyloid A2	SEQ ID NOS: 11611-11616
SAA4	Serum amyloid A4, constitutive	SEQ ID NO: 11617
SAP30	Sin3A-associated protein, 30 kDa	SEQ ID NO: 11618
SAR1A	Secretion associated, Ras related GTPase 1A	SEQ ID NOS: 11619-11625
SARAF	Store-operated calcium entry-associated regulatory factor	SEQ ID NOS: 11626-11636
SARM1	Sterile alpha and TIR motif containing 1	SEQ ID NOS: 11637-11640
SATB1	SATB homeobox 1	SEQ ID NOS: 11641-11653
SAXO2	Stabilizer of axonemal microtubules 2	SEQ ID NOS: 11654-11658
SBSN	Suprabasin	SEQ ID NOS: 11659-11661
SBSPON	Somatomedin B and thrombospondin, type 1 domain containing	SEQ ID NO: 11662
SCARF1	Scavenger receptor class F, member 1	SEQ ID NOS: 11663-11667
SCG2	Secretogranin II	SEQ ID NOS: 11668-11670
SCG3	Secretogranin III	SEQ ID NOS: 11671-11673
SCG5	Secretogranin V	SEQ ID NOS: 11674-11678
SCGB1A1	Secretoglobin, family 1A, member 1 (uteroglobin)	SEQ ID NOS: 11679-11680
SCGB1C1	Secretoglobin, family 1C, member 1	SEQ ID NO: 11681
SCGB1C2	Secretoglobin, family 1C, member 2	SEQ ID NO: 11682
SCGB1D1	Secretoglobin, family 1D, member 1	SEQ ID NO: 11683
SCGB1D2	Secretoglobin, family 1D, member 2	SEQ ID NO: 11684
SCGB1D4	Secretoglobin, family 1D, member 4	SEQ ID NO: 11685
SCGB2A1	Secretoglobin, family 2A, member 1	SEQ ID NO: 11686
SCGB2A2	Secretoglobin, family 2A, member 2	SEQ ID NOS: 11687-11688
SCGB2B2	Secretoglobin, family 2B, member 2	SEQ ID NOS: 11689-11690
SCGB3A1	Secretoglobin, family 3A, member 1	SEQ ID NO: 11691
SCGB3A2	Secretoglobin, family 3A, member 2	SEQ ID NOS: 11692-11693
SCN1B	Sodium channel, voltage gated, type I beta subunit	SEQ ID NOS: 11694-11699
SCN3B	Sodium channel, voltage gated, type III beta subunit	SEQ ID NOS: 11700-11704
SCPEP1	Serine carboxypeptidase 1	SEQ ID NOS: 11705-11712
SCRG1	Stimulator of chondrogenesis 1	SEQ ID NOS: 11713-11714
SCT	Secretin	SEQ ID NO: 11715

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
SCUBE1	Signal peptide, CUB domain, EGF-like 1	SEQ ID NOS: 11716-11719
SCUBE2	Signal peptide, CUB domain, EGF-like 2	SEQ ID NOS: 11720-11726
SCUBE3	Signal peptide, CUB domain, EGF-like 3	SEQ ID NO: 11727
SDC1	Syndecan 1	SEQ ID NOS: 11728-11732
SDF2	Stromal cell-derived factor 2	SEQ ID NOS: 11733-11735
SDF2L1	Stromal cell-derived factor 2-like 1	SEQ ID NO: 11736
SDF4	Stromal cell derived factor 4	SEQ ID NOS: 11737-11740
SDHAF2	Succinate dehydrogenase complex assembly factor 2	SEQ ID NOS: 11741-11748
SDHAF4	Succinate dehydrogenase complex assembly factor 4	SEQ ID NO: 11749
SDHB	Succinate dehydrogenase complex, subunit B, iron sulfur (Ip)	SEQ ID NOS: 11750-11752
SDHD	Succinate dehydrogenase complex, subunit D, integral membrane protein	SEQ ID NOS: 11753-11762
SEC14L3	SEC14-like lipid binding 3	SEQ ID NOS: 11763-11769
SEC16A	SEC16 homolog A, endoplasmic reticulum export factor	SEQ ID NOS: 11770-11776
SEC16B	SEC16 homolog B, endoplasmic reticulum export factor	SEQ ID NOS: 11777-11780
SEC22C	SEC22 homolog C, vesicle trafficking protein	SEQ ID NOS: 11781-11793
SEC31A	SEC31 homolog A, COPII coat complex component	SEQ ID NOS: 11794-11823
SECISBP2	SECIS binding protein 2	SEQ ID NOS: 11824-11828
SECTM1	Secreted and transmembrane 1	SEQ ID NOS: 11829-11836
SEL1L	Sel-1 suppressor of lin-12-like (<i>C. elegans</i>)	SEQ ID NOS: 11837-11839
SELM	Selenoprotein M	SEQ ID NOS: 11847-11849
SELO	Selenoprotein O	SEQ ID NOS: 11854-11855
SEMA3A	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3A	SEQ ID NOS: 11862-11866
SEMA3B	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3B	SEQ ID NOS: 11867-11873
SEMA3C	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3C	SEQ ID NOS: 11874-11878
SEMA3E	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3E	SEQ ID NOS: 11879-11883
SEMA3F	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3F	SEQ ID NOS: 11884-11890
SEMA3G	Sema domain, immunoglobulin domain (Ig), short basic domain, secreted, (semaphorin) 3G	SEQ ID NOS: 11891-11893
SEMA4A	Sema domain, immunoglobulin domain (Ig), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 4A	SEQ ID NOS: 11894-11902
SEMA4B	Sema domain, immunoglobulin domain (Ig), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 4B	SEQ ID NOS: 11903-11913
SEMA4C	Sema domain, immunoglobulin domain (Ig), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 4C	SEQ ID NOS: 11914-11916
SEMA4D	Sema domain, immunoglobulin domain (Ig), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 4D	SEQ ID NOS: 11917-11930
SEMA4F	Sema domain, immunoglobulin domain (Ig), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 4F	SEQ ID NOS: 11931-11939
SEMA4G	Sema domain, immunoglobulin domain (Ig), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 4G	SEQ ID NOS: 11940-11947

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
SEMA5A	Sema domain, seven thrombospondin repeats (type 1 and type 1-like), transmembrane domain (TM) and short cytoplasmic domain, (semaphorin) 5A	SEQ ID NOS: 11948-11949
SEMA6A	Sema domain, transmembrane domain (TM), and cytoplasmic domain, (semaphorin) 6A	SEQ ID NOS: 11950-11957
SEMA6C	Sema domain, transmembrane domain (TM), and cytoplasmic domain, (semaphorin) 6C	SEQ ID NOS: 11958-11963
SEMA6D	Sema domain, transmembrane domain (TM), and cytoplasmic domain, (semaphorin) 6D	SEQ ID NOS: 11964-11977
SEMG1	Semenogelin I	SEQ ID NO: 11978
SEMG2	Semenogelin II	SEQ ID NO: 11979
SEPN1	Selenoprotein N, 1	SEQ ID NOS: 11850-11853
SEPP1	Selenoprotein P, plasma, 1	SEQ ID NOS: 11856-11861
SEPT15	15 kDa selenoprotein	SEQ ID NOS: 11840-11846
SEPT9	Septin 9	SEQ ID NOS: 11980-12016
SERPINA1	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 1	SEQ ID NOS: 12017-12033
SERPINA10	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 10	SEQ ID NOS: 12034-12037
SERPINA11	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 11	SEQ ID NO: 12038
SERPINA12	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 12	SEQ ID NOS: 12039-12040
SERPINA3	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 3	SEQ ID NOS: 673-12047
SERPINA4	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 4	SEQ ID NOS: 12048-12050
SERPINA5	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 5	SEQ ID NOS: 12051-12062
SERPINA6	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 6	SEQ ID NOS: 12063-12065
SERPINA7	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 7	SEQ ID NOS: 12066-12067
SERPINA9	Serpin peptidase inhibitor, clade A (alpha-1 antiproteinase, antitrypsin), member 9	SEQ ID NOS: 12068-12074
SERPINB2	Serpin peptidase inhibitor, clade B (ovalbumin), member 2	SEQ ID NOS: 12075-12079
SERPINC1	Serpin peptidase inhibitor, clade C (antithrombin), member 1	SEQ ID NOS: 12080-12081
SERPIND1	Serpin peptidase inhibitor, clade D (heparin cofactor), member 1	SEQ ID NOS: 12082-12083
SERPINE1	Serpin peptidase inhibitor, clade E (nexin, plasminogen activator inhibitor type 1), member 1	SEQ ID NO: 12084
SERPINE2	Serpin peptidase inhibitor, clade E (nexin, plasminogen activator inhibitor type 1), member 2	SEQ ID NOS: 12085-12091
SERPINE3	Serpin peptidase inhibitor, clade E (nexin, plasminogen activator inhibitor type 1), member 3	SEQ ID NOS: 12092-12095
SERPINF1	Serpin peptidase inhibitor, clade F (alpha-2 antiplasmin, pigment epithelium derived factor), member 1	SEQ ID NOS: 12096-12104
SERPINF2	Serpin peptidase inhibitor, clade F (alpha-2 antiplasmin, pigment epithelium derived factor), member 2	SEQ ID NOS: 12105-12109
SERPING1	Serpin peptidase inhibitor, clade G (C1 inhibitor), member 1	SEQ ID NOS: 12110-12120
SERPINH1	Serpin peptidase inhibitor, clade H (heat shock protein 47), member 1, (collagen binding protein 1)	SEQ ID NOS: 12121-12135
SERPINI1	Serpin peptidase inhibitor, clade I (neuroserpin), member 1	SEQ ID NOS: 12136-12140
SERPINI2	Serpin peptidase inhibitor, clade I (pancpin), member 2	SEQ ID NOS: 12141-12147
SETD8	SET domain containing (lysine methyltransferase) 8	SEQ ID NOS: 7589-7592

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
SEZ6L2	Seizure related 6 homolog (mouse)-like 2	SEQ ID NOS: 12148-12154
SFRP1	Secreted frizzled-related protein 1	SEQ ID NOS: 12155-12156
SFRP2	Secreted frizzled-related protein 2	SEQ ID NO: 12157
SFRP4	Secreted frizzled-related protein 4	SEQ ID NOS: 12158-12159
SFRP5	Secreted frizzled-related protein 5	SEQ ID NO: 12160
SFTA2	Surfactant associated 2	SEQ ID NOS: 12161-12162
SFTPA1	Surfactant protein A1	SEQ ID NOS: 12163-12167
SFTPA2	Surfactant protein A2	SEQ ID NOS: 12168-12172
SFTPB	Surfactant protein B	SEQ ID NOS: 12173-12177
SFTPD	Surfactant protein D	SEQ ID NOS: 12178-12179
SFXN5	Sideroflexin 5	SEQ ID NOS: 12180-12184
SGCA	Sarcoglycan, alpha (50 kDa dystrophin-associated glycoprotein)	SEQ ID NOS: 12185-12192
SGSH	N-sulfoglucosamine sulfohydrolase	SEQ ID NOS: 12193-12201
SH3RF3	SH3 domain containing ring finger 3	SEQ ID NO: 12202
SHBG	Sex hormone-binding globulin	SEQ ID NOS: 12203-12221
SHE	Src homology 2 domain containing E	SEQ ID NOS: 12222-12224
SHH	Sonic hedgehog	SEQ ID NOS: 12225-12228
SHKBP1	SH3KBP1 binding protein 1	SEQ ID NOS: 12229-12244
SIAE	Sialic acid acetyltransferase	SEQ ID NOS: 12245-12247
SIDT2	SID1 transmembrane family, member 2	SEQ ID NOS: 12248-12257
SIGLEC10	Sialic acid binding Ig-like lectin 10	SEQ ID NOS: 12258-12266
SIGLEC6	Sialic acid binding Ig-like lectin 6	SEQ ID NOS: 12267-12272
SIGLEC7	Sialic acid binding Ig-like lectin 7	SEQ ID NOS: 12273-12277
SIGLECL1	SIGLEC family like 1	SEQ ID NOS: 12278-12283
SIGMAR1	Sigma non-opioid intracellular receptor 1	SEQ ID NOS: 12284-12287
SIL1	SIL1 nucleotide exchange factor	SEQ ID NOS: 12288-12296
SIRPB1	Signal-regulatory protein beta 1	SEQ ID NOS: 12297-12309
SIRPD	Signal-regulatory protein delta	SEQ ID NOS: 12310-12312
SLAMF1	Signaling lymphocytic activation molecule family member 1	SEQ ID NOS: 12313-12315
SLAMF7	SLAM family member 7	SEQ ID NOS: 12316-12324
SLC10A3	Solute carrier family 10, member 3	SEQ ID NOS: 12325-12329
SLC15A3	Solute carrier family 15 (oligopeptide transporter), member 3	SEQ ID NOS: 12330-12335
SLC25A14	Solute carrier family 25 (mitochondrial carrier, brain), member 14	SEQ ID NOS: 12336-12342
SLC25A25	Solute carrier family 25 (mitochondrial carrier; phosphate carrier), member 25	SEQ ID NOS: 12343-12349
SLC2A5	Solute carrier family 2 (facilitated glucose/fructose transporter), member 5	SEQ ID NOS: 12350-12358
SLC35E3	Solute carrier family 35, member E3	SEQ ID NOS: 12359-12360
SLC39A10	Solute carrier family 39 (zinc transporter), member 10	SEQ ID NOS: 12361-12367
SLC39A14	Solute carrier family 39 (zinc transporter), member 14	SEQ ID NOS: 12368-12378
SLC39A4	Solute carrier family 39 (zinc transporter), member 4	SEQ ID NOS: 12379-12381

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
SLC39A5	Solute carrier family 39 (zinc transporter), member 5	SEQ ID NOS: 12382-12388
SLC3A1	Solute carrier family 3 (amino acid transporter heavy chain), member 1	SEQ ID NOS: 12389-12398
SLC51A	Solute carrier family 51, alpha subunit	SEQ ID NOS: 12399-12403
SLC52A2	Solute carrier family 52 (riboflavin transporter), member 2	SEQ ID NOS: 12404-12414
SLC5A6	Solute carrier family 5 (sodium/multivitamin and iodide cotransporter), member 6	SEQ ID NOS: 12415-12425
SLC6A9	Solute carrier family 6 (neurotransmitter transporter, glycine), member 9	SEQ ID NOS: 12426-12433
SLC8A1	Solute carrier family 8 (sodium/calcium exchanger), member 1	SEQ ID NOS: 12434-12445
SLC8B1	Solute carrier family 8 (sodium/lithium/calcium exchanger), member B1	SEQ ID NOS: 12446-12456
SLC9A6	Solute carrier family 9, subfamily A (NHE6, cation proton antiporter 6), member 6	SEQ ID NOS: 12457-12468
SLCO1A2	Solute carrier organic anion transporter family, member 1A2	SEQ ID NOS: 12469-12481
SLIT1	Slit guidance ligand 1	SEQ ID NOS: 12482-12485
SLIT2	Slit guidance ligand 2	SEQ ID NOS: 12486-12494
SLIT3	Slit guidance ligand 3	SEQ ID NOS: 12495-12497
SLITRK3	SLIT and NTRK-like family, member 3	SEQ ID NOS: 12498-12500
SLPI	Secretory leukocyte peptidase inhibitor	SEQ ID NO: 12501
SLTM	SAFB-like, transcription modulator	SEQ ID NOS: 12502-12515
SLURP1	Secreted LY6/PLAUR domain containing 1	SEQ ID NO: 12516
SMARCA2	SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 2	SEQ ID NOS: 12517-12562
SMG6	SMG6 nonsense mediated mRNA decay factor	SEQ ID NOS: 12563-12574
SMIM7	Small integral membrane protein 7	SEQ ID NOS: 12575-12591
SMOC1	SPARC related modular calcium binding 1	SEQ ID NOS: 12592-12593
SMOC2	SPARC related modular calcium binding 2	SEQ ID NOS: 12594-12598
SMPDL3A	Sphingomyelin phosphodiesterase, acid-like 3A	SEQ ID NOS: 12599-12600
SMPDL3B	Sphingomyelin phosphodiesterase, acid-like 3B	SEQ ID NOS: 12601-12605
SMR3A	Submaxillary gland androgen regulated protein 3A	SEQ ID NO: 12606
SMR3B	Submaxillary gland androgen regulated protein 3B	SEQ ID NOS: 12607-12609
SNED1	Sushi, nidogen and EGF-like domains 1	SEQ ID NOS: 12610-12616
SNTB1	Syntrophin, beta 1 (dystrophin-associated protein A1, 59 kDa, basic component 1)	SEQ ID NOS: 12617-12619
SNTB2	Syntrophin, beta 2 (dystrophin-associated protein A1, 59 kDa, basic component 2)	SEQ ID NOS: 12620-12624
SNX14	Sorting nexin 14	SEQ ID NOS: 12625-12636
SOD3	Superoxide dismutase 3, extracellular	SEQ ID NOS: 12637-12638
SOST	Sclerostin	SEQ ID NO: 12639
SOSTDC1	Sclerostin domain containing 1	SEQ ID NOS: 12640-12641
SOWAHA	Sosondowah ankyrin repeat domain family member A	SEQ ID NO: 12642
SPACA3	Sperm acrosome associated 3	SEQ ID NOS: 12643-12645
SPACA4	Sperm acrosome associated 4	SEQ ID NO: 12646
SPACA5	Sperm acrosome associated 5	SEQ ID NOS: 12647-12648
SPACA5B	Sperm acrosome associated 5B	SEQ ID NO: 12649

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
SPACA7	Sperm acrosome associated 7	SEQ ID NOS: 12650-12653
SPAG11A	Sperm associated antigen 11A	SEQ ID NOS: 12654-12662
SPAG11B	Sperm associated antigen 11B	SEQ ID NOS: 12663-12671
SPARC	Secreted protein, acidic, cysteine-rich (osteonectin)	SEQ ID NOS: 12672-12676
SPARCL1	SPARC-like 1 (hevin)	SEQ ID NOS: 12677-12686
SPATA20	Spermatogenesis associated 20	SEQ ID NOS: 12687-12700
SPESP1	Sperm equatorial segment protein 1	SEQ ID NO: 12701
SPINK1	Serine peptidase inhibitor, Kazal type 1	SEQ ID NOS: 12702-12703
SPINK13	Serine peptidase inhibitor, Kazal type 13 (putative)	SEQ ID NOS: 12704-12706
SPINK14	Serine peptidase inhibitor, Kazal type 14 (putative)	SEQ ID NOS: 12707-12708
SPINK2	Serine peptidase inhibitor, Kazal type 2 (acrosin-trypsin inhibitor)	SEQ ID NOS: 12709-12714
SPINK4	Serine peptidase inhibitor, Kazal type 4	SEQ ID NOS: 12715-12716
SPINK5	Serine peptidase inhibitor, Kazal type 5	SEQ ID NOS: 12717-12722
SPINK6	Serine peptidase inhibitor, Kazal type 6	SEQ ID NOS: 12723-12725
SPINK7	Serine peptidase inhibitor, Kazal type 7 (putative)	SEQ ID NOS: 12726-12727
SPINK8	Serine peptidase inhibitor, Kazal type 8 (putative)	SEQ ID NO: 12728
SPINK9	Serine peptidase inhibitor, Kazal type 9	SEQ ID NOS: 12729-12730
SPINT1	Serine peptidase inhibitor, Kunitz type 1	SEQ ID NOS: 12731-12738
SPINT2	Serine peptidase inhibitor, Kunitz type, 2	SEQ ID NOS: 12739-12746
SPINT3	Serine peptidase inhibitor, Kunitz type, 3	SEQ ID NO: 12747
SPINT4	Serine peptidase inhibitor, Kunitz type 4	SEQ ID NO: 12748
SPOCK1	Sparc/osteonectin, cwcv and kazal-like domains proteoglycan (testican) 1	SEQ ID NOS: 12749-12752
SPOCK2	Sparc/osteonectin, cwcv and kazal-like domains proteoglycan (testican) 2	SEQ ID NOS: 12753-12756
SPOCK3	Sparc/osteonectin, cwcv and kazal-like domains proteoglycan (testican) 3	SEQ ID NOS: 12757-12782
SPON1	Spondin 1, extracellular matrix protein	SEQ ID NO: 12783
SPON2	Spondin 2, extracellular matrix protein	SEQ ID NOS: 12784-12793
SPP1	Secreted phosphoprotein 1	SEQ ID NOS: 12794-12798
SPP2	Secreted phosphoprotein 2, 24 kDa	SEQ ID NOS: 12799-12801
SPRN	Shadow of prion protein homolog (zebrafish)	SEQ ID NO: 12802
SPRYD3	SPRY domain containing 3	SEQ ID NOS: 12803-12806
SPRYD4	SPRY domain containing 4	SEQ ID NO: 12807
SPTY2D1-AS1	SPTY2D1 antisense RNA 1	SEQ ID NOS: 12808-12813
SPX	Spexin hormone	SEQ ID NOS: 12814-12815
SRGN	Serglycin	SEQ ID NO: 12816
SRL	Sarcolumenin	SEQ ID NOS: 12817-12819
SRP14	Signal recognition particle 14 kDa (homologous Alu RNA binding protein)	SEQ ID NOS: 12820-12823
SRPX	Sushi-repeat containing protein, X-linked	SEQ ID NOS: 12824-12827
SRPX2	Sushi-repeat containing protein, X-linked 2	SEQ ID NOS: 12828-12831
SSC4D	Scavenger receptor cysteine rich family, 4 domains	SEQ ID NO: 12832
SSC5D	Scavenger receptor cysteine rich family, 5 domains	SEQ ID NOS: 12833-12836
SSPO	SCO-spondin	SEQ ID NO: 12837

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
SSR2	Signal sequence receptor, beta (translocon-associated protein beta)	SEQ ID NOS: 12838-12847
SST	Somatostatin	SEQ ID NO: 12848
ST3GAL1	ST3 beta-galactoside alpha-2,3-sialyltransferase 1	SEQ ID NOS: 12849-12856
ST3GAL4	ST3 beta-galactoside alpha-2,3-sialyltransferase 4	SEQ ID NOS: 12857-12872
ST6GAL1	ST6 beta-galactosamide alpha-2,6-sialyltransferase 1	SEQ ID NOS: 12873-12888
ST6GALNAC2	ST6 (alpha-N-acetyl-neuraminyl-2,3-beta-galactosyl-1,3)-N-acetylgalactosaminide alpha-2,6-sialyltransferase 2	SEQ ID NOS: 12889-12893
ST6GALNAC5	ST6 (alpha-N-acetyl-neuraminyl-2,3-beta-galactosyl-1,3)-N-acetylgalactosaminide alpha-2,6-sialyltransferase 5	SEQ ID NOS: 12894-12895
ST6GALNAC6	ST6 (alpha-N-acetyl-neuraminyl-2,3-beta-galactosyl-1,3)-N-acetylgalactosaminide alpha-2,6-sialyltransferase 6	SEQ ID NOS: 12896-12903
ST8SIA2	ST8 alpha-N-acetyl-neuraminide alpha-2,8-sialyltransferase 2	SEQ ID NOS: 12904-12906
ST8SIA4	ST8 alpha-N-acetyl-neuraminide alpha-2,8-sialyltransferase 4	SEQ ID NOS: 12907-12909
ST8SIA6	ST8 alpha-N-acetyl-neuraminide alpha-2,8-sialyltransferase 6	SEQ ID NOS: 12910-12911
STARD7	StAR-related lipid transfer (START) domain containing 7	SEQ ID NOS: 12912-12913
STATH	Statherin	SEQ ID NOS: 12914-12916
STC1	Stanniocalcin 1	SEQ ID NOS: 12917-12918
STC2	Stanniocalcin 2	SEQ ID NOS: 12919-12921
STMND1	Stathmin domain containing 1	SEQ ID NOS: 12922-12923
STOML2	Stomatin (EPB72)-like 2	SEQ ID NOS: 12926-12929
STOX1	Storkhead box 1	SEQ ID NOS: 12930-12934
STRC	Stereocilin	SEQ ID NOS: 12935-12940
SUCLG1	Succinate-CoA ligase, alpha subunit	SEQ ID NOS: 12941-12942
SUDS3	SDS3 homolog, SIN3A corepressor complex component	SEQ ID NO: 12943
SULF1	Sulfatase 1	SEQ ID NOS: 12944-12954
SULF2	Sulfatase 2	SEQ ID NOS: 12955-12959
SUMF1	Sulfatase modifying factor 1	SEQ ID NOS: 12960-12964
SUMF2	Sulfatase modifying factor 2	SEQ ID NOS: 12965-12978
SUSD1	Sushi domain containing 1	SEQ ID NOS: 12979-12984
SUSD5	Sushi domain containing 5	SEQ ID NOS: 12985-12986
SVEP1	Sushi, von Willebrand factor type A, EGF and pentraxin domain containing 1	SEQ ID NOS: 12987-12989
SWSAP1	SWIM-type zinc finger 7 associated protein 1	SEQ ID NO: 12990
SYAP1	Synapse associated protein 1	SEQ ID NO: 12991
SYCN	Syncollin	SEQ ID NO: 12992
TAC1	Tachykinin, precursor 1	SEQ ID NOS: 12993-12995
TAC3	Tachykinin 3	SEQ ID NOS: 12996-13005
TAC4	Tachykinin 4 (hemokinin)	SEQ ID NOS: 13006-13011
TAGLN2	Transgelin 2	SEQ ID NOS: 13012-13015
TAPBP	TAP binding protein (tapasin)	SEQ ID NOS: 13016-13021
TAPBPL	TAP binding protein-like	SEQ ID NOS: 13022-13023
TBL2	Transducin (beta)-like 2	SEQ ID NOS: 13024-13036

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
TBX10	T-box 10	SEQ ID NO: 13037
TCF12	Transcription factor 12	SEQ ID NOS: 13038-13051
TCN1	Transcobalamin I (vitamin B12 binding protein, R binder family)	SEQ ID NO: 13052
TCN2	Transcobalamin II	SEQ ID NOS: 13053-13056
TCTN1	Tectonic family member 1	SEQ ID NOS: 13057-13075
TCTN3	Tectonic family member 3	SEQ ID NOS: 13076-13080
TDP2	Tyrosyl-DNA phosphodiesterase 2	SEQ ID NOS: 13081-13082
TEK	TEK tyrosine kinase, endothelial	SEQ ID NOS: 13097-13101
TEPP	Testis, prostate and placenta expressed	SEQ ID NOS: 13102-13103
TEX101	Testis expressed 101	SEQ ID NOS: 13104-13105
TEX264	Testis expressed 264	SEQ ID NOS: 13106-13117
TF	Transferrin	SEQ ID NOS: 13121-13127
TFAM	Transcription factor A, mitochondrial	SEQ ID NOS: 13128-13130
TFF1	Trefoil factor 1	SEQ ID NO: 13131
TFF2	Trefoil factor 2	SEQ ID NO: 13132
TFF3	Trefoil factor 3 (intestinal)	SEQ ID NOS: 13133-13135
TFPI	Tissue factor pathway inhibitor (lipoprotein-associated coagulation inhibitor)	SEQ ID NOS: 13136-13145
TFPI2	Tissue factor pathway inhibitor 2	SEQ ID NOS: 13146-13147
TG	Thyroglobulin	SEQ ID NOS: 13148-13157
TGFB1	Transforming growth factor, beta 1	SEQ ID NOS: 13158-13159
TGFB2	Transforming growth factor, beta 2	SEQ ID NOS: 13160-13161
TGFB3	Transforming growth factor, beta 3	SEQ ID NOS: 13162-13163
TGFB1	Transforming growth factor, beta-induced, 68 kDa	SEQ ID NOS: 13164-13171
TGFBR1	Transforming growth factor, beta receptor 1	SEQ ID NOS: 13172-13181
TGFBR3	Transforming growth factor, beta receptor III	SEQ ID NOS: 13182-13188
THBS1	Thrombospondin 1	SEQ ID NOS: 13189-13190
THBS2	Thrombospondin 2	SEQ ID NOS: 13191-13193
THBS3	Thrombospondin 3	SEQ ID NOS: 13194-13198
THBS4	Thrombospondin 4	SEQ ID NOS: 13199-13200
THOC3	THO complex 3	SEQ ID NOS: 13201-13210
THPO	Thrombopoietin	SEQ ID NOS: 13211-13216
THSD4	Thrombospondin, type I, domain containing 4	SEQ ID NOS: 13217-13220
THY1	Thy-1 cell surface antigen	SEQ ID NOS: 13221-13226
TIE1	Tyrosine kinase with immunoglobulin-like and EGF-like domains 1	SEQ ID NOS: 13227-13228
TIMMDC1	Translocase of inner mitochondrial membrane domain containing 1	SEQ ID NOS: 13229-13236
TIMP1	TIMP metalloproteinase inhibitor 1	SEQ ID NOS: 13237-13241
TIMP2	TIMP metalloproteinase inhibitor 2	SEQ ID NOS: 13242-13246
TIMP3	TIMP metalloproteinase inhibitor 3	SEQ ID NO: 13247
TIMP4	TIMP metalloproteinase inhibitor 4	SEQ ID NO: 13248
TINAGL1	Tubulointerstitial nephritis antigen-like 1	SEQ ID NOS: 13249-13251

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
TINF2	TERF1 (TRF1)-interacting nuclear factor 2	SEQ ID NOS: 13252-13261
TLL2	Tolloid-like 2	SEQ ID NO: 13262
TLR1	Toll-like receptor 1	SEQ ID NOS: 13263-13268
TLR3	Toll-like receptor 3	SEQ ID NOS: 13269-13271
TM2D2	TM2 domain containing 2	SEQ ID NOS: 13272-13277
TM2D3	TM2 domain containing 3	SEQ ID NOS: 13278-13285
TM7SF3	Transmembrane 7 superfamily member 3	SEQ ID NOS: 13286-13300
TM9SF1	Transmembrane 9 superfamily member 1	SEQ ID NOS: 13301-13311
TMCO6	Transmembrane and coiled-coil domains 6	SEQ ID NOS: 13312-13319
TMED1	Transmembrane p24 trafficking protein 1	SEQ ID NOS: 13320-13326
TMED2	Transmembrane p24 trafficking protein 2	SEQ ID NOS: 13327-13329
TMED3	Transmembrane p24 trafficking protein 3	SEQ ID NOS: 13330-13333
TMED4	Transmembrane p24 trafficking protein 4	SEQ ID NOS: 13334-13336
TMED5	Transmembrane p24 trafficking protein 5	SEQ ID NOS: 13337-13340
TMED7	Transmembrane p24 trafficking protein 7	SEQ ID NOS: 13341-13342
TMED7-TICAM2	TMED7-TICAM2 readthrough	SEQ ID NOS: 13343-13344
TMEM108	Transmembrane protein 108	SEQ ID NOS: 13345-13353
TMEM116	Transmembrane protein 116	SEQ ID NOS: 13354-13365
TMEM119	Transmembrane protein 119	SEQ ID NOS: 13366-13369
TMEM155	Transmembrane protein 155	SEQ ID NOS: 13370-13373
TMEM168	Transmembrane protein 168	SEQ ID NOS: 13374-13379
TMEM178A	Transmembrane protein 178A	SEQ ID NOS: 13380-13381
TMEM179	Transmembrane protein 179	SEQ ID NOS: 13382-13387
TMEM196	Transmembrane protein 196	SEQ ID NOS: 13388-13392
TMEM199	Transmembrane protein 199	SEQ ID NOS: 13393-13396
TMEM205	Transmembrane protein 205	SEQ ID NOS: 13397-13410
TMEM213	Transmembrane protein 213	SEQ ID NOS: 13411-13414
TMEM25	Transmembrane protein 25	SEQ ID NOS: 13415-13431
TMEM30C	Transmembrane protein 30C	SEQ ID NO: 13432
TMEM38B	Transmembrane protein 38B	SEQ ID NOS: 13433-13437
TMEM44	Transmembrane protein 44	SEQ ID NOS: 13438-13447
TMEM52	Transmembrane protein 52	SEQ ID NOS: 13448-13452
TMEM52B	Transmembrane protein 52B	SEQ ID NOS: 13453-13455
TMEM59	Transmembrane protein 59	SEQ ID NOS: 13456-13463
TMEM67	Transmembrane protein 67	SEQ ID NOS: 13464-13475
TMEM70	Transmembrane protein 70	SEQ ID NOS: 13476-13478
TMEM87A	Transmembrane protein 87A	SEQ ID NOS: 13479-13488
TMEM94	Transmembrane protein 94	SEQ ID NOS: 13489-13504
TMEM95	Transmembrane protein 95	SEQ ID NOS: 13505-13507

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
TMIGD1	Transmembrane and immunoglobulin domain containing 1	SEQ ID NOS: 13508-13509
TMPRSS12	Transmembrane (C-terminal) protease, serine 12	SEQ ID NOS: 13510-13511
TMPRSS5	Transmembrane protease, serine 5	SEQ ID NOS: 13512-13523
TMUB1	Transmembrane and ubiquitin-like domain containing 1	SEQ ID NOS: 13524-13530
TMX2	Thioredoxin-related transmembrane protein 2	SEQ ID NOS: 13531-13538
TMX3	Thioredoxin-related transmembrane protein 3	SEQ ID NOS: 13539-13546
TNC	Tenascin C	SEQ ID NOS: 13547-13555
TNFAIP6	Tumor necrosis factor, alpha-induced protein 6	SEQ ID NO: 13556
TNFRSF11A	Tumor necrosis factor receptor superfamily, member 11a, NFkB activator	SEQ ID NOS: 13557-13561
TNFRSF11B	Tumor necrosis factor receptor superfamily, member 11b	SEQ ID NOS: 13562-13563
TNFRSF12A	Tumor necrosis factor receptor superfamily, member 12A	SEQ ID NOS: 13564-13569
TNFRSF14	Tumor necrosis factor receptor superfamily, member 14	SEQ ID NOS: 13570-13576
TNFRSF18	Tumor necrosis factor receptor superfamily, member 18	SEQ ID NOS: 13577-13580
TNFRSF1A	Tumor necrosis factor receptor superfamily, member 1A	SEQ ID NOS: 13581-13589
TNFRSF1B	Tumor necrosis factor receptor superfamily, member 1B	SEQ ID NOS: 13590-13591
TNFRSF25	Tumor necrosis factor receptor superfamily, member 25	SEQ ID NOS: 13592-13603
TNFRSF6B	Tumor necrosis factor receptor superfamily, member 6b, decoy	SEQ ID NO: 13604
TNFSF11	Tumor necrosis factor (ligand) superfamily, member 11	SEQ ID NOS: 13605-13609
TNFSF12	Tumor necrosis factor (ligand) superfamily, member 12	SEQ ID NOS: 13610-13611
TNFSF12-TNFSF13	TNFSF12-TNFSF13 readthrough	SEQ ID NO: 13612
TNFSF15	Tumor necrosis factor (ligand) superfamily, member 15	SEQ ID NOS: 13613-13614
TNN	Tenascin N	SEQ ID NOS: 13615-13617
TNR	Tenascin R	SEQ ID NOS: 13618-13620
TNXB	Tenascin XB	SEQ ID NOS: 13621-13627
TOMM7	Translocase of outer mitochondrial membrane 7 homolog (yeast)	SEQ ID NOS: 13634-13637
TOP1MT	Topoisomerase (DNA) I, mitochondrial	SEQ ID NOS: 13638-13652
TOR1A	Torsin family 1, member A (torsin A)	SEQ ID NO: 13653
TOR1B	Torsin family 1, member B (torsin B)	SEQ ID NOS: 13654-13655
TOR2A	Torsin family 2, member A	SEQ ID NOS: 13656-13662
TOR3A	Torsin family 3, member A	SEQ ID NOS: 13663-13667
TPD52	Tumor protein D52	SEQ ID NOS: 13668-13680
TPO	Thyroid peroxidase	SEQ ID NOS: 13681-13691
TPP1	Tripeptidyl peptidase I	SEQ ID NOS: 13692-13709
TPSAB1	Tryptase alpha/beta 1	SEQ ID NOS: 13710-13712
TPSB2	Tryptase beta 2 (gene/pseudogene)	SEQ ID NOS: 13713-13715
TPSD1	Tryptase delta 1	SEQ ID NOS: 13716-13717
TPST1	Tyrosylprotein sulfotransferase 1	SEQ ID NOS: 13718-13720
TPST2	Tyrosylprotein sulfotransferase 2	SEQ ID NOS: 13721-13729

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
TRABD2A	TraB domain containing 2A	SEQ ID NOS: 13730-13732
TRABD2B	TraB domain containing 2B	SEQ ID NO: 13733
TREH	Trehalase (brush-border membrane glycoprotein)	SEQ ID NOS: 13734-13736
TREM1	Triggering receptor expressed on myeloid cells 1	SEQ ID NOS: 13737-13740
TREM2	Triggering receptor expressed on myeloid cells 2	SEQ ID NOS: 13741-13743
TRH	Thyrotropin-releasing hormone	SEQ ID NOS: 13744-13745
TRIM24	Tripartite motif containing 24	SEQ ID NOS: 13746-13747
TRIM28	Tripartite motif containing 28	SEQ ID NOS: 13748-13753
TRIO	Trio Rho guanine nucleotide exchange factor	SEQ ID NOS: 13754-13760
TRNP1	TMF1-regulated nuclear protein 1	SEQ ID NOS: 13761-13762
TSC22D4	TSC22 domain family, member 4	SEQ ID NOS: 13763-13766
TSHB	Thyroid stimulating hormone, beta	SEQ ID NOS: 13767-13768
TSHR	Thyroid stimulating hormone receptor	SEQ ID NOS: 13769-13776
TSKU	Tsukushi, small leucine rich proteoglycan	SEQ ID NOS: 13777-13781
TSLP	Thymic stromal lymphopoietin	SEQ ID NOS: 13782-13784
TSPAN3	Tetraspanin 3	SEQ ID NOS: 13785-13790
TSPAN31	Tetraspanin 31	SEQ ID NOS: 13791-13797
TSPEAR	Thrombospondin-type laminin G domain and EAR repeats	SEQ ID NOS: 13798-13801
TTC13	Tetratricopeptide repeat domain 13	SEQ ID NOS: 13802-13808
TTC19	Tetratricopeptide repeat domain 19	SEQ ID NOS: 13809-13814
TTC9B	Tetratricopeptide repeat domain 9B	SEQ ID NO: 13815
TTLL11	Tubulin tyrosine ligase-like family member 11	SEQ ID NOS: 13816-13820
TTR	Transthyretin	SEQ ID NOS: 13821-13823
TWSG1	Twisted gastrulation BMP signaling modulator 1	SEQ ID NOS: 13824-13826
TXNDC12	Thioredoxin domain containing 12 (endoplasmic reticulum)	SEQ ID NOS: 13827-13829
TXNDC15	Thioredoxin domain containing 15	SEQ ID NOS: 13830-13836
TXNDC5	Thioredoxin domain containing 5 (endoplasmic reticulum)	SEQ ID NOS: 13837-13838
TXNRD2	Thioredoxin reductase 2	SEQ ID NOS: 13839-13851
TYRP1	Tyrosinase-related protein 1	SEQ ID NOS: 13852-13854
UBAC2	UBA domain containing 2	SEQ ID NOS: 13855-13859
UBALD1	UBA-like domain containing 1	SEQ ID NOS: 13860-13868
UBAP2	Ubiquitin associated protein 2	SEQ ID NOS: 13869-13875
UBXN8	UBX domain protein 8	SEQ ID NOS: 13876-13882
UCMA	Upper zone of growth plate and cartilage matrix associated	SEQ ID NOS: 13883-13884
UCN	Urocortin	SEQ ID NO: 13885
UCN2	Urocortin 2	SEQ ID NO: 13886
UCN3	Urocortin 3	SEQ ID NO: 13887
UGGT2	UDP-glucose glycoprotein glucosyltransferase 2	SEQ ID NOS: 13888-13893
UGT1A10	UDP glucuronosyltransferase 1 family, polypeptide A10	SEQ ID NOS: 13894-13895
UGT2A1	UDP glucuronosyltransferase 2 family, polypeptide A1, complex locus	SEQ ID NOS: 13896-13900

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
UGT2B11	UDP glucuronosyltransferase 2 family, polypeptide B11	SEQ ID NO: 13901
UGT2B28	UDP glucuronosyltransferase 2 family, polypeptide B28	SEQ ID NOS: 13902-13903
UGT2B4	UDP glucuronosyltransferase 2 family, polypeptide B4	SEQ ID NOS: 13904-13907
UGT2B7	UDP glucuronosyltransferase 2 family, polypeptide B7	SEQ ID NOS: 13908-13911
UGT3A1	UDP glycosyltransferase 3 family, polypeptide A1	SEQ ID NOS: 13912-13917
UGT3A2	UDP glycosyltransferase 3 family, polypeptide A2	SEQ ID NOS: 13918-13921
UGT8	UDP glycosyltransferase 8	SEQ ID NOS: 13922-13924
ULBP3	UL16 binding protein 3	SEQ ID NOS: 13925-13926
UMOD	Uromodulin	SEQ ID NOS: 13927-13938
UNC5C	Unc-5 netrin receptor C	SEQ ID NOS: 13939-13943
UPK3B	Uropod 3	SEQ ID NOS: 13944-13946
USP11	Ubiquitin specific peptidase 11	SEQ ID NOS: 13947-13950
USP14	Ubiquitin specific peptidase 14 (tRNA-guanine transglycosylase)	SEQ ID NOS: 13951-13957
USP3	Ubiquitin specific peptidase 3	SEQ ID NOS: 13958-13973
UTS2	Urotensin 2	SEQ ID NOS: 13984-13986
UTS2B	Urotensin 2B	SEQ ID NOS: 13987-13992
UTY	Ubiquitously transcribed tetratricopeptide repeat containing, Y-linked	SEQ ID NOS: 13993-14005
UXS1	UDP-glucuronate decarboxylase 1	SEQ ID NOS: 14006-14013
VASH1	Vasohibin 1	SEQ ID NOS: 14014-14016
VCAN	Versican	SEQ ID NOS: 14017-14023
VEGFA	Vascular endothelial growth factor A	SEQ ID NOS: 14024-14049
VEGFB	Vascular endothelial growth factor B	SEQ ID NOS: 14050-14052
VEGFC	Vascular endothelial growth factor C	SEQ ID NO: 14053
VGF	VGF nerve growth factor inducible	SEQ ID NOS: 14055-14057
VIP	Vasoactive intestinal peptide	SEQ ID NOS: 14058-14060
VIPR2	Vasoactive intestinal peptide receptor 2	SEQ ID NOS: 14061-14064
VIT	Vitrin	SEQ ID NOS: 14065-14072
VKORC1	Vitamin K epoxide reductase complex, subunit 1	SEQ ID NOS: 14073-14080
VLDLR	Very low density lipoprotein receptor	SEQ ID NOS: 14081-14083
VMO1	Vitellogenesis membrane outer layer 1 homolog (chicken)	SEQ ID NOS: 14084-14087
VNN1	Vanin 1	SEQ ID NO: 14088
VNN2	Vanin 2	SEQ ID NOS: 14089-14102
VNN3	Vanin 3	SEQ ID NOS: 14103-14114
VOPP1	Vesicular, overexpressed in cancer, prosurvival protein 1	SEQ ID NOS: 14115-14127
VPREB1	Pre-B lymphocyte 1	SEQ ID NOS: 14128-14129
VPREB3	Pre-B lymphocyte 3	SEQ ID NOS: 14130-14131
VPS37B	Vacuolar protein sorting 37 homolog B (<i>S. cerevisiae</i>)	SEQ ID NOS: 14132-14134
VPS51	Vacuolar protein sorting 51 homolog (<i>S. cerevisiae</i>)	SEQ ID NOS: 14135-14146
VSIG1	V-set and immunoglobulin domain containing 1	SEQ ID NOS: 14147-14149

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
VSIG10	V-set and immunoglobulin domain containing 10	SEQ ID NOS: 14150-14151
VSTM1	V-set and transmembrane domain containing 1	SEQ ID NOS: 14152-14158
VSTM2A	V-set and transmembrane domain containing 2A	SEQ ID NOS: 14159-14162
VSTM2B	V-set and transmembrane domain containing 2B	SEQ ID NO: 14163
VSTM2L	V-set and transmembrane domain containing 2 like	SEQ ID NOS: 14164-14166
VSTM4	V-set and transmembrane domain containing 4	SEQ ID NOS: 14167-14168
VTN	Vitronectin	SEQ ID NOS: 14169-14170
VWA1	Von Willebrand factor A domain containing 1	SEQ ID NOS: 14171-14174
VWA2	Von Willebrand factor A domain containing 2	SEQ ID NOS: 14175-14176
VWA5B2	Von Willebrand factor A domain containing 5B2	SEQ ID NOS: 14177-14178
VWA7	Von Willebrand factor A domain containing 7	SEQ ID NO: 14179
VWC2	Von Willebrand factor C domain containing 2	SEQ ID NO: 14180
VWC2L	Von Willebrand factor C domain containing protein 2-like	SEQ ID NOS: 14181-14182
VWCE	Von Willebrand factor C and EGF domains	SEQ ID NOS: 14183-14187
VWDE	Von Willebrand factor D and EGF domains	SEQ ID NOS: 14188-14193
VWF	Von Willebrand factor	SEQ ID NOS: 14194-14196
WDR25	WD repeat domain 25	SEQ ID NOS: 14197-14203
WDR81	WD repeat domain 81	SEQ ID NOS: 14204-14213
WDR90	WD repeat domain 90	SEQ ID NOS: 14214-14221
WFDC1	WAP four-disulfide core domain 1	SEQ ID NOS: 14222-14224
WFDC10A	WAP four-disulfide core domain 10A	SEQ ID NO: 14225
WFDC10B	WAP four-disulfide core domain 10B	SEQ ID NOS: 14226-14227
WFDC11	WAP four-disulfide core domain 11	SEQ ID NOS: 14228-14230
WFDC12	WAP four-disulfide core domain 12	SEQ ID NO: 14231
WFDC13	WAP four-disulfide core domain 13	SEQ ID NO: 14232
WFDC2	WAP four-disulfide core domain 2	SEQ ID NOS: 14233-14237
WFDC3	WAP four-disulfide core domain 3	SEQ ID NOS: 14238-14241
WFDC5	WAP four-disulfide core domain 5	SEQ ID NOS: 14242-14243
WFDC6	WAP four-disulfide core domain 6	SEQ ID NOS: 14244-14245
WFDC8	WAP four-disulfide core domain 8	SEQ ID NOS: 14246-14247
WFIKKN1	WAP, follistatin/kazal, immunoglobulin, kunitz and netrin domain containing 1	SEQ ID NO: 14248
WFIKKN2	WAP, follistatin/kazal, immunoglobulin, kunitz and netrin domain containing 2	SEQ ID NOS: 14249-14250
WIF1	WNT inhibitory factor 1	SEQ ID NOS: 14255-14257
WISP1	WNT1 inducible signaling pathway protein 1	SEQ ID NOS: 14258-14262
WISP2	WNT1 inducible signaling pathway protein 2	SEQ ID NOS: 14263-14265
WISP3	WNT1 inducible signaling pathway protein 3	SEQ ID NOS: 14266-14273
WNK1	WNK lysine deficient protein kinase 1	SEQ ID NOS: 14274-14287
WNT1	Wingless-type MMTV integration site family, member 1	SEQ ID NOS: 14288-14289
WNT10B	Wingless-type MMTV integration site family, member 10B	SEQ ID NOS: 14290-14294

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
WNT11	Wingless-type MMTV integration site family, member 11	SEQ ID NOS: 14295-14297
WNT16	Wingless-type MMTV integration site family, member 16	SEQ ID NOS: 14298-14299
WNT2	Wingless-type MMTV integration site family member 2	SEQ ID NOS: 14300-14302
WNT3	Wingless-type MMTV integration site family, member 3	SEQ ID NO: 14303
WNT3A	Wingless-type MMTV integration site family, member 3A	SEQ ID NO: 14304
WNT5A	Wingless-type MMTV integration site family, member 5A	SEQ ID NOS: 14305-14308
WNT5B	Wingless-type MMTV integration site family, member 5B	SEQ ID NOS: 14309-14315
WNT6	Wingless-type MMTV integration site family, member 6	SEQ ID NO: 14316
WNT7A	Wingless-type MMTV integration site family, member 7A	SEQ ID NO: 14317
WNT7B	Wingless-type MMTV integration site family, member 7B	SEQ ID NOS: 14318-14322
WNT8A	Wingless-type MMTV integration site family, member 8A	SEQ ID NOS: 14323-14326
WNT8B	Wingless-type MMTV integration site family, member 8B	SEQ ID NO: 14327
WNT9A	Wingless-type MMTV integration site family, member 9A	SEQ ID NO: 14328
WNT9B	Wingless-type MMTV integration site family, member 9B	SEQ ID NOS: 14329-14331
WSB1	WD repeat and SOCS box containing 1	SEQ ID NOS: 14332-14341
WSCD1	WSC domain containing 1	SEQ ID NOS: 14342-14351
WSCD2	WSC domain containing 2	SEQ ID NOS: 14352-14355
XCL1	Chemokine (C motif) ligand 1	SEQ ID NO: 14356
XCL2	Chemokine (C motif) ligand 2	SEQ ID NO: 14357
XPNPEP2	X-prolyl aminopeptidase (aminopeptidase P) 2, membrane-bound	SEQ ID NOS: 14358-14359
XXbac-BPG116M5.17		SEQ ID NOS: 679-680
XXbac-BPG181M17.5		SEQ ID NO: 681
XXbac-BPG32J3.20		SEQ ID NO: 682
XXYL1	Xyloside xylosyltransferase 1	SEQ ID NOS: 14360-14365
XYLT1	Xylosyltransferase I	SEQ ID NO: 14366
XYLT2	Xylosyltransferase II	SEQ ID NOS: 14367-14372
ZFYVE21	Zinc finger, FYVE domain containing 21	SEQ ID NOS: 14373-14377
ZG16	Zymogen granule protein 16	SEQ ID NO: 14378
ZG16B	Zymogen granule protein 16B	SEQ ID NOS: 14379-14382
ZIC4	Zic family member 4	SEQ ID NOS: 14383-14391
ZNF207	Zinc finger protein 207	SEQ ID NOS: 14392-14402
ZNF26	Zinc finger protein 26	SEQ ID NOS: 14403-14406
ZNF34	Zinc finger protein 34	SEQ ID NOS: 14407-14410
ZNF419	Zinc finger protein 419	SEQ ID NOS: 14411-14425
ZNF433	Zinc finger protein 433	SEQ ID NOS: 14426-14435
ZNF449	Zinc finger protein 449	SEQ ID NOS: 14436-14437
ZNF488	Zinc finger protein 488	SEQ ID NOS: 14438-14439
ZNF511	Zinc finger protein 511	SEQ ID NOS: 14440-14441
ZNF570	Zinc finger protein 570	SEQ ID NOS: 14442-14447
ZNF691	Zinc finger protein 691	SEQ ID NOS: 14448-14455

TABLE 9-continued

Exemplary therapeutic proteins (and proteins to enhance CAR-T efficacy).		
Gene Name	Gene Description	Protein SEQ ID NO
ZNF98	Zinc finger protein 98	SEQ ID NOS: 14456-14459
ZBPB	Zona pellucida binding protein	SEQ ID NOS: 14460-14463
ZBPB2	Zona pellucida binding protein 2	SEQ ID NOS: 14464-14467
ZSCAN29	Zinc finger and SCAN domain containing 29	SEQ ID NOS: 14468-14474

Cas-Clover

The disclosure provides a composition comprising a guide RNA and a fusion protein or a sequence encoding the fusion protein wherein the fusion protein comprises a dCas9 and a Clo051 endonuclease or a nuclease domain thereof.

Small Cas9 (SaCas9)

The disclosure provides compositions comprising a small, Cas9 (Cas9) operatively-linked to an effector. In certain embodiments, the disclosure provides a fusion protein comprising, consisting essentially of or consisting of a DNA localization component and an effector molecule, wherein the effector comprises a small, Cas9 (Cas9). In certain embodiments, a small Cas9 construct of the disclosure may comprise an effector comprising a type IIS endonuclease.

Amino acid sequence of *Staphylococcus aureus* Cas9 with an active catalytic site.

15 Inactivated, Small Cas9 (dSaCas9)

The disclosure provides compositions comprising an inactivated, small, Cas9 (dSaCas9) operatively-linked to an effector. In certain embodiments, the disclosure provides a fusion protein comprising, consisting essentially of or consisting of a DNA localization component and an effector molecule, wherein the effector comprises a small, inactivated Cas9 (dSaCas9). In certain embodiments, a small, inactivated Cas9 (dSaCas9) construct of the disclosure may comprise an effector comprising a type IIS endonuclease.

dSaCas9 Sequence: D10A and N580A mutations (bold, capitalized, and underlined) inactivate the catalytic site.

(SEQ ID NO: 18040)

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1 mkrnyilgld igitsvggygi idyetrdivd agvrlfkean vennegrrsk rgarrlkrrr
61 rhriqrkvkl lfdynlltdh selsginpye arvkglsqkl seeefsaall hlakrrgvhn
121 vneveedtgn elstkeqisr nskaleekyv aelqlerlkk dgevrgsinr fktsdyvkea
181 kqlkqvqkay hqldqsfdt yidlletrtrt yyegpgegsp fgwkdikewy emlmghctyf
241 peelsrvkya ynadlynaln dlennlvitrd enekleyyek fqiienvfkq kkkptlkqia
301 keilvneedi kgyrvstgk peftnlkvyh dikditarke iienaelldq iakiltiyqs
361 sediqeeltn lnseltqeei eqisnlkgyt gthnlslkai nlildelwht ndnqiaifnr
421 lklvpkkvdl sqqkeipttl vddfilspvv krsfiqsikv inaiikkygl pndiielar
481 eknskdqkm inemqkrnrq tnerieeir ttgkenakyl iekiklhdmq egkclyslea
541 ipledllnnp fnyevdhiip rsysfdnsfn nkvlvkeqn skkgnrtpfq ylsssdskis
601 yetfkkhiln lakggrisk tkkeylleer dinrfsvqkd finrnlvdtr yatrglmnl
661 rsyfrvnld kvksinggf tsflrrkwkf kknkgykh haedaliian adfifkewk
721 ldkakkvmen qmfeekqaes mpeieteqey keifitphqi khikdfkdyk yshrvdtkpn
781 relindtlys trkddkgntl ivnnlnglyd kdndklklli nkspekllmy hhdptqyqkl
841 klimeqygde knplykyee tgnyltkysk kdngpvikki kyygnkl nah lditddypns
901 rnkvvklslk pyrfdvldn gvykfvtkn ldvikkeny evnskyeea kklkkisnqa
961 efiasfynd likingelyr vigvndlln rievnmidit yreylenmnd krppriikti
1021 asktqsikky stdilgnlye vskkhpqi kkg.

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(SEQ ID NO: 18041)

1 mkrnyilglA igitsvggygi idyetrdivd agvrlfkean vennegrrsk rgarrlkrrr
61 rhriqrkvkl lfdynltdh selsginpye arvkglsqkl seeefsaall hlakrrgvhn
121 vneveedtgn elstkeqisr nskaleekyv aelqlerlkk dgevrsgsinr fktedyvkea
181 kqllkvqkay hqldqsfidt yidlletrrt yyegpggesp fgwkdkikewy emlmghctyf
241 peelrsvkya ynadlynaln dlennlvitrd enekleyek fqiienvfkq kkkptlkqia
301 keilvneedi kgyrvststgk peftnlkvvh dikditarke iienaelldq iakiltiyqs
361 sediqeeltn lnseltqeei eqisnlkgyt gthnlslkai nlildelwht ndnqiaifnr
421 lklvpkkvdl sqkpepttl vddfilspvv krsfiqsikv inaiikkygl pndiielar
481 eknskdqkm inemqkrnrq tnerieeir ttgkenakyl iekiklhdmq egkclyslea
541 ipledllnnp fnyevdhiip rsysfdnsfn nkvlvkqeeA skkgnrtfpq ylsdsdskis
601 yetfkkhiln lakgkgrisk tkkeylleer dinrfsvqkd finrnlvdtr yatrglmnl
661 rsyfrvnnd vkvksinggf tsflrrkwkf kkernkgykh haedaliian adfifkewkk
721 ldkakkvmen qmfeekqaes mpeietegey keifitphqi khikdfkdyk yshrvdkkpn
781 relindtlys trkddkngtl ivnnlnglyd kdndklkli nkspekllmy hhdptqyql
841 klimegygde knplykyee tgnyltkysk kdngpvikki kyygnkl nah lditddypns
901 rnkvvklslk pyrfdvylnd gvykftvkn ldvikenyy evnskcyee kklkksnqa
961 efiasfynnd likingelyr vigvnndlln rievnmidit yreylennmd krppriikti
1021 asktsikky stdilgnlye vskkhpqi kkg.

Inactivated Cas9 (dCas9)

The disclosure provides compositions comprising an inactivated Cas9 (dCas9) operatively-linked to an effector. In certain embodiments, the disclosure provides a fusion protein comprising, consisting essentially of or consisting of a DNA localization component and an effector molecule, wherein the effector comprises an inactivated Cas9 (dCas9). In certain embodiments, an inactivated Cas9 (dCas9) construct of the disclosure may comprise an effector comprising a type IIS endonuclease.

In certain embodiments, the dCas9 of the disclosure comprises a dCas9 isolated or derived from *Staphylococcus pyogenes*. In certain embodiments, the dCas9 comprises a dCas9 with substitutions at positions 10 and 840 of the amino acid sequence of the dCas9 that inactivate the catalytic site. In certain embodiments, these substitutions are D10A and H840A. In certain embodiments, the amino acid sequence of the dCas9 comprises the sequence of:

(SEQ ID NO: 18042)

1 XDKKYSIGLA IGTNSVGWAV ITDEYKVPSK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE
61 ATRLKRTARR RYTRKRNIC YLQEIFSNEK AKVDDSPFHR LEESFLVEED KKHHRHPIFG
121 NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGDLPNDNSD
181 VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKINGLFGN
241 LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI
301 LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI PFDQSKNGYA
361 GYIDGGASQE EFYKFIKPIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH
421 AILRRQEDFY PFLKDNREKI EKILTFRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE
481 VVDKGASQAS FIERMTNFDK NLPNEKVLKP HSLLYEYFTV YNELTKVKYV TEGMRKPAPL
541 SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK KIECFDSVEI SGVEDRFNAS LGTYHDLKLI
601 IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKYA HLFDDKVMKQ LKRRRYTGWG
661 RLSRKLINGI RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGDSL
721 HEHIANLAGS PAIKKGILQT VKVVDLVKV MGRHKPENIV IEMARENQTT QKGQKNSRER
781 MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYDVDA

-continued

841 IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKMK NYWRQLLNAK LITQRKFDNL
 901 TKAERGG LSE LDKAGFIK RQ LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS
 961 KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVG TALIKK YPKLESEFVY GDYKVYDVRK
 1021 MIAKSEQEIG KATAKYFFYS NIMNFFKTEI TLANGEIRKR PLIETNGETG EIVWDKGRDF
 1081 ATVRKVL S MP QVNIVKKTEV QTGGFSKESI LPKRNSDKLI ARKDWDPKK YGGFDSPTVA
 1141 YSVLVVAKVE KGSKKLKSV KELLGITIME RSSFEKNPID FLEAKGYKEV KKD LIIKLPK
 1201 YSLFELENGR KRMLASAGEL QKGNELALPS KYVNFLYLAS HYEKLKGSPE DNEQKQLFVE
 1261 QHKHYLDEII EQISEFSKRV ILADANLDKV LSAYNKH RDK PIREQAENII HLFTLTNLGA
 1321 PAAFKYFDTT IDRKYTSTK EVLDATLIHQ SITGLYETRI DLSQLGGD.

In certain embodiments, the amino acid sequence of the dCas9 comprises the sequence of: KIRSGEMTIEELERAMFNNSEFILKY (SEQ ID NO: 18044).

(SEQ ID NO: 18043)

1 MDKKYSIGLA IGTVSGWAV ITDEYKVPSK KFKVLGNTDR HSIKKNLIGA LLFDSGETAE
 61 ATRLKRTARR RYTRRNKRIC YLQEIFS NEM AKVDDSFH R LEESFLVEED KKHERHPIFG
 121 NIVDEVAYHE KYPTIYHLRK KLVDSTDKAD LRLIYLALAH MIKFRGHFLI EGD LNP D NSD
 181 VDKLFIQLVQ TYNQLFEENP INASGVDAKA ILSARLSKSR RLENLIAQLP GEKKNGLFGN
 241 LIALSLGLTP NFKSNFDLAE DAKLQLSKDT YDDDLNLLA QIGDQYADLF LAAKNLSDAI
 301 LLSDILRVNT EITKAPLSAS MIKRYDEHHQ DLTLLKALVR QQLPEKYKEI PFDQSKNGYA
 361 GYIDGGASQE EFKYFIKPIL EKMDGTEELL VKLNREDLLR KQRTFDNGSI PHQIHLGELH
 421 AILRRQEDFY PFLKDNREKI EKILTRIPY YVGPLARGNS RFAWMTRKSE ETITPWNFEE
 481 VVDKGASQS FIERMTNFDK NLPNEKVL PK HSLLYEYFTV YNELTKVKYV TEGMRKPAFL
 541 SGEQKKAIVD LLFKTNRKVT VKQLKEDYFK KIECFDSVEI SGVEDRFNAS LGTYHDL LKI
 601 IKDKDFLDNE ENEDILEDIV LTLTLFEDRE MIEERLKTYA HLFDDKVMKQ LKRRRYTGWG
 661 RLSRKLINGI RDKQSGKTIL DFLKSDGFAN RNFMQLIHDD SLTFKEDIQK AQVSGQGD SL
 721 HEHIANLAGS PAIKKGILQT VKVDELVKV MGRHKPENIV IEMARENQTT QKGQKNSRER
 781 MKRIEEGIKE LGSQILKEHP VENTQLQNEK LYLYYLQNGR DMYVDQELDI NRLSDYDVDA
 841 IVPQSFLKDD SIDNKVLTRS DKNRGKSDNV PSEEVVKMK NYWRQLLNAK LITQRKFDNL
 901 TKAERGG LSE LDKAGFIK RQ LVETRQITKH VAQILDSRMN TKYDENDKLI REVKVITLKS
 961 KLVSDFRKDF QFYKVRINN YHHAHDAYLN AVVG TALIKK YPKLESEFVY GDYKVYDVRK
 1021 MIAKSEQEIG KATAKYFFYS NIMNFFKTEI TLANGEIRKR PLIETNGETG EIVWDKGRDF
 1081 ATVRKVL S MP QVNIVKKTEV QTGGFSKESI LPKRNSDKLI ARKDWDPKK YGGFDSPTVA
 1141 YSVLVVAKVE KGSKKLKSV KELLGITIME RSSFEKNPID FLEAKGYKEV KKD LIIKLPK
 1201 YSLFELENGR KRMLASAGEL QKGNELALPS KYVNFLYLAS HYEKLKGSPE DNEQKQLFVE
 1261 QHKHYLDEII EQISEFSKRV ILADANLDKV LSAYNKH RDK PIREQAENII HLFTLTNLGA
 1321 PAAFKYFDTT IDRKYTSTK EVLDATLIHQ SITGLYETRI DLSQLGGD.

Clo051 Endonuclease

An exemplary Clo051 nuclease domain may comprise, consist essentially of or consist of, the amino acid sequence of: EGIKSNISLLKDELRGQISHISHEYLSLIDLAFD-SKQNRLFEMKVLELLVNEYGFKGRH LGGSRKPDG-IVYSTLTLEDNFGIIVDTKAYSEGYSLPISQADEM-ERYVRENSNRDEEVN PNKWWENFSEEVKKYFVFI-SGSFKGKFEEQLRRLSMTTG VNGSAVNVNLLLGAE

Cas-Clover Fusion Protein

In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 1) may comprise, consist essentially of or consist of, the amino acid sequence of (Clo051 sequence underlined, linker bold italics, dCas9 sequence (*Streptococcus pyogenes*) in italics):

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(SEQ ID NO: 18045)

MAPKKRKVEGIKSNISLLKDELRGQISHISHEYLSLIDLAFDSKQNRLF

EMKVLELLVNEYGFKGRHLGGSRKPDGIVYSTTLEDNFGIIVDTKAYSEG

YSLPISQADEMERYVRENSNRDEEVNPNKWWENFSEEVKKYFVFIGSGF

KGKFEEQLRRLSMTTGVNGSAVNVNLLGAEKIRSGEMTIEELERAMFN

NSEFILKYGGGSDKKYSIGLAIGTNSVGWAVITDEYKVPSKKFKVLGNT

DRHSIKKNLIGALLFDSGETAEATRLKRTARRRYTRRKNRICYLQEIFSN

EMAKVDSFFHRLEESFLVEEDKKHERHPIFGNIVDEVAYHEKYPTIYHL

RKKLVDSSTDKADRLRIYLALAHMIKFRGHFLIEGDLNPDNSDVKLFILQ

VQTYNQLFEEINPINSAGVDAKAIL SARLSKSRLENLIAQLPGEKKNGLF

GNLIALSLGLTPNFKSNFDLAEDAKLQLSKDQYDDDLNLLAQLIGDQYAD

LFLAAKNLSDAILLSDILRVNTEITKAPLSASMIKRYDEHHQDLTLKAL

VRQQLPEKYKEIFFDQSKNGYAGYIDGGASQEYFYKFIKPILEKMDGTEE

LLVKLNREDLLRKQRTFDNGSIPHQIHLGELHAILRRQEDFYFPLKDNRE

KIEKILTFRIPIYVGPLARGNSRFAMWTRKSEETITPWNFEVVDKGASA

QSFIERMTNFDKNLPNEKVLPHKSLLYEYFTVYNELTKVKYVTEGMRKPA

FLSGEQKKAIVDLLFKTNRKVTVKQLKEDYFKKIECFDSVEISGVEDRFN

ASLGTYHDLKLIKDKDFLDNEENEDILEDIVLTTLTFEDREMIEERLKT

468

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YAHLFDDKVMKQLKRRRYTGWGRLSRKLINGIRDKQSGKTIILDFLKSDGF

ANRNFMLIHDDSLTFKEDIQKAQVSGQGDSLHEHIANLAGSPAIAKKGIL

QTVKVVDLVKVMGRHKPENI VIEMARENQTTQKQKNSRERMKRIEEGI

KELGSQILKEHPVENTQLQNEKLYLYLQNGRDMYVDQELDINRLSDYDV

DAIVPQSFLKDDSIDNKVLTRSDKNRGKSDNPSEEVVKMKNYWRQLLN

AKLITQRKFDNLTKAERGGLSELDKAGFIKRQLVETRQITKHVAQILDSR

MNTKYDENDKLIREVKVITLKSCLVSDFRKDFQFYKREINNYHHAHDAY

LNAVVGITALIKKYPKLESEFVYGDYKVYDVRKMIKSEQEIGKATAKYFF

YSNIMNFFKTEITLANGETIRKPLIETNGETGEI VWDKGRDFATVRKVL

MPQVNI VVKTEVQTGGFSKESILPKRNSDKLIARKKDWDPKKYGGDFSPT

VAYSVLVVAKEVKGSKKLKSVKELLGITIMERSSFEKNPIDFLEAKGYK

EVKCDLIIKLPKYSLFELENGRKRMLASAGELQKGNELALPSKYVNFYLY

ASHYEKLKGGPEDNEQKQLFVEQHKHYLDEIEEQISEFSKRVIADANLD

KVLSAYNKHRDKPIREQAENI IHLFTLTNLGAPAFKYFDTTIDRKRYTS

TKEVLDTLHQSITGLYETRIDLSQLGGDGSPPKKRKRVS.

In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 1) may comprise, consist essentially of or consist of, the nucleic acid sequence of (dCas9 sequence derived from *Streptococcus pyogenes*):

(SEQ ID NO: 18046)

1 atggcaccaa agaagaaaag aaaagtgag ggcacaaagt caaacatcag cctgctgaaa

61 gacgaactgc ggggacagat tagtcacatc agtcacagat acctgtcact gattgatctg

121 gccttcgaca gcaagcagaa tagactgttt gagatgaaag tgctggaact gctgggtcaac

181 gagtatggct tcaagggcag acatctgggc gggctctagga aacctgacgg catcggtgac

241 agtaccacac tggaagacaa ctctcggaatc attgtcgata ccaaggctta ttccgagggc

301 tactctctgc caattagtcg ggcagatgag atggaaaggt acgtgcgaga aaactcaaat

361 agggacgagg aagtcaaccc caataagtgg tgggagaatt tcagcgagga agtgaagaaa

421 tactacttcg tctttatctc aggcagcttc aaaggaagt ttgaggaaca gctgcggaga

481 ctgtccatga ctaccggggt gaacggatct gctgtcaacg tgggtcaatct gctgctgggc

541 gcagaaaaga tcaggtccgg ggagatgaca attgaggaac tggaacgcgc catgttcaac

601 aattctgagt ttatcctgaa gtatggaggc gggggaagcg ataagaaata ctccatcgga

661 ctggccattg gcaccaatc cgtgggctgg gctgtcatca cagacagta caaggtgcca

721 agcaagaagt tcaaggtcct gggaacacc gatcgccaca gtatcaagaa aaatctgatt

781 ggagccctgc tgctcgactc aggcgagact gctgaagcaa cccgactgaa ggcgactgct

841 aggcgcgat ataccggag aaaaaatcgg atctgctacc tgcaggaat tttagcaac

901 gagatggcca aggtggagca tagtttcttt caccgcctgg aggaatcatt cctgggtggag

961 gaagataaga aacacgagcg gcatccatc tttggcaaca ttgtggagca agtcgcttat

1021 cagcagaagt accctactat ctatcatctg aggaagaaac tgggtggactc caccgataag

1081 gcagacctgc gctgatcta tctggccctg gctcacatga tcaagtccg ggggcatttt

1141 ctgatcgagg gagatctgaa ccctgacaat tctgatgtgg acaagctgtt catccagctg

1201 gtccagacat acaatcagct gtttgaggaa aaccaatta atgcctcagg cgtggagcga

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1261 aaggccatcc tgagcgccag actgtccaaa tctagcgccc tggaaaaact gatcgctcag
1321 ctgccaggag agaagaaaaa cggcctgttt gggaatctga ttgactgtc cctgggcctg
1381 acaccaact tcaagtataa ttttgatctg gccgaggacg ctaagctgca gctgtccaaa
1441 gacacttatg acgatgacct ggataacctg ctggctcaga tcggcgatca gtacgcagac
1501 ctgttccttg ccgctaagaa tctgagtgc gccatcctgc tgtcagatat tctgcgcgtg
1561 aacacagaga ttactaaggc cccactgagt gcttcaatga tcaaaagata tgacgagcac
1621 catcaggatc tgacctgct gaaggctctg gtgaggcagc agctgccga gaaatacaag
1681 gaaatcttct ttgatcagag caagaatgga tacgccggct atattgacgg cggggcttcc
1741 caggaggagt tctacaagtt catcaagccc attctggaaa agatggacgg caccgaggaa
1801 ctgctggtga agctgaatcg ggaggacctg ctgagaaaac agaggacatt tgataacgga
1861 agcatccctc accagattca tctgggcgaa ctgcacgcca tcttgcgacg gcaggaggac
1921 ttctacccat ttctgaagga taaccgcgag aaaatcgaaa agatcctgac cttcagaatc
1981 cctactatg tggggcctct ggcacgggga aatagtagat ttgcctggat gacaagaaag
2041 tcagagggaa ctatcacccc ctggaacttc gaggaagtgg tcgataaagg cgtagcgca
2101 cagtccttca ttgaaaggat gacaaathtt gacaagaacc tgccaaatga gaagtgctg
2161 cccaaacaca gcctgtgta cgaatatttc acagtgtata acgagctgac taaagtgaag
2221 tacgtcaccg aagggtgctg caagcccga tctctgtccg gagagcagaa gaaagccatc
2281 gtggacctgc tgtttaagac aaatcggaat gtgactgtca aacagctgaa ggaagactat
2341 ttcaagaaaa ttgagtgttt cgattcagtg gaaatcagcg gcgtcgagga caggtttaac
2401 gcctcccttg ggacctacca cgatctgctg aagatcatca aggataagga cttctggac
2461 aacgagggaa atgaggacat cctggaggac attgtgctga cactgactct gtttgaggat
2521 cgcgaaatga tcgaggaacg actgaagact tatgcccac tgttcgatga caaagtgatg
2581 aagcagctga aaagaaggcg ctacaccgga tggggacgcc tgagccgaaa actgatcaat
2641 gggattagag acaagcagag cggaaaaact atcctggact ttctgaagtc cgatggcttc
2701 gccaacagga acttcatgca gctgattcac gatgactctc tgaccttcaa ggaggacatc
2761 cagaaagcac aggtgtcttg ccagggggac agtctgcacg agcatatcgc aaacctggcc
2821 ggcagccccg ccatcaagaa agggattctg cagaccgtga aggtggtgga cgaactggtc
2881 aaggctcatg gacgacacaa acctgagaac atcgtgattg agatggcccg cgaatatcag
2941 acaactcaga agggccagaa aaacagtcga gaacggatga agagaatcga ggaaggcatc
3001 aaggagctgg ggtcacagat cctgaaggag catcctgttg aaaacactca gctgcagaat
3061 gagaaactgt atctgtacta tctgcagaat ggacgggata tgtacgtgga ccaggagctg
3121 gatattaaca gactgagtga ttatgacgtg gatgccatcg tccctcagag cttcctgaag
3181 gatgactcca ttgacaacaa ggtgctgacc aggtccgaca agaaccgcgg caaatcagat
3241 aatgtgccaa gcgaggaagt ggtcaagaaa atgaagaact actggaggca gctgctgaat
3301 gccaaagtga tcacacagcg gaaatttgat aacctgacta aggcagaaaag aggaggcctg
3361 tctgagctgg acaaggcccg cttcatcaag cggcagctgg tggagacaag acagatcact
3421 aagcacgtcg ctcagattct ggatagcaga atgaacacaa agtacgatga aaacgacaag
3481 ctgatcaggg aggtgaaaat cattactctg aaatccaagc tgggtgtctga ctttagaaag
3541 gatttccagt ttataaaat caggagatc aacaactacc accatgctca tgacgcatac
3601 ctgaacgcag tggtcgggac cgcctgatt aagaaatacc ccaagctgga gtccgagttc
3661 gtgtacggag actataaagt gtacgatgtc cggaagatga tcgccaaatc tgagcaggaa

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3721 attggcaagg ccaccgctaa gtatttcttt tacagtaaca tcatgaattt cttaagacc
 3781 gaaatcacac tggcaaatgg ggagatcaga aaaaggcctc tgattgagac caacggggag
 3841 acaggagaaa tctgtgtgga caagggaagg gattttgcta cctgtcgcaa agtctgtcc
 3901 atgccccaa gtaattattgt caagaaaact gaagtgcaga cggggggatt ctctaaggag
 3961 agtattctgc ctaagcgaaa ctctgataaa ctgatcgccc ggaagaaaga ctgggacccc
 4021 aagaagtatg gcgggttcga ctctccaaca gtggcttaca gtgtcctggt ggtcgcaaag
 4081 gtggaaaagg ggaagtocaa gaaactgaag tctgtcaaa agctgctggg aatcactatt
 4141 atggaacgca gctccttcga gaagaatcct atcgattttc tggaagccaa gggctataaa
 4201 gaggtgaaga aagacctgat cattaagctg ccaaaatact cactgtttga gctggaaaac
 4261 ggacgaaagc gaatgctggc aagcgccgga gaactgcaga agggcaatga gctggccctg
 4321 cctcccaaat acgtgaactt cctgtatctg gctagccact acgagaaact gaagggtcc
 4381 cctgaggata acgaacagaa gcagctgttt gtggagcagc acaaacatta tctggacgag
 4441 atcattgaac agatttcaga gttcagcaag agagtgatcc tggctgacgc aaatctggat
 4501 aaagtctga gcgcatacaa caagcacga gacaaacaa tccgggagca ggccgaaaat
 4561 atcattcatc tgttcaccct gacaaactg ggcccccctg cagccttcaa gtattttgac
 4621 accacaatcg atcggaagag atacacttct accaaagagg tgctggatgc tacctgatc
 4681 caccagagta ttaccggcct gtatgagaca cgcacgacc tgtcacagct gggaggcgat
 4741 gggagcccca agaaaaagcg gaaggtgtct agttaa.

In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 1) of the disclosure may comprise a DNA. In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 1) of the disclosure may comprise an RNA.

In certain embodiments, an exemplary dCas9-Clo051 fusion protein (embodiment 2) may comprise, consist essentially of or consist of, the amino acid sequence of (Clo051 sequence underlined, linker bold italics, dCas9 sequence (*Streptococcus pyogenes*) in italics):

(SEQ ID NO: 18047)

1 MPKKKRKVEG IKSNISLLKD ELRGQISHIS HEYLSLIDLA FDSKQNRLE MKVLELLVNE
 61 YGFKGRHLGG SRKPDGIVYS TTLEDNFGII VDTKAYSEGY SLPISQADEM ERYVRENSNR
 121 DEEVNPNKWW ENFSEEVKKY YFVFISGSFK GKFEQLRRL SMTTGVNGSA VNVNLLLGA
 181 EKIRSGEMTI EELERAMFNN SEFILKYGGS ***GS***DKKYSIGL AIGTNSVGWA VITDEYKVPS
 241 KFKVLGNTD RSHIKKNLIG ALLFDSGETA EATRLKRTAR RRYTRKNRI CYLQEIFSNE
 301 MAKVDDSFPH RLEESFLVEE DKKHERHPIF GNIVDEVAYH EKYPTIYHLR KKLVDSTDKA
 361 DLRLIYLALA HMIKFRGHFL IEGDLNPDNS DVDKLFQLV QTYNQLFEEN PINASGVDAK
 421 AILSARLSKS RRLLENLIAQL PGEKKNGLFG NLIALSLGLT PNFKSNFDLA EDAKLQLSKD
 481 TYDDDLNLL AQIGDQYADL FLAAKNLSDA ILLSDILRVN TEITKAPLSA SMIKRYDEHH
 541 QDLTLLKALV RQQLPEKYKE IFFDQSKNGY AGYIDGGASQ EEFYKFIKPI LEKMDGTEEL
 601 LVKLNREDLL RKQRTFDNGS IPHQIHLGEL HAILRRQEDF YPFLKDNREK IEKILTRFIP
 661 YYVGPLARGN SRPAWMTRKS EETITPWNFE EVVDKGASAQ SFIERMTNED KNLPNKVLPL
 721 KHSLLYEFYFT VYNELTKVKY VTEGMRKPAF LSGEQKKAIV DLLFKTNRKV TVKQLKEDYF
 781 KKIECFDSVE ISGVEDRFNA SLGTYHDLK IIKDKDFLDN EENEDILEDI VLTTLTFEDR
 841 EMIEERLKTY AHLFDDKVMK QLKRRRYTGW GLSRKLING IRDKQSGKTI LDFLKSDGFA
 901 NRNFMQLIHD DSLTFKEDIQ KAQVSGQGDS LHEHIANLAG SPAIKKGILQ TVKVVDLVK

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961 VMGRHKPENI VIEMARENQT TQKGQKNSRE RMKRIEEGIK ELGSQILKEH PVENTQLQNE
 1021 KLYLYYLQNG RDMYVDQELD INRLSDYDVD AIVPQSFLKD DSIDNKVLTR SDKNRGKSDN
 1081 VPSEEVVKKM KNYWRQLLNA KLITQRKFDN LTKAERGGLS ELDKAGFIKR QLVETRQITK
 1141 HVAQILDSRM NTKYDENDKL IREVKVITLK SKLVSDFRKD FQFYKVREIN NYHHAHDAYL
 1201 NAVVGTALIK KYPKLESEFV YGDYKVYDVR KMIakseQEI GKATAKYFFY SNIMNFFKTE
 1261 ITLANGEIRK RPLIETNGET GEIVWDKGRD FATVRKVLMS PQVNIVKTE VQTGGFSKES
 1321 ILPKRNSDKL IARKRDWDPK KYGGFDSPTV AYSVLVVAKV EKGKSKKLKS VKELLGITIM
 1381 ERSSEFEKNPI DFLEAKGYKE VRKDLIIKLP KYSLFELENG RKRMLASAGE LQKGNELALP
 1441 SKYVNFLYLA SHYEKLKGSF EDNEQKQLFV EQHKHYLDEI IEQISEFSKR VILADANLKD
 1501 VLSAYNKHRD KPIREQAENI IHLFTLTNLG APAAFKYFDT TIDRKRYTST KEVLDTLIH
 1561 QSITGLYETR IDLSQLGGDG SPKKKRKV.

In certain embodiments, an exemplary dCas9-Clo051²⁰ fusion protein (embodiment 2) may comprise, consist essentially of or consist of, the nucleic acid sequence of (dCas9 sequence derived from *Streptococcus pyogenes*):

(SEQ ID NO: 18048)

1 atgcctaaga agaagcggaa ggtggaaggc atcaaaagca acatctccct cctgaaagac
 61 gaactccggg ggcagattag ccacattagt cacgaatacc tctccctcat cgacctggct
 121 ttcatagca agcagaacag gctctttgag atgaaagtgc tggaactgct cgtcaatgag
 181 tacgggttca agggctgaca cctcggcgga tctaggaaac cagacggcat cgtgtatagt
 241 accacactgg aagacaactt tgggatcatt gtggatacca aggcatactc tgagggttat
 301 agtctgcccc tttcacaggc cgacgagatg gaacggtacg tgcgcgagaa ctcaaataga
 361 gatgaggaag tcaaccctaa caagtggtag gagaacttct ctgaggaagt gaagaaatac
 421 tacttctgct ttatcagcgg gtccttcaag ggtaaatttg aggaacagct caggagactg
 481 agcatgacta ccggcgtgaa tggcagcgcc gtcaacgtgg tcaatctgct cctgggcgct
 541 gaaaagattc ggagcggaga gatgaccatc gaagagctgg agagggcaat gtttaataat
 601 agcgagttta tcctgaaata cgggtggcgg gtatccgata aaaagtattc tattggttta
 661 gccatcggca ctaattccgt tggatgggct gtcataaccg atgaatacaa agtaccttca
 721 aagaaattta aggtgttggg gaacacagac cgtcattcga ttaaaaagaa tcttatcggt
 781 gccctcctat tcgatagtgg cgaaacggca gaggcgactc gectgaaacg aaccgctcgg
 841 agaaggtata cacgtcgcaa gaaccgaata tgttacttac aagaaatttt tagcaatgag
 901 atggccaaag ttgacgattc tttctttcac cgtttggaag agtccttcct tgtcgaagag
 961 gacaagaaac atgaacggca ccccatcttt ggaacatag tagatgaggt ggcataatcat
 1021 gaaaagtacc caacgattta tcacctcaga aaaaagctag ttgactcaac tgataaagcg
 1081 gacctgaggt taatctactt ggctcttgcc catatgataa agttccgtgg gcactttctc
 1141 attgaggggt atctaaatcc ggacaactcg gatgtcgaca aactgttcat ccagttagta
 1201 caaacctata atcagttggt tgaagagaac cctataaatg caagtggcgt ggatgcgaag
 1261 gctattctta gcgcccgcct ctctaaatcc cgacggctag aaaacctgat cgcacaatta
 1321 cccggagaga agaaaaatgg gttgttcggt aaccttatag cgctctcact aggcctgaca
 1381 ccaaatttta agtcgaactt cgacttagct gaagatgcca aattgcagct tagtaaggac
 1441 acgtacgatg acgatctcga caatctactg gcacaaattg gagatcagta tgcggactta
 1501 tttttggctg ccaaaaacct tagcgatgca atcctcctat ctgacatact gagagttaat

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1561 actgagatta ccaaggcgcc gttatccgct tcaatgatca aaaggtaacga tgaacatcac
1621 caagacttga cactttctaa ggccttagtc cgtcagcaac tgctgagaa atataaggaa
1681 atattctttg atcagtcgaa aaacgggtac gcagggtata ttgacggcgg agcgagtcaa
1741 gaggaattct acaagtttat caaaccata ttagagaaga tggatgggac ggaagagttg
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1861 attccacatc aaatccactt aggcgaattg catgctatac ttagaaggca ggaggatttt
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1981 tactatgttg gacccctggc ccgagggaaac tctcggttcg catggatgac aagaaagtcc
2041 gaagaaacga ttactccatg gaattttgag gaagtgtgctg ataaagggtgc gtcagctcaa
2101 tcgttcacgc agaggatgac caactttgac aagaatttac cgaacgaaaa agtattgcct
2161 aagcacagtt tactttacga gtatttcaca gtgtacaatg aactcacgaa agttaagtat
2221 gtcactgagg gcatgcgtaa acccgccctt ctaagcggag aacagaagaa agcaatagta
2281 gatctgttat tcaagaccaa ccgcaaagtg acagttaagc aattgaaaga ggactacttt
2341 aagaaaattg aatgcttcga ttctgtcgag atctccgggg tagaagatcg atttaatgcg
2401 tcacttggtg cgtatcatga cctcctaag ataattaaag ataaggactt cctggataac
2461 gaagagaatg aagatatctt agaagatata gtgttgactc ttaccctctt tgaagatcgg
2521 gaaatgattg aggaaagact aaaaacatac gtcacctgt tcgacgataa ggttatgaaa
2581 cagttaaaga ggcgtcgcta tacgggctgg ggacgattgt cgcggaaact tatcaacggg
2641 ataagagaca agcaaaagtgg taaaactatt ctcgattttc taaagagcga cggcttcgcc
2701 aataggaact ttatgcagct gatccatgat gactctttaa ccttcaaaga ggatatacaa
2761 aaggcacagg ttccggaca aggggactca ttgcacgaac atattgcgaa tcttgctggt
2821 tcgccagcca tcaaaaaggg catactccag acagtcaaag tagtggatga gctagttaag
2881 gtcatgggac gtcacaaacc ggaaaacatt gtaatcgaga tggcacgcga aaatcaaacg
2941 actcagaagg ggcaaaaaaa cagtcgagag cggatgaaga gaatagaaga ggggtattaa
3001 gaactgggca gccagatctt aaaggagcat cctgtggaaa ataccaatt gcagaacgag
3061 aaactttacc tctattacct acaaaatgga agggacatgt atgttgatca ggaactggac
3121 ataaaccgtt tatctgatta cgacgtcgat gccattgtac cccaatcctt tttgaaggac
3181 gattcaatcg acaataaagt gcttacacgc tcggataaga accgagggaa aagtgacaat
3241 gttccaagcg aggaagtctg aaagaaaatg aagaactatt ggcggcagct cctaattgcg
3301 aaactgataa cgcaaaagaaa gttcgataac ttaactaaag ctgagagggg tggcttgtct
3361 gaacttgaca aggcgggatt tattaacgt cagctcgtgg aaaccgccca aatcacaaag
3421 catgttgac agatactaga ttcccgaatg aatacgaat acgacgagaa cgataagctg
3481 attcgggaag tcaagtaat cactttaag tcaaaattgg tgtcggactt cagaaaggat
3541 tttcaattct ataaagttag ggagataaat aactaccacc atgcgcacga cgcttatctt
3601 aatgccgtcg tagggaccgc actcatlaag aaataccga agctagaaag tgagtttgtg
3661 tatggtgatt acaaaattta tgacgtccgt aagatgatcg cgaaaagcga acaggagata
3721 ggcaaggcta cagccaaata cttcttttat tctaacatta tgaatttctt taagacggaa
3781 atcactctgg caaacggaga gatacgcaa cgacctttaa ttgaaaccaa tggggagaca
3841 ggtgaaatcg tatgggataa gggccgggac ttcgcgacgg tgagaaaagt tttgtccatg
3901 ccccaagtca acatagtaaa gaaaactgag gtgcagaccg gagggttttc aaaggaatcg
3961 attcttccaa aaaggaatag tgataagctc atcgctcgta aaaaggactg ggaccgaaa

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4021 aagtacggtg gcttcgatag ccttacagtt gcctattctg tcttagtagt ggcaaaagtt
 4081 gagaaggga aatccaagaa actgaagta gtcaaagaat tattggggat aacgattatg
 4141 gagcgctcgt cttttgaaaa gaaccccatc gacttccttg aggcgaaagg ttacaaggaa
 4201 gtaaaaaagg atctcataat taaactacca aagtatagtc tgtttgagtt agaaaatggc
 4261 cgaaaacgga tgttggttag cgccggagag cttcaaaagg ggaacgaact cgactaccg
 4321 tctaaatagc tgaatttcct gtatttagcg tccattacg agaagttgaa aggttcacct
 4381 gaagataacg aacagaagca actttttgtt gagcagcaca aacattatct cgacgaaatc
 4441 atagagcaaa tttcggaatt cagtaagaga gtcattcctag ctgatgcaa tctggacaaa
 4501 gtattaagcg catacaacaa gcacagggat aaaccctacg gtgagcagcg ggaaaatatt
 4561 atccatttgt ttactcttac caacctcggc gtcacagccg cattcaagta ttttgacaca
 4621 acgatagatc gcaaacgata cacttctacc aaggaggtgc tagacgcgac actgattcac
 4681 caatccatca cgggattata tgaaactcgg atagatttgt cacagcttgg gggtagcgga
 4741 tcccccaaga agaagaggaa agtctga.

In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 2) of the disclosure may comprise a DNA. In certain embodiments, the nucleic acid sequence encoding a dCas9-Clo051 fusion protein (embodiment 2) of the disclosure may comprise an RNA.

EXAMPLES

Example 1: Knock Down Efficiency of Checkpoint Signaling Proteins on Armored T-Cells

To create armored T-cells that have enhanced therapeutic potential, genetic modifications may be made in order to render the T-cells less sensitive to immunologic and/or metabolic checkpoints. One mechanism to produce armored T-cells is to inhibit checkpoint signaling is to knockout various checkpoint receptors. The Cas-CLOVER™ platform was used to target and knockout the checkpoint receptors PD-1, TGFβR2, LAG-3, Tim-3, and CTLA-4 in resting (or quiescent) primary pan T cells. As measured by flow cytometry, gene editing resulted in 30-70% loss of protein expression at the cell surface (FIG. 11). These results show that Cas-CLOVER™ is able to efficiently target the knockout of these genes resulting in loss of target protein expression on the T-cell surface. Knockout efficiency can significantly be increased by further optimization of guide RNA pairs, or by using additional guide RNA pairs targeting the same gene and/or regulators or promoters of the target gene.

Example 2: Strategies for the Expression of Null or Switch Intracellular Signaling Proteins on Armored T-Cells

Another strategy to produce armored T-cells is to reduce or inhibit endogenous checkpoint signaling by expressing various modified/chimeric checkpoint receptors that have an altered or absent intracellular signaling domain. Checkpoint signals that could be targeted using this strategy include PD-1 or TGFβRII of T-cells, which bind to the PD-L1 ligand and TGFβ cytokine, respectively. FIG. 12 shows a schematic diagram of various strategies for producing decoy/null/dominant negative receptor (Null receptors) for two different

inhibitory receptors (PD-1 (top panel) and TGFβRII (bottom panel)). To design Null receptors, the intracellular domain (ICD) of PD1 or TGFβRII can be mutated (mutated null) or deleted (truncated null). As a result, binding of the cognate ligand(s) of the null receptor does not result in delivery of the checkpoint signal to the T-cells. Furthermore, since the Null receptor competes with wildtype receptors for binding of the endogenous ligand(s), any binding by the Null receptor sequesters endogenous ligand(s) from binding the wild-type receptor. This results in dilution of the overall level of checkpoint signaling effectively delivered to the T-cell, thus, reducing or blocking checkpoint inhibition. FIG. 12 also shows switch receptor design strategies for the inhibitory receptors PD-1 (top panel) and TGFβRII (bottom panel). In switch receptors, wildtype ICD is replaced with the ICD from either an immuno-stimulatory molecule (Co-stimulatory switch) or a different inhibitory molecule (Inhibitory switch). Immuno-stimulatory molecules include but are not limited to CD3z, CD28, 4-1BB and the examples listed in Table 1. Inhibitory molecules include but are not limited to CTLA4, PD1, Lag3 and the examples listed in Tables 1 and 9. In the former case, binding of the endogenous ligand by the modified switch receptor results in the delivery of a positive signal to the T-cells, thereby helping to enhance stimulation of the T-cell, facilitating continuation of tumor targeting and killing. In the latter case, binding of the endogenous ligand by the modified switch receptor results in the delivery of a negative signal to the T-cells, thereby helping to reduce stimulation and activity of the T-cell.

Example 3: Enhancing Surface Expression of PD1 and TGFβRII Null or Switch Intracellular Signaling Proteins on Armored T-Cells

To create armored T-cells, a number of truncated null receptors expressing alternative signal peptides (SP) and transmembrane domains (TM) designed and tested for maximal expression on the surface of modified T-cells. FIG. 13 shows schematic diagrams of several null receptor constructs for PD-1 (top) and TGFβRII (bottom). Extracellular domains (ECD) of these proteins were modified such that the wildtype signal peptide (SP) and/or the transmembrane domains (TM) were replaced with that from the human T

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cell CD8 α receptor (red arrows). Each of the six truncated null constructs shown in FIG. 13 were DNA synthesized and then subcloned into an mRNA IVT DNA vector (pRT). High quality mRNA was produced via IVT for each. Transfection of mRNA encoding each of the six molecules was performed using electroporation (EP) delivery into primary human T cells and FACS analysis was performed 24 hours post-EP to evaluate expression level of each construct on the cell surface (FIG. 14). By flow cytometry, replacement of the WT SP with the alternative CD8 α (02.8aSP-PD-1 and 02.8aSP-TGF β R2) resulted in the highest level of expression at the T cell surface. 02.8aSP-PD-1 Null receptor exhibited an MFI of 43,680, which is 177-fold higher than endogenous T cell PD-1 expression and 2.8-fold higher than the WT PD-1 Null receptor. 02.8aSP-TGF β R2 Null receptor exhibited an MFI of 13,809, which is 102-fold higher than endogenous T cell TGF β R2 expression and 1.8-fold higher than the WT TGF β R2 Null receptor. These results show that replacement of wildtype SP with the alternative CD8 α SP for both PD1 and TGF β R2 inhibitory proteins leads to enhanced surface expression of the Null or Switch receptor. This in turn will maximize checkpoint inhibition or co-stimulation, respectively, upon binding of the natural ligand(s).

Example 4: Design of NF-KB Inducible Vectors for Expression in Modified T-Cells

Two T cell activation NF-KB inducible vectors were developed (FIGS. 15A and 15B); one with the gene expression system (GES) in the forward orientation (A) and the other in the complementary direction (B), both preceding the constitutive EF1a promoter. These vectors also direct expression of a CAR molecule and a DHFR selection gene, separated by a T2A sequence. Both the conditional NF-KB inducible system and the EF1a directed genes are a part of a piggyBac transposon, which can be permanently integrated into T cells using EP. Once integrated into the genome, the T cells constitutively express the CAR on the membrane surface and the DHFR within the cell, while expression of the NF-KB inducible gene, GFP, will be expressed to the highest level only upon T cell activation.

Example 5: NF-KB Inducible Vectors for GFP Expression in Modified T-Cells

T cells were nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mutein gene under control of an EF1a promoter along with the absence (No gene expression system (GES) control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFkB-GFP forward) or reverse orientation (pNFkB-GFP reverse). Cells were cultured in the presence of methotrexate selection until the cells were almost completely resting (Day 19) and GFP expression was assessed at Day 5 and Day 19. At Day 5, all T cells are proliferating and highly stimulated, with cells harboring the NF-KB inducible expression cassette producing high levels of GFP due to strong NFkB activity (see FIG. 16). The No GES control cells did not express detectable levels of GFP. By Day 19, the GES T cells were almost fully resting and GFP expression was significantly lower than Day 5 ($\sim 1/8$ MFI), since NFkB activity is lower. GFP expression is still observed at Day 19, which may due to the long half-life of GFP protein (~ 30 hr), or, basal level of NFkB activity

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through, for example, a TCR, a CAR, a cytokine receptor, or a growth factor receptor signal.

Example 6: NF-KB Inducible Vectors for Anti-BCMA CAR-Mediated GFP Expression in Modified T-Cells

T cells were either unmodified (Mock T cells) or nucleofected with a piggyBac vector expressing an anti-BCMA CAR and a DHFR mutein gene under control of an EF1a promoter along with the absence (No GES control) or presence of an NF-KB inducible expression system driving GFP expression in either the forward (pNFkB-GFP forward) or reverse orientation (pNFkB-GFP reverse). All cells were cultured for 22 days, either with or without methotrexate selection (Mock T cells), until the cells were almost completely resting. Cells were then stimulated for 3 days in the absence (No stimulation) or presence of BCMA-(K562), BMCA+(RPMI 8226), or positive control anti-CD3 anti-CD28 activation reagent (CD3/28 stimulation). GFP expression was undetectable under all conditions with the No GES control or Mock T cells. However, while pNFkB-GFP forward- and reverse-transposed cells exhibited little GFP expression over the No stimulation control when cultured with BCMA- K562 cells, they both demonstrated dramatic upregulation of gene expression either in the presence of BCMA+ tumor cells or under positive control conditions (FIG. 17). Little difference in GFP expression was observed between the pNFkB-GFP forward- and reverse-transposed cells that were cocultured with BCMA+ tumor cells.

Example 7: Control of Anti-BCMA CAR-Mediated Expression in Modified T-Cells

The expression level of inducible gene can be regulated by the number of response elements upstream or preceding the inducible promoter. T cells were nucleofected with a piggyBac vector encoding an anti-BCMA CARTyrin followed by a selection gene, both under control of a human EF1a promoter (FIG. 18). Further, vectors either additionally encoded the conditional NF-KB inducible gene expression system driving expression of a truncated CD19 protein (dCD19) and included a number of NFkB response elements (RE) varying from 0-5, no GES (No GES), or received an electroporation pulse but no piggyBac nucleic acid (Mock). Data are shown for only the GES in the reverse (opposite) direction/orientation. All cells were cultured for 18 days and included selection for piggyBac-modified T cells using methotrexate addition. Cells were then stimulated for 3 days using anti-CD3 anti-CD28 bead activation reagent and dCD19 surface expression was assessed by FACS at Days 0, 3 and 18, and data are shown as FACS histograms and MFI of target protein staining. Surface dCD19 expression was detected at low levels at Day 0 in all T cells transposed with vectors encoding the GES. At 3 days post-stimulation, dramatic upregulation of dCD19 expression was observed for all T cells expressing the GES, with a greater fold increase in surface expression in those with higher numbers of REs. Thus, surface dCD19 expression was directly proportional with the number of REs encoded in the GES. No dCD19 was detected on the surface of T cells that did not harbor the GES: No GES and Mock controls.

INCORPORATION BY REFERENCE

Every document cited herein, including any cross referenced or related patent or application is hereby incorporated

herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

While particular embodiments of the disclosure have been illustrated and described, various other changes and modifications can be made without departing from the spirit and scope of the disclosure. The scope of the appended claims includes all such changes and modifications that are within the scope of this disclosure.

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12415844B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A cell comprising a chimeric antigen receptor (CAR) comprising:

- (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH that specifically binds to B-Cell Maturation Antigen (BCMA);
- (b) a transmembrane domain, and
- (c) an endodomain comprising at least one costimulatory domain;

wherein the VH comprises:

- a complementarity determining region 1 (CDR1) comprising an amino acid sequence of SEQ ID NO: 18056;
- a complementarity determining region 2 (CDR2) comprising an amino acid sequence of SEQ ID NO: 18060; and
- a complementarity determining region 3 (CDR3) comprising an amino acid sequence of SEQ ID NO: 18064.

2. The cell of claim 1, wherein the VH comprises or consists of a recombinant or chimeric sequence.

3. The cell of claim 1, wherein the VH comprises or consists of a human or humanized sequence.

4. The cell of claim 1, wherein the ectodomain of (a) further comprises a signal peptide.

5. The cell of claim 1, wherein the ectodomain of (a) further comprises a hinge between the antigen recognition region and the transmembrane domain.

6. The cell of claim 4, wherein the signal peptide comprises an amino acid sequence of a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR signal peptide.

7. The cell of claim 6, wherein the signal peptide comprises an amino acid sequence of a human CD8a signal peptide.

8. The cell of claim 7, wherein the human CD8a signal peptide comprises an amino acid sequence of SEQ ID NO: 18012.

9. The cell of claim 1, wherein the transmembrane domain comprises an amino acid sequence of a human CD2, CD3δ, CD3ε, CD3γ, CD3ζ, CD4, CD8α, CD19, CD28, 4-1BB or GM-CSFR transmembrane domain.

10. The cell of claim 9, wherein the transmembrane domain comprises an amino acid sequence of a human CD8a transmembrane domain.

11. The cell of claim 10, wherein the human CD8a transmembrane domain comprises an amino acid sequence of SEQ ID NO: 18014.

12. The cell of claim 1, wherein the endodomain comprises an amino acid sequence of a human CD3ζ endodomain.

13. The cell of claim 1, wherein the at least one costimulatory domain comprises an amino acid sequence of a human 4-1BB, CD28, CD40, ICOS, MyD88, OX-40 intracellular segment, or any combination thereof.

14. The cell of claim 13, wherein the at least one costimulatory domain comprises an amino acid sequence of a human 4-1BB costimulatory domain.

15. The cell of claim 12, wherein the human CD35 endodomain comprises an amino acid sequence of SEQ ID NO: 18016.

16. The cell of claim 14, wherein the human 4-1BB costimulatory domain comprises an amino acid sequence of SEQ ID NO: 18018.

17. The cell of claim 5, wherein the hinge comprises an amino acid sequence of a human CD8α, IgG4, CD4 hinge or any combination thereof.

18. The cell of claim 17, wherein the hinge comprises an amino acid sequence of a human CD8α hinge.

19. The cell of claim 18, wherein the human CD8α comprises an amino acid sequence of SEQ ID NO: 18020.

20. A composition comprising the cell of claim 1 and at least one pharmaceutically acceptable carrier.

21. The cell of claim 1, wherein the cell is an immune cell.

22. The cell of claim 21, wherein the immune cell is a T-cell, a Natural Killer (NK) cell, a Natural Killer (NK)-like cell, a Cytokine Induced Killer (CIK) cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell or an umbilical cord blood (UCB) derived T-cell.

23. The cell of claim 1, wherein the cell is autologous.

24. The cell of claim 1, wherein the cell is allogeneic.

25. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 1.

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26. The cell of claim 1, wherein the VH comprises an amino acid sequence of SEQ ID NO: 18051.

27. The cell of claim 1, wherein the CAR comprises comprises amino acids 22-362 of the amino acid sequence of SEQ ID NO: 18006.

28. A cell comprising a chimeric antigen receptor (CAR) comprising:

- (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH that specifically binds to BCMA and wherein the VH comprises an amino acid sequence of SEQ ID NO: 18051;
- (b) a hinge domain comprising a human CD8 α hinge domain;
- (c) a transmembrane domain comprising a human CD8 α transmembrane domain; and
- (d) an endodomain comprising a human 4-1BB costimulatory domain and a human CD3 ζ endodomain.

29. The cell of claim 28, wherein the human CD8 α hinge domain comprises the amino acid sequence of SEQ ID NO: 18020, wherein the human CD8 α transmembrane domain comprises the amino acid sequence of SEQ ID NO: 18014, wherein the human 4-1BB costimulatory domain comprises the amino acid sequence of SEQ ID NO: 18018, and wherein the human CD3 ζ endodomain comprises the amino acid sequence of SEQ ID NO: 18016.

30. The cell of claim 28, wherein the cell is an immune cell.

31. The cell of claim 30, wherein the immune cell is a T-cell, a Natural Killer (NK) cell, a Natural Killer (NK)-like cell, a Cytokine Induced Killer (CIK) cell, a hematopoietic progenitor cell, a peripheral blood (PB) derived T cell or an umbilical cord blood (UCB) derived T-cell.

32. The cell of claim 31, wherein the immune cell is a T cell.

33. The cell of claim 32, wherein the immune cell is allogeneic.

34. The cell of claim 27, wherein the cell is a T cell.

35. The cell of claim 34, wherein the T cell is a memory stem T cell (T_{SCM}).

36. The cell of claim 35, wherein the T_{SCM} is allogeneic.

37. The cell of claim 27, wherein the cell further comprises an exogenous gene that confers resistance to a deleterious compound.

38. The cell of claim 37, wherein the gene is a Dihydrofolate Reductase (DHFR) gene.

39. The cell of claim 27, wherein the cell further comprises an exogenous inducible proapoptotic polypeptide.

40. The cell of claim 39, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

41. A composition comprising the cell of claim 29 and at least one pharmaceutically acceptable carrier.

42. A composition comprising the cell of claim 33 and at least one pharmaceutically acceptable carrier.

43. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 29.

44. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 33.

45. The cell of claim 1, wherein the CAR comprises an amino acid sequence of SEQ ID NO: 18006.

46. The cell of claim 28, wherein the ectodomain further comprises a human CD8 α signal peptide.

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47. The cell of claim 46, where in the human CD8 α signal peptide comprises the amino acid sequence of SEQ ID NO: 18012.

48. The cell of claim 27, wherein the cell is an allogeneic T-cell.

49. The cell of claim 48, wherein the cell further comprises an exogenous gene that confers resistance to a deleterious compound.

50. The cell of claim 49, wherein the gene is a Dihydrofolate Reductase (DHFR) gene.

51. The cell of claim 48, wherein the cell further comprises an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

52. The cell of claim 50, wherein the cell further comprises an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

53. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 50.

54. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 52.

55. A composition comprising a population of cells, wherein a plurality of cells of the population comprises the cell of claim 49.

56. The cell of claim 49, wherein the DHFR gene comprises the sequence of SEQ ID NO: 17012.

57. The cell of claim 48, wherein the truncated caspase 9 polypeptide comprises the sequence of SEQ ID NO: 18028.

58. The cell of 50, wherein the truncated caspase 9 polypeptide comprises the sequence of SEQ ID NO: 18028.

59. A nucleic acid encoding a chimeric antigen receptor (CAR) comprising:

- (a) an ectodomain comprising an antigen recognition region, wherein the antigen recognition region comprises at least one VH that specifically binds to B-Cell Maturation Antigen (BCMA);
- (b) a transmembrane domain, and
- (c) an endodomain comprising at least one costimulatory domain;

wherein the VH comprises:

- a complementarity determining region 1 (CDR1) comprising an amino acid sequence of SEQ ID NO: 18056;
- a complementarity determining region 2 (CDR2) comprising an amino acid sequence of SEQ ID NO: 18060; and

- a complementarity determining region 3 (CDR3) comprising an amino acid sequence of SEQ ID NO: 18064.

60. The nucleic acid of claim 59, wherein the ectodomain of (a) further comprises a signal peptide.

61. The nucleic acid of claim 59, wherein the ectodomain of (a) further comprises a hinge between the antigen recognition region and the transmembrane domain.

62. The nucleic acid of claim 60, wherein the signal peptide is a human CD8 α signal peptide.

63. The nucleic acid of claim 59, wherein the transmembrane domain comprises a human CD8 α transmembrane domain.

64. The nucleic acid of claim 59, wherein the endodomain comprises a human CD3 ζ endodomain.

65. The nucleic acid of claim 59, wherein the at least one costimulatory domain comprises an amino acid sequence of a human 4-1 BB costimulatory domain.

66. The nucleic acid of claim 59, wherein the hinge comprises an amino acid sequence of a human CD8 α hinge.

67. The nucleic acid of claim 59, wherein the VH comprises the amino acid sequence of SEQ ID NO: 18051.

68. The nucleic acid of claim 59, wherein the CAR comprises the amino acid sequence of SEQ ID NO: 18006.

69. The nucleic acid of claim 59 comprising the sequence of SEQ ID NO: 18007.

70. A transposon comprising the nucleic acid of claim 59.

71. The transposon of claim 70, wherein the transposon is a piggyBac transposon.

72. The nucleic acid of claim 68, wherein the polynucleotide further comprises an exogenous gene that confers resistance to a deleterious compound.

73. The nucleic acid of claim 72, wherein the gene is a Dihydrofolate Reductase (DHFR) gene.

74. The nucleic acid of claim 68, wherein the polynucleotide further encodes an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

75. The nucleic acid of claim 73, wherein the polynucleotide further encodes an exogenous inducible proapoptotic polypeptide, wherein the inducible proapoptotic polypeptide comprises a truncated caspase 9 polypeptide.

76. A transposon comprising the nucleic acid of claim 73.

77. The transposon of claim 76, wherein the transposon is a piggyBac transposon.

78. A transposon comprising the nucleic acid of claim 75.

79. The transposon of claim 78, wherein the transposon is a piggyBac transposon.

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